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Life Cycle Inventories of Building Products

Data v2.0 (2007)



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Comments on this version 2.0

This part corresponds to the original report (Werner et al. 2003) that has been updated with the changes reported in (Frischknecht et al. 2006) and augmented with various building products and components such as window frames, doors, claddings. The report consists of the following parts:

Part I	Gravel and Sand Products and Processes
Part II	Cement Products and Processes
Part III	Concrete Products and Processes
Part IV	Clay Products and Processes
Part V	Refractory Bricks
Part VI	Ceramics
Part VII	Lime Products and Processes
Part VIII	Gypsum Products and Processes
Part IX	Fibre Cement Products and Processes
Part X	Sand-Lime Brick
Part XI	Flat Glass Products and Processes
Part XII	Glass Products and Processing for Solar Collectors
Part XIII	Glass Fibre Reinforced Plastic Products
Part XIV	Insulation Products and Processes
Part XV	Elastomere Tube Insulation
Part XVI	Cork
Part XVII	Mortar and Plaster Products and Processes
Part XVIII	Additional Products and Processes
Part XIX	Bentonite Products and Processes
Part XX	Synthetic rubber (EPDM)
Part XXI	Lightweight Products and its Preliminary Products and Processes
Part XXII	Natural Stone Plate
Part XXIII	Bituminous Products and Processes
Part XXIV	Titanium Zinc Plate
Part XXV	Tin plating
Part XXVI	Glazing, Window frames, Claddings and Doors

Part I

Gravel and Sand Products and Processes

Data v2.0 (2007)

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Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

This part deals with the mining process and its products sand and round gravel and with the products crushed gravel and unspecified gravel. Unspecified gravel is modelled as a consumption mix of round and crushed gravel. Silica sand, mainly used as raw material for the flat glass production is included as well. For the infrastructure of all products an average gravel/sand mine has been inventoried. The data are provided by four representative Swiss gravel pits.

Round gravel and sand are co-produced in the same process. Burdens are allocated by mass, thus the inventories are the same. Crushed gravel has a higher energy demand and higher emissions because additional to the quarrying process a crushing process is included.

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1 Introduction

The study Künniger et al. (2001) summarises the results of an environmental Life Cycle Analysis carried out for gravel, cement and concrete produced and used in Switzerland. Results are based on production data from 1997 including four gravel plants, eight cement works and six concrete plants. The goal of the study was an inventory and assessment of all in- and output-material flows from and to nature which are related to the production of gravel, cement and concrete

The report of Künniger et al. (2001) has been used in 4 reports in this study: “Gravel products and processes”, “Cement products and processes”, “Concrete products and processes” and “Refractory Bricks”. The present report describes the mining of gravel and its related processes.

The following introduction is based on EPA (1998a) and adapted to the Swiss situation:

Deposits of sand and gravel, the unconsolidated granular materials resulting from the natural disintegration of rock or stone, are generally found in near-surface alluvial deposits and in subterranean and sub aqueous beds. Sand and gravel are products of the erosion of rock and occur in nature as unconsolidated or poorly consolidated materials. They consist of siliceous and calcareous components.

Construction sand and gravel are made up of varying amounts of different rock types and are, therefore of varying chemical compositions. Silica, or silicon dioxide (SiO_2), is the major constituent of commercial sands. Further contents are feldspar, mica, iron oxides, and heavy minerals. The difference of industrial sand, often called silica sand and industrial gravel to construction sand and gravel is the high percentages (typically 95 to 99 percent) of quartz or silica in industrial materials.

The results of the process mining/digging of sand are on the one hand round gravel and on the other hand sand and therefore it is handled as a multi-output process.

Silica sand is special sand with a high percentage of quartz.

2 Reserves and Resources

The most important resources from nature for the extraction of sand and gravel in Switzerland are post glacial and late quaternary river deposits. Also other types of occurrences are suitable for mining, such as debris of streams or landslide deposits. Depending on the use and type of unconsolidated deposits, an expensive preparation and a great amount of mud from washing can result.

About 400 gravel pits with a yearly mining amount of 35 Mio tons of gravel and brash existed in Switzerland in 2000 (Künniger et al. (2001)). This amount includes gravel, brash, recycling and import materials. It's equal to about 20 Mio. m^3/a . The economical value is approximately 1.2 billion CHF. This production level represents about 80% of the annual consumption in Switzerland. The remaining 20% are imported from the European neighbouring countries (Germany, France, Italy, and Austria).

About 55% of the 400 gravel pits are small with a yearly production volume of less than 50'000 m^3 of gravel. Only about 6 to 7% can be called “big” as they have a yearly production volume of more than 200'000 m^3 .

Construction sand and gravel is produced in every cantonal state. The leading cantons are, in descending order of mining capacity: Zurich (3.1 Mio. m^3/a), Argovia (2.7 Mio. m^3/a), St. Gall (2 Mio. m^3/a), Vaud, Valais and Fribourg (1.5 Mio. m^3/a), Grisons (1.2 Mio. m^3/a), Geneva, Neuchâtel and Solothurn (0.8 Mio. m^3/a), Thurgovia, Nidwalden and Zug (0.5 Mio. m^3/a) (Kündig et al. (1997)).

Important silica sand deposits are found in Germany: Niedersachsen, Westfalen and Niederrhein (Bach et al. (1999)).

For the perspective of the mining industry resources can be regarded as quarryable if mining is possible with actual technology, the enterprise realisable with supportable financial and temporal expenditure and if resource exploitation does not collide with other interests. This last point causes a lot of problems in Switzerland as in its intensely populated part (middle Lands) all different kinds of use of natural resources collide. These are for example water protection, settlement area etc. This circumstance reduces the real quarryable area substantially.¹

For larger countries as Germany, France or Italy the mentioned scenario can also be applicable but for longer time period.

3 Characterisation

Construction sand and gravel are composed of varying amounts of different rock types and are therefore of varying chemical composition. Silica, or silicon dioxide (SiO_2), is the major constituent of commercial sands and gravel.

To define the chemical quality it is often easier to describe the components with their effects. For sand and gravel the type and quantity of components, which can be harmful for the further use are known and some examples listed in Tab. 3.1:

Tab. 3.1 Harmful components in gravel and sand, depending on type of use

Components	Effect
Humus	Hinders the hydration of the cement in concrete or mortar.
Chlorides	Have a corroding effect on steel reinforcements.
Coal	Change colour
Mica, clay, wood, vegetable matter	Chemical and physical effects

The grain shape, the surface and the grain size are the most important characterisation indicators for the further use of gravel and sand (see Tab. 3.2). The distribution of grain size can strongly vary in natural sand and gravel deposits.

The distribution of grain size, humidity and bulk density can vary strongly in natural sand and gravel deposits. The bulk density ranges between 1'300 and 2'000 kg/m³, depending on the water content which is between 0 and 17% (w/w) (Kündig et al. (1997)). The maximum content of water means that the sand is actually soaked.

Silica sand is special sand with a SiO_2 (Quarz) -content of more than 85% (w/w)². If the SiO_2 -content is higher than 98% the silica sand is called pure silica sand or industrial sand. Pure silica sand with a SiO_2 -content over 99% and heavy crop content (especially iron compounds) under 0.05% is called "vitreous" sand.

¹ http://www.sgtk.ethz.ch/100jahresgtk/meilensteine/abbaubare_rohstoffe.html, access date: March 2003

² <http://www.euroquarz.de/>, access date: 30.05.03

Tab. 3.2 Characterisation of gravel and sand by grain form and size ³

Material	Description
Sand	Size ≤ 4 mm
Crushed	Edged, rough, Size 0-4 mm
Round	Round, smooth, Size 0-4 mm
Superfine	Size ≤ 0.25 mm
Gravel	Size ≥ 4 mm
Stone chips (grit)	Edged, rough, Size 4-32 mm
Crushed stone (ballast)	Edged, rough, Size > 32 mm
Gravel crushed	Edged, rough, Size ≥ 4 mm
Gravel round	Round, smooth, Size ≥ 4 mm

4 Use

The largest use of construction sand and gravel is as aggregate for the production of concrete. The second largest use is as foundation material for highways, railways, runways, etc. Other major uses include aggregate for hot mix asphalt, and filler for highway, dam, and other back-filling. Smaller volumes of construction sand and gravel are used in plaster and gunite sands, snow and ice control, and railroad ballast.

The use of silica sand is subdivided in two categories (Bach et al. (1999)):

1. **Building sand:** Used as aggregate for concrete, mortar, plaster, dumping, non-freezing, used for the production of sand-lime bricks and as paring and brake sand.
2. **Industrial sand:** Glass-, grinding-, enamel-, oil-, moulding- and filter sand

Beside the use in the building sector silica sand is also used in the beverage industry, as raw material for the chemical industry, for the production of refractories, as filling material or for the covering of playing fields.

5 Process Description

The description of EPA (1998b) has been adapted to the situation in Switzerland.

Sand and gravel are typically mined in moist or wet conditions by open pit. Open pit excavation is carried out with power shovels, draglines, front end loaders and bucket wheel excavators.

After mining, the materials are transported to the processing plant by earth mover, truck, belt conveyors or other means. Although significant amounts of sand and gravel are used for fill, bedding, sub base, and base course without processing, most domestic sand and gravel is processed prior to use. The processing of sand and gravel for a specific market involves the use of different combinations of washers, screens, and classifiers to segregate particle sizes; crushers to reduce oversized material; and storage and loading facilities. A process flow diagram for construction sand and gravel processing is presented in Fig. 5.1. The following paragraphs describe the process in more detail.

³ http://www.ch.skw-mbt.com/german/lexikon/index_p.htm, access date: March 2003

After being transported to the processing plant, the wet sand and gravel (raw feed) is stockpiled or emptied directly into a hopper, which is typically covered with a "grizzly" of parallel bars to screen out large cobbles and boulders. From the hopper, the material is transported to fix or vibrating scalping screens by gravity, belt conveyors, or bucket elevators. The scalping screens separate the oversize material from the smaller sizes. The oversize material may be directed to a crusher for size reduction, to produce crushed aggregate or to produce manufactured sands. Crushing generally is carried out in one or two stages, although three-stage crushing may also be performed. Following crushing, the material is returned to the screening operation for sizing. Alternatively, oversize material may be used for erosion control, reclamation, or other uses. The material that passes through the scalping screen is fed into a battery of sizing screens, which generally consist of horizontal or sloped, single or multi-deck vibrating screens. Water is sprayed onto the material throughout the screening process for dust control. After screening, the sized gravel is transported to stockpiles, storage silos, or, in some cases, to crushers by belt conveyors. Log washers free the sand from clay and organic impurities. After scrubbing, the sand is typically sized by water classification. Wet and dry screening is rarely used to size the sand. After classification, the sand is dewatered using screws, separatory cones, or hydroseparators. After processing, the sand is transported to storage silos or stockpiles by belt conveyors.

Fig. 5.1 describes the mining process in form of a flow scheme and Fig. 5.2 shows the system boundary for this study.

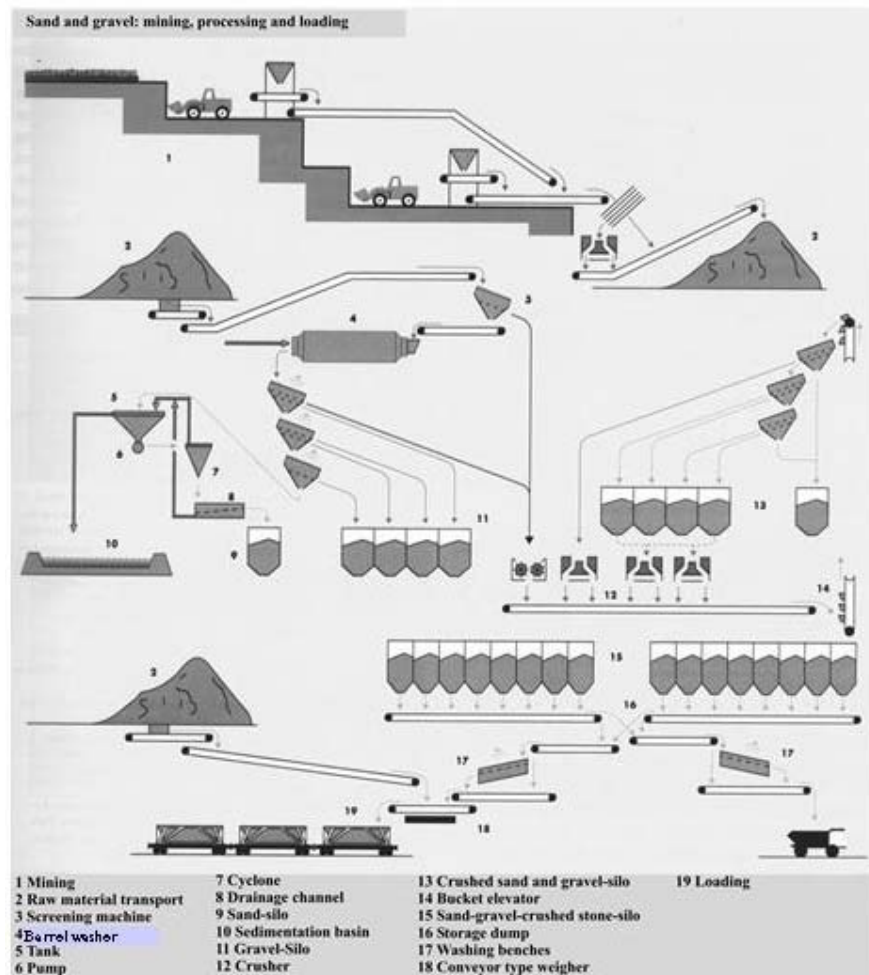


Fig. 5.1 Flow scheme of sand and gravel production (Kündig et al. (1997))

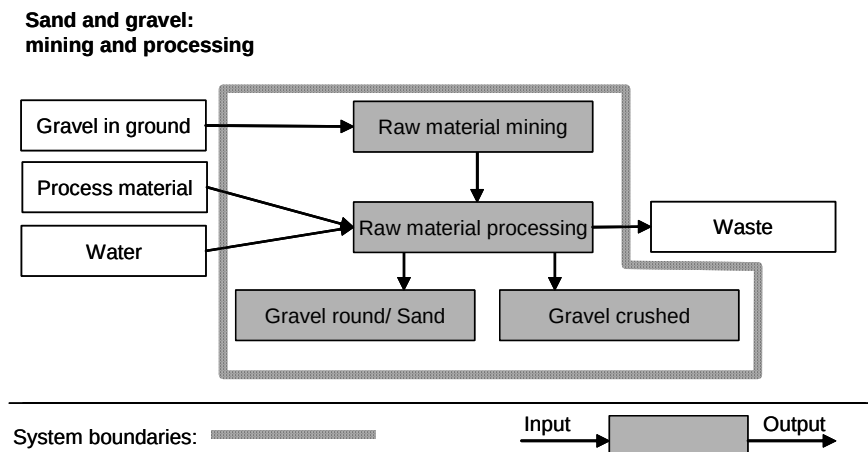


Fig. 5.2 System boundaries of the sand and gravel production for this study

Some general comments on the emissions are made in EPA (1998b):

Emissions from the production of sand and gravel consist primarily of particulate matter (PM) and PM less than 10 micrometers (PM-10) in aerodynamic diameter that are emitted by many operations at sand and gravel processing plants, such as conveying, screening, crushing, and storing operations. Generally, these materials are kept wet or moist when handled, and process emissions are often negligible. A substantial portion of these emissions may consist of heavy particles that settle out within the plant. Other potentially significant sources of PM and PM-10 emissions are haul roads. Emissions from dryers include PM and PM-10, as well as typical combustion products including CO, CO₂, and NO_x. In addition, dryers may be sources of volatile organic compounds (VOC) or sulphur oxides (SO_x) emissions depending on the type of fuel used to fire the dryer.

Generally there are no differences in the quarrying process of silica sand compared to the “normal” sand described above. An important difference is that silica sand is dried⁴. It is assumed that the content of moisture before drying is about 5 % (compare chapter 3) and afterwards around 1%. Another difference is based on the high quota of SiO₂, which makes the raw material rarer; therefore, additional conveyor belt between mine and processing plant are assumed.

6 Life cycle Inventories

Ecoinvent meta information (tables), flows and their representation in the ecoinvent database are shown in (figures):

Gravel/sand mine	Tab. 6.1 and Fig. 6.3
Mining of gravel/sand (products: “sand, at mine”, “gravel, round, at mine”)	Tab. 6.2 and Fig. 6.4
Gravel, crushed, at mine	Tab. 6.3 and Fig. 6.5
Gravel, unspecified, at mine	Tab. 6.4 and Fig. 6.6
Silica sand, at plant	Tab. 6.5 and Fig. 6.7

⁴ <http://www.glaskompetenzzentrum.de/inf020702.htm>, access date: 30.05.03

The data in this study of the process “mining of gravel/sand” with the two output modules “gravel, round, at mine” and “gravel, crushed, at mine” are based on inquiries of two medium and two big gravel pits in Switzerland. The difference in production of the two kinds of gravel is the crushing process which is very energy intensive. The previous washing and sieving process and the subsequent transports do not differ significantly. The energy consumption for the production of crushed gravel is about two and a half times as high as for the round gravel production.

The data are based on the investigation in 4 representative Swiss gravel pits. However data can vary considerably depending on mining place, size of plant and other conditions. This fact has been considered in the uncertainties.

The infrastructure is estimated on the basis of the process description in Fig. 6.1

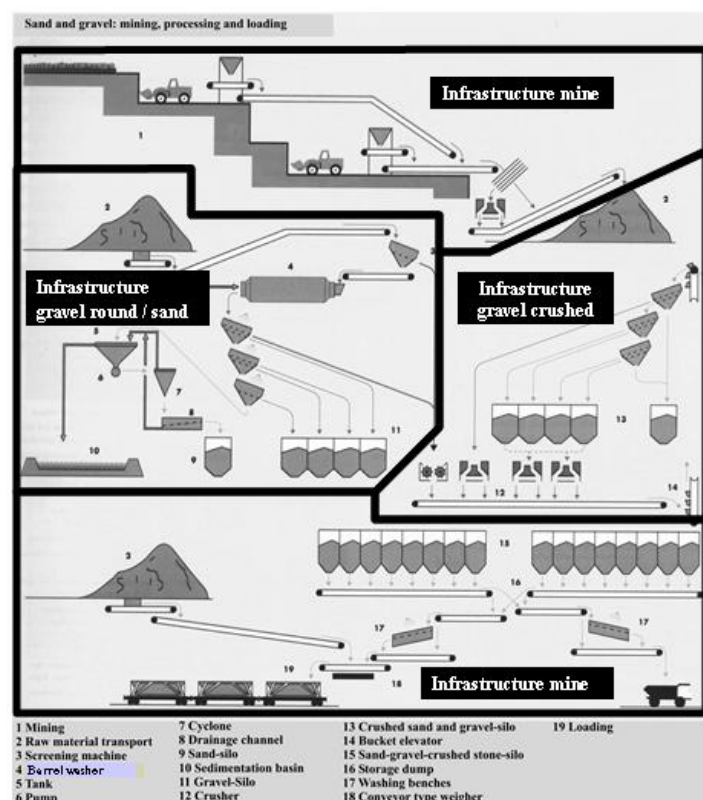


Fig. 6.1 Systems boundary for estimation of the different infrastructures (Kündig et al. (1997))

Künniger et al. (2001) treat crushed and round gravel as two single processes. Although they mainly have the same process steps (supply with raw material, material discharge, covering and recultivation, digging of gravel, internal transports). The difference is the washing and rounding process for round gravel and sand and the crushing process for crushed gravel. It is assumed that the four gravel plants in Switzerland offering their data are representative for the Swiss production share of round and crushed gravel. 79% of that total gravel production (without sand!) is round gravel and 21% is crushed gravel. The two products gravel round ($\geq 4\text{mm}$) and sand ($\leq 4\text{mm}$) are separated by a screening process. The total output gravel round and sand are allocated by weight with an estimated share of 35% sand and 65% gravel round⁵. With the total amount of gravel produced in 1997 in Switzerland, Fig. 6.2 show the masses calculated on the basis of the shares.

⁵ Estimation by T. Künniger, EMPA Dübendorf

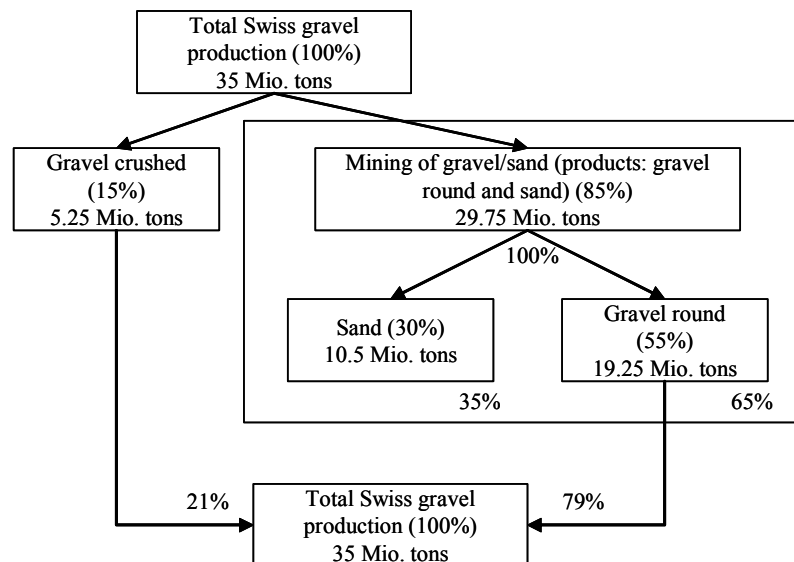


Fig. 6.2 Swiss production share of gravel round, gravel crushed and sand for the year 1997 ⁶

6.1 Mine gravel/sand

The mine includes the infrastructure excavation part of the mine and is therefore used for all gravel/sand modules (compare Fig. 6.1).

The life span of the portrayed mine is estimated to be about 50 years and its yearly production amount in the range 400'000 to 500'000 tons (--> large operation).

Machines in the Gravel/Sand Mine (excluding production specific machines):

As no exact information is available the weight of the different machines mostly stationed in the building halls are estimated based on the flow scheme (Fig. 6.1) and summed up:

1 sieve (assumption each 5t):	5t
20 „small“ silos (assumption each 7t):	140t
diverse machines:	25t
Total weight of machines:	170t

The service life for all machines is assumed to be 25 years.

The ecoinvent database meta information for “mine gravel/sand” includes all important and necessary information to use the data correctly. Fig. 6.3 shows all in- and output flows of the product “mine gravel/sand”, the allocation to the modules in the ecoinvent database, the sources (mainly Künniger et al. (2001)) and the standard deviation for each datum.

⁶ Estimation by T. Künniger, EMPA Dübendorf

Tab. 6.1 Ecoinvent database meta information for the product "mine, gravel/sand"

Name	mine, gravel/sand
Name	mine, gravel/sand
Location	CH
Infrastructure Process	1
Unit	unit
Data Set Version	2.0
Included Processes	Includes the area of the infrastructure (buildings, paved roads, etc.) and the machines used for the gravel mining which are in common for gravel and sand products. It does not include the area of the actual mine (quarry), which is accounted for in mining.
Amount	1
Local Name	Mine, Kies/Sand
Synonyms	pit//Grube//
General Comment to reference function	Data are estimated based on an air-picture for one mine and sketches of the process. The average yearly production volume per unit (mine) is about 400'000 to 500'000 t/a and the life time is assumed to be 50 years
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	The data are based on the investigation on 4 representative Swiss gravel pits; for some exchanges RER modules have been used as proxy
Technology text	Average Swiss gravel/sand mine
Representativeness [%]	100
Production Volume	Unknown
Sampling Procedure	Measured data of four Swiss plants
Extrapolations	See geography

General Flow information					Representation in ecoinvent 2000							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
infrastructure machines	→	gravel/ sand mine			Machines used in mine by all types of products (gravel round and crushed an sand); weight assumption: 170t; assumption service life: 25years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	3.40E+05	kg	estimation EMPA by air-picture and sketch of process	1	3.10	(3,5,1,2,1,5);
infrastructure: building hall				estimated area: 3'000m2; estimated servic life: 50 years	construction processes	buildings	Yes	CH	building, hall, steel construction	3.00E+03	m2	estimation EMPA by air-picture and sketch of process	1	3.10	(3,5,1,1,1,5);	
infrastructure: building multistore				estimated area: 520m2; estimated highth: 6m -> volume: 3'120m3; estimated servic life: 50 years	construction processes	buildings	Yes	RER	building, multi-storey	3.12E+03	m3	estimation EMPA by air-picture and sketch of process	1	3.10	(3,5,1,2,1,5);	
total land use (transformation)	→			sum of green part, raod part and building part of built area	resource	land			Transformation, from unknown	6.09E+04	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);	
land occupation for green part of build area	→			assumption: life span 50 years, average land-use: 42'667 m2	resource	land			Occupation, industrial area, vegetation	2.13E+06	m2a	Künniger et al. (2001)	1	1.57	(2,3,1,1,1,5);	
land transformation to green part of build area	→			average land-use: 42'667 m2	resource	land			Transformation, to industrial area, vegetation	4.27E+04	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);	
land occupation for road part of build area	→			assumption: life span 50 years, average land-use: 14'675 m2	resource	land			Occupation, traffic area, road network	7.34E+05	m2a	Künniger et al. (2001)	1	1.57	(2,3,1,1,1,5);	
land transformation to road part of build area	→			average land-use: 14'675 m2	resource	land			Transformation, to traffic area, road network	1.47E+04	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);	
land occupation for building part of build area	→			assumption: life span 50 years, average land-use: 3'515 m2	resource	land			Occupation, industrial area, built up	1.76E+05	m2a	Künniger et al. (2001)	1	1.57	(2,3,1,1,1,5);	
land transformation to building part of build area	→			average land-use: 3'515 m2	resource	land			Transformation, to industrial area, built up	3.52E+03	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);	
		→	gravel/sand mining plant		assumption: life span 50 years, production amount about 30 Mio. tons for the year 1997	construction materials	additives	Yes	CH	mine, gravel/sand	1.00E+00	unit				

Fig. 6.3 Flow for “mine gravel/sand” and their representation in the ecoinvent database

6.2 Mining of gravel/sand

As shown in Fig. 6.2 the part of round gravel and sand on the total amount of dug gravel is 85%. This is about 30 Mio. tons for the year 1997.

Machines used in the Production of Gravel/Sand:

Since no detailed information could be gathered, a rough estimation of the machines used for the production of gravel and sand is attempted here. As described in chapter 6.1, the part of the machines which belongs to the mining process itself is included in the module “gravel/sand mine”.

3 sieves (assumption each 5 t):	15t
5 „small“ silos (assumption each 7 t):	35t
1 barrel washer:	20t
Other machines	30t
Total weight of the machines:	100t

Length of conveyor belt: 1'000m

The service life of all these machines is assumed to be 25 years. Wearing parts and lubricating oil is taken into account. The total amount of material treated by these machines over 25 years is therefore about 743.75 Mio tons.

The total area of the mine is 12'000m², of which 3'000m² are included in the module “gravel/sand mine” and the remaining 9'000m² are accounted here.

For the process of recultivation the module “recultivation, limestone mine” has been used as proxy. The amount of diesel used for the recultivation is copied from the recultivation process of a limestone mine as it is assumed that it is similar.

No dust emissions are taken into account as the excavation is a wet process. Wastewater resulting from this process is not taken into account as the main content is particulate matter which is superfine sand and is only displaced. Therefore no negative effect on the ground water and soil is expected.

The ecoinvent database meta information of “mining of gravel/sand” and thus also for it's allocated co-products “sand, at mine” and “gravel, round at mine” (not “gravel, crushed, at mine”) include all important and necessary information to use the data correctly. Fig. 6.4 shows all in- and output flows of the process “mining of gravel/sand” with its output-modules “sand at mine” and “gravel, round, at mine”, the allocation to the modules in the ecoinvent database, the sources (mainly Künniger et al. (2001)) and the standard deviation for each datum.

Tab. 6.2 Ecoinvent database meta information for "mining of gravel/sand"

Name	mining, gravel / sand
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process, internal processes (transport, etc.) and infrastructure. The land-use of the mine (incl. unpaved roads) is included directly, the land-use of the paved roads and buildings are included in the module "mine, gravel/sand". Recultivation is taken into account. No loads from administration are included.
Amount	Includes the whole manufacturing process for digging of gravel round and sand (no crushed gravel), internal processes (transport, etc.), and infrastructure for the operation (machinery). The land-use of the mine (incl. unpaved roads) is included directly, while the land-use of the paved roads and buildings are included in the module "mine, gravel/sand". Recultivation of closed mines is taken into account. No environmental burdens from administration are included. No dust included because it is mostly a "wet" process and no wastewater included because process water is not polluted (only sand and gravel) and therefore directly seeped.
Local Name	1
Synonyms	Abbau Kies / Sand
General Comment to reference function	Rundkies//Sand//Zuschlagstoff//gravel round//sand//aggregate
Start Date	
End Date	1997
Data Valid For Entire Period	2001
Other Period Text	1
Geography text	
Technology text	For some exchanges RER-modules have been used as proxy
Representativeness [%]	Typical technology for Swiss production
Production Volume	100
Sampling Procedure	29750000 t/a
Extrapolations	Measured data of four Swiss plants
Uncertainty Adjustments	See geography

General Flow information					Representation in ecoinvent 2000							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
gravel, in ground	→	mining of gravel/sand (Input, excl. land-use)			4% of losses in process water, goes back in mine	resource	in ground			Gravel, in ground	1.04E+00	kg	Künniger et al. (2001)	1	1.14	(2,3,1,1,1,4);
water	→				process water	resource	in water			Water, unspecified natural origin	1.38E-03	m3	Künniger et al. (2001)	1	1.14	(2,3,1,1,1,4);
diesel, burned in building machine	→				excavation, mining sand gravel, covering-recultivation, internal transport	construction processes	machinery	No	GLO	diesel, burned in building machine	1.47E-02	MJ	Künniger et al. (2001)	1	1.14	(2,3,1,2,1,4);
transport, lorry 16t	→				transport of process materials (50 km) (compare remarks on other in- and outputs) and removal of waste (20 km)	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	8.80E-07	tkm	Künniger et al. (2001)	1	2.03	(3,3,1,1,1,4);
transport, lorry 28t	→				primary fuel 50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	1.72E-05	tkm	calculated	1	2.02	(2,3,1,1,1,4);
light fuel oil	→				heating of production plant	oil	heating systems	No	CH	heat, light fuel oil, at boiler 10kW, non-modulating	2.44E-03	MJ	Künniger et al. (2001)	1	1.14	(2,3,1,1,1,4);
lubricating grease and oil	→				process material; lubricating grease and oil	chemicals	organics	No	RER	lubricating oil, at plant	1.85E-06	kg	Künniger et al. (2001)	1	1.14	(2,3,1,2,1,4);
electricity, swiss supply mix	→				electricity required by the complete production	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.72E-03	kWh	Künniger et al. (2001)	1	1.14	(2,3,1,1,1,4);
transport, van <3.5t	→				internal transport with van (0.00008 Fzkm with fuel in van with catalytic converter and 0.00002 Fzkm in conventional van)	transport systems	road	No	CH	transport, van <3.5t	1.55E-05	tkm	Künniger et al. (2001)	1	2.02	(2,3,1,1,1,4);
steel low-alloyed	→				process material; wearing parts (metal)	metals	extraction	No	RER	steel, low-alloyed, at plant	1.30E-05	kg	Künniger et al. (2001)	1	1.14	(2,3,1,2,1,4);
synthetic rubber, at plant	→				process material; wearing parts (no metal)	plastics	polymers	No	RER	synthetic rubber, at plant	2.00E-06	kg	Künniger et al. (2001)	1	1.14	(2,3,1,2,1,4);
tap water, at user	→					water supply	production	No	RER	tap water, at user	1.01E-02	kg	Künniger et al. (2001)	1	1.14	(2,3,1,2,1,4);
conveyor belt	→				estimated total length: 1'000m; 85% for transportation of crushed gravel, estimated service life: 25years, average yearly production amount: 85% of 420'645 tons	construction processes	machinery	Yes	RER	conveyor belt, at plant	9.51E-08	m	estimation EMPA	1	3.28	(5,4,1,2,1,5);
additional machines	→				Infrastructure of specific machines; weight assumption: 100t; assumption service life: 25years, average yearly yield of mine: 85% of 420'645 t/a	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.12E-05	kg		1	3.10	(3,5,1,2,1,5);
infrastructure	→				85% of total unit mine for sand / gravel round	construction materials	additives	Yes	CH	mine, gravel/sand	4.75E-11	unit	Künniger et al. (2001)	1	3.05	(1,3,1,1,1,5);
additional infrastructure: building hall	→				estimated total area: 12'000m2; already included in infrastructure module "mine gravel/sand": 3'000m2, rest: 9'000m2; estimated service life: 50 years, average yearly yield of mine: 85% of 420'645 t/a	construction processes	buildings	Yes	CH	building, hall, steel construction	5.03E-07	m2		1	3.10	(3,5,1,1,1,5);

Fig. 6.4 Flows for “mining of gravel/sand” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent 2000							Uncertainty information						
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
land occupation mineral extraction site	→	mining of gravel / sand (Input only land-use and Output)				assumption: After 10 years the used land is recultivated, average land-use: 0.288 m2/t	resource	land			Occupation, mineral extraction site	2.88E-03	m2a	Künniger et al. (2001)	1	1.57	(2,3,1,1,1,5);	
land transformation to mineral extraction site	→					average land-use:0.288 m2/t	resource	land			Transformation, to mineral extraction site	2.88E-04	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);	
land transformation mine from unknown	→					average land-use: 0.288 m2/t	resource	land			Transformation, from unknown	2.88E-04	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);	
land transformation to mud pool	→					average land-use: 0.063 m2/t	resource	land			Transformation, to water bodies, artificial	6.27E-05	m2	Künniger et al. (2001)	1	1.57	(2,3,1,1,1,5);	
land transformation mud pool from unknown	→					average land-use:0.063 m2/t	resource	land			Transformation, from unknown	6.27E-05	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);	
land occupation mud pool	→					assumption: After 10 years the used land is recultivated, average land-use: 0.063 m2/t	resource	land			Occupation, water bodies, artificial	6.27E-04	m2a	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);	
recultivation [m2]	→					“recultivation, limestone mine” has been used as proxy, an important difference is the amount of transformation to a certain area which is just left to nature, diesel consumption for recultivation is included in module “diesel burned in building machine”	construction materials	additives	No	CH	recultivation, limestone mine	8.48E-05	m2	Künniger et al. (2001)	1	2.11	(3,3,1,1,3,5);	
						Same amount as electricity used	air	unspecified			Heat, waste disposal, used mineral oil, 10% water, to hazardous waste incineration	9.77E-03	MJ		1	1.14	(2,3,1,2,1,4);	
	→			Waste heat used oil to municipal incineration				waste management	hazardous waste incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.85E-06	kg	Künniger et al. (2001)	1	1.14	(2,3,1,1,1,4);
	→			waste to municipal incineration			different waste as synthetic rubber, plastics, etc.; disposal of municipal solid waste used as proxy	waste management	municipal incineration	No	CH		2.77E-06	kg	Künniger et al. (2001)	1	1.14	(2,3,1,1,1,4);
	→		sand, at mine			Allocation factor 35%	construction materials	additives	No	CH	sand, at mine	3.50E-01	kg	Künniger et al. (2001)				
	→		gravel, round, at mine			Allocation factor 65%	construction materials	additives	No	CH	gravel, round, at mine	6.50E-01	kg	Künniger et al. (2001)				

Fig. 6.4 Flows for "mining of gravel/sand" and their representation in the ecoinvent database (cont)

6.3 Gravel, crushed, at mine

The ecoinvent database meta information of “gravel, crushed, at mine” includes all important and necessary information to use the data correctly. Fig. 6.5 shows all in- and output flows of the product “gravel, crushed, at mine”, the allocation to the modules in the ecoinvent database, the sources (mainly Künniger et al. (2001)) and the standard deviation for each datum.

As shown in Fig. 6.2 the part of crushed gravel on the total amount of dug gravel is 15%. This is about 5.25 Mio. tons for the year 1997.

Machines used for “gravel, crushed, at mine”:

Since no detailed information could be gathered, a rough estimation of the machines used for the production of crushed gravel is attempted here. As described in chapter 6.1, the part of the machines which belongs to the mining process itself is included in the module “gravel/sand mine”.

4 crushers (each 25t):	100t
3 sieves (assumption each 5t):	15t
5 „small“ silos (assumption each 7t):	35t
Total weight of the machines:	150t
Length of conveyor belt:	1'000m

The service life of all these machines is assumed to be 25 years. Wearing parts and lubricating oil is taken into account. The total amount of material treated by these machines over 25 years is therefore about 131.25 Mio tons.

The total area of the mine is 12'000m², of which 3'000m² are included in the module “gravel/sand mine” and the remaining 9'000m² are accounted here.

The data collected on the dust production refers to total release. The partition to the different categories has been overtaken from the process “mining of limestone”. This distribution is used as the basis for most mining/digging processes as it is assumed that differences among them are marginal. The total of the dust emissions for excavating and processing crushed gravel is smaller than the total of the limestone mining process because no blasting is needed.

No wastewater has been taken into account because it has been assumed that the content is mainly superfine sand which has no negative effect on the ground water and soil.

Tab. 6.3 Ecoinvent database meta information for the product “gravel, crushed, at mine”

Name	gravel, crushed, at mine
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process, internal processes (transport, etc.) and infrastructure (machinery) specifically used for this product. No administration is included. Dust emissions included. No wastewater included because process water is not polluted (only sand and gravel) and therefore directly seeped. Recultivation of closed mines is taken into account.
Amount	1
Local Name	Kies gebrochen, ab Abbau
Synonyms	Kies gebrochen//Zuschlagstoff//gravel crushed//aggregate
General Comment to reference function	From the total amount (100%) of mined gravel round, crushed and sand, about 15% is crushed gravel. Dust emission size shares (for particulates >PM10, 2.5<PM<10, and <PM2.5) are assumed equal to "mining limestone" process.
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Typical technology for Swiss production
Representativeness [%]	100
Production Volume	5250000 t/a
Sampling Procedure	Measurement data of four Swiss plants
Extrapolations	See geography
Uncertainty Adjustments	None

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
gravel, in ground	→	gravel, crushed, at mine (part 1)				resource	in ground			Gravel, in ground	1.04E+00 kg		Künniger et al. (2001)	1	1.22	(2,3,1,2,1,5);	
water	→					process water	resource	in water			Water, unspecified natural origin	1.35E-03 m3		Künniger et al. (2001)	1	1.22	(2,3,1,1,1,5);
diesel, burned in building machine	→				includes mining of sand and gravel, covering-recultivation and internal transport	construction processes	machinery	No	GLO	diesel, burned in building machine	1.43E-02 MJ		Künniger et al. (2001)	1	1.22	(2,3,1,2,1,5);	
transport, lorry 16t	→				process materials 50 km, transport (removal) of waste 20 km	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	2.92E-06 tkm		Künniger et al. (2001)	1	2.06	(3,3,1,1,1,5);	
transport, lorry 28t	→				primary fuel 50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	1.67E-05 tkm		calculated	1	2.05	(2,3,1,1,1,5);	
light fuel oil	→				heating of production plant	oil	heating systems	No	CH	heat, light fuel oil, at boiler 10kW, non-modulating	4.91E-03 MJ		Künniger et al. (2001)	1	1.22	(2,3,1,1,1,5);	
lubricating grease and oil	→				process material; lubricating grease and oil	chemicals	organics	No	RER	lubricating oil, at plant	2.50E-06 kg		Künniger et al. (2001)	1	1.22	(2,3,1,2,1,5);	
electricity, swiss supply mix	→				complete production	electricity	supply mix	No	CH	electricity, medium voltage, at grid	9.06E-03 kWh		Künniger et al. (2001)	1	1.22	(2,3,1,1,1,5);	
transport, van <3.5t	→				internal transport with van (0.00008 Fzkm with fuel in van with catalytic converter and 0.00002 Fzkm in conventional van)	transport systems	road	No	CH	transport, van <3.5t	1.55E-05 tkm		Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);	
steel low-alloyed	→				process material; wearing parts (metal)	metals	extraction	No	RER	steel, low-alloyed, at plant	5.10E-05 kg		Künniger et al. (2001)	1	1.22	(2,3,1,2,1,5);	
synthetic rubber, at plant	→				process material; wearing parts (no metal)	plastics	polymers	No	RER	synthetic rubber, at plant	4.00E-06 kg		Künniger et al. (2001)	1	1.22	(2,3,1,2,1,5);	
tap water, at user	→						water supply	production	No	RER	tap water, at user	1.22E-02 kg		Künniger et al. (2001)	1	1.22	(2,3,1,2,1,5);
conveyor belt					estimated total length: 1'000m; 15% for transportation of crushed gravel, estimated service life: 25years, average yearly production amount: 15% of 420'645 tons	construction processes	machinery	Yes	RER	conveyor belt, at plant	9.51E-08 m		estimation EMPA	1	3.28	(5,4,1,2,1,5);	
recultivation [m2]	→				"recultivation, limestone mine" has been used as proxy, an important difference is the amount of transformation to a certain area which is just left to nature, diesel consumption for recultivation is included in module "diesel burned in building machine"	construction materials	additives	No	CH	recultivation, limestone mine	8.48E-05 m2		Künniger et al. (2001)	1	2.06	(3,3,1,1,1,5);	

Fig. 6.5 Flows for “gravel, crushed, at mine” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent 2000							Uncertainty information						
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
land occupation mine	→	gravel, crushed, at mine (part 2)				assumption: After 10 years the used land is recultivated, average land-use: 0.288 m2/t	resource	land			Occupation, mineral extraction site	2.88E-03	m2a	Künniger et al. (2001)	1	1.57	(2,3,1,1,1,5);	
land occupation mud pool	→					assumption: After 10 years the used land is recultivated, average land-use: 0.063 m2/t	resource	land			Occupation, water bodies, artificial	6.27E-04	m2a	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);	
land transformation to mineral extraction site	→					average land-use: 0.288 m2/ t	resource	land			Transformation, to mineral extraction site	2.88E-04	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);	
land transformation from unknown (to mineral extraction)	→					average land-use: 0.288 m2/ t	resource	land			Transformation, from unknown	2.88E-04	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);	
land transformatin from unknown (to mud pool)	→					average land-use: 0.063 m2/ t	resource	land			Transformation, from unknown	6.27E-05	m2	Künniger et al. (2001)	1	1.57	(2,3,1,1,1,5);	
land transformation to mud pool	→					average land-use: 0.063 m2/ t	resource	land			Transformation, to water bodies, artificial	6.27E-05	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);	
infrastructure	→					15% of total unit mine for gravel crushed	construction materials	additives	Yes	CH	mine, gravel/sand	4.75E-11	unit	Künniger et al. (2001)	1	3.05	(1,3,1,1,1,5);	
additional machines	→					weight of additional machines: 150t	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	9.51E-05	kg		1	3.10	(3,5,1,2,1,5);	
additional infrastructure: building hall	→					estimated total area: 12'000m2; already included in infrastructure module "mine gravel/sand": 3'000m2, rest: 9'000m2; estimated service life: 50 years, average yearly yield of mine: 15% of 420'645 t/a Same amount as electricity used	construction processes	buildings	Yes	CH	building, hall, steel construction	2.85E-06	m2		1	3.10	(3,5,1,1,1,5);	
	→			waste heat			air	unspecified			Heat, waste	3.26E-02	MJ		1	1.14	(2,3,1,2,1,4);	
	→			used oil to municipal incineration				waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	2.50E-06	kg	Künniger et al. (2001)	1	1.14	(2,3,1,1,1,4);
	→			waste to municipal incineration				waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	2.12E-06	kg	Künniger et al. (2001)	1	1.14	(2,3,1,1,1,4);
	→			dust to air [kg]				air	low population density			Particulates, > 10 um	5.60E-09	kg	Künniger et al. (2001)	1	1.83	(2,3,1,1,4,5);
	→			dust to air [kg]				air	low population density			Particulates, < 2.5 um	4.00E-10	kg	Künniger et al. (2001)	1	3.28	(2,3,1,1,4,5);
	→			dust to air [kg]				air	low population density			Particulates, > 2.5 um, and < 10um	2.00E-09	kg	Künniger et al. (2001)	1	2.28	(2,3,1,1,4,5);
	→			gravel, crushed, at mine				construction materials	additives	No	CH	gravel, crushed, at mine	1.00E+00	kg	Künniger et al. (2001)			

Fig. 6.5 Flows for "gravel, crushed, at mine" and their representation in the ecoinvent database (cont.)

6.4 Gravel, unspecified, at mine

The ecoinvent database meta information of “gravel, unspecified, at mine” includes all important and necessary information to use the data correctly. Fig. 6.6 shows all in- and output flows of the product “gravel, unspecified, at mine”, the allocation to the modules in the ecoinvent database, the sources (mainly Künniger et al. (2001)) and the standard deviation for each datum.

The module “gravel, unspecified, at mine” describes the Swiss mix of gravel types’ production. The ratio of “gravel, round, at mine” to “gravel, crushed, at mine” is based on the production amounts of the four Swiss quarries: 79% round gravel and 21% crushed gravel.

Tab. 6.4 Ecoinvent database meta information for the product “gravel, unspecified, at mine”

Name	gravel, unspecified, at mine
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	The module merges gravel crushed and gravel round to give average gravel. Includes the whole manufacturing process, internal processes (transport, etc.) and infrastructure. No administration is included. Recultivation of closed mines is taken into account.
Amount	1
Local Name	Kies, unspezifisch, ab Abbau
Synonyms	Kies//Zuschlagstoff//gravel//aggregate
General Comment to reference function	Mix of round and crushed gravel, allocation based on production of 4 swiss gravel plants: 79% round gravel and 21% crushed gravel
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Typical technology for Swiss production
Representativeness [%]	100
Production Volume	24500000 t/a
Sampling Procedure	Measured data of four Swiss plants
Extrapolations	See geography
Uncertainty Adjustments	None

6.5 Silica sand, at mine

The ecoinvent meta information of “silica sand, at mine” includes all important and necessary information to use the data correctly. Fig. 6.7 shows all in- and output flows of the product “silica sand”, the allocation to the modules in the ecoinvent database, the sources (Kasser & Pöll (1998)) and the standard deviation for each datum.

Tab. 6.5 Ecoinvent meta information for the product "silica sand, at mine"

Name	silica sand, at plant
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the raw material "sand, at plant", a certain additional amount of conveyor belt and the energy for drying the sand. No requirements for administration are included.
Amount	Includes the raw material "sand, at plant", a certain additional amount of conveyor belt and the energy for drying the sand. No requirement for administration is included.
Local Name	1
Synonyms	Quarzsand, ab Werk
General Comment to reference function	
Start Date	
End Date	1998
Data Valid For Entire Period	2001
Other Period Text	1
Geography text	
Technology text	The amount of energy used for drying sand is a Swiss data.
Representativeness [%]	Typical technology for Swiss production
Production Volume	
Sampling Procedure	Unknown
Extrapolations	Literature and estimations
Uncertainty Adjustments	See geography

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
gravel, round, at mine	→	gravel, unspecified, at plant				allocation based on total production of 4 Swiss gravel plants	construction materials	additives	No	CH	gravel, round, at mine	7.90E-01	kg	Künniger et al. (2001)	1	1.22	(2,3,1,1,1,5);
gravel, crushed, at mine	→					allocation based on total production of 4 Swiss gravel plants	construction materials	additives	No	CH	gravel, crushed, at mine	2.10E-01	kg	Künniger et al. (2001)	1	1.22	(2,3,1,1,1,5);
mine								construction materials	additives	No	CH	gravel, unspecified, at mine	1.00E+00	kg	Künniger et al. (2001)		

Fig. 6.6 Flows for “gravel, unspecified, at mine” and their representation in the ecoinvent database

General Flow information						Representation in ecoinvent 2000							Uncertainty information			
Input		Process Name		Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
sand, at mine	→	silica sand				construction materials	additives	No	CH	sand, at mine	1.04E+00	kg		1	1.63	(5,5,1,3,3,3);
heating energy	→			for drying of sand from 5 to 1% of water	oil	heating systems	No	CH	heat, light fuel oil, at industrial furnace 1MW	2.00E-01	MJ	Kasser & Pöll (1998)	1	1.23	(2,3,1,1,3,3);	
						construction materials	additives	No	DE	silica sand, at plant	1.00E+00	kg				

Fig. 6.7 Flows for “silica sand, at mine” and their representation in the ecoinvent database

7 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

Concerning the excavation and treatment processes producing “sand, at mine”, “gravel, round, at mine” and “gravel, crushed, at mine”, all flows except “conveyor belt” are based on Künniger et al. (2001). Therefore the only adaptations which were made apply to the geographical correlation between the location of the assigned ecoinvent module and the Swiss location of the gravel/sand modules.

Because of lack of information, most additional data for silica sand are based on estimations. Since mining is almost the same as quarrying “normal” sand, only the additional data have a quite high uncertainty, the overall quality can be assumed to be fair. Nevertheless the module should not be used to compare different materials among themselves. In a larger inventorying context and if the importance of these products is not too high compared to other inventoried items the data may be used as such.

8 Abbreviations

TFB	Technical Research and Consulting on Cement and Concrete (Technische Forschung und Beratung für Zement und Beton)
EMPA	Swiss Federal Laboratories for Materials Testing and Research (Eidgenössische Materialprüfungs- und Forschungsanstalt)
ETHZ	Swiss Federal Institute of Technology Zurich (Eidgenössische Technische Hochschule Zürich)

9 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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Part II

Cement Products and Processes

Data v2.0 (2007)

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Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

This part deals with five different cement products and unspecified cement which is composed of the five specific cements based on a actual consumption mix in Switzerland. Besides cement, clinker as main raw material, cement plant and the packing process are inventoried in this report. Other materials which are dealt with in this report are calcareous marl as important raw material for clinker and hydraulic lime as main raw material for the production of plastering mortar.

The most energy intense sub-process in the cement production is clinkering. The energy is covered by different fossil fuels and secondary fuels like meat-and-bone meal, animal fat, used oil and tyres etc. These secondary fuels enter the system without environmental burdens since they are declared waste from other processes. The used amounts and the heating values are reported but not included in the inventory. The emissions are included as total because the provided data did not allow a separation.

Besides the main primary raw materials as limestone, calcareous marl and clay which are needed for the clinker production, also secondary raw materials are used. These are for example decarbonation sludge, lime mud from limestone production, diverse ashes from paper production plants, etc. These materials too enter the system without environmental burdens since they are declared waste.

The results of the cumulated energy demand (CED) for cement are dominated by the energy needed in the clinkering process. An important part of the emissions is dust emitted by the limestone mining process.

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1 Introduction

The study Künniger et al. (2001) summarises the results of an environmental Life Cycle Analysis carried out for gravel, cement and concrete produced and used in Switzerland which are based on production data from 1997 including four gravel plants, eight cement works and six concrete plants. The goal of the study was an inventory and assessment of all in- and output-material flows from and to nature which are related to the production of gravel, cement and concrete

The report of Künniger et al. (2001) has been separated for this study in 4 reports: “Gravel products and processes”, “Cement products and processes”, “Concrete products and processes” and “Refractory Bricks”. The present report describes cement products and processes.

Routschka & Granitzki (1997) describe cement as a hydraulic binder. It is a finely ground, non-metallic, inorganic material. When mixed with water it forms a paste that sets and hardens by hydration. After hardening, it retains its strength and stability even under water. The hydraulic hardening is primarily due to the formation of calcium silicate hydrates. Accordingly, cements consist of those substances, which react with water to form calcium silicate hydrates. Aluminous cements harden hydraulically by forming calcium aluminate hydrates.

Beside cement, this report also deals with hydraulic lime because the chemical composition of the raw materials and the production process is quite similar. However, the firing temperature of hydraulic lime is lower than that of cement-clinker. Thus the raw material is calcined but in contrast to cement-clinker not sintered. Hydraulic lime is used as masonry cement or mortar by adding sand and water.

2 Reserve, Resources and Raw Materials

From a geological point of view Switzerland is rich on raw material for cement production (Kündig et al. (1997)). The whole “Jura” mountains, the northern pre-Alps region as well as the southern Alps contain limestone, clay and marl (Compare part VII, lime products and processes, part IV, clay products and processes). The mining amount is strongly determined by the available infrastructure, the demand and the environmental protection aspects.

From the perspective of the mining industry resources can be regarded as quarryable if mining is possible with actual technology, the enterprise realisable with supportable financial and temporal expenditure and if resource exploitation does not collide with other interests. This last point causes a lot of problems in Switzerland as in its intensely populated part (middle Lands) all different kinds of utilisation of natural resources collide. These are for example water protection, settlement area etc. This reduces the quarryable area substantially. This scenario is valid for all groups of mineral resources in Switzerland.¹

For larger countries as Germany, France or Italy the mentioned scenario can also be applicable but for longer time period.

The cement industry is making great efforts to produce resource-efficiently and therefore uses suitable secondary raw materials for the clinker production wherever possible. This leads to substitution of conventional fuel and raw material. The quantity of secondary fuel (waste) varies considerably for the eight Swiss cement plants. The percentage of energy from waste relative to the total energy consumed for clinker firing is about 36% in this study (Tab. 6.3).

¹ http://www.sgtek.ethz.ch/100jahresgtek/meilensteine/abbaubare_rohstoffe.html, access March 2003

The raw materials for hydraulic lime and cement production are for example the geological deposits “effinger”-layer in the region “Jura”, the “Couches Rouges” in the “präalpine” covering or the “See-
wer”-slate and “Drusberg”-layer in the “Helveticum”.

The amount of consumed hydraulic lime in Switzerland and the production volume are decreasing since about 1980. In 1992 the three Swiss production plants produced about 19'000t hydraulic lime. About 9'600t were imported. In total about 28'600t were used in Switzerland. The amount of exported hydraulic lime is negligible. 1997 only two plants producing hydraulic lime were still operational in Switzerland.

Cement cast plaster floor is a mixture of cement, aggregate (sand, pumice, expanded clay or expanded perlite) and water. Therefore the availability of the basic components controls the availability of the product.

3 Characterisation

Within the binding agent industry in Switzerland two industry branches play a major role: The cement industry with its main product portland cement and the lime industry with its main products quicklime, hydrated lime and hydraulic lime.

3.1 Components of Cement

3.1.1 Clinker

The main component of all silicate cements is portland cement clinker, also known as cement clinker or just clinker. Crystallised calcium silicates that are rich in calcium oxide and can rapidly react with the mixing water make up at least two-thirds of the commercial portland cement clinker.

The raw materials required for the production of cement clinker must have a composition leading to the final mineralogical components (Tab. 3.1) and the chemical composition (Tab. 3.2). These chemical components are found in limestone, chalk, or clay and in naturally occurring mixtures, i.e., calcareous marl. The mineralogical composition of these raw materials is strongly depending on the location and has a wide range. Therefore no exact composition can be presented.

Tab. 3.1 Required mineralogical components of clinker (Kündig et al. (1997))

Mineral phase	Name of mineral	Formula	Short notation	Percent
Tri-calcium silicates	Alit	3 CaO SiO ₂	C3S	45-70
Di-calcium silicates	Berlit	2 CaO SiO ₂	C2S	15-25
Tri-calcium aluminates	Aluminates	3 CaO Al ₂ O ₃	C3A	3-12
Tetra-calcium aluminate ferrite	Ferrite	4 CaO Al ₂ O ₃ Fe ₂ O ₃	C4AF	6-12

The calcium silicate phases Alit and Berlit are important for the quality of the cement. Their part on the total of all clinker components is about 75%. Aluminates and Ferrite are important for the sintering process itself.

As shown in Tab. 3.1 and Tab. 3.2 mainly the four elements CaO , SiO_2 , Al_2O_3 and Fe_2O_3 are decisive for the clinker mineralogy.

Tab. 3.2 Representative chemical composition of clinker (Kündig et al. (1997))

Element	Percentage of the clinker [%]
SiO_2	18 – 25
Al_2O_3	4 - 8
Fe_2O_3	2 – 5
CaO	60 – 67
MgO	1 - 2
SO_3	0.3 - 1.2
K_2O	0.3 - 1.2
Na_2O	0.2 - 0.4
Cl	0.01 - 0.02
loss on ignition	0.3 - 0.5
rest	0.05 - 1

To get a clinker with a positive mineralogy the chemical composition must be very precise (Compare Tab. 3.2). This requires a precise selection of the different forms of the raw materials limestone, chalk or clay.

3.1.2 Blast Furnace Slag

Another widely diffused hydraulically reactive material is glassy blast-furnace slag. It is produced by rapid cooling of the molten slag from the production of pig iron in a blast furnace with water and fine dispersing of the material (granulation). The glass content, which is decisive for the hydraulic properties of slag, must be kept as high as possible. At least two-thirds of the glassy blast furnace slag consists of calcium oxide, magnesium oxide, and silicon dioxide. It can also react with water to form calcium silicate hydrates, but this reaction proceeds very slowly. If the slag is mixed with cement clinker or calcium hydroxide, the reaction proceeds relatively rapidly. The glassy blast-furnace slag is known as a latent hydraulic material.

3.1.3 Other Natural/Synthetic Substances

There are other natural or synthetic substances, which are also capable of hydraulic hardening. They have a high content of silicon dioxide and can react with calcium hydroxide in the presence of water to form calcium silicate hydrates. Materials having these properties are e.g. certain types of volcanic ash and the fly ash from coal power plants. They are called pozzolanic materials after the town Pozzuoli near Naples, where the Romans first registered the occurrence of volcanic ash. Ash that accumulates during the firing of oil shale can also be regarded as a component of cement. Depending on its composition, it can either harden hydraulically on its own or by forming compounds with other components of cement clinker.

3.1.4 Fillers

Cements may also contain small amounts of fillers. These substances are inert or are slightly hydraulic, latent hydraulic, or pozzolanic. As a result of their grain size distribution, they are capable of improving the physical properties of cement, especially its workability. Calcium sulphate in the form of gypsum ($\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}$) or anhydrite (CaSO_4) is an auxiliary component of cements, used to control setting.

3.2 Types of Cement

Cement has been produced in Switzerland for more than 100 years. In the past the production was almost entirely Portland cement in compliance with the previous Swiss Standard SIA 215. The present standard SIA 215.002, which complies with the European standard EN 197-1, defines 150 different mixing cements.

The following classification parameters of cement in Tab. 3.3 have been taken from MBT (Master Builder Technology) Switzerland Ltd.²; five main types of cement exist in SIA 215.002:

Tab. 3.3 Main types of cement under SIA standard 215.002

Type of cement	Name	Amount of clinker [weight-%]	Type	Additional components [weight-%]
CEM I	Portland cement	95 - 100	-	0 - 5
CEM II	Portland composite cement	65 - 94	Limestone, granulated blast furnace slag, silica sand (max. 10%), pozzolan, flue ash or burned slate	0 - 5
CEM III	Blast furnace slag cement	5 - 64	Granulated blast furnace slag	0 - 5
CEM IV	Pozzolan cement	45 - 89	Pozzolan, flue ash	0 - 5
CEM V	Composite cement	20 - 64	Granulated blast furnace slag and pozzolan and/or flue ash	0 - 5

To produce mixing cements, materials like limestone, fly ash or blast furnace slag are added. In Switzerland primarily limestone and small quantities of silica dust, fly ash and granulated blast furnace slag are added, usually less than 20%, rarely up to 70%.

Cements are also categorised in function of compressive strength, shown in Tab. 3.4.

The finer the cement is milled the faster is the development of strength and the faster the standard compressive strength after 28 days is reached.

Tab. 3.5 shows a list of cement types and quantities sold by the Swiss cement industry in the late nineties. For the cements described in this inventory the ingredients and amounts shown in Tab. 3.6 were taken into account.

² <http://www.meynadier.ch/german/lexikon/zem.htm>

Tab. 3.4 Strength classes under SIA standard 215.002

Class of strength	Compressive strength (N/mm ²), initial strength	Compressive strength (N/mm ²), standard strength
32.5	2 days: not defined 7 days: > 16	28 days: > 32.5 and <52.5
32.5 R	2 days: > 10 7 days: not defined	28 days: > 32.5 and <52.5
42.5	2 days: > 10 7 days: not defined	28 days: > 42.5 and <62.5
42.5 R	2 days: > 20 7 days: not defined	28 days: > 42.5 and <62.5
52.5	2 days: > 20 7 days: not defined	28 days: > 52.5
52.5 R	2 days: > 30 7 days: not defined	28 days: > 52.5

Tab. 3.5 Cement types and quantities sold by the Swiss cement industry in the late nineties

Cement type	1995	1997	1998	1999
Portland cement (CEM I)	3'763'000 t	2'820'000 t	2'362'000 t	2'031'000 t
Portland calcareous cement (CEM II A-L)	226'000 t	550'000 t	991'000 t	1'336'000 t
Portland silica dust cement (CEM II/A-L)	4'600 t	86'000 t	92'000 t	136'000 t
Portland composite cement (CEM II/A-M)				
Flue ash cement (CEM II/B-V)	1'800 t	1'700 t	1'700 t	2 t
Blast furnace slag cement (CEM III)				
Total	3'995'400 t	3'457'700 t	3'446'700 t	3'503'002 t

Tab. 3.6 Cement types and their ingredients per kg cement

Cement type	Clinker [kg]	Gypsum (CaSO ₄ * 2H ₂ O) [kg]	Limestone (CaCO ₃) [kg]	granulated blast furnace slag, dry [kg]
Portland cement (CEM I 42.5)	0.903	0.052	0.045	-
Portland cement (CEM I 52.5)	0.912	0.063	0.025	-
Portland calcareous cement (CEM II/A-L 32.5)	0.787	0.054	0.159	-
Portland slag sand cement (CEM II/A-S 42.5)	0.764	0.046	0.050	0.140
Blast furnace slag cement (CEM III/B 32.5)	0.459	0.040	0.040	0.460

The term portland cement generally refers to cement which completely or predominantly consists of cement clinker. Calcium sulphate, which is present in practically all cements, is ignored when defining the composition. White portland cement is made from a cement clinker that does not contain any coloured components, e.g., calcium aluminoferrite.

Cement consisting of at least 60 wt-% cement clinker and a ground additive is classified as portland slag sand cement, portland pozzolana cement, or portland fly-ash cement. Portland slag cement is called iron portland cement in Germany. Pozzolanic cement is of substantial economic importance, especially in Italy. It generally contains up to 40 wt-% pozzolana as a ground additive. This amount is not limited, but the cement must pass the pozzolanicity test, proofing the capability of the pozzolana to bind lime. The probability of passing this test increases with the reactivity of the pozzolana and with the pozzolana content of the cement. The trass cement in Germany corresponds to the portland pozzolana cement and to pozzolanic cement.

Blast-furnace cement refers to cement containing about 35 – 80 wt % glassy blast-furnace slag. Other substances with lower hydraulic reactivity may only be added in quantities up to about 40% in weight.

3.3 Hydraulic Lime

Hydraulic lime is a binder, which is produced from marl lime or calcareous marl by calcination below the sintering point (temperature of calcination: around 1'050°C, max. 1'200°C). The basic material must have a share of 65 to 75% in calcium carbonate. The calcined hydraulic lime is slaked and milled. To regulate the cementation, gypsum is added during the grinding process. Compared to cement, the SiO₂ content of hydraulic lime is varying strongly (5 - 30%). No content of Alit (tricalcium silicate), a high content of Belit (dicalcium silicate) and a higher amount of lime (>5%) are typical for hydraulic lime. Further significant differences to cement are better working properties but worse hydraulic characteristics and lower strength (1 – 5 N/mm²).

The main raw material is lime marl or marl lime with a content of 65 – 75% of CaCO₃. The composition of the raw material should fulfil the requirements in Tab. 3.7:

Tab. 3.7 Chemical composition of the raw materials for hydraulic lime (Kündig et al. (1997))

Criteria	Order of magnitude [%]
CaCO ₃	65 – 75
(CaO)	(35 – 42)
Al ₂ O ₃	3 – 8.5
Fe ₂ O ₃	0 – 4
SiO ₂	15 – 25
Ratio of CaO to SiO ₂	1.25 – 2.5
Ratio of SiO ₂ to (Al ₂ O ₃ + Fe ₂ O ₃)	2 – 4
Ratio of Al ₂ O ₃ to Fe ₂ O ₃	≥ 2

The raw material parameters are usually pre-determined by the type of the “parent” material (requirements of the quarry). The mining and beneficiation of the raw material is comparable to the mining of limestone. Usually the pre-homogenised and crushed material can be put directly in the kiln.

“The chemical-physical characterisation of the raw material and its reactivity are crucial for the choice of the type of kiln. In most cases either rotary or shaft kilns are used. For each of these the beneficiation of the raw material is different. Shaft kilns accept coarse material (size between 30 and 100 mm) whereas the rotary kilns accept only crushed material (size smaller than 30 mm). The fuel used in Switzerland is either coal or oil. The energy demand per kilogram hydraulic lime is between 3'000 kJ and 4'000 kJ. Firing time and firing temperature is depending on the characteristics of the raw material. The firing temperature is usually between 900°C and 1'050°C (maximum 1'200°C) and must be lower than the sintering point of the material. During the firing process different mineral phases are build up. Parts of the calcined product include certain amount of not activated oxides as SiO₂, Al₂O₃,

Fe₂O as well as CaO (lime) and MgO. Dicalciumsilicat (belite) and lime have the biggest influence on the compressive strength development. Based on the following formula, lime is recarbonated:



Calciumaluminate in general increases the low starting compressive strength. Gypsum is used as setting regulation agent but cannot improve the starting compressive strength. Gehlenite, a Calciumaluminosilicate has no influence on the compressive strength development. A good quality for hydraulic lime is guaranteed when the amount of both components which are strength relevant (lime and belite) is large enough. In the case of lime, the improvement of compressive strength is caused by carbonation, whereas in the case of belite is hydration. For the hydration of the calcined product the hydraulic lime is mixed with a certain amount of water. The necessary amount of water is depending on the amount of lime (CaO) remaining in the product and must therefore be individually fixed for each plant. A too small amount of water leads to incomplete hydration. This is the reason for dilatation and cracking if the remaining lime in the end product during the hardening process can form calcium hydroxide. A too large amount of water causes stickiness of the product and therefore problems within milling. Furthermore, the achievable end- strength gets decreased. If the amount of lime is low no hydration is necessary. To reach a complete hydration the damped product is stored about three weeks. Afterwards the hydrated hydraulic lime is milled together with 3 to 4% of gypsum.” (Kündig et al. (1997), translation by author)

3.4 Calcareous Marl

A short description of calcareous marl is given in Kündig et al. (1997) which is reported in the following: “The raw materials for the cement production are a combination of one component rich on limestone, CaCO₃ as supplier of calcium and one component rich on clay, as supplier of silicon, aluminium and iron (in from of SiO₂, Al₂O₃ and Fe₂O₃) and if necessary a third component, as supplier of substances to adjust the mixture. The share is about three parts. limestone and one part clay. This share is used in this report Usually in nature no pure limestone and clay are available. All possible mixtures can be found in nature. This is advantageous in so far as the natural sediments (marly limestone and marl) are more useful than the pure ones as they have already a pre-homogenized mixture of minerals. Thus it is easier to prepare them.” (Kündig et al. (1997))

3.5 Cement Cast Plaster Floor

The cement cast plaster floor represented in this study requires 0.186 kg cement, 0.740 kg sand and 0.074 kg water per kg product.

The physical characteristics of cement cast plaster floors are: (Starzner & Wurmer-Weiss (2000)):

- Gross density: 2'000 to 2'400 kg/m³.
- Thermal conductivity: about 1.4 W m⁻¹ K⁻¹ (at ≤ 600 kg/m³)

4 Use

Different economic and industrial developments, sources of raw materials, and climatic conditions have led to different developments in building technology and construction materials and, also, to different types of cement. Thus, there are substantial differences in the standards of cement composition, designation, and minimum requirements in different countries.

Cements having special properties are produced for specific purposes. Cements with a high sulphate resistance are used for concrete constructions exposed to sulphate solutions. Portland cement with low tricalcium aluminate content and blast furnace cement with high blast-furnace slag content belong to this category. Indeed, the cement standards in different countries specify definite limits for the components of these cements. The content of tricalcium aluminate in highly sulphate-resistant portland cement is limited to 3 – 5 weight-percent, and the slag content in blast-furnace cement with a high sulphate resistance must be at least 65 weight-percent.

Cements with a low heat of hydration, used in the production of massive concrete constructions, are either portland cements containing small amounts of tricalcium aluminate and tricalcium silicate or blast furnace cements containing a large amount of slag. According to various national cement standards the heat of hydration developed in the first 7 or 28 days should not exceed 250 – 290 J/g as measured by the heat-of-solution method. It may not exceed 290 J/g after 28 days.

Hydraulic lime is primarily used for the production of plastering mortar (hydraulic lime + water + two to three parts of sand). It is appreciated for its good working properties and adhesiveness. Further it is used for the production of mortar for brick masonry, hardened cement paste or sand-lime brick. To improve the working properties and pump ability of concrete, occasionally hydraulic lime is added (10% – 15% of the cement weight).

Calcareous marl is an important raw material for the clinker production process. This is again an important part of the cement composition.

Cement cast plaster floors are in general used as bond floor pavements, as floor pavement on a separation layer or as floating floor pavement, depending on the connection to the load-bearing subsurface.

5 Process Description

5.1 Cement Production Process

Routschka & Granitzki (1997) describe the production of cement as illustrated in Fig. 5.1, which starts with isolation and preparation of the raw materials and continue with firing of the raw material mixture to give cement clinker, preparation of the other components of the cement, and grinding of the cement components with calcium sulphate to regulate setting. These continuous processes demand the monitoring of large material flows with respect to amount and required composition. This procedure is necessary to compensate for variations in the raw materials and in the production procedure during the preparation of cement clinker and the grinding of cement.

Each of these process steps is described briefly below (based on EPA (1998), adapted to the Swiss situation). The focus of the discussion is on pyroprocessing operations, which constitute the core of a Portland cement plant.

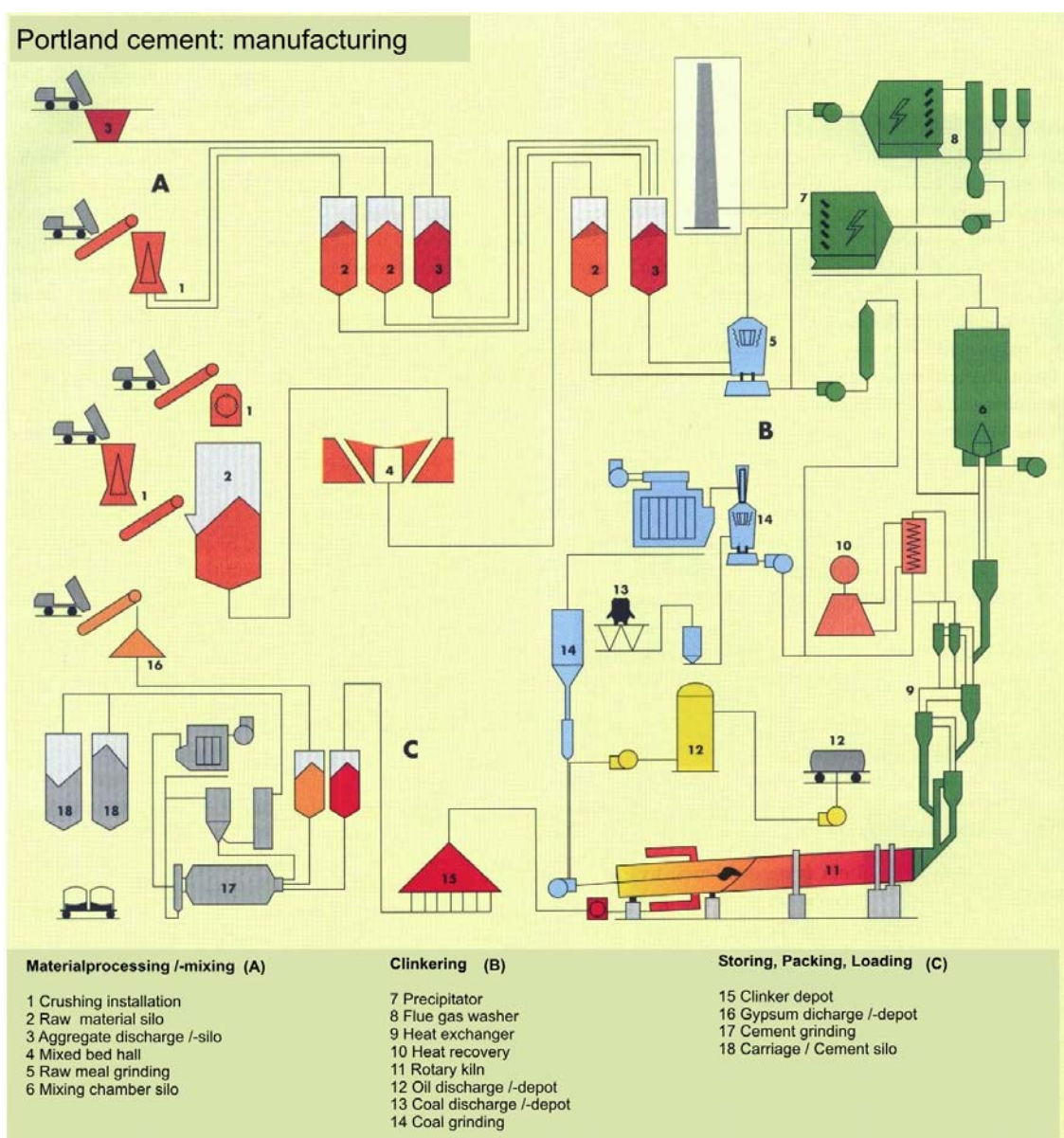


Fig. 5.1 Flow scheme of Portland cement production (Kündig et al. (1997))

5.1.1 Raw Materials

The initial production step in Portland cement manufacturing is raw materials acquisition. More than 30 raw materials are known that can be used to manufacture Portland cement. Calcium, the element of highest concentration in Portland cement, is obtained from a variety of calcareous raw materials, including limestone, chalk, marl, sea shells, aragonite, and an impure limestone known as "natural cement rock". Typically, these raw materials are obtained from quarries. Raw materials vary from facility to facility. Portland cement plants in Switzerland are located in close proximity to a source of raw material. The Portland cement manufacturing industry is relying increasingly on replacing virgin materials as described above with waste materials or by-products from other manufacturing operations, to the extent that such replacement can be implemented without adversely affecting plant operations or product quality. Materials that have been used include fly ash, mill scale, metal smelting slag, pyrite cinder lime sludge, building bulky goods, sludge and moulding sand.

5.1.2 Raw Mix Preparation

The second step in Portland cement manufacturing is preparing the raw mix or kiln feed for the pyroprocessing operation (clinkering). Raw material preparation includes a variety of blending and sizing operations that are designed to provide a feed with appropriate chemical and physical properties. The raw material processing operations somewhat differ for wet and dry processes, as described in the paragraphs below.

Cement raw materials are received with an initial moisture content varying from one to more than fifty%. If the facility uses dry process kilns this moisture is usually reduced to less than 1% before or during grinding. Drying alone can be accomplished in impact dryers, drum dryers, paddle-equipped rapid dryers, air separators, or autogenous mills. However, drying can also be accomplished during grinding in ball-and-tube mills or roller mills. While thermal energy from exhaust gases from separate, direct-fired coal, oil, or gas burners can supply drying, the most efficient and widely used source of heat for drying is the hot exit gases from the pyroprocessing system. All facilities in Switzerland use this set-up. All published data in the studied inventory relate to the dry process kiln.

Material transport for raw milling systems can be accomplished by a variety of machines, including screw conveyors, belt conveyors, drag conveyors, bucket elevators, air slide conveyors and pneumatic conveying systems. The dry raw mix is pneumatically blended and stored in specially constructed silos until it is fed to the pyroprocessing system.

In the wet process, water is added to the raw mill during the grinding of the raw material in ball or tube mills, thereby producing a pumpable slip or slurry of approximately 65% solids. The slurry is agitated, blended, and stored in various kinds and sizes of cylindrical tanks or slurry basins until it is fed to the pyroprocessing system.

5.1.3 Pyroprocessing System (Clinkering)

The heart of the Portland cement manufacturing process is the pyroprocessing system. This system transforms the raw mix into clinkers, which are grey, glass-hard, spherically shaped nodules that range from 0.32 to 5.1 centimetres (0.125 to 2.0 inches) in diameter. The chemical reactions and physical processes that constitute the transformation are quite complex, but they can be viewed conceptually as the following sequential steps:

1. Evaporation of free water;
2. Evolution of combined water in the argillaceous components;
3. Calcination of the calcium carbonate (CaCO_3) to calcium oxide (CaO);
4. Reaction of CaO with silica to form dicalcium silicate;
5. Reaction of CaO with the aluminium and iron-bearing constituents to form the liquid phase;
6. Formation of the clinker nodules;
7. Evaporation of volatile constituents (e.g., sodium, potassium, chlorides, and sulphates);
8. Reaction of excess CaO with dicalcium silicate to form tricalcium silicate.

This sequence of events may be conveniently divided into four stages, as a function of location and temperature of the materials in the rotary kiln.

9. Evaporation of uncombined water from raw materials as material temperature increases to 100°C ;
10. Dehydration as the material temperature increases from 100°C to approximately 400°C to form oxides of silicon, aluminium, and iron;
11. Calcination, during which carbon dioxide (CO_2) is evolved, between 900°C and 982°C , to form CaO ;
12. Reaction of the oxides in the firing zone of the rotary kiln to form cement clinker at temperatures of approximately 1510°C .

The rotary kilns are long, cylindrical, slightly inclined furnaces that are lined with refractory to protect the steel shell and retain heat within the kiln. The raw material mix enters the kiln at the elevated end, and the combustion fuels generally are introduced into the lower end of the kiln in a counter current manner. The materials are continuously and slowly moved to the lower end by rotation of the kiln. As they move down the kiln, the raw materials are changed to cementitious metal oxides by the direct heat exchange. The most commonly used kiln fuels are coal, natural gas, and occasionally oil. Many cement plants currently burn coal, but use of supplemental fuels such as waste solvents, scrap rubber, and petroleum coke has expanded in recent years. This is the case for the studied plants in Switzerland.

For the lining of the cement kiln three types out of a huge amount of different types of refractory bricks are used (compare report “Refractory Bricks”): fireclay refractory, high aluminium oxide refractory and basic refractory.

Five different processes are used in the Portland cement industry to accomplish the pyroprocessing step: the wet process, the dry process (long dry process), the semidry process, the dry process with a pre-heater, and the dry process with a pre-heater/precalciner. Each of these processes accomplishes the physical/chemical steps identified above. However, the processes vary with respect to equipment design, method of operation, and fuel consumption. Generally, fuel consumption decreases in the order of the processes listed above. The paragraphs below briefly describe the processes, starting with the wet process and then noting differences in the other processes.

In the wet process and long dry process, the entire pyroprocessing activity occurs in the rotary kiln. Depending on the process type, kilns have length-to-diameter ratios in the range 15:1 to 40:1. While some wet process kilns may be as long as 210 m, many wet process kilns and all dry process kilns are shorter. Wet process and long dry process pyroprocessing systems consist solely of the simple rotary kiln. Usually, a system of chains is provided at the feed end of the kiln in the drying or preheating zones to improve heat transfer from the hot gases to the solid materials. As the kiln rotates, the chains are raised and exposed to the hot gases. Further kiln rotation causes the hot chains to fall into the cooler materials at the bottom of the kiln, thereby transferring the heat to the load.

Dry process pyroprocessing systems (used in the studied plants in Switzerland) have been improved in thermal efficiency and productive capacity through the addition of one or more cyclone-type pre-heater vessels in the gas stream after the rotary kiln. This system is called the pre-heater process. The vessels are arranged vertically, in series, and are supported by a structure known as the pre-heater tower. Hot exhaust gases from the rotary kiln pass counter currently through the downward-moving raw materials in the pre-heater vessels. Compared to the simple rotary kiln, the heat transfer rate is significantly increased, the degree of heat utilisation is more complete, and the process time is markedly reduced owing to the intimate contact of the solid particles with the hot gases. The improved heat transfer allows the length of the rotary kiln to be reduced. The still hot gases from the pre-heater tower are often used as a source of heat for drying raw materials in the raw material mill. Because the catch from the mechanical collectors, fabric filters, and/or electrostatic precipitators (ESP's) that follow the raw mill is returned to the process, these devices are considered to be production machines as well as pollution control devices.

Additional thermal efficiency and productivity gains have been achieved by diverting some fuel to a calciner vessel at the base of the pre-heater tower. This system is called the pre-heater /pre-calciner process. While a substantial amount of fuel is used in the pre-calciner, at least 40% of the thermal energy is required in the rotary kiln. The amount of fuel that is introduced to the calciner is determined by the availability and source of the oxygen for combustion in the calciner. Calciner systems sometimes use lower-quality fuels (e.g., less-volatile matter) as a means of improving process economics.

Pre-heater and pre-calciner kiln systems often have a bypass system between the feed end of the rotary kiln and the pre-heater tower to remove the undesirable volatile constituents. Otherwise, the volatile constituents condense in the pre-heater tower and subsequently re-circulate to the kiln. Build up of

these condensed materials can restrict process and gas flows. In a bypass system, a portion of the kiln exit gas stream is withdrawn and quickly cooled by air or water to condense the volatile constituents to fine particles. The solid particles, which are removed from the gas stream by fabric filters and ESP's, are then returned to the process.

The semidry process is a variation of the dry process. In the semidry process, the water is added to the dry raw mix in a pelletizer to form moist nodules or pellets. The pellets then are conveyed on a moving grate pre-heater before being fed to the rotary kiln. The pellets are dried and partially calcined on the moving grate through which hot exhaust gases from kiln pass.

Regardless of the type of pyroprocess used, the last component of the pyroprocessing system is the clinker cooler. This process step recoups up to 30% of the heat input to the kiln system, locks in desirable product qualities by freezing mineralogy, and makes it possible to handle the cooled clinker with conventional conveying equipment. The more common types of clinker coolers are reciprocating grate, planetary, and rotary. In these coolers, the clinker is cooled from about 1100°C to 93°C by ambient air that passes through the clinker and into the rotary kiln for use as combustion air. However, in the reciprocating grate cooler, lower clinker discharge temperatures are achieved by passing an additional quantity of air through the clinker. Because this additional air cannot be utilised in the kiln for efficient combustion, it is vented to the atmosphere, used for drying coal or raw materials, or used as a combustion air source for the pre-calciner.

5.1.4 Cement Manufacturing

The final step in Portland cement manufacturing involves a sequence of blending and grinding operations that transform clinker to finished Portland cement. Up to 5% gypsum or natural anhydrite is added to the clinker during grinding to control the cement setting time, and other speciality chemicals are added as needed to impart specific product properties. This finish milling is accomplished almost exclusively in ball or tube mills. Typically, finishing is conducted in a closed- circuit system with product sizing via air separation.

5.2 Emissions

A general description of emissions from cement production is found in EPA (1998). Particulate matter (PM and PM-10), nitrogen oxides, sulphur dioxide, carbon monoxide, and carbon dioxide are the primary emissions in the manufacture of Portland cement. Small quantities of volatile organic compounds (VOC), ammonia, chlorine, and hydrogen chloride may be also emitted. Emissions may also include residual materials from the fuel and raw materials or products of incomplete combustion that are considered to be hazardous. Some facilities burn waste fuels, particularly spent solvents in the kiln and these systems also may emit small quantities of additional hazardous organic pollutants. Raw material feeds and fuels typically contain trace amounts of heavy metals that may be emitted as particles or vapour.

Sources of PM at cement plants include quarrying and crushing of raw material, raw material storage, grinding and blending (in the dry process only), clinker production, finish grinding, packaging and loading. The largest emission source of PM within cement plants is the pyroprocessing system that includes the kiln and clinker cooler exhaust stacks. Emissions from kilns are determined by several factors, including differences in convective patterns, material movement patterns, burner locations and insertion lengths, heat transfer mechanisms, and the type of clinker cooler that supplies secondary air to the kiln for combustion. In addition, operators can vary the degree of calcination taking place within a pre-heater or pre-calciner. Often, dust from the kiln is collected and recycled into the kiln thereby producing clinker from the dust. However, if the alkali content of the raw materials is too high, some or all of the dust is discarded, or leached before returned to the kiln. In many instances, the maximum

allowable cement alkali content of 0.6% (calculated as sodium oxide) restricts the amount of dust that can be recycled. Bypass systems sometimes have a separate exhaust stack. Additional sources of dust are raw material storage piles, conveyors, storage silos, and unloading facilities.

Oxides of nitrogen are generated during fuel combustion by oxidation of chemically bound nitrogen in the fuel and by thermal fixation of nitrogen in the combustion air. In the cement manufacturing process, NO_x is generated in the firing zone of the kiln and the firing zone of a pre-calcining vessel. As flame temperature increases, the amount of thermally generated NO_x increases. The amount of NO_x generated from fuel increases with the quantity of nitrogen in the fuel. However, natural gas combustion with a high flame temperature and low fuel nitrogen generates a larger quantity of NO_x than does oil or coal, which has higher fuel nitrogen but burn with lower flame temperatures. Types of fuels used vary across the industry. Historically, a combination of coal, oil, and natural gas was used, but over the last 15 years, most plants switched to coal, which generates less NO_x than oil or gas. However, in recent years a number of plants have switched to systems that burn a combination of coal and waste fuel. The effect of waste fuel use on NO_x emissions is not clearly established. Sulphur dioxide may be generated both from the sulphur compounds in the raw materials and from sulphur in the fuel, differing from plant to plant and with geographic location. However, the alkaline nature of the cement provides for direct absorption of SO_2 into the product, thereby mitigating the quantity of SO_2 emissions in the exhaust stream.

The CO_2 emissions from Portland cement manufacturing are generated by two mechanisms. One is combustion of fossil or waste. Substantial quantities of CO_2 (55%-65% of the total CO_2 emission) also are generated through calcining of limestone or other calcareous material. This calcining process thermally decomposes CaCO_3 to CaO and CO_2 . Typically, Portland cement contains about 63.5% CaO . Consequently about 1.135 units of CaCO_3 are required to produce 1 unit of cement, and the amount of geogenic CO_2 released in the calcining process is about 500 kilograms per tonne of Portland cement produced.

In addition to CO_2 emissions, fuel combustion at Portland cement plants can emit a wide range of pollutants in smaller quantities. If the combustion reactions do not reach completion, CO and volatile organic pollutants, which are typically measured as total organic compounds (TOC), VOC, or condensable organic particulate, can be emitted. Incomplete combustion can also lead to emissions of specific highly hazardous organic air pollutants, although these pollutants are generally emitted at substantially lower levels than CO or TOC.

Emissions of metal compounds from Portland cement kilns can be grouped into three general classes: volatile metals, including mercury (Hg) and thallium (Tl); semi-volatile metals, including antimony (Sb), cadmium (Cd), lead (Pb), selenium (Se), zinc (Zn), potassium (K), and sodium (Na); and refractory or non-volatile metals, including barium (Ba), chromium (Cr), arsenic (As), nickel (Ni), vanadium (V), manganese (Mn), copper (Cu), and silver (Ag). Although the partitioning of these metal groups is affected by kiln operating conditions, the refractory metals tend to concentrate in the clinker, while the volatile and semi-volatile metals tend to be discharged via the primary exhaust stack and the by-pass stack, respectively.

5.3 Air Pollution Control Technology

Fugitive dust sources in the industry include quarrying and mining operations, vehicular traffic during mineral extraction and at the manufacturing site, raw materials storage piles, and clinker storage piles. The measures used to control emissions from these fugitive dust sources are comparable to those used throughout the mineral products industry. Vehicular traffic dust controls include paving and road wetting. Controls that are applied to other open dust sources include water sprays with and without surfactants, chemical dust suppressants, wind screens, and process modifications to reduce drop heights or enclose storage operations.

Process particle emission sources include material handling and transfer, raw milling operations in dry process facilities, and finish milling operations. Typically, emissions from these processes are captured by a ventilation system and collected in fabric filters. Some facilities use an air pollution control system comprising one or more mechanical collectors with a fabric filter in series. Because the dust from these units is returned to the process, they are considered to be process units as well as air pollution control devices. The industry uses shaker, reverse air, and pulsejet filters as well as some cartridge units, but most new facilities use pulsejet filters. For process fugitive operations, the different systems are reported to achieve typical outlet PM loadings below 45 milligrams per cubic meter.

Within pyroprocessing units, PM emissions are controlled by fabric filters (reverse air, pulse jet, or pulse plenum) and ESP's. Typical control measures for the kiln exhaust are reverse air fabric filters and ESP's. These systems are reported to achieve outlet PM loadings of 45 mg/m³. Clinker cooler systems are controlled most frequently with pulse jet or pulse plenum fabric filters. Also a few gravel bed filters have been used to control clinker cooler emissions. Typical outlet PM loadings are identical to those reported for kilns.

Cement kiln systems have highly alkaline internal environments that can absorb up to 95% of potential SO₂ emissions. However, in systems that have sulphides (pyrites) in the kiln feed (e.g. coal), the sulphur absorption rate may be as low as 50% without specific design considerations or adjustments in the composition of raw materials. The cement kiln system itself has been determined to provide substantial SO₂ control. Fabric filters on cement kilns are also reported to absorb SO₂. Generally, substantial control is not achieved. An absorbing reagent (e.g., CaO) must be present in the filter cake for SO₂ capture to occur effectively. Without the presence of water, which is undesirable in the operation of a fabric filter, CaCO₃ is not an absorbing reagent. It has been observed that as much as 50% of the SO₂ can be removed from the pyroprocessing system exhaust gases when this gas stream is used in a raw mill for heat recovery and drying. In this case, moisture and calcium carbonate are simultaneously present for sufficient time to accomplish the chemical reaction with SO₂.

5.4 Hydraulic Lime

The production process is similar to the production of cement clinker (Kündig et al. (1997)). The two main differences are the lower calcining temperature ($\leq 1'250^{\circ}\text{C}$ with average about $1'050^{\circ}\text{C}$ whereas for clinker it is about $1'500^{\circ}\text{C}$) and the hydration process following the firing process. Caused by the lower calcining temperature, the hydraulic characteristic and the compressive strength of hydraulic lime are inferior then for cement. The requirements on the raw material beneficiation for the production of hydraulic lime are much lower than for the cement production.

5.5 Calcareous Marl

Calcareous marl is present in nature in greatly variable combinations of the containing minerals; therefore, specific data are not easy to find. Kündig et al. (1997) describes the ratio of mineral components of about one part of pure clay and three parts of highly pure limestone. Therefore the module in this study is build up being a mix of one fourth "clay, at mine" and three fourth "limestone, crushed, at mine".

5.6 System Boundary

The flow scheme in Fig 5.2 shows the sub-processes of the cement production and the system boundary of this unit process. Data for the cement production are based on eight Swiss cement factories.

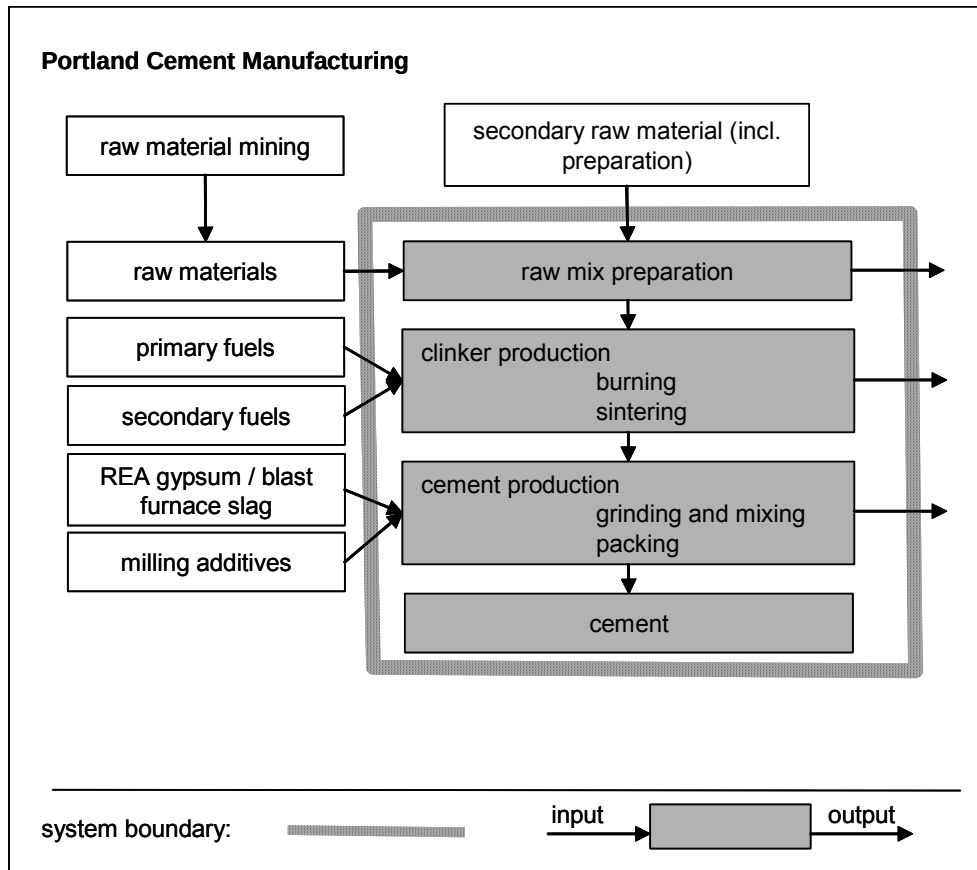


Fig 5.2 System boundary for Portland cement production (Künniger et al. (2001))

5.7 Cement Cast Plaster Floor

Cement cast plaster floor can either be mixed on the building site or in a plant. If mixed in a factory noise and dust emissions can be better controlled. As no data on dust emissions are available they are not taken into account in this study. After mixing the raw materials (portland cement, sand and water), the cast plaster floor is placed either by pump or by bucket. Shaking and stamping helps to ram the wet cement cast plaster floor.

6 Life Cycle Inventories

Ecoinvent database meta information (tables), flows and their representation in the ecoinvent database are shown in (figures):

Clinker, at plant	Tab. 6.2 and Fig. 6.1
Lime hydraulic, at plant	Tab. 6.6 and Fig. 6.2
Calcareous marl, at plant	Tab. 6.8 and Fig. 6.3
Cement plant	Tab. 6.9 and Fig. 6.4
Blast furnace slag cement, at plant	Tab. 6.10 and Fig. 6.5
Portland calcareous cement, at plant	Tab. 6.11 and Fig. 6.6
Portland cement, strength class Z 42.5, at plant	Tab. 6.12 and Fig. 6.7
Portland cement, strength class Z 52.5, at plant	Tab. 6.13 and Fig. 6.8
Portland slag sand cement, at plant	Tab. 6.14 and Fig. 6.9
Cement, unspecified, at plant	Tab. 6.15 and Fig. 6.10
Cement cast plaster floor, at plant	Tab. 6.16 and Fig. 6.11
Packing, cement	Tab. 6.17 and Fig. 6.12

The product “clinker, at plant” is usually not available on the market, as it is the main pre-product of all different cement products.

Any cement product is usually sold in a packed form. Therefore the module “Packing and loading of 1 kg cement” has been defined.

The amount of imported cement is very small and varies in a range of some few percents. Therefore it has not been taken into account in this study.

The module “Cement, unspecified, at plant” is an aggregation of most important cements offered on the Swiss market. It has been defined on the basis of the individual modules weighted by the production shares shown in Tab. 6.1.

Tab. 6.1 Cement types (with Swiss abbreviations) and the Swiss production share³

Type of cement	Ecoinvent module name	Share [%]
CEM II/A-L 32.5R including in-house mixture in concrete factories	Portland calcareous cement, at plant	50
CEM I 42.5	Portland cement, strength class Z 42.5, at plant	40
CEM I 52.5	Portland cement, strength class Z 52.5, at plant	6
CEM II/A-S, CEM III, diverse concrete for special purposes	Portland slag sand cement, at plant	4

³ personal communication K. Moser, EMPA Dübendorf, August 2002

6.1 Clinker, at plant

The ecoinvent database meta information of “clinker, at plant” (Tab. 6.2) includes all important and necessary information to use the data correctly.

Tab. 6.2 Ecoinvent database meta information for the product “clinker, at plant”

Name	clinker, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce clinker (raw material provision, grinding, and mixing; rotary kiln process), internal processes (transport, etc.). For the infrastructure, only the rotary kiln is taken into account (as material consumption). No administration is included. Waste (as secondary fuel and raw material) enter the system without environmental burdens from upstream processes. Only its transports to the cement plant and the emissions stemming from its burning are included.
Amount	1
Local Name	Klinker, ab Werk
Synonyms	Klinker//Drehrohrofen//Zement//clinker//rotary kiln//cement
General Comment to reference function	Special inputs are waste materials which are used as fuel (e.g. dried sludge, used oil, etc.) and as secondary raw materials (e.g. pyrite ash, building bulky goods, sludges, moulding sand, etc). The weight of the secondary raw materials and the energy of the secondary fuels have been assessed according to specifications of the 8 cement factories in CH for 1997.
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	typical technology for Swiss production
Representativeness [%]	100
Production Volume	2551000t/a
Sampling Procedure	Measured data of all Swiss plants (8 plants)
Extrapolations	See geography
Uncertainty Adjustments	none

Fig 6.1 shows the process “clinker production” with its output-module “clinker, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent database, the sources (Künniger et al. (2001) and EPA (1998)), and the standard deviation for each data.

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
bauxite, at mine	→	Clinker, at plant (Inputs: primary raw material, fuel and electricity)				primary raw material	metals	extraction	No	GLO	bauxite, at mine	1.20E-04	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
lime, hydrated, loose, at plant	→						construction materials	binder	No	CH	lime, hydrated, loose, at plant	3.92E-03	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
limestone, at mine	→						construction materials	additives	No	CH	limestone, milled, loose, at plant	8.41E-01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
calcareous marl, at plant	→						construction materials	additives	No	CH	calcareous marl, at plant	4.66E-01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
sand, at plant	→						construction materials	additives	No	CH	sand, at mine	9.26E-03	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
clay, at mine	→						construction materials	additives	No	CH	clay, at mine	3.31E-01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
natural gas to plant	→					primary fuel	natural gas	fuels	No	CH	natural gas, high pressure, at consumer	6.81E-03	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
light fuel oil, at regional storage CH	→						oil	fuels	No	CH	light fuel oil, at regional storage	3.74E-04	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
heavy fuel oil, at regional storage CH	→						oil	fuels	No	CH	heavy fuel oil, at regional storage	2.55E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
hard coal from regional store	→						hard coal	fuels	No	WEU	hard coal, at regional storage	3.54E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
petroleum coke Hu 32.5, at refinery	→					primary fuel; net calorific value of input: 32.5 MJ/kg; net calorific value of allocated module: 35 MJ/kg; value has been adjusted	oil	fuels	No	RER	petroleum coke, at refinery	3.91E-03	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
electricity	→						electricity	supply mix	No	CH	electricity, medium voltage, at grid	5.80E-02	kWh	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);

Fig 6.1 Flows for "clinker, at plant " and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent 2000							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
secondary raw material (recycled): Aluminiumhydroxide	Clinker, at plant (Inputs: secondary raw materials)			not balanced as they have no environmental burdens from upstream processes						4.54E-04	kg	Künniger et al. (2001)			
secondary raw material (recycled): Building bulky goods, fine fraction										1.25E-03	kg	Künniger et al. (2001)			
secondary raw material (recycled): diverse ashes from paper production plant										1.03E-02	kg	Künniger et al. (2001)			
secondary raw material (recycled): Decarbonation sludge										2.00E-02	kg	Künniger et al. (2001)			
secondary raw material (recycled): Moulding sand										1.93E-03	kg	Künniger et al. (2001)			
secondary raw material (recycled): Lime mud, limestone										6.25E-02	kg	Künniger et al. (2001)			
secondary raw material (recycled): Oil smudged soil										6.35E-03	kg	Künniger et al. (2001)			
secondary raw material (recycled): Pyrite ash										4.54E-03	kg	Künniger et al. (2001)			
secondary raw material (recycled): Sludges										3.06E-03	kg	Künniger et al. (2001)			
secondary raw material (recycled): Diverse others										3.29E-03	kg	Künniger et al. (2001)			

Fig 6.1 Flows for "clinker, at plant " and their representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent 2000							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Dried sludge as secondary fuel	Clinker, at plant (Inputs: secondary fuels)			secondary fuel according to data of the 8 cement factories in CH, are not balanced as they have no environmental burdens from upstream processes						7.51E-02 MJ		Künniger et al. (2001)			
Used oil as secondary fuel										5.84E-01 MJ		Künniger et al. (2001)			
Solvents as secondary fuel										1.64E-01 MJ		Künniger et al. (2001)			
Used tyre as secondary fuel										1.36E-01 MJ		Künniger et al. (2001)			
Distillation residue as secondary fuel										2.25E-02 MJ		Künniger et al. (2001)			
Plastics as secondary fuel										9.92E-02 MJ		Künniger et al. (2001)			
Rubber meal as secondary fuel										2.55E-03 MJ		Künniger et al. (2001)			
Crap coke from coke filter as secondary fuel										8.32E-03 MJ		Künniger et al. (2001)			
Meat-and-bone meal as secondary fuel										4.66E-02 MJ		Künniger et al. (2001)			
Animal fat as secondary fuel										3.42E-02 MJ		Künniger et al. (2001)			
CSS as secondary fuel				CSS: substitute solid combustibles; secondary fuel according to data of the 8 cement factories in CH, are not balanced as they have no environmental burdens from upstream processes						5.04E-02 MJ		Künniger et al. (2001)			

Fig 6.1 Flows for "clinker, at plant " and their representation in the ecoinvent database (cont.)

General Flow information						Representation in ecoinvent 2000							Uncertainty information				
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
ammoniacal gas liquor (25%), at plant	→	Clinker, at plant (Inputs: ancillary material and transports)				total weight of the diluted ammoniacal gas liquor (including 25% of ammonia): 0.00363 kg: ancillary material for nitrogen oxide removal during burning process	chemicals	inorganics	No	CH	ammonia, liquid, at regional storehouse	9.08E-04	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
water for diluting the ammonia	→				weight of the diluted ammoniacal gas liquor (75% water): 0.00363 kg	resource	in water				Water, unspecified natural origin	2.72E-06	m3	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
transport, freight, rail	→				according to data of the 8 cement factories in CH	transport systems	train	No	CH	transport, freight, rail	1.77E-02	tkm	Künniger et al. (2001)	1	2.06	(3,3,1,1,1,5);	
transport, lorry 16t	→				according to data of the 8 cement factories in CH	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	8.61E-05	tkm	Künniger et al. (2001)	1	2.06	(3,3,1,1,1,5);	
transport, lorry 28t	→				according to data of the 8 cement factories in CH	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	2.68E-03	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);	
transport, lorry 40t	→				according to data of the 8 cement factories in CH	transport systems	road	No	CH	transport, lorry >28t, fleet average	2.11E-03	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);	
transport, barge	→				according to data of the 8 cement factories in CH	transport systems	ship	No	RER	transport, barge	7.22E-03	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,2,1,5);	
diesel, burned in building machine	→				internal transport; the energy used for mining of limestone (0.018 MJ) is not included here because already inventoried in the modules of each raw material.	construction processes	machinery	No	GLO	diesel, burned in building machine	1.34E-02	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);	
transport hard coal from regional store to cement plant	→				estimated distance: 200km	transport systems	train	No	RER	transport, freight, rail	7.09E-03	tkm		1	2.05	(2,1,1,2,1,5);	
transport, van <3.5t	→				internal transport van (0.00033 vkm with fuel with converter and 0.000035 vkm conventional fuel)	transport systems	road	No	CH	transport, van <3.5t	7.09E-05	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,2,1,5);	

Fig 6.1 Flows for "clinker, at plant " and their representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
fireproof brick, alkaline brick, at plant	→	Clinker, at plant (Inputs: ancillaries and infrastructure)				process material: wearing parts in cement rotary kiln	construction materials	bricks	No	DE	refractory, basic, packed, at plant	1.90E-04	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
fireproof brick, high alumina brick, at plant	→					process material: wearing parts in cement rotary kiln	construction materials	bricks	No	DE	refractory, high aluminium oxide, packed, at plant	1.37E-04	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
fireproof brick, fireclay brick, at plant	→					process material: wearing parts in cement rotary kiln	construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	8.21E-05	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
lubricating oil, at plant	→					process material: lubricating grease /-oil	chemicals	organics	No	RER	lubricating oil, at plant	4.71E-05	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
steel high-alloyed	→					process material: wearing parts (metal)	metals	extraction	No	RER	chromium steel 18/8, at plant	5.86E-05	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
process and cooling water	→					miscellaneous water consumption	resource	in water			Water, unspecified natural origin	1.62E-03	m3	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
tap water	→						water supply	production	No	RER	tap water, at user	3.40E-01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
infrastructure kiln	→					attention: Infrastructure includes only round kiln; weight assumption: 300t; assumption service life: 25years, rest of infrastructure in "cement, plant" for cement production included	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	3.76E-05	kg	estimated EMPA	1	3.10	(4,4,1,1,1,5);
cement plant	→					production volume over 50 years: about 16 Mio. t, assumed that 10% of total plant used for cement production	construction materials	others	Yes	CH	cement plant	6.27E-12	unit	estimated EMPA	1	3.05	(2,1,1,1,1,5);

Fig 6.1 Flows for "clinker, at plant " and their representation in the ecoinvent database (cont.)

General Flow information						Representation in ecoinvent 2000							Uncertainty information			
Input		Process Name		Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Clinker, at plant (Outputs: waste, waste heat, particulate matters and carbon dioxide/monoxide)	→			waste heat	Total amount (incl. electricity consumption)	air	unspecified			Heat, waste	3.62E+00	MJ		1	1.22	(2,1,1,1,1,5);
	→			waste to inert material landfill	primarily used fireproof bricks	waste management	inert material landfill	No	CH	disposal, inert waste, 5% water, to inert material landfill	8.00E-05	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			waste to municipal incineration	different materials, "disposal, municipal solid waste" used as proxy	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	4.50E-05	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			dust to air [kg]	from combustion: PM>10 um = 0.15*TSP	air	low population density			Particulates, > 10 um	5.66E-06	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);
	→			dust to air [kg]	from combustion: PM<2.5 um = 0.64*TSP	air	low population density			Particulates, < 2.5 um	2.41E-05	kg	Künniger et al. (2001)	1	3.05	(2,1,1,1,1,5);
	→			dust to air [kg]	from combustion: PM<10, >2.5 = 0.21*TSP	air	low population density			Particulates, > 2.5 um, and < 10um	7.92E-06	kg	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
	→			carbon dioxide, fossil	from combustion of fossil fuel	air	unspecified			Carbon dioxide, fossil	2.07E-01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			carbon dioxide, fossil from burning of waste	fossil CO2 from combustion of waste	air	unspecified			Carbon dioxide, fossil	8.96E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			carbon dioxide, geogenic	geogenic carbon dioxide (from minerals)	air	unspecified			Carbon dioxide, fossil	5.43E-01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			carbon dioxide, regenerativ from waste	regenerativ CO2 from combustion of waste	air	unspecified			Carbon dioxide, biogenic	1.51E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			carbon monoxide, fossil		air	unspecified			Carbon monoxide, fossil	4.72E-04	kg	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);

Fig 6.1 Flows for "clinker, at plant " and their representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent 2000								Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
Clinker, at plant (Outputs: diverse emissions)		→ arsenic		air	unspecified			Arsenic	1.20E-08	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ beryllium		air	unspecified			Beryllium	3.00E-09	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ cadmium		air	unspecified			Cadmium	7.00E-09	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ methane, fossil		air	unspecified			Methane, fossil	8.88E-06	kg	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);	
		→ cobalt		air	unspecified			Cobalt	4.00E-09	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ chromium III	chromium III is 72.5% of total chromium	air	unspecified			Chromium	1.45E-09	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ chromium VI	chromium VI is 27.5% of total chromium	air	unspecified			Chromium VI	5.50E-10	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ copper		air	unspecified			Copper	1.40E-08	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ hydrogen chloride		air	unspecified			Hydrogen chloride	6.31E-06	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ mercury		air	unspecified			Mercury	3.30E-08	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ ammonia		air	unspecified			Ammonia	2.29E-05	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ nickel		air	unspecified			Nickel	5.00E-09	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ NMVOC, non-methane volatile organic compounds, unspecified origin		air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	5.64E-05	kg	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);	
		→ nitrogen oxides		air	unspecified			Nitrogen oxides	1.08E-03	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ lead		air	unspecified			Lead	8.50E-08	kg	Künniger et al. (2001)	1	5.06	(2,1,1,1,1,5);	
		→ antimony		air	unspecified			Antimony	2.00E-09	kg	Künniger et al. (2001)	1	5.06	(2,1,1,1,1,5);	
		→ selenium		air	unspecified			Selenium	2.00E-09	kg	Künniger et al. (2001)	1	5.06	(2,1,1,1,1,5);	
		→ tin		air	unspecified			Tin	9.00E-09	kg	Künniger et al. (2001)	1	5.06	(2,1,1,1,1,5);	
		→ sulfur dioxide		air	unspecified			Sulfur dioxide	3.55E-04	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin		air	unspecified			Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin	9.60E-13	kg	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);	
		→ thallium		air	unspecified			Thallium	1.30E-08	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ vanadium		air	unspecified			Vanadium	5.00E-09	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ zinc		air	unspecified			Zinc	6.00E-08	kg	Künniger et al. (2001)	1	1.56	(2,1,1,1,1,5);	
		→ clinker, at plant	yearly production volume: 318'875 t	construction materials	binder	No	CH	clinker, at plant	1.00E+00	kg	Künniger et al. (2001)				

Fig 6.1 Flows for "clinker, at plant " and their representation in the ecoinvent database (cont.)

6.1.1 Secondary Fuel

For the calculation of the specific consumption of alternative fuels, the mix feed for the Swiss cement industry in 1997 has been taken into account, as illustrated in Tab. 6.1 (Künniger et al. (2001)). The shares for the secondary fuel are subject to strong fluctuations, depending for example on offer and demand (waste wood) and on political situations (meat-and-bone meal and animal fat has been more easily available after the BSE crisis) as can be seen in Tab. 6.3.

Tab. 6.3 Share of secondary fuel for the clinker production in the year 1997 und 2001

Secondary fuel	Share 1997 [weight-%]	Share 2001 [weight-%]
Dried sludge*	16	18
Used oil	32.8	20
Solvent	12.7	10.6
Used tyre	10.8	8.7
Distillation residue	1.4	
Plastics	8.2	11.5
Rubber meal	0.3	
Crap coke from coke filter	0.8	
Meat-and-bone meal*	6.7	23
Animal fat*	2.0	
CSS (Substitute Solid Combustibles)**	8.2	
Others		8.2
Total	100	100

* Waste is assigned to regenerative resource.

** Assumption that 50% is from regenerative and 50% from fossil resources.

The amount of each secondary fuel used in 1997, the heating values, and the CO₂-emissions of all different fuels (primary and secondary) used for the firing of clinker are listed in Tab. 6.4.

All secondary fuels are assumed to enter the system without environmental burdens since they are declared waste. The burdens are allocated 100% to the product or process from which the secondary fuels are by-products, as assumed in this project (Frischknecht et al. (2007b)). Their use in cement kilns is thus preferred to waste incineration. Based on the source, CO₂ emissions are categorised in “regenerative from waste”, “fossil from waste” or “fossil from fossil fuel”. The fossil CO₂ emission from waste is allocated to “carbon dioxide, fossil”.

Tab. 6.4 Thermal values and CO₂-emissions for different primary and secondary fuels (Künniger et al. (2001))

Fuel	Primary or secondary fuel	C-fraction	CO ₂ -Emission in t/t fuel	Classification of CO ₂ -emission	Heating value in MJ / t fuel	Share 1997 (weight %)	Heating value in MJ/kg clinker	Kg fuel per kg clinker	CO ₂ -Emission in kg per kg clinker
Wood (absolute dry)	S	0.5	1.83	reg. from waste	17'100	0.0	0.0000	0.0000	0.0000
Paper, cardboard	S	0.4	1.46	reg. from waste	10'000	0.0	0.0000	0.0000	0.0000
Dried sludge	S	0.23	0.84	reg. from waste	9'700	16.0	0.0751	0.0077	0.0065
Used oil	S	0.88	3.22	fossil from waste	36'800	32.8	0.5839	0.0159	0.0511
Solvent	S	0.6	2.2	fossil from waste	26'600	12.7	0.1637	0.0062	0.0135
Used tyre	S	0.55	2.01	fossil from waste	26'200	10.8	0.1363	0.0052	0.0105
Straight-Run residue	S	0.68	2.49	fossil from waste	33'300	1.4	0.0225	0.0007	0.0017
Plastics	S	0.65	2.38	fossil from waste	25'000	8.2	0.0992	0.0040	0.0094
Rubber meal	S	0.45	1.65	fossil from waste	17'300	0.3	0.0025	0.0001	0.0002
Crap coke from coke filter	S	0.87	3.18	fossil from waste	21'000	0.8	0.0083	0.0004	0.0013
Meat-and-bone meal	S	0.35	1.28	reg. from waste	14'400	6.7	0.0466	0.0032	0.0041
Animal fat	S	0.72	2.64	reg. from waste	35'500	2.0	0.0342	0.0010	0.0025
CSS	S	0.26	0.95	50% fossil from waste 50% reg. from waste	12'700	8.2	0.0504	0.0040	0.0019
Heating oil heavy / medium	P	0.87	3.18	fossil from fossil fuel	40'600		1.0337	0.0255	0.0810
Heating oil light	P	0.86	3.15	fossil from fossil fuel	42'700		0.0160	0.0004	0.0012
Hard coal	P	0.85	3.11	fossil from fossil fuel	29'300		1.0381	0.0354	0.1102
Natural gas	P	0.8	2.93	fossil from fossil fuel	28'760		0.0068	0.0002	0.0007
Petroleum coke, thermal value of different companies	P	0.97	3.55	fossil from fossil fuel	25'000		0.0978	0.0039	0.0139
					30'200				
					32'500				
Total secondary fuel						100	1.2227	0.04832	0.1047
Total primary fuel							2.1924	0.0654	0.2069
Total							3.4151		

CO2 regenerative from waste	0.0151
CO2 fossil from waste	0.0896
CO2 fossil from primary fuel	0.2069

6.1.2 Secondary Raw Material

Each secondary raw material and the shares in 1997 are listed in Tab. 6.5. The mix has been calculated based on the information from eight Swiss plants. The data for each plant are confidential and therefore no breakdown is provided here. However the shares vary quite strongly from year to year.

Tab. 6.5 Shares of secondary raw materials for clinker production in Switzerland in 1997

Material	Share [%]
Aluminiumhydroxide	0.4 %
Building bulky goods, fine fraction	1.1 %
Diverse ashes from paper production plants	9.1 %
Decarbonation sludge	17.6 %
Moulding sand	1.7 %
Lime mud from limestone production	55.1 %
Oil smudged soil	5.6 %
Pyrite ash	4.0 %
Sludge	2.7 %
Others	2.9 %

As for wastes used as fuels, all secondary raw materials enter the system without burdens as they are declared waste. The burdens are allocated 100% to the product or process from which the secondary materials are by-products. Hence, though their flow is included in Fig 6.1, they are no input to the “clinker, at plant” module.

6.1.3 Emissions

The share of the particle emissions from the clinker firing process to the three different categories used in this project are taken from (EPA (1998)): 64% smaller than 2.5 μ m, 21% between 2.5 and 10 μ m, and 15% larger than 10 μ m.

The dust from mining included in Künniger et al. (2001) is not included in the present study because it is already inventoried in the input-module of the raw materials.

6.1.4 Infrastructure

Only the kiln has been taken into account. The lifespan is assumed to be 25 years. The land-use of the factory is not included in the clinker module. The whole land-use is included in the infrastructure module “cement plant”.

6.1.5 Data Check

A rough mass balance shows that for one kilogram clinker the total weight of raw materials needed (bauxite, hydrated lime, limestone, calcareous marl, sand and clay) is 1.77 kg, which is separated in 1.65 kg primary and 0.114 kg secondary raw materials. The main emission coming from these minerals is geogenic CO₂ with a weight of 0.543 kg per kg clinker. The rest is evaporated water included in some raw materials like sand and clay.

6.2 Lime hydraulic, at plant

The ecoinvent database meta information of “lime, hydraulic, at plant” (Tab. 6.6) includes all important and necessary information to use the data correctly.

Tab. 6.6 Ecoinvent database meta information for the product “lime, hydraulic, at plant”

Name	lime, hydraulic, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to burn hydraulic lime (raw material provision, grinding and mixing; rotary kiln process), internal processes (transport, etc.). For the infrastructure only the rotary kiln is taken into account (as material consumption). No administration is included.
Amount	1
Local Name	Kalk, hydraulisch, ab Werk
Synonyms	Klinker//Drehrohrofen//Zement//clinker//rotary kiln//cement
General Comment to reference function	Contrary to clinker production, in this module no waste materials as fuel or secondary raw materials are included. Accordingly, no CO ₂ -emissions from burning of waste are included. Other air emissions, assumed proportional to the fuel use, are estimated to be 88% of the emissions from the clinker calcination process, because the energy consumption is assumed to be 88% of the uses for clinker making.
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Copy of the clinker production technology with mainly adjustments of the energy amount and the emissions. Additionally some water for the hydration has been included.
Representativeness [%]	70
Production Volume	22952 t/a
Sampling Procedure	Data adapted from module "clinker, at plant" based on literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.2 shows the module “lime, hydraulic, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each data.

General Flow information						Representation in ecoinvent 2000							Uncertainty information				
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
bauxite, at mine	→	Hydraulic lime, at plant (Part 1)				amount copied from clinker-production module; primary raw material	metals	extraction	No	GLO	bauxite, at mine	1.20E-04	kg	assumption EMPA	1	5.00	estimation
lime, hydrated, loose, at plant	→					amount copied from clinker-production module; primary raw material	construction materials	binder	No	CH	lime, hydrated, loose, at plant	3.92E-03	kg	assumption EMPA	1	5.00	estimation
limestone or lime marl (with an amount of CaCO3 between 65 - 75%)	→					amount of primary raw material adjusted as assumed that compared to clinker production, no secondary raw material is used and it is assumed that the total amount of raw material is the same. Therefore includes the total amount of secondary raw material from clinker production.	construction materials	additives	No	CH	limestone, milled, loose, at plant	9.54E-01	kg	assumption EMPA	1	5.00	estimation
calcareous marl (with an amount of CaCO3 between 65 - 75%)	→					amount copied from clinker-production module; primary raw material, limestone used as proxy for calcareous marl	construction materials	additives	No	CH	limestone, milled, loose, at plant	4.66E-01	kg	assumption EMPA	1	5.00	estimation
sand, at plant	→					amount copied from clinker-production module; primary raw material	construction materials	additives	No	CH	sand, at mine	9.26E-03	kg	assumption EMPA	1	5.00	estimation
clay, at mine	→					amount copied from clinker-production module; raw material	construction materials	additives	No	CH	clay, at mine	3.31E-01	kg	assumption EMPA	1	5.00	estimation
heavy fuel oil, at regional storage CH	→					primary fuel	oil	fuels	No	CH	heavy fuel oil, at regional storage	3.69E-02	kg	assumption EMPA	1	5.00	estimation
hard coal from regional store	→					primary fuel	hard coal	fuels	No	WEU	hard coal, at regional storage	5.12E-02	kg	assumption EMPA	1	5.00	estimation
electricity	→					amount copied from clinker-production module	electricity	supply mix	No	CH	electricity, medium voltage, at grid	5.80E-02	kWh	assumption EMPA	1	5.00	estimation

Fig. 6.2 Flows for "lime hydraulic, at plant" and their representation in the ecoinvent database

General Flow information						Representation in ecoinvent 2000							Uncertainty information				
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
ammoniacal gas liquor (25%), at plant	→	Hydraulic lime, at plant (Part 2)				amount copied from clinker production process; total weight of the diluted ammoniacal gas liquor (including 25% of ammonia): 0.00363 kg; ancillary material for nitrogen oxide removal during burning process	chemicals	inorganics	No	CH	ammonia, liquid, at regional storehouse	9.08E-04	kg	assumption EMPA	1	5.00	estimation
transport, freight, rail	→				amount copied from clinker production process; according to data of the 8 cement factories in CH	transport systems	train	No	CH	transport, freight, rail	1.77E-02	tkm	assumption EMPA	1	5.00	estimation	
transport, lorry 16t	→				amount copied from clinker production process; according to data of the 8 cement factories in CH	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	8.61E-05	tkm	assumption EMPA	1	5.00	estimation	
transport, lorry 28t	→				amount adjusted from clinker production process by subtracting the transports for secondary raw material ; according to data of the 8 cement factories in CH	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	2.81E-04	tkm	assumption EMPA	1	5.00	estimation	
diesel, burned in building machine	→				amount copied from clinker-production module; internal transport; the energy used for mining of limestone (0.018 MJ) is not included here because already inventoried in the modules of each raw material.	construction processes	machinery	No	GLO	diesel, burned in building machine	1.34E-02	MJ	assumption EMPA	1	5.00	estimation	
transport hard coal from regional store to cement plant	→				amount copied from clinker-production module; estimated distance: 200km	transport systems	train	No	RER	transport, freight, rail	7.09E-03	tkm	assumption EMPA	1	5.00	estimation	
transport, van <3.5t	→				amount copied from clinker-production module; internal transport van (0.00033 Fzkm with fuel with converter and 0.000035 Fzkm conventional fuel)	transport systems	road	No	CH	transport, van <3.5t	7.09E-05	tkm	assumption EMPA	1	5.00	estimation	

Fig. 6.2 Flows for "lime hydraulic, at plant" and their representation in the ecoinvent database (cont.)

General Flow information						Representation in ecoinvent 2000							Uncertainty information				
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
fireproof brick, alkaline brick, at plant	→	Hydraulic lime, at plant (Part 3)				amount copied from clinker-production module; wearing parts in cement rotary kiln	construction materials	bricks	No	DE	refractory, basic, packed, at plant	1.90E-04	kg	assumption EMPA	1	5.00	estimation
fireproof brick, high alumina brick, at plant	→					amount copied from clinker-production module; wearing parts in cement rotary kiln	construction materials	bricks	No	DE	refractory, high aluminium oxide, packed, at plant	1.37E-04	kg	assumption EMPA	1	5.00	estimation
fireproof brick, fireclay brick, at plant	→					amount copied from clinker-production module; wearing parts in cement rotary kiln	construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	8.21E-05	kg	assumption EMPA	1	5.00	estimation
lubricating oil, at plant	→					amount copied from clinker-production module; lubricating grease /-oil	chemicals	organics	No	RER	lubricating oil, at plant	4.71E-05	kg	assumption EMPA	1	5.00	estimation
steel high-alloyed	→					amount copied from clinker-production module; wearing parts (metal)	metals	extraction	No	RER	chromium steel 18/8, at plant	5.86E-05	kg	assumption EMPA	1	5.00	estimation
process and cooling water	→					amount copied from clinker-production module; miscellaneous water consumption	resource	in water			Water, unspecified natural origin	1.62E-03	m3	assumption EMPA	1	5.00	estimation
tap water, at user	→					amount copied from clinker-production module	water supply	production	No	RER	tap water, at user	3.40E-01	kg	assumption EMPA	1	5.00	estimation
water consumption for hydration process	→					copied from hydration process for hydrated lime	resource	in water			Water, well, in ground	6.02E-01	m3	assumption EMPA	1	5.00	estimation
water for diluting the ammoniac	→					amount copied from clinker production process; weight of the diluted ammoniacal gas liquor (75% water): 0.00363 kg	resource	in water			Water, unspecified natural origin	2.72E-06	m3	assumption EMPA	1	5.00	estimation
infrastructure kiln	→					amount copied from clinker-production module; atention: Infrastructure includes only round kiln; weight assumption: 300t; assumption service life: 25years, rest of infrastructure in "cement, plant" for cement production included	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	3.76E-05	kg	assumption EMPA	1	5.00	estimation
cement plant	→				production volume over 50 years: about 16 Mio. t, assumed that 10% of total plant used for cement production	construction materials	others	Yes	CH	cement plant	6.27E-12	unit	assumption EMPA	1	5.00	estimation	

Fig. 6.2 Flows for "lime hydraulic, at plant" and their representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent 2000							Uncertainty information			
Input		Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Hydraulic lime, at plant (Part 4)	→		waste heat	Total amount (incl. electricity consumption)	air	unspecified			Heat, waste	3.21E+00	MJ	assumption EMPA	1	5.00	estimation
	→		waste to inert material landfill	amount copied from clinker-production module; primarily used fireproof bricks	waste management	inert material landfill	No	CH	disposal, inert waste, 5% water, to inert material landfill	8.00E-05	kg	assumption EMPA	1	5.00	estimation
	→		waste to municipal incineration	amount copied from clinker-production module; different materials, "disposal, municipal solid waste" used as proxy	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	4.50E-05	kg	assumption EMPA	1	5.00	estimation
	→		dust to air [kg]	amount copied and adjusted with factor 0.88 from clinker-production module; from combustion: PM>10 um = 0.15*TSP, adjusted by a factor of 0.88 in accordance to the lower energy consumption	air	low population density			Particulates, > 10 um	4.98E-06	kg	assumption EMPA	1	5.00	estimation
	→		dust to air [kg]	amount copied and adjusted with factor 0.88 from clinker-production module; from combustion: PM>2.5 um = 0.64*TSP, adjusted by a factor of 0.88 in accordance to the lower energy consumption	air	low population density			Particulates, < 2.5 um	2.12E-05	kg	assumption EMPA	1	5.00	estimation
	→		dust to air [kg]	amount copied and adjusted with factor 0.88 from clinker-production module; from combustion: PM<10, >2.5 = 0.21*TSP, adjusted by a factor of 0.88 in accordance to the lower energy consumption	air	low population density			Particulates, > 2.5 um, and < 10um	6.97E-06	kg	assumption EMPA	1	5.00	estimation
	→		carbon dioxide, fossil	from combustion of fossil fuel	air	unspecified			Carbon dioxide, fossil	2.77E-01	kg	assumption EMPA	1	5.00	estimation
	→		carbon dioxide, geogen	amount copied and adjusted with factor 0.88 from clinker-production module; geogenic carbon dioxide (from minerals)	air	unspecified			Carbon dioxide, fossil	4.78E-01	kg	assumption EMPA	1	5.00	estimation
	→		carbon monoxide, fossil	amount copied from clinker-production process and adjusted by a factor of 0.88 in accordance to the lower energy consumption	air	unspecified			Carbon monoxide, fossil	4.15E-04	kg	assumption EMPA	1	5.00	estimation

Fig. 6.2 Flows for "lime hydraulic, at plant" and their representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent 2000							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Hydraulic lime, at plant (Part 5)	→	arsenic	amount copied from clinker-production process and adjusted by a factor of 0.88 in accordance to the lower energy consumption	air	unspecified			Arsenic	1.06E-08	kg	assumption EMPA	1	5.00	estimation
		beryllium		air	unspecified			Beryllium	2.64E-09	kg	assumption EMPA	1	5.00	estimation
		cadmium		air	unspecified			Cadmium	6.16E-09	kg	assumption EMPA	1	5.00	estimation
		methane, fossil		air	unspecified			Methane, fossil	7.81E-06	kg	assumption EMPA	1	5.00	estimation
		cobalt		air	unspecified			Cobalt	3.52E-09	kg	assumption EMPA	1	5.00	estimation
		copper		air	unspecified			Copper	1.23E-08	kg	assumption EMPA	1	5.00	estimation
		hydrogen chloride		air	unspecified			Hydrogen chloride	5.55E-06	kg	assumption EMPA	1	5.00	estimation
		mercury		air	unspecified			Mercury	2.90E-08	kg	assumption EMPA	1	5.00	estimation
		ammonia		air	unspecified			Ammonia	2.01E-05	kg	assumption EMPA	1	5.00	estimation
		nickel		air	unspecified			Nickel	4.40E-09	kg	assumption EMPA	1	5.00	estimation
		NMVOC, non-methane volatile organic compounds, unspecified origin	amount copied from clinker-production process and adjusted by a factor of 0.88 in accordance to the lower energy consumption	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	4.96E-05	kg	assumption EMPA	1	5.00	estimation
		nitrogen oxides		air	unspecified			Nitrogen oxides	9.50E-04	kg	assumption EMPA	1	5.00	estimation
		chromium III		air	unspecified			Chromium	1.28E-09	kg	assumption EMPA	1	5.00	estimation
		chromium VI		air	unspecified			Chromium VI	4.84E-10	kg	assumption EMPA	1	5.00	estimation
		lead		air	unspecified			Lead	7.48E-08	kg	assumption EMPA	1	5.00	estimation
		antimony		air	unspecified			Antimony	1.76E-09	kg	assumption EMPA	1	5.00	estimation
		selenium		air	unspecified			Selenium	1.76E-09	kg	assumption EMPA	1	5.00	estimation
		tin		air	unspecified			Tin	7.92E-09	kg	assumption EMPA	1	5.00	estimation
		sulfur dioxide		air	unspecified			Sulfur dioxide	3.13E-04	kg	assumption EMPA	1	5.00	estimation
		dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin	amount copied from clinker-production process and adjusted by a factor of 0.88 in accordance to the lower energy consumption	air	unspecified			Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin	8.44E-13	kg	assumption EMPA	1	5.00	estimation
		thallium		air	unspecified			Thallium	1.14E-08	kg	assumption EMPA	1	5.00	estimation
		vanadium		air	unspecified			Vanadium	4.40E-09	kg	assumption EMPA	1	5.00	estimation
		zinc		air	unspecified			Zinc	5.28E-08	kg	assumption EMPA	1	5.00	estimation
		hydraulic lime, at plant	yearly production volume assumed to be the same as for clinker production: 318'875 t	construction materials	binder	No	CH	lime, hydraulic, at plant	1.00E+00	kg	assumption EMPA			

Fig. 6.2 Flows for "lime hydraulic, at plant" and their representation in the ecoinvent database (cont.)

The data are in general directly taken from the clinker firing process. Also the same annual production has been assumed for the infrastructure modules. The following input and output flows are not taken into account in the module or they have been adjusted based on these (sometimes just assumed) reasons:

- It is assumed that no secondary raw material is used because the quality of hydraulic lime presupposes a clear raw material composition (Tab. 3.7). In order for the present module to reach consistency with the module describing the clinker calcination process, and to have a correct mass balance, the secondary raw material is assigned to limestone.
- The transport requirements for secondary fuels and raw materials are not included.
- Instead of the different fuel types (including secondary fuel) used for the clinker production, the calcining process of hydraulic lime uses only coal and oil (Kündig et al. (1997)). It is assumed that the energy share is about 50% each. The total amount used in this study is based on the energy need of 3'000 to 4'000 kJ per kilogram hydraulic lime (Kündig et al. (1997)). The energy demand for the clinker production is about 3'400 kJ/kg clinker. Since calcining temperature is lower, it is assumed that the energy demand of hydraulic lime is smaller and therefore 3'000 kJ/kg is chosen.
- The amount of water for the hydration process is the same as for the hydration process of quicklime.
- In general all emissions are adjusted by a factor of 0.88 in accordance to the lower energy consumption for calcining.
- Fossil and regenerative CO₂-emissions from combustion of wastes are excluded, since no waste is burned. The CO₂-emissions from coal and oil are newly calculated as shown in Tab. 6.7.

Tab. 6.7 Thermal value and weight of fuel for the calcination process of hydraulic lime

Fuel	Primary or secondary fuel	C-amount	CO ₂ -Emission in t/t fuel	CO ₂ -emission revealed as	Thermal value in MJ / t fuel	share 1997 (weight %)	Thermal value in MJ/kg hydraulic lime	kg fuel per kg hydraulic lime	CO ₂ -Emission in kg per kg
Heating oil heavy / medium	P	0.87	3.18	fossil from fossil fuel	40'600		1.5000	0.0369	0.1175
Hard coal	P	0.85	3.11	fossil from fossil fuel	29'300		1.5000	0.0512	0.1592
Total secondary fuel							0.0000	0.0000	0.0000
Total primary fuel							3.0000	0.08814	0.2767
Total							3.0000		

Based on the above list, a quite high uncertainty has been defined for all in- and output products and processes.

The Swiss production in 1996 was 15'900t. The amount of imported hydraulic lime was 7'052t. If it is assumed that the stock balance is zero, the total consumption in Switzerland in 1996 was 22'952 t. Therefore the share of domestic production for market supply is 70%.

6.3 Calcareous marl, at plant

The ecoinvent database meta information of “calcareous marl, at plant” (Tab. 6.8) includes all important and necessary information to use the data correctly.

Tab. 6.8 Ecoinvent database meta information for the product “calcareous marl, at plant”

Name	calcareous marl, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes a mix of 3/4 pure limestone and 1/4 clay
Amount	1
Local Name	Kalkmergel, ab Werk
Synonyms	clay//lime//Kalk//Mergel//mergeliger
General Comment to reference function	
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	
Technology text	composed from pure limestone and clay
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	share of literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.3 shows the module “calcareous marl, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each data.

General Flow information					Representation in ecoinvent 2000										Uncertainty information		
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
pure limestone, crushed	→	mining of calcareous marls				3/4 of total amount	construction materials	additives	No	CH	limestone, crushed, washed	7.50E-01	kg	Kündig et al. (1997)	1	1.22	(4,2,1,1,1,3);
clay	→				1/4 of total amount	construction materials	additives	No	CH	clay, at mine	2.50E-01	kg	Kündig et al. (1997)	1	1.22	(4,2,1,1,1,3);	
						construction materials	additives	No	CH	calcareous marl, at plant	1.00E+00	kg	Kündig et al. (1997)				
				calcareous marl													

Fig. 6.3 Flows for “calcareous marl, at plant” and their representation in the ecoinvent database

6.4 Cement plant

The ecoinvent database meta information of “cement plant” (Tab. 6.9) includes all important and necessary information to use the data correctly.

Tab. 6.9 Ecoinvent database meta information for the product “mine, gravel/sand”

Name	cement plant
Location	CH
Infrastructure Process	1
Unit	unit
Data Set Version	2.0
Included Processes	includes the infrastructure (buildings, paved roads, etc.) and the land-use for the cement production, rotary kiln excluded (see 'clinker, at plant')
Amount	1
Local Name	Zementwerk
Synonyms	Zementwerk//Cement plant
General Comment to reference function	The lifespan of the infrastructure and the land occupation are assumed to be 50 years. The module is valid for a yearly production volume of about 340'000 t/a
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Average Swiss cement plant
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	most data measured by all Swiss plants (8 plants)
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.4 shows the module “cement plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each data.

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
infrastructure machines (exkl. clinker burning kiln)	→	cement plant				Infrastructure of all machines; weight assumption: 200t; assumption service life: 25years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	4.00E+05	kg	estimation EMPA	1	3.10	(3,5,1,2,1,5);
infrastructure: building hall						estimated area: 70% of total building area (33'600m2)	construction processes	buildings	Yes	CH	building, hall, steel construction	2.35E+04	m2	Künniger et al. (2001)	1	3.10	(3,5,1,1,1,5);
infrastructure: building multi storey						estimated area: 30% of total building area (33'600m2), assumed average hight: 12m	construction processes	buildings	Yes	RER	building, multi-storey	1.21E+05	m3	Künniger et al. (2001)	1	3.10	(3,5,1,2,1,5);
total land use (transformation)	→						resource	land			Transformation, from unknown	1.48E+05	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);
land occupation for green part of built area	→					assumption: life span 50 years, area about 53'000m2	resource	land			Occupation, industrial area, vegetation	2.64E+06	m2a	Künniger et al. (2001)	1	1.57	(2,3,1,1,1,5);
land transformation to green part of built area	→					area about 53'000m2	resource	land			Transformation, to industrial area, vegetation	5.28E+04	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);
land occupation for sealed part of built area	→					assumption: life span 50 years, area about 61'400m2	resource	land			Occupation, traffic area, road network	3.07E+06	m2a	Künniger et al. (2001)	1	1.57	(2,3,1,1,1,5);
land transformation to sealed part of built area	→					area about 61'400m2	resource	land			Transformation, to traffic area, road network	6.14E+04	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);
land occupation for building part of built area	→					assumption: life span 50 years, area about 33'600m2	resource	land			Occupation, industrial area, built up	1.68E+06	m2a	Künniger et al. (2001)	1	1.57	(2,3,1,1,1,5);
land transformation to building part of built area	→					total building area: about 33'600m2	resource	land			Transformation, to industrial area, built up	3.36E+04	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);
infrastructure conveyor belt	→				Assumed length 500 m; width 1.5 m; service life 20 years. The module in ecoinvent has a width of 3 m, which is here taken into account by reducing the used length to half.	construction processes	machinery	Yes	RER	conveyor belt, at plant	7.50E+02	m	estimation EMPA	1	3.10	(3,5,1,2,1,5);	
			→	cement plant		assumption: life span 50 years	construction materials	others	Yes	CH	cement plant	1.00E+00	unit				

Fig. 6.4 Flow for “cement plant” and their representation in the ecoinvent database

The life span of the portrayed plant is estimated to be about 50 years and the yearly production about 340'000 tonnes cement.

Machines used in cement plant:

As no exact information is available the weight of the most important machines stationed in the building halls is roughly estimated and summed up:

2 filtering installation (assumption each 10 t):	20 t
10 big silos (assumption each 10 t):	100 t
Diverse materials:	80 t
Total weight of machines:	200 t
Length of conveyor belt:	500 m
Width of conveyor belt:	1.5 m

The general module “conveyor belt” is used to describe the conveyor belt of the modelled cement plant. The width of the conveyor belt in the ecoinvent module is 3 m. This circumstance has been taken into account by shortening the input length from 500 to 250 m. Its service life is assumed to be 20 years. Within the assumed service life of 50 years of the cement plant, three replacements are necessary.

The infrastructure of the kiln for firing clinker is included in the module “clinker, at plant”. Several smaller machines as lifts, pumps, motors, etc. are not taken into account. Different buildings (halls and multi-storey) have been included (compare Fig. 6.4)

The service life for all machines in general is assumed to be 25 years.

6.5 Blast furnace slag cement, at plant

The ecoinvent database meta information of “blast furnace slag cement, at plant” (Tab. 6.10) includes all important and necessary information to use the data correctly.

Tab. 6.10 Ecoinvent database meta information for the product “blast furnace slag cement, at plant”

Name	blast furnace slag cement, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the manufacturing processes mixing and grinding, internal processes (transport, etc.) and infrastructure (specific machines and plant). No administration and no packing are included. Special inputs: blast furnace slag, not balanced as it is waste from iron production
Amount	1
Local Name	Hochofenzement, ab Werk
Synonyms	Zement//Portlandzement//cement//portland cement//CEM III B 32.5
General Comment to reference function	Part of total Swiss cement production: 2%. Composition: gypsum 4%, blast furnace slag 50%, clinker 46%
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Typical technology for Swiss production
Representativeness [%]	100
Production Volume	53766.06 t/a
Sampling Procedure	most data measured by all Swiss plants (8 plants)
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.5 shows the module “blast furnace slag cement, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each data.

The Swiss identifier for this type of cement is “CEM III B 32.5”. The most import part of the raw material for producing blast furnace slag cement is dry granulated blast furnace slag. It’s assumed that the raw material has no associated environmental burdens, as it is waste. To have it in the form usable for cement production it has to be mixed and grinded. Therefore a certain amount of electricity and oil has been inventoried in this module to take into account the energy requirements for these preparatory operations. Another important part of the raw material of blast furnace slag cement is clinker.

No dust emissions are taken into account as no data are available and as it is assumed that dust is “fully” recycled.

General Flow information						Representation in ecoinvent 2000							Uncertainty information				
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
blast furnace slag	→	blast furnace slag cement, at plant				dry granulated blast furnace slag, not balanced as it is waste from iron production, transportation is neglected						5.00E-01	kg	Künniger et al. (2001)			
electricity for grinding blast furnace slag	→					dry granulated blast furnace slag, electricity used for grinding blast furnace slag	electricity	supply mix	No	CH	electricity, medium voltage, at grid	4.99E-03	kWh	Marmé, Seeberger 1983	1	1.57	(2,1,5,2,1,5);
energy from heavy fuel oil for drying blast furnace slag	→					dry granulated blast furnace slag, energy used for drying blast furnace slag	oil	heating systems	No	CH	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	1.03E-01	MJ	Marmé, Seeberger 1983	1	1.57	(2,1,5,1,1,5);
clinker, at plant	→						construction materials	binder	No	CH	clinker, at plant	4.60E-01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
gypsum REA	→					not balanced as burden free						4.00E-02	kg	Künniger et al. (2001)			
transport, lorry 16t	→					materials, 50km	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	2.70E-02	tkm	Künniger et al. (2001)	1	2.06	(3,3,1,1,1,5);
electricity	→					energy for grinding and packing	electricity	supply mix	No	CH	electricity, medium voltage, at grid	7.33E-02	kWh	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
ethylene glycol, at plant	→					ancillary product for grinding	chemicals	organics	No	RER	ethylene glycol, at plant	5.50E-04	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
steel	→					wearing parts (metal)	metals	extraction	No	RER	steel, low-alloyed, at plant	1.10E-04	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
cement plant	→					production volume over 50 years: about 17 Mio. tonnes (50* 336'038 t), assumed that 90% of total plant used for cement production	construction materials	others	Yes	CH	cement plant	5.36E-11	unit	Künniger et al. (2001)	1	3.05	(2,1,1,1,1,5);
						total average yearly Swiss cement production amount of one factory: 336'038 tonnes	air	unspecified			Heat, waste	2.82E-01	MJ		1	1.22	(2,1,1,1,1,5);
							construction materials	binder	No	CH	blast furnace slag cement, at plant	1.00E+00	kg	Künniger et al. (2001)			

Fig. 6.5 Flows for "blast furnace slag cement" and their representation in the ecoinvent database

6.6 Portland calcareous cement, at plant

The ecoinvent database meta information of “portland calcareous cement, at plant” (Tab. 6.11) includes all important and necessary information to use the data correctly.

Tab. 6.11 Ecoinvent database meta information for the product “portland calcareous cement, at plant”

Name	portland calcareous cement, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the manufacturing processes mixing and grinding, internal processes (transport, etc.) and infrastructure (specific machines and plant). No administration and no packing are included. Special inputs: additional milling substances for example dust from the cement rotary kiln, fly ash, silica dust, limestone; these are not balanced as they are wastes to which no burdens are allocated within ecoinvent.
Amount	1
Local Name	Portlandkalksteinzement, ab Werk
Synonyms	Zement//Portlandzement//cement//portland cement//CEM II A-L 32.5
General Comment to reference function	Part of total Swiss cement production: 50%. Composition: gypsum 5%, additional milling substances 16%, clinker 79%
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Typical technology for Swiss production
Representativeness [%]	100
Production Volume	1344151.5 t/a
Sampling Procedure	most data measured by all Swiss plants (8 plants)
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.6 shows the module “portland calcareous cement, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each data.

The Swiss identifier for this type of cement is “CEM II A-L 32.5”. The most important raw material used for portland calcareous cement is clinker. Other important raw materials for portland calcareous cement making are different additional milling substances as dust from the rotary kiln, fly ash, silica dust or limestone. These raw materials have no burdens as they are declared as waste.

No dust emissions are taken into account as no data are available and as it is assumed that dust is “fully” recycled.

General Flow information						Representation in ecoinvent 2000							Uncertainty information				
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
clinker, at plant	→	portland calcareous cement, at plant					construction materials	binder	No	CH	clinker, at plant	7.87E-01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
gypsum, REA	→			not balanced as burden free								5.40E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
additional milling substances	→			for example dust from the cement rotary kiln, fly ash, silica dust, limestone, not balanced as it is waste without burdens								1.59E-01	kg	Künniger et al. (2001)			
transport, lorry 16t	→			distance could not be found out	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	1.07E-02	tkm	Künniger et al. (2001)	1	2.06	(3,3,1,1,1,5);		
electricity	→			energy for grinding and packing	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.86E-02	kWh	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);		
ethylene glycol	→			process material for grinding	chemicals	organics	No	RER	ethylene glycol, at plant	2.10E-04	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);		
steel	→			wearing parts (metal)	metals	extraction	No	RER	steel, low-alloyed, at plant	4.00E-05	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);		
cement plant	→			production volume over 50 years: about 17 Mio. tonnes (50* 336'038 t), assumed that 90% of total plant used for cement production	construction materials	others	Yes	CH	cement plant	5.36E-11	unit	Künniger et al. (2001)	1	3.05	(2,1,1,1,1,5);		
				Same amount as electricity used	air	unspecified			Heat, waste	1.03E-01	MJ		1	1.22	(2,1,1,2,1,5);		
	→			total average yearly Swiss cement production amount of one factory: 336'038 tonnes	construction materials	binder	No	CH	portland calcareous cement, at plant	1.00E+00	kg	Künniger et al. (2001)					
</																	

Fig. 6.6 Flows for "portland calcareous cement, at plant" and their representation in the ecoinvent database

6.7 Portland cement, strength class Z 42.5, at plant

The ecoinvent database meta information of “portland cement, strength class Z 42.5, at plant” (Tab. 6.12) includes all important and necessary information to use the data correctly.

Tab. 6.12 Ecoinvent database meta information for the product “portland cement, strength class Z 42.5, at plant”

Name	portland cement, strength class Z 42.5, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the manufacturing processes mixing and grinding, internal processes (transport, etc.) and infrastructure (specific machines and plant). No administration and no packing are included. Special inputs: additional milling substances for example dust from the cement rotary kiln, fly ash, silica dust, lime-stone; these are not balanced as they are wastes to which no burdens are allocated within ecoinvent.
Amount	1
Local Name	Portlandzement, Festigkeitsklasse Z 42.5, ab Werk
Synonyms	Zement//Portlandzement//cement//portland cement//CEM I 42.5
General Comment to reference function	Part of total Swiss cement production: 40%. Composition: gypsum 5%, additional milling substances 5%, clinker 90%
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Typical technology for Swiss production
Representativeness [%]	100
Production Volume	1075321.2 t/a
Sampling Procedure	most data measured by all Swiss plants (8 plants)
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.7 shows the module “portland cement, strength class Z 42.5, at plant”. It shows all in- and out-put flows, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each data.

The strength class Z 42.5 means (as described in chapter 3.2) that after 28 days the compressive strength is between 42.5 and 62.5 N/mm².

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
clinker, at plant	→	portland cement, strength class Z 42.5, at plant					construction materials	binder	No	CH	clinker, at plant	9.03E-01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
gypsum, REA	→			not balanced as burden free								5.20E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
additional milling substances	→			for example dust from the cement rotary kiln, fly ash, silica dust, limestone, not balanced as it is waste without burdens								4.50E-02	kg	Künniger et al. (2001)			
transport, lorry 16t	→			distance could not be found out	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	4.86E-03	tkm	Künniger et al. (2001)	1	2.06	(3,3,1,1,1,5);		
electricity	→			energy for grinding and packing	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.92E-02	kWh	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);		
ethylene glycol, at plant	→			process material for grinding	chemicals	organics	No	RER	ethylene glycol, at plant	1.90E-04	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);		
steel	→			wearing parts (metal)	metals	extraction	No	RER	steel, low-alloyed, at plant	5.00E-05	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);		
cement plant	→			production volume over 50 years: about 17 Mio. tonnes (50* 336'038 t), assumed that 90% of total plant used for cement production	construction materials	others	Yes	CH	cement plant	5.36E-11	unit	Künniger et al. (2001)	1	3.05	(2,1,1,1,1,5);		
				Same amount as electricity used	air	unspecified			Heat, waste	1.05E-01	MJ		1	1.22	(2,1,1,2,1,5);		
				total average yearly Swiss cement production amount of one factory: 336'038 tonnes	construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	1.00E+00	kg	Künniger et al. (2001)					
			waste heat														
			portland cement, strength class Z 42.5, at plant														

Fig. 6.7 Flows for "portland cement, strength class Z 42.5, at plant" and their representation in the ecoinvent database

The Swiss identifier for this type of cement is “CEM I 42.5”. The most important raw materials used for portland cement with the strength class Z 42.5 is clinker. Different additional milling substances are added in a small amount. These are for example dusts from the rotary kiln, fly ash, silica dust or limestone. It's assumed that these additional milling substances have no associated environmental burdens, as they are waste.

No dust emissions are taken into account as no data are available and as it is assumed that dust is “fully” recycled.

6.8 Portland cement, strength class Z 52.5, at plant

The ecoinvent database meta information of “portland cement, strength class Z 52.5, at plant” (Tab. 6.13) includes all important and necessary information to use the data correctly.

Tab. 6.13 Ecoinvent database meta information for the product “portland cement, strength class Z 52.5, at plant”

Name	portland cement, strength class Z 52.5, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the manufacturing processes mixing and grinding, internal processes (transport, etc.) and infrastructure (specific machines and plant). No administration and no packing are included. Special inputs: additional milling substances for example dust from the cement rotary kiln, fly ash, silica dust, limestone; these are not balanced as they are wastes to which no burdens are allocated within ecoinvent.
Amount	1
Local Name	Portlandzement, Festigkeitsklasse Z 52.5, ab Werk
Synonyms	Zement//Portlandzement//cement//portland cement//CEM I 52.5
General Comment to reference function	Part of total Swiss cement production: 6%. Composition: gypsum 6%, additional milling substances 3%, clinker 91%
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Typical technology for Swiss production
Representativeness [%]	100
Production Volume	161298.18t/a
Sampling Procedure	most data measured by all Swiss plants (8 plants)
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.8 shows the module “portland cement, strength class Z 52.5, at plant”. It shows all in- and out-put flows, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each data.

The strength class Z 52.5 means that after 28 days the compressive strength is bigger than 52.5 N/mm² (compare chapter 3.2).

General Flow information						Representation in ecoinvent 2000							Uncertainty information				
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
clinker, at plant	→	portland cement, strength class Z 52.5, at plant					construction materials	binder	No	CH	clinker, at plant	9.12E-01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
gypsum, REA, at plant	→			not balanced as burden free								6.30E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
additional milling substances	→			for example dust from the cement rotary kiln, fly ash, silica dust, limestone, not balanced as it is waste without burdens								2.50E-02	kg	Künniger et al. (2001)			
transport, lorry 16t	→			process materials 50 km	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	4.40E-03	tkm	Künniger et al. (2001)	1	2.06	(3,3,1,1,1,5);		
electricity	→			energy for grinding and packing	electricity	supply mix	No	CH	electricity, medium voltage, at grid	4.85E-02	kWh	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);		
ethylene glycol, at plant	→			process material for grinding	chemicals	organics	No	RER	ethylene glycol, at plant	3.50E-04	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);		
steel	→			wearing parts (metal)	metals	extraction	No	RER	steel, low-alloyed, at plant	6.00E-05	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);		
cement plant	→			production volume over 50 years: about 17 Mio. tonnes (50* 336'038 t), assumed that 90% of total plant used for cement production	construction materials	others	Yes	CH	cement plant	5.36E-11	unit	Künniger et al. (2001)	1	3.05	(2,1,1,1,1,5);		
				Same amount as electricity used	air	unspecified			Heat, waste	1.75E-01	MJ		1	1.22	(2,1,1,2,1,5);		
				total average yearly Swiss cement production amount of one factory: 336'038 tonnes	construction materials	binder	No	CH	portland cement, strength class Z 52.5, at plant	1.00E+00	kg	Künniger et al. (2001)					
			waste heat														
			portland cement, strength class Z 52.5, at plant														

Fig. 6.8 Flows for "portland cement, strength class Z 52.5, at plant" and their representation in the ecoinvent database

The Swiss identifier for this type of cement is "CEM I 52.5". The most important raw material used for portland cement with the strength class Z 42.5 is clinker. Different additional milling substances are added in a small amount. These are for example dusts from the rotary kiln, fly ash, silica dust or limestone. These additional milling substances have no burdens as they are declared as waste.

No dust emissions are taken into account as no data are available and as it is assumed that dust is "fully" recycled.

6.9 Portland slag sand cement, at plant

The ecoinvent database meta information of “portland slag sand cement, at plant” (Tab. 6.14) includes all important and necessary information to use the data correctly.

Tab. 6.14 Ecoinvent database meta information for the product “portland slag sand cement, at plant”

Name	portland slag sand cement, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the manufacturing processes mixing and grinding, internal processes (transport, etc.) and infrastructure (specific machines and plant). No administration and no packing are included. Special inputs: blast furnace slag, not balanced as it is waste from iron production
Amount	1
Local Name	Portlandhüttenzement, ab Werk
Synonyms	Zement//Portlandzement//cement//portland cement//CEM II A-S 42.5
General Comment to reference function	Part of total Swiss cement production: 2%. Composition: gypsum 5%, blast furnace slag 19%, clinker 76%
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Typical technology for Swiss production
Representativeness [%]	100
Production Volume	53766.06 t/a
Sampling Procedure	most data measured by all Swiss plants (8 plants)
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.9 shows the module “portland slag sand cement, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each data.

The Swiss identifier for this type of cement is “CEM II A-S 42.5”. Most important part of the raw material of slag sand cement is clinker. Another important part of the raw material of slag sand cement is blast furnace slag. This raw material has no burdens as it is waste. A certain additional amount of electricity and oil has been taken into account to make the material suitable for the use in cement (grinding and drying).

No dust emissions are taken into account as no data are available and as it is assumed that dust is “fully” recycled.

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
blast furnace slag	→	portland slag sand cement, at plant				dry granulated blast furnace slag, not balanced as it is waste from iron production, transportation is neglected						1.90E-01	kg	Künniger et al. (2001)			
electricity for grinding blast furnace slag	→					electricity	supply mix	No	CH	electricity, medium voltage, at grid	6.82E-03	kWh	Marmé, Seeberger 1983	1	1.57	(2,1,5,1,1,5);	
energy from heavy fuel oil for drying blast furnace slag	→					oil	heating systems	No	CH	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	3.91E-02	MJ	Marmé, Seeberger 1983	1	1.57	(2,1,5,1,1,5);	
clinker, at plant	→					construction materials	binder	No	CH	clinker, at plant	7.64E-01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
gypsum, REA, at plant	→										4.60E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
transport, lorry 16t	→					transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	1.18E-02	tkm	Künniger et al. (2001)	1	2.06	(3,3,1,1,1,5);	
electricity	→					electricity	supply mix	No	CH	electricity, medium voltage, at grid	4.13E-02	kWh	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
ethylene glycol, at plant	→					chemicals	organics	No	RER	ethylene glycol, at plant	3.10E-04	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);	
steel	→					metals	extraction	No	RER	steel, low-alloyed, at plant	6.00E-05	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);	
cement plant	→					construction materials	others	Yes	CH	cement plant	5.36E-11	unit	Künniger et al. (2001)	1	3.05	(2,1,1,1,1,5);	
	→					air	unspecified			Heat, waste	1.73E-01	MJ		1	1.22	(2,1,1,2,1,5);	
	→			waste heat portland slag sand cement, at plant		construction materials	binder	No	CH	portland slag sand cement, at plant	1.00E+00	kg	Künniger et al. (2001)				

Fig. 6.9 Flows for "portland slag sand cement, at plant" and their representation in the ecoinvent database

6.10 Cement, unspecified, at plant

The ecoinvent database meta information of “cement, unspecified, at plant” (Tab. 6.15) includes all important and necessary information to use the data correctly.

Tab. 6.15 Ecoinvent database meta information for the product “cement, unspecified, at plant”

Name	cement, unspecified, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	mix of different types of cement, based on CH statistics: 2% blast furnace slag cement, 50% portland calcareous cement, 40% portland cement, resistance class Z 42.5, 6% portland cement, resistance class Z 52.5, 2% portland slag sand cement
Amount	1
Local Name	Zement, unspezifisch, ab Werk
Synonyms	Fabrik//Betrieb//company//firmmineralisch//Isolation//Dämm//matte//platte//mineral//insulation//mat//board
General Comment to reference function	
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Typical technology for Swiss production
Representativeness [%]	100
Production Volume	2688303 t/a
Sampling Procedure	most data measured by all Swiss plants (8 plants)
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.10 shows the module “cement, unspecified, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each data.

This module is to be used if it is not clear which type of cement is needed. It represents the Swiss production mix (compare Tab. 6.1 in the introduction of chapter 5.7).

General Flow information						Representation in ecoinvent 2000							Uncertainty information				
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
blast furnace slag cement, at plant	→	cement, unspecified, at plant				2% of the Swiss market	construction materials	binder	No	CH	blast furnace slag cement, at plant	2.00E-02	kg	EMPA	1	1.11	(3,na,1,1,1,na);
portland calcareous cement, at plant	→					50% of the Swiss market	construction materials	binder	No	CH	portland calcareous cement, at plant	5.00E-01	kg	EMPA	1	1.11	(3,na,1,1,1,na);
portland cement, strength class Z 42.5, at plant	→					40% of the Swiss market	construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	4.00E-01	kg	EMPA	1	1.11	(3,na,1,1,1,na);
portland cement, strength class Z 52.5, at plant	→					6% of the Swiss market	construction materials	binder	No	CH	portland cement, strength class Z 52.5, at plant	6.00E-02	kg	EMPA	1	1.11	(3,na,1,1,1,na);
portland slag sand cement, at plant	→					2% of the Swiss market	construction materials	binder	No	CH	portland slag sand cement, at plant	2.00E-02	kg	EMPA	1	1.11	(3,na,1,1,1,na);
					cement, unspecified, at plant			construction materials	binder	No	CH	cement, unspecified, at plant	1.00E+00	kg	EMPA		

Fig. 6.10 Flows for "cement, unspecified, at plant" and their representation in the ecoinvent database

6.11 Cement cast plaster floor, at plant

The ecoinvent database meta information of “cement cast plaster floor, at plant” (Tab. 6.16) includes all important and necessary information to use the data correctly.

Tab. 6.16 Ecoinvent database meta information for the product “cement cast plaster floor, at plant”

Name	Cement cast plaster floor, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	It includes mixing of the raw materials (portland cement, sand and water), casting (either by pump or by bucket), ramming and finishing. Some transports and infrastructure are also included
Amount	1
Local Name	Zement Unterlagsboden, ab Werk
Synonyms	Estrich
General Comment to reference function	No losses and no waste water have been taken into account. The used cement is "portland cement, strength class Z 42.5, at plant"
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH- and RER-modules have been used as proxy
Technology text	typical technology for Swiss production
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.11 shows the module “cement cast plaster floor, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly Hutter et al. (1999)) and the standard deviation for each data.

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
portland cement	→	production of a cement cast plaster floor					construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	1.86E-01	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
aggregate: sand	→			aggregate can also be pumice and expanded clay or expanded perlite	construction materials	additives	No	CH	sand, at mine	7.40E-01	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);		
water	→				water supply	production	No	RER	tap water, at user	7.40E-02	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);		
electricity consumption	→			for proportioning, mixing and transporting	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.00E-03	kWh	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);		
transportation of raw materials to plant by lorry	→			Energy: 0.123MJ/kg, operation lorry 32 t -> 1.635MJ energy from diesel per tkm	transport systems	road	No	RER	transport, lorry >16t, fleet average	7.52E-02	tkm	Hutter et al. (1999)	1	2.34	(3,5,5,3,1,5);		
infrastructure (including building, machines and land-use)	→			amount copied from module "sand-lime brick"	construction materials	additives	Yes	CH	mine, clay	1.67E-10	unit	estimation	1	5.00	estimation		
packing	→			packing cement products used as proxy	construction materials	others	No	CH	packing, cement	1.00E+00	kg	estimation	1	5.00	estimation		
	→			same amount as electricity consumption	air	unspecified			Heat, waste	7.20E-03	MJ	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);		
	→			no data													
	→		cement cast plaster floor I (Type I means: 340 kg cement per m3)	construction materials	concrete	No	CH	cement cast plaster floor, at plant	1.00E+00	kg							
	→		waste heat														
	→		waste water														
	→		cement cast plaster floor														

Fig. 6.11 Flows for "packing, cement" and their representation in the ecoinvent database

6.12 Packing, cement

The ecoinvent database meta information of “packing, cement” (Tab. 6.17) includes all important and necessary information to use the data correctly.

Tab. 6.17 Ecoinvent database meta information for the product “packing, cement”

Name	packing, cement
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the process of packing, transportation within the plant and loading of the bags on paletts. The machines for these processes are taken into account. The waste treatment after use by the end-user and transports of the packing materials are included. Energy demand for packing not included as it is included in the different concrete types.
Amount	1
Local Name	Verpacken, Zement
Synonyms	
General Comment to reference function	The packing and loading process is normalized with an annual production volume of 1'500'000 kg/a. The estimated lifespan of the machines is 25 years.
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are from one company (packing, lime products) in Switzerland, adapted for cement.
Technology text	Assumption: same process as for packing of lime products
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Measured data of one Swiss plant for packing of lime products
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.12 shows the module “packing, cement”. It shows all in- and output flows, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each data.

This module is based on the packing and loading module of lime products adapted to cement products. The weight of the machines has been assumed to be 50 t.

The loading weight and geometry on a pallet does not differ much compared to the lime products and therefore the data used for lime are used directly here.

The electricity consumption for packing and loading has not been taken into account, as it is already included in the different cement modules, as the total was given by the companies.

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
infrastructure packing and loading of lime products	→	packing of cement (without energy consumption)				estimated total weight of machines: 50t, estimated lifespan: 25years, yearly estimated yearly packed amount: 1'500'000t	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.33E-06	kg	EMPA	1	1.24	(1,4,1,2,1,5);
conveyor belt	→				estimated length of conveyor belt: 10m, lifespan 20 years	construction processes	machinery	Yes	RER	conveyor belt, at plant	3.33E-10	m	EMPA	1	1.21	(4,na,na,na,na,na);	
pallets, at plant	→				40 bags à 25 kg on one pallet: total weight of load: 1 t, weight of pallet: 22kg; pallet used 10 times	wooden materials	processing	No	RER	EUR-flat pallet	1.00E-04	unit	Information Frantschach per Email 05.06.02	1	1.21	(4,na,na,na,na,na);	
paperbag, at plant	→					paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	1.57E-03	kg	Information Frantschach per Email 05.06.02	1	2.23	(5,na,na,na,na,na);	
plastic foil, at plant	→				circumference of pallet: 4m; height: 1m, 2 layer of plastic with a weight of 0.0123 kg/m2, loading weight 1000kg	plastics	processing	No	RER	extrusion, plastic film	1.23E-04	kg	http://www.dm-folien.de/hpage.htm, access Oct. 2003	1	1.50	(5,na,na,na,na,na);	
plastic foil, at plant	→				circumference of pallet: 4m; height: 1m, 2 layer of plastic with a weight of 0.0123 kg/m2, loading weight 1000kg	plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	1.23E-04	kg	http://www.dm-folien.de/hpage.htm, access Oct. 2003	1	2.23	(5,na,na,na,na,na);	
transport of pallets, paper bags and plastic bags to plant by train	→					standard distance: transport by train with distance of: 200km,	transport systems	train	No	CH	transport, freight, rail	4.74E-03	tkm	EMPA	1	1.50	(5,na,na,na,na,na);
transport of paper bags and plastic bags to municipal incineration						assumption: transports all by lorry 16t and total of transportdistance: 10km	transport systems	road	No	RER	transport, lorry 3.5-16t, fleet average	3.44E-05	tkm	EMPA	1	1.24	(1,4,1,2,1,5);
transport of pallets, paper bags and plastic bags to plant by lorry	→					standard distance: transport by lorry 28t with distance of: 50km,	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	3.59E-04	tkm	EMPA	1	1.21	(4,na,na,na,na,na);
	→			pallets, to disposal			waste management	municipal incineration	No	CH	disposal, wood untreated, 20% water, to municipal incineration	1.83E-03	kg	EMPA	1	1.32	(1,4,1,2,3,5);
	→		paperbag, in municipal waste incineration			waste management	municipal incineration	No	CH	disposal, paper, 11.2% water, to municipal incineration	3.32E-03	kg	EMPA	1	1.32	(1,4,1,2,3,5);	
	→		plastic foil, in municipal waste incineration			waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	1.23E-04	kg	EMPA	1	1.32	(1,4,1,2,3,5);	
	→		process of packing of 1 kg cement			construction materials	others	No	CH	packing, cement	1.00E+00	kg					

Fig. 6.12 Flows for "packing, cement" and their representation in the ecoinvent database

7 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007b)). For detailed results compare the different flows and the ecoinvent database meta information.

Most data of the clinker module are based on Künniger et al. (2001) which is based on the data from eight Swiss cement factories. The factories have a long tradition of collecting data. The clinker module included in this study represents an average clinker based on the total Swiss production in 1997. For these reasons, the data for clinker are of very good overall quality, although the infrastructure is roughly estimated. This fact has been considered in the uncertainties.

Most of the data of the cement modules are based on Künniger et al. (2001) which is based on the data from the same eight Swiss cement factories. The module for average cement represents the Swiss production share in 1997. For these reasons, the data for all cement modules are of very good overall quality although the infrastructure is only roughly estimated.

As no detailed data for the hydraulic lime production are available the data of the clinker production are used as basis. The adaptation is based on a series of assumptions, which results in a higher uncertainty. The data therefore are of poor overall quality.

The data of the module “packing, cement” are based on the module “packing, lime products”. Some adjustments due to differences in the density of products are made. As the difference between the two packing modules is quite small and the data quality of the module “packing, lime products” is quite good, the overall quality of the module “packing, cement” and is presumably also good.

The module “calcareous marl, at plant” is based on the assumption of one fourth of clay and three fourth of limestone. Whereas several combinations of these and additional components for calcareous marl exist, this module might not precisely describe these differences.

As different mixtures of raw material for different qualities of cement cast plaster floors exist, care has to be taken when using the ecoinvent module. The data is of good overall quality for the specific mixture considered in this study.

8 Abbreviations

BSE	“Mad cow disease” (Bovine Spongiform Encephalopathy)
CED	Cumulated Energy Demand
CSS	Substitute solid combustibles (in French: Combustibles solides de substitution)
EMPA	Swiss Federal Laboratories for Materials Testing and Research (Eidgenössische Materialprüfungs- und Forschungsanstalt)
EN	European Standard
ESP	Electrostatic Precipitators
ETHZ	Swiss Federal Institute of Technology Zurich (Eidgenössische Technische Hochschule Zürich)
MBT	Master Builder Technology Switzerland Ltd
PM	Particulate Matter
SIA	Association of Swiss Engineers and Architects (Schweizerischer Ingenieur- und Architektenverein)
TFB	Technical Research and Consulting on Cement and Concrete (Technische Forschung und Beratung für Zement und Beton)

9 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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Part III

Concrete Products and Processes

Data v2.0 (2007)

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Summary

This part deals with the five main concrete types used in Switzerland and an average mixing plant. The five types are poor and normal concrete, concrete for sole plates and foundations, pretentious concrete and pretentious concrete with de-icing salt contact.

Besides the concrete types also prefabricated products as concrete block, concrete roof tile and quarry tiles are dealt with in this report.

In general the production of concrete requires the raw materials cement, sand/gravel, admixture and water. In a ready-mixing plant the fresh concrete is produced by mixing these raw materials.

For all types of concrete, the main part of the energy demand and the emissions (incl. CO₂) come from the clinker production. Water emissions at the mixing process are modelled in a separate wastewater treatment module.

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1 Introduction

The following general information is taken from (Routschka & Granitzki (1997)).

Concrete is an artificial stone in which natural stones of well-graded sizes are bound together by a cement matrix, thus providing strength. Although this principle of manufacturing building materials can be traced back at least to the Roman Empire, it was later neglected for centuries. Concrete regained importance in the 19th century, when methods for large-scale production of cement were developed.

Concrete is a brittle material. Steel bars (reinforced concrete) or prestressed tendons (prestressed concrete) have greatly expanded its applicability. It is now one of the most important building materials because of its structural, physical and chemical properties and its economics. Fresh concrete can easily be shaped and most concrete is poured in place on the construction site. However, there is increasing use of concrete elements or concrete products that have been prefabricated in plants.

In its simplest composition, concrete is a mixture of cement, aggregate (gravel and sand) and water, as illustrated in Fig. 1.1:

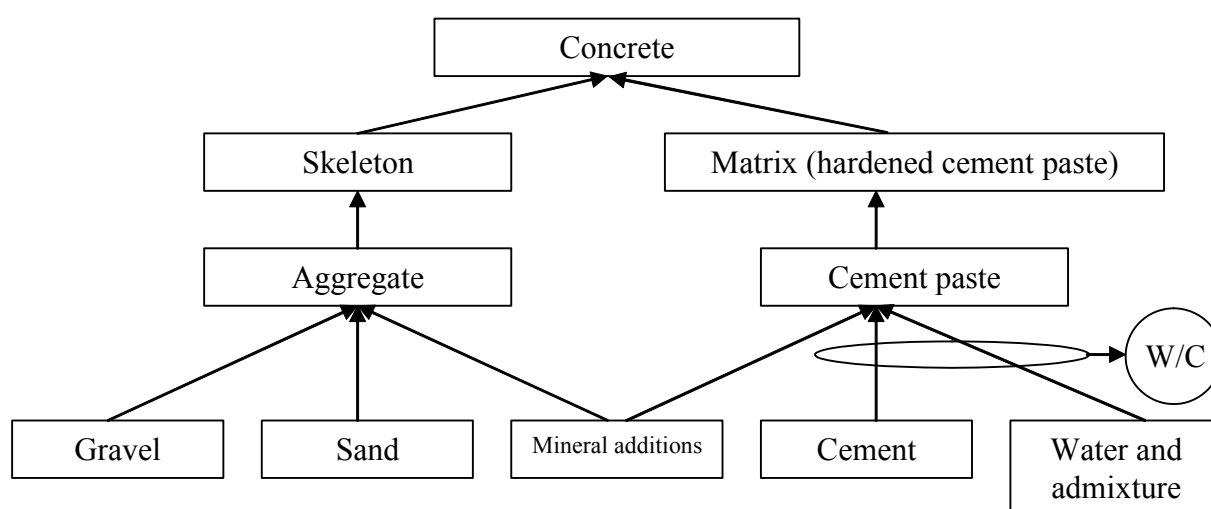


Fig. 1.1 Composition of concrete ¹

In 2000, ca. $1'560 \times 10^6$ tonnes of cement were produced world-wide to manufacture ca. 5×10^9 m³ of concrete for the construction industry². Seven countries – the People's Republic of China, the Soviet Union, Japan, the United States, Italy, Spain, and the Federal Republic of Germany – consumed more than half of the world production of cement and concrete.

Concrete is one of the less energy-intensive building materials. Concrete based on portland cement has a total energy content of only ca. 1250 MJ/m³, and even less when waste materials or industrial by-products such as fly ash, silica fume or granulated slags are incorporated.

The supply of natural concrete aggregates of good quality is not guaranteed world-wide, due to continuous consumption and because of nature conservation and landscape protection, a shortage of suitable material has been observed in some regions. Independently from the development of more sophisticated methods of recovery or treatment, the use of low-quality aggregates for special applications as

¹ <http://www.meynadier.ch/german/lexikon/index.htm> (Catchword: Beton/Festigkeitsklassen), access Jan. 2003

² http://www.ecosmart.ca/enviro_statistics.cfm; access Nov. 2003

well as recycling of old concrete, other building materials, or waste products must be considered in the future.

The properties of structural concrete depend to a great extent on its age and curing history. Physical and chemical properties must be tested by standardised procedures both before and during the construction process. National standards, as well as international standards or recommendations given by the International Organisation for Standardisation (ISO), the Comité Européen de Normalisation (CEN), the Comité Européen du Béton (CEB), or the Réunion Internationale des Laboratoires d'Essais et de Recherche sur les Matériaux et les Constructions (RILEM), cover quality control of concrete raw materials, as well as the concrete itself. The purchaser may specify additional guidelines.

2 Reserves and Resources

Basic materials for the production of concrete are cement, aggregates, and water. Up to certain limits, additional compounds such as concrete admixtures may be used to control some properties of fresh or hardened concrete. These materials must meet specifications set forth in standards or in approval documents. The compatibility of all compounds used must also be considered.

The reserves of the basic or raw material for the production of concrete are described in

- Part 1: Sand and Gravel and
- Part 2: Cement products and processes

3 Characterisation

This chapter deals with the concrete in its hardened state, not the ready-mixed and still wet concrete. Most of the physical properties of hardened concrete are controlled by its composition, mainly by the type and amount of aggregates used and the water content of the fresh or hardened concrete (Routschka & Granitzki (1997)).

3.1 General

3.1.1 Materials

This chapter deals with the basic materials for the production of concrete (cement, aggregates, and water).

Cement

The hydraulic binder cement reacts with water to form a highly dispersed cement paste matrix in which the aggregates are embedded. This cement matrix normally constitutes ca. 25 % of the volume of concrete. Different types of cement can be distinguished according to standard compressive strength, chemical composition, kinetics and heat of hydration, or resistance to chemical or physical attack. Economic considerations may influence the choice of the type of cement, depending on the local supply of raw materials for cement production or concrete aggregates.

Aggregates

The remaining 75 % of the volume of concrete is made of aggregates, which act as filler and give the concrete the properties needed in a structural building material (e.g., volume stability, limited crack

development, modulus of elasticity, wear resistance, ductility). The mechanical and physical properties of the aggregates may differ considerably from those of the cement paste. They are reflected in the properties of the hardened concrete but not necessarily in proportion to the concentrations of the aggregates. Concrete aggregates can be categorised according to their origin as natural or artificial aggregates; according to their density as light, normal, or heavyweight concrete aggregates or according to their composition as mineral, metallic, or organic materials. The last two types are used only for special concretes.

Hypes of aggregates used for producing high-quality concrete have a particle-size distribution in the range 0.01 mm – 0.1 m. The finer fraction is classified as sand, which comprises particles up to 4 – 5 mm in diameter; larger particles are called coarse aggregates. As a general rule, the size distribution of aggregates is selected to maximise their concentration in the concrete while maintaining sufficient workability of fresh concrete with low water content. Therefore, the total surface area of the aggregates should be kept as small as possible.

The particle-size distribution of aggregates is determined by sieve analysis, which subdivides a sample into fractions. Within each fraction, particles of sizes between a lower and an upper limit are present, depending on the mesh sizes of two consecutive sieves. For two consecutive sieves, these mesh openings are normally increased by a factor of 2. Specifications for sieves, as well as procedures for sieve analyses, are given in various national and international standards or recommendations.

After separation of the aggregates into fractions, grading is based on the specific surface area, water requirement factor, average particle size, fineness modulus, etc. The grading curve is a graphic presentation of the cumulative percentage retained on or passing through successive sieves.

In general, specifications of limiting grading curves for the aggregates as a whole, or for the sand only, are commonly used. The maximum aggregate size is usually specified. Concrete aggregates must not contain compounds that inhibit the hardening of the cement paste or that are vulnerable to chemical or physical degradation. Therefore, the permissible contents of organic or expansive compounds, soluble sulphates, silt, and clay are limited. Aggregates to be used in reinforced or pre-stressed concrete must not contain soluble compounds, such as chlorides, that attack steel. Aggregates containing reactive silica, e.g. opal, demand special, low-alkali cements to prevent expansive alkali–silica reactions in the hardened concrete.

Other characteristics of aggregates, such as modulus of elasticity, hardness, density, porosity, thermal conductivity, thermal expansion, and resistance to wear and frost action, must be considered in many applications.

Water

The mixing water added in the preparation of fresh concrete serves to hydrate the cement and assures the workability of the fresh concrete. Although only 20 – 40 % of the mass of cement must be water for the chemical reaction with cement, additional water is needed to achieve “complete” hydration, as well as good workability.

Impurities in the mixing water, such as sugar, organic compounds, oil, carbonic acid, sulphates, or other salts, may affect the hardening of the concrete or reduce the concrete strength. The concentration of dissolved chlorides is restricted for reinforced and pre-stressed concrete. Doubtful cases should be subjected to chemical analysis, but normally hints of excessive amounts of impurities can be derived from the colour, odour, taste, turbidity, or formation of bubbles or foam.

As a general rule, potable water is suitable as mixing water. In case of doubt, a trial mix with the water should be prepared and the development of concrete strength compared to that of a control mix. A trial mix is also important because the effect of impurities on concrete properties may depend on the type and amount of cement used.

Admixtures

Compounds may be added to control specific properties of the fresh or hardened concrete. These admixtures are either water-soluble (mainly organic) compounds (= admixtures) in low concentration or finely divided mineral material in higher concentration. The classification of admixtures with respect to their effect is arbitrary, because in most cases several concrete properties are influenced simultaneously. The majority of admixtures serve to control setting and hardening, workability, or concrete porosity.

- a.) *Admixtures controlling Setting and Hardening:* In some instances, the setting and hardening development of concrete must be accelerated or retarded to meet the needs of a specific construction. Set-retarding agents are used to prolong the period of workability during hot weather, to avoid discontinuities in subsequent placements of concrete, or to reduce crack formation due to deformations in an early stage. Accelerators are used when concrete is placed under cold weather conditions or when there is need to strip the formwork earlier than normal. Use of high concentrations of accelerators in shotcrete leads to rapid development of strength only a few seconds after mixing.
- b.) *Admixtures controlling Workability:* Many types of admixtures exert a significant influence on concrete workability while also affecting other properties such as setting time. The main objective of using such admixtures is to improve concrete workability. These plasticizers or superplasticizers reduce the amount of water needed and are the most commonly used admixtures, because they facilitate the placing and compacting of the concrete. Because they permit use of less water/cement ratios, strength and durability of concrete are improved.
- c.) *Admixtures controlling Porosity:* The resistance of concrete to frost and de-icing salts can be improved substantially by the use of air-entraining admixtures that form spherical pores with diameters considerably less than 0.5 mm. To prevent frost damage, these pores must be closely spaced and uniformly distributed in the cement paste matrix of the concrete. The necessary volume of entrained air depends on the mix proportion of the concrete and is normally 2 – 6 % by volume of the concrete. Foam-forming admixtures are used to make concrete with fine aggregates and a cellular structure. Such special lightweight concretes are primarily used for thermal insulation.

Defoaming agents are also available for controlling porosity, and are often added to plasticizers.

- d.) *Other Admixtures:* Other admixtures are used for special applications, for example, expansion-producing agents; corrosion inhibitors; fungicidal, insecticidal, or germicidal agents; damp-proofing and permeability-reducing admixtures; bonding agents; or pigments. Their total consumption, however, is low.

Much experience and careful control of concrete properties is required when admixtures are used because their effects may depend on many parameters, such as concrete composition, type of cement, and temperature. The simultaneous use of more than one admixture may lead to problems because of interferences and undesired concrete properties. In such situations a trial batch is mandatory.

- e.) *Fine-grained Material and Polymers:*
 - *Fine-grained Material:* The main groups of fine-grained mineral solids used as admixtures are powdered unreactive rock, such as limestone or quartz, which are either natural or industrial by-products. Unreactive rock material is an inert filler that can improve the workability of the fresh concrete, especially when the available aggregate lack fines.
 - *Reactive Minerals:* Natural pozzolanas include volcanic ashes, opaline shales, and cherts that contain reactive compounds of silicon and/or aluminium. Among the synthetic pozzolanas, pulverised fly ash is the most important admixture and is used in large quantities. Pozzolanas

may increase the strength and reduce the permeability of the hardened concrete because of chemical reactions with hydration products of the cement. These reactions consume the calcium hydroxide formed during the hydration of portland cement. Accordingly, excessive amounts of pozzolanas may lower the alkaline reserve of the concrete. Aside from improving certain concrete properties, the use of fly ash can reduce the amount of energy-expensive cement and save natural resources by recycling waste material. Its requirements as a concrete admixture are specified in standards or approval documents.

- *Silica Fume*: New developments in concrete technology have shown that silica fume – a pozzolanic waste material from silicon and ferrosilicon industries – opens new areas of application for advanced cement-based materials. Owing to its small particle sizes (ca. 0.1 μm), silica fume acts as a microfiller in the cement paste, thus reducing the total porosity considerably. Compressive strengths exceeding 250 MPa have been reported for concretes containing silica fume. For special applications, this new material can be used as substitute for cast iron, other metals, ceramics, or polymers.
- *Metal Oxides*: Colouring admixtures are generally alkali-resistant oxides of iron, titanium, or chromium. A uniform colouring effect is best achieved when such admixtures are used in combination with white cement.
- *Polymers*: A considerable increase in concrete strength and durability under severe conditions is achieved by the addition of certain polymers (polymer cement concrete, PCC). However, the properties of polymer-modified concrete may deteriorate on continuous exposure to high humidity.

3.1.2 Fresh Concrete

Workability

The term workability describes several characteristics of fresh concrete which are of particular importance for its placement, consolidation and finishing. These characteristics include flowability, resistance to plastic deformation, compactibility, cohesiveness and tendency to bleed or segregate. There is no single test method for evaluating all aspects of the workability of fresh concrete at once. Accordingly, test methods commonly determine either the flowability, expressed in terms of consistency, or the compactibility, which may be described by different compaction factors.

A widely used method for evaluating the consistency of fresh concrete is the slump test, which is specified (with minor differences) in various standards. In principle, in this test a frustum of a cone, which serves as a mold, is filled with concrete in a standard manner. After the mold is removed, the subsidence of the concrete cone is measured to determine the slump.

For tests concerning the compactibility of fresh concrete in general, a mold is filled with concrete in such a manner that any uncontrolled compaction is avoided. After the mold is filled, the fresh concrete is consolidated by vibration. Compaction factors are then derived from density measurements or a comparison of the original volume (height) of the fresh concrete to the volume (height) after compaction.

There is no strict correlation of results obtained from different methods for evaluating the workability of fresh concrete, and no single test method is universally applicable. Although plastic concrete mixes can be characterised by the slump or spread, the compacting tests are more appropriate for stiff and dry mixes.

Placement, Consolidation, and Finish

After mixing, the concrete must be placed in its final position, consolidated, and eventually surface-finished within 1 – 2 h and before the initial set. The method of placing concrete into forms depends

on its consistency and the equipment available at the site. Separate placement of single batches by buckets or buggies, as well as continuous placement by concrete pumps and hoses or conveyor belts, is successful if care is taken to avoid separation of the concrete constituents. To prevent cold joints, a subsequent layer must be placed while the first layer is still plastic and while knitting of the two layers can be achieved by vibration.

Consolidation of fresh concrete after casting removes entrapped air. Allowing for a monolithic and densely packed concrete structure, the reinforcement must be embedded thoroughly to provide a good bond with the concrete and to be protected from corrosion.

Fresh concrete is consolidated by vibration, which is usually provided by vibrators inserted into the concrete. Vibrators mounted on vertical formworks of thin concrete cross-sections or surface vibrators are also used. The vibration energy necessary for complete compaction decreases as the workability of the fresh concrete increases. Plasticizing admixtures may substantially reduce the consolidation work, and concrete containing superplasticizers may not require any consolidation (self compacting concrete). However, excess vibration can cause segregation.

Concrete may benefit from a second vibration after the initial setting to repair cracks that have developed due to settlement, early shrinkage, or deformations of the formwork.

The upper surface of fresh concrete can be troweled or screeded. Vacuum processing of the fresh concrete surface increases the durability of horizontal concrete surfaces, such as pavements and parking garage decks. Partial withdrawal of the mixing water results in lower porosity and higher strength of the surface region.

Curing

The development of concrete strength and impermeability requires favourable conditions for hydration of the cement. In addition, crack formation should be avoided. Curing of concrete, therefore, comprises all measures undertaken to protect the hardening concrete from drying and to control the temperature across a section.

Early loss of water from the young concrete may prevent further hydration of the cement, especially in the surface region, resulting in insufficient strength development and high permeability. Furthermore, premature drying causes extensive shrinkage, which will lead to random cracks in the surface layer. Protection against moisture loss is normally achieved by covering the concrete surface with water-absorbent materials such as burlap or straw, which are kept wet with damp-proof plastic sheets, or by the application of membrane-forming curing compounds. The last-named compounds contain wax or resins and are sprayed or brushed on the concrete surface; however, they do not provide sufficient protection for hot weather concrete curing.

Control of concrete temperature may be necessary when concrete is placed at low ambient temperature. Because the hydration of cement proceeds only slowly or even ceases at temperatures $< 5^{\circ}\text{C}$, thermal insulation of the concrete surface is used to prevent rapid loss of the heat of hydration. Thermal insulation also prevents excessive temperature gradients in a concrete cross-section and protects the young concrete from freezing. Thermal insulation is particularly important for massive concrete sections.

The required curing period depends primarily on the ambient climatic conditions and on the hydration kinetics of the cement used; it may vary from two days for rapidly hardening cement under moderate climatic conditions up to 14 days for massive concrete structures.

Interior sections of a concrete member are less affected by curing than surface regions. Adequate curing is, therefore, essential for achieving the required durability of a concrete structure.

The accelerating effect of elevated temperature on the hardening of concrete is used in the prefabrication of concrete elements; however, curing temperatures exceeding ca. 60°C may reduce the final strength of the concrete and also durability problems are possible (so called “delayed ettringite forma-

tion”). Steam curing at atmospheric or high pressure is applied in the preparation of certain concrete products such as pavement units, masonry units and precast elements.

Special Methods

A variety of special methods have been developed to simplify or increase the efficiency of preparing, transporting, placing, consolidating, and curing concrete.

For repair work, for strengthening existing structures, or for special applications, shotcrete may be superior to conventional concrete placing practice. In shotcrete, concrete is transferred pneumatically in a hose and projected through a nozzle onto a rigid surface at high velocity; consolidation is readily achieved by impact. Therefore the use of special shotcrete accelerators is necessary.

In construction of large concrete elements such as dams or for concrete placed under water, grout is injected into the voids of replaced coarse aggregates to produce concrete.

Special procedures also exist for the production of concrete products such as masonry units, concrete pavements, or concrete pipes.

3.1.3 Mechanical Properties

The most important design characteristics of concrete needed for structural analysis are the compressive strength and the modulus of elasticity of the hardened concrete; in many instances the tensile strength is not a design parameter. For concrete sections subjected to tensile stresses, a fracture mechanics approach is recommended because the tensile strength of concrete is controlled by such material defects as voids, pores, or microcracks. Structural design also requires information on dimensional changes due to creep, moisture migration, or temperature variation.

Crack Development

Defects or microcracks, which are caused by the nature of the hydrated cement paste or which are generated during the hardening of concrete, control the mechanical behaviour of concrete. The cement hydrates continuously, thereby dissicating the matrix and causing it to shrink. Additional shrinkage is due to evaporation of water from the concrete surface. Because the stiff aggregates restrain the shrinkage of the cement paste, microscopic cracks are formed in the interfaces between mortar and coarse aggregates. Crack propagation starts from these defects when the concrete is under stress. Tensile stresses and compressive stresses cause different types of crack propagation.

When concrete is compressed, microcracks start to propagate when the stress exceeds ca. 40 % of the maximum stress. With an increasing stress level, crack growth proceeds mainly along the interfaces between coarse aggregates and the matrix. At ca. 80 % of the maximum stress, the cracks start to extend into the mortar matrix. Frequently cracks are arrested at aggregates, which results in branch cracking. As a consequence, the nonlinearity of the stress–strain diagram increases substantially. The predominant orientation of the cracks is then parallel to the axis of loading. At high stress, microcracks join to form longer cracks. Thus, the cracks reach a critical length, and unstable crack growth can only be avoided by reduction of the stress. As long as crack propagation remains stable, the strains continue to increase and the stress–strain relationship shows a descending part (stress s = force/area, strain e = change in length/original length).

If the load is kept constant at a strain greater than the strain at maximum stress, a continuous fracture plane is formed, causing the complete collapse of the concrete.

Concrete loaded in tension behaves like a brittle material. Initiation of crack growth takes place at pre-existing microcracks, but multiple crack growth and branch cracking are much less pronounced than when under compression. Crack propagation is usually limited to one section. Nevertheless, a process zone is formed in front of a propagating crack, thus increasing the energy dissipation. At tensile failure stress, unstable crack growth forms a fracture surface perpendicular to the axis of applied tensile

stress. Sophisticated testing procedures can detect some stable crack propagation, which is more pronounced for larger amounts and sizes of aggregates.

Strength

Compressive Strength: The compressive strength of normal concrete is in the range 10 – 100 MPa (MN/m^2 or N/mm^2), depending on the composition and preparation procedures. The compressive strength of most concrete placed, however, is in the range 25 – 50 MPa (N/mm^2). The most important cement content parameter controlling concrete strength is the water to cement ratio. Other factors are the type of cement, curing conditions, moisture content, degree of hydration of the cement, and the grading and shape of aggregates. If sufficient water is provided, the hydration of cement will continue over a period of several years, and concrete strength may increase continuously during this time. It is customary to classify concrete according to the compressive strength attained after a standardised curing of 28 days. In certain cases, however, the compressive strength after shorter or longer curing periods may be specified.

Because of inevitable differences in the properties of concrete components, preparation procedures, handling, consolidation, and curing, the strength of any particular type of concrete varies within a range. Therefore for the classification of concrete strength, a characteristic strength f_{ck} is defined, below which only a certain percentage, usually 5 %, of the entire population of data falls. This value of f_{ck} defines a 5 % defective concrete. Assuming a normal frequency distribution of the individual strength values, the mean strength, f_{cm} required to reach a certain 5 % defective for a given standard deviation s can be estimated.

$$f_{cm} = f_{ck} + 1.64s$$

When good quality control is maintained, the standard deviation s is ca. 5 MPa. If a sufficient number of individual strength data are available, the standard deviation can be computed from the observed frequency distribution of individual strength results.

In general, the strength of concrete is determined by testing separately cast companion specimens, which are either cured under standard laboratory conditions or are exposed to the same curing conditions as the ultimate structure. Because the test results depend on the size and slenderness (i.e., height/thickness) of the concrete specimens tested, standard test specimens must be used. The most common specimens are cubes with a side length of 150 or 200 mm or cylinders with a diameter of 150- mm and 300 mm high. Specimen preparation and testing procedures are specified in the corresponding standards.

The use of drilled cores for destructive laboratory testing is the most reliable method for testing the strength of an existing concrete structure. Non-destructive methods such as ultrasonic pulse velocity, rebound, or penetration resistance require calibration.

The sustained load strength of concrete is controlled by two counteracting mechanisms. While a continuous increase of strain under a sustained load may weaken the integrity of the concrete, a strengthening effect is exerted by the continuing hydration of the cement, which is more pronounced for younger concrete than for older concrete. The sustained load strength of concrete loaded at an age of 28 days is ca. 80% of its short-time strength, which is less influenced by creep.

The fatigue strength of concrete decreases with an increasing number of stress cycles and an increasing stress range.

Tensile Strength: The tensile strength of normal concrete is approximately one-tenth the compressive strength. Detailed analyses predict that the tensile strength should not increase in proportion to the compressive strength. In general, the tensile strength of concrete is determined by testing beams subjected to flexure or cylinders that are subjected to diametrically opposed axial line loads, giving the so-called tensile splitting strength. The direct tensile strength can be determined by axial tension. Differ-

ent test methods yield different results because of differences in stress state, stress distribution, and probability of failure.

Fracture Mechanics: The application of linear elastic fracture mechanics to concrete is limited because of the heterogeneity of the material and because of the development of a process zone, i.e., a region of microcracks near major crack.

Accordingly, specimens with minimum dimension at least 20 times the maximum aggregate size should be used when the conventional fracture mechanics characteristics of a concrete are determined.

The fracture toughness K_{IC} of cement paste, mortar, and concrete are as follows:

hydrated cement paste	$K_{IC} = 5 - 15 \text{ N/mm}^{3/2}$
mortar	$K_{IC} = 10 - 20 \text{ N/mm}^{3/2}$
concrete	$K_{IC} = 15 - 50 \text{ N/mm}^{3/2}$

K_{IC} increases with increasing degree of hydration and decreasing water to cement ratio. Because aggregates act as crack arresters and cause multiple crack growth, K_{IC} increases with increasing content and size of aggregates.

Fracture mechanics concepts can be applied successfully in concrete technology and design, e.g., for the description of fracture processes, for estimating notch sensitivity, and for the design of large not reinforced concrete members.

More recently, the fracture energy G_F has been introduced as a fracture parameter. It corresponds to the total energy required to separate a concrete section loaded in tension. Since G_F is less dependent on size than K_{IC} , it can be used to analyse smaller concrete members.

Deformation Characteristics

The deformations of concrete under load can be separated into elastic and time-dependent deformations. In structural analysis, time-dependent strains are of considerable importance in estimating the serviceability of constrained members or prestressed structures in which a loss of prestress may occur. All time-dependent deformations are markedly influenced by the concrete composition and by the prevailing climatic conditions.

Relationship of Stress to Strain: Stress–strain diagrams of normal concretes with different strengths are presented for compression and tension in Fig. 3.1. The diagrams were recorded at a constant strain rate. Under compression, these diagrams are non-linear even at low stress levels. With increasing stress, the nonlinearity increases. The strain at maximum compressive stress is almost independent of the strength of the concrete and amounts to ca. 2×10^{-3} . The stress–strain relationships are also influenced by the strain rate. A reduction in strain rate causes a slight reduction in strength and a pronounced increase in the ductility of the concrete.

The stress–strain relationship for concrete loaded in tension is almost linear up to failure, which is normally brittle. Failure strains range from ca. 0.1×10^{-3} for a concrete loaded in axial tension to 0.3×10^{-3} for concrete subjected to flexure.

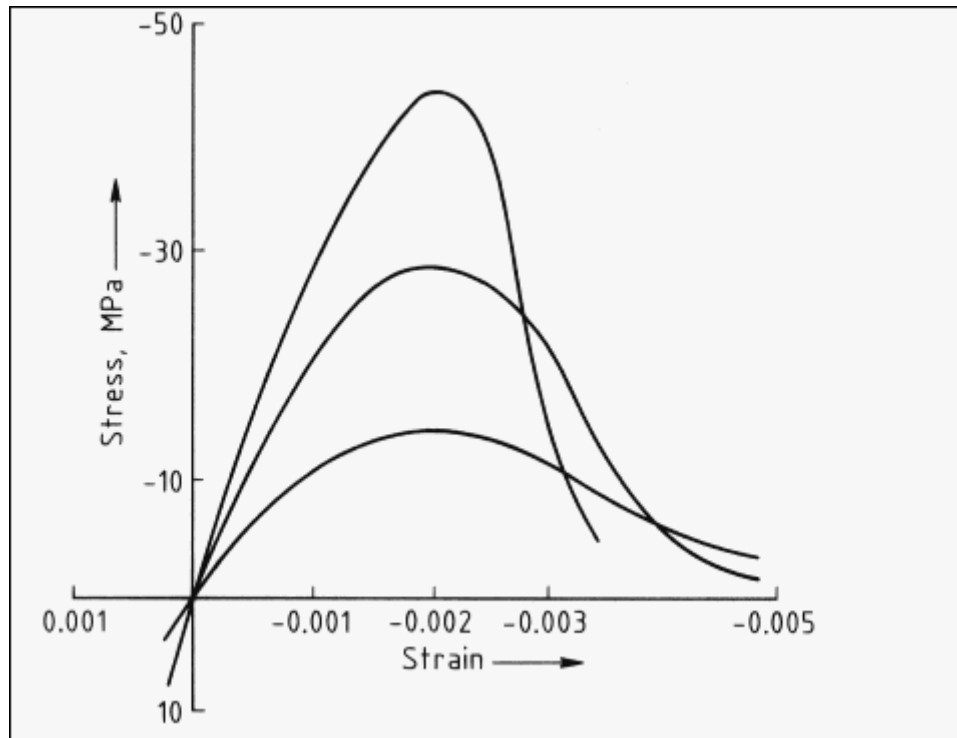


Fig. 3.1 Typical stress-strain relationship for concretes of different strength

Modulus of Elasticity and Poisson's Ratio: The data in Fig. 3.1 show that there is no constant modulus of elasticity for concrete in any part of the stress-strain curve.

The modulus of elasticity E_c for normal concrete is in the range 20000 – 40000 MPa. A higher compressive strength of the concrete leads to a higher modulus of elasticity. This correlation is expressed empirically as follows:

$$E_c = 9500 f_{cm}^{1/3}$$

where f_{cm} represents the mean compressive strength of a concrete cylinder.

The modulus of elasticity can also be deduced from the natural frequency of concrete specimens subjected to longitudinal, transverse, or torsional vibration as the dynamic modulus of elasticity. These non-destructive tests yield higher values of E_c .

Poisson's ratio ν of concrete is about 0.2 as long as the stress level s does not exceed $0.8 f_c$. At higher stress levels, values of Poisson's ratio exceeding 0.5 are observed due to multiple crack formation.

Creep: Creep is defined as the time-dependent increase of load-dependent strain under constant stress. It is often expressed in terms of the creep coefficient, ϕ_c , which is the ratio of creep strain, ϵ_{cc} , to initial strain, ϵ_{ce} . For the range of working loads, the creep coefficient of concrete ranges from ca. 1 to 4. Creep is partially reversible. The part of creep that can be recovered on unloading is referred to as delayed elastic strain, whereas the irreversible part is generally described as flow.

The water content of concrete exerts an important influence on creep deformation. Basic creep is defined as the creep of concrete in a state of moisture equilibrium. Basic creep is smaller as the water content of the concrete prior to load application decreases. Drying creep is defined as the difference between basic creep and the creep of concrete that is allowed to dry under load. Drying creep is thus proportional to moisture loss. Creep deformation increases because of microcracking, particularly at higher stress levels.

Creep of concrete increases with an increase in the amount of cement paste, of the water content of the concrete prior to loading, and of the water to cement ratio. Creep increases at low relative humidity and with decreasing member size. A higher degree of hydration at the time of loading reduces creep.

There are several mathematical methods for estimating the time development of creep. In these formulas, which may incorporate viscoelastic models, terms or functions represent the characteristics of the concrete, geometry of the member, and nature of the environment.

Shrinkage and Swelling: Shrinkage and swelling are load- and temperature-independent volume changes essentially caused by hydration (changes in the moisture content) of the concrete. In case of shrinkage the volume of the products is smaller than the volume of the educts. As a first approximation, shrinkage is proportional to the moisture loss. When the concrete is rewetted, swelling occurs, although some irreversible shrinkage may remain.

The total shrinkage of hydrated cement paste may amount to several points percent. Aggregates restrain shrinkage so that shrinkage of concrete is considerably less and is ca. 0.1×10^{-3} to 1×10^{-3} . Rapid drying of concrete generates shrinkage-induced tensile stresses in the surface zones that may lead to cracks.

3.1.4 Physical Properties

Density

Concrete is classified according to its density:

lightweight concrete class 1	$\rho \leq 1'850 \text{ kg/m}^3$
lightweight concrete class 2	$\rho \leq 2'000 \text{ kg/m}^3$
normal-weight concrete	$\rho \leq 2'800 \text{ kg/m}^3$
heavyweight concrete	$\rho > 2'800 \text{ kg/m}^3$

Structural lightweight concrete can be reinforced or prestressed like normal concrete. Heavyweight concrete is used mainly as a radiation-shielding material.

Porosity

The total porosity of concrete comprises gel pores and capillary pores in the hydrated cement paste, the porosity of aggregates and aggregate interfaces, and voids due to incomplete consolidation. Additionally, a protective pore system may be introduced into the matrix by air-entraining admixtures. The size of pores ranges from small gel pores with diameters of smaller than $0.03 \cdot 10^{-6} \text{ mm}$ up to several millimetres for the voids. In normal concrete with dense aggregates, the total porosity is 10 – 15 % by volume. Air-entrained concrete contains additional 2 – 6 % pores. Lightweight concrete with porous aggregates, lightweight concrete with a honeycomb structure, or cellular lightweight concrete may have a total porosity of 60 – 90 % by volume. Such types of concrete have low strength and are used only for thermal insulation.

Thermal Conductivity

The thermal conductivity of concrete increases with decreasing porosity and increasing moisture content. Crystalline aggregates give higher conductivity than amorphous aggregates. For ordinary concrete the thermal conductivity is $1.5 - 3.7 \text{ W m}^{-1} \text{ K}^{-1}$. The thermal conductivity is shown as a function of the density of dry concrete in following Fig. 3.2:

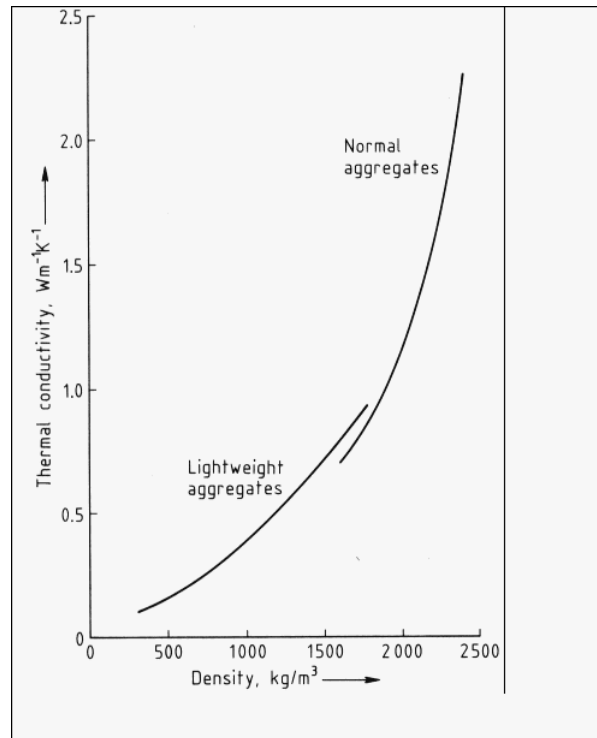


Fig. 3.2 Thermal conductivity of concrete as a function of concrete density (Routschka & Granitzki (1997))

Electrical Conductivity

Dissolved ions present in the pore solution of hardened concrete are primarily responsible for the electrical conductivity. For dry concrete, specific ohmic resistance is $10^{11} \Omega \text{ cm}$, whereas for concrete saturated with water it is ca. $10^4 \Omega \text{ cm}$. Concrete contaminated with soluble salts has even lower resistance. Because the water in the pores is an electrolyte, conductivity increases with increasing water to cement ratio.

Permeability

An important parameter for concrete durability is its permeability, which describes the ability of gases or liquids to penetrate into and more through concrete sections. The movement of gases or liquids in concrete occurs primarily in the capillary pores of the cement paste matrix. Because the volume of capillary pores increases with an increasing water to cement ratio, permeability increases correspondingly. High values of permeability can be expected when the capillaries form an interconnected continuous network caused by a low degree of hydration. For water : cement ratios exceeding ca. 0.7, continuity of capillaries must be assumed even after complete hydration.

For normal concrete mixes with dense aggregates, the *coefficient of water permeability* is 10^{-11} to 10^{-13} m/s . A lot of measures regarding *gas permeability* have been made at the EMPA in Dübendorf (Switzerland) by Mr. J. Kaufmann. Coefficients of gas permeability of 10^{-15} to 10^{-17} m/s have been reported.

Moisture movement is important in the drying of concrete. The diffusion of water molecules in the vapour phase or in condensed films depends strongly on the local moisture concentration and can be 10^{-8} to $10^{-11} \text{ m}^2/\text{s}$ for concretes of normal compositions at any relative humidity.

Thermal Expansion

The thermal expansion of concrete is controlled by the thermal expansion of the cement paste and the aggregates. Therefore, different aggregates lead to different coefficients of thermal expansion of the concrete. Restraint of the matrix by coarse aggregates or the formation of microcracks, however,

causes a total deformation of concrete that is not equal to the sum of the individual deformations of matrix and aggregates weighted by their volume concentrations. The coefficient of thermal expansion for concrete made with quartzitic aggregates is ca. $12 \times 10^{-6} \text{ K}^{-1}$. For concrete made with calcareous aggregates, it is ca. $6 \times 10^{-6} \text{ K}^{-1}$.

Changing temperatures always cause a moisture movement in the matrix, which may lead to shrinkage or swelling. These deformations are added to the thermal dimensional changes.

Shielding Properties

Concrete is an economical material for shielding against high-energy X-rays, gamma rays, or neutrons because it combines structural properties with good shielding characteristics. Because of its heterogeneity, concrete contains a variety of different elements for attenuation or absorption of the different types of radiation. Although these elements exist in normal concrete, special designs can improve the shielding properties.

Neutrons are classified according to their energy as fast, intermediate, and slow (thermal) neutrons. The attenuation of fast and intermediate neutrons by elastic scattering by light elements, such as hydrogen or boron, and inelastic scattering by heavy elements lead to slow neutrons that are finally absorbed by heavy elements. Gamma rays are emitted in the steps of neutron attenuation. Because the attenuation of X-rays and gamma rays is proportional to the density of the material penetrated, heavy elements are required for effective shielding.

Special aggregates for shielding concrete can be natural heavy-element aggregates like baryte, iron ores, or granulated iron. Light elements can be added by using aggregates that contain large amounts of crystal water such as serpentine or by the introduction of insoluble boron compounds.

In the structural design of a biological shield, one should keep in mind that the attenuation of radiation generates heat that causes stress. Furthermore, high-energy radiation causes crystal defects in the aggregates that may reduce the strength of the concrete.

3.1.5 Durability

Structural concrete is subjected to a variety of physical and physicochemical influences that result from environmental conditions or other types of attack. In most instances, concrete needs no particular protective measures. In reinforced or pre-stressed concrete structures, the concrete must prevent corrosion of the reinforcement by providing a high alkalinity to passivate the surface of steel and by a thick enough coverage of the steel.

Concrete aggregates are inert to most types of external attack. However, the hydrated cement paste may interact with the environment because of its interconnected pore structure. Water, ionic solutions, or gases may penetrate into the concrete through the capillary pores or microcracks to react chemically with constituents of the cement paste, thus causing deterioration. A durable concrete, therefore, should have a dense structure with a low total porosity. Low water to cement ratios and sufficient curing of the concrete are mandatory.

Another important factor in durability is the moisture content of the concrete. Because most damage mechanisms require a certain amount of water, dry concrete exhibits a high durability. Continuous wetting of concrete may increase the depth of penetration of dissolved compounds, and repeated drying–wetting cycles can accelerate the deterioration.

Chemical Attack

Chemical attack on hydrated cement paste involves either partial or complete dissolution of certain hydration products or formation of new compounds, accompanied by an increase in solid volume or in expansive gels.

Water that does not contain excessive amounts of ions or industrial waste pollutants does not dissolve cement paste compounds to a significant degree. However, continuing exposure to distilled water may leach the calcium hydroxide from the paste. Natural waters containing considerable amounts of carbonic acid will attack the cement paste and calcareous aggregates, forming soluble calcium hydrogen carbonate. Efflorescence of CaCO_3 is often observed.

Organic and inorganic acids dissolve the compounds of hydrated cement paste, and the rate of dissolution increases pH of the environment decreases. Moderate attack can be seen when the pH of the surrounding solution drops below 6.5 – 6.8.

The formation of expansive phases in interior concrete sections can be detected by the formation of a network of surface cracks, through which white efflorescence or penetration of a gel-like material can occur.

Sulphates penetrating into the matrix will react with either calcium hydroxide or calcium aluminates to form gypsum ($\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}$) or ettringite ($3 \text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3 \text{CaSO}_4 \cdot 32 \text{H}_2\text{O}$). The disruptive effect of ettringite formation is more pronounced because it involves a considerable increase in solid volume.

Magnesium compounds, either contained in the aggregates or penetrating from outside, can be decomposed to form an expansive gel of magnesium hydroxide.

Alkali-silica reactions occur between the alkali materials of the cement and amorphous silica of some siliceous aggregates. When moisture is available, an expansive gel is formed that causes a continuous deterioration of the concrete. Questionable aggregates can be tested for the presence of reactive silica. Their use as concrete aggregates may require special cements with low effective alkali contents.

The activity of natural waters and soils is evaluated in standards, where recommendations are given for the composition of concrete to be used in contact with them. Concrete that is subject to strong chemical attack may require protective coatings of organic materials on the surface.

Depending on the power of the chemical attack concrete with a certain quality must be chosen (capacity of cement, w/c-ratio, if necessary coating)

Physical Attack

a.) *Admixtures controlling Setting and Hardening:* In some instances, the setting and hardening development of concrete must be accelerated or retarded to meet the needs of a specific construction. Set-retarding agents are used to prolong the period of workability during hot weather, to avoid discontinuities in subsequent placements of concrete, or to reduce crack formation due to deformations in an early stage. Accelerators are used when concrete is placed under cold weather conditions or when there is need to strip the formwork earlier than normal. Use of high concentrations of accelerators in shotcrete leads to rapid development of strength only a few seconds after mixing.

a.) *Frost Action:* The mechanism of freezing of water in concrete is rather complex and not entirely clear. The following description is highly simplified.

Water contained in the larger capillary pores of concrete starts to freeze at temperatures below ca. -8°C . If the concrete is critically saturated, i.e., more than ca. 90 % of its pores are water-filled, the expansion of ice may have a disruptive effect if free expansion of ice or migration of the remaining liquid water into free spaces cannot take place. Thus, important parameters for controlling frost damage are the amount of freezable water in the concrete and the rate of ice formation.

The application of de-icing salts to a frozen concrete surface may cause further damage. While the ice on the concrete surface will melt, a sudden drop in the temperature of the underlying concrete layer occurs. Furthermore, de-icing salts can increase the degree of saturation and the

number of freeze–thaw cycles. Additional damaging effects due to osmotic pressure or pressure from salt crystallisation have not been clarified.

Preparing concrete with high resistance to freezing and de-icing agents requires aggregates with a high resistance to freezing and a dense pore structure of the cement paste that is provided by low water to cement ratios and sufficient curing. Furthermore, when critical saturation is likely to occur, a protective system of small, spherical, closely spaced pores must be introduced into the mortar matrix by air-entraining admixtures. Because these pores do not fill with water under normal conditions, they provide space for the expansion of ice or water.

- b.) *Elevated Temperature:* The resistance of concrete to elevated temperatures depends on the mineral composition of the aggregates and the water content of the concrete. The mortar matrix in a concrete that is allowed to dry while hot will exhibit large shrinkage deformations, whereas the aggregates expand as the temperature increases. These opposing deformations lead to the formation of cracks, thus reducing concrete strength. The cement paste decomposes at temperatures $> 250\text{ }^{\circ}\text{C}$.

Concrete exposed to temperatures $> 100\text{ }^{\circ}\text{C}$ in a moist state undergoes phase transformations in the hydrated cement paste. In the absence of finely divided siliceous material, these phase transformations increase the total porosity of the paste and cause a significant reduction in the concrete strength. If fine siliceous aggregates or siliceous admixtures are present, new compounds are formed that can compensate for the loss of strength.

- c.) *Radiation:* Concrete subjected to high-energy radiation may lose strength. While there is no damaging effect of radiation on the hydrated cement paste, lattice defects can be formed in some types of crystalline rocks used as aggregates. Heterogeneous rock material with crystalline compounds such as granites undergoes a substantial loss in strength.
- d.) *Abrasion and Wear:* The surface regions of concrete structures can be subjected to abrasion and wear due to traffic loads, sliding, or scraping of hard objects, or repeated impact. Damage is often encountered on pavements, walls of silos or bunkers, and in pipes and channels for water that carries solids (erosion). Water flowing at a high velocity can cause cavitation. Concrete with a high resistance to wear requires a low water to cement ratio and careful curing. Vacuum processing of the fresh concrete and the use of hard aggregates or admixtures can substantially improve the resistance to wear.

Corrosion of the Reinforcement

The surface zones of reinforced concrete structures can be damaged by corrosion of the embedded reinforcement. The formation of voluminous iron hydroxides generates tensile stresses that cause spalling of the concrete cover. The most common causes of corrosion of the reinforcement are carbonation of the concrete or intrusion of reactive ions such as chlorides.

Carbon dioxide in the atmosphere penetrates the surface zones of concrete and reacts with the alkaline hydration products of the cement. In this diffusion-controlled reaction, insoluble calcium carbonates are formed and the pH of the pore solution of the matrix drops from more than 12 to ca. 8.7. In this environment, an embedded steel reinforcement is no longer protected by passivation because the passive layers become unstable, and anodic dissolution of the iron occurs when oxygen and moisture is available.

The passive layer of steel in concrete can also be destroyed by chloride ions. The major sources of chloride contamination of concrete are de-icing agents and seawater.

Although some chloride can be immobilised chemically or physically by the hydration products of the cement, additional chloride ion greater than a threshold concentration will destroy the passive layers of the steel reinforcement locally, and corrosion occurs. Structures exposed to seawater may, therefore, require an increased concrete cover of the reinforcement.

Carbonation and intrusion of active ions are controlled by diffusion. A dense pore structure of the cement paste matrix, especially in the surface zones, reduces the penetration rate and thus improves durability. Under very severe conditions, additional protective measures such as surface coating of the concrete or the steel reinforcement may be required.

3.1.6 Special Concretes

The basic principles of concrete technology for normal concrete are generally valid for other types of concrete, which may differ from normal concrete with respect to composition or properties tailored for special applications. Preparation, handling, and testing of such concretes are often specified in separate standards or guidelines. The most important types of concrete that differ from normal concrete are lightweight and heavyweight concrete, mass concrete, fibre-reinforced concrete, and polymer concrete.

Lightweight Concrete

Lightweight concrete has a lower bulk density than normal concrete, which can be achieved in several ways. Structural lightweight concrete with a dense structure is prepared with lightweight aggregates, such as tuff, expanded shales, or calcined waste products. Structural lightweight concrete may be reinforced or pre-stressed like normal concrete.

Lightweight concrete with a honeycomb structure is prepared with only coarse aggregates, which are enveloped by a thin shell of mortar, thus providing a large volume of interstitial voids. Such concretes can be used for masonry units or drainage purposes. Cellular lightweight concrete can be prepared by introducing foam-forming admixtures into a mortar mix. Because of its low compressive strength, cellular lightweight concrete is used only for making masonry units or for thermal insulation.

The basic relations between the water to cement ratio and concrete strength are valid for structural lightweight concrete, although with limitations because the strength of the matrix may be higher than the strength of the aggregates. Furthermore, the definition of a water to cement ratio is uncertain because the porous aggregates can absorb various amounts of the water added to the mix. Preparing lightweight concrete, therefore, always requires trial mixes. In general, compressive strength increases with increasing bulk density. For densities of $1.0 - 2.0 \text{ kg/dm}^3$, the compressive strength is $10 - 60 \text{ MPa}$. Compared with normal concrete, lightweight concrete has a lower compressive strength, lower modulus of elasticity, lower thermal conductivity, and lower thermal expansion, but higher creep and shrinkage deformations.

Heavyweight Concrete

The major applications of heavyweight concrete are as shielding against nuclear radiation, as ballast, and as anchor bodies. Heavyweight aggregates like baryte, ores with densities of $3.6 - 4.6 \text{ kg/dm}^3$, slags, or iron can be used to achieve bulk densities of the concrete exceeding 2.8 kg/dm^3 . The mix design for heavyweight concrete follows the principles for normal concrete except that additional care must be taken to ensure the specified concrete density. Segregation must be avoided in handling heavyweight concrete. Therefore, comparatively dry mixes are preferred.

Massive Concrete

The term massive concrete refers not to a particular composition but rather to a concrete structure of such size that the heat of hydration of the cement must be considered. In interior parts of large concrete cross-sections, the heat of hydration may generate temperatures $> 80^\circ\text{C}$, causing thermal stresses that can lead to crack formation in the concrete during early hydration. The temperature increase in the concrete member can be kept low by using cements with a low heat of hydration, addition of finely divided pozzolanas, and reduction of the cement content of the mix. The size of the aggregates may be as great as $125 - 150 \text{ mm}$. The curing of massive concrete may require thermal insulation of the sur-

faces to avoid steep temperature gradients over the cross-section. Otherwise cracking of the concrete is likely to occur.

Fibre-reinforced Concrete

Short fibres randomly oriented and homogeneously distributed can reinforce the mortar matrix of concrete. The diameter of these fibres may vary from several micrometers to ca. 1 mm and the length from a few millimetres to 50 or 60 mm. High-strength steel fibres are used in fibre-reinforced shotcrete, and alkali-resistant glass fibres are sometimes used in thin shells or cladding elements. New types of organic fibres are being tested as substitutes for asbestos.

Fibre-reinforced concrete has good ductility in both compression and tension. Problems in manufacturing fibre-reinforced concrete are the limited workability of the fresh concrete and the diminished effectiveness of the fibres because of insufficient bonding between fibre and matrix.

Polymer Concrete

In polymer concrete, organic resins form an essential part of the binding material. Polymer concrete has high strength in tension and compression, and exhibits a superior resistance to chemical attack, freezing, and thawing as compared to normal concrete. Disadvantages may result from the high cost of the resins and reduced thermal stability compared to that of normal concrete. The polymers may be added to the fresh mix as the only binding material, e.g., epoxy concrete; they can be combined with water and cement, as in polymer cement concrete (PCC); or an already hardened concrete can be impregnated with monomers for subsequent polymerisation as in polymer impregnated concrete (PIC).

3.1.7 Mortar

Mortar is distinguished from concrete by the maximum size of its aggregates. In general, mortar contains only sand with a maximum grain size of 2 – 4 mm. Other binding materials such as lime or hydraulic lime can be used in addition to cement.

In concrete construction, cement mortars are used for connecting pre-cast elements, as injection mortars in ducts of posttensioned structures with bonded tendons, and for the repair of old concrete surfaces with areas of deterioration.

Because of the small maximum size of aggregates, mortars often contain higher water to cement ratios than are normally encountered in concrete technology. Therefore, they exhibit lower compressive strength and volume stability. These disadvantages can be offset by the use of water-reducing and expansion-producing admixtures.

The composition and strength requirements for masonry mortars are specified in national standards. Lightweight aggregate mortars with reduced thermal conductivity have been developed to improve the thermal insulation properties of a masonry structure. Polymer-modified cement-based mortars with fine sand as aggregate make possible bed joints only ca. 3 mm thick (thin-layer mortar). Centrally mixed mortars may contain retarding admixtures to allow placement of the mortar up to 48 h after mixing.

A variety of polymer-modified mortars designed for the repair of concrete structures with deteriorated surface zones is available. Basic requirements of mortar for such applications are the ability to form strong bonds with old concrete and a low shrinkage strain.

3.2 Types of Concrete Addressed in this Study

This inventory includes the five most important concrete types used in Switzerland. The selection is based on the Swiss market share:

Tab. 3.1 Most important types of concrete on the Swiss market ³

Type of concrete	Cement proportioning [kg/m ³]	Aggregate	Dry bulk density [kg/m ³]	Share [%]
Poor concrete	150	alluvial and/or crushed gravel	around 2'200	insignificant
Ordinary concrete	300 - 325	alluvial and crushed gravel	2'380 – 2'420	55
Concrete for sole plate and foundation	300	alluvial gravel	2'390 - 2'400	2
Exacting concrete with de-icing salt contact	300 - 330	alluvial gravel	2'380 – 2'450	28
Exacting concrete	350 - 375	alluvial gravel	2'400 – 2'440	15

The composition of these five types of concrete and specially the density in the wet form are listed in Tab. 3.2 (Classification based on Standard SIA 162):

Tab. 3.2 Composition of the different concrete types studied in Ecoinvent (based on Swiss nomenclature)

Name	Type	Cement		Water	Aggregate			w/c ¹⁾	Density ⁴⁾	Comment
		Type	[kg/m ³]	[kg/m ³]	[kg/m ³]	Type	Particle size		[kg/m ³]	
Poor concrete	Poor concrete	CEM II/A-L 32.5	150	109.5	1930	round	0/32	0.73	2190	
Ordinary Concrete	B 35/25	CEM I 42.5	300	186	1891	round	0/32	0.62	2377	
Concrete for sole plate and foundation	B 35/25	CEM III/B 32.5	325	178.8	1883	round	0/32	0.55	2387	
Exacting concrete with de-icing salt contact	B 45/35 FT	CEM I 42.5	330	148.5	1966	round	0/33	0.45	2445	admixture ²⁾
Exacting concrete	B 60/50	CEM I 42.5	375	150	1880	round	0/32	0.40	2405	mineral addition ³⁾ , admixture

¹⁾ w/c: ratio between amount of water and amount of cement

²⁾ Reagent: 1.32 kg air-entraining agent (Sodium salt of natural fatty acids and resins) and 3.96 kg high-performance plasticizer (naphthalene sulfonates condensate)

³⁾ Admixtures: 35 kg Silica dust and 2.8 kg high-performance plasticizer (naphthalene sulfonates condensate)

⁴⁾ The density of wet and dry concrete is differing marginally and therefore not distinguished⁴

An explanation of the way of describing the type of concrete in Switzerland according to the Standard SIA 162 by means of an example⁵.

³ E-mail communication from Mr. K. Moser of EMPA Dübendorf on 22.08.2002

⁴ Personal communication from Mr. R. Steiger of EMPA Dübendorf in August 2003

Concrete type B25/35:

- B: Concrete with a density of between 2'000 and 2'800 kg/m³ (The other category is LB: light weight concrete with density <2'000 kg/m³)
- 35: Mean value of compressive strength in N/mm² (starting from demanded minimum value under consideration of a standard deviation of 5 N/mm²)
- 25: Minimum value of compressive strength in N/mm² (if a large number of tests has been made more than 98% of the results must reach or exceed this minimum value)

The way of describing the cement types is explained in the report no. 2, Cement Products and Processes.

4 Use

In 2000, about 250 plants in Switzerland produced about 8.6 million m³ of ready-mixed concrete. These are about 80% of the total Swiss production of concrete.

The product for every different concrete modules described in this report is ready-mixed concrete. Concrete in its final form is combined with reinforcement steel. The steel has mainly the function to reduce the tension and the hardened concrete to absorb the compression forces.

Following applications are applied to ready-mixed concretes (unless differently noted, the information is taken from Wendehorst & Vollenschaar (1998)):

- **Poor concrete ⁶:** Poor concrete has a low part of cement. It is mainly used for fillings which have to be highly compression proof, for granular subbases, for levelling out inequalities in the subsoil, as protection layer over/under sealings, and for hardening and filling.
- **Ordinary Concrete:** The most common and used type of concrete
- **Concrete for sole plate and foundation:** Concrete which is rarely used and only for sole plate and foundation. The used cement is "blast furnace slag cement".
- **Concrete, exacting, with de-icing salt contact:** Concrete which is strongly exposed to different environmental impacts as freeze-thaw-changes, penetration of deicing salts etc. Therefore a special composition is necessary. The application is mainly for constructions in shore protection installations, harbour dam, bridges in frost regions, etc.
- **Concrete, exacting:** Concrete which is strongly exposed to different environmental impacts as for example sudden freeze-thaw-changes in moisture penetrated conditions or chemical attacks. Also the need of a high wearing resistance or just simply a massive construction deduces exacting concrete. Possible usages are railway bridges, bridge piers without salt contact, barrages, etc.

⁵ <http://www.meynadier.ch/german/lexikon/index.htm> (Catchword: Beton/Festigkeitsklassen), access date: March 2003

⁶ http://www.baumarkt.de/b_markt/service/magerbet.htm

Specific product made of concrete are concrete blocks, roof tiles and quarry tiles. Concrete blocks are not reinforced and can be used as pavement concrete blocks, promenade tiles, kerbstones or bricks. Concrete roof tiles are used for roofing and concrete quarry tiles are mainly used in landscape architecture.

5 Process Description

The following concrete production process description is copied from (Locher & Kropp (1997)).

5.1 Concrete Production Process

5.1.1 Definition of Concrete Properties

Advances in concrete technology have established the basic rules for making concrete with desired properties in both the fresh and hardened conditions. The desired properties of fresh concrete must be deduced from the working conditions and equipment available at a particular construction site, whereas the requirements for hardened concrete are determined by the expected service conditions. Strength, physical properties, and durability must be considered. A careful analysis of all required concrete properties is necessary before the selection of materials and their proportions is made.

5.1.2 Proportioning of Components

Basic concrete materials and admixtures must be proportioned in such a manner that all desired properties of the concrete are achieved. There is no general approach, but rather empirical methods for determining the proper mix proportions, based on either volume or density. The volume or mass of the individual concrete constituents is determined in consecutive steps. One of several possible approaches to obtain a first estimate of the mix proportions is presented below.

The concrete quality is specified by its characteristic compressive strength, which normally is determined after 28 days on cubes or cylinders. Because of variations in the strength of concrete, the mean strength f_{cm} of a set of test samples must exceed the required characteristic strength f_{ck} by a certain amount, which can be derived by statistical methods.

For normal aggregates and a given type of cement, the strength of completely compacted concrete is a function of the water to cement ratio. Approximate values of this ratio for achieving different strength levels are given in Fig. 5.1 for non-air-entrained and air-entrained concretes; these values can be obtained from diagrams for different types of cement, such as shown in Fig. 5.2 for non-air-entrained concrete.

The content of mixing water required to achieve a certain consistency of the fresh concrete depends mainly on the maximum size and the grading of the aggregates. As a general rule, more sand or fine particles and smaller maximum aggregate size require more mixing water. For angular coarse aggregates, Fig. 5.3 gives maximum quantities of mixing water as a function of aggregate size and consistency as described by the slump (settlement allowance) of the fresh concrete.

The amount of water required for a concrete of given consistency may be estimated from the parameter k , which is defined as the sum of the percentages of aggregate retained on 9 standard sieves (0.25; 0.5; 1.0; 2.0; 4.0; 8.0; 16.0; 31.5; and 63 mm mesh), divided by 100:

$$k = 1/100 \sum R_i$$

Where R_i is the percentage of material retained on each of nine sieves, each of which is charged with the complete range of aggregate sizes.

In Fig. 5.4 the water content of concrete with rounded aggregates is shown as a function of the water requirement factor for various ranges of consistency as described by the compaction factor.

Compressive strength, MPa	Water : cement ratio	
	Non-air-entrained	Air-entrained
45	0.38	
40	0.43	
35	0.48	0.40
30	0.55	0.46
25	0.62	0.53
20	0.70	0.61
15	0.80	0.71

Fig. 5.1 Relationship between water to cement ratio and compressive strength for concrete cylinders

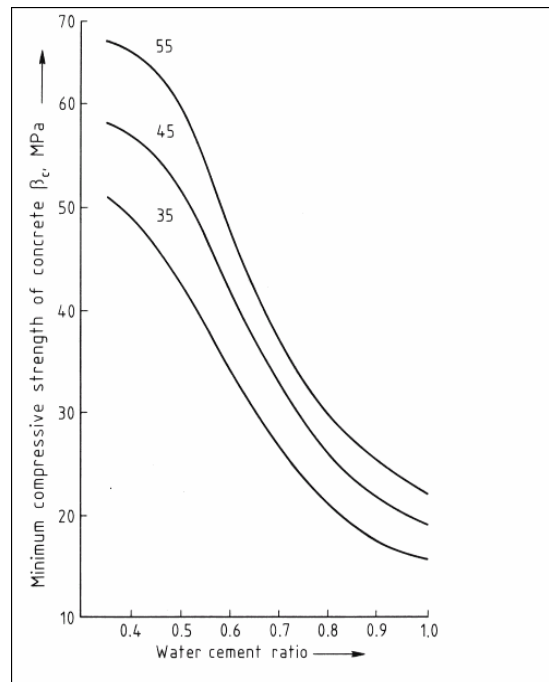


Fig. 5.2 Min. compressive strength β_c of concrete cubes for cements with different standard compressive strengths

Slump, mm	Water content, kg/m ³				
	Maximum diameter of aggregate, mm				
	10	12.5	20	25	40
30–50	205	200	185	180	160
80–100	225	215	200	195	175
150–180	240	230	210	205	185

Fig. 5.3 Approximate water content of fresh concrete

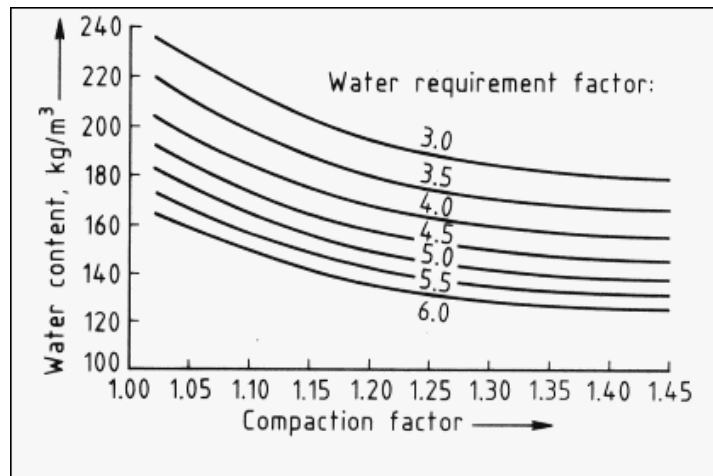


Fig. 5.4 Water content of fresh concrete as function of the compaction factor for different water requirement factors

The amount of water added to the mix may need to be corrected for the moisture content of the aggregates, particularly sand. After the water content is estimated, the cement content of the mix is determined from the water to cement ratio. The amount of aggregates can be calculated by considering the absolute volume of the various concrete constituents contained in 1 m³ of compacted fresh concrete.

In this example, no restrictions on the water to cement ratio or cement content is considered. Furthermore, the results obtained by any method of mix design will give only a first estimate of the concrete composition. Necessary corrections to the individual parameters are determined by trial batches. Trial batches are mandatory whenever admixtures are used because the effects of the admixtures cannot be reliably predicted and may depend on the temperature, intensity of mixing, and composition of the concrete.

5.1.3 Production Process

For the production of uniform concrete of high quality, adequate storage facilities for the individual components must be provided. The cement should be protected from moisture to prevent partial pre-hydration. Fine and coarse aggregates stored in separate fractions must be handled carefully to avoid segregation. Sand, if not stored in protected bins, should be drained to avoid excessive water. Liquid admixtures must be protected from freezing and thoroughly mixed before use.

All solid materials are measured by mass; liquids can be measured by volume. All measuring devices should be calibrated regularly to operate accurately within a margin of 1 – 3%. On large construction

sites or in ready-mixed concrete plants, measurement of the components may be automated at the bin discharge. These components are carried to the mixing station by conveyor belts. Automatic sensing of the surface moisture of the aggregates allows for the correction of the amount of water to be added to the mix. Solid admixtures are added to the mix with the cement, whereas liquid admixtures are mixed with the water. Superplasticizers are effective for only a limited period of time. Therefore, they are added to the fresh concrete during a second mixing immediately before placing the concrete.

To achieve a thoroughly mixed and homogeneous fresh concrete, the materials are charged into mixers which generally consist of revolving drums equipped with blades, stationary or revolving pans in which blades rotate on vertical shafts, or horizontal drums in which spiral blades operate on a rotating horizontal shaft. The necessary mixing time depends on the intensity of mixing, as well as on the batch size, and normally ranges from 30 s to 120 s. Truck-mixed concrete may require up to 100 revolutions of the truck-mounted drum rotating at mixing speed.

Fig. 5.5 gives an overview of a ready-mix concrete plant.

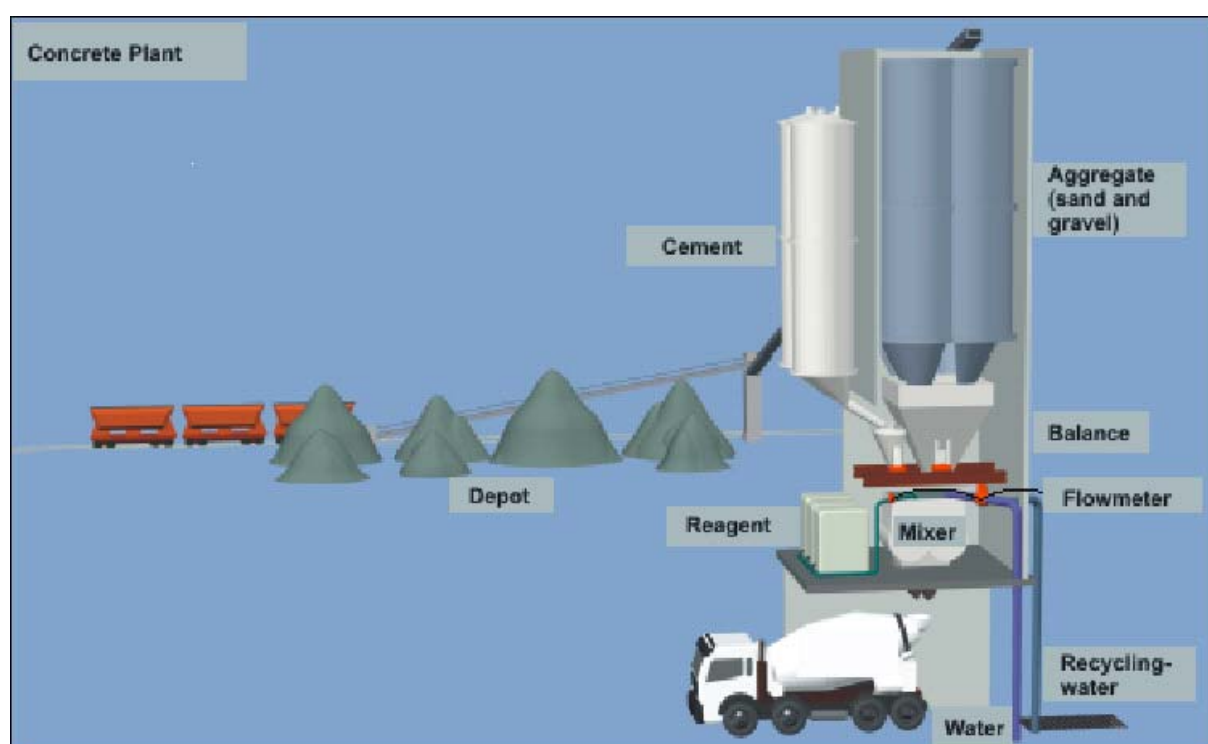


Fig. 5.5 Ready-mixed Concrete Plant (Künniger et al. (2001))

5.2 Emissions

In general, particulate matter consisting primarily of cement and pozzolan dust but also including some aggregate and sand dust emissions, is the primary pollutant of concern (EPA (1998)). In addition, there are emissions of metals that are associated with this particulate matter. All but one of the emission points is fugitive in nature. The only point sources are the transfer of cement and pozzolan material to silos, and these are usually vented to a fabric filter or "sock". Fugitive sources include the transfer of sand and aggregate, truck loading, mixer loading, vehicle traffic, and wind erosion of sand and aggregate storage piles. The amount of fugitive emissions generated during the transfer of sand and aggregate depends primarily on the surface moisture content of these materials. The extent of fugitive emission control varies widely from plant to plant.

Types of dust controls used may include water sprays, enclosures, hoods, curtains, shrouds, movable and telescoping chutes, and the like. A major source of potential emissions, the movement of heavy trucks over unpaved or dusty surfaces in and around the plant, can be controlled by good maintenance and wetting of the road surface.

Künniger et al. (2001) do not consider any dust emissions due to lack of data.

Despite the usually closed water cycle in concrete mixing plants some wastewater has to be lead in the sewerage system. Künniger et al. (2001) presume the same amount of wastewater for all six concrete plants. The calculated weight average is: 0.006 litre waste water per kg concrete. Hence depending on the density of the different concrete types the amount of wastewater per m³ differs, as shown in Fig. 5.1.

Tab. 5.1 Amount of waste water as function of concrete density

Name	Density "wet"	Waste water amount per m ³ concrete
	[kg/m ³]	[m ³ /m ³]
Poor concrete	2190	0.0131
Ordinary Concrete	2380	0.0143
Concrete for sole plate and foundation	2390	0.0143
Concrete for high demand; de-icing salt contact	2450	0.0147
Concrete for high demand	2440	0.0146

The pollution burdens of the wastewater used in this study are based on analyses of German ready-mix concrete plants. This inspection includes 12 samples of waste water from 11 plants. The average chemical quality of wastewater shown in Tab. 5.2 has been calculated and included in the study of (Künniger et al. (2001)).

Tab. 5.2 Typical waste water composition from concrete production (Künniger et al. (2001))

Emission to water	average amount [kg per litre wastewater]
chloride sum	2.97E-05
Ion arsenic	1.00E-09
Ion lead	1.00E-09
Ion chromium-VI	2.13E-08
Ion potassium	3.18E-04
Ion sodium	8.28E-05
Ion zinc	3.46E-08
nitrate	1.40E-05
phosphate (PO ₄)	2.10E-07
sulphate (SO ₄ 2-)	3.01E-06
TOC	1.22E-05

The average pH value of the wastewater is around 13. However in Switzerland it has to be neutralised to a pH value of 6.5 to 9.5 before releasing it in the sewage system. The neutralisation is not considered in the Output-module "treatment, unspecified concrete prod. effluent, to wastewater treatment, class 3" due to lack of data (Doka (2007)). The addition of a certain amount of hydrochloric acid would change insignificantly the composition of the wastewater (increase the amount of chloride).

5.3 System Boundary for the Ecoinvent Data

The flow scheme in Fig. 5.2 shows the sub-processes of the concrete production and the considered system boundary. The gravel/sand extraction and the cement production are not included in this boundary (compare part 1 and 2). The data for all modelled concrete mixing processes are based on six of total about 250 Swiss concrete factories.

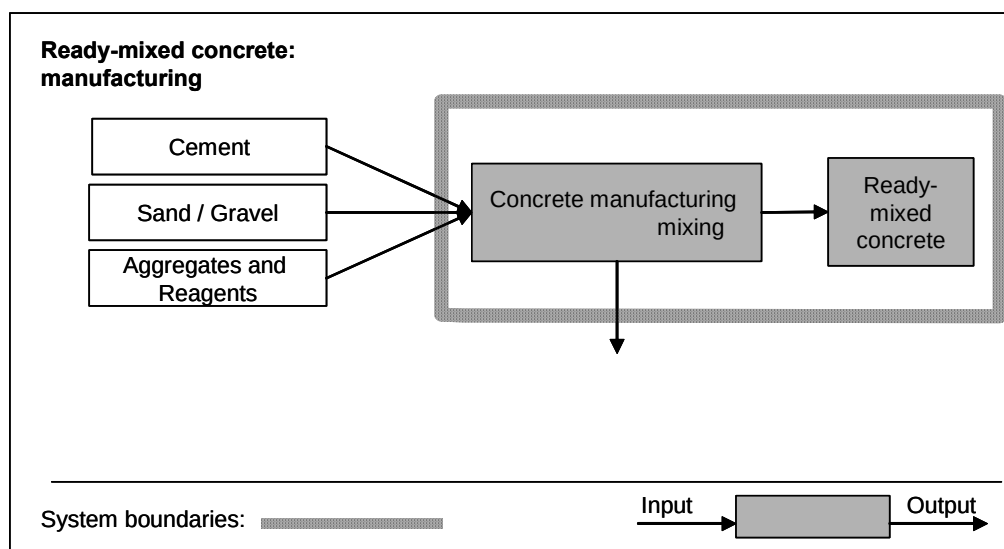


Fig. 5.2 System boundaries of Ready-mixed Concrete manufacturing (Künniger et al. (2001))

5.4 Concrete Block, Roof and Quarry Tile Production Process

The raw material is normal wet concrete which is poured into a mould and pressed. After air-hardening the raw moulded concrete is finished by abrading. The ready concrete block is then packed.

6 Life Cycle Inventories

Ecoinvent database meta information (tables), flows and their representation in the ecoinvent database are shown in (figures):

Poor concrete, at plant	Tab. 6.1 and Fig. 6.1
Concrete, normal, at plant	Tab. 6.2 and Fig. 6.2
Concrete, sole plate and foundation, at plant	Tab. 6.3 and Fig. 6.3
Concrete, exacting, with de-icing salt contact	Tab. 6.4 and Fig. 6.4
Concrete, exacting, at plant	Tab. 6.5 and Fig. 6.5
Concrete mixing plant	Tab. 6.6 and Fig. 6.6
Concrete block, at plant	Tab. 6.7 and Fig. 6.7
Concrete roof tile, at plant	Tab. 6.8 and Fig. 6.8
Quarry tile, at plant	Tab. 6.9 and Fig. 6.9

The following sections start with background information included in the specific ecoinvent database meta information. This includes essential information for the understanding, which is not picked up again in the report.

The total Swiss production amount of concrete in 1998 was 8.6 million m³. The average Swiss production amount per plant in 1998 was 104'146.17 m³.

6.1 Poor concrete, at plant

The ecoinvent database meta information for “poor concrete, at plant” (Tab. 6.1) includes all important and necessary information to use the data correctly.

Fig. 6.1 shows all in- and output flows of the product “poor concrete, at plant”, the allocation to the modules in ecoinvent database, the sources (Künniger et al. (2001)) and the standard deviation for each data.

Poor concrete is composed of portland limestone cement, water for the cement past matrix, and gravel for the aggregate

Tab. 6.1 Ecoinvent database meta information for the product “poor concrete, at plant”

Name	poor concrete, at plant
Location	CH
Infrastructure Process	0
Unit	m3
Data Set Version	2.0
Included Processes	Includes the whole manufacturing processes to produce ready-mixed concrete, other internal processes (transport, etc.), and infrastructure (machines). No administration is included. Wastewater and some solid wastes are considered. No dust is accounted for.
Amount	1
Local Name	Magerbeton, ab Werk
Synonyms	Transportbeton//Magerbeton//ready mixed concrete//poor concrete
General Comment to reference function	Part of total Swiss concrete production: insignificant. Density: 2'190 kg/m ³ . Ingredients: Cement 150 kg, Water 110 kg, Gravel 1'930 kg. Wastewater is an average from data on 11 German concrete mixing plant.
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Typical technology for average Swiss production
Representativeness [%]	100
Production Volume	Unknown
Sampling Procedure	Measured data of 6 Swiss plants
Extrapolations	See geography
Uncertainty Adjustments	None

General Flow information					Representation in ecoinvent 2000							Uncertainty information						
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
portland calcareous cement, at plant	→	poor concrete, at plant					construction materials	binder	No	CH	portland calcareous cement, at plant	1.50E+02	kg	Künniger et al. (2001)	1	1.30	(4,2,1,1,1,5);	
gravel, round, at mine	→						construction materials	additives	No	CH	gravel, round, at mine	1.93E+03	kg	Künniger et al. (2001)	1	1.30	(4,2,1,1,1,5);	
water	→						water supply	production	No	CH	tap water, at user	1.10E+02	kg	Künniger et al. (2001)	1	1.30	(4,2,1,1,1,5);	
diesel, burned in building machine	→			internal transports			construction processes	machinery	No	GLO	diesel, burned in building machine	2.09E+01	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);	
natural gas	→						natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100KW	1.07E+00	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);	
light fuel oil	→						oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	1.22E+01	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
heavy fuel oil, burned in industrial furnace	→						oil	heating systems	No	CH	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	2.85E+00	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
electricity	→						electricity	supply mix	No	CH	electricity, medium voltage, at grid	4.01E+00	kWh	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
lubricating oil, at plant	→						lubricating grease /-oil	chemicals	organics	No	RER	lubricating oil, at plant	1.09E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
transport, lorry 16t	→						process materials and removal of waste	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	9.20E-01	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
transport, lorry 28t	→						primary fuel, cement and aggregates according to data of 6 concrete plants in CH	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	8.69E+00	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
transport, barge	→						cement and aggregates according to data of 6 concrete plants in CH	transport systems	ship	No	RER	transport, barge	4.53E+01	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,2,1,5);
transport, freight, rail	→						cement and aggregates according to data of 6 concrete plants in CH	transport systems	train	No	CH	transport, freight, rail	6.28E+00	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
steel	→						wearing parts (metal)	metals	extraction	No	RER	steel, low-alloyed, at plant	2.19E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
synthetic rubber, at plant	→						wearing parts (no metal)	plastics	polymers	No	RER	synthetic rubber, at plant	6.57E-03	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
concrete plant	→						production volume over 50 years: 5.2 Mio. tons (yearly average: 104'146 t), density: 2.19 t/m3	construction materials	others	Yes	CH	concrete mixing plant	4.21E-07	unit	Künniger et al. (2001)	1	3.05	(2,1,1,1,1,5);
	→			waste heat			Same amount as electricity used	air	unspecified			Heat, waste	1.45E+01	MJ		1	1.22	(2,1,1,2,1,5);
	→			waste to inert landfill			concrete spilling on the ground when not working properly	waste management	inert material landfill	No	CH	disposal, concrete, 5% water, to inert material landfill	1.55E+01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			waste to municipal incineration			different waste, module "disposal, municipal waste" has been used as proxy	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	8.76E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			treatment, concrete production effluent, to wastewater treatment, class 3				waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	1.31E-02	m3	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			poor concrete, at plant				construction materials	concrete	No	CH	poor concrete, at plant	1.00E+00	m3	Künniger et al. (2001)			

Fig. 6.1 Flows for “poor concrete, at plant” and their representation in the ecoinvent database

6.2 Concrete, normal, at plant

The ecoinvent database meta information for “concrete, normal, at plant” (Tab. 6.2) includes all important and necessary information to use the data correctly.

Tab. 6.2 Ecoinvent database meta information for the product “concrete, normal, at plant”

Name	concrete, normal, at plant
Location	CH
Infrastructure Process	0
Unit	m3
Data Set Version	2.0
Included Processes	Includes the whole manufacturing processes to produce ready-mixed concrete, other internal processes (transport, etc.), and infrastructure (machines). No administration is included. Wastewater and some solid wastes are considered. No dust is accounted for.
Amount	1
Local Name	Beton, normal, ab Werk
Synonyms	Transportbeton//ready mixed concrete//B 35/25 with CEM I 42.5
General Comment to reference function	Part of total Swiss concrete production: 55%. Density: 2'380 kg/m3. Ingredients: Cement 300 kg, Water 190 kg, Gravel 1'890 kg. Wastewater is an average from data on 11 German concrete mixing plant.
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Typical technology for average Swiss production
Representativeness [%]	100
Production Volume	4730000 t/a
Sampling Procedure	Measured data of 6 Swiss plants
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 6.2 shows all in- and output flows of the product “concrete, normal, at plant”, the allocation to the modules in ecoinvent database, the sources (Künniger et al. (2001)) and the standard deviation for each data.

This module represents the most used concrete which is assumed to be composed of portland cement (portland cement, resistance class Z 42.5), water for the cement past matrix and gravel for the aggregate.

General Flow information						Representation in ecoinvent 2000							Uncertainty information				
Input		Process Name		Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
portland cement, strength class Z 42.5, at plant	→	concrete, normal, at plant				construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	3.00E+02 kg		Künniger et al. (2001)	1	1.30	(4,2,1,1,1,5);	
gravel, round, at mine	→					construction materials	additives	No	CH	gravel, round, at mine	1.89E+03 kg		Künniger et al. (2001)	1	1.30	(4,2,1,1,1,5);	
water	→					water supply	production	No	CH	tap water, at user	1.86E+02 kg		Künniger et al. (2001)	1	1.30	(4,2,1,1,1,5);	
diesel, burned in building machine	→				internal transports	construction processes	machinery	No	GLO	diesel, burned in building machine	2.27E+01 MJ		Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);	
electricity	→					electricity	supply mix	No	CH	electricity, medium voltage, at grid	4.36E+00 kWh		Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
natural gas	→					natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	1.16E+00 MJ		Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);	
light fuel oil	→					oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	1.33E+01 MJ		Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
heavy fuel oil, burned in industrial furnace	→					oil	heating systems	No	CH	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	3.09E+00 MJ		Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
lubricating oil, at plant	→					lubricating grease /-oil	chemicals	organics	No	RER	lubricating oil, at plant	1.19E-02 kg		Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
transport, lorry 16t	→					process materials and removal of waste	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	9.98E-01 tkm		Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
transport, lorry 28t	→					primary fuel, cement and aggregates according to data of 6 concrete plants in CH	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	9.44E+00 tkm		Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
transport, barge	→					cement and aggregates according to data of 6 concrete plants in CH	transport systems	ship	No	RER	transport, barge	4.92E+01 tkm		Künniger et al. (2001)	1	2.05	(2,1,1,2,1,5);
transport, freight, rail	→					cement and aggregates according to data of 6 concrete plants in CH	transport systems	train	No	CH	transport, freight, rail	6.82E+00 tkm		Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
steel	→					wearing parts (metal)	metals	extraction	No	RER	steel, low-alloyed, at plant	2.38E-02 kg		Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
synthetic rubber, at plant	→					wearing parts (no metal)	plastics	polymers	No	RER	synthetic rubber, at plant	7.13E-03 kg		Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
concrete plant	→					production volume over 50 years: 5.2 Mio. tons (yearly avarage: 104'146 t), density: 2.38 t/m3	construction materials	others	Yes	CH	concrete mixing plant	4.57E-07 unit		Künniger et al. (2001)	1	3.05	(2,1,1,1,1,5);
	→			waste heat		Same amount as electricity used	air	unspecified			Heat, waste	1.57E+01 MJ			1	1.22	(2,1,1,2,1,5);
	→			waste to inert landfill		concrete spilling on the ground when not working properly	waste management	inert material landfill	No	CH	disposal, concrete, 5% water, to inert material landfill	1.69E+01 kg		Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			waste to municipal incineration		different waste, module "disposal, municipal waste" has been used as proxy	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	9.51E-02 kg		Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			treatment, concrete production effluent, to wastewater treatment, class 3			waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	1.43E-02 m3		Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			concrete, normal, at plant			construction materials	concrete	No	CH	concrete, normal, at plant	1.00E+00 m3		Künniger et al. (2001)			

Fig. 6.2 Flows for “concrete, normal, at plant” and their representation in the ecoinvent database

6.3 Concrete, sole plate and foundation, at plant

The ecoinvent database meta information for “concrete, sole plate and foundation, at plant” (Tab. 6.3) includes all important and necessary information to use the data correctly.

Tab. 6.3 Ecoinvent database meta information for the product “concrete, sole plate and foundation, at plant”

Name	concrete, sole plate and foundation, at plant
Location	CH
Infrastructure Process	0
Unit	m3
Data Set Version	2.0
Included Processes	Includes the whole manufacturing processes to produce ready-mixed concrete, other internal processes (transport, etc.), and infrastructure (machines). No administration is included. Wastewater and some solid wastes are considered. No dust is accounted for.
Amount	1
Local Name	Beton, Bodenplatten und Fundamente, ab Werk
Synonyms	Transportbeton//ready mixed concrete//B 35/25 with CEM III/B 32.5
General Comment to reference function	Part of total Swiss concrete production: 2%. Density: 2'385 kg/m3. Ingredients: Cement 325 kg, Water 180 kg, Gravel 1'880 kg. Wastewater is an average from data on 11 German concrete mixing plant.
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Typical technology for average Swiss production
Representativeness [%]	100
Production Volume	172000 t/a
Sampling Procedure	Measured data of 6 Swiss plants
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 6.3 shows all in- and output flows of the product “concrete, sole plate and foundation, at plant”, the allocation to the modules in ecoinvent database, the sources (Künniger et al. (2001)) and the standard deviation for each data.

This type of concrete is usually used for sole plates and foundations. It is composed of “blast furnace slag cement”, water for the cement past matrix, and gravel for the aggregate.

General Flow information						Representation in ecoinvent 2000							Uncertainty information				
Input		Process Name		Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
blast furnace slag cement, at plant	→	concrete, sole plate and foundation, at plant				construction materials	binder	No	CH	blast furnace slag cement, at plant	3.25E+02	kg	Künniger et al. (2001)	1	1.30	(4,2,1,1,1,5);	
gravel, round, at mine	→					construction materials	additives	No	CH	gravel, round, at mine	1.88E+03	kg	Künniger et al. (2001)	1	1.30	(4,2,1,1,1,5);	
water	→					water supply	production	No	CH	tap water, at user	1.79E+02	kg	Künniger et al. (2001)	1	1.30	(4,2,1,1,1,5);	
diesel, burned in building machine	→				internal transports	construction processes	machinery	No	GLO	diesel, burned in building machine	2.27E+01	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);	
electricity	→					electricity	supply mix	No	CH	electricity, medium voltage, at grid	4.38E+00	kWh	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
natural gas	→					natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	1.17E+00	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);	
light fuel oil	→					oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	1.33E+01	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
heavy fuel oil, burned in industrial furnace	→					oil	heating systems	No	CH	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	3.10E+00	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
lubricating oil, at plant	→					lubricating grease /-oil	chemicals	organics	No	RER	lubricating oil, at plant	1.19E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
transport, lorry 16t	→					process materials and removal of waste	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	1.00E+00	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
transport, lorry 28t	→					primary fuel, cement and aggregates according to data of 6 concrete plants in CH	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	9.48E+00	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
transport, barge	→					cement and aggregates according to data of 6 concrete plants in CH	transport systems	ship	No	RER	transport, barge	4.94E+01	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,2,1,5);
transport, freight, rail	→					cement and aggregates according to data of 6 concrete plants in CH	transport systems	train	No	CH	transport, freight, rail	6.85E+00	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
steel	→					wearing parts (metal)	metals	extraction	No	RER	steel, low-alloyed, at plant	2.39E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
synthetic rubber, at plant	→					wearing parts (no metal)	plastics	polymers	No	RER	synthetic rubber, at plant	7.16E-03	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
concrete plant	→					production volume over 50 years: 5.2 Mio. tons (yearly avarage: 104'146 t), density: 2.39 t/m3	construction materials	others	Yes	CH	concrete mixing plant	4.59E-07	unit	Künniger et al. (2001)	1	3.05	(2,1,1,1,1,5);
	→			waste heat		Same amount as electricity used	air	unspecified			Heat, waste	1.58E+01	MJ		1	1.22	(2,1,1,2,1,5);
	→			waste to inert landfill		concrete spilling on the ground when not working properly	waste managemement	inert material landfill	No	CH	disposal, concrete, 5% water, to inert material landfill	1.69E+01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			waste to municipal incineration		different waste, module "disposal, municipal waste" has been used as proxy	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	9.55E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			treatment, concrete production effluent, to wastewater treatment, class 3			waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	1.43E-02	m3	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			concrete, sole plate and foundation, at plant			construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	1.00E+00	m3	Künniger et al. (2001)			

Fig. 6.3 Flows for “Concrete, sole plate and foundation, at plant” and their representation in the ecoinvent database

6.4 Concrete, exacting, with de-icing salt contact, at plant

The ecoinvent database meta information for “concrete, exacting, with de-icing salt contact, at plant” (Tab. 6.4) includes all important and necessary information to use the data correctly.

Tab. 6.4 Ecoinvent database meta information for the product “concrete, exacting, with de-icing salt contact, at plant”

Name	concrete, exacting, with de-icing salt contact, at plant
Location	CH
Infrastructure Process	0
Unit	m3
Data Set Version	2.0
Included Processes	Includes the whole manufacturing processes to produce ready-mixed concrete, internal processes (transport, etc.), and infrastructure. No administration is included. Special inputs: high-performance plasticizer for concrete (solids content 40%), based on naphthalenesulfonic acid condensate; air-entraining additive that produces micro air bubbles to raise frost resistance based on resin soaps or synthetic surfactants. Wastewater and some solid wastes are considered. No dust is accounted for.
Amount	1
Local Name	Beton, hohe Anforderungen, Streusalzkontakt, ab Werk
Synonyms	Transportbeton//ready mixed concrete//B 45/35 with CEM I 42.5
General Comment to reference function	Part of total Swiss concrete production: 28%. Density: 2'455 kg/m3. Ingredients: Cement 330 kg, Water 150 kg, Gravel 1'970 kg, air-entraining additive 1.32 kg, high-performance plasticizer 3.96 kg. Wastewater is an average from data on 11 German concrete mixing plant.
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Typical technology for average Swiss production
Representativeness [%]	100
Production Volume	2408000 t/a
Sampling Procedure	Measured data of 6 Swiss plants
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 6.4 shows all in- and output flows of the product “concrete, exacting, with de-icing salt contact, at plant”, the allocation to the modules in ecoinvent database, the sources (von Arx (1995), Künniger et al. (2001)) and the standard deviation for each data

This type of concrete is usually used for constructions with de-icing salt contact. It is assumed to be composed of ordinary cement (portland cement, resistance class Z 42.5), water for the cement past matrix, and gravel for the aggregate. To improve the resistance of concrete to frost and de-icing salt, an air-entraining admixture is added. To improve concrete workability, strength and durability, a high-performance plasticizer is used. This plasticizer reduces the amount of water needed and permits the use of a lower water to cement ratio. The composition of the used air-entraining admixture and the high-performance plasticizer is listed in Fig. 6.4.

General Flow information					Representation in ecoinvent 2000							Uncertainty information			
Input		Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StdV 95%	General Comment
portland cement, strength class Z 42.5, at plant	→	concrete, exacting, with de-icing salt contact, at plant (Input: raw materials)			construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	3.30E+02	kg	Künninger et al. (2001)	1	1.30	(4,2,1,1,1,5);
gravel, round, at mine	→				construction materials	additives	No	CH	gravel, round, at mine	1.97E+03	kg	Künninger et al. (2001)	1	1.30	(4,2,1,1,1,5);
colophonium, balsamic turpentine oil as part of the air-entraining agent	→			air-entraining agent: for production of concrete admixtures (resin soaps or synthetic surfactants), calculated; total: 1.32 kg	chemicals	organics	No	GLO	chemicals organic, at plant	3.30E-02	kg	Von Arx (1995)	1	1.69	(4,5,2,5,4,5);
caustic soda hydrated (NaOH) as part of the air-entraining agent	→				chemicals	inorganics	No	RER	sodium hydroxide, 50% in H ₂ O, production mix, at plant	2.64E-02	kg	Von Arx (1995)	1	1.69	(4,5,2,5,4,5);
water, completely softened as part of the air-entraining agent	→				water supply	production	No	RER	water, completely softened, at plant	1.27E+00	kg	Von Arx (1995)	1	1.69	(4,5,2,5,4,5);
natural gas, burned in industrial furnace for production of high-performance plasticizer	→				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NO _x >100kW	2.15E-01	MJ	estimated EMPA 1999	1	1.26	(3,4,1,2,1,5);
Formaldehyd (37%ig) for production of high-performance plasticizer, 37% of total weight: formaldehyd	→			used for the production of high-performance plasticizer for concrete (solids content 40%), total: 3.98kg	chemicals	organics	No	RER	formaldehyde, production mix, at plant	1.49E-01	kg	personal communication by SIKAG Zürich 1999	1	1.24	(2,4,1,2,1,5);
water for diluting formaldehyd to concentration of 37% for production of high-performance plasticizer, 63% of total weight: water	→				water supply	production	No	RER	water, completely softened, at plant	2.53E-01	kg	personal communication by SIKAG Zürich 1999	1	1.24	(2,4,1,2,1,5);
naphthalin for production of high-performance plasticizer	→				chemicals	organics	No	GLO	chemicals organic, at plant	6.45E-01	kg	personal communication by SIKAG Zürich 1999	1	1.24	(2,4,1,2,1,5);
sodium hydroxide, 50% in H ₂ O, for production of high-performance plasticizer	→				chemicals	inorganics	No	RER	sodium hydroxide, 50% in H ₂ O, production mix, at plant	5.28E-01	kg	personal communication by SIKAG Zürich 1999	1	1.24	(2,4,1,2,1,5);
sulphuric acid, liquid, for production of high-performance plasticizer	→				chemicals	inorganics	No	RER	sulphuric acid, liquid, at plant	6.42E-01	kg	personal communication by SIKAG Zürich 1999	1	1.24	(2,4,1,2,1,5);
water, completely softened, for production of high-performance plasticizer	→				water supply	production	No	RER	water, completely softened, at plant	2.38E+00	kg	personal communication by SIKAG Zürich 1999	1	1.24	(2,4,1,2,1,5);
water	→				water supply	production	No	CH	tap water, at user	1.49E+02	kg	Künninger et al. (2001)	1	1.30	(4,2,1,1,1,5);

Fig. 6.4 Flows for “concrete, exacting, with de-icing contact at plant” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent 2000							Uncertainty information						
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
diesel, burned in building machine	→	concrete, exacting, with de-icing salt contact, at plant (Input: energy, transports and diverse materials and output)				internal transports	construction processes	machinery	No	GLO	diesel, burned in building machine	2.33E+01	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);	
electricity	→						electricity	supply mix	No	CH	electricity, medium voltage, at grid	4.49E+00	kWh	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
natural gas	→						natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100KW	1.20E+00	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);	
light fuel oil, burned in industrial furnace	→						oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	1.37E+01	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
heavy fuek oil, burned in industrial furnace	→						oil	heating systems	No	CH	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	3.18E+00	MJ	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
lubricating oil, at plant	→						lubricating grease /-oil	chemicals	organics	No	RER	lubricating oil, at plant	1.22E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
transport, lorry 16t	→						process materials and removal of waste	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	1.03E+00	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
transport, lorry 28t	→						primary fuel, cement and aggregates according to data of 6 concrete plants in CH	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	9.73E+00	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
transport, barge	→						cement and aggregates according to data of 6 concrete plants in CH	transport systems	ship	No	RER	transport, barge	5.07E+01	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,2,1,5);
transport, freight, rail	→						cement and aggregates according to data of 6 concrete plants in CH	transport systems	train	No	CH	transport, freight, rail	7.03E+00	tkm	Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
steel	→						wearing parts (metal)	metals	extraction	No	RER	steel, low-alloyed, at plant	2.45E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
synthetic rubber, at plant	→						wearing parts (no metal)	plastics	polymers	No	RER	synthetic rubber, at plant	7.35E-03	kg	Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
concrete plant	→						production volume over 50 years: 5.2 Mio. tons (yearly avarage: 104'146 t), density: 2.45 t/m3	construction materials	others	Yes	CH	concrete mixing plant	4.70E-07	unit	Künniger et al. (2001)	1	3.05	(2,1,1,1,1,5);
	→			waste heat			Same amount as electricity used	air	unspecified			Heat, waste	1.62E+01	MJ		1	1.22	(2,1,1,2,1,5);
	→			waste to inert landfill			concrete spilling on the ground when not working properly	waste management	inert material landfill	No	CH	disposal, concrete, 5% water, to inert material landfill	1.74E+01	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→			waste to municipal incineration treatment, concrete production effluent, to wastewater treatment, class 3			different waste, module "disposal, municipal waste" has been used as proxy	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	9.80E-02	kg	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→							waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	1.47E-02	m3	Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→							construction materials	concrete	No	CH	concrete, exacting, with de-icing salt contact, at plant	1.00E+00	m3	Künniger et al. (2001)			
			concrete, exacting, with de-icing salt contact, at plant															

Fig. 6.4 Flows for “concrete, exacting, with de-icing contact at plant“ and their representation in the ecoinvent database (cont.)

6.5 Concrete, exacting, at plant

The ecoinvent database meta information for “concrete, exacting, at plant” (Tab. 6.5) includes all important and necessary information to use the data correctly.

Tab. 6.5 Ecoinvent database meta information for the product “concrete, exacting, at plant”

Name	concrete, exacting, at plant
Location	CH
Infrastructure Process	0
Unit	m3
Data Set Version	2.0
Included Processes	Includes the whole manufacturing processes to produce ready-mixed concrete, internal processes (transport, etc.) and infrastructure. No administration is included. Special inputs: high-performance plasticizer for concrete (solids content 40%), based on naphthalenesulfonic acid condensate. Wastewater and some solid wastes are considered. No dust is accounted for.
Amount	1
Local Name	Beton, hohe Anforderungen, ab Werk
Synonyms	Transportbeton//ready mixed concrete//B 60/50 m. CEM I 42.5
General Comment to reference function	Part of total Swiss concrete production: 15%. Density: 2'440 kg/m3. Ingredients: Cement 375 kg, Water 150 kg, Gravel 1'880 kg, high-performance plasticizer 2.8 kg, additives (Silicadust) 35 kg. Wastewater is an average from data on 11 German concrete mixing plant.
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	Typical technology for average Swiss production
Representativeness [%]	100
Production Volume	860000 t/a
Sampling Procedure	Measured data of 6 Swiss plants
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 6.5 shows all in- and output flows of the product “concrete, exacting, at plant”, the allocation to the modules in ecoinvent database, the sources (Künniger et al. (2001)) and the standard deviation for each data.

This type of concrete is usually used for buildings with a high demand of strength and durability. It is composed of ordinary cement (portland cement, resistance class Z 42.5), water for the cement past matrix, and gravel for the aggregate. To improve concrete workability, strength and durability, a high-performance plasticizer is added. This plasticizer reduces the amount of water needed and permits the use of lower water to cement ratio. The composition of the high-performance plasticizer is listed in Fig. 6.5. To reduce the amount of energy-expensive cement and to save natural resources by recycling waste material, fine grained materials as hard coal fly ash or silica dust are added. These admixtures enter the system without burdens as they are declared as waste. The burdens are allocated 100% to the product or process from which the waste comes from (Frischknecht et al. (2007)).

General Flow information						Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
portland cement, strength class Z 42.5, at plant	→	concrete, exacting, at plant (Input: raw materials)				used for the production of high-performance plasticizer for concrete (solids content 40%).	construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	3.75E+02	kg	Künniger et al. (2001)	1	1.30	(4,2,1,1,1,5);	
gravel, round, at mine	→						construction materials	additives	No	CH	gravel, round, at mine	1.88E+03	kg	Künniger et al. (2001)	1	1.30	(4,2,1,1,1,5);	
natural gas, burned in industrial furnace for production of high-performance plasticizer	→						natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	1.52E-01	MJ	calculation by EMPA	1	1.26	(3,4,1,2,1,5);	
Formaldehyd (37%ig) for production of high-performance plasticizer, 37% of total weight: formaldehyd	→						chemicals	organics	No	RER	formaldehyde, production mix, at plant	1.05E-01	kg	personal communication by SIKA AG Zürich 1999	1	1.26	(3,4,1,2,1,5);	
water for diluting formaldehyd to concentration of 37% for production of high-performance plasticizer, 63% of total weight: water	→						water supply	production	No	RER	water, completely softened, at plant	1.79E-01	kg	personal communication by SIKA AG Zürich 1999	1	1.26	(3,4,1,2,1,5);	
Naphthalin for production of high-performance plasticizer	→						chemicals	organics	No	GLO	chemicals organic, at plant	4.56E-01	kg	personal communication by SIKA AG Zürich 1999	1	1.26	(3,4,1,2,1,5);	
sodium hydroxide, 50% in H2O, for production of high-performance plasticizer	→						chemicals	inorganics	No	RER	sodium hydroxide, 50% in H2O, production mix, at plant	3.74E-01	kg	personal communication by SIKA AG Zürich 1999	1	1.26	(3,4,1,2,1,5);	
sulphuric acid, liquid, for production of high-performance plasticizer	→						chemicals	inorganics	No	RER	sulphuric acid, liquid, at plant	4.54E-01	kg	personal communication by SIKA AG Zürich 1999	1	1.26	(3,4,1,2,1,5);	
water, completely softened, for production of high-performance plasticizer	→						water supply	production	No	RER	water, completely softened, at plant	1.68E+00	kg	personal communication by SIKA AG Zürich 1999	1	1.26	(3,4,1,2,1,5);	
concrete additives, at plant	→						Substances which do change the quality of concrete; mostly substances which result as waste in other processes, for example hard coal fly ash or silica dust, substances which change the quality of concrete; not balanced					3.50E+01	kg	Künniger et al. (2001)	1	1.26	(3,4,1,2,1,5);	
water	→							water supply	production	No	CH	tap water, at user	1.50E+02	kg	Künniger et al. (2001)	1	1.30	(4,2,1,1,1,5);

Fig. 6.5 Flows for “concrete, exacting, at plant” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent 2000							Uncertainty information						
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
diesel, burned in building machine	→	concrete, exacting, at plant (Input: energy, transports and diverse materials and output)				internal transports	construction processes	machinery	No	GLO	diesel, burned in building machine	2.33E+01 MJ		Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);	
electricity	→						electricity	supply mix	No	CH	electricity, medium voltage, at grid	4.48E+00 kWh		Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
natural gas, burned in industrial furnace	→						natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	1.20E+00 MJ		Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);	
light fuel oil, burned in industrial furnace	→						oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	1.36E+01 MJ		Künniger et al. (2001)	1	1.26	(3,4,1,2,1,5);	
heavy fuel oil, burned in undustrial furnace	→						oil	heating systems	No	CH	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	3.18E+00 MJ		Künniger et al. (2001)	1	1.26	(3,4,1,2,1,5);	
lubricating oil, at plant	→						lubricating grease /-oil	chemicals	organics	No	RER	lubricating oil, at plant	1.22E-02 kg		Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
transport, lorry 16t	→						process materials and removal of waste	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	1.03E+00 tkm		Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
transport, lorry 28t	→						primary fuel, cement and aggregates according to data of 6 concrete plants in CH	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	9.70E+00 tkm		Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
transport, barge	→						cement and aggregates according to data of 6 concrete plants in CH	transport systems	ship	No	RER	transport, barge	5.05E+01 tkm		Künniger et al. (2001)	1	2.05	(2,1,1,2,1,5);
transport, freight, rail	→						cement and aggregates according to data of 6 concrete plants in CH	transport systems	train	No	CH	transport, freight, rail	7.01E+00 tkm		Künniger et al. (2001)	1	2.05	(2,1,1,1,1,5);
steel high-alloyed	→						wearing parts (metal)	metals	extraction	No	RER	steel, low-alloyed, at plant	2.44E-02 kg		Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
synthetic rubber, at plant	→						wearing parts (no metal)	plastics	polymers	No	RER	synthetic rubber, at plant	7.33E-03 kg		Künniger et al. (2001)	1	1.22	(2,1,1,2,1,5);
concrete plant	→						production volume over 50 years: 5.2 Mio. tons (yearly average: 104146 t), density: 2.44 t/m3	construction materials	others	Yes	CH	concrete mixing plant	4.69E-07 unit		Künniger et al. (2001)	1	3.05	(2,1,1,1,1,5);
	→			waste heat			Same amount as electricity used	air	unspecified			Heat, waste	1.61E+01 MJ			1	1.22	(2,1,1,2,1,5);
	→			waste to inert landfill			primarily concrete	waste management	inert material landfill	No	CH	disposal, concrete, 5% water, to inert material landfill	1.73E+01 kg		Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);
	→		waste to municipal incineration			primarily wooden pallets and used oil, plastics and rubber	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	9.77E-02 kg		Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
	→		incineration treatment, concrete production effluent, to wastewater treatment, class 3				waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	1.46E-02 m3		Künniger et al. (2001)	1	1.22	(2,1,1,1,1,5);	
	→		concrete, exacting, at plant				construction materials	concrete	No	CH	concrete, exacting, at plant	1.00E+00 m3		Künniger et al. (2001)				

Fig. 6.5 Flows for “concrete, exacting, at plant” and their representation in the ecoinvent database (cont.)

The usage of ashes and dust in the building and cement industry increases the decay activity of the products as exacting concrete. This can lead to slightly raised indoor activity concentration (United Nations Scientific Committee on the Effects of Atomic Radiation et al. (1988)). This activity of the recycling is not taken into account but should be studied and included in ecoinvent in a future edition.

6.6 Concrete mixing plant

The ecoinvent database meta information for “concrete plant” (Tab. 6.6) includes all important and necessary information to use the data correctly.

Tab. 6.6 Ecoinvent database meta information for the product “concrete plant”

Name	concrete mixing plant
Location	CH
Infrastructure Process	1
Unit	unit
Data Set Version	2.0
Included Processes	Includes the area for the infrastructure (buildings, roads, etc.) and all silos and machines used for the ready-mixed concrete production
Amount	1
Local Name	Betonmischwerk
Synonyms	
General Comment to reference function	The lifespan of the infrastructure and the land occupation is assumed to be 50 years. The average yearly production is about 100'000 t/a
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER-modules have been used as proxy
Technology text	The data of the infrastructure corresponds to high level of technology
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Measured data of 6 Swiss plants
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 6.6 shows the module “concrete plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent database, the sources (Künniger et al. (2001)) and the standard deviation for each data.

The life span of the portrayed plant is estimated to be about 50 years (default value for plants and mines) and the yearly production amount about 100'000 tonnes (about 42'000 m³) of ready-mix concrete.

Machines used in the concrete plant:

Since no exact information is available, the weights of the different machines and the building halls are estimated based on photos and figures:

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
infrastructure machines	→	concrete mixing plant				Infrastructure of all machines; weight assumption: 80t; assumption service life: 25years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.60E+05	kg	estimation EMPA	1	3.10	(3,5,1,2,1,5);
infrastructure: building hall	→					estimated area: 70% of total building area (2'00m2)	construction processes	buildings	Yes	CH	building, hall, steel construction	1.40E+03	m2	Künniger et al. (2001)	1	3.10	(3,5,1,1,1,5);
infrastructure: building multistore	→					estimated area: 30% of total building area (2'000m2), assumed average height: 12m	construction processes	buildings	Yes	RER	building, multi-storey	7.19E+03	m3	Künniger et al. (2001)	1	3.10	(3,5,1,2,1,5);
total land use (transformation)	→						resource	land			Transformation, from unknown	7.37E+03	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);
land occupation for green part of build area	→					assumption: life span 50 years, area about 24'000m2	resource	land			Occupation, industrial area, vegetation	1.19E+05	m2a	Künniger et al. (2001)	1	1.57	(2,3,1,1,1,5);
land transformation to green part of build area	→					area about 24'000m2	resource	land			Transformation, to industrial area, vegetation	2.37E+03	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);
land occupation for road part of build area	→					assumption: life span 50 years, area about 3'000m2	resource	land			Occupation, traffic area, road network	1.50E+05	m2a	Künniger et al. (2001)	1	1.57	(2,3,1,1,1,5);
land transformation to road part of build area	→					area about 3'000m2	resource	land			Transformation, to traffic area, road network	3.01E+03	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);
land occupation for building part of build area	→					assumption: life span 50 years, area about 2'000m2	resource	land			Occupation, industrial area, built up	9.99E+04	m2a	Künniger et al. (2001)	1	1.57	(2,3,1,1,1,5);
land transformation to building part of build area	→					total building area: about 2'000m2	resource	land			Transformation, to industrial area, built up	2.00E+03	m2	Künniger et al. (2001)	1	2.05	(2,3,1,1,1,5);
infrastructure conveyor belt	→				Assumed length 500 m; width 1.5 m; service life 20 years. The module in ecoinvent has a width of 3 m, which is here taken into account by reducing the used length to half.	construction processes	machinery	Yes	RER	conveyor belt, at plant	7.50E+02	m	estimation EMPA	1	3.10	(3,5,1,2,1,5);	
	→		concrete plant			assumption: life span 50 years; production volume: 5 Mio. tons	construction materials	others	Yes	CH	concrete mixing plant	1.00E+00	unit				

Fig. 6.6 Flows for “concrete mixing plant” and their representation in the ecoinvent database

6 big silos (assumption each 10 t):	60 t
diverse material:	20 t
<i>Total weight of machines:</i>	<i>80 t</i>
Length of conveyor belt:	500 m
Width of conveyor belt:	1.5 m

The general module “conveyor belt” is used to describe the conveyor belt of the modeled cement plant. The width of the conveyor belt in the ecoinvent module is 3 m. This circumstance has been taken into account by shortening the input length from 500 to 250m. The service life is assumed to be 20 years. Within the assumed service life of 50 years concerning the total cement plant, three replacements are necessary.

The service life for all other machines is assumed to be 25 years.

6.7 Concrete block, at plant

The ecoinvent database meta information for “concrete block, at plant” (Tab. 6.7) includes all important and necessary information to use the data correctly.

Fig. 6.7 shows the module “concrete block, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly Hutter et al. (1999)) and the standard deviation for each data.

Tab. 6.7 Ecoinvent database meta information for the product “concrete block, at plant”

Name	concrete block, at plant
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the raw material normal concrete which is poured into a mould, air-dried and packed. Some transports and infrastructure are also included.
Amount	1
Local Name	Betonstein, ab Werk
Synonyms	
General Comment to reference function	No losses and no waste water have been taken into account. The used concrete is "normal concrete"
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH-, RER-, and GLO-modules have been used as proxy
Technology text	typical technology for German production
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

General Flow information						Representation in ecoinvent 2000								Uncertainty information			
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
wet concrete	→	production of concrete block, at plant				Density: 2'380 kg/m3, assumed that no losses occur	construction materials	concrete	No	CH	concrete, normal, at plant	4.20E-04	m3	estimation	1	1.50	estimation
diesel in building machine	→					for internal transports, module "diesel, burnded in building machine" used as proxy, amount: 0.0006 l/kg	construction processes	machinery	No	GLO	diesel, burned in building machine	2.16E-02	MJ	Hutter et al. (1999)	1	1.34	(3,5,3,1,1,5);
electricity consumption	→					for vibration ramming (assumption based on electricity consumption for cement cast plaster floor production)	electricity	supply mix	No	DE	electricity, medium voltage, at grid	2.00E-03	kWh	estimation	1	5.00	estimation
infrastructure (including building, machines and land-use)	→					copied from module "sand-lime brick"	construction materials	additives	Yes	CH	mine, clay	1.67E-10	unit	estimation	1	5.00	estimation
packing	→					packing clay products used as proxy	construction materials	others	No	CH	packing, clay products	1.00E+00	kg	estimation	1	5.00	estimation
						no data											
			→	waste water			construction materials	concrete	No	DE	concrete block, at plant	1.00E+00	kg				
			→	concrete block													

Fig. 6.7 Flows for "concrete block, at plant" and their representation in the ecoinvent database

6.8 Concrete roof tile, at plant

The ecoinvent database meta information for “concrete roof tile, at plant” (Tab. 6.7) includes all important and necessary information to use the data correctly.

Tab. 6.8 Ecoinvent database meta information for the product “concrete block, at plant”

Name	concrete roof tile, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the raw material “normal concrete” which is poured into a mould, air-dried and packed. Some transports and infrastructure are also included.
Amount	1
Local Name	Betonziegel, ab Werk
Synonyms	
General Comment to reference function	No losses and no waste water are taken into account.
Start Date	2004
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH-, RER-, and GLO-datasets have been used as proxy
Technology text	typical technology for Swiss production
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.7 shows the module “concrete roof tile, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly Vogel et al. (2003)) and the standard deviation for each data.

General Flow information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
cement	concrete roof tile, at plant			construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	2.00E-01	kg	Vogel et al. (2003)	1	1.62	(2,5,5,1,1,5);
iron oxide			iron ore with 65% Fe has been used as proxy	metals	extraction	No	GLO	iron ore, 65% Fe, at beneficiation	4.90E-03	kg		1	1.62	(2,5,5,1,1,5);
water				water supply	production	No	CH	tap water, at user	8.00E-02	kg		1	1.62	(2,5,5,1,1,5);
sand				construction materials	additives	No	CH	sand, at mine	7.00E-01	kg		1	1.62	(2,5,5,1,1,5);
latex paint				paintings	production	No	RER	acrylic dispersion, 65% in H ₂ O, at plant	1.50E-02	kg		1	1.62	(2,5,5,1,1,5);
diesel in building machine			for internal transports, dataset "diesel, burned in building machine" used as proxy, amount: 0.0006 l/kg	construction processes	machinery	No	GLO	diesel, burned in building machine	2.16E-02	MJ	copied from dataset "concrete block, at plant"	1	1.34	(3,5,3,1,1,5);
electricity consumption			for vibration ramming (assumption based on electricity consumption for cement cast plaster floor production)	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.00E-03	kWh	copied from dataset "cement cast plaster floor, at plant"	1	5.00	estimation
infrastructure (including building, machines and land-use)			copied from dataset "sand-lime brick"	construction materials	additives	Yes	CH	mine, clay	1.67E-10	unit	copied from dataset "sand-lime brick"	1	5.00	estimation
packing			packing clay products used as proxy	construction materials	others	No	CH	packing, clay products	1.00E+00	kg	estimation	1	5.00	estimation
			same amount as electricity consumption	air	unspecified			Heat, waste	7.20E-03	MJ		1	5.00	estimation
		waste heat												
		waste water												
		concrete roof tile		construction materials	coverings	No	CH	concrete roof tile, at plant	1.00E+00	kg				

Fig. 6.8 Flows for “concrete roof tile, at plant” and their representation in the ecoinvent database

6.9 Quarry tile, at plant

The ecoinvent database meta information for “quarry tile, at plant” (Tab. 6.7) includes all important and necessary information to use the data correctly.

Tab. 6.9 Ecoinvent database meta information for the product “quarry tile, at plant”

Name	quarry tile, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the raw material “normal” concrete which is poured into a mould, air-dried and packed. Some transports and infrastructure are also included.
Amount	1
Local Name	Kunststeinplatte, ab Werk
Synonyms	
General Comment to reference function	No losses and no waste water have been taken into account.
Start Date	2004
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH-, RER-, and GLO-datasets are used as proxy
Technology text	Typical technology for Swiss production
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.7 shows the module “quarry tile, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly Vogel et al. (2003)) and the standard deviation for each data.

General Flow information				Representation in ecoinvent								Uncertainty						
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
wet concrete	→	quarry tile, at plant				Density: 2'380 kg/m3, assumed that no losses occur	construction materials	concrete	No	CH	concrete, normal, at plant	4.20E-04	m3	estimation	1	1.50	estimation	
raising of amount of cement in normal concrete	→					correction of cement amount from 12.5% to 25%	construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	1.25E-01	kg	Vogel et al. (2003)	1	5.00	estimation	
reduction of amount of sand in normal concrete	→					reduction of sand amount from 80% to 67.5%	construction materials	additives	No	CH	gravel, round, at mine	-1.25E-01	kg		1	5.00	estimation	
diesel in building machine	→					for internal transports, dataset "diesel, burnded in building machine" used as proxy, amount: 0.0006 l/kg	construction processes	machinery	No	GLO	diesel, burned in building machine	2.16E-02	MJ	copied from dataset "concrete block, at plant"	1	1.34	(3,5,3,1,1,5);	
transportation for delivery of concrete	→					assumed distance: 20km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	2.00E-02	tkm	Vogel et al. (2003)	1	5.00	estimation	
electricity consumption	→					for vibration ramming (assumption based on electricity consumption for cement cast plaster floor production)	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.00E-03	kWh	estimation	1	5.00	estimation	
infrastructure (including building, machines and land-use)	→					copied from dataset "sand-lime brick"	construction materials	additives	Yes	CH	mine, clay	1.67E-10	unit		1	5.00	estimation	
packing	→					packing clay products used as proxy	construction materials	others	No	CH	packing, clay products	1.00E+00	kg		1	5.00	estimation	
				→	waste heat		same amount as electricity consumption	air	unspecified			Heat, waste	7.20E-03	MJ	calculated	1	5.00	estimation
				→	waste water		no data											
			→	quarry tile			construction materials	others	No	CH	quarry tile, at plant	1.00E+00	kg					

Fig. 6.9 Flows for "quarry tile, at plant" and their representation in the ecoinvent database

7 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed information compare the different flow charts and the ecoinvent database meta information.

Most of the data from the different types of concrete are based on information from six Swiss ready-mix concrete factories. The different factories have a long tradition of collecting data referring energy and material inputs. Data concerning emissions and plant infrastructure are not collected yet and are therefore based on assumptions with high uncertainty.

The data for the air-entraining agent concern only one type of agent. For those concrete types requiring an input on air-entraining agents this module has been used as proxy.

The data of the high performance plasticizer also concern only one type of plasticizer. For those concrete types requiring an input on plasticizer this module has been used as proxy.

For all above-mentioned reasons the data for all different concrete types are estimated of good overall quality although the infrastructure has been only roughly estimated.

The data for concrete blocks are from a German study (Hutter et al. (1999)) which seems comprehensively made. Unfortunately there is no information on the validity area and period area. Nevertheless it is assumed that the data are of good overall quality.

8 Abbreviations

CEN	Comité Européen de Normalisation
CEB	Comité Européen du Béton
ISO	International Organisation for Standardisation
LB	light weight concrete
PCC	polymer cement concrete
PIC	polymer impregnated concrete
RILEM	Réunion Internationale des Laboratoires d'Essais et de Recherche sur les Matériaux et les Constructions
SIA	Association of Swiss Engineers and Architects (Schweizerischer Ingenieur- und Architektenverein)
W/C	water to cement ratio

9 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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Part IV

Clay Products and Processes

Data v2.0 (2007)

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Summary

This part deals with the mining of the raw material clay and two resulting products; brick and roof tile. Infrastructure and packing are treated in separate modules called “mine, clay” and “packing, clay products”.

The data used for the inventory are from a study which covers twelve factories in Germany, Austria and Switzerland. The most energy intense production step is the firing process. The energy is covered by different fossil fuels. The emissions are mainly caused by burning these fuels.

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1 Introduction

Bricks and structural tiles are mass-produced ceramic products requiring large amounts of raw materials for manufacture. Brick and tile making is mostly a low-value industry. Therefore transport costs have an important part which results in as short as possible distances between mine and manufacturing facility. The increasing quality standards and the competition from other building materials necessitate new costly but still economical manufacturing techniques. (Hauck et al. (1997))

This report focuses on clay products- Brick and roof tile production processes and related modules are included. The roof tile production module has been derived from the brick production module. The modules include processes from mining the raw material to fabrication of brick and tiles, ready to be delivered. The developed modules are:

- Brick, at plant (one kg of brick)
- Roof tile, at plant (one kg of roof tile)
- Clay, at mine (one kg of excavated clay)

The data of brick production are based on 12 brick production plants evaluated within a life cycle inventory of plants in Germany, Austria and Switzerland and are published in (Bruck (1996)). The data are operational data and therefore determined by measurements (concentration of air emissions in flue gas, consumption of electricity and natural gas, etc.) or from the business management recordings (raw material, production amounts, process materials, spare parts, transport distances and masses, etc.). Following categories of data were identified as important and therefore published in form of yearly amounts:

Tab. 1.1 Categories of data published in Hauck et al. (1997)

Input	Output
Energy	Products
Raw materials and additives	Emissions to air
Water	
Process materials	
Ancillary transports	

Hauck et al. (1997) does not include water emissions, solid waste, noise and sealing of the soil as it is declared to be negligible. Also data concerning the infrastructure necessary for the production process are not taken into account in the study of Hauck et al. (1997). Therefore in this study the module infrastructure “mine, clay” is based on an evaluation of the Ziegelei Rapperswil-Bern as an exemplary plant with an excavated amount of 80'000 t/year. It is assumed that the size and volume of the buildings, the weight of the production machines and the land-use (build part, not mine) depending on the production volume is about the same all over Western Europe.

2 Reserves and Resources

The yearly amount of clay digged in the Swiss clay pits is between 100 and 100'000m³. The average amount is about 25'000m³. For the production of clay products in total about 1.4 Mio m³ raw material (loose volume) has been mined in the year 1991. The guaranteed amount of reserves of raw material (consolidated volume) in Switzerland is about 35 Mio. m³. This meets the Swiss demand for the next 35 to 40 years. It is assumed that about 70 Mio. m³ of raw material near the existing mines is an additionally available but not guaranteed reserve. In addition Switzerland has a huge potential on raw ma-

terial specially molasse marl and opalinus clay. For other raw materials as for example loess clay the reserves are very small. (Kündig et al. (1997))

The Swiss Geotechnical Commission made the following general comment on resources of primary materials in Switzerland¹.

The resources used from the non-metallic minerals industry can be regarded as quarryable when their mining is possible with actual technology, the usage realisable with supportable financial and temporal expenditure and if the utilisation does not collide with other interests. This last point causes a lot of problems in Switzerland as in the intensely populated landscape (Middle Lands) all different kind of utilizations collide. These are for example water protection, settlement area etc. This circumstance reduces the real quarryable area fundamentally.

3 Composition

3.1.1 Primary Raw Material

Ullmann's encyclopaedia of industrial chemistry describes the situation referring to the primary raw material in full range.

“A great variety of clay sediments are used, ranging from loose to compact in consistency. The principal raw materials are loams, clays, marls, mudstone, shale, and slate together with smaller quantities of boulder clay, loess, and sand. They can contain all types of suitable clay minerals (layered silicates), such as illite, sericite, kaolinite, chlorite, and montmorillonite. Quartz is an important component and is always present, and there can also be considerable amounts of calcite, dolomite, and iron compounds. The composition and particle size are the two principal factors that determine the nature of the products.” (Hauck et al. (1997))

Wendehorst & Vollenschaar (1998) write that the main component of the raw material clay (and therefore brick) is Kaolin. Kaolin approaches the formula of the mineral kaolinite ($\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$). As all other clays it is weathered feldspar rock containing several impurities like quartz, mica, residual feldspar, calcite, organic substances and/or Fe-oxides. The various geological and weathering conditions determine the type and composition of clay beds.

Kaolins are primary clays formed by local decomposition of feldspar. They are mainly found in small depressions rather than in vast stratified layers and they are relatively free from mineral impurities like iron. Their grain is coarse and they are not plastic, compared to most sedimentary clays. Pure kaolins are very refractory and their melting point exceeds 1760 °C. They are very difficult to use in genuine state because of their low plasticity and high melting point. Consequently other materials are added to make Kaolin more plastic and to lower its melting point.

“Some constituents used as raw material for the production of clay products are harmful. These are mainly coarse pyrites and limestone inclusions because they can cause spalling and sulphur compounds, which can cause efflorescence. Other harmful constituents are large carbonaceous particles and plant remains.

Whereas common bricks may be manufactured from fairly low-quality clays, large hollow bricks require adequate plasticity of the clay body. For hollow bricks, clays and marls are used primarily, sometimes mixed with loam and shale. Other products with a porous structure are produced from these raw materials, e.g., face bricks, paving bricks, roofing tiles, drainage pipes, and cable conduits.

¹ http://www.sgk.ethz.ch/100jahresgk/meilensteine/abbaubare_rohstoffe.html

Clinker, with its impervious or nearly impervious structure is produced mainly from shale and stoneware clays. Shale clay tends to be used for the manufacture of clinker face bricks, high-compressive-strength solid or hollow bricks, paving bricks, and sewer channel bricks. Stoneware clays are the first choice for high-quality face bricks, for split clinker flags and flooring tiles” (see part 6 “Ceramic Products and Processes”, the module “ceramic tiles, at plant”).

3.1.2 Additives

Ullmann's encyclopaedia of industrial chemistry describes the different additives, which are generally used in the brick and roofing tile industry:

“Additives have increased in importance in recent years, improving both the manufacturing process and the products. A variety of combustible organic materials are used, such as foamed polystyrene, sawdust, powdered straw, powdered brown coal, and materials with organic content including oily filtration aids (fuller's earth), oily metallic hydroxide slimes and coal washery refuse. These additives may increase the porosity and thus the insulating quality of the brick. Furthermore, the amount of fuel required for firing may be reduced.

Powdered limestone or chalk may improve firing properties, increase porosity, and lighten the colour. Powdered slags from blast furnaces or power stations are opening agents that add strength, while making the plastic mass leaner; glass powder is also a suitable additive. Iron oxide powder and various sludges provide colour.

As a consequence of ever more stringent quality requirements, some high-quality clay is commonly used for brick manufacture”. (Hauck et al. (1997)) d

4 Classification

The next two sections describe the characterisation of the most common brick products (Brick and roofing tile) and the third some special products (Hauck et al. (1997)):

4.1 Bricks

Ordinary bricks constitute the largest fraction of total brick production. They have a porous body. If there are no special requirements with regard to appearance or resistance to weathering, they are called inner-wall bricks, or backing bricks, and can be solid or hollow. For hollow bricks there are requirements for compression strength (4 – 28 MPa) and brick bulk density of 1.0 – 2.2 kg/dm³. (The brick bulk density is calculated by dividing the mass of the dried brick by its volume including all channels, finger holes, and mortar pockets, or frogs.)

One special variety of brick is the *insulating brick* used to reduce heat flow. For these the strength specification is between 2 and 28 MPa. The brick density must be only between 0.6 and 1.0 kg/dm³. Adding materials that burn during firing increases the porosity. Insulating brick can be as large as 49 × 30 × 23.8 cm.

Progress is being made in producing room high building components. These are known as *brick planks* and are produced as monolithic pieces with an extrusion press.

If the brick has surface and colour suitable for facing and it is able to withstand frost, the brick is classified as *face brick*, or *facing brick*. If the average compression strength is ≥ 45 MPa, the brick is ranked as *extra strong*.

4.2 Roofing Tiles

Roofing tiles are manufactured to produce their particular shape either with natural colour or with a slip coating (engobed) or a glazed finish. They are manufactured by extrusion or stamping. They usually have a porous, unsintered structure, and before being weatherproof they must be properly shaped, water impermeable, and able to bear sufficient load.

4.3 Special Products

4.3.1 Clinker Brick

This product is not produced in Switzerland. Nevertheless it is part of the classification and therefore described in this chapter:

To achieve clinker classification, the bricks must have a specified degree of sintering (water absorption below ca. 7 wt %), average compression strength > 35 MPa, and a body density $> 1'900$ kg/m³. (The body density is calculated by dividing the mass of the dried brick by its volume excluding all channels, finger holes, and mortar pockets.) Extra strong clinker bricks must have average compression strength > 45 MPa.

Ceramic clinker bricks, which are manufactured from high-quality nonporous sintering clays, are face bricks having a water absorption capacity ≤ 6 %. The average net density must be $\leq 2'000$ kg/m³, and the average compression strength must be > 75 MPa. Furthermore, the surface must have a specified abrasive hardness and colourfastness in light. The clinker materials also include bricks for sewer construction, paving stones, and sintered flagstones, for which the water absorption must be below 3 to 6 %, depending on the intended use.

4.3.2 Others

There are several porous products other than bricks and roofing tiles: wall tiles, floor tiles, drainpipe, and cable conduits.

Specially to be mentioned is the expanded clay. It is a product usually consisting of spherical particles with a diameter of 1 – 16 mm, having a densely sintered surface and an expanded finely porous structure. The quality of this product depends critically on the packing density of the particles, the density of the outer skin and the uniformity of the pore structure. The bulk density varies from 300 to 800 kg/m³, permitting use as aggregate in light concrete or as insulator for walls and floors.

5 Use

The main product of the Swiss industry is brick. It is usually plastered. If it is used for a visible brick wall aesthetic criterias as colour and structure of the surface have to be fulfilled and a good weathering resistance is necessary.

Three main types of roof tiles are produced in Switzerland. The most used tile is called roof tile and is offered in different shapes, colours, strength, consistencies and weathering resistances. Another type is the plain tile, which is flat and produced in the billet method. They are specially used for steep or round roofs (country houses and houses put under preservation order). Double welt tiles are also produced in the billet method but used mainly for less steep but large roofs.

6 System Characterisation

The U.S. Environmental Protection Agency describes a very sound process in “Emission Factor Documentation for AP-42” (EPA (1998)). This system characterisation is reported here, with some modifications to adapt to the Swiss situation. Fig. 6.1 illustrates the course of manufacture and shows which steps are included in the Life Cycle Inventory.

The manufacture of brick and structural clay products involves mining, grinding, screening and blending of the raw materials followed by forming, cutting or shaping, drying, firing, cooling, storage, and shipping of the final product.

The raw materials used in the manufacture of brick and other structural clay products include surface clays and shales, which are mined in open pits. Most facilities in Switzerland have onsite mining operations, which are subsequently restored after excavation. The raw material is typically loaded into a primary crusher for initial size reduction by a lorry or front-end loader. The material is then conveyed to an area for wet processing, where, adding water to the raw material, the clay is grinded, mixed and plasticized. In several grinding mills and banks of screens, fine material suitable for forming brick is produced. From the grinding room the material is conveyed to a storage room, where it matures around two months before further processing.

The plastic and ductile material is then introduced into a continuous extrusion process and cut into form by wire. Following forming and stacking, the brick-laden kiln cars enter a predryer or a holding area and are then loaded into the dryer. Dryers typically are heated to about 200°C using waste heat from the cooling zone of the kiln. However, some plants might heat dryers with gas or other fuels. Dryers may be in line or separate from the kiln. The most common type of kiln for firing brick is the tunnel kiln. A typical tunnel kiln ranges from about 100m to 150m in length and includes a preheating zone, a firing zone, and a cooling zone. The firing zone is typically maintained at a maximum temperature of about 1000°C. After firing the bricks enter the cooling zone, where they are cooled to near ambient temperature before leaving the kiln. The bricks are then packaged in bunch on pallets and usually wrapped with shrinking PE-foil and shipped by truck to their destination.

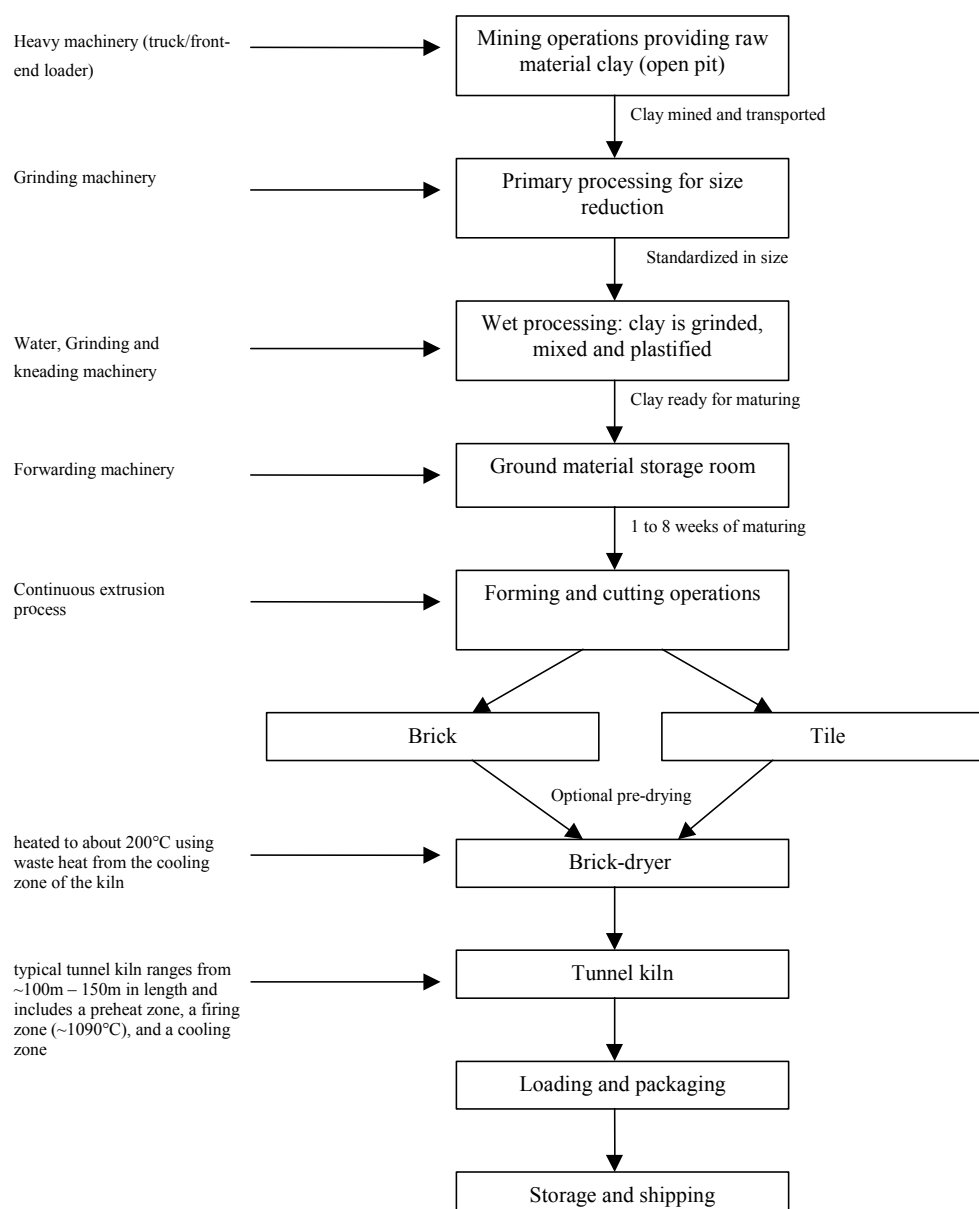


Fig. 6.1 Flow scheme of brick and roof tile manufacture (on the left: additional information to each production step)

7 Life Cycle Inventories

Ecoinvent database meta information (tables), flows and their representation in the ecoinvent database are shown in (figures):

Mine, clay	Tab. 7.1 and Fig. 7.1
Clay, at mine	Tab. 7.2 and Fig. 7.2
Brick, at plant	Tab. 7.3 and Fig. 7.3
Roof tile, at plant	Tab. 7.9 and Fig. 7.4
Packing, clay products	Tab. 7.10 and Fig. 7.5

7.1 Mine, clay

The ecoinvent database meta information of “mine, clay” includes all-important and necessary information to use the data correctly.

Tab. 7.1 Ecoinvent database meta information for the product “mine, clay”

Name	mine, clay
Location	CH
Infrastructure Process	1
Unit	unit
Data Set Version	2.0
Included Processes	Includes the area of the "build" infrastructure (buildings, roads, etc.) used for brick and roof tile production
Amount	1
Local Name	Mine, Ton
Synonyms	
General Comment to reference function	Data are estimated based on an air-picture and sketches of the process. The average yearly production volume is about 100'000 t, Total land occupation: 40'000m; about 20'000m ² buildings (1'000m ² multi storey and 19'000m ² hall), 15'000m ² paved roads, 5'000m ² vegetation. Assumed weight of machines: 2'000t and service life of 25 years.
Start Date	1992
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER modules have been used as proxy
Technology text	The data of the infrastructure corresponds to the high level of technology
Representativeness	
Production Volume	Unknown
Sampling Procedure	The estimation of land use and infrastructure relates to one company in Switzerland (Ziegelei Gasser AG in Rapperswil BE)
Extrapolations	
Uncertainty Adjustments	None

Fig. 7.1 shows the system boundary of the module “mine, clay”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (impressions of visit and air photos) and the standard deviation for each data.

As this is an infrastructure module, the described land-use does not include the mine which is included in the module “brick, at plant” or “roof tile, at plant” as it is a part of the production. Considering that the brick factory is practically attached to the mine and that the land exploitation develops in parallel to brick making operations, the land use of the excavated area is attributed to this module

The required amount of infrastructure referring to the machines is based on the assumption that the total mass of all machines used for the production (including burning kiln, excluding the lining with refractory bricks) come to about 2'000 tons. The service life is assumed to be about 25 years.

General Flow information					Representation in ecoinvent							Uncertainty					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
land occupation for build area	→	infrastructure clay plant				assumption from air photo: 40'000m2, assumption time occupation: 50 years	resource	land			Occupation, industrial area, built up	2.00E+06	m2a	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	1.83	(5,5,1,1,1,1);
land transformation to build area	→					total area with roof: 20'000m2	resource	land			Transformation, to industrial area, built up	2.00E+04	m2	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	2.28	(5,5,1,1,1,1);
land transformation to roads	→					total area with roads: 15'000m2	resource	land			Transformation, to traffic area, road network	1.50E+04	m2	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	2.28	(5,5,1,1,1,1);
land transformation to green area	→					total green area: 5'000m2	resource	land			Transformation, to industrial area, vegetation	5.00E+03	m2	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	2.28	(5,5,1,1,1,1);
land transformation from unknown	→					former use of the land unknown	resource	land			Transformation, from unknown	4.00E+04	m2	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	2.28	(5,5,1,1,1,1);
buildings multi storage	→					5% of roof area: 1'000m2 , assumption avarage hight: 6m	construction processes	buildings	Yes	RER	building, multi-storey	6.00E+03	m3	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	3.27	(5,5,1,1,1,1);
buildings hall	→					95% of roof area: 19'000m2	construction processes	buildings	Yes	CH	building, hall, steel construction	1.90E+04	m2	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	3.32	(5,5,1,1,3,1);
Infrastructure (machines)	→					assumption of total weight of infrastructure: 2'000 t; assumed service life: 25 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	4.00E+06	kg	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	3.32	(5,5,1,1,1,5);
				→	clay plant		this module applies to a plant with a total produced amount per unit (50 years) of 4'000'000'000 kg; yearly production: 80'000'000ka/a	construction materials	additives	Yes	CH	mine, clay	1.00E+00	unit			

Fig. 7.1 Flows for "mine, clay" and their representation in the ecoinvent database

7.2 Clay, at mine

The ecoinvent database meta information of “clay, at mine” includes all-important and necessary information to use the data correctly.

Tab. 7.2 Ecoinvent database meta information for the product “clay, at mine”

Name	clay, at mine
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes excavation by digger, transportation to first grinding machine, the land-use of the mining and the recultivation area
Amount	1
Local Name	Ton, ab Abbau
Synonyms	Lehm//loam
General Comment to reference function	Assumed thickness of clay layer in nature: 30m. No electricity consumption as transports are made by heavy diesel machines and further treatment (including water consumption) is included in "brick, at plant" or "roof tile, at plant". Different types of transformation in Europe, therefore "transformation, to unknown" has been chosen.
Start Date	1992
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	"recultivation, bauxite mine" has been used as Proxy
Technology text	Typical technology for Swiss clay mining
Representativeness	
Production Volume	Unknown
Sampling Procedure	
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 7.2 shows the system boundary of the module “clay, at mine”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (Bruck (1996) and impressions of visit and air photos) and the standard deviation for each data.

The calculation of the land transformation is based on the following assumptions:

The volume of the excavation amount of clay is based on an estimated thickness of the clay layer of 30m. Per m² the total volume is therefore 30m³ clay. The density is about 2'000 kg/m³, which gives a weight of 60'000 kg of excavated clay per m². Thus per kg clay the land used is 1/60'000 m². The soil removed before digging is very small (maximum 0.5m) compared to the layer of clay and therefore neglected.

The calculation of the land occupation is based on the following assumptions:

The transition from mining to recultivation is fluid. There is no exact time when a certain area is exploited and being recultivated. To have a certain statement regarding time to calculate the land occupa-

tion the crossover from mining to recultivation is modulated: After 10 years a certain mineable area is exploited and clay is digged in a nearby area. The exhausted area is recultivated.

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
clay	excavation of clay from nature			resource	in ground			Clay, unspecified, in ground	1.00E+00	kg	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	1.05	(1,1,1,1,1,1);
land use (mine) [m2a]			The timeperiod until the open part of the mine will be exhausted is assumed to be 10 years: -> occupation: $10 \cdot 1.667E-5 = 1.667E-4$ m2a. Reclamation is assumed to start immediately after that.	resource	land			Occupation, mineral extraction site	1.67E-04	m2a	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	1.79	(5,2,3,1,1,1);
land transformation (mine) [m2]			per m2 totally 30m3 clay (with a density of 2000 kg/m3 -> 60'000kg) is excavated -> per kg clay 1/60'000 m2 land is used -> transformation: $1.667E-5$ m2	resource	land			Transformation, to mineral extraction site	1.67E-05	m2	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	2.25	(5,2,3,1,1,1);
land transformation (mine) [m2]				resource	land			Transformation, from unknown	1.67E-05	m2	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	2.25	(5,2,3,1,1,1);
recultivation [m2]			"recultivation, bauxite mine" has been used as Proxy	metals	extraction	No	GLO	recultivation, bauxite mine	1.67E-05	m2	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	2.29	(5,2,3,1,3,1);
land transformation (renaturierung) [m2]			There are so many different types of recultivating in Europe that "transformation, to unknown" is chosen	resource	land			Transformation, to unknown	1.67E-05	m2	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	2.25	(5,2,3,1,1,1);
diesel (mine)			diese consumption concerning excavation	construction processes	machinery	No	GLO	diesel, burned in building machine	2.97E-02	MJ	Bruck 1996	1	1.07	(1,3,1,1,1,1);
			no electricity consumption, as there is no treatment and no transport by conveyor belt of the clay in this process											
			no water consumption, as there is no treatment of the clay in this process											
		clay, at mine		construction materials	additives	No	CH	clay, at mine	1.00E+00	kg				

Fig. 7.2 Flows for "clay, at mine" and their representation in the ecoinvent database

7.3 Brick, at plant

The ecoinvent database meta information of “brick, at plant” includes all-important and necessary information to use the data correctly.

Tab. 7.3 Ecoinvent database meta information for the product “brick, at plant”

Name	brick, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes first grinding process, wet process (includes second grinding, mixing and plastifying), storage, forming (extruding moulding method) and cutting, drying, firing, loading, packing and storage
Amount	1
Local Name	Backstein, ab Werk
Synonyms	Ziegel//clay//Ton
General Comment to reference function	not included: charge of the wastewater and solid waste, it's assumed that the waste heat is totally reused for drying, no hard coal coke is used within these 12 plants
Start Date	1992
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	certain exchanges are proxies (CH for RER), specially for the petrol consumption the proxy "operation, passenger car" has been used; data stem from Switzerland, Germany and Austria
Technology text	Mix of different technologies (different firing fuels) in A, D, CH.
Representativeness	
Production Volume	unknown
Sampling Procedure	The data relates to 12 brick production plants in Germany, Austria and Switzerland; The estimation of infrastructure relates to one company in Switzerland (Ziegelei Gasser AG in Rapperswil BE),
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 7.3 shows the system boundary of the module “brick, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly Bruck (1996), Frischknecht et al. (1996) and Berdowski et al. (1995)) and the standard deviation for each data.

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
electricity	Energy consumption (originally in Wh/kg product)		electricity from electric supply company; one plant produces one part of it's consumption by itself (assumption: by heat-generation from natural gas; not considered)	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.94E-02 kWh		Bruck (1996)	1	1.64	(5,5,3,3,1,5);
natural gas			combustible (Assumption: data is "net calorific value", efficiency: 0.87)	natural gas	fuels	No	RER	natural gas, high pressure, at consumer	1.20E+00 MJ		Bruck (1996), Frischknecht et al. (1996)	1	1.64	(5,5,3,3,1,5);
propan			combustible (Assumption: data is "net calorific value", efficiency: 0.87)	natural gas	fuels	No	RER	natural gas, high pressure, at consumer	4.00E-02 MJ		Bruck (1996), Frischknecht et al. (1996)	1	1.64	(5,5,3,3,1,5);
heating oil light			combustible (Assumption: data is "net calorific value", efficiency: 0.85)	oil	fuels	No	RER	light fuel oil, at regional storage	5.41E-03 kg		Bruck (1996), Frischknecht et al. (1996)	1	1.64	(5,5,3,3,1,5);
heating oil heavy			combustible (Assumption: data is "net calorific value", efficiency: 0.85)	oil	fuels	No	RER	heavy fuel oil, at regional storage	3.81E-04 kg		Bruck (1996), Frischknecht et al. (1996)	1	1.64	(5,5,3,3,1,5);
diesel (general)			fuel (not included: diesel for mining as it is included in module "clay, at mine")	construction processes	machinery	No	GLO	diesel, burned in building machine	2.97E-02 MJ		Bruck (1996)	1	1.64	
petrol				transport systems	road	No	RER	transport, passenger car	1.66E-02 pkm		Bruck (1996)	1	1.64	
clay, at pit	slug mass, clay, medium to make it porous, aggregate, water (originally in g/kg product)		Swiss module as proxy	construction materials	additives	No	CH	clay, at mine	1.35E+00 kg		Bruck (1996)	1	1.64	(5,5,3,3,1,5);
sand			Swiss module as proxy	construction materials	additives	No	CH	sand, at mine	1.47E-02 kg		Bruck (1996)	1	1.64	(5,5,3,3,1,5);
saw dust (new)			substance to make brick porous: Swiss module as proxy; dried matter content : 188.6 kg/m3 -> 40% humidity: density: 1.4*188.6 kg/m3	wood energy	fuels	No	RER	wood chips, mixed, from industry, u=40%, at plant	5.30E-05 m3		Bruck (1996)	1	1.64	(5,5,3,3,1,5);
saw dust (recycling)			substance to make brick porous: not balanced, as it is a reused waste product						1.37E-02 kg		Bruck (1996)	1	1.64	(5,5,3,3,1,5);
polystyrol (new)			substance to make brick porous	plastics	polymers	No	RER	polystyrene, expandable, at plant	3.52E-04 kg		Bruck (1996)	1	1.64	(5,5,3,3,1,5);
polystyrol (recycling)			substance to make brick porous: not balanced, as it is a reused waste product						3.28E-04 kg		Bruck (1996)	1	1.64	(5,5,3,3,1,5);
Papierfangstoff (recycling)			substance to make brick porous: not balanced, as it is a reused waste product						3.04E-02 kg		Bruck (1996)	1	1.64	(5,5,3,3,1,5);
coal dust			substance to make brick porous: net calorific value: 10.51 MJ/kg	lignite	fuels	No	DE	pulverised lignite, at plant	2.45E-02 MJ		Bruck (1996)	1	1.64	(5,5,3,3,1,5);
natural stone flour			Assumed that in most cases limestone and therefore allocated to limestone (Swiss module as proxy)	construction materials	others	No	CH	limestone, milled, packed, at plant	2.39E-02 kg		Bruck (1996)	1	1.64	(5,5,3,3,1,5);
granulated lime			Swiss module as proxy	construction materials	others	No	CH	limestone, milled, packed, at plant	2.20E-05 kg		Bruck (1996)	1	1.64	(5,5,3,3,1,5);
water (tab)			water is supplied from public water supply	water supply	production	No	RER	tap water, at user	2.72E-02 kg		Bruck (1996)	1	1.64	(5,5,3,3,1,5);
water (well)			water is pumped on the company's ground, Swiss module as proxy	resource	in water			Water, well, in ground	7.36E-05 m3		Bruck (1996)	1	1.64	(5,5,3,3,1,5);
Infrastructure and land-use [unite]			Input includes infrastructure needed at production place (buildings, access roads,...) over 50 years	construction materials	additives	Yes	CH	mine, clay	2.00E-10 unit		Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	3.38	(3,5,5,1,3,5);

Fig. 7.3 Flows for "brick, at plant" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
PE foil	process materials: foil, lubricants, steel parts, paletts, lime granulate (originally in mg/kg product)			plastics	processing	No	RER	packaging film, LDPE, at plant	5.42E-04	kg	Bruck (1996)	1	1.64	(5,5,3,3,1,5);
steel band: raw material (steel)			used for packing of bricks on pallet	metals	extraction	No	RER	steel, low-alloyed, at plant	1.57E-05	kg	Bruck (1996)	1	1.64	(5,5,3,3,1,5);
steel band: manufacturing process			used for packing of bricks on pallet	metals	processing	No	RER	sheet rolling, steel	1.57E-05	kg	Bruck (1996)	1	1.64	(5,5,3,3,1,5);
plastic band				plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	8.58E-07	kg	Bruck (1996)	1	1.64	(5,5,3,3,1,5);
lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	7.00E-06	kg	Bruck (1996)	1	1.64	(5,5,3,3,1,5);
gear lubricant and hydraulic oil			lubricating oil used as proxy	chemicals	organics	No	RER	lubricating oil, at plant	3.38E-06	kg	Bruck (1996)	1	1.64	(5,5,3,3,1,5);
lubricating grease			lubricating oil used as proxy	chemicals	organics	No	RER	lubricating oil, at plant	2.84E-06	kg	Bruck (1996)	1	1.64	(5,5,3,3,1,5);
granulated lime			Swiss module as proxy	construction materials	others	No	CH	limestone, crushed, for mill	3.96E-04	kg	Bruck (1996)	1	1.64	(5,5,3,3,1,5);
wooden paletts			weight of one pallet: 22 kg	wooden materials	processing	No	RER	EUR-flat pallet	1.61E-05	unit	weight of palett: email Thomas Ulitzsch, Gütegemeinschaft Paletten of 03.09.02	1	1.64	(5,5,3,3,1,5);
mould (or die reliefs) (chromium-plated): steel			used for extruding (forming) the brick; material: steel, low-alloyed, process chromium-plating	metals	extraction	No	RER	steel, low-alloyed, at plant	1.58E-07	kg	Bruck (1996)	1	1.64	(5,5,3,3,1,5);
forming of mould (or die reliefs) (chromium plated): rolling steel	purveyor transports		process of steel forming	metals	processing	No	RER	sheet rolling, chromium steel	1.58E-07	kg	Bruck (1996)	1	1.64	(5,5,3,3,1,5);
roll sleeve			steel, low-alloyed used as proxy	metals	extraction	No	RER	steel, low-alloyed, at plant	1.06E-05	kg	Bruck (1996)	1	1.64	(5,5,3,3,1,5);
other steel parts			steel, low-alloyed used as proxy	metals	extraction	No	RER	steel, low-alloyed, at plant	4.16E-06	kg	Bruck (1996)	1	1.64	(5,5,3,3,1,5);
fireclay refractory			Not balanced as no data is available								Bruck (1996)			
lorry 28t			25% of total tkm with lorry 28t (in report "Bruck 1996" allocated to 16t lorry; based on information of H. Schmid: 28t lorry). Swiss module used as proxy.	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	4.68E-03	tkm	Bruck (1996) and personal communication with an expert of the clay	1	2.34	(5,5,3,3,1,5);
lorry 40t			Swiss module as proxy; 75% of total tkm with lorry 40t	transport systems	road	No	CH	transport, lorry >28t, fleet average	1.40E-02	tkm	Bruck (1996)	1	2.34	(5,5,3,3,1,5);
fright train				transport systems	train	No	RER	transport, freight, rail	9.00E-05	tkm	Bruck (1996)	1	2.34	(5,5,3,3,1,5);

Fig. 7.3 Flows for “brick, at plant” and their representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
emissions to air, product (originally in mg/kg product)		→ Waste heat	Same amount as electricity used	air	unspecified			Heat, waste	1.42E-01	MJ		1	1.24	(2,4,1,2,1,5);
		→ dust pariculates > 10 um	total measured emissions within the system boundary; share based on EPA-report (share based on EPA-report: about 25%)	air	unspecified			Particulates, > 10 um	4.68E-06	kg	Bruck (1996) and EPA (1998): Brick and Structural Clay Product Manufacturing	1	1.89	(5,5,3,3,1,5);
		→ dust pariculates < 2.5 um	total measured emissions within the system boundary; assumed that the amount is 100% from combustion and therefore the share is calculated based on the emissions from the combustion (share based on EPA-report: assumption total of the 75% of the particulates < 2.5um)	air	unspecified			Particulates, < 2.5 um	1.40E-05	kg	Bruck (1996) and EPA (1998): Brick and Structural Clay Product Manufacturing	1	3.33	(5,5,3,3,1,5);
		→ nitrogen oxide (as NO2)	total measured emissions within the system boundary (mainly from combustion of the firing process)	air	unspecified			Nitrogen oxides	2.60E-04	kg	Bruck (1996)	1	1.89	(5,5,3,3,1,5);
		→ sulphur oxide (as SO2)	total measured emissions within the system boundary (mainly from combustion of the firing process)	air	unspecified			Sulfur dioxide	9.98E-05	kg	Bruck (1996) and Berdowski et al. (1995)	1	2.34	(5,5,3,3,1,5);
		→ hydrogen fluoride	total measured emissions within the system boundary (mainly from combustion of the firing process)	air	unspecified			Hydrogen fluoride	1.06E-05	kg	Bruck (1996) and Berdowski et al. (1995)	1	1.89	(5,5,3,3,1,5);
		→ carbon monoxide	total measured emissions within the system boundary (mainly from combustion of the firing process)	air	unspecified			Carbon monoxide, fossil	3.91E-04	kg	Bruck (1996) and Berdowski et al. (1995)	1	1.64	(5,5,3,3,1,5);
		→ carbon dioxide	total measured emissions within the system boundary (mainly from combustion of the firing process)	air	unspecified			Carbon dioxide, fossil	1.80E-01	kg	Bruck (1996)	1	1.64	(5,5,3,3,1,5);
		→ hydrogen chloride	total measured emissions within the system boundary (mainly from combustion of the firing process)	air	unspecified			Hydrogen chloride	1.22E-05	kg	Bruck (1996)	1	1.89	(5,5,3,3,1,5);
		→ organic mediums (totale C)	total measured emissions within the system boundary assumption that an avarage NMVOC includes about 80 weight% of C	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	7.63E-05	kg	Bruck (1996)	1	2.34	(5,5,3,3,1,5);
		→ benzen	total measured emissions within the system boundary	air	unspecified			Benzene	2.96E-06	kg	Bruck (1996)	1	2.34	(5,5,3,3,1,5);
		→ formaldehyde	total measured emissions within the system boundary	air	unspecified			Formaldehyde	1.64E-05	kg	Bruck (1996)	1	2.34	(5,5,3,3,1,5);
		→ phenol	total measured emissions within the system boundary	air	unspecified			Phenol	1.30E-07	kg	Bruck (1996)	1	2.34	(5,5,3,3,1,5);
		→ brick, at plant		construction materials	bricks	No	RER	brick, at plant	1.00E+00	kg	Bruck (1996)	1		

Fig. 7.3 Flows for “brick, at plant” and their representation in the ecoinvent database (cont.)

7.3.1 General Comment

The LCI data published in Bruck (1996) show several inconsistencies which have been corrected and updated in co-operation with an expert of the clay production industry².

The general problems and adaptations:

- The calculation of the average data from the data of all different companies reported in Bruck (1996) is not properly done as the production amount of each company has not been taken into account. That was not possible, as the author could not offer the detailed production amounts³.
- All data relate to 1 kg burned product. It is assumed that the brick fragments from loading are not taken into account. This fact is considered in the uncertainties.
- Several companies have not indicated data for certain in- and outputs. Either these in- or outputs do not exist (indicated with "0" in the tables) or they are not reported (indicated with "no data" in the tables). In another case the data are obviously not realistic and wrong and therefore not taken into account (indicated "not considered" in the tables). In co-operation with a deputy of the Swiss brick industry all data and lacks have been discussed and decided which case could be correct. Depending on the case the data of the company has been taken into the calculation of the average amount or not.

7.3.2 Fuel used for Firing Bricks

The data used in this study cover 12 brick making companies (Bruck 1996). Nine companies cover their energy (heat) consumption 100% with natural gas, one plant combines natural gas with heavy fuel oil and one plant combines natural gas with light fuel oil. Another company covers its energy consumption by burning propane, heavy and light fuel oil.

For the year 2001 the share of fuel for firing bricks in Europe is about 20% heavy fuel oil and 80% natural gas. In general the tendency in Europe is using less oil and more natural gas².

The data refer to the year 1992/1993. The sum of all different energy consumptions and the average over all companies show that the amount is too high. Nevertheless the data are used for this study.

The energy data include a certain amount of petrol. It is assumed that petrol is used for different transportation processes. This can be petrol operated fork lift trucks or other internal transports with petrol operated vehicles. In this study the amount of petrol is converted into the operation of a passenger car. The amount therefore does not say anything about the true distance, which is driven by passenger cars for one kg of brick or roof tile. It just takes into account the effect of using petrol in a combustion-engined vehicle as a proxy for different kind of vehicles.

² personal communication Mr. H. Schmid (former director of "ZZ Ziegeleien" Switzerland)

³ E-mail from M. Feller from "Kanzlei Dr. Manfred Bruck" from 2.10.2002

Tab. 7.4 Adapted energy consumption data for the brick production from (Bruck (1996)) with remarks

Input	Process Name	Company 1	Company 2	Company 3	Company 4	Company 5	Company 6	Company 7	Company 8	Company 9	Company 10	Company 11	Company 12	data used in this study	Remarks
electricity	Endenergy / kWh/t product	49.52	34.96	36.54	36.38	38.06	39.81	35.90	44.00	45.00	45.13	24.64	41.21	39.43	Average amount over all 12 companies; it is assumed that more companies than only nr. 12 generate their power themselves
self-power generation		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.90	0.00	1.41	
natural gas		322.28	315.14	343.57	451.44	0.00	349.78	318.07	366.69	277.00	162.71	587.41	500.05	332.85	Average amount over all 12 companies; in total the share of oil to gas for the year 2001 in europe is about 20 to 80%. This is depicted with these data from the 12 companies relating to the year 1992. In general the trend is away from heavy heating oil.
propan		0.00	0.00	0.00	0.00	133.38	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.12	
heating oil light		0.00	0.00	0.00	0.00	31.46	0.00	0.00	0.00	0.00	0.00	0.00	8.10	3.30	
heating oil heavy		0.00	0.00	0.00	0.00	284.75	0.00	0.00	0.00	0.00	484.08	0.00	0.00	64.07	Average amount only over 8 companies as no data (n.d.) supplied from 4 companies.
diesel (for movements in general but without mine)		0.99	4.18	5.43	5.10	0.49	n.d.	n.d.	n.d.	n.d.	1.19	6.15	11.38	4.36	
diesel (for movements within mine)		3.10	4.38	7.56	8.35	6.80	n.d.	n.d.	n.d.	n.d.	2.77	14.88	18.11	8.24	
petrol		n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	1.03	1.03	Amount of one company used as average for all companies as it is assumed that the others also use petrol but did not supply data (n.d.).
saw dust (new)		135.60	127.57	94.35	0.00	n.c.	51.70	0.00	0.00	46.94	32.57	0.00	n.c.	48.87	
saw dust (recycling)		86.71	0.00	55.41	174.75	n.c.	51.70	0.00	0.00	46.94	0.00	60.97	n.c.	47.65	Company Nr. 12 is not considered (n.c.) as no data at all for additional fuels are supplied. Company Nr. 5 is not considered as the amount supplied is much too small (not realistic). For this study the average is made over 10 companies.
polystyrol (new)		0.00	0.00	0.00	0.00	n.c.	0.00	5.77	23.46	0.00	0.00	5.73	n.c.	3.50	
polystyrol (recycling)		0.00	0.00	0.00	3.34	n.c.	0.00	5.77	23.46	0.00	0.00	0.00	n.c.	3.26	
coal dust (brown coal)		0.00	68.04	0.00	0.00	n.c.	0.00	0.00	0.00	0.00	0.00	0.00	n.c.	6.80	
Papierfangstoff (recycling)		0.00	0.00	0.00	0.00	n.c.	145.00	82.48	0.00	108.07	0.00	58.60	n.c.	39.42	The overall amount concerning the energy demand is in general too high. For europe (EU) it is smaller than 400 kWh/t, for Switzerland it is smaller than 500 kWh/t (the share in lime is higher)
Total kWh/t		598.20	554.27	544.86	679.36	494.94	637.99	447.99	457.61	523.95	728.45	775.28	579.88	615.29	

7.3.3 Raw Materials, Process Materials and Additives

The data for the raw materials, process materials and additives used in this study and the explanation of the differences to the original study are given in Tab. 7.6.

Most of the additives used as air entraining agents are also energy carriers as they burn almost completely. The amount of heat released corresponds to the thermal values. Bruck (1996) uses the amounts illustrated in Tab. 7.5:

Tab. 7.5 Air-entraining additives in the brick production (Bruck (1996))

Additive	Thermal value	Unit
Saw dust (new and recycling)	3.49	kWh/kg
Polystyrol (new and recycling)	9.93	kWh/kg
Paper degasifying matter	1.29	kWh/kg
Coal dust	2.92	kWh/kg

,The energy content of these additives is not taken into account.

Tab. 7.6 Adapted data for the raw and process materials and the additives in the brick production from Bruck (1996) with remarks

Input	Process Name	Company 1	Company 2	Company 3	Company 4	Company 5	Company 6	Company 7	Company 8	Company 9	Company 10	Company 11	Company 12	data used in this study	Remarks
clay	slug mass, clay, medium to make it porous, aggregate, water, kg/t product	1400.67	1318.00	1470.00	1232.24	1055.56	1320.00	1409.99	1389.99	1420.14	1233.33	1723.24	1216.00	1349.10	Average amount over all 12 companies
sand		0.00	0.00	1.59	0.00	0.00	174.79	0.00	0.00	0.00	0.00	0.00	0.00	14.70	Average amount over all 12 companies as it is assumed that the companies with no data use 100% clay.
saw dust (new)		38.86	36.55	27.03	0.00	n.c.	14.82	0.00	0.00	13.45	9.33	0.00	n.c.	14.00	Company Nr. 12 is not considered (n.c.) as no data at all for process materials are supplied. Company Nr. 5 is not considered as today the data for saw dust is "Papierfangstoff" and the amount supplied is very small (not realistic). Therefore the average for this study is made over 10 companies.
saw dust (recycling)		24.85	0.00	15.88	50.07	n.c.	14.82	0.00	0.00	13.45	0.00	17.47	n.c.	13.65	
polystyrol (new)		0.00	0.00	0.00	0.00	n.c.	0.00	0.58	2.37	0.00	0.00	0.57	n.c.	0.35	
polystyrol (recycling)		0.00	0.00	0.00	0.33	n.c.	0.00	0.58	2.37	0.00	0.00	0.00	n.c.	0.33	
Papierfangstoff (recycling)		0.00	0.00	0.00	0.00	n.c.	116.00	65.98	0.00	86.46	0.00	35.94	n.c.	30.44	
coal dust		0.00	23.30	0.00	0.00	n.c.	0.00	0.00	0.00	0.00	0.00	0.00	n.c.	2.33	
natural stone flour		0.00	0.00	1.59	21.11	n.c.	0.00	0.00	0.00	0.00	0.00	0.00	n.c.	2.27	
perlite		0.00	0.00	0.00	0.00	n.c.	0.00	0.00	0.00	0.00	0.00	0.00	n.c.	0.00	
granulated lime		0.22	0.00	0.00	0.00	n.c.	0.00	0.00	0.00	0.00	0.00	0.00	n.c.	0.02	
water (tab)		0.00	0.00	0.00	0.00	92.00	0.00	0.00	84.09	0.00	0.00	0.00	150.26	27.20	Based on oral information of Mr. H. Schmid the amount of water used in company Nr. 5 is not ground water but tab water (is changed)
water (well)		100.11	121.17	193.05	93.46	0.00	89.04	105.57	0.00	93.75	29.33	57.61	0.00	73.59	
pvc foil	process materials: foil, lubricants, steel parts, pallets, lime granulate, g/t product	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	n.c.	n.c.	n.c.	0.00	Company Nr. 10 and 12 is not considered (n.c.) as no data at all for any process material is supplied, which seems not to be possible. Company Nr. 11 is not considered as the declared amount is much too high. If it can be supposed that any process material is used although the company has not declared it is marked with "n.d." (no data) and not taken into calculation.
pe foil		828.77	667.36	826.62	671.88	602.78	336.04	298.83	334.07	314.51	n.c.	n.c.	n.c.	542.32	
steel band		n.d.	55.43	n.d.	13.41	n.d.	6.68	5.94	6.64	6.25	n.c.	n.c.	n.c.	15.73	
plastic band		0.00	0.00	7.72	0.00	0.00	0.00	0.00	0.00	0.00	n.c.	n.c.	n.c.	0.86	
lubricating oil		12.18	5.87	9.99	12.02	0.00	6.01	5.35	5.98	5.63	n.c.	n.c.	n.c.	7.00	
gear lubricant and hydraulic oil		6.63	10.07	n.d.	2.40	n.d.	1.19	1.06	1.18	1.11	n.c.	n.c.	n.c.	3.38	
lubricating grease		8.79	2.33	1.64	2.97	4.17	1.48	1.32	1.48	1.39	n.c.	n.c.	n.c.	2.84	
granulated lime		215.02	n.d.	n.d.	741.77	n.d.	370.99	329.91	368.81	347.22	n.c.	n.c.	n.c.	395.62	
wooden pallets		455.57	417.10	692.57	433.94	n.d.	217.03	193.00	215.75	203.13	n.c.	n.c.	n.c.	353.51	
die reliefs (chromium-plated)		0.20	0.09	0.10	0.30	n.d.	0.15	0.13	0.15	0.14	n.c.	n.c.	n.c.	0.16	Amount which is declared by company 5 is to not considered as it is much too high and it has to be assumed that it is a sum of all steel parts.
roll sleeve		0.59	16.77	6.76	20.77	n.d.	10.39	9.24	10.33	9.72	n.c.	n.c.	n.c.	10.57	
other steel parts		0.13	4.38	4.54	8.31	n.c.	4.16	3.70	4.13	3.89	n.c.	n.c.	n.c.	4.16	

7.3.4 Emissions

The data for the emissions used in this study are shown in Tab. 7.7.

Tab. 7.7 Adapted data for the emissions in the brick production from Bruck (1996) with remarks

Process Name	Output 1 / ev. Input	Company 1	Company 2	Company 3	Company 4	Company 5	Company 6	Company 7	Company 8	Company 9	Company 10	Company 11	Company 12	data used in this study	Remarks
emissions to air, g/t product	dust particulates	3.73	3.54	3.57	4.01	9.31	29.23	27.91	27.96	2.01	n.d.	90.37	4.09	18.70	As it is not possible that a company does not emit one of the emissions the lacks are filled with "n.d." (no data) and not taken into the calculation of the "average".
	nitrogen oxide (as NO ₂)	320.08	57.79	133.21	43.91	223.06	n.d.	64.73	464.70	n.d.	n.d.	147.80	885.34	260.07	
	sulphur oxide (as SO ₂)	264.11	198.53	21.24	38.57	42.22	159.60	21.21	199.44	146.32	66.67	9.37	30.88	99.85	
	hydrogen fluoride	0.29	2.98	0.39	0.59	6.67	0.15	92.00	2.58	1.25	6.67	4.17	9.02	10.56	
	carbon monoxide	101.62	708.37	129.34	1023.65	271.25	n.d.	n.d.	n.d.	n.d.	n.d.	154.04	348.40	390.95	
	carbon dioxide	269.34	140.51	83.49	204.25	194.44	186.77	155.06	189.05	226.14	n.d.	145.17	190.37	180.42	
	hydrogen chloride	1.52	2.14	3.19	4.60	n.d.	31.76	18.48	18.44	8.82	n.d.	20.96	n.d.	12.21	
	organic mediums (total C)	18.66	53.13	18.34	74.18	0.69	49.71	33.85	270.56	20.69	n.d.	70.87	n.d.	61.07	
	benzen	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	2.96	n.d.	2.96	
	formaldehyde	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	16.41	n.d.	16.41	
	phenol	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	0.13	n.d.	0.13	

The emissions from burning of brick, described in (Bruck (1996)) originate mainly from the combustion of the firing process. This includes the burning of the combustible, burning of eventually existing organic aggregates, raw material reactions as the transformation of CaCO_3 to CaO and CO_2 , and the burning of organic materials in the raw clay. Some minor emissions come from the combustion of combustible in the boiler house and the transportation traffic within the company fences. Due to the different fuels, the different compositions of the raw material (clay) used in Europe and the reactions between the raw material and the emissions, which can't be estimated the measured emissions are not completed by further emissions from the combustion. The emissions indicated are CO_2 , CO , dust, NO_2 , SO_2 , HF, HCl, org. C, C_6H_6 (benzene), CH_2O (formaldehyde) and $\text{C}_6\text{H}_5\text{O}$ (phenol). There is no information on the share of the CO_2 emission to the different sources (e.g. geogenic (from mineral raw material), fossil (from burning of fuel) or biogenic (from additive (e.g. saw dust))

The share of the particulates is based on EPA (1998) and declares 75% to be PM_{10} (particulates $< 10\mu\text{m}$) and 25% to be bigger than $10\mu\text{m}$. No data are reported smaller than $2.5\mu\text{m}$. As in Europe a large part of the heating energy is produced with natural gas (about 80%) and for gas furnaces almost 100% of the TSP emission is $\text{PM}_{2.5}$ (Berdowski et al. (1995)), in this study 100% of PM_{10} is allocated to $\text{PM}_{2.5}$. No particulates between $2.5\mu\text{m}$ and $10\mu\text{m}$ are to be expected.

7.3.5 Transports

The data for the transports used in this study is shown in Tab. 7.8.

Tab. 7.8 Adapted data for the transports in the brick production from Bruck (1996) with remarks

Input	Process Name	Company 1	Company 2	Company 3	Company 4	Company 5	Company 6	Company 7	Company 8	Company 9	Company 10	Company 11	Company 12	data used in this study
lorry	m purveyor transports, tkm/t purveyor	0.04	0.01	0.03	0.06	0.00	0.03	0.02	0.01	0.01	0.00	0.00	n.c.	0.02
fright train		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	n.c.	0.00

The size of the lorry reported in the study of Bruck (1996) with partly 16 and partly 40t is adapted by arrangement with an expert of the clay production industry⁴ to 28t.

7.3.6 Infrastructure

The required amount of infrastructure (excluding machines) and land-use is based on direct information from a producer⁴. The data are based on:

Yearly produced bricks:	80'000'000 kg
Yearly produced roof tiles:	3.5 Mio pieces
Presumed weight per tile: 4 kg, total weight:	14'000'000 kg
Total amount produced yearly, including earthenware slabs:	about 100'000'000 kg

Several different transformation possibilities are employed for recultivation of exploited mines all over Europe: e.g. Transformation to former land-use (e.g. forest, cultivated area, etc.), to forest, to biotopes or just leaving over to the nature. Therefore in this study the type of transformation chosen is "to unknown".

⁴ Personal communication H. Gasser from Gasser Ziegeleien in Rapperswil, Bern at February 28th, 2002.

7.4 Roof tile, at plant

The ecoinvent database meta information of “roof tile, at plant” includes all-important and necessary information to use the data correctly.

Tab. 7.9 Ecoinvent database meta information for the product “roof tile, at plant”

Name	roof tile, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes first grinding process, wet process (includes second grinding, mixing and plastifying), storage, forming (compression moulding method) and cutting, drying, firing, loading, packing and storage.
Amount	1
Local Name	Dachziegel, ab Werk
Synonyms	
General Comment to reference function	Not included: charge of the wastewater and solid waste. It's assumed that the waste heat is totally reused for drying. No hard coal coke is used by any of these 12 plants. In order to obtain "impermeable" tiles, no media is included which may make it porous (as made for bricks). Not included is the use of engobe to colour the tiles.
Start Date	1992
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Certain exchanges are proxies (CH for RER), specially for the petrol consumption the proxy "operation, passenger car" and for gypsum moulding the module "gypsum fibre board" has been used; data stem from Switzerland, Germany and Austria
Technology text	Mix of different technologies (different firing fuels) in A, D, CH. Share of heavy oil to natural gas about 20% to 80%
Representativeness	
Production Volume	Unknown
Sampling Procedure	
Extrapolations	Extrapolated from "brick, at plant" as it is basically the same production process. The heating fuel consumption and most emissions have been increased by the factor 1.5 (because of need to e.g. increase frost resistance compared to brick). The emissions HCl and HF have been increased by the factor 5. The electricity consumption has been approximately doubled.
Uncertainty Adjustments	None

Fig. 7.4 shows the system boundary of the module “roof tile, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly EPA (1998), Bruck (1996), Frischknecht et al. (1996) and Weibel & Stritz (1995)) and the standard deviation for each data.

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
electricity	Energy consumption (originally in Wh/kg product)		electricity from electric supply company; information rather low (uncertainty adjusted)	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	8.00E-02	kWh	Personal communication with an expert of the clay production industry (Mr. H. Schmid) Bruck (1996),	1	1.69	(5,5,3,3,3,5);
natural gas			combustible (Assumption: data is "net calorific value", efficiency: 0.87)	natural gas	fuels	No	RER	natural gas, high pressure, at consumer	1.80E+00	MJ	Frischknecht et al. (1996), Weibel & Stritz (1995)	1	1.69	(5,5,3,3,3,5);
propan			combustible (Assumption: data is "net calorific value", efficiency: 0.87)	natural gas	fuels	No	RER	natural gas, high pressure, at consumer	6.00E-02	MJ	Bruck (1996), Frischknecht et al. (1996), Weibel & Stritz (1995)	1	1.69	(5,5,3,3,3,5);
heating oil light			combustible (Assumption: data is "net calorific value", efficiency: 0.85)	oil	fuels	No	RER	light fuel oil, at regional storage	8.12E-03	kg	Bruck (1996), Frischknecht et al. (1996), Weibel & Stritz (1995)	1	1.69	(5,5,3,3,3,5);
heating oil heavy			combustible (Assumption: data is "net calorific value", efficiency: 0.85)	oil	fuels	No	RER	heavy fuel oil, at regional storage	5.72E-04	kg	Bruck (1996), Frischknecht et al. (1996), Weibel & Stritz (1995)	1	1.69	(5,5,3,3,3,5);
diesel (general)			fuel (not included: diesel for mining as it is included in module "clay, at mine")	construction processes	machinery	No	GLO	diesel, burned in building machine	2.97E-02	MJ	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
petrol			fuel: proxy transport passenger car; fuel consumption: 2.82 MJ/km, load: 1.59 persons per car	transport systems	road	No	RER	transport, passenger car	1.66E-02	pkm	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
clay, at pit	slug mass, clay, medium to make it porous, aggregate, water (originally in g/kg product)		Swiss module as proxy	construction materials	additives	No	CH	clay, at mine	1.35E+00	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
sand			Swiss module as proxy	construction materials	additives	No	CH	sand, at mine	1.47E-02	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
natural stone flour			Assumed that in most cases limestone and therefore allocated to limestone (Swiss module as proxy)	construction materials	others	No	CH	limestone, milled, packed, at plant	2.39E-02	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
engobe			not unproblematic; 98% clay mass and 2% different mediums for colouring the tiles; no information about the substance.						3.60E-03	kg	Personal communication with an expert of the clay production industry (Mr. H. Schmid)	1		(,,,,);
moulding gypsum			Swiss average data from one company for the years 1985 to 2002	construction materials	binder	No	CH	stucco, at plant	1.53E-03	kg	Personal communication with an expert of the clay production industry (Mr. H. Schmid)	1	1.69	(5,5,3,3,3,5);
water (tap)			water is supplied from public water supply	water supply	production	No	RER	tap water, at user	2.72E-02	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
water (well)			water is pumped on the company's ground, Swiss module as proxy	resource	in water	Yes	CH	Water, well, in ground	7.36E-05	m3	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
infrastructure and land-use [unite]			Input includes infrastructure needed at production place (buildings, access roads,...) over 50 years	construction materials	additives	Yes	CH	mine, clay	2.00E-10	unit	Impressions from visit and air photo of brickworks Gasser in Rapperswil BE	1	3.38	(3,5,5,1,3,5);

Fig. 7.4 Flows for "roof tile, at plant" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
PE foil	process materials: foil, lubricants, steel parts, paletts, lime granulate (originally in mg/kg product)			plastics	processing	No	RER	packaging film, LDPE, at plant	5.42E-04	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
steel band: raw material (steel)			used for packing of bricks on pallet	metals	extraction	No	RER	steel, low-alloyed, at plant	1.57E-05	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
steel band: manufacturing process			used for packing of bricks on pallet	metals	processing	No	RER	sheet rolling, steel	1.57E-05	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
plastic band				plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	8.58E-07	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	7.00E-06	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
gear lubricant and hydraulic oil			lubricating oil used as proxy	chemicals	organics	No	RER	lubricating oil, at plant	3.38E-06	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
lubricating grease			lubricating oil used as proxy	chemicals	organics	No	RER	lubricating oil, at plant	2.84E-06	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
granulated lime			Swiss module as proxy	construction materials	others	No	CH	limestone, crushed, for mill	3.96E-04	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
wooden pallets			weight of one palett: 22 kg	wooden materials	processing	No	RER	EUR-flat pallet	1.61E-05	unit	weight of palett: email Thomas Ullitzsch, Gütegemeinschaft Paletten of 03.09.02	1	1.69	(5,5,3,3,3,5);
mould (or die reliefs) (chromium-plated): steel			used for extruding (forming) the brick; material: steel, low-alloyed, process chromium-plating	metals	extraction	No	RER	steel, low-alloyed, at plant	1.58E-07	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
forming of mould (or die reliefs) (chromium plated): rolling steel	purveyor transports		process of steel forming	metals	processing	No	RER	sheet rolling, chromium steel	1.58E-07	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
roll sleeve			steel, low-alloyed used as proxy	metals	extraction	No	RER	steel, low-alloyed, at plant	1.06E-05	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
other steel parts			steel, low-alloyed used as proxy	metals	extraction	No	RER	steel, low-alloyed, at plant	4.16E-06	kg	Bruck (1996)	1	1.69	(5,5,3,3,3,5);
fireclay refractory			Not balanced as no data is available								Bruck (1996)			
lorry 28t			25% of total tkm with lorry 28t (in report "Bruck 1996" allocated to 16t lorry; based on information of H. Schmid: 28t lorry). Swiss module used as proxy.	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	4.68E-03	tkm	Bruck (1996), personal communication with an expert of the clay production industry (Mr. H. Schmid)	1	2.34	(5,5,3,3,1,5);
lorry 40t			Swiss module "lorry, 28t" used as proxy; 75% of total tkm with lorry 40t	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	1.40E-02	tkm	Bruck (1996)	1	2.38	(5,5,3,3,3,5);
fright train				transport systems	train	No	RER	transport, freight, rail	9.00E-05	tkm	Bruck (1996)	1	2.38	(5,5,3,3,3,5);

Fig. 7.4 Flows for "roof tile, at plant" and their representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
emissions to air, product (originally in mg/kg product)		→ Waste heat	Same amount as electricity used	air	unspecified			Heat, waste	2.88E-01	MJ		1	1.24	(2,4,1,2,1,5);
		→ dust pariculates > 10 um	total measured emissions within the system boundary; share based on EPA-report (share based on EPA-report: about 25%)	air	unspecified			Particulates, > 10 um	7.01E-06	kg	Bruck (1996), EPA (1998): Brick and Structural Clay Product Manufacturing and Weibel & Stritz (1995)	1	1.94	(5,5,3,3,3,5);
		→ dust pariculates < 2.5 um	total measured emissions within the system boundary; assumed that the amount is 100% from combustion and therefore the share is calculated based on the emissions from the combustion (share based on EPA-report: assumption total of the 75% of the particulates < 2.5um)	air	unspecified			Particulates, < 2.5 um	2.10E-05	kg	Bruck (1996), EPA (1998): Brick and Structural Clay Product Manufacturing and Weibel & Stritz (1995)	1	3.38	(5,5,3,3,3,5);
		→ nitrogen oxide (as NO2)	total measured emissions within the system boundary (mainly from combustion of the firing process)	air	unspecified			Nitrogen oxides	3.90E-04	kg	Bruck (1996), Weibel & Stritz (1995)	1	1.94	(5,5,3,3,3,5);
		→ sulphur oxide (as SO2)	total measured emissions within the system boundary (mainly from combustion of the firing process)	air	unspecified			Sulfur dioxide	1.50E-04	kg	Bruck (1996), Weibel & Stritz (1995)	1	2.38	(5,5,3,3,3,5);
		→ hydrogen fluoride	total measured emissions within the system boundary (mainly from combustion of the firing process)	air	unspecified			Hydrogen fluoride	5.28E-05	kg	Bruck (1996), Weibel & Stritz (1995)	1	1.94	(5,5,3,3,3,5);
		→ carbon monoxide	total measured emissions within the system boundary (mainly from combustion of the firing process)	air	unspecified			Carbon monoxide, fossil	5.86E-04	kg	Bruck (1996), Weibel & Stritz (1995)	1	1.69	(5,5,3,3,3,5);
		→ carbon dioxide	total measured emissions within the system boundary (mainly from combustion of the firing process)	air	unspecified			Carbon dioxide, fossil	2.71E-01	kg	Bruck (1996), Weibel & Stritz (1995)	1	1.69	(5,5,3,3,3,5);
		→ hydrogen chloride	total measured emissions within the system boundary (mainly from combustion of the firing process)	air	unspecified			Hydrogen chloride	6.11E-05	kg	Bruck (1996), Weibel & Stritz (1995)	1	1.94	(5,5,3,3,3,5);
		→ organic mediums (totale C)	total measured emissions within the system boundary assumption that an avarage NMVOC includes about 80 weight% of C	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	1.15E-04	kg	Bruck (1996), Weibel & Stritz (1995)	1	2.38	(5,5,3,3,3,5);
		→ benzen	total measured emissions within the system boundary	air	unspecified			Benzene	4.44E-06	kg	Bruck (1996), Weibel & Stritz (1995)	1	2.38	(5,5,3,3,3,5);
		→ formaldehyde	total measured emissions within the system boundary	air	unspecified			Formaldehyde	2.46E-05	kg	Bruck (1996), Weibel & Stritz (1995)	1	2.38	(5,5,3,3,3,5);
		→ phenol	total measured emissions within the system boundary	air	unspecified			Phenol	1.95E-07	kg	Bruck (1996), Weibel & Stritz (1995)	1	2.38	(5,5,3,3,3,5);
		→ roof tile, at plant		construction materials	coverings	No	RER	roof tile, at plant	1.00E+00	kg	Bruck (1996)			

Fig. 7.4 Flows for “roof tile, at plant” and their representation in the ecoinvent database (cont.)

The time for firing dried roof tiles is about 1.5 time longer than for bricks, according to (Weibel & Stritz (1995)). An expert of the clay production industry⁵ is very sceptical concerning this factor. He says that in nowadays tiles are burned in fast firing furnaces. As no better data are available nevertheless this factor has in general been used to increase the energy consumption and emission rates for brick production in order to adapt the values to tile production. The heating modules affected: natural gas, propane, light and heavy oil.

The electricity consumption for the production of 1kg roof tile in Europe is between 70 and 130Wh⁵. For the study 80kW/kg is chosen. This assumption is rather optimistic. Anyway in relation to the consumption for brick it is about 5 to 10 times higher. This can be partly explained with the high-energy consumption for the flue gas cleaning plant as the cleaning time is about 1.5 times longer but no further explanation has been found.

A corresponding probing by an expert of the clay production industry⁵ has shown that the factor of 1.5 for CO₂, CO, dust, NO₂, SO₂, and org. C is realistic (between 0.3 and 1.7). It must be stated that the emission of hydrogen fluoride and hydrochloride acid is about 5 times higher than for the brick production. The source is based on the composition of the raw material clay which differs from the raw material which is used for brick.

A further difference to bricks is that roof tiles have to be more leek- and freeze-proven. Therefore pore forming media are not included in tile-production.

To have the roof tiles coloured often engobe⁶ is used. The amount of one company in Switzerland is about 3.6g/kg burned tile. The composition of engobe is confidential and therefore it can not be taken into account in this study. More than 50% of the waste water from cleaning of the engobe equipment is usually reused as mixing water for the wet process to plastify the clay. Solid waste is deposited in the mine and waste water is supplied purified to the municipal waste water treatment. As no data are available it is not taken into account.

⁵ personal communication Mr. H. Schmid (former director of “ZZ Ziegeleien” Switzerland)

⁶ Engobe is a coloured slip used in decorating ceramics. They have several distinctive attributes, but are also excellent alternatives to glazes because they are less expensive and less time consuming. Engobes are typically made by mixing water with a claybody in use, then mixing in one or more colorants (e.g. oxides) -- in a proportion of dry ingredients to water that is about 1:2 by volume

7.5 Packing, clay products

The ecoinvent meta information of “packing, clay products” (Tab. 7.10) includes all important and necessary information to use the data correctly.

Tab. 7.10 Ecoinvent meta information for the product “packing, clay products”

Name	packing, clay products
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the process of packing, transportation of packed products to storage and loading of the bags on pallets. The machines used for these processes are taken into account. The waste treatment of the packaging materials is included.
Amount	1
Local Name	Verpacken, Tonprodukte
Synonyms	
General Comment to reference function	The packing and loading process is normalized with an annual production volume of 1'000'000 kg/a and a density of 1'600 m ³ /kg. The estimated life-span of the machines is assumed to be 25 years.
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are from one company (packing, lime products) in Switzerland and adjusted for clay products.
Technology text	Assumption that the same process is used as for packing lime products
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Measured data of one Swiss plant
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 7.5 shows the module “packing, clay products”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources and the standard deviation for each data. This module is based on the module “packing, lime products”. The flows are extrapolated using the ratio of the average densities for lime and clay products (about 800 kg/m³ bulk density for average lime product and 1'600 kg/m³ for clay products).

General Flow information					Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
Infrastructure packing and loading	→	packing and loading of 1 kg clay product		infrastructure weight, copied from module "packing, lime product": estimated total weight of machines: 30t, estimated lifespan: 20years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.50E-03	kg		1	3.45	(5,na,na,na,4,na);	
energy consumption packing and loading	→		as no data available, the energy consumption is copied from "packing, lime product"	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.09E-03	kWh		1	1.51	(1,2,1,1,4,1);		
pallets, at plant	→		Assumptions: average density (over all lime products): 1600kg/m3; loading volume: 1m3; pallets used 10 times; weight of pallet: 22 kg	wooden materials	processing	No	RER	EUR-flat pallet	6.25E-05	unit	Information Frantschach per Email 05.06.02	1	1.68	(4,5,1,1,4,5);		
plastic foil extrusion	→		circumference of pallet: 4 m; height: 2 m, 2 layer of plastic with a weight of 0.0123 kg/m2	plastics	processing	No	RER	extrusion, plastic film	6.15E-05	kg	http://www.dm-folien.de/hpage.htm, access Oct. 2003	1	1.56	(4,na,na,na,4,na);		
plastic foil	→		circumference of pallet: 4 m; height: 2 m, 2 layer of plastic with a weight of 0.0123 kg/m2	plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	6.15E-05	kg	http://www.dm-folien.de/hpage.htm, access Oct. 2003	1	1.56	(4,na,na,na,4,na);		
transports of pallets and plastic foil to plant by train	→		standard distance: transport by train with distance of: 200km,	transport systems	train	No	CH	transport, freight, rail	1.39E-03	tkm		1	2.46	(5,na,na,na,4,na);		
transports of pallets and plastic foil to plant by lorry	→		standard distance: transport by lorry 28t with distance of: 50km,	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	1.39E-03	tkm		1	2.46	(5,na,na,na,4,na);		
transports of plastic foil to municipal incineration	→		assumption: transports all by lorry 16t and total of transportdistance: 10km	transport systems	road	No	RER	transport, lorry 3.5-16t, fleet average	6.15E-07	tkm		1	1.78	(5,na,na,na,4,na);		
	→		waste heat		Same amount as electricity used	air	unspecified			Heat, waste	1.11E-02	MJ		1	1.57	(1,3,1,1,4,5);
	→		pallets, to disposal			waste management	municipal incineration	No	CH	disposal, wood untreated, 20% water, to municipal incineration	1.83E-03	kg	EMPA	1	1.32	(1,4,1,2,3,5);
	→	plastic foil, in municipal waste incineration			waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	6.15E-05	kg		1	1.56	(4,na,na,na,4,na);	
	→	packing of 1 kg of clay product in general			construction materials	others	No	CH	packing, clay products	1.00E+00	kg					

Fig. 7.5 Flows for "packing, clay products" and their representation in the ecoinvent database

8 Data Quality

The uncertainties have been assessed with the help of the pedigree matrix (see Dones et al. (2007)). For detailed results compare the different flows and the ecoinvent meta information.

Most data are based on the report “Life Cycle Assessment Brick” Bruck (1996) which portrayed 12 companies in Germany, Switzerland and Austria. No information concerning the production volume of each company could have been found out and therefore an unweighted mean value is used.

The data of the module “packing, clay products” are extrapolated from the module “packing, lime products”. Since the difference in the production process between the two packing modules is quite small and the data quality of the module “packing, lime products” is quite good, the overall quality of the module “packing, clay products” can be assumed to be also good.

9 Abbreviations

A	Austria
AG	Joint Stock Company (Aktiengesellschaft)
BE	Canton Berne
CH	Switzerland
D	Germany

10 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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Part V

Refractory Bricks

Data v2.0 (2007)

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Kellenberger D., Althaus H.-J., Jungbluth N. and Künniger T. (2007) Life Cycle Inventories of Building Products. Final report ecoinvent Data v2.0 No. 7. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.org.

Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

This part deals with three different kinds of refractory bricks. These are fireclay refractory, high aluminium oxide refractory and basic refractory. The main raw materials for fireclay refractory are chamotte, for high aluminium oxide refractory bauxite and for basic refractory magnesia.

The main application of all three types of refractory bricks is in furnace constructions to prevent damages. The production process is mainly mixing of the raw materials, forming and baking at 1200 to 1800°C.

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1 Introduction

The study Künniger et al. (2001)) summarises the results of an environmental Life Cycle Analysis carried out on gravel, cement and concrete made and used in Switzerland which is based on production data from 1997 including four gravel quarries, eight cement factories and six concrete production plants.

The report of Künniger et al. (2001)) has been used in four ecoinvent reports: “Gravel products and processes”, “Cement products and processes”, “Concrete products and processes” and “Refractory Bricks”. The present report describes three refractory bricks (and their production processes) for the lining of cement kilns: fireclay refractory, high aluminium oxide refractory and basic refractory. Many other types of refractory bricks exist, but they are not described here.

2 Characterisation

2.1 Fireclay Refractory

Routschka & Granitzki (1997) state that fireclay bricks have an Al_2O_3 content of 10 – 45 % according to ISO 1109. They are manufactured from chamotte and bonding-clay mixtures. The limit of 45 % Al_2O_3 represents the maximum Al_2O_3 content of the clay mineral with aluminium oxide (i.e., ignited kaolinite). The amounts of bonding clay and added water vary a factor that largely determines both the subsequent forming process (plastic, semidry, dry shaping) and the properties of the resulting fired bricks. Low-aluminium oxide or siliceous fireclay bricks may contain as additives granular SiO_2 raw materials. Variation of the clays, grogs, and grain sizes permits the production of numerous brick types. Fireclay bricks contain up to ca. 3 % Fe_2O_3 (for low-iron bricks, <1%), ca. 3 % TiO_2 , <1% CaO + MgO , and alkali oxides, mainly K_2O (c. 3.5%) and Na_2O (usually <0.25%). High-quality bricks include <0.5% Na_2O + K_2O .

Mullite, cristobalite, and/or quartz and a glassy phase (composition: ca. 80% SiO_2 , 10% Al_2O_3 , the rest fluxes, mainly alkalis) are the usual components of fireclay bricks. The mullite content is generally proportional to the Al_2O_3 content of the bricks. The amount of cristobalite (0 – 25%) and of the glassy phase (20 – 60%) depends largely on the content of alkali oxides, which can be deduced from the phase diagram Al_2O_3 – SiO_2 – K_2O . With rising temperature, cristobalite dissolves in the glassy phase, increasing the amount of melt. On cooling, the cristobalite again crystallises to a greater or lesser extent from the melt. The amount of residual quartz depends on the firing temperature and the grain size of the quartz.

The resulting highly viscous melt enhances both sintering and densification of the brick texture on firing. A high firing temperature is necessary with a low alkali content to produce a low-porosity brick. Otherwise the glassy phase begins to soften at about 950 °C, imparting a wide softening range under load.

Attack by slag on fireclay bricks normally occurs in relatively small reaction zones; wear progresses by erosion of the resulting highly viscosity reaction zone. In contact with glass melts, alkali migrates into the brick texture and the bricks are slowly dissolved in the glass melt. Other forms of damage include alkali bursting (for bricks with > ca. 35% Al_2O_3), CO disintegration, and thermal spalling (dense bricks).

2.2 High Aluminium Oxide Refractory

The description of high aluminium oxide refractory is taken from (Routschka & Granitzki (1997)). Bricks with $\geq 45\%$ Al_2O_3 are required for increased thermal and chemical stresses. These bricks are produced using various high-aluminium oxide raw materials (sillimanite group, mullitized materials, sintered and fused mullite and corundum, sintered bauxite), alone, in combination, or together with other raw material types, mainly chamotte.

A small amount of clay ($<15\%$) is normally used to produce ceramic-bonded fired bricks. The use of a mixture of clay and fine aluminium oxide leads to a mullite-bonded corundum brick. A chemical bond on an aluminium phosphate base has also proven successful. This brick type is supplied either fired ($1000 - 1500^\circ\text{C}$) or heat-treated ($150 - 800^\circ\text{C}$). The chemical bond gives the brick higher resistance to attacking melts, slag, and alkali oxides. A pitch or resin bond is used for applications in the steel industry. Coking at about 300°C causes a slag-rejecting carbon bond to develop. In special cases fired bricks are impregnated with pitch or resin.

Addition of Cr_2O_3 powder ($5 - 15\%$) to corundum bricks, either alone or in combination with zirconium silicate, leads to a highly corrosion-resistant corundum chrome brick.

The chemical and mineralogical compositions of high-aluminium oxide bricks vary over wide ranges because firing of the bricks normally fails to produce phase equilibrium, and significant changes may occur in the properties of the bricks in service. Bauxite bricks, which constitute the majority of high-aluminium oxide bricks, show more or less after-expansion in service caused by reactions between clay bond and corundum and resulting in mullitization. Compared with fireclay refractories, the alkali oxide content and the extent of glassy phase are low ($<20\%$). The main mineralogical components are corundum and mullite. Sillimanite and andalusite bricks normally contain residual amounts of these minerals.

The corrosion resistance of such bricks to glass, slag, and iron and steel melts is good. They are prone to alkali attack: high-aluminium oxide bricks with low Al_2O_3 content react to produce low-melting alkali aluminium silicate phases or glasses. With higher aluminium oxide content alkali bursting occurs (due to crystallisation of, for example, nepheline), as does the formation of $\text{b-Al}_2\text{O}_3$.

2.3 Basic Refractory (Magnesia Bricks)

Routschka & Granitzki (1997)) state, that magnesia bricks according to ISO 1109 contain at least 80% MgO . A variety of magnesia raw materials have been used to manufacture magnesia and magnesia-containing products (i.e., natural magnesite sinters, seawater magnesia, and fused magnesia with various chemical and mineralogical compositions). Magnesia contains varying amounts of Fe_2O_3 ($<8\%$), Al_2O_3 ($<0.3\%$), CaO ($<4\%$), SiO_2 ($<4\%$), and B_2O_3 ($<0.2\%$). Beside the main mineralogical component periclase, silicates and ferrites occur as secondary components. The CaO/SiO_2 ratio and the B_2O_3 content (formation of low melting components) are especially important in determining the mineral compositions and properties (hot strength, corrosion resistance) of magnesia bricks. Monticellite ($\text{CaO} \cdot \text{MgO} \cdot \text{SiO}_2$) with a CaO/SiO_2 mass ratio of 0.94 has the lowest melting point (1495°C) of the silicates.

High-performance bricks are highly fired, have a low iron content ($<1.5\%$ Fe_2O_3), and contain a dicalcium silicate ($2 \text{ CaO} \cdot \text{SiO}_2$, melting point: 2130°C) or forsterite ($2 \text{ MgO} \cdot \text{SiO}_2$, melting point: 1890°C) bond. These bricks show first subsidence at temperatures $\geq 1600^\circ\text{C}$. Magnesia bricks display high thermal expansion and are prone to thermal shock and thermally induced stress in furnace linings. Thermal shock resistance can be increased by a coarse grain distribution or the addition of Al_2O_3 (spinel formation on firing, microcrack formation) or chrome ore (texture elasticity). Special products include magnesia – forsterite (olivine) bricks and zircon- or zirconia- containing magnesia bricks.

High refractoriness, good resistance to attack by basic slags and alkali resistance characterise magnesia refractories. Slagging causes diffusion and migration of silicate or ferritic melts deep into the brick, producing a zoned structure prone to peeling (structural spalling). A pitch, resin, or carbon bond reduces the infiltration of metal and slags under reducing-atmosphere service conditions. Magnesia bricks are sensitive to hydration (humid atmosphere the range 40 – 120°C), SO₃ (magnesium sulphate) formation, and SiO₂ attack (forsterite bursting). Excessive growth of magnesia crystals during service, promoted by acquired impurities and high temperature, can also loosen brick texture.

3 Use

The main applications of **fireclay refractory** are general furnace constructions, blast furnaces and hot-blast stoves, foundries, furnaces for the non-ferrous-metal industry, coke ovens, glass industry, and cement industry.

The main applications of **high aluminium oxide refractory** are the iron and steel industry, non-ferrous industry, glass and cement industry, ceramic industry (also as high temperature kiln furniture).

The main applications of **basic refractory** (magnesia brick) are steel melting and refining furnaces and vessels, regenerators for glass melting furnaces, cement and lime furnaces.

4 Process Description

One company in Germany has gathered the data concerning the production of the three studied refractory brick products. The data were communicated directly and then published in the report of the gravel, cement and concrete study of Künniger et al. (2001). The corresponding description of the production process is reproduced in translation.

4.1 Fireclay Refractory

For the production of one ton of fireclay refractory 1.1 ton of chamotte (clay is used as proxy) and additionally 10% of clay (0.1 ton) is required. This results in a total amount of 1.2 kg clay for 1 kg of fireclay refractory. The raw materials are fractionated in different coarse size groups and one or several flour components. The grains and flours are mixed in a certain share and a fixing agent (lime milk) is added. The mixture is pressed to bricks, which are baked at 1200 to 1800°C.

Chamotte is burned and milled clay. Since no data is available for this burning process, the energy consumption given for the fireclay refractory has been doubled.

Unlike high aluminium oxide refractory and basic refractory, fireclay refractory does not have a diesel input. The reason is unclear. The diesel is assumed to be used for different transportation services within the plant.

4.2 High Aluminium Oxide Refractory

For the production of one ton of high aluminium oxide refractory 1.1 ton of bauxite (95%) and clay (5%) are required. The raw materials are fractionated in different coarse size groups and one or several flour components. The grains and flours are mixed in a certain share and a fixing agent (lime milk) is added. The mixture is pressed to bricks, which are baked at 1200 to 1800°C.

4.3 Basic Refractory

For the production of one ton of basic refractory 0.95 tons of magnesite (sinter of magnesium oxide) and 0.1 ton of chromite ore is required. The raw materials are fractionated in different coarse size groups and one or several flour components. The grains and flours are mixed in a certain share and a fixing agent (lime milk) is added. The mixture is pressed to bricks, which are baked at 1200 to 1800°C.

5 Life Cycle Inventories

Ecoinvent database meta information (tables), flows and their representation in the ecoinvent database (figures) are shown in:

Refractory, fireclay, packed, at plant	Tab. 5.1 and Fig. 5.1
Refractory, high aluminium oxide, packed, at plant	Tab. 5.2 and Fig. 5.2
Refractory, basic, packed, at plant	Tab. 5.3 and Fig. 5.3

5.1 Fireclay Refractory

The ecoinvent database meta information of “refractory, fireclay, packed, at plant” (Tab. 5.1) includes all important and necessary information to use the data correctly.

Tab. 5.1 Ecoinvent database meta information for the product “refractory, fireclay, packed, at plant”

Name	refractory, fireclay, packed, at plant
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process, internal processes (transport, etc.), packing and infrastructure. No administration is included.
Amount	1
Local Name	Feuerfeste Steine, Schamottesteine, verpackt, ab Werk
Synonyms	Keramik//Ofenausmauerung//ceramics//lining
General Comment to reference function	
Start Date	1999
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	German, European and global modules used as proxy. Manufacturing data from Germany for German module
Technology text	typical technology for German production
Representativeness Boustead)	
Production Volume	unknown
Sampling Procedure	Measured data of one German plant
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 5.1 shows the module “refractory, fireclay, packed, at plant”. It shows all in- and output flows, the attribution to the modules in the ecoinvent database, the sources (mainly data from one German company, Bruck (1996) and estimations) and the standard deviation for each data. The data are based on the information of one German company.

General Flow information						Representation in ecoinvent							Uncertainty information				
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
clay	→	refractory, fireclay, packed, at plant				main ressource, 1.089 kg clay for production of fire-brick (chamotte) + 0.11 kg regional clay	construction materials	additives	No	CH	clay, at mine	1.20E+00	kg	One company in Germany, 1999	1	1.24	(2,4,1,3,1,5);
lime, hydrated	→						construction materials	binder	No	CH	lime, hydrated, packed, at plant	1.50E-02	kg	One company in Germany, 1999	1	1.24	(2,4,1,3,1,5);
heat from natural gas for baking	→						natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	1.14E+01	MJ	One company in Germany, 1999	1	1.24	(2,4,1,2,1,5);
diesel , burned in building machine	→					assumed that used for interal tranports and as not data available and it is assumed that the amount is between the amount of "high aluminium oxide refractory" and "basic refractory"	construction processes	machinery	No	GLO	diesel , burned in building machine	2.00E+00	MJ	estimation EMPA	1	1.24	(2,4,1,2,1,5);
electricity	→						electricity	supply mix	No	DE	electricity, medium voltage, at grid	2.39E-01	kWh	One company in Germany, 1999	1	1.24	(2,4,1,1,1,5);
transport, van <3.5t	→					internal transports, 0.9 l diesel per t fireclay brick	transport systems	road	No	CH	transport, van <3.5t	2.37E-03	tkm	One company in Germany, 1999	1	2.06	(2,4,1,3,1,5);
transport, lorry 16t	→					transport (removal) of waste 50 km	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	2.50E-04	tkm	transport distance estimated by EMPA	1	2.06	(2,4,1,3,1,5);
transport, lorry 28t	→					transport of regional clay 50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	5.50E-03	tkm	transport distance estimated by EMPA	1	2.06	(2,4,1,3,1,5);
transport, freight, rail	→					transport of chamotte 500 km	transport systems	train	No	RER	transport, freight, rail	5.45E-01	tkm	transport distance estimated by EMPA	1	2.06	(2,4,1,2,1,5);
packing of fireproof clay	→					packing of 1 kg product	construction materials	others	No	CH	packing, clay products	1.00E+00	kg		1	1.12	(3,n.A.,n.A.,3,1,n.A.);
lubricating oil	→					Process material: Data from module "brick, at plant"	chemicals	organics	No	RER	lubricating oil, at plant	6.77E-06	kg	Bruck (1996)	1	1.62	(1,5,1,2,4,5);
gear lubricant and hydraulic oil	→					Process material: Data from module "brick, at plant"	chemicals	organics	No	RER	lubricating oil, at plant	3.72E-06	kg	Bruck (1996)	1	1.62	(1,5,1,2,4,5);
lubricating grease	→					Process material: Data from module "brick, at plant"	chemicals	organics	No	RER	lubricating oil, at plant	2.59E-06	kg	Bruck (1996)	1	1.62	(1,5,1,2,4,5);
water	→					process and cooling water	resource	in water			Water, cooling, unspecified natural origin	2.00E-05	m3	One company in Germany, 1999	1	1.24	(2,4,1,2,1,5);
fireproof brick	→					used to lay out the inside of the furnace; not balanced, as no data is available	construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	-	kg		1		(,...);
land-use and buildings	→					"mine, clay" used as proxy; the amount has been copied from land-use "brick, at plant", assumed production volume: 1'000'000 kg/a and production time: 50 years	construction materials	additives	Yes	CH	mine, clay	2.00E-08	unit		1	3.32	(1,5,1,3,4,5);
infrastructure machines	→					Infrastructure of all machines; assumed production volume: 10'000'000 kg/a, weight assumption machines: 50t; assumption service life: 30years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.67E-04	kg		1	3.28	(5,4,1,2,1,5);
	→			waste heat		Same amount as electricity used	air	unspecified			Heat, waste	8.60E-01	MJ		1	1.24	(2,4,1,2,1,5);
	→			waste to municipal incinaration		primarily wood pallets (input is taken into account within the packing module) and used oil	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	5.00E-03	kg	One company in Germany, 1999	1	1.24	(2,4,1,3,1,5);
	→			refractory, fireclay, packed, at plant			construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	1.00E+00	kg				

Fig. 5.1 Flows for the module "refractory, fireclay, packed, at plant" and their representation in the ecoinvent database

5.2 High Aluminium Oxide Refractory

The ecoinvent database meta information of “refractory, high aluminium oxide, packed, at plant” (Tab. 5.2Tab. 5.) includes all important and necessary information to use the data correctly.

Tab. 5.2 Ecoinvent database meta information for the product “refractory, high aluminium oxide, packed, at plant”

Name	Refractory, high aluminium oxide, packed, at plant
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process, internal processes (transport, etc.), packing and infrastructure. No administration is included.
Amount	1
Local Name	Feuerfeste Steine, Hochtonerdesteine, verpackt, ab Werk
Synonyms	Keramik//Ofenausmauerung//ceramics//brick lining of a furnace
General Comment to reference function	
Start Date	1999
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	German, European and global modules used as proxy. Manufacturing data from Germany for German module. Switzerland is not producing any high alumina refractories.
Technology text	typical technology for German production
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Measured data of one German plant
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 5.1 shows the module “refractory, high aluminium oxide, packed, at plant”. It shows all in- and output flows, the attribution to the modules in the ecoinvent database, the sources (mainly data from one German company and Bruck (1996)) and the standard deviation for each data. The data are based on the information of one German company.

[illegible]

Fig. 5.2 Flows for “refractory, high aluminium oxide, packed, at plant” and their representation in the ecoinvent database

5.3 Basic Refractory

The ecoinvent database meta information of “refractory, basic, packed, at plant” (Tab. 5.3) includes all important and necessary information to use the data correctly.

Tab. 5.3 Ecoinvent database meta information for the product “refractory, basic, packed, at plant”

Name	refractory, basic, packed, at plant
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process, internal processes (transport, etc.), packing and infrastructure. No administration is included.
Amount	1
Local Name	Feuerfeste Steine, basische Steine, verpackt, ab Werk
Synonyms	Keramik//Ofenausmauerung//ceramics//brick lining of a furnace
General Comment to reference function	
Start Date	1999
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	German, European and global modules used as proxy. Manufacturing data from Germany for German module. Switzerland is not producing any basic refractories.
Technology text	typical technology for German production
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Measured data of one German plant
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 5.3 shows the system boundary of the module “refractory, basic, packed, at plant”. It shows all in- and output flows, the attribution to the modules in the ecoinvent database, the sources (mainly data from one German company, Bruck (1996) and estimations) and the standard deviation for each data. The data are based on the information of one German company.

The data are based on information of one German company. The amount of carbon dioxide emission is based on the natural gas consumption of the company and on the produced amount of refractory bricks. That reported amount is considerably smaller than the calculated amount from the burning process. It is assumed that some of the carbon dioxide is fixed in the raw material (hydrated lime or limestone in the clay) or that the amount declared by the company is too low.

General Flow information						Representation in ecoinvent							Uncertainty information				
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
chromite ore	→	refractory, basic, packed, at plant				Raw material	metals	extraction	No	GLO	chromite, ore concentrate, at beneficiation	1.00E-01	kg	One company in Germany, 1999	1	1.24	(2,4,1,2,1,5);
magnesite	→					Raw material	chemicals	inorganics	No	RER	magnesium oxide, at plant	9.50E-01	kg	One company in Germany, 1999	1	1.24	(2,4,1,3,1,5);
lime, hydrated	→					temporary binder, lime-wash (water + hydrated lime)	construction materials	binder	No	CH	lime, hydrated, packed, at plant	1.50E-02	kg	One company in Germany, 1999	1	1.24	(2,4,1,3,1,5);
water	→						resource	in water			Water, unspecified natural origin	2.00E-05	m3	One company in Germany, 1999	1	1.24	(2,4,1,3,1,5);
diesel, burned in building machine	→					assumed that diesel is used for internal transports	construction processes	machinery	No	GLO	diesel, burned in building machine	1.16E+00	MJ	One company in Germany, 1999	1	1.24	(2,4,1,2,1,5);
heat from natural gas for baking	→						natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	1.38E+01	MJ	One company in Germany, 1999	1	1.24	(2,4,1,2,1,5);
electricity	→						electricity	supply mix	No	DE	electricity, medium voltage, at grid	1.20E-01	kWh	One company in Germany, 1999	1	1.24	(2,4,1,1,1,5);
transport, freight ship	→					transport of magnesite (and chromite ore) 10'000 km	transport systems	ship	No	OCE	transport, transoceanic freight ship	1.05E+01	tkm	estimation EMPA	1	2.06	(2,4,1,2,1,5);
transport, van <3.5t	→					internal transports, 0.9 l diesel per t alkaline brick	transport systems	road	No	CH	transport, van <3.5t	2.37E-03	tkm	estimation EMPA	1	2.06	(2,4,1,3,1,5);
transport, lorry 16t	→					transport (removal) of waste 50 km	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	2.50E-04	tkm	estimation EMPA	1	2.06	(2,4,1,3,1,5);
transport, freight, rail	→					transport of magnesite (and chromium) 500 km	transport systems	train	No	RER	transport, freight, rail	5.25E-01	tkm	estimation EMPA	1	2.06	(2,4,1,2,1,5);
packing of fireproof clay	→					packing of 1 kg product	construction materials	others	No	CH	packing, clay products	1.00E+00	kg		1	1.12	(3,n.A.,n.A.,3,1,n.A.);
lubricating oil	→					data copied directly from module "brick, at plant"	chemicals	organics	No	RER	lubricating oil, at plant	6.77E-06	kg	Bruck (1996)	1	1.62	(1,5,1,2,4,5);
gear lubricant and hydraulic oil	→					data copied directly from module "brick, at plant"	chemicals	organics	No	RER	lubricating oil, at plant	3.72E-06	kg	Bruck (1996)	1	1.62	(1,5,1,2,4,5);
lubricating grease	→					data copied directly from module "brick, at plant"	chemicals	organics	No	RER	lubricating oil, at plant	2.59E-06	kg	Bruck (1996)	1	1.62	(1,5,1,2,4,5);
fireproof brick	→					used to lay out the inside of the furnace; not balanced, as no data is available "mine, clay" used as proxy; the amount has been copied from land-use "brick, at plant", production volume: 1'000'000 kg/a and production time: 50 years	construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	-	kg				
land-use and buildings	→					Infrastructure of all machines; assumed production volume: 10'000'000 kg/a and production time: 50 years	construction materials	additives	Yes	CH	mine, clay	2.00E-08	unit		1	3.32	(1,5,1,3,4,5);
infrastructure machines	→					assumption machines: 50t; assumption service life: 30years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.67E-04	kg		1	3.28	(5,4,1,2,1,5);
	→			waste heat		Same amount as electricity used	air	unspecified			Heat, waste	4.32E-01	MJ		1	1.24	(2,4,1,2,1,5);
	→			waste to municipal incineration		primarily wood pallets and used oil	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	5.00E-03	kg	One company in Germany, 1999	1	1.24	(2,4,1,3,1,5);
	→			refractory, fireclay, packed, at plant			construction materials	bricks	No	DE	refractory, basic, packed, at plant	1.00E+00	kg				

Fig. 5.3 Flows for “refractory, basic, packed, at plant” and their representation in the ecoinvent database

6 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

Most of the data concerning the refractory bricks are based on information from one company in Germany. The data on the electricity consumption varies very strongly. It seems that different system boundaries are used for the modules. Also the available data include only a part of the information needed and therefore they had to be completed. In most cases they have been copied from other processes. For the process materials lubricating oil and grease the data for normal brick production have been used as proxy. Data for the different refractory bricks are of poor overall quality.

7 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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Part VI

Ceramics

Data v2.0 (2007)

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Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

This part deals with the production of ceramic tiles and sanitary ceramic products. Data are taken from a LCA study of tile production in Italy and from an environmental report of one producer in Germany.

Data are not representative for engineering ceramics and they are not meant to be used for direct comparison of ceramics to other materials.

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1 Introduction

This part deals with the production of ceramic tiles and sanitary ceramic products.

The materials and technologies used for producing technical ceramics can be quite different. If one of the inventories is used as a proxy for sanitary ceramics care has to be taken with the interpretation of the results.

2 Reserves and Resources of Materials

For information about reserves and resources of ceramics see the information on the raw materials (mainly clay, feldspars and sand).

3 Characterisation and use of the materials

This information is based on EPA (1998).

Ceramics are defined as a class of inorganic, non-metallic solids that are subjected to high temperature in manufacture and/or use. The most common ceramics are composed of oxides, carbides, and nitrides. Silicides, borides, phosphides, tellurides, and selenides also are used to produce ceramics.

Ceramic processing generally involves high temperatures, and the resulting materials are heat resistant or refractory.

Traditional ceramics refers to ceramic products that are produced from unrefined clay and combinations of refined clay and powdered or granulated non-plastic minerals. Often, traditional ceramics is used to refer to ceramics in which the clay content exceeds 20 percent. The general classification of traditional ceramics is:

- Pottery is sometimes used as a generic term for ceramics that contain clay and are not used for structural, technical, or refractory purposes.
- Whiteware refers to ceramic ware that is white, ivory, or light gray in color after firing. Whiteware is further classified as earthenware, stoneware, chinaware, porcelain, and technical ceramics.
 - Earthenware is defined as glazed or unglazed non-vitreous (porous) clay-based ceramic ware. Applications for earthenware include artware, kitchenware, ovenware, tableware, and tile.
 - Stoneware is vitreous or semi-vitreous ceramic ware of fine texture, made primarily from non-refractory fire clay or some combination of clays, fluxes, and silica that, when fired, has properties similar to stoneware made from fire clay. Applications for stoneware include artware, chemicalware, cookware, drainpipe, kitchenware, tableware, and tile.
 - Chinaware is vitreous ceramic ware of zero or low absorption after firing that are used for non-technical applications. Applications for chinaware include artware, ovenware, sanitaryware, and tableware.
 - Porcelain is defined as glazed or unglazed vitreous ceramic ware used primarily for technical purposes. Applications for porcelain include artware, ball mill balls, ball mill liners, chemicalware, insulators, and tableware.
 - Technical ceramics include vitreous ceramic whiteware used for such products as electrical insulation, or for chemical, mechanical, structural, or thermal applications.

Ceramic products that are made of highly refined natural or synthetic compounds and designed to have special properties are referred to as advanced or technical ceramics. Advanced ceramics can be classified according to application as electrical, magnetic, optical, chemical, thermal, mechanical, biological, and nuclear.

Most ceramic products (including sanitary ceramics and ceramic tiles) are clay-based and are made from a single clay or one or more clays mixed with mineral modifiers such as quartz and feldspar. The types of commercial clays used for ceramics are primarily kaolin and ball clay.

4 System Characterisation

4.1 Process description

This chapter is mainly based on EPA (1998).

The basic steps of ceramic production include raw material procurement, possibly beneficiation, mixing, forming, green machining, drying, presinter thermal processing, glazing, firing, final processing, and packaging. Procurement and (if necessary) beneficiation of raw material are dealt with in the respective reports. The following paragraphs describe the remaining operations in detail.

Sanitary ceramic and ceramic tiles are produced by slurry processing. No powder processing is inventoried in ecoinvent. The information is given for comprehensibility reasons only.

4.1.1 Mixing

The purpose of mixing or blunging is to combine the constituents of a ceramic powder or slurry to produce a more chemically and physically homogenous material for forming. Pug mills often are used for mixing ceramic materials. Several processing aids may be added to the ceramic mix during the mixing stage. Binders and plasticizers are used in dry powder and plastic forming. In slurry processing, deflocculants, surfactants, and antifoaming agents are added to improve processing. Liquids also are added in plastic and slurry processing. Binders are polymers or colloids that are used to impart strength to green or unfired ceramic bodies. Water is the most commonly used liquid in plastic and slurry processing. Organic liquids such as alcohols may also be used in some cases. Deflocculants also are used in slurry processing to improve dispersion and dispersion stability. Surfactants are used in slurry processing to aid dispersion, and antifoams are used to remove trapped gas bubbles from the slurry.

4.1.2 Forming

In the forming step, dry powders, plastic bodies, pastes, or slurries are consolidated and moulded to produce a cohesive body of the desired shape and size. Dry forming consists of the simultaneous compacting and shaping of dry ceramic powders in a rigid die or flexible mold. Jiggering is widely used in the manufacture of small, simple, axially symmetrical whiteware ceramic such as cookware, fine china, and electrical porcelain. Slurry forming of ceramics generally is accomplished using slip casting, gelcasting, or tape casting. In slip casting, a ceramic slurry, which has a moisture content of 20 to 35 percent, is poured into a porous mould, often made of gypsum. Capillary suction of the mould draws the liquid from the mould, thereby consolidating the cast ceramic material. After a fixed time the excess slurry is drained, and the cast is dried. Slip casting is widely used in the manufacture of sinks and other sanitaryware, figurines, porous thermal insulation, fine china, and structural ceramics with complex shapes.

4.1.3 Green Machining

After forming, the ceramic shape often is machined to eliminate rough surfaces and seams or to modify the shape. The methods used to machine green ceramics include surface grinding to smooth surfaces, blanking and punching to cut the shape and create holes or cavities, and laminating for multi-layer ceramics.

4.1.4 Drying

After forming, ceramics must be dried. Drying must be carefully controlled to strike a balance between minimising drying time and avoiding differential shrinkage, warping, and distortion. The most commonly used method of drying ceramics is by convection, in which heated air is circulated around the ceramics. Air drying often is performed in tunnel kilns, which typically use heat recovered from the cooling zone of the kiln. Periodic kilns or dryers operating in batch mode are also used. Convection drying is carried out in divided tunnel dryers, which include separate sections with independent temperature and humidity controls. An alternative to air drying is radiation drying in which microwave or infrared radiation is used to enhance drying.

4.1.5 Presinter Thermal Processing

Prior to firing, ceramics often are heat-treated at temperatures well below firing temperatures. The purpose of this thermal processing is to provide additional drying, to vaporize or decompose organic additives and other impurities, and to remove residual, crystalline, and chemically bound water. Presinter thermal processing can be applied as a separate step, which is referred to as bisque firing, or by gradually raising and holding the temperature in several stages.

This process step is not commonly done for sanitary ceramics or ceramic tiles.

4.1.6 Glazing

For traditional ceramics, glaze coatings are often applied to dried or bisque-fired ceramic ware prior to sintering. Glazes consist primarily of oxides and can be classified as raw glazes or frit glazes. In raw glazes, the oxides are in the form of minerals or compounds that melt readily and act as solvents for the other ingredients. Some of the more commonly used raw materials for glazes are quartz, feldspars, carbonates, borates, and zircon. A frit is a prereacted glass. To prepare glazes, the raw materials are ground in a ball mill or attrition mill. Glazes generally are applied by spraying or dipping. Depending on their constituents, glazes mature at temperatures of 600 ° to 1500 °C.

4.1.7 Firing

Firing is the process by which ceramics are thermally consolidated into a dense, cohesive body comprised of fine, uniform grains. This process is also referred to as sintering or densification. Firing results are depending on material properties and on properties of the green ceramics as well as on firing parameters.

Material properties that affect firing include particle size (powder), material surface energy, diffusion coefficients, fluid viscosity, and bond strength.

Properties of the green ceramics affecting firing are the density and the shape.

Parameters that affect firing include firing temperature, time, pressure, and atmosphere. A short firing time results in a product that is porous and has a low density; a short to intermediate firing time results in fine-grained, high-strength products; and long firing times result in a coarse-grained products that are more creep resistant.

Applying pressure decreases firing time and makes it possible to fire materials that are difficult to fire using conventional methods. Oxidizing or inert atmospheres are used to fire oxide ceramics to avoid reducing transition metals and degrading the finish of the product.

Conventional firing is accomplished by heating the green ceramic to approximately two-thirds of the melting point of the material at ambient pressure and holding it for a specified time in a periodic or tunnel kiln. Periodic kilns are heated and cooled according to prescribed schedules. The heat for periodic kilns generally is provided by electrical element or by firing with gas or oil.

4.1.8 Final Processing

After firing, some ceramic products are processed further to enhance their characteristics or to meet dimensional tolerances. Ceramics can be machined by abrasive grinding, chemical polishing, electrical discharge machining, or laser machining. Annealing at high temperature, followed by gradual cooling can relieve internal stresses within the ceramic and surface stresses due to machining. In addition, surface coatings are applied to many fired ceramics. Surface coatings are applied to traditional clay ceramics to create a stronger, impermeable surface and for decoration. Coatings also may be applied to improve strength, and resistance to abrasion and corrosion. Coatings can be applied dry, as slurries, by spraying, or by vapour deposition.

4.2 Infrastructure

The infrastructure needed for ceramic production consists basically of a building, storage for raw materials, mills, mixers and kilns.

4.3 Emissions

4.3.1 Air

Besides the emissions from heat generation, only dust emissions from the handling of the raw materials are of relevance. Nicoletti et al. (2002) mention a relevant dust emission from grinding of the raw materials for tiles but they don't give any quantitative information. Thus the total of the values given in EPA (1998) for "comminution" and "ceramic glass spray booth" (controlled) is inventoried ($8.71\text{E-}03$ kg/kg). Since the ceramic powder has a grain size around $1\text{ }\mu\text{m}$ and fabric filters or wet scrubbers are utilised to abate dust emissions, the particles are inventoried as PM 2.5.

If organic plasticizers are used (which usually is not the case for sanitary ceramics and ceramic tiles), emissions from burning them are relevant too.

4.4 Waste

4.4.1 Solid waste

Wastes to disposal are often oil-contaminated. A major part of the non contaminated production waste can be recycled in the process.

4.4.2 Waste water

If ceramics are produced in slurry processing emissions of the raw materials to water occur. This water is usually pre-treated in house and disposed to a municipal wastewater treatment plant. For one plant in Austria measurements of the wastewater composition (after internal pre-treatment) are available. They are presented in Tab. 4.1.

Tab. 4.1 Waste water composition (after internal treatment) of sanitary ceramic production in Gmunden (ÖSPAG (2002))

element	measured value	value used in this study	unit
Zn	0.012	1.200E-02	g/m ³
Ni	< 0.05	2.500E-02	g/m ³
Pb	< 0.05	2.500E-02	g/m ³
Cu	< 0.01	5.000E-03	g/m ³
Cr	< 0.05	2.500E-02	g/m ³
Co	< 0.05	2.500E-02	g/m ³
Cd	< 0.016	8.000E-03	g/m ³
Ba	0.48	4.800E-01	g/m ³

4.4.3 Ceramics production in ecoinvent

Sanitary ceramic and ceramic tiles are produced by slurry processing.

The modules "sanitary ceramics, at regional storage" and "ceramic tiles, at regional storage " are based on information from a major producer of sanitary ceramics in Europe (ÖSPAG (2002)) and on a LCA case study with inventory data for one producer in Italy (Nicoletti et al. (2002)) respectively. Because the data are meant to represent consumption in Switzerland, the electricity and other products consumed are not inventoried with the specific modules for the country where the data come from but with European average modules since the production for Swiss consumption is not limited to one country.

The input and output data for the sanitary ceramics production stem from the environmental report of a company (ÖSPAG (2002)). They refer to the years 2000 and 2001 and to two factories. A weighted mean value of the flows from the different years and factories is calculated. Detailed information on production waste is reported. This degree of detail is not available for input material. Thus some waste fractions without corresponding input are inventoried.

The standard distances from to Frischknecht et al. (2007)) for the transportation of the metal inputs are used. The transport of the product to the consumption in Switzerland is inventoried with 600 km rail freight.

The infrastructure is calculated and estimated based on information from ÖSPAG (2002) (see Tab. 4.2).

Tab. 4.2 Estimation on infrastructure

Factory	Wilhelmsburg		Gmuden		Unit	Category	Remarks
Year	2000	2001	2000	2001			
total area	88'673	88'673	48'200	48'200	m2	land use	ÖSPAG (2002)
sealed area	55'635	55'635	19'200	19'200	m2		ÖSPAG (2002)
vegetation area	33'038	33'038	29'000	29'000	m2		ÖSPAG (2002)
built up area	44'508	44'508	15'360	15'360	m2		estimation based on photography: 80% of sealed area is built up
paved parking lot	11'127	11'127	3'840	3'840	m2		
building hall	40'057	40'057	13'824	13'824	m2	buildings	estimation based on photography: 90% built up area
building multi-storey	35'606	35'606	12'288	12'288	m3		estimation based on photography: 10% built up area, 3 storeys
machines	100'000	100'000	50'000	50'000	kg	machines	rough estimation
Product output	4'597'000	5'231'000	3'159'000	4'798'000	kg		

The lifetime of the buildings is assumed 50 years, that of the machines 25 years. Thus the machines are inventoried twice for the total operation time of the plant.

The ecoinvent meta-information for the ceramic production is given in Tab. 4.3 and the input and output flows for "sanitary ceramics, at regional storage", "ceramic tiles, at regional storage" and "ceramic plant" are given in Fig. 4.1, Fig. 4.2 and Fig. 4.3, respectively.

4.4.4 Data quality

Data uncertainty is derived using the Pedigree matrix as described in (Frischknecht et al. (2007)).

All data are from sources of unknown representativeness. Both inventories are based on information from one single company (except particle emissions). On the other hand, the sources used are fairly detailed and comprehensible. Thus the overall quality must be described as mediocre.

Tab. 4.3 ecoinvent meta information for the ceramic production processes

Name	sanitary ceramics, at regional storage	ceramic tiles, at regional storage	ceramic plant
Location	CH	CH	CH
Infrastructure Process	0	0	1
Unit	kg	kg	unit
DataSet Version	2.0	2.0	2.0
Included Processes	Gate to gate production of sanitary ceramics, including transports of raw materials to factory and of product to Switzerland	Gate to gate production of ceramic tiles, including transports of raw materials to factory and of product to Switzerland	Includes land use and materials used in buildings and machinery as well as their disposal
Amount	1	1	1
Local Name	Sanitärkeramik, ab Regionallager	Keramikplatten, ab Regionallager	Keramikwerk
Synonyms	China //Chinaware//Porzellan	Steinzeug//stoneware	
General Comment to reference function	Data from one producer only. Composition of ceramic might differ from case to case. This dataset refers to oxidic ceramics.	Data from one producer only. Composition of ceramic might differ from case to case. This dataset refers to oxidic ceramics.	Life time of 50 years for buildings and of 25 years for machines is assumed. Dataset refers to a factory with yearly output of 5'000 t of ceramic products.
StartDate	1998	1998	2001
EndDate	2002	2002	2002
Data Valid For Entire Period	1	1	1
Other Period Text			
Geography text	Data from two factories of one producer in Austria	Data from Italy	Data from two factories of one producer in Austria
Technology text	Large scale production in Europe. Gas fired kiln	Large scale, single fired production in Europe. Gas fired kiln	Large scale production plant in Europe.
Representativeness [%]			
Production Volume	Unknown	Unknown	Unknown
Sampling Procedure	Environmental report	Publication	Environmental report
Extrapolations	See Geography	See Geography	See Geography
Uncertainty Adjustments	None	None	None

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
feldspar	→	sanitary ceramics, at regional storage; inputs	for ceramic mass	construction materials	others	No	RER	feldspar, at plant	3.79E-01	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,3)
recycled production waste	→		for ceramic mass (from in-house production)						8.97E-02	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,3)
kaolin	→		for ceramic mass	chemicals	inorganics	No	RER	kaolin, at plant	4.08E-01	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,3)
clay	→		for ceramic mass	construction materials	additives	No	CH	clay, at mine	4.25E-01	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,3)
silica meal, porcelain meal, refractories	→		for ceramic mass	construction materials	additives	No	DE	silica sand, at plant	2.20E-01	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,3)
ancillary materials	→		for ceramic mass	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	2.75E-03	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,3)
oxydic minerals	→		for glazing	construction materials	additives	No	DE	silica sand, at plant	3.17E-02	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,3)
pigments	→		for glazing	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	2.87E-04	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,3)
kaolin refractories, silica meal	→		for glazing	chemicals	inorganics	No	RER	kaolin, at plant	3.63E-02	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,3)
gypsum	→		for mould	construction materials	binder	No	CH	stucco, at plant	1.25E-01	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,4)
plastic forms	→		for mould	plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	1.51E-03	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,4)
natural gas	→		for heating	natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	2.41E+01	MJ	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,1)
electricity	→		total consumption	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	8.78E-01	kWh	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,2)
well water	→		total consumption	resource	in water			Water, well, in ground	1.06E-02	m3	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,4)
tap water	→		total consumption	water supply	production	No	RER	tap water, at user	5.41E-01	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,4)
transport	→		of final product to Switzerland: 600 km assumed	transport systems	train	No	CH	transport, freight, rail	6.00E-01	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
transport	→		of raw materials to fabrication: 50 km assumed	transport systems	road	No	RER	transport, lorry >16t, fleet average	8.14E-02	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
infrastructure	→		Life time: 50 a; production volume: 5000000 kg/a	construction materials	others	Yes	CH	ceramic plant	4.00E-09	unit	ÖSPAG (2002)	1	3.02	(1,3,1,3,1,4,9)

Fig. 4.1 Flows for "sanitary ceramics, at regional storage" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
sanitary ceramics, at regional storage; outputs, part 1	→	waste ceramic (not sanded)	internally recycled (substitutes feldspar)						8.97E-02	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	paper	recycled						7.59E-03	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	waste ceramic (sanded)	contamination from sanding. Cannot be recycled	waste management	sanitary landfill	No	CH	disposal, inert material, 0% water, to sanitary landfill	9.40E-02	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	rumble		waste management	sanitary landfill	No	CH	disposal, inert material, 0% water, to sanitary landfill	3.19E-02	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	filter material (inert)		waste management	inert material landfill	No	CH	disposal, inert waste, 5% water, to inert material landfill	1.64E-03	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	gypsum	recycled in cement plants						1.33E-01	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	clay suspension	recycled in brick fabrication						1.85E-01	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	municipal waste		waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	3.86E-02	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	metal containers (soiled)		waste management	underground deposit	No	DE	disposal, hazardous waste, 0% water, to underground deposit	6.85E-05	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	refrigerators	neglected (for consistency)						5.49E-05	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	lead batteries	neglected (for consistency)						2.97E-04	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	fluorescent lamps	neglected (for consistency)						2.06E-05	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	developing agent		waste management	hazardous waste incineration	No	CH	disposal, solvents mixture, 16.5% water, to hazardous waste incineration	2.39E-06	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	used oil		waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	1.88E-04	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	fat		waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	1.58E-06	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	oil-water mixtures		waste management	hazardous waste incineration	No	CH	disposal, bilge oil, 90% water, to hazardous waste incineration	6.61E-05	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)

Fig. 4.1 Flows for "sanitary ceramics, at regional storage" and its representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
sanitary ceramics, at regional storage; outputs, part 2	→	waste from oil interceptor		waste management	hazardous waste incineration	No	CH	disposal, bilge oil, 90% water, to hazardous waste incineration	1.11E-04	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	used oil binding agents		waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	6.33E-06	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	oil contaminated ancillary materials		waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	8.71E-05	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	used paints		waste management	hazardous waste incineration	No	CH	disposal, emulsion paint remains, 0% water, to hazardous waste incineration	1.58E-07	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	uncured resins		waste management	municipal incineration	No	CH	disposal, plastics, mixture, 15.3% water, to municipal incineration	2.01E-05	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	plastic containers with rest content		waste management	municipal incineration	No	CH	disposal, plastics, mixture, 15.3% water, to municipal incineration	2.32E-05	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	chemical wastes		waste management	hazardous waste incineration	No	CH	disposal, solvents mixture, 16.5% water, to hazardous waste incineration	6.69E-05	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	other hazardous wastes		waste management	underground deposit	No	DE	disposal, hazardous waste, 0% water, to underground deposit	3.30E-05	kg	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	waste water		waste management	wastewater treatment	No	CH	treatment, ceramic production effluent, to wastewater treatment, class 3	9.83E-04	m3	ÖSPAG (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	dust emissions from grinding	no information in ÖSPAG (2002)	air	unspecified			Particulates, < 2.5 um	8.71E-03	kg	EPA (1998)	1	2.06	(1,3,1,5,1,5,26)
	→	waste heat		air	unspecified			Heat, waste	3.16E+00	MJ	calculated	1	1.13	(1,3,1,3,1,4,13)
	→	Sanitary ceramics		construction materials	others	No	CH	sanitary ceramics, at regional storage	1.00E+00	kg	ÖSPAG (2002)			

Fig. 4.1 Flows for "sanitary ceramics, at regional storage" and its representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
clay	ceramic tiles, at regional storage; inputs, part 1		for ceramic body	construction materials	additives	No	CH	clay, at mine	4.89E-01	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
feldspars			for ceramic body	construction materials	others	No	RER	feldspar, at plant	2.72E-01	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
soldering materials (feldspars)			for ceramic body; 62.5% of total soldering materials	construction materials	others	No	RER	feldspar, at plant	1.02E-01	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
soldering materials (limestone)			for ceramic body; 37.5% of total soldering materials	construction materials	others	No	CH	limestone, milled, packed, at plant	6.10E-02	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
silicious and feldspar sands			for ceramic body	construction materials	additives	No	DE	silica sand, at plant	1.56E-01	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
ground fired waste			for ceramic body; reused process waste						6.67E-03	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
ground raw waste			for ceramic body; reused process waste						2.78E-02	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
exhausted lime			for ceramic body; reused process waste						2.22E-04	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
boron concentrate			for ceramic body; reused from water treatment						6.67E-03	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
zinc oxide			for frit; only metal content of oxide is inventoried as metal	metals	extraction	No	RER	zinc, primary, at regional storage	3.39E-03	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
zirconium powders			for frit; only metal content of oxide is inventoried as metal	chemicals	inorganics	No	RER	titanium dioxide, production mix, at plant	1.83E-02	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
colemanite			for frit	construction materials	others	No	CH	limestone, milled, packed, at plant	7.22E-03	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
dolomite			for frit	construction materials	others	No	CH	limestone, milled, packed, at plant	9.06E-03	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
penta-hydrate borax			for frit	washing agents	bleaches	No	RER	sodium perborate, tetrahydrate, powder, at plant	6.11E-03	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
quarz and feldspar sands			for frit	construction materials	additives	No	DE	silica sand, at plant	1.39E-02	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
feldspars			for frit	construction materials	others	No	RER	feldspar, at plant	1.23E-02	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)

Fig. 4.2 Flows for "ceramic tiles, at regional storage" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
aluminium oxide	ceramic tiles, at regional storage; inputs, part 2		for frit; only metal content of oxide is inventoried as metal	chemicals	inorganics	No	RER	aluminium oxide, at plant	3.89E-04	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
lead oxide			for frit; only metal content of oxide is inventoried as metal	metals	extraction	No	RER	lead, at regional storage	6.65E-03	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
barium oxide			for frit	chemicals	inorganics	No	RER	barite, at plant	5.56E-04	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
tin oxide			for frit; only metal content of oxide is inventoried as metal	metals	extraction	No	RER	tin, at regional storage	4.90E-05	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
titanium oxide			for frit	chemicals	inorganics	No	RER	titanium dioxide, production mix, at plant	1.11E-04	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,3)
electricity			general consumption	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.13E-01	kWh	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,2)
heat			given as primary energy; natural gas assumed no information about material for moulds in Nicoletti et al. (2002) --> half the amount of sanitary ceramic production assumed	natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	5.83E+00	MJ	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,1)
gypsum				construction materials	binder	No	CH	stucco, at plant	6.23E-02	kg	assumption, based on ÖSPAG (2002)	1	1.24	(1,3,1,3,3,4,4)
water			general consumption	water supply	production	No	RER	tap water, at user	6.67E-01	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,4)
recycled water			internal recycling						6.67E-04	m3	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,4)
transport			of final product to Switzerland: 600 km assumed	transport systems	train	No	CH	transport, freight, rail	6.00E-01	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
transport			of raw materials to fabrication: 50 km assumed	transport systems	road	No	RER	transport, lorry >16t, fleet average	6.10E-02	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
infrastructure			Life time: 50 a; production volume: 5000000 kg/a	construction materials	others	Yes	CH	ceramic plant	4.00E-09	unit	ÖSPAG (2002)	1	3.02	(1,3,1,3,1,4,9)

Fig. 4.2 Flows for "ceramic tiles, at regional storage" and its representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
ceramic tiles, at regional storage; outputs	→	waste water to internal recycling	recycled to wet grinding						6.67E-04	m3	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	waste water after internal pre-treatment	to external treatment	waste management	wastewater treatment	No	CH	treatment, ceramic production effluent, to wastewater treatment, class 3	2.00E-05	m3	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	sludge from waste water treatment to external recycling	not inventoried						2.22E-03	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	sludge from waste water treatment to landfill	assumed to be hazardous waste because it can not be recycled	waste management	underground deposit	No	DE	disposal, hazardous waste, 0% water, to underground deposit	1.11E-03	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	wastes to internal recycling	not inventoried						3.47E-02	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	wastes to external recycling	not inventoried						3.29E-02	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	wastes to disposal	assumed to be hazardous wastes because they can not be recycled	waste management	underground deposit	No	DE	disposal, hazardous waste, 0% water, to underground deposit	8.50E-03	kg	Nicoletti et al. (2002)	1	1.13	(1,3,1,3,1,4,6)
	→	dust emissions to air	Value for comminution and glaze spray booth (controlled)	air	unspecified			Particulates, < 2.5 um	8.71E-03	kg	EPA (1998), size assumed	1	1.32	(1,3,1,5,1,5,28)
	→	waste heat		air	unspecified			Heat, waste	1.13E+00	MJ	calculated	1	1.13	(1,3,1,3,1,4,13)
	→	tiles (product)		construction materials	coverings	No	CH	ceramic tiles, at regional storage	1.00E+00	kg				

Fig. 4.2 Flows for "ceramic tiles, at regional storage" and its representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
total area transformed	ceramic plant			resource	land			Transformation, from unknown	7.69E+04	m2	ÖSPAG (2002)	1	2.02	(1,3,1,3,1,4,8)
transformed to vegetation area				resource	land			Transformation, to industrial area, vegetation	3.59E+04	m2	ÖSPAG (2002)	1	2.02	(1,3,1,3,1,4,8)
transformed to built up area				resource	land			Transformation, to industrial area, built up	3.28E+04	m2	ÖSPAG (2002)	1	2.03	(3,3,1,3,1,4,8)
transformed to paved parking lot				resource	land			Transformation, to traffic area, road network	8.20E+03	m2	ÖSPAG (2002)	1	2.03	(3,3,1,3,1,4,8)
occupation as vegetation area				resource	land			Occupation, industrial area, vegetation	1.80E+06	m2a	ÖSPAG (2002)	1	1.54	(3,3,1,3,1,4,7)
occupation as built up area				resource	land			Occupation, industrial area, built up	1.64E+06	m2a	ÖSPAG (2002)	1	1.54	(3,3,1,3,1,4,7)
occupation as paved parking lot				resource	land			Occupation, traffic area, road network	4.10E+05	m2a	ÖSPAG (2002)	1	1.54	(3,3,1,3,1,4,7)
building hall				construction processes	buildings	Yes	CH	building, hall	2.95E+04	m2	assumption, based on ÖSPAG (2002)	1	3.03	(3,3,1,3,1,4,9)
building multi-storey				construction processes	buildings	Yes	RER	building, multi-storey	2.63E+04	m3	assumption, based on ÖSPAG (2002)	1	3.03	(3,3,1,3,1,4,9)
machines				construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.68E+05	kg	assumption	1	3.23	(5,n.A.,n.A.,n.A.,n.A.,n.A.,9)
		machines to recycling							1.68E+05	kg	assumption	1	3.23	(5,n.A.,n.A.,n.A.,n.A.,n.A.,9)
		ceramic plant	Life time: 50 a; production volume: 5000000 kg/a	construction materials	others	Yes	CH	ceramic plant	1.00E+00	unit				

Fig. 4.3 Flows for "ceramic plant" and its representation in the ecoinvent database

5 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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Part VII

Lime Products and Processes

Data v2.0 (2007)

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Summary

This part deals with the lime production process from limestone from nature to hydrated lime. Included are the mining process with the product “limestone at mine”, the mineral preparation process with the main product milled limestone, the calcining process with the main product quicklime, the milling process and the hydration process with its product hydrated lime. Several auxiliary products as intermediate products are additionally offered. Infrastructure is treated in the module “mine, limestone” and for each main product a specific packing module is included.

Data for this study are provided by the only company in Switzerland producing hydrated lime. The data refer to the year 2001.

The most energy intense production step is the calcination process. The fuel used is heavy fuel oil.

The main emissions for quicklime and hydrated lime are caused by the firing process. The rather high amount of particulates is mainly due to the high dust emissions at the mining process.

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1 Introduction

This part of the “Life Cycle Inventories of Building Products“ addresses different lime products used in the building industry.

The Ullmann’s encyclopaedia of industrial chemistry (Oates (1997)) describes lime and lime products as follows:

“Limestone is a naturally occurring mineral that consists principally of calcium carbonate but may also contain magnesium carbonate as a secondary component. It is found in many forms and is classified in terms of its origin, chemical composition, structure, and geological formation. Limestone occurs widely throughout the world and is an essential raw material for many industries.

Quicklime is produced by the thermal decomposition of limestone. It consists mainly of calcium oxide. Its quality depends on many factors, including physical properties, degree of sintering and chemical composition. As the most readily available and cost effective alkali, quicklime plays an essential part in a wide variety of industrial processes.

Hydrated lime is produced by reacting quicklime with water; it consists mainly of calcium hydroxide. In general, hydrated lime refers to a dry calcium hydroxide powder. Hydrated lime is widely used in aqueous systems as a low-cost alkali.

The term lime refers to quicklime and, less frequently, to hydrated lime. It is, however, sometimes used incorrectly to describe limestone, which is a frequent cause of confusion.”

The data basis for the lime products quicklime and hydrated lime is provided by Kalkfabrik Netstal AG (KFN), situated in the canton Glarus of Switzerland. It is the only factory in Switzerland still producing burnt lime products.

KFN’s production is divided into about 75% of quicklime (lump lime and ground quicklime containing over 96% of CaO and about 25% of slaked lime containing over 97% of calcium hydroxide (Ca(OH)₂). Besides these products, the factory produces unburned material (aggregate) which is not included in the data, as the aggregate production is a distinctly separated production branch. Aggregate will be presented in Part 1, “Gravel and Sand”.

The cement industry uses limestone, marl and clay for producing clinker. Therefore the raw material mixture is calcined in a rotary kiln. Clinker is then used as an important component for the production of cement (Compare part 2, Cement Products and Processes). The difference between clinker and quicklime used for products in the binding agent industry is mainly the calcination temperature which is a maximum of 1’450°C for clinker and a maximum of 1’200°C for quicklime.

2 Reserves and Resources

Limestone is widely distributed throughout the world in deposits of varying sizes and degrees of purity. The situation for Switzerland is described by Kündig et al. (1997) (translation by the author):

“Pure limestone as basis for quicklime is mainly found in Switzerland in the upper oolite formation of the Jura-mountains and the helvetic covering (“helvetische Decke”) as well as partly in the cretaceous stage of the last mentioned tectonic unit.”

Kalkfabrik Netstal plans to go on mining for 65 years. They count on more than 160’000 m³ of mineable limestone on site. The total resources in Switzerland and world wide could not be found out in the available literature.

A general comment on resources of primary materials in Switzerland has been made by the Swiss Geotechnical Commission:¹

“For the perspective of the mining industry resources can be regarded as quarryable if mining is possible with actual technology, the enterprise realisable with supportable financial and temporal expenditure and if resource exploitation does not collide with other interests. This last point causes a lot of problems in Switzerland as in its intensely populated part (middle Lands) all different kinds of utilisation of natural resources collide. These are for example water protection, settlement area etc. This circumstance reduces the real quarryable area substantially.”

3 Characterisation

3.1 Limestone

The physical and chemical properties of limestone (Oates (1997)) vary widely as a result of the origin of the deposit, the microstructure, and the impurities. The information given below is typical for most commercially exploited deposits or, where indicated, specific for the Kalkfabrik Netstal AG (KFN).

Colour:	Pure calcite is white. Chalk and marble are also generally white, although impurities in the latter can produce a variety of colours and patterns. Many lime stones, however, contain carbonaceous material, which produces various shades of grey. Iron compounds can introduce a red colour. Some impurities produce a surface coloration with weathering. The limestone mined at KFN is dark grey.
Odour:	Any odour from limestone generally originates from its content of carbonaceous material. The smell is musty or earthy.
Structure:	All lime stones are crystalline. The grain size increases with the amount of recrystallisation that has occurred during formation of the deposit. Thus, shell limestone has a grain size of ca. 1 mm, marl and chalk from 2 to 5 mm, dense limestone from 5 to 250 mm, and marble and calcite spar from 250 to 1000 mm. Calcite and dolomite have rhombohedral crystal structures, and aragonite is orthorhombic.
Porosity:	The porosity of limestone decreases with increasing level of recrystallisation. Thus, marl has 20 – 50 % porosity, chalk 10 – 20 %, dense limestone 1 – 10 %, and calcite spar <1 %.
Density:	The density of pure calcite is 2'710 kg/m ³ at 20°C; aragonite and dolomite densities are 2.93 g/cm ³ and ca. 3.0 g/cm ³ , respectively. The porosity of limestone results in apparent densities of 2.1 – 2.5 g/cm ³ for chalk, 2.5 – 2.7 g/cm ³ for high-calcium limestone, and 2.75 – 2.9 g/cm ³ for dolomitic limestone.
Bulk Density:	The bulk density depends primarily on the apparent density of the limestone and its particle size distribution. Crushed, screened limestone with a 2 : 1 size ratio generally has a bulk density of 1'300 – 1'450 kg/m ³ . Crushed limestone, including the fines, has a bulk density of 1'600 – 1'750 kg/m ³ . Ground limestone of KFN has a bulk density of 1'100 kg/m ³ .
Impurities:	Magnesium carbonate is not generally regarded as an impurity. Impurities may be dispersed homogeneously as a result of their presence during the recrystallization

¹ http://www.sgk.ethz.ch/100jahresgk/meilensteine/abbaubare_rohstoffe.html, access data: October 2002

process. Alternatively they may be present heterogeneously in features such as faults, bedding planes, or nodules. Silica and alumina, in the form of clay, silt, and sand, are commonly found as homogeneous impurities, but they also occur as heterogeneous impurities. Similarly, iron can exist homogeneously as iron carbonate and heterogeneously as pyrite and limonite. Sulphur from sulphates and organic remains generally exist as a homogeneous impurity. Other trace impurities such as lead are often found in the vicinity of faults where mineralization has occurred.

Hardness: Pure calcite has a hardness of 3 Mohs. Naturally occurring limestone lies in the range 2 – 4 Mohs.

Strength: The compressive strength of limestone varies from 10 MPa for some marls and chalks to 200 MPa for some marbles.

Specific Heat: The specific heat of high-calcium limestone and dolomite at 20 °C is 817 and 880 J/kgK, respectively. The specific heat increases with temperature.

Solubility: The solubility of calcite in distilled water free of carbon dioxide has been reported as ca. 15 mg/l at 20 °C. The solubility in distilled water in equilibrium with atmospheric carbon dioxide is approximately double. Dolomite has been reported to be somewhat more soluble than calcite.

Reaction with Acids:

Calcium carbonate reacts with acids with evolution of carbon dioxide and heat (reported to be about 19 kJ/mol in hydrochloric acid).

pH: Both limestone and dolomite are essentially neutral products, giving pHs between 8.5 and 9.0.

Thermal Dissociation:

The heat of dissociation of calcium carbonate is 3180 kJ/kg of CaO at 25°C, and 3010 kJ/kg at 900 °C (760 kcal/kg and 720 kcal/kg, respectively). The corresponding values for magnesium carbonate are 3010 kJ/kg MgO at 25 °C, and 2850 kJ/kg at 700 °C

Chemical composition: see Tab. 3.1

Tab. 3.1 Chemical composition of ground (milled) limestone in KFN ²

Constituent	percent
CaCO ₃ (corresponds to ca. 38 % Ca)	94.5
MgCO ₃ (corresponds to ca. 0.8 % Mg)	2.7
Al ₂ O ₃	0.2
Fe ₂ O ₃	0.1
SiO ₂	0.8
unspecified	1.6

Humidity: 0.2%

Sieving analyse tailing: see Tab. 3.2

² Technical specifications for nekafill[®] 15 and nekafor[®] 15 of KFN, date of edition: 31.01.2001, version 2.1

Tab. 3.2 Amount of residue after sieving of ground (milled) limestone

Diameter of mash size	percent
0.5 mm	0.0
0.09 mm	14.6

3.2 Quicklime (CaO)

The physical and chemical properties of quicklime are (Oates (1997)):

- Colour:** Most quicklime is white. Impurities can result in grey, brown, or yellow tints. When quicklime is produced by solid fuel firing, a grey surface contamination is produced.
- Odour:** Quicklime has a slight earthy odour.
- Porosity:** The porosity of commercially produced quicklime can be as high as 55 %, if highly porous limestone is lightly burned. Exposure to elevated temperature results in sintering which can reduce the porosity to below 25 %. Dead-burned dolomite has a porosity of ca. 10 %.
- Density:** The true (apparent) density of calcium oxide is ca. 3.3 g/cm³. The apparent density of lumps can be as low as 1.4 g/cm³ for highly porous quicklime. This can rise to more than 2.3 g/cm³ after sintering. Calcined dolomite is generally denser than high-calcium quicklime, if given the same heat treatment. Dead-burned dolomite has an apparent density of ca. 3.2 g/cm³.
- Bulk Density:** The bulk density depends on the mean apparent density of the particles and on the voidage between them. The latter is related to the particle size distribution and the particle shape. Most screened quicklime commercial products have compact bulk densities of 900 – 1'200 kg/m³. Inclusion of fines can increase the bulk density by 30 %. One piece of Quicklime at KFN has an average density of 1'200 kg/m³. After having the piece ground the bulk density is about 900 kg/m³.
- Hardness:** Most commercial quicklime products have a hardness of 2 – 3 Mohs. The value for dead-burned dolomite is in the range 3 to 5 Mohs.
- Melting Point:** The melting points for calcium oxide and magnesium oxide are 2570°C and 2'800°C, respectively, with calcined dolomite being intermediate.
- Specific Heat:** The mean specific heats for high-calcium and dolomitic quicklime at 20°C are 760 and 830 J kg⁻¹ K⁻¹, respectively. They increase with temperature.
- Angle of Repose:** The angle of repose for cubical, well-graded pebble quicklime is about 35°. This increases, however, as the fines content increases. For bunker design, valley angles of not less than 60° are recommended.
- Heat of Hydration:** The heat liberated by the reaction of quicklime with water is 1140 kJ/kg of CaO. The value for dolomite is 880 kJ/kg of CaO·MgO.
- Reactivity:** Reactivity to water can be measured by the rate of release of the heat of hydration or by the rate at which an aqueous suspension reacts with hydrochloric acid. The various reactivity tests are interrelated. They are also related to the mean apparent density of quicklime produced from a given source stone. It should be noted that the reactivity of quicklime could be markedly affected by impurities in the water. For this reason, distilled water should be used as a refer-

ence standard.

The low reactivity of calcined dolomite probably arises mainly from the low solubility of $\text{Mg}(\text{OH})_2$ in water. Sintering of MgO , which forms at lower temperatures than CaO may also be a factor.

Affinity for Water:

Quicklime has a high affinity for water and is a more efficient desiccant than silica gel. Because of its high affinity for water vapour and (after partial hydration) for carbon dioxide, care should be taken to minimise exposure of quicklime to the atmosphere. Relatively small amounts of air slaking (less than 1.5 % of combined water) can reduce the reactivity significantly.

The reaction of quicklime with water is associated with an increase in volume. This can cause the expansion of products that contain some lime that is not fully hydrated.

Reaction with Carbon Dioxide:

In the absence of calcium hydroxide, quicklime only reacts with dry carbon dioxide between ca. 300°C and 800°C, depending on the carbon dioxide pressure. Partial recarbonization in the absence of calcium hydroxide can occur in limekilns under certain conditions. It can produce an unexpectedly low reactivity and markedly affects the normal relationship between reactivity and mean apparent specific gravity.

Acid Neutralisation:

Quicklime is used extensively to neutralise acidic oxides, such as SiO_2 , Al_2O_3 , and Fe_2O_3 , in high-temperature non-aqueous systems (e.g., in steel making). Temperatures are generally selected so that the resulting calcium and magnesium salts produce molten slag.

Reaction with Carbon:

Above 1700 °C, carbon reduces calcium oxide to produce calcium carbide and carbon monoxide.

Chemical composition: see Tab. 3.3

Tab. 3.3 Chemical composition of lump and ground quicklime³

Constituent	lump [percent]	ground [percent]
CaO total	96	96.4
MgO	0.6	0.6
Al_2O_3	< 0.1	< 0.1
Fe_2O_3	0.03	0.02
SiO_2	0.1	0.01
SO_3	0.25	0.17
Other indicators		
Loss on ignition	3	< 0.3
H_2O total	1.0	0.55
CO_2	2.0	1.5

³ Technical specification for nekafer[®], nekafin[®] 2 and nekasol[®] 10 of KFN, date of edition: 31.01.2001, version 2.1

- Granulometry:** Ground quicklime has a sieve residue approximately 2% (nekafin 2) and 10% (nekasol 10), respectively being larger than 0.09 mm
For the lump quicklime one commercial article has granulation between 5 and 15 mm and another between 30 and 80 mm
- Reactivity:** For the ground quicklime: t_{60} -value: 0.9 min
For the lump quicklime: t_{60} -value: 0.54 min

3.3 Hydrated and Slaked Lime ($\text{Ca}(\text{OH})_2$)

The differences between hydrated and slaked lime are (Oates (1997)):

Slaked lime generally describes an aqueous suspension of mainly calcium hydroxide. It includes milk of lime and lime putties. The term slaked lime is sometimes used to describe hydrated lime.

Hydrated lime is a dry powder produced by reacting quicklime with a controlled amount of water.

The physical and chemical properties of hydrated and slaked lime are (Oates (1997)):

Colour: Most dry hydrated lime is white. High levels of impurity can result in a grey or buff colour.

Density: High-calcium hydrated lime has a density of about 2.3 g/cm^3 . The values for partially and fully hydrated dolomitic lime are about 2.7 and 2.5 g/cm^3 , respectively.

Bulk Density: The compacted bulk density is in the range 550 to 650 kg/m^3 . In the as-poured state, it can be as low as 400 kg/m^3 , owing to air entrainment. Very fine-grained and just fine-grained hydrated lime of KFN have a bulk density of 500 kg/m^3 .

Specific surface area

It can be measured by air permeability, or by surface absorption of nitrogen (BET surface area). The former method generally gives areas of $1000 - 2000 \text{ m}^2/\text{kg}$; the latter gives some 20 to 30 times greater values.

Angle of Repose In the fluidised state, the angle of repose is 0° . In the compacted state, and particularly with 1 % or more excess water, the angle of repose can be over 80° . With bunker aeration, valley angles of 70° have proved to be satisfactory.

Hardness This is between 2 and 3 Mohs.

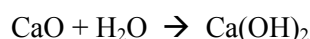
Specific Heat The specific heat of calcium hydroxide rises from 1130 J/kgK at 0°C to 1550 J/kgK at 400°C . The specific heat of dolomitic hydrate is about 5 % higher.

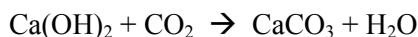
Solubility It decreases from about $1.85 \text{ g Ca}(\text{OH})_2/\text{l}$ water at 0°C to 0.71 g/l at 100°C for commercial calcium hydroxide, depending on impurity levels.

Some inorganic compounds affect the solubility. Calcium sulphate is of particular interest; a 2 g/l solution reduces the solubility to $0.06 \text{ g Ca}(\text{OH})_2/\text{l}$. Organic compounds can increase the "solubility" of calcium hydroxide. Sugar has the greatest effect, as a result of the formation of saccharate lime.

Magnesium hydroxide is only sparingly soluble in water (ca. 0.01 g/l).

Carbon Dioxide Calcium hydroxide reacts readily with carbon dioxide in the absence of water at temperatures below its dissociation point (ca. 540°C). The reaction of quicklime with carbon dioxide below 500°C only proceeds in the presence of some water vapour via formation of calcium hydroxide.



**Acid Neutralisation:**

Hydrated lime (Ca(OH)_2 , $\text{Ca(OH)}_2 \text{ MgO}$, or $\text{Ca(OH)}_2 \text{ Mg (OH)}_2$) reacts readily with acids and acidic gases. The rate of reaction depends in part on the particle size of the hydrated lime.

Silica and Alumina:

Hydrated lime reacts with pozzolans (materials containing reactive silica and alumina) in the presence of water to produce hydrated calcium silicates and aluminates. The reactions may take months to proceed to completion at ambient temperatures (e.g., in mortars and soil stabilisation), but proceed within hours at elevated temperature and water vapour pressures (e.g., at 180°C and steam pressure of 1 MPa). This pozzolanic reaction is the basis of the strength generated by hydraulic quicklime.

pH:

Because calcium hydroxide is a strong base, a concentration of 0.1 g Ca(OH)_2 /l gives a pH of about 11.3 at 25°C. A saturated solution at 25°C, containing 1.8 g/l, gives a pH of 12.7. Magnesium hydroxide gives a pH of about 10.5, owing to its lower solubility. The pH of dolomitic hydrate is dominated by its Ca(OH)_2 content and is similar to that of calcium hydroxide.

Thermal Dissociation:

Calcium hydroxide decomposes at about 540°C to quicklime and water. The quicklime so produced has an exceptionally high reactivity to water.

Causticization: Hydrated lime reacts with soluble metal carbonates to produce insoluble calcium carbonate and metal hydroxide. This reaction is used to produce caustic soda.

Chemical composition: see Tab. 3.4

Tab. 3.4 Chemical composition of fine- and very fine-grained hydrated lime⁴

Constituent	Fine-grained [percent]	Very fine-grained [percent]
Ca(OH)_2	94.9	97.2
MgO	0.4	0.4
Al_2O_3	< 0.1	< 0.1
Fe_2O_3	0.02	0.02
SiO_2	0.25	0.3
SO_3	0.2	0.2
Other indicators		
annealing loss	24.7	24.9
humidity	0.3	0.3
fixed water	23.1	24
CO_2	1.3	1

⁴ Technical specification for nekapur[®] 5 and nekapur[®] 2 of KFN, date of edition: 31.01.2001, version 2.1

Sieving analysis: see Tab. 3.5

Tab. 3.5 Amount of residue after sieving of fine- and very fine-grained hydrated lime

Diameter of sieve holes	Fine-grained [percent]	Very fine-grained [percent]
0.2 mm	0	0
0.09 mm	5.7	2.1

5.7% of the fine- and 2.1% of the very fine-grained hydrated lime is between 0.09 and 0.2 mm, 94.3% and 97.9% respectively are smaller than 0.09 mm.

Brightness: For the very fine-grained hydrated lime: 96.5%

4 Use

Limestone is generally used for construction, cement, quicklime, agriculture, metal refining and flue gas desulphurisation (Oates (1997)):

Construction: Limestone is the most widely used crushed rock, although it is generally sold in form of sand and gravel.

Cement: The production of cement is the main use for chalk, and a major use for dense, high-calcium limestone. Approximately 1.1 t of limestone is required for 1 t of cement.

Quicklime: Approximately 1.8 t of limestone is required per 1.0 t of quicklime.

Agriculture: Many crops grow best under neutral to slightly acid conditions (i.e., pH between 6 and 7). Thus soils which are more acidic than pH 6 generally benefit from the application of limestone.

Metal refining: When limestone is used in metal refining it is initially converted to quicklime, which reacts with acidic oxides (e.g., SiO_2 , Al_2O_3 , and Fe_2O_3) to produce molten slag.

Flue gas desulphurisation:

A growing use of limestone is in the treatment of flue gas from combustion processes to remove sulphur dioxide.

The market shares of lime products (quicklime and hydrated lime) for various uses in Switzerland are displayed in table Tab. 4.1 (Kündig et al. (1997)):

Tab. 4.1 Market share of lime products (quicklime and hydrated lime) in Switzerland 1992 (Kündig et al. (1997))

Use	Market share, %
Iron- and steel industry: Addition of lump lime to the melting process to extract the impurity (slag)	35
Environmental protection: water conditioning, neutralisation and sludge drying in sewage treatment plants, flue gas desulfurization in waste incineration plants, cement and other industries, treatment of cooling tower water in nuclear power stations.	24
Building materials industry: production of sand-lime bricks and plaster	22
Construction: Use of stabilising lime for road construction (stabilisation of soil containing clay and loam)	12
Other industries for example in the chemical industry, sugar production, cellulose mill, galvanic factories, zinc coating plants, or in pharmaceutical industry.	7

As shown, the applicability and the applications of quicklime and hydrated lime are very wide. The product requirements are different, depending on the end use. The main application fields for quicklime and hydrated lime in Switzerland are presented in the Tab. 4.2(Kündig et al. (1997)):

Tab. 4.2 Application of quicklime and hydrated lime in Switzerland

Industry	Quicklime	Hydrated lime
Building industry		Masonry and plastering mortar, limewash
Road construction industry	Soil stabilisation	Asphalt improvement
Building material industry	Sand-lime bricks, porous concrete	Ready-mix, dry mortar, insulation construction elements
Chemical and other industries	Neutralisation, process materials for reactions, sugar industry (usually they calcinate themselves)	Neutralisation (paper industry, tanneries), cooling water softening
Iron and steel industry	Additive for formation of slag in electric furnace	Lubricant
Agriculture		Lime fertiliser, acid soil improvement and agro-chemistry
Environmental protection	Waste water purification, neutralisation, sludge stabilisation, flue gas dust collector	Neutralisation, waste water purification, sludge stabilisation, flue gas washing machines

5 Process Description

5.1 Overview

The basic processes in lime production are raw limestone crushing and screening the limestone for the kilns by crushing and sizing, limestone calcining, lime processing by hydration and miscellaneous transport, storage, and handling operations.

The production process of KFN for the year 2001 is displayed in Fig. 5.1. Each major step is explained in the following chapters.

5.2 Limestone Mining

The description by (Oates (1997)) has been adapted to KFN in Switzerland.

Limestone is mined by opencast quarrying. The first operation is to remove the overburden (i.e., the soil, clay, and bulk rock overlying the deposit). Various techniques and types of earth moving equipment are used.

The next stage is generally to drill the bedrock. Rotary and percussive drills are used. The spacing between the drilled holes and the burden (distance between the holes and the quarry face) is carefully controlled. The diameter of the holes varies from 5 – 25 cm, depending on the design of the blast.

The holes are filled with the explosive: The foot is filled with Tovex and the shaft with Amolit. A mixture of ammonium nitrate and fuel oil (ANFO) is widely used. In a typical blast, 5'000 – 20' 000 t of stone is fragmented with about 90 g of explosive per 1 t limestone.

It is important to control the blast. Too little fragmentation produces oversize boulders, which have to be broken using secondary blasting or a hydraulic hammer. Too much fragmentation produces an excessive amount of fine particles and increases the risk of throwing rock away from the quarry face. Optimum fragmentation is obtained by selection of the correct diameter and spacing of holes, time delay between holes, thickness of burden, and quantity of explosive.

After blasting, the rock is loaded into dump trucks, generally by front-end shovels or hydraulic excavators and transported to the crushing and screening plant.

5.3 Mineral Preparation

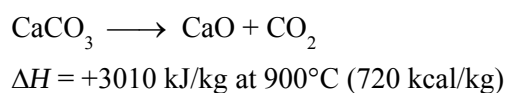
The description from Oates (1997) has been adapted to the situation at KFN in Switzerland.

In the crushing and screening plant, the larger lumps of rock are broken in a primary jaw crusher to a maximum of 200 mm. After having reached the desired size, the rock is conveyed to the washing and classifying installation. By passing different conveyor belts and shaking screens with different mesh apertures, the crushed rock is washed and stored.

The water used for washing is ground water. After the washing process the muddy water is clarified in a sewage plant with sludge flocculation. About 80% of the pumped water is being recycled. The residual 20% are leaving the loop with the wet aggregate products or is being released to the quarry with the dewatered sludge for recultivation purposes. As no further data are available and no environmental impact is expected in this regard, no emissions are taken into account in this study.

5.4 Calcination

The chemical reaction (EPA (1998)) for the decomposition of calcium carbonate by heat is:



The partial pressure of carbon dioxide above calcium carbonate is increasing logarithmically with the temperature. The pressure is 101 kPa at ca. 900 °C. Thus, although surface calcination can proceed at lower temperatures, complete calcination only occurs above 900 °C.

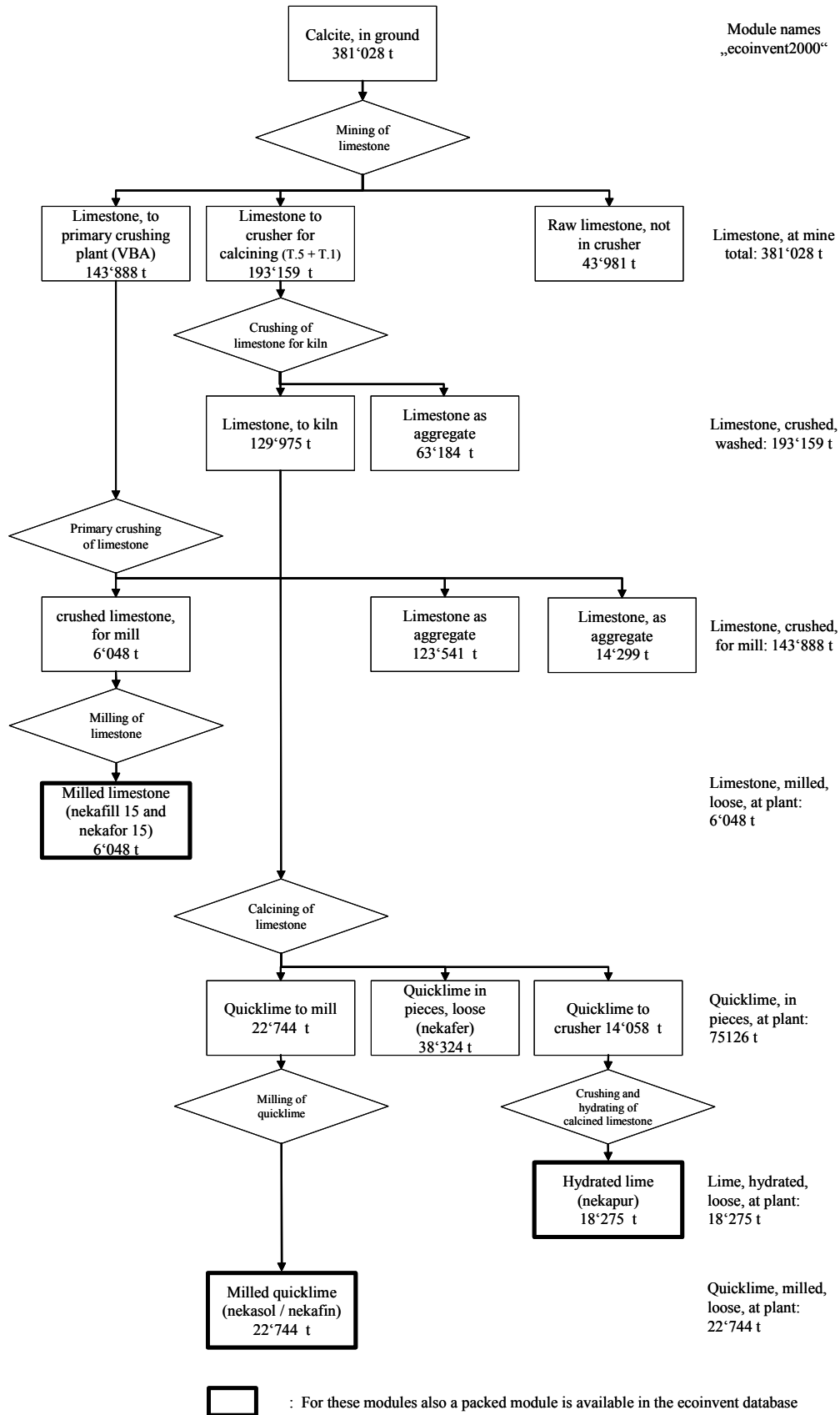


Fig. 5.1 Flow diagram: manufacturing of lime products

The calcination mechanism is complex and involves several stages: heat transfer to the surface of the particle and through the outer layer of lime. The heat is absorbed by the chemical reaction at the lime–limestone interface. The carbon dioxide so generated migrates to the surface of the particle counter to the heat flow and then diffuses from the surface into the kiln gases.

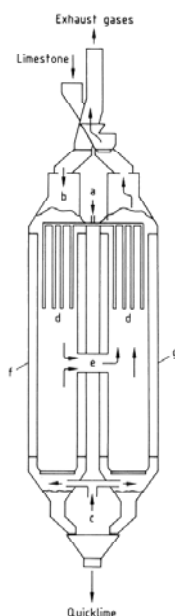
The rate-determining stage in the above process depends on the particle size, temperature, amount of calcination that has already taken place, and the composition of the kiln gases. Although attempts have been made to produce a mathematical model to account for observed effects in practical lime burning, none has proved applicable over wide ranges of the above variables.

Most of the kilns currently used are based on either the vertical shaft or on the rotary design. There are a few other kilns based on different principles. All these designs incorporate the concept of the three zones. In some kilns, the zones are merged into one unit, in others they exist as separate units.

The lime plant in Netstal (KFN) uses two kinds of kiln: a parallel-flow regenerative kiln (see Fig. 5.2) and a vertical shaft kiln which has been transformed into a multiple-inclined shaft kiln. Both are fired by heavy fuel oil. The main kiln is the parallel-flow regenerative (or Maerz) kiln consisting of two interconnected vertical shafts. The operation consists of two equal stages, which last about 12 minutes each.

In the first stage, fuel is injected through lances into shaft 1 and burns with combustion air blown down the shaft. The heat released is partly absorbed by calcining the limestone in shaft 1. Air is blown into the base of each shaft to cool the lime. The air in shaft 1 mixes with the combustion gases, including the carbon dioxide from calcination. The mixture passes through the cross-duct into shaft 2, at about 1050 °C. In shaft 2, the gases from shaft 1 mix with the cooling air blown into shaft 2 and pass upwards. In so doing, they heat the stone in the pre-heating zone of that shaft.

During the second stage of the operation, the converse applies. The same amounts of fuel and combustion air are added to shaft 2. The combustion gases plus cooling air pass upward in shaft 1, heating the stone in the pre-heating zone of that shaft.



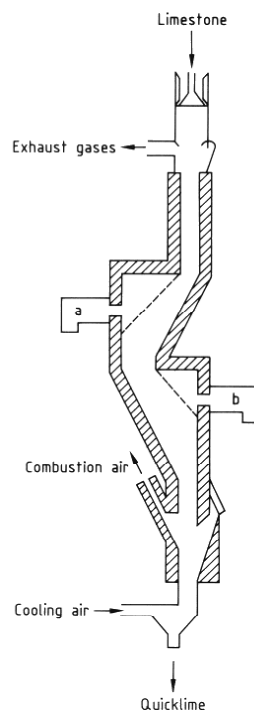
(a) fuel; (b) combustion air; (c) cooling air; (d) lances; (e) cross-duct; (f) shaft 1; (g) shaft 2.

Fig. 5.2 Flow scheme of parallel-flow regenerative (or Maerz) kiln

The two key principles of the above operation are:

1. The stone-packed preheating zone in each shaft acts as a regenerative heat exchanger in addition to pre-heating the stone to calcining temperature. The surplus heat in the gases is transferred to the stone in shaft 2 during the first stage. It is then transferred from the stone to the combustion air in the second stage. As a result, the combustion air is pre-heated to about 800 °C. The net heat usage of the kiln is very low at below 3'600 kJ/kg (860 kcal/kg) of quicklime.
2. The calcination of the quicklime is completed at the level of the cross-duct at a moderate temperature of about 1100 °C. This favours the production of highly reactive quicklime, which may, if required, be produced with a low CaCO₃ content. The kiln accepts stone with a maximum size of 5 – 12 cm, and it can be fired with gas, oil, or solid fuel. KFN feeds its Maerz kiln mainly with 40 – 90 mm inserting few layers of 15 – 40 mm stone and fires with heavy fuel oil.

The second kiln used at KFN is a vertical shaft kiln which has been transformed into a multiple-inclined shaft kiln, illustrated in Fig. 5.3.



(a) upper burner; (b) lower burner

Fig. 5.3 Flow scheme of double-inclined shaft kiln

The figure shows a double-inclined kiln. The construction of the multiple-inclined kiln of KFN functions with the same principle but instead of two sloping sections, several are connected in series. The stone moves downwards under gravity passing several hearths, which are inclined at about 60°. Opposite to each hearth the burners are mounted underneath the arches. The products of fuel combustion and calcination travel upwards through the stone and are removed by an exhaust fan. Most of the combustion air is drawn through the cooling zone.

The multiple-inclined kiln usually accepts stones with a minimum size of 20 mm and a maximum of 100 mm. KFN uses 15 to 40 mm. These kilns produce reactive, low-carbonate quicklime at a net heat usage of about 4'200 kJ/kg (1'000 kcal/kg). Although various fuels can be used, they should be prop-

erly selected to avoid excessive build-ups caused by fuel ash and calcium sulphate deposits. KFN fires with heavy fuel oil.

KFN produces two grades of lump lime: 5 – 15 mm and 30 – 80 mm. These are manufactured using a combination of crushing the raw lime in a jaw crusher, an impact crusher and screening. The fines from this process are being ground to fine quicklime by a roller mill, and slaked to hydrated lime in the slaking plant, respectively.

5.5 Hydration

The normal hydration process is carried out under atmospheric pressure at about 100°C (EPA (1998)).

In KFN, the quicklime is mixed rapidly with something less than twice the stoichiometric amount of water in the first stage of the hydration plant consisting of two counter-rotating shafts. The so produced hydrated lime powder has a much lower density than the quicklime, and thus can pass over a weir to enter into the second slaking stage being only slightly agitated. In the second stage hydration is completed, and the product is dried.

The raw hydrated lime is fed into a first air separator circuit in which an air stream removes the finest particles. These fines are recovered using a cyclone, and then transferred to the storage and bagging silos by a pneumatic transportation system. The hydrated lime is discharged at about 100°C and contains less than 1% excess water.

The coarse particles from the first circuit are conducted to a second circuit there being ground in a hammer mill, and recovered by a similar air separator and cyclone combination. This product is sold as a less fine hydrated lime.

5.6 Packing and Loading of Lime Products

The process of packing the product into paper bags is fully automatic. Once filled, the bags are transported to a robot by conveyor belt. The robot piles them up on Euro-flat pallets. To ensure moisture protection, and for a better transportation the pallets are wrapped in a plastic foil.

The pallet is assumed to be used 10 times before being disposed. Thus only 1/10th of the demand is inventoried.

A loaded pallet has a rectangular circumference of 4 m, a height of 1 m and a loading weight of 1'000 kg. If two layers of plastic foil (including an overlay of 2 m) with a weight of 0.0123 kg/m² are wrapped around, the amount of plastic per kg product is 0.000123 kg. To have the foil correctly taken into account a module „HDPE, granulate“ for the material and a module „extrusion“ for the preparations are included.

The transport distances have been estimated based on the ecoinvent standard distances in Frischknecht et al. (2007)).

6 Life Cycle Inventories

Ecoinvent meta information (tables), flows and their representation in the ecoinvent database are shown in (figures):

Limestone, at mine	Tab. 6.2 and Fig. 6.1
Mine, limestone	Tab. 6.3 and Fig. 6.2
Recultivation, mine limestone	Tab. 6.4 and Fig. 6.3
Limestone, crushed, washed	Tab. 6.7 and Fig. 6.4
Limestone, crushed for mill	Tab. 6.8 and Fig. 6.5
Limestone, milled, loose, at plant	Tab. 6.9 and Fig. 6.6
Quicklime, in pieces, loose, at plant	Tab. 6.15 and Fig. 6.7
Quicklime, milled, loose, at plant	Tab. 6.16 and Fig. 6.8
Lime, hydrated, loose, at plant	The service life is as-
sumed to be 25 years. The total amount of material treated by the machines over this period is about 450'000 t (Compare Fig. 5.1).	
Tab. 6.17 and Fig. 6.9	
Packing, lime products	Tab. 6.18 and Fig. 6.10
Limestone, milled, packed, at plant	Tab. 6.19 and Fig. 6.11
Quicklime, milled, packed, at plant	Tab. 6.20 and Fig. 6.12
Lime, hydrated, packed, at plant	Tab. 6.21 and Fig. 6.13

The following sections, each describing a process or a product, start showing the specific background information (ecoinvent database meta information).

The flow chart with all processes and products given in Fig. 5.1 provides a useful overview.

The energy used for space heating is allocated to the different modules depending on the process and its production amount. Only the heating expenditure for administration is allocated to all modules (see Tab. 6.1). In the year 2001 the consumption of heating oil for the production buildings was 10'323 litre and 13'548 litre for other buildings. With a fuel oil density of 860 kg/m³ and a net calorific value of 42.6 MJ/kg the total heating energy for production buildings was 378'193 MJ, respectively for diverse other buildings 496'345 MJ.

Tab. 6.1 Allocation of the total heating energy to the different modules

Name of module	Definition of KFN / explanation	Production amount [t]	Production place	Share of heating energy to process buildings [%]	Share of heating energy to buildings of production [MJ]	Accounted share of heating energy to administration [%]	Share of heating energy to administration [MJ]	Total heating energy per module [MJ]
limestone, at mine		381'028	Outdoor activity			46	226'158	226'158
limestone, crushed, washed		193'159	Production indoor: Crushing and washing	46	175'233	23	114'649	289'882
limestone, crushed, for mill		129'589	Production indoor: Crushing	31	117'562	15	76'917	194'480
limestone, milled, loose, at plant	nekafill und nekafor (milled limestone)	6'048	Production indoor: Milling	1	5'487	1	3'590	9'076
quicklime, in pieces, loose, at plant	calcinated lime	38'324	Production indoor: Calcination	9	34'767	5	22'747	22'747
lime, hydrated, loose, at plant	nekapur (hydrated lime)	18'275	Production indoor: Hydrating	4	16'579	2	10'847	27'426
quicklime, milled, loose, at plant	nekasol und nekafin (milled and calcinated lime)	22'744	Production indoor: Milling	5	20'633	3	13'500	34'133
limestone, milled, packed, at plant	packing of "limestone, milled, loose, at plant"	6'048	Production indoor: Packing	1	5'487	1	3'590	9'076
quicklime, milled, packed, at plant	packing of "quicklime, milled, loose, at plant"	22'744	Production indoor: Packing	5	20'633	3	13'500	34'133
lime, hydrated, packed, at plant	packing of "lime, hydrated, loose, at plant"	18'275	Production indoor: Packing	4	16'579	2	10'847	27'426
total production indoor		416'882		100	378'193	100	496'345	
total production indoor and outdoor		836'234						

(*) it is zero because the building is heated by waste heat

6.1 Limestone, at mine

The ecoinvent meta information for "limestone, at plant" includes all important and necessary information to use the data correctly. Fig. 6.1 shows all in- and output flows of the product "limestone, at plant", the allocation to the modules in ecoinvent database, the sources (mostly from Berger & Durner (1992), from management ratio of KFN and EAA (2000)) and the standard deviation for each datum.

Most data are based on yearly amounts, subsequently normalized by the yearly production. The total amount of material transported within open pit mine and from mine to processing plant of KFN in the year 2001 was 381'028 t.

Land use

The calculation of the land use is based on the report (Berger & Durner (1992)) which was commissioned by KFN. This report describes five mining stages each with its area and volume. KFN plans to mine 10'280'000 m³ over 65 years. The yearly average amount of mined material is therefore about 160'000 m³/a. With a density⁵ of 2'300 kg/m³ the amount is 368'000'000 kg/a. The amount of limestone mined in 2001, which is used in this study for consistency with the data on material and energy consumption, is very close to the above average, thus justifying its validity for the lifetime of the mine. The total area to be mined over 65 years is 156'000 m² or 2'400 m² per year. The total amount of material excavated per m² for the year 2001 is therefore about 160'000 kg. It is assumed that the mining duration for each stage lasts about 10 years. After that period of time the next stage is tackled and the last is recultivated.

Blasting

KFN provided the data for the amount of explosives for blasting in 2001. About half of the used explosives is „Tovex“ which is used at the foot of the drilled hole; the upper half is filled with „Amolit“.

⁵ Density of limestone in general: <http://userpage.fu-berlin.de> and the same density for KFN, confirmed by Mr. Huser from KFN in spring 2002 (personal communication)

In general, the types of explosive used in mining can change quickly⁶, hence the module „explosive, Tovex, at plant“ is used for the total amount used in the process, as good approximation.

Infrastructure

The life span of the infrastructure of the plant could not be assessed. A default value of 50 years was used.

Dust from mining

The total dust production for limestone mining is 0.16g/kg (BUWAL 2001). The distribution of particle sizes is taken from the process „bauxite mining“ (50% of the total dust are > 10µm, 5% < 2.5µm and 45% between 2.5µm and 10µm) as no specific information is available (see Classen et al. (2007)).

Tab. 6.2 Ecoinvent database meta information for the product “limestone, at plant”

Name	limestone, at mine
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes blasting, all transports within the mine (for mining and recultivation), a part of the heating energy for "administration" buildings and the infrastructure. The land-use of the mine is included, the land-use of the paved roads and buildings within the mine are included in the module "mine, limestone". Output of overburden material neglected as it is only about 0.26% of mined limestone.
Amount	1
Local Name	Kalkstein, ab Abbau
Synonyms	
General Comment to reference function	Land-use for recultivation has been included; size partition of dust emissions adapted from bauxite mining. The value for the module are normalized using a capacity of about 380'000 tons of product per year.
Start Date	1992
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are only from one company in Switzerland (KFN), for some exchanges GLO-modules have been used as proxy
Technology text	The company works on a technically high level; heavy machinery are operated electrically, building machines with diesel. Explosives used: Tovex at bottom and Amolith in shaft of drilled hole
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Survey
Extrapolations	Data which are only available for GLO have been used as proxy for CH
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Limestone (resource)	mining of limestone		total amount of mined limestone 2001: 381'028 t; the losses by dust emissions have been neglected.	resource	in ground			Calcite, in ground	1.00E+00	kg	data of KFN for the year 2001	1	1.33	(3,3,1,1,3,5);
land occupation, mine [m2a]			assumption: after 13 years the used land is recultivated	resource	land			Occupation, mineral extraction site	8.48E-05	m2a	Berger & Durner (1992)	1	1.52	(1,2,3,1,1,1);
land transformation (to mineral extraxtion site)				resource	land			Transformation, to mineral extraction site	6.52E-06	m2	Berger & Durner (1992)	1	2.02	(2,2,3,1,1,1);
land transformation (from forest)				resource	land			Transformation, from forest	6.52E-06	m2	Berger & Durner (1992)	1	2.02	(2,2,3,1,1,1);
recultivation				construction materials	additives	No	CH	recultivation, limestone mine	6.52E-06	m2	Berger & Durner (1992)	1	2.02	(2,2,3,1,1,1);
blasting [kg explosive]			Foot of drilled hole filled with Tovex und shank filled with Amolit	construction processes	civil engineering	No	RER	blasting	7.73E-05	kg	data of KFN for the year 2001	1	1.05	
total machines used in mine for drilling, mining and transports				construction processes	machinery	No	GLO	diesel, burned in building machine	1.80E-02	MJ	data of KFN for the year 2001	1	2.00	
share of energy consumption for administration			includes the total amount of fuel used for heating of mostly administration buildings	oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	5.94E-04	MJ	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
Infrastructure (mine) [unit]			Input includes infrastructure needed at mine (buildings, access roads,...) (exclusive building machines); Service life assumption: 50 years, total produced products over 50 years: 19'051'400'000 kg	construction materials	additives	Yes	CH	mine, limestone	5.25E-11	unit	data of KFN for the year 2001	1	3.00	(1,2,1,1,1,1);
		→ Dust to air [kg]	TPM: 0.16 g, thereof 5%<2.5um	air	low population density			Particulates, < 2.5 um	8.00E-06	kg	BUWAL, 2001	1	3.28	(1,5,2,3,4,3);
		→ Dust to air [kg]	TPM: 0.16 g, thereof 10um>25%>2.5um	air	low population density			Particulates, > 2.5 um, and < 10um	4.00E-05	kg	BUWAL, 2001	1	2.28	(1,5,2,3,4,3);
		→ Dust to air [kg]	TPM: 0.16 g, thereof 70%>10um	air	low population density			Particulates, > 10 um	1.12E-04	kg	BUWAL, 2001	1	1.83	(1,5,2,3,4,3);
		→ overburden material (topsoil)	neglected. assumed thickness of topsoil layer: 0.3m, density: 1'400 kg/m3, weight per m2: 420kg compared to 160'000kg mined limestone						2.60E-03	kg	calculated			
		→ limestone, to primary crushing plant	used for limestone crushed, gravel or brash; amount 2001: 143'888t	construction materials	additives	No	CH	limestone, at mine	3.78E-01	kg	data of KFN for the year 2001	1		(,,,,);
		→ raw limestone, not in crusher	used as rocks and boulders; amount 2001: 43'981 t	construction materials	additives	No	CH	limestone, at mine	1.15E-01	kg	data of KFN for the year 2001	1		(,,,,);
		→ limestone, to crusher for burning	used for calcination; amount 2001: 193'159 t	construction materials	additives	No	CH	limestone, at mine	5.07E-01	kg	data of KFN for the year 2001	1		(,,,,);

Fig. 6.1 Flows for "limestone, at plant" and their representation in the ecoinvent database

6.2 Infrastructure limestone mine

The ecoinvent meta information for “mine, limestone” includes all important and necessary information to use the data correctly. Fig. 6.2 shows all in- and output flows of the product “mine, limestone”, the allocation to the modules in ecoinvent database, the sources (assumptions) and the standard deviation for each datum.

The land-use included in the infrastructure module „mine, limestone“ is the land used for buildings and paved roads. The land where mining takes place is included in the module „limestone mining“. The values are derived from an air-picture. The land use during recultivation is included in the module “recultivation, mine limestone” (compare Fig. 6.3)

Tab. 6.3 Ecoinvent database meta information for the infrastructure “mine, limestone”

Name	mine, limestone
Location	CH
Infrastructure Process	1
Unit	unit
Data Set Version	2.0
Included Processes	Includes the "land-use" of the built part of the mine and therefore the production places and the paved roads
Amount	1
Local Name	Mine, Kalkstein
Synonyms	quarry//Steinbruch
General Comment to reference function	Data are estimated from air-picture. The mine has an annual yield of about 380'000 t
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are from only one company in Switzerland (KFN), for some exchanges RER-modules have been used as proxy
Technology text	Medium size Swiss limestone mine
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Survey
Extrapolations	Data which are only available for RER have been used as proxy for CH
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
land use for roads in mine area	Infrastructure limestone mine		Is balanced in land-use for mine									1		(,,,,);
land use for buildings (occupation)			Assumption from air-picture, it's assumed that mine is in use for 50 years	resource	land			Occupation, industrial area, built up	2.00E+04	m2a	assumption from air-picture	1	1.56	(4,2,1,1,1,1);
land use for buildings (transformation to industrial area)			Assumption from air-picture	resource	land			Transformation, to industrial area, built up	5.80E+02	m2	assumption from air-picture	1	2.05	(4,2,1,1,1,1);
land use for buildings (transformation from forest)			Before mostly forest	resource	land			Transformation, from forest	5.80E+02	m2	assumption from air-picture	1	2.05	(4,2,1,1,1,1);
land use for paved roads (transformation to paved roads)			Assumption from air-picture	resource	land			Transformation, to traffic area, road network	4.00E+02	m2	assumption from air-picture	1	2.05	(4,2,1,1,1,1);
land use for paved roads (transformation from forest)			Before mostly forest	resource	land			Transformation, from forest	4.00E+02	m2	assumption from air-picture	1	2.05	(4,2,1,1,1,1);
building multi-storey			Assumption from air-picture: area of 80m2 and average height of 6m	construction processes	buildings	Yes	RER	building, multi-storey	4.80E+02	m3	assumption from air-picture	1	3.05	(4,2,1,2,1,1);
building hall			Assumption from air-picture	construction processes	buildings	Yes	CH	building, hall	5.00E+02	m2	assumption from air-picture	1	3.05	(4,2,1,1,1,1);
		mine, limestone		construction materials	additives	Yes	CH	mine, limestone	1.00E+00	unit		1		(,,,,);

Fig. 6.2 Flows for "mine, limestone" and their representation in the ecoinvent database

6.3 Recultivation of Limestone Mine

The ecoinvent meta information for “recultivation, mine limestone” includes all important and necessary information to use the data correctly. Fig. 6.3 shows all in- and output flows of the product “recultivation, mine limestone”, the allocation to the modules in ecoinvent database, the sources (mainly Berger & Durner (1992)) and the standard deviation for each datum.

After having removed the topsoil and mined the limestone the topsoil is brought out in an assumed average thickness of about 0.3 m. Its density is assumed⁷ to be 1400 kg/m³. The weight of the topsoil per m² is therefore about 420 kg. The ratio of topsoil to extracted limestone is 0.26%. The diesel fuel used for covering the closed mine is thus calculated using this ratio compared to the consumption during operation of the mine.

Since the mine is entirely located within a forest, it shall be restored to the same status (Berger & Durner (1992)). After having mined one area for about ten years, the phase of recultivation starts. This guarantees that the active area, will never be disturbed for longer than a decade.

Tab. 6.4 Ecoinvent database meta information for the infrastructure “recultivation, mine limestone”

Name	recultivation, limestone mine
Location	CH
Infrastructure Process	0
Unit	m2
Data Set Version	2.0
Included Processes	includes only transformation from mineral extraction site to forest and the diesel consumption.
Amount	1
Local Name	Rekultivierung, Kalkmine
Synonyms	Renaturierung
General Comment to reference function	The ratio of topsoil (assumed density: 1'400 kg/m3, thickness: 0.3 m, weight: 420 kg/m2,) to extracted limestone (density: 2'300 kg/m3, thickness: 70m, weight: 160'000 kg/m2) is about 0.26%. The diesel fuel used for covering the closed mine is thus calculated using this ratio compared to its requirement during operation of the mine
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are only from one company in Switzerland (KFN)
Technology text	Recultivation topsoil thickness 30 cm for a medium-size Swiss limestone mine.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Survey
Extrapolations	Data which are only available for GLO have been used as proxy for CH
Uncertainty Adjustments	None

⁷ <http://www.geo.unizh.ch/~shofstet/site/home/labor/boden/node7.html>

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
land transformation from mineral extraction site	recultivation of limestone mine			resource	land			Transformation, from mineral extraction site	1.00E+00	m2	Berger & Durner (1992)	1	2.01	(3,2,1,1,1,1);
land transformation to forest				resource	land			Transformation, to forest	1.00E+00	m2	Berger & Durner (1992)	1	2.01	(3,2,1,1,1,1);
machines used for recultivation			0.26% of the total yearly diesel consumption is used for transports concerning recultivation. The yearly recultivated area is about 2'400m2	construction processes	machinery	No	GLO	diesel, burned in building machine	7.44E+00	MJ	estimated by EMPA	1	3.09	(4,3,1,1,1,5);
		recultivation		construction materials	additives	No	CH	recultivation, limestone mine	1.00E+00	m2				

Fig. 6.3 Flows for “recultivation, mine limestone” and their representation in the ecoinvent database

6.4 Mineral Preparation

The datasets resulting from primary crushing and crushing of limestone, grinding of limestone and grinding of quicklime are “limestone, crushed, washed”, “limestone, crushed for mill” and “limestone, milled, loose, at plant”. The ecoinvent database meta information of these modules include all important and necessary information to use the data correctly. Fig. 6.4 shows all in- and output flows of the product “limestone crushed, washed”, Fig. 6.5 shows all in- and output flows of the product “limestone, crushed for mill” and Fig. 6.6 shows all in- and output flows of the product “limestone, milled, loose, at plant”. Beside the in- and output flows, the allocation to the modules in ecoinvent database, the sources (management ratio of KFN, EPA (1998) and estimations) and the standard deviation for each datum are indicated.

Emissions from crusher, mills, and transfer operations

The particulate emission from crusher, mills, and transfer operations (e.g. by conveyor belt) is calculated based on the summary description in (EPA (1998)) illustrated in . Tab. 6.5. As no specific information on particle size distribution is provided in this source, an assumption is used here, explained after Tab. 6.5.

Tab. 6.5 Summary of average emission factors for dust from crushing, screening, transporting in the lime manufacturing (EPA (1998))

Source of emission	Emission factor for dust (kg/t of material processed)
Primary crushing	$8.3 \cdot 10^{-3}$
Primary screening	$3.0 \cdot 10^{-4}$
Crushed material conveyor transfer	$4.4 \cdot 10^{-5}$
Secondary and tertiary screening	$6.5 \cdot 10^{-5}$
Total particles > 10 µm	$8.7 \cdot 10^{-3}$ ⁽¹⁾

⁽¹⁾ Assumption

If TSP is known the dust distribution in this study is generally assumed to be: 50% < 10µm (PM10), 45% between 10 µm and 2.5 µm, 5% <2.5 µm.

The dust emissions calculated based on above estimations are used for “limestone crushed, washed” and “limestone, crushed, for mill” and listed in Tab. 6.6:

Tab. 6.6 Dust emissions used in study for “limestone crushed, washed” and “limestone, crushed, for mill”

Size of dust particles	Emission factor (kg/kg of material processed)
Total particles > 10 µm (= 50% of total)	$8.7 \cdot 10^{-6}$
Total particles < 10 µm (= 50% of total)	$8.7 \cdot 10^{-6}$
Total particles > 2.5 and < 10 µm (= 45% of total)	$7.8 \cdot 10^{-6}$
Total particles < 2.5 µm (= 5% of total)	$8.7 \cdot 10^{-7}$
Total particles	$17.4 \cdot 10^{-6}$

The module „milling of limestone“ does not include dust emissions since these emissions are included in the estimation for the dust emissions from mining and because KFN has installed an efficient closed-loop system.

Equipment for crushing limestone to feed the kiln”:

As no specific information is available the weight of the machines is estimated and summed up:

2 crushers (each 25 t):	50 t
2 sieves (assumption each 5 t):	10 t
2 „small“ silos (assumption each 7 t):	14 t
Total weight of equipment:	74 t
Length of conveyor belt:	100 m

The service life for all machines including conveyor belt is assumed to be 25 years. The total amount of material handled by the machines over this period is about 5 Mio. t (compare Fig. 5.1).

Equipment for primary crushing of limestone:

As no specific information is available the weight of the machines is estimated and summed up:

1 crusher:	25 t
1 sieve:	5 t
1 „small“ silo:	7 t
Total weight of equipment:	37 t
Length of conveyor belt:	100 m

The service life is assumed to be 25 years. The total amount of material treated by the machines over this period is about 3.6 Mio. t (Compare Fig. 5.1).

Equipment for grinding of limestone:

As no specific information is available, the weight of the machines is estimated and summed up:

1 crusher:	25 t
1 sieve:	5 t
2 „small“ silos (assumption each 7 t):	14 t
1 filter:	10 t
Total weight of equipment:	54 t

The service life is assumed to be 25 years. The total amount of material treated by the machines over this period is about 150'000 t (Compare Fig. 5.1).

Tab. 6.7 Ecoinvent database meta information for the product "limestone crushed, washed"

Name	limestone, crushed, washed
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the processes: crushing, washing, and transportation by conveyor belt. One part of the total heating energy for "production" and "administration" is included. Equipment included in the infrastructure: 2 crushers, 2 sieves, and 2 small silos.
Amount	1
Local Name	Kalkstein, gebrochen, gewaschen
Synonyms	
General Comment to reference function	Infrastructure data are estimated based on a tour and sketches of the process. The value of the infrastructure is normalized with a annual production capacity of about 195'000 tons of product per year. The estimated lifespan of the machines is 25 years. Assumed size partition of dust emissions for crushing, screening and transport processes: 50% >10µm (PM10), 45% between 10 µm and 2.5 µm, 5%<2.5 µm. Limestone, crushed, washed is a pre-product for the milling process. The module can also be used for gravel products in general. 50% of the electricity consumption is covered by the grid and 50% by a hydropower plant nearby. The infrastructure for electricity transportation from power plant to KFN is not included. 50% of the electricity consumption is covered by the grid and 50% by a hydropower plant nearby. The infrastructure for electricity transportation from power plant to KFN is not included.
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are from only one company in Switzerland (KFN), for some exchanges RER-modules have been used as proxy
Technology text	The company works on a technically high level; heavy machines (excl. building machines) are operated electrically; air recirculated in closed loop to avoid dust emissions; ground water used for washing.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Survey
Extrapolations	Data which are only available for RER have been used as proxy for CH
Uncertainty Adjustments	None

Tab. 6.8 Ecoinvent database meta information for the product "limestone, crushed for mill"

Name	limestone, crushed, for mill
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the processes: primary crushing, washing, and transportation by conveyor belt. One part of the total heating energy for "production" and "administration" is included. Equipment included in the infrastructure: 1 crusher, 1 sieve and 1 small silo.
Amount	1
Local Name	Kalkstein, gebrochen für mahlen
Synonyms	
General Comment to reference function	Infrastructure data are estimated based on a tour and sketches of the process. The value of the infrastructure is normalized with a annual production capacity of about 145'000 tons of product per year. The estimated lifespan of the machines is 25 years. Assumed size partition of dust emissions for crushing, screening and transport processes: 50% >10µm (PM10), 45% between 10 µm and 2.5 µm, 5%<2.5 µm. The module can also be used for limestone as screened ballast or as gravel for road bed construction. 50% of the electricity consumption is covered by the grid and 50% by a hydropower plant nearby. The infrastructure for electricity transportation from power plant to KFN is not included.
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are from only one company in Switzerland (KFN), for some exchanges RER-modules have been used as proxy
Technology text	The company works on a technically high level; heavy machines (excl. building machines) are operated electrically; air is recirculated in closed loop to avoid dust emissions
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Survey
Extrapolations	Data which are only available for RER have been used as proxy for CH
Uncertainty Adjustments	None

Tab. 6.9 Ecoinvent database meta information for the product "limestone, milled, loose, at plant"

Name	limestone, milled, loose, at mine
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the processes: milling, sieving, filtering and storing. One part of the total heating energy for "production" and "administration" is included. Equipment included in the infrastructure: 1 crusher, 1 sieve, and 2 small silos and 1 filter.
Amount	1
Local Name	Kalkstein, gemahlen, lose, ab Werk
Synonyms	
General Comment to reference function	Infrastructure data are estimated based on a tour and sketches of the process. The value of the infrastructure is normalized with a annual production capacity of about 6'000 tons of product per year. The estimated lifespan of the machines is 25 years. There are no dust emissions taken into account as the mill is a closed system. 50% of the electricity consumption is covered by the grid and 50% by a hydropower plant nearby. The infrastructure for electricity transportation from power plant to KFN is not included.
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are from only one company in Switzerland (KFN), for some exchanges RER-modules have been used as proxy
Technology text	The company works on a technically high level; heavy machines (excl. building machines) are operated electrically; air is recirculated in closed loop to avoid dust emissions
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Survey
Extrapolations	Data which are only available for RER have been used as proxy for CH
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
limestone, to crusher	crushing and washing of limestone for kiln		Total amount 2001 of raw limestone prepared:193'159t; losses are neglected as very small in form of dust.	construction materials	additives	No	CH	limestone, at mine	1.00E+00	kg	data of KFN for the year 2001	1	1.33	(3,3,1,1,3,5);
machines for crushing, washing, sieving and storage			machines concerning crushing process; total weight 44t, assumed life time: 25 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	9.11E-06	kg	estimation	1	3.09	(4,3,1,1,1,5);
conveyor belt			assumption: length of conveyor belt = 100m, assumed life time: 25 years	construction processes	machinery	Yes	RER	conveyor belt, at plant	2.07E-08	m	estimation	1	3.09	(4,3,1,1,1,5);
water consumption			total water consumption 2001 for hydration: 11'010 m3; total amount of pumped water 2001: 19'450 m3; Half of rest for this process and other half of rest for primary crushing of limestone (for mill)	resource	in water			Water, well, in ground	2.18E-05	m3	data of KFN for the year 2001	1	1.12	(3,2,1,1,1,1);
half of electricity consumption for crusher			at grid	electricity	supply mix	No	CH	electricity, medium voltage, at grid	3.60E-04	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
half of electricity consumption for crusher			at grid, from Elggis Kraft AG	hydro power	power plants	No	CH	electricity, hydropower, at run-of-river power plant	3.60E-04	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
share of energy consumption for production and administration			includes the total amount of fuel used for heating of mostly administration buildings	oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.50E-03	MJ	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
		Waste heat	Same amount as electricity used	air	low population density			Heat, waste	2.59E-03	MJ		1	1.06	(1,2,1,2,1,1);
		dust to air [kg]		air	low population density			Particulates, > 10 um	8.71E-06	kg	EPA (1998), Lime Manufacturing	1	2.29	(1,5,1,5,4,3);
		dust to air [kg]		air	low population density			Particulates, < 2.5 um	8.71E-07	kg	EPA (1998), Lime Manufacturing	1	2.29	(1,5,1,5,4,3);
		dust to air [kg]		air	low population density			Particulates, > 2.5 um, and < 10um	7.84E-06	kg	EPA (1998), Lime Manufacturing	1	2.29	(1,5,1,5,4,3);
		limestone, as aggregate	used as brash; amount 2001: 63'184 t	construction materials	additives	No	CH	limestone, crushed, washed	3.27E-01	kg	data of KFN for the year 2001	1		(,,,,);
		limestone, to kiln	Total amount 2001 of prepared limestone to both vertical kiln: 129'975 t	construction materials	additives	No	CH	limestone, crushed, washed	6.73E-01	kg	data of KFN for the year 2001	1		(,,,,);

Fig. 6.4 Flows for "limestone crushed, washed" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
limestone, to primary crushing plant	primary crushing of limestone for mill		Total amount 2001 of raw limestone prepared: 143'888 t, losses by dust emission is neglected in the input amount	construction materials	additives	No	CH	limestone, at mine	1.00E+00	kg	data of KFN for the year 2001	1	1.33	(3,3,1,1,3,5);
machines for primary crushing			Infrastructure crusher VBA; total weight 22t, assumed life time: 25 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	6.12E-06	kg	estimation	1	3.09	(4,3,1,1,1,5);
conveyor belt			Infrastructure, assumption: length of conveyor belt = 100m, assumed life time: 25 years	construction processes	machinery	Yes	RER	conveyor belt, at plant	2.78E-08	m	estimation	1	3.09	(4,3,1,1,1,5);
water consumption			total water consumption 2001 for hydration: 11'010 m3; total amount of pumped water 2001: 19'450 m3; Half of rest for this process and other half of rest for crushing of limestone (for kiln)	resource	in water			Water, well, in ground	2.93E-05	m3	data of KFN for the year 2001	1	1.12	(3,2,1,1,1,1);
half of electricity consumption for crusher			at grid	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.55E-04	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
half of electricity consumption for crusher			at grid, from Elggis Kraft AG	hydro power	power plants	No	CH	electricity, hydropower, at run-of-river power plant	2.55E-04	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
share of energy consumption for production and administration			includes the total amount of fuel used for heating of mostly administration buildings	oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.50E-03	MJ	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
		waste heat	Same amount as electricity used	air	low population density			Heat, waste	1.84E-03	MJ		1	1.06	(1,2,1,2,1,1);
		dust to air [kg]		air	low population density			Particulates, > 10 um	8.71E-06	kg	EPA (1998), Lime Manufacturing	1	2.29	(1,5,1,5,4,3);
		dust to air [kg]		air	low population density			Particulates, < 2.5 um	8.71E-07	kg	EPA (1998), Lime Manufacturing	1	2.29	(1,5,1,5,4,3);
		dust to air [kg]		air	low population density			Particulates, > 2.5 um, and < 10um	7.84E-06	kg	EPA (1998), Lime Manufacturing	1	2.29	(1,5,1,5,4,3);
		limestone, as aggregate	used as brash for different products; amount 2001: 123'541 t	construction materials	others	No	CH	limestone, crushed, for mill	8.59E-01	kg	data of KFN for the year 2001	1		(,,,,);
		limestone as aggregate	after second crushing used as special brash (Bergschotter); amount 2001: 14'299 t	construction materials	others	No	CH	limestone, crushed, for mill	9.94E-02	kg	data of KFN for the year 2001	1		(,,,,);
		limestone to mill	Total amount 2001 of prepared limestone to mill: 6'048 t	construction materials	others	No	CH	limestone, crushed, for mill	4.20E-02	kg	data of KFN for the year 2001	1		(,,,,);

Fig. 6.5 Flows for "limestone, crushed for mill" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
limestone, to mill	milling of limestone		the total amount 2001: 6'048 t; dust emission is neglected in the input amount	construction materials	others	No	CH	limestone, crushed, for mill	1.00E+00	kg	data of KFN for the year 2001	1	1.33	(3,3,1,1,3,5);
machines for sieve, storage, roller mill and filter			infrastructure roller mill, filter, sieve and two silos; total weight 34t, assumed life time: 25 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	2.25E-04	kg	estimation	1	3.09	(4,3,1,1,1,5);
half of electricity consumption for roller mill			at grid	electricity	supply mix	No	CH	electricity, medium voltage, at grid	1.60E-02	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
half of electricity consumption for roller mill			at grid, from Elggis Kraft AG	hydro power	power plants	No	CH	electricity, hydropower, at run-of-river power plant	1.60E-02	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
energy for grinding drying			oil consumption 2001 for grinding drying: 2.45 l/t = 0.0000245 m ³ /kg, density: 860 kg/m ³ -> 0.002107 kg/kg, net calorific value: 42.6 MJ/kg -> 0.089758 MJ/kg	oil	heating systems	No	CH	heat, light fuel oil, at industrial furnace 1MW	8.98E-02	MJ	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
share of energy consumption for production and administration			includes the total amount of fuel used for heating of production and administration buildings	oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.50E-03	MJ	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
		waste heat	Same amount as electricity used	air	low population density			Heat, waste	1.15E-01	MJ		1	1.06	(1,2,1,2,1,1);
		milled limestone	nekafill 15 und nekafor 15; total 2001: 6'048t	construction materials	additives	No	CH	limestone, milled, loose, at plant	1.00E+00	kg	data of KFN for the year 2001			

Fig. 6.6 Flows for "limestone, milled, loose, at plant" and their representation in the ecoinvent database

6.5 Quicklime (CaO)

The product from calcination of crushed limestone is “quicklime, in pieces, loose, at plant”. After grinding quicklime the result is “quicklime, milled, loose, at plant”. The ecoinvent database meta information concerning these products include all important and necessary information to use the data correctly. Fig. 6.7 shows all in- and output flows of the product “quicklime, in pieces, loose, at plant”, Fig. 6.8 shows all in- and output flows of the product “quicklime, milled, loose, at plant”. Additionally the allocation to the modules in ecoinvent database, the sources (mainly management ratio of KFN, Anonymous (2001) and estimations) and the standard deviation for each datum are indicated.

Heavy oil for heating of kiln

A comparison of the energy consumption of the kiln in the factory Netstal and the data from (BUWAL (1995)) is provided in Fig. 6.11.

Tab. 6.10 Comparison of the energy consumption for the two heavy-oil burned kilns „Fiedler“ and „Maerz“ at the lime plant Netstal and the data from (BUWAL (1995))

Energy consumption	BUWAL (1995)	KFN Netstal, kilns “Fiedler” and “Maerz”	Place & year
	[MJ/kg]	[MJ/kg]	
fuel mix ⁸	4.7		CH 1990
heavy fuel		3.742	CH 2001

Based on information from the operators of KFN, the reason for its better total efficiency lies in the higher combustion efficiency of its 2 chamber-system with use of the waste heat for preheating.

Emissions to air

The following table shows the air emissions from KFN measured by the “Technische Forschung und Beratung für Zement und Beton” (Anonymous (2001)). The choice of the parameters is based on the clean-air regulations. The total yearly output of emissions are illustrated in Tab. 6.11

The emission profile from the combustion of heavy oil as assessed in Jungbluth (2003) presents more species than reported in Anonymous (2001). However, the list of emitted species assumed here has not been integrated with missing substances, because it is not clear what kind of reactions occur during the considered process (e.g. SO_x emissions react with the feed material to gypsum). Only the CO₂-emissions have been adapted:

The CO₂-emission is caused partly by the calcination of limestone and partly by the fuel combustion.

The comparison with the process emissions described in “Ökoinventare für Energiesysteme” (Frischknecht et al. (1996)), which are based on BUWAL (1995), results in Tab. 6.12:

⁸ Fuel mix: coke: 67%; heavy fuel oil: 21%; light fuel oil: 12% (BUWAL (1995)BUWAL (1995) Emissionsfaktoren für stationäre Quellen; Ausgabe 1995. In: *Vollzug Umwelt*. Bundesamt für Umwelt, Wald und Landschaft (BUWAL), Bern.)

Tab. 6.11 Overview of the emissions for the two heavy-oil burned kilns „Fiedler“ and „Maerz“ and the installation of hydration in the lime plant Netstal (Anonymous (2001))

Parameter	Unit	Kiln „Fiedler“ SOFi	Kiln „Maerz“ SOM	Installation for hydration
Production output	t/h	2.800	8.200	11.000
Production output	%	25	75	100
CO ₂	kg/h	2000.000	8000.000	
	kg/t output	714.286	975.610	
	Total kg/kg output	0.90909		
NO _x (as NO ₂)	kg/h	1.700	2.000	
	kg/t output	0.607	0.244	
	Total kg/kg output	0.00034		
SO ₂	kg/h	0.100	3.600	
	kg/t output	0.036	0.439	
	Total kg/kg output	0.00034		
CO	kg/h	3.400	49.100	
	kg/t output	1.214	5.988	
	Total kg/kg output	0.00477		
VOC	kg/h	0.100	0.400	
	kg/t output	0.036	0.049	
	Total kg/kg output	0.00005		
Particulates	kg/h	0.680	0.070	0.010
	kg/t output	0.243	0.009	0.001
	Total kg/kg output	0.00007		0.000001

Tab. 6.12 Overview of the emissions for the calcination process in general

Process	particulates, > 10µm	CO ₂ -Emissions	Place & year	Source
	[kg/kg]	[kg/kg]		
Calcination (direct)	-	0.37	CH / 1990	BUWAL (1995)
Calcination (from heating-mix)	0.00016	0.42	CH / 1990	BUWAL (1995)
Calcination (total)	0.00016	0.79	CH / 1990	BUWAL (1995)
Calcination KFN	0.000075	1.00	CH / 2001	Anonymous (2001)

The difference between the two amounts of particulate emissions depending on the source is based on the different fuels and different technologies used, whereas the difference of CO₂ emissions depends on the different total efficiencies of the two processes (cf. Tab. 6.10).

The distribution of the dust emissions to the different categories of sizes are made based on the information from (Berdowski et al. (1995)).

Tab. 6.13 Overview of the emissions from the CEPMEIP-database (Berdowski et al. (1995))

	Particles, < 2.5 µm (PM2.5)	PM 10	Particles, > 2.5 µm and < 10µm	Particles, > 10 µm	TSP
As shown in [CEPMEIP Data- base]: production of lime	30 g/t lime	150 g/t lime			300 g/t lime
Distribution over all categories	30 g/t lime		120 g/t lime	150 g/t lime	300 g/t lime
	10 %		40 %	50 %	
Adaptation based on assumption (*)	20 %		40 %	40 %	
Measured data calci- nation KFN total					75 g/t lime
Calcination KFN dis- tribution	15 g/t lime		30 g/t lime	30 g/t lime	

(*): The adaptation is based on the fact that in general the amount of PM2.5 emitted increases when TSP is reduced by filters, which are most effective in retaining the particulates of larger size.

The two kilns used in KFN (SOM and SOFi) have different energy consumptions because of different efficiencies. Tab. 6.14 shows data for each kiln and total for KFN.

Tab. 6.14 Heat and electricity consumption of the two kilns (SOM and SOFi) at KFN for 2001

	Vertical kiln 1 (SOM)		Vertical kiln 2 (SOFi)	
Capacity in t/day	189.4	t/day	50.6	t/day
Share of capacity in %	79	%	21	%
Specific heat consumption (2001)	3'629	kJ/kg	4'180	kJ/kg
Total heat consumption (weighted average)	3'742 kJ/kg			
Electricity consumption (2001) for calcination process	0.0490	kWh/kg	0.0432	kWh/kg
Total electricity consumption (weighted average)	0.0478 kWh/kg			

Equipment for grinding quicklime:

As no exact information is available the weight of the machines has been estimated and summed up:

1 crusher:	25 t
1 roller mill:	25 t
1 plant for dedusting:	10 t
1 cyclone:	10 t
1 „small“ silo:	7 t
Total weight of equipment:	77 t
Length of conveyor belt:	50 m

The service life is assumed to be 25 years. The total amount of material treated by the machines over this period is about 550'000 to 600'000 t (Compare Fig. 5.1).

Tab. 6.15 Ecoinvent database meta information for the product "quicklime, in pieces, loose, at plant"

Name	quicklime, in pieces, loose, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the calcination process. Also included is the electricity consumption for preheating of the heavy fuel oil and one part of the total heating energy for "production" and "administration". Equipment included in the infrastructure: 2 vertical kilns; not included is the use of fireproof bricks since no data are available. Only the measured emissions are included.
Amount	1
Local Name	Kalk, gebrannt, in Stücke, lose, ab Werk
Synonyms	burnt//caustic//quick//lime//Calciumhydroxid//Calcium hydroxide
General Comment to reference function	Infrastructure data are estimated based on a tour and sketches of the process. The value of the infrastructure is normalized with a annual production capacity of about 75'000tons of product per year. The estimated lifespan of the machines is 25 years. Only measured emissions are taken into account. The distribution of the dust emissions to the different categories of sizes are made based on the information of the CEPMEIP-Database and adapted. This Module describes a pre-product for the milling process, and therefore for module "quicklime, milled, loose, at plant".
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are from only one company in Switzerland (KFN), for some exchanges RER-modules have been used as proxy
Technology text	The company KFN works on a technically high level. Two vertical kilns are in use. The fuel for the calcination process is heavy fuel oil.
Representativeness [%]	50
Production Volume	150'000t
Sampling Procedure	Survey
Extrapolations	Data which are only available for RER have been used as proxy for CH
Uncertainty Adjustments	None

Tab. 6.16 Ecoinvent database meta information for the product "quicklime, milled, loose, at plant"

Name	quicklime, milled, loose, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the processes: crushing, milling, filtering (cyclone), dedusting, transportation, and storing. One part of the total heating energy for "production" and "administration" is included. Equipment included in the infrastructure: 1 crusher, 1 roller mill, 1 plant for dedusting, 1 cyclone, and 1 small silo.
Amount	1
Local Name	Kalk, gebrannt, gemahlen, lose, ab Werk
Synonyms	burnt//caustic//quick//lime//Calciumhydroxid//Calcium hydroxide
General Comment to reference function	Infrastructure data are estimated based on a tour and sketches of the process. The value of the infrastructure is normalized with a annual production capacity of about 23'000 tons of product per year. The estimated lifespan of the machines is 25 years. There are no significant dust emissions as a dust control system is installed. This product exists also packed.
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are from only one company in Switzerland (KFN), for some exchanges RER-modules have been used as proxy
Technology text	The company KFN works on a technically high level. Effective dust control systems are installed.
Representativeness [%]	50
Production Volume	150'000t
Sampling Procedure	Survey
Extrapolations	Data which are only available for RER have been used as proxy for CH
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
limestone, to kiln	calcination of limestone (Inputs)		Total amount 2001: 129975 t (two types of kiln called SOM + SOFI)	construction materials	additives	No	CH	limestone, crushed, washed	1.73E+00	kg	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
infrastructure of two kilns			estimation of total weight of the two types of kiln (SOM + SOFI): 30 t, assumed life time: 25 years; amount of Input-material has been taken into account	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	9.23E-06	kg	estimation	1	3.09	(4,3,1,1,1,5);
heavy oil for heating of kiln				oil	fuels	No	CH	heavy fuel oil, at regional storage	9.14E-02	kg	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
half of electricity consumption in connection with calcination from grid			at grid (half of total; total is 0.0478 kWh/kg)	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.39E-02	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
half of electricity consumption in connection with calcination from hydropower			at grid, from Elggis Kraft AG (half of total; total is 0.0478 kWh/kg)	hydro power	power plants	No	CH	electricity, hydropower, at run-of-river power plant	2.39E-02	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
half of electricity consumption for preparation of heavy fuel oil from grid			at grid (half of total; total is 0.046 kWh/kg)	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.30E-02	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
half of electricity consumption for preparation of heavy fuel oil from hydropower			at grid, from Elggis Kraft AG (half of total; total is 0.046 kWh/kg)	hydro power	power plants	No	CH	electricity, hydropower, at run-of-river power plant	2.30E-02	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
share of energy consumption for administration			includes the total amount of fuel used for heating of mostly administration buildings	oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	5.94E-04	MJ	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);

Fig. 6.7 Flows for "quicklime, in pieces, loose, at plant" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
calcination of limestone (Outputs)	→ waste heat	Total heat production (incl. electricity)	air	low population density				Heat, waste	9.38E-02	MJ		1	1.06	(1,2,1,2,1,1);
	→ CO ₂ -emissions	total: geogenic and from burning of heavy fuel oil	air	low population density				Carbon dioxide, fossil	9.09E-01	kg	Anonymous (2001), Emissions-Messung Kalkfabrik Netstal	1	1.21	(1,2,1,1,1,5);
	→ NO _x -emissions		air	low population density				Nitrogen oxides	3.36E-04	kg	Anonymous (2001), Emissions-Messung Kalkfabrik Netstal	1	1.21	(1,2,1,1,1,5);
	→ SO ₂ -emissions		air	low population density				Sulfur dioxide	3.36E-04	kg	Anonymous (2001), Emissions-Messung Kalkfabrik Netstal	1	1.21	(1,2,1,1,1,5);
	→ CO-emissions		air	low population density				Carbon monoxide, fossil	4.77E-03	kg	Anonymous (2001), Emissions-Messung Kalkfabrik Netstal	1	1.21	(1,2,1,1,1,5);
	→ VOC-emissions		air	low population density				NM VOC, non-methane volatile organic compounds, unspecified origin	4.55E-05	kg	Anonymous (2001), Emissions-Messung Kalkfabrik Netstal	1	1.21	(1,2,1,1,1,5);
	→ dust emissions		air	low population density				Particulates, < 2.5 µm	1.36E-05	kg	Anonymous (2001), Emissions-Messung Kalkfabrik Netstal	1	1.21	(1,2,1,1,1,5);
	→ dust emissions		air	low population density				Particulates, > 2.5 µm, and < 10µm	2.73E-05	kg	Anonymous (2001), Emissions-Messung Kalkfabrik Netstal	1	1.21	(1,2,1,1,1,5);
	→ dust emissions		air	low population density				Particulates, > 10 µm	2.73E-05	kg	Anonymous (2001), Emissions-Messung Kalkfabrik Netstal	1	1.22	(1,3,1,1,1,5);
	→ quicklime, in pieces, loose, at plant	amount 2001: 75'126 t	construction materials	additives	No	CH		quicklime, in pieces, loose, at plant	1.00E+00	kg	data of KFN for the year 2001			

Fig. 6.7 Flows for "quicklime, in pieces, loose, at plant" and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
lime burnt, to mill	milling of quicklime		Total amount 2001 of quicklime to mill: 22'744 t, possible dust emissions are neglected (no mass losses)	construction materials	additives	No	CH	quicklime, in pieces, loose, at plant	1.00E+00	kg	data of KFN for the year 2001	1	1.24	(3,2,1,1,1,5);
machines: sieve, storage, roller mill, cyclone and dedusting			Infrastructure total, weight: 62t, life time: 25 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.09E-04	kg	estimation	1	3.09	(4,3,1,1,1,5);
conveyor belt			Infrastructure, estimated length of conveyor belt: 50m, life time: 25 years	construction processes	machinery	Yes	RER	conveyor belt, at plant	8.79E-08	m	estimation	1	3.09	(4,3,1,1,1,5);
half of electricity consumption for roller mill			at grid	electricity	supply mix	No	CH	electricity, medium voltage, at grid	1.35E-02	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
half of electricity consumption for roller mill			at grid, from Elggis Kraft AG	hydro power	power plants	No	CH	electricity, hydropower, at run-of-river power plant	1.35E-02	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
share of energy consumption for production and administration			includes the total amount of fuel used for heating of production and administration buildings	oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.50E-03	MJ	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
		waste heat	Same amount as electricity used	air	low population density			Heat, waste	9.72E-02	MJ		1	1.05	(1,2,1,1,1,1);
		quicklime, milled, loose, at plant	nekafin 2 und nekasol 10: amount 2001: 22'744 t	construction materials	additives	No	CH	quicklime, milled, loose, at plant	1.00E+00	kg	data of KFN for the year 2001			

Fig. 6.8 Flows for "quicklime, milled, loose, at plant" and their representation in the ecoinvent database

6.6 Hydrated Lime (Ca(OH)₂)

The ecoinvent database meta information concerning crushing and hydrating of quicklime include all important and necessary information to use the data correctly. Fig. 6.9 shows all in- and output flows of the product "Lime hydrated, loose, at plant", the allocation to the modules in ecoinvent database, the sources (Landfield & Karra (2000), management ratio of KFN and estimations) and the standard deviation for each datum.

Equipment for the process "lime hydrated, loose, at plant":

The weight of the machines has been estimated and summed up:

1 swing-hammer crusher:	25 t
2 cyclone dust catchers (assumption each 5 t):	10 t
1 lime slaking plant:	10 t
1 silo:	9 t
2 conveyor worms (each 3 t):	6t
Total weight of equipment:	60 t

The service life is assumed to be 25 years. The total amount of material treated by the machines over this period is about 450'000 t (Compare Fig. 5.1).

Tab. 6.17 Ecoinvent database meta information for the product "lime hydrated, loose, at plant"

Name	lime, hydrated, loose, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the processes: slaking, crushing, dust abatement (cyclone), transportation and storing. A part of the total heating energy for "production" and "administration" is included. Equipment included in the infrastructure: 1 swing hammer crusher, 2 cyclone dust catchers, 1 lime slaking plant, 2 conveyor worms, and 1 silo.
Amount	1
Local Name	Kalk, gelöscht, lose, ab Werk
Synonyms	slaked//hydriert//hydrate
General Comment to reference function	Infrastructure data are estimated based on a tour and sketches of the process. The value of the infrastructure is normalized with a annual production capacity of about 20'000 tons of product per year. The estimated lifespan of the machines is 25 years. There are no significant dust emissions as a dust control system is installed. This product exists also packed.
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are from only one company in Switzerland (KFN), for some exchanges RER-modules have been used as proxy
Technology text	The company KFN works on a technically high level. Effective dust control systems are installed.
Representativeness [%]	50
Production Volume	37'000t
Sampling Procedure	Survey
Extrapolations	Data which are only available for RER have been used as proxy for CH
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
quicklime, milled, loose, to hydration	hydration of burnt and milled lime		amount 2001: 14'058 t	construction materials	additives	No	CH	quicklime, in pieces, loose, at plant	7.69E-01	kg	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
machines for slaking, conveyor worm, sieving (two cyclone dust catcher), swing-hammer crusher and storing			Infrastructure machines, estimated weight: 40 t, life time: 25 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	8.76E-05	kg	estimation	1	3.09	(4,3,1,1,1,5);
half of electricity consumption for hydration			at grid, energy consumption: 11 - 12 kWh/t	electricity	supply mix	No	CH	electricity, medium voltage, at grid	6.00E-03	kWh		1	1.05	(1,2,1,1,1,1);
half of electricity consumption for hydration			at grid, from Elggis Kraft AG	hydro power	power plants	No	CH	electricity, hydropower, at run-of-river power plant	6.00E-03	kWh		1	1.05	(1,2,1,1,1,1);
half of electricity consumption for crusher			estimation: electricity consumption of module "crushing rock"	electricity	supply mix	No	CH	electricity, medium voltage, at grid	3.58E-04	kWh	Landfield & Karra (2000)	1	1.05	(1,2,1,1,1,1);
half of electricity consumption for crusher			at grid, from Elggis Kraft AG	hydro power	power plants	No	CH	electricity, hydropower, at run-of-river power plant	3.58E-04	kWh	Landfield & Karra (2000)	1	1.05	(1,2,1,1,1,1);
water consumption			total water consumption 2001 for hydration: 11'010 m3	resource	in water			Water, well, in ground	6.02E-04	m3	data of KFN for the year 2001	1	1.12	(3,2,1,1,1,1);
share of energy consumption for production and administration			includes the total amount of fuel used for heating of production and administration buildings	oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.50E-03	MJ	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
		waste heat	Same amount as electricity used, total waste heat from hydration process is included in calcination process	air	low population density			Heat, waste	4.58E-02	MJ		1	1.05	(1,2,1,1,1,1);
		lime, hydrated, loose, at plant	amount 2001: 18'275 t (nekapur)	construction materials	binder	No	CH	lime, hydrated, loose, at plant	1.00E+00	kg	data of KFN for the year 2001			

Fig. 6.9 Flows for "lime hydrated, loose, at plant" and their representation in the ecoinvent database

6.7 Packing and Loading of Lime Products

This process is used as basis to compose the products' modules: "limestone, milled, packed, at plant", "quicklime, milled, packed, at plant" and "lime, hydrated, packed, at plant". The ecoinvent database meta information concerning the basis process "packing, lime products" and the according products include all important and necessary information to use the data correctly. Fig. 6.10 shows all in- and output flows of the process "packing, lime products", the allocation to the modules in ecoinvent database, the sources (management ratio of KFN, personal communication or communication by e-mail and estimations) and the standard deviation for each datum. Fig. 6.11 to Fig. 6.13 show the implementation of the packing module on all different lime products.

Tab. 6.18 Ecoinvent database meta information for the process "packing, lime products"

Name	packing, lime products
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the process of packing, transportation, and loading of the bags on pallets. The machines, the waste treatment after use by the end-user of the lime products and the transports of the packing materials are included.
Amount	1
Local Name	Verpacken, Kalkprodukte
Synonyms	
General comment to reference function	The value of the packing and loading modules are normalized with a annual production capacity of about 50'000 tons/year. The estimated lifespan of the machines is 25 years.
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are only from one company in Switzerland (KFN), for some exchanges RER-modules have been used as proxy
Technology text	The company KFN works on a technically high level. The packing and loading process is fully automated.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Survey
Extrapolations	Data which are only available for RER have been used as proxy for CH
Uncertainty Adjustments	None

Tab. 6.19 Ecoinvent database meta information for the product "limestone, milled, packed, at plant"

Name	limestone, milled, packed, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the packing and one part of the total heating energy for "production" and "administration"
Amount	1
Local Name	Kalkstein, gemahlen, verpackt, ab Werk
Synonyms	
General comment to reference function	Packed lime products do all use the module "packing, lime products"
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are from only one company in Switzerland (KFN)
Technology text	The company KFN works on a technically high level.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Survey
Extrapolations	None
Uncertainty Adjustments	None

Tab. 6.20 Ecoinvent database meta information for the product "quicklime, milled, packed, at plant"

Name	quicklime, milled, packed, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the packing and one part of the total heating energy for "production" and "administration"
Amount	1
Local Name	Kalk, gebrannt, gemahlen, verpackt, ab Werk
Synonyms	burnt//caustic//quick//lime//Calciumhydroxid//Calcium hydroxide
General comment to reference function	Packed lime products do all use the module "packing, lime products"
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are from only one company in Switzerland (KFN)
Technology text	The company KFN works on a technically high level.
Representativeness [%]	50
Production Volume	150'000t
Sampling Procedure	Survey
Extrapolations	None

Uncertainty Adjustments	None
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Tab. 6.21 Ecoinvent database meta information for the product "lime, hydrated, packed, at plant"

Name	lime, hydrated, packed, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the packing and one part of the total heating energy for "production" and "administration"
Amount	1
Local Name	Kalk, gelöscht, verpackt, ab Werk
Synonyms	slaked//hydriert//hydrate
General comment to reference function	Packed lime products do all use the module "packing, lime products"
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are from only one company in Switzerland (KFN)
Technology text	The company KFN works on a technically high level.
Representativeness [%]	50
Production Volume	37'000t
Sampling Procedure	Survey
Extrapolations	None
Uncertainty Adjustments	None

Fig. 6.11 shows all in- and output flows of the process "limestone, milled, packed, at plant", the allocation to the modules in ecoinvent database, the sources and the standard deviation for each datum.

Fig. 6.12 shows all in- and output flows of the process "quicklime, milled, packed, at plant", the allocation to the modules in ecoinvent database, the sources and the standard deviation for each datum.

Fig. 6.13 shows all in- and output flows of the process "Lime, hydrated, packed, at plant", the allocation to the modules in ecoinvent database, the sources and the standard deviation for each datum.

General Flow information				Representation in ecoinvent							Uncertainty						
Input		Process Name		Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
infrastructure packing and loading of lime products	→	packing and loading of lime products (including energy consumption)			estimated total weight of machines: 30t, estimated lifespan: 25years; total amount of packed material (2001): 47067 t/a	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	2.55E-05	kg	estimation based on visiting tour	1	3.09	(4,3,1,1,1,5);	
conveyor belt	→				assumed length of conveyor belt: 10m, life time: assumed 25years	construction processes	machinery	Yes	RER	conveyor belt, at plant	8.50E-09	m	estimation based on visiting tour	1	3.09	(4,3,1,1,1,5);	
half of electricity consumption for packing and loading	→				at grid	electricity	supply mix	No	CH	electricity, medium voltage, at grid	1.55E-03	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);	
half of electricity consumption for packing and loading	→				at grid, from Elggis Kraft AG	hydro power	power plants	No	CH	electricity, hydropower, at run-of-river power plant	1.55E-03	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);	
pallets, at plant	→				assumed average density (over all lime products): 800kg/m3, assumed loading volume: 1m2 area, 1m loading hight: 1m3, total weight of load per pallet: 0.8 t, weight of palett:22 kg, number of applications: 10	wooden materials	processing	No	RER	EUR-flat pallet	1.25E-04	unit	Information from company "Frantschach" by Email at 05.06.02	1	1.44	(4,5,1,1,3,5);	
paperbag, at plant	→					paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	4.25E-03	kg	Information from company "Frantschach" by Email at 05.06.02	1	1.24	(1,4,1,2,1,5);	
plastic foil, at plant	→				circumference of palett: 4m; height: 1m, 2 layer of plastic with a weight of 0.0123 kg/m2, loading weight 800kg	plastics	processing	No	RER	packaging film, LDPE, at plant	1.23E-04	kg	http://www.dm-folien.de/hpage.htm, access Oct. 2003	1	1.38	(4,3,1,1,3,5);	
transports of pallets, paper bags and plastic bags to plant by train	→				standard distance: transport by train with distance of: 200km,	transport systems	train	No	CH	transport, freight, rail	5.25E-03	tkm	standard distance	1	2.33	(5,3,1,1,3,5);	
transports of pallets, paper bags and plastic bags to plant by lorry	→				standard distance: transport by lorry 28t with distance of: 50km,	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	1.31E-03	tkm	standard distance	1	2.33	(5,3,1,1,3,5);	
transports of paper bags and plastic bags to municipal incineration	→				assumption: transports all by lorry 16t and total of transportdistance: 10km	transport systems	road	No	RER	transport, lorry 3.5-16t, fleet average	4.37E-05	tkm	data of KFN for the year 2001	1	1.62	(5,3,1,1,3,5);	
	→		waste heat		Same amount as electricity used	air	low population density				Heat, waste	1.11E-02	MJ		1	1.05	(1,2,1,1,1,1);
	→		pallets, to recycling			waste management	municipal incineration	No	CH		disposal, wood untreated, 20% water, to municipal incineration	3.66E-03	kg		1	1.24	(1,4,1,2,1,5);
	→		paperbag, in municipal waste incineration			waste management	municipal incineration	No	CH		disposal, paper, 11.2% water, to municipal incineration	4.25E-03	kg	data of KFN for the year 2001	1	1.24	(1,4,1,2,1,5);
	→		plastic foil, in municipal waste incineration			waste management	municipal incineration	No	CH		disposal, polyethylene, 0.4% water, to municipal incineration	1.23E-04	kg	data of KFN for the year 2001	1	1.38	(4,3,1,1,3,5);
	→		packing and loading of 1 kg lime product			construction materials	others	No	CH		packing, lime products	1.00E+00	kg				

Fig. 6.10 Flows for "packing, lime products" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
limestone, milled, loose, at plant	packing and loading of milled and loose limestone			construction materials	additives	No	CH	limestone, milled, loose, at plant	1.00E+00	kg	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
packing and loading of 1 kg lime product				construction materials	others	No	CH	packing, lime products	1.00E+00	kg		1	3.09	(4,3,1,1,1,5);
share of energy consumption for production and administration			includes the total amount of fuel used for heating of production and administration buildings	oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.50E-03	MJ		1	1.05	(1,2,1,1,1,1);
		limestone, milled, packed, at plant	products called "nekafill 15" and "nekafor 15"	construction materials	others	No	CH	limestone, milled, packed, at plant	1.00E+00	kg	data of KFN for the year 2001			

Fig. 6.11 Flows for "limestone, milled, packed, at plant" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
quicklime, milled, loose, at plant	packing and loading of milled, loose quicklime			construction materials	additives	No	CH	quicklime, milled, loose, at plant	1.00E+00	kg	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
packing and loading of 1 kg lime product				construction materials	others	No	CH	packing, lime products	1.00E+00	kg		1	3.09	(4,3,1,1,1,5);
share of energy consumption for production and administration			includes the total amount of fuel used for heating of production and administration buildings	oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.50E-03	MJ		1	1.05	(1,2,1,1,1,1);
		milled and packed quicklime	packed products called "nekafin 2" und "mekasol 10"	construction materials	additives	No	CH	quicklime, milled, packed, at plant	1.00E+00	kg	data of KFN for the year 2001			

Fig. 6.12 Flows for "quicklime, milled, packed, at plant" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
lime, hydrated, loose, at plant	packing and loading of hydrated and loose lime			construction materials	binder	No	CH	lime, hydrated, loose, at plant	1.00E+00	kg	data of KFN for the year 2001	1	1.11	(1,4,1,1,1,1);	
packing and loading of 1 kg lime product				construction materials	others	No	CH	packing, lime products	1.00E+00	kg		1	3.28	(5,4,1,1,1,5);	
share of energy consumption for production and administration			includes the total amount of fuel used for heating of production and administration buildings	oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.50E-03	MJ		1	1.05	(1,2,1,1,1,1);	
		hydrated and packed lime	packed nekafin 2 und nekasol 10	construction materials	binder	No	CH	lime, hydrated, packed, at plant	1.00E+00	kg	data of KFN for the year 2001				

Fig. 6.13 Flows for "lime, hydrated, packed, at plant" and their representation in the ecoinvent database

7 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed information compare the different flow charts and the ecoinvent database meta information.

Most information was made available by the only Swiss company (KFN) still producing quicklime. The data from this company refer to the year 2001. The land-use is based on a report (Berger & Durner (1992)) specifically made for KFN to secure the mining of further raw material reserves. The data are based on the status in 1992 and describe the mining stages for about 65 years.

All facilities have been roughly estimated based on sketches describing the process steps, on visual impressions and/or on air-picture.

The quicklime production from KFN covers only about half of the Swiss market. The rest is imported from neighbour countries. The fuel used in KFN for calcining is heavy fuel oil. The other companies supplying the Swiss market use different fuel types. This fact is not reflected in the inventory, which focuses on KFN. The representativeness has been adjusted and therefore the uncertainty has increased.

From the above, - the data for the different lime products can be considered of good overall quality.

8 Abbreviations

EPAL	European Pallet Association
KFN	Lime factory Netstal Ltd. (Kalkfabrik Netstal AG)
SOM	Regenerative vertical kiln (Doppel-Schachtofen) called „Maerz“
SOFi	Vertical kiln (Schachtofen) called „Fiedler“
TSP	Total Suspended Particulates

9 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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Part VIII

Gypsum Products and Processes

Data v2.0 (2007)

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Summary

This part deals with the mining and production of gypsum and anhydrite. The term “gypsum” is often used for quite different materials with the main chemical composition of CaSO_4 but with different amounts of crystal water.

The products inventoried are the natural gypsum and the natural anhydrite at mine, stucco (burned gypsum) and burned anhydrite at plant and a mix of natural and burned anhydrite.

A considerable amount of gypsum used in Europe stems from the desulphurisation of flue gas. The desulphurisation of the flue gas in ecoinvent is allocated to 100% to the energy conversion process. Thus flue gas gypsum would carry no burdens from upstream processes. Since flue gas gypsum is not produced nor consumed in relevant quantities in Switzerland, it is not included in the inventories presented here.

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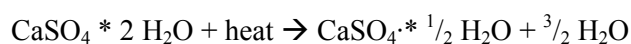
1 Introduction

Ullmann's encyclopaedia of industrial chemistry (Wirsching (1997)), describes shortly and pointedly the basis of the gypsum technology:

“Both gypsum (3397-24-5) $\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}$ (0101-41-4) and anhydrite (4798-04-0) CaSO_4 (778-18-9) are widely distributed in the earth's crust. Only in volcanic regions are gypsum and anhydrite rock completely absent. In addition, large quantities of gypsum and anhydrite are obtained as a by-product of industrial processes when flue gases are desulphurized or calcium salts are reacted with sulphuric acid. In these cases calcium sulphate is usually obtained as a moist, fine powder.

Gypsum is useful as an industrial material because it readily loses its water of hydration when heated (dehydration) producing partially or totally dehydrated calcined gypsum, and when water is added to this calcined gypsum (rehydration), it reverts to the original dihydrate – the set and hardened gypsum product. These two phenomena, dehydration and rehydration, are the basis of gypsum technology:

Dehydration



Rehydration



The apparent density of the hydrated calcium sulphate is much lower, 0.5 – 1.5 t/m³ than the density of the calcium sulphate rock, 2.3 t/m³.

Gypsum and anhydrite are non-toxic.”

Fig. 1.1 gives an overview of the different gypsum products and their connections. Anhydrite plaster is finely ground natural anhydrite rock mixed with activators that is used as raw material for hard finish plaster, cast and for the cement industry. Calcined products are either “plaster of Paris” and “stucco” or “overburnt plaster”. “Plaster of Paris” and “stucco” are used either directly in the building industry or as basis for prefabricated components or they are mixed with “overburnt plaster” to get a “multiphase plaster” which is used to finish interior walls and ceilings.

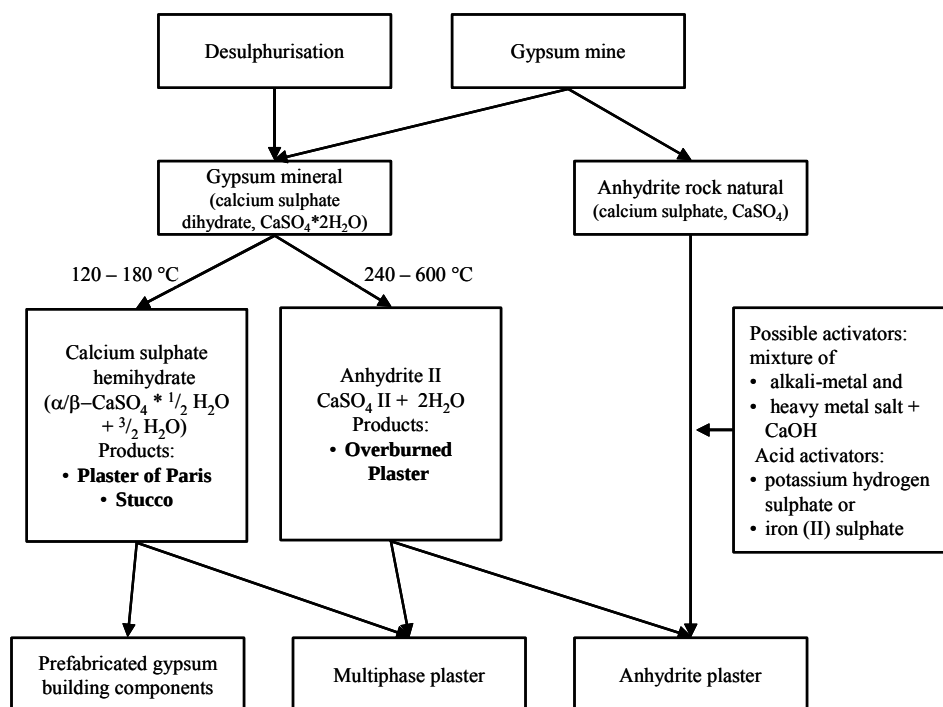


Fig. 1.1 Overview of gypsum products and processes

2 Reserves and Resources of Materials

2.1 Natural Gypsum and Anhydrite

A general description of the natural occurrence of gypsum and anhydrite rock in Europe is found in Wirsching (1997):

“Gypsum and anhydrite deposits are found in many countries. They originated from supersaturated aqueous solutions in shallow seas, which evaporated and deposited first carbonates, then sulphates, and finally chlorides, i.e., in order of increasing solubility.

Gypsum and anhydrite are almost always found in the Permian, Triassic, and Tertiary formations. The Permian includes the Upper Permian (Zechstein deposits, Germany). The Triassic consists of Lower (Buntsandstein), Middle (Muschelkalk), and Upper Triassic (Keuper). The Tertiary deposits include the Eocene (Spain, Persia, Near East), the Oligocene (Paris), and the Miocene deposits (Mediterranean area, North Africa). The Jurassic and Cretaceous formations, which lie between the Triassic and Tertiary, are almost devoid of useful gypsum deposits, as are the older pre-Permian formations – at least in Europe – as, for example, the Carboniferous formation.

The currently accepted view is that gypsum, $\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}$, was formed upon initial crystallization. The best known deposits of primary gypsum are those of the Paris Basin and the areas around the Mediterranean. Under increased pressure and temperature, anhydrite could have developed locally from the original gypsum. However, uptake of surface water can convert the anhydrite back into gypsum. This gypsum is therefore a secondary rock, especially in formations older than the Middle Tertiary.

The various gypsum and anhydrite deposits differ in purity, structure, and colour. The major impurities are calcium carbonate (limestone), dolomite, marl, clay; less frequently, silica, bitumen,

glauberite, syngenite, and polyhalite. If these are present in the gypsum and anhydrite from its formation, they are called primary impurities. A distinction is drawn between carbonate-sulphate deposits, mainly found in the Upper Permian (Zechstein), and the clay-sulphate deposits, which predominate in the Triassic. Secondary impurities are formed during exposure to materials flushed into cracks and leached cavities, but may also be introduced into the rock as waste material during mining. Gypsum is readily soluble in water, even if it is only slightly soluble, and therefore gypsum rock is leached by surface water.

Workable gypsum and anhydrite rock are structurally quite distinct from one another. The most important types of native gypsum are sparry gypsum, also called gypsum spar or flaky gypsum (selenite), fibrous gypsum, alabaster (grainy gypsum), gypsum rock (common gypsum, also massive gypsum), porphyritic gypsum, earthy gypsum (gypsite in the United States), and gypsum sand (contaminated with Glauber's salt $\text{Na}_2\text{SO}_4 \cdot 10 \text{H}_2\text{O}$). Anhydrite, always crystalline, can either be sparry (anhydrite spar), coarse to close grained, or even rod shaped.

Gypsum can be pure white. If it contains iron oxide, it is reddish to yellowish. If clay and/or bitumen is present, it is grey to black. Very pure anhydrite is bluish white, but usually it is gray with a bluish tinge. The white veins sometimes found on the boundary between gypsum and anhydrite consist largely of glauberite ($\text{CaSO}_4 \cdot \text{Na}_2\text{SO}_4$) or Glauber's salt ($\text{Na}_2\text{SO}_4 \cdot 10 \text{H}_2\text{O}$).'' (Wirsching (1997))

The quarryable¹ gypsum resources in Switzerland amount to about 10 million tons. At the present consumption level the resources will last for another 30 years (Coutalides (1998)). As in the highly populated area of the Swiss mid-land all different kind of land utilisation (e.g. agriculture, forestry, water protection areas and settlements) compete, it is not expected that new quarryable resources will be available.

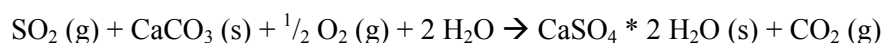
For larger countries in Europe the same is true but less dramatically.

2.2 Flue-Gas Gypsum

This chapter is mainly based on Wirsching (1997).

Flue-gas gypsum is a by-product of energy conversion processes that is only produced because the legal constraints force the companies to abate sulphur emissions in order to prevent health and environmental damage. Flue-gas gypsum is obtained from the desulphurisation with limestone of combustion gases of fossil fuels, such as anthracite, bituminous coal, lignite, and oil, in large combustion plants, especially power stations.

The net reaction of the flue-gas gypsum formation is:



The large compact gypsum crystals are separated from the aqueous gypsum suspension in hydrocyclones and vacuum drum filters or centrifuges. The product is a moist, fine, fairly pure powder with free water content below 10 %.

About 5.4 tons of gypsum are produced per tonne of sulphur in the fuel.

In 1983 Japan produced about 2.5 million tons per year of flue-gas gypsum while the Federal Republic of Germany and the United States produced about 0.3 million tons per year each. Since then, the figures for Germany and the United States have increased considerably because of increasing flue gas

¹ quarryable in the sense of technically, economically and socially feasible (according to http://www.sgtk.ethz.ch/100jahresgtk/meilensteine/abbaubare_rohstoffe.html, accessed 28.05.2003)

desulphurisation. Today an estimated 50% of the total gypsum consumption in Germany are estimated to stem from sulphur abatement.

Characteristic of flue-gas gypsum is its high percentage of calcium sulphate and its almost pure white colour. The fineness and the crystal structure of the gypsum are important factors that determine its usefulness in the gypsum industry. Apart from small quantities used in cement making, all flue-gas gypsum currently produced is consumed by the gypsum industry.

No flue-gas gypsum is used in Swiss gypsum production. The gypsum plaster boards that are imported from Germany contain maximum 1% flue-gas gypsum (Coutalides (1998)).

3 Characterisation of product/material

3.1 General

The $\text{CaSO}_4 \cdot \text{H}_2\text{O}$ system is characterized by five solid phases: Calcium sulphate dihydrate, calcium sulphate hemihydrate, anhydrite III, and anhydrite II exist at room temperature; the fifth phase, anhydrite I, exists only at temperatures above 1180 °C (see Tab. 3.1).

Calcium sulphate dihydrate ($\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}$, CAS 13397-24-5) is both the raw material before dehydration and the final product after rehydration. Calcium sulphate hemihydrate ($\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$, CAS 10034-76-1) occurs in two different forms, α and β , representing two limiting states. They differ from each other in their application characteristics, heat of hydration, and method of preparation. The α -hemihydrate consists of compact, well-formed, transparent, large primary particles. The β -hemihydrate forms flaky, rugged secondary particles made up of extremely small crystals. For this study only β -hemihydrate is relevant.

Tab. 3.1 Phases and physical properties in the $\text{CaSO}_4\text{-H}_2\text{O}$ system (Wirsching (1997))

Characteristic	Gypsum mineral (Calcium sulphate dihydrate)	Plaster of Paris (β -calcium sulphate hemihydrate)	Overburned plaster (Anhydrite II)
Formula	$\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}$	$\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$	CaSO_4
Molecular mass	172.17	145.15	136.14
Thermodynamic stability (°C)	< 40	Metastable in air saturated with water vapour	40 - 1180
Synthesis conditions: Temperature, °C, and atmosphere	< 40	45 – 200 in dry air	200 - 1180
Production temperature	< 40	120 - 180	300 - 900
Water of crystallisation (weight-%)	20.92	6.21	0
Density, g/cm ³	2.31	2.619 – 2.637	2.93 – 2.97
Hardness, Mohs	1.5		3 - 4
Solubility in water at 20°C, g per 100g solution	0.21	0.88	(0.27)
Lattice symmetry	Monoclinic	Rhombohedral	Rhombic

3.2 Gypsum boards

There are two different types of gypsum board that are very similar in properties and use: gypsum plaster board and gypsum fibre board.

Gypsum plaster board is a sandwich construction with a gypsum core and a cardboard layer on each side. The weight of the cardboard is about 5% of the board. Gypsum fibre board consists of gypsum with 15-20% cellulose fibres.

Both types of gypsum board have a density of 1000 – 1250 kg/m³ and a thermal conductivity of 0.36 W/mK. Gypsum boards have good sound-absorbing properties and because of the constitutional water they show a good fire resistance and consequently often are used where these properties are required.

4 Use / Application of Material

The use of calcium sulphate and its hydrates is enormously multifarious in industry and technique. Some examples are given in Kündig et al. (1997)):

- Dihydrate ($\text{CaCO}_4 \cdot 2\text{H}_2\text{O}$)
Used as raw material in the gypsum industry, in the cement industry for regulation of the solidification and in the paper industry as filling material
- Hemihydrate ($\text{CaCO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$)
Used as binding agent (plaster) in the building industry, as raw material for the gypsum board industry, for moulds in the ceramic industry and for gypsum bandage or for moulds in surgery and dental surgery.
- Anhydrite II, burned ($\text{CaSO}_4 \text{ II}$)
Used for cast plaster floors and artificial marble in the building industry or as filling material in the paper industry.
- Anhydrite II, mineral (natural anhydrite; $\text{CaSO}_4 \text{ II}$)
Used for cast plaster floors and plaster in the building industry and as raw material in the cement industry.
- Anhydrite III, ($\text{CaSO}_4 \text{ III}$)
Used as desiccant in the chemical industry.

Some typical gypsum products are (Kündig et al. (1997)):

- Gypsum plaster
Gypsum as binding agent is the most used base material for interior finishing.
- Solid gypsum board
The structural members are produced semi-automatically from stucco. Only additive are glass fibres to raise the strength and resistance to fire.
- Gypsum plaster board
Gypsum plaster boards are made of a gypsum core with varying thickness, which is wrapped in a special cardboard. The main application is in lightweight construction. This product is not fabricated in Switzerland (import mainly from Germany).

Gypsum boards are used for non-load bearing walls, ceilings, fire retarding constructions and so on.

5 System Characterisation

5.1 Process description

5.1.1 Mining and crushing

In Switzerland all natural gypsum comes from open pit mines. There are mines in Zeglingen, Ennetmoos, Leissingen, Bex and Granges (Coutalides (1998)). The gypsum layers are about 30 – 40 m thick. The output of the mine is crushed and sorted. The intermediate products are composed of about 65% natural gypsum ($\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}$), 34% natural anhydrite rock (CaSO_4) and 1% soil and gravel that are deposited back in the mine.² The material is gained by blasting and digging.

A part of the output is sold to the cement industry.

5.1.2 Calcining and milling

The natural gypsum is calcinated (outside Switzerland possibly together with flue-gas gypsum) in different types of kilns for different times at different temperatures depending on the desired product (hemihydrate or anhydrite II). Tab. 5.1 shows some production data for different products made in Switzerland.

Tab. 5.1 Examples of production data for different gypsum products (Coutalides (1998))

Kiln type		rotary kiln	kettle calciner	conveyor kiln
Product		β -hemihydrate	β -hemihydrate	anhydrite II
Output capacity	t/d	600	760	1200
Theoretical heat consumption	MJ/t	599	584	777
Effective heat consumption	MJ/t	945	808	1118
Thermal efficiency	%	63	72	69
Free water content of raw material	% (w/w)	1.5	1.5	1.6
Purity of raw material	% (w/w)	90	90	94
Constitutional water in raw material	% (w/w)	18.8	18.8	19.7
Constitutional water in product	% (w/w)	3.0	5.5	1.0

The products are finely ground after calcination and stored in silos

5.2 Gypsum in ecoinvent

5.2.1 Gypsum

Fig. 5.1 gives an overview of the gypsum products as they are modelled in this study. Tab. 5.2 shows what the different modules represent and for what products they can be used:

² personal communication Mr. Buchheim, RiGips AG, 23.01.2002

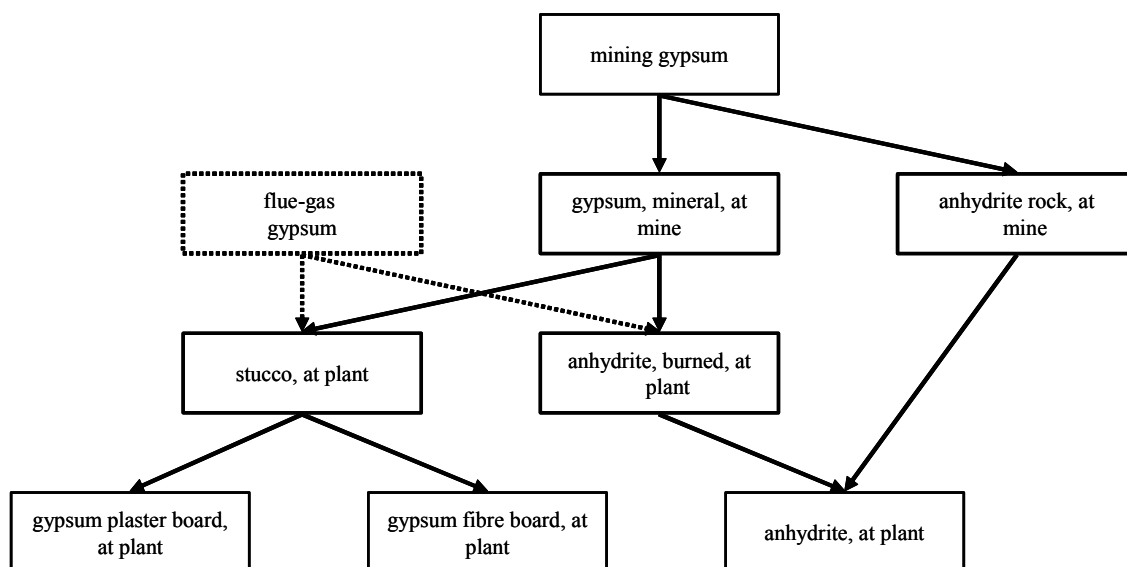


Fig. 5.1 Unit processes of gypsum production.

Since “flue-gas gypsum” is a by-product of other processes that is not used in Switzerland, it is not included in the database.

Tab. 5.2 Module names for gypsum products and processes in ecoinvent and how they can be applied

Module name	representation	use	remarks
mining gypsum	Mining process	Mining gypsum describes a process that yields two products. Only the products of mining can be used as inputs to other processes.	
gypsum, mineral, at mine	natural $\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}$, milled	Cement industry; Filler in paper industry; Raw material for (burned) anhydrite and stucco.	first output of “mining gypsum”
anhydrite rock, at mine	natural CaSO_4 , milled	Cement industry; has to be finely ground for use as cast plaster floor.	second output of “mining gypsum”
anhydrite, burned, at plant	natural CaSO_4 , calcinated from gypsum, milled	Cement industry; cast plaster floor.	
anhydrite, at plant	mix of anhydrite rock and anhydrite burned (consumer mix)	Cement industry; cast plaster floor.	
stucco, at plant	natural $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$, calcinated from gypsum, milled	plaster; moulds; raw material for gypsum boards.	

The land use of the mines is calculated with the average thickness of the gypsum layers (30 m) and the density of the products (2300 kg/m^3) assuming that the open part of the mine is recultivated 10 years after it was opened. The mining activities are partly calculated and estimated based on the information on limestone mining (see Part II of this report). The gypsum production is assumed to produce the same dust emissions as the limestone mining. The infrastructure for the mining, burning and milling (per kg product) is assumed to be the same as for the respective operations of limestone winning.

No flue-gas gypsum is used for the stucco and burned anhydrite production in Switzerland. However in other European countries the share of flue-gas gypsum is high. The highest shares are found in Germany (estimated 50%) and in the UK. Anyway, the imports of gypsum products from Germany

contain maximum 1% flue-gas gypsum (Coutalides (1998)). Thus for the Swiss consumption no flue-gas gypsum is inventoried.

The mean of the energy consumption values for the rotary kiln and the kettle calciner in Tab. 5.1 are taken for production of stucco and the value for the conveyor kiln is used for production of anhydrite. Since no measured emission data are available the "heat, at..." modules are used to account for the energy consumption including the emissions. A mix of 80% natural gas and 20% heavy fuel oil is assumed. The dust emissions from grinding are estimated (together with dust from mining) whereas the water use and other emission from the calcining process are neglected.

5.2.2 Gypsum board

Gypsum plaster boards are produced continuously. The gypsum is mixed with water and additives and evenly spread on the lower cardboard. The upper cardboard is then applied and the edge profile is formed. When the gypsum is set, the surplus water is removed by drying for about one hour in a kiln. Finally the boards are cut and packed.

Gypsum fibre boards are produced in a similar way as gypsum plaster boards. The gypsum is mixed with water, fibres and additives and evenly spread on the belt of a calender. The board is pressed and when the gypsum is set, the edges are cut and the surplus water is removed by drying for about one hour in a kiln. Finally the boards are cut and packed.

Both types of gypsum board use starch and small amounts of glass fibres, siloxanes and surfactants as additives.

Most LCI information is taken from Coutalides (1998). Since the demand of gypsum is not given, it was calculated from the yearly production, the yearly demand of ancillaries and the stoichiometric demand of water ($1.5 \text{ mol H}_2\text{O} / \text{mol stucco}$) for the setting. Since the source indicates only the water and energy consumption for the production of solid gypsum board, these values are used to calculate the values for the other types of gypsum board assuming a linear relation of the water consumption and of the energy needed for drying to the amount of gypsum input.

The amount of fibres in the gypsum fibre board is taken from Starzner & Wurmer-Weiss (2000).

Gypsum plaster board for the Swiss market is almost exclusively produced in Germany (Coutalides (1998)). Thus the German electricity mix is used and the transports to Switzerland are included. It is assumed that the situation for gypsum fibre board is similar.

5.2.3 Data

The ecoinvent meta information is shown in Tab. 5.3. The In- and Output flows and their representation in the ecoinvent database are shown in following figures:

- mining, gypsum with the products:
 - gypsum, mineral, at mine Fig. 5.2
 - anhydrite rock, at mine Fig. 5.3
- stucco, at plant Fig. 5.4
- anhydrite, burned, at plant Fig. 5.5
- anhydrite, at plant Fig. 5.6
- gypsum plaster board, at plant Fig. 5.7
- gypsum fibre board, at plant

5.3 Data quality

The uncertainties have been estimated with the help of the pedigree matrix (Frischknecht et al. (2007)). For detailed information compare the flows and the ecoinvent meta information.

The mining data are based on expert's estimates and on measurements in a limestone mine. The energy consumption for calcining is measured. However, the representation of the directly fired calcining furnaces by the module "heat at industrial furnace..." is rather crude. The overall data quality is fair.

Tab. 5.3 Ecoinvent meta information for gypsum production

Name	mining, gypsum	stucco, at plant
Location	CH	CH
Infrastructure Process	0	0
Unit	kg	kg
Data Set Version	2.0	2.0
Included Processes	mining and crushing of gypsum and anhydrite	burning of gypsum mineral and milling of stucco
Amount	1	1
Local Name	Abbau, Gips	Stuckgips, ab Werk
Synonyms	Gipsstein//anhydrite//calcium sulphate dihydrate//Kalziumsulfat dihydrat	gips//gypsum//plaster of Paris//calcium sulphate hemihydrate//Kalziumsulfat halbhydrat
General Comment to reference function	The multi-output-process 'mining, gypsum' delivers the co-products 'gypsum, mineral, at mine' and 'anhydrite rock, at mine'. Allocation is done by weight. In some mines anhydrite is not exploited but left in the mine. Since the electricity consumption has a relevant share on the total energy consumption, the choice of the energy mix is important. For gypsum mined in countries with high fossil share in the electricity mix this module has to be adopted accordingly.	used for plaster, moulds, and gypsum boards; the dust emission is included in the estimated emission inventoried in the "mining Gypsum" process
Start Date	1997	1997
End Date	2003	2003
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	data for Switzerland	data for Switzerland
Technology text	open pit mining; composition of products: 65% gypsum, 34% anhydrite, 1% others.	Assumed average technology for Switzerland
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	environmental report, literature	environmental report, personal communication
Extrapolations	none	none
Uncertainty Adjustments	none	none

Tab. 5.3 Ecoinvent meta information for gypsum production (cont.)

Name	anhydrite, burned, at plant	anhydrite, at plant	gypsum plaster board, at plant	gypsum fibre board, at plant
Location	CH	CH	CH	CH
Infrastructure Process	0	0	0	0
Unit	kg	kg	kg	kg
Data Set Version	2.0	2.0	2.0	2.0
Included Processes	burning of gypsum mineral and milling of anhydrite	mix of burned and natural anhydrite; milling of natural anhydrite	production of board (incl. drying)	production of board (incl. drying)
Amount	1	1	1	1
Local Name	Anhydrit, gebrannt, ab Werk	Anhydrit, ab Werk	Gipskartonplatte, ab Werk	Gipsfaserplatte, ab Werk
Synonyms	calcium sulphate//Kalziumsulfat	calcium sulphate//Kalziumsulfat		
General Comment to reference function	used in cement industry and cast plaster floor; the dust emission is included in the estimated emission inventoried in the "mining Gypsum" process	used in cement industry and cast plaster floor	produced 100% from natural gypsum (no FGD). Energy consumption is extrapolated from production of solid gypsum board	produced 100% from natural gypsum (no FGD). Energy consumption is extrapolated from production of solid gypsum board
Start Date	1997	1997	1997	1997
End Date	2003	2003	2003	2003
Data Valid For Entire Period	1	1	1	1
Other Period Text				
Geography text	data for Switzerland	data for Switzerland	data for Switzerland	data for Switzerland
Technology text	Assumed average technology for Switzerland	Mix of products	Average technology for consumption in Switzerland	Average technology for consumption in Switzerland
Representativeness [%]				
Production Volume	unknown	unknown	unknown	unknown
Sampling Procedure	environmental report, personal communication	environmental report, personal communication	environmental report	environmental report
Extrapolations	none	none	none	none
Uncertainty Adjustments	none	none	none	none

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
Gypsum (resource)	mining, gypsum, part 1			resource	in ground			Gypsum, in ground	1.00E+00	kg	definition	1	1.00	definition
land occupation, mine [m2a]			assumption: After 10 years the used land is recultivated	resource	land			Occupation, mineral extraction site	1.85E-04	m2a	calculated	1	1.63	(4,4,1,1,1,5,7)
land transformation mine [m2]			thickness of deposit: 30 m, density 1800 kg/m3	resource	land			Transformation, to mineral extraction site	1.85E-05	m2	calculated	1	2.11	(4,4,1,1,1,5,8)
land transformation (from unknown to mine)				resource	land			Transformation, from unknown	1.85E-05	m2	calculated	1	2.11	(4,4,1,1,1,5,8)
recultivation				construction materials	additives	No	CH	recultivation, limestone mine	1.85E-05	m2	calculated	1	2.11	(4,4,1,1,1,5,8)
blasting [kg explosive]			includes explosive and emissions from blasting	construction processes	civil engineering	No	RER	blasting	7.73E-05	kg	Anonymous 2001	1	1.32	(1,4,1,1,3,5,4)
total diesel used in machines				construction processes	machinery	No	GLO	diesel, burned in building machine	1.80E-02	MJ	Anonymous 2001	1	2.11	(1,4,1,1,3,5,5)
energy consumption for crusher and conveyer belt			0.000716 kWh/kg for crushing, 0.0002 kWh/kg for conveyer belt	electricity	supply mix	No	CH	electricity, medium voltage, at grid	9.16E-04	kWh	Anonymous 2001	1	1.34	(3,4,1,1,3,5,2)
machines for primary crushing			Weight of crushers 44 t, life time 25 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	4.63E-06	kg	estimation, based on Anonymous 2001	1	3.12	(3,4,1,1,3,5,9)

Fig. 5.2 Flows for “mining, gypsum” and their representation in the ecoinvent database

Part VIII: Gypsum Products and Processes

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
conveyor belt	mining, gypsum, part 2		length of conveyor belt = 100m; life time 25 years; annual production = 380'000 t	construction processes	machinery	Yes	RER	conveyor belt, at plant	1.05E-08	m	estimation	1	3.33	(5,4,1,1,3,5,9)
Infrastructure (mine) [unite]			limestone mine as proxy: annual output 380'000 t, life time 50 years	construction materials	additives	Yes	CH	mine, limestone	5.26E-11	unit		1	3.12	(3,4,1,1,3,5,9)
		Waste heat	Same amount as electricity used	air	low population density			Heat, waste	3.30E-03	MJ		1	1.34	(3,4,1,1,3,5,13)
		Dust to air [kg]	includes emissions from mining and further processing	air	low population density			Particulates, < 2.5 um	8.00E-06	kg	estimation, based on limestone mining	1	3.28	(1,5,2,3,4,3,20)
		Dust to air [kg]	includes emissions from mining and further processing	air	low population density			Particulates, > 2.5 um, and < 10um	4.00E-05	kg	estimation, based on limestone mining	1	2.28	(1,5,2,3,4,3,19)
		Dust to air [kg]	includes emissions from mining and further processing	air	low population density			Particulates, > 10 um	1.12E-04	kg	estimation, based on limestone mining	1	1.83	(1,5,2,3,4,3,18)
		overburden	used for recultivation --> not inventoried						1.00E-02	kg				
		gypsum	Total production	construction materials	others	No	CH	gypsum, mineral, at mine	6.50E-01	kg				
		anhydrite rock		construction materials	others	No	CH	anhydrite rock, at mine	3.40E-01	kg				

Fig. 5.2 Flows for “mining, gypsum” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
gypsum mineral	stucco, at plant			construction materials	others	No	CH	gypsum, mineral, at mine	1.16E+00	kg	calculation based on Coutalides 1998	1	1.31	(4,5,2,1,1,3,3)
heat from natural gas		80% of total heat		natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace >100kW	7.01E-01	MJ	calculation based on Coutalides 1998	1	1.09	(2,1,2,1,1,3,1)
heat from heavy fuel oil		20% of total heat		oil	heating systems	No	CH	heat, heavy fuel oil, at industrial furnace 1MW	1.75E-01	MJ	calculation based on Coutalides 1998	1	1.09	(2,1,2,1,1,3,1)
electricity		for milling		electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.70E-02	kWh	Anonymous 2001	1	1.34	(3,4,1,1,3,5,2)
infrastructure milling		Assumed weight 230 t at 75'000 t annual production, life time 25 years.		construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.23E-04	kg	estimation, based on Anonymous 2001	1	3.12	(3,4,1,1,3,5,9)
infrastructure furnace		Assumed weight 30 t at 75'000 t annual production, life time 25 years.		construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.60E-05	kg	estimation, based on Anonymous 2001	1	3.12	(3,4,1,1,3,5,9)
		Waste heat	Same amount as electricity used	air	low population density			Heat, waste	9.72E-02	MJ	calculated	1	1.34	(3,4,1,1,3,5,13)
		Water (vapour to air)	not inventoried						1.61E-01	kg				
		Dust to air	is included in estimation of dust from mining											
		stucco		construction materials	binder	No	CH	stucco, at plant	1.00E+00	kg				

Fig. 5.3 Flows for “stucco, at plant” and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
gypsum mineral	anhydrite, burned, at plant			construction materials	others	No	CH	gypsum, mineral, at mine	1.20E+00	kg	calculation based on Coutalides 1998	1	1.31	(4,5,2,1,1,3,3)
heat from natural gas				natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace >100kW	8.94E-01	MJ	calculation based on Coutalides 1998	1	1.09	(2,1,2,1,1,3,1)
heat from heavy fuel oil				oil	heating systems	No	CH	heat, heavy fuel oil, at industrial furnace 1MW	2.24E-01	MJ	calculation based on Coutalides 1998	1	1.09	(2,1,2,1,1,3,1)
electricity			for milling	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.70E-02	kWh	Anonymous 2001	1	1.34	(3,4,1,1,3,5,2)
infrastructure milling			Assumed weight 230 t at 75'000 t annual production, life time 25 years.	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.23E-04	kg	estimation, based on Anonymous 2001	1	3.12	(3,4,1,1,3,5,9)
infrastructure furnace			Assumed weight 30 t at 75'000 t annual production, life time 25 years.	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.60E-05	kg	estimation, based on Anonymous 2001	1	3.12	(3,4,1,1,3,5,9)
		Waste heat	Same amount as electricity used	air	low population density			Heat, waste	9.72E-02	MJ	calculated	1	1.34	(3,4,1,1,3,5,13)
		Water (vapour to air)	not inventoried						2.03E-01	kg				
		Dust to air	is included in estimation of dust from mining											
		anhydrite		construction materials	binder	No	CH	anhydrite, burned, at plant	1.00E+00	kg				

Fig. 5.4 Flows for “anhydrite, burned, at plant” and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
burned anhydrite	→ anhydrite, at plant		10% of input assumed	construction materials	binder	No	CH	anhydrite, burned, at plant	1.00E-01	kg	estimation, based on personal communication	1	1.32	(4,4,1,1,1,5,3)
anhydrite rock			90% of input assumed	construction materials	others	No	CH	anhydrite rock, at mine	9.00E-01	kg	estimation, based on personal communication	1	1.32	(4,4,1,1,1,5,3)
energy for milling of anhydrite rock			Burned anhydrite is already milled --> only for anhydrite rock	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.43E-02	kWh	estimation, based on Anonymous 2001	1	1.34	(3,4,1,1,3,5,2)
infrastructure for milling anhydrite rock			Assumed weight 230 t at 75'000 t annual production, life time 25 years.	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.10E-04	kg	estimation, based on Anonymous 2001	1	3.12	(3,4,1,1,3,5,9)
		→ Waste heat	Same amount as electricity used	air	low population density			Heat, waste	8.75E-02	MJ	calculated	1	1.06	(1,2,1,2,1,1,2)
		→ Dust to air	is included in estimation of dust from mining											
		→ anhydrite, mix		construction materials	binder	No	CH	anhydrite, at plant	1.00E+00	kg				

Fig. 5.5 Flows for “anhydrite, at plant” and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
stucco	gypsum plaster board, at plant		calculation from other inputs incl. the part of water bound in the product.	construction materials	binder	No	CH	stucco, at plant	8.11E-01	kg	calculation based on Coutalides 1998	1	1.09	(2,1,2,1,1,3,3)
cardboard				paper & cardboard	cardboard & corrugated board	No	RER	whiteline chipboard, WLC, at plant	4.84E-02	kg	calculation based on Coutalides 1998	1	1.09	(2,1,2,1,1,3,3)
water			1/3 ground, 2/3 tap water. Original publication gives only the water consumption for solid gypsum board production (without cardboard). Assumption made that input correlates to stucco input.	resource	in water			Water, unspecified natural origin	1.82E-04	m3	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,4)
water				water supply	production	No	RER	tap water, at user	3.64E-01	kg	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,4)
glass fibres				glass	construction	No	RER	glass fibre, at plant	1.61E-04	kg	calculation based on Coutalides 1998	1	1.09	(2,1,2,1,1,3,3)
siloxanes / stearates			for hydrophobing	chemicals	inorganics	No	RER	silicone product, at plant	1.29E-04	kg	calculation based on Coutalides 1998	1	1.09	(2,1,2,1,1,3,3)
surfactants				washing agents	tensides	No	RER	alkylbenzene sulfonate, linear, petrochemical, at plant	9.68E-06	kg	calculation based on Coutalides 1998	1	1.09	(2,1,2,1,1,3,3)
starch				agricultural production	plant production	No	DE	potato starch, at plant	2.90E-03	kg	calculation based on Coutalides 1998	1	1.09	(2,1,2,1,1,3,3)
proteins			not inventoried						9.68E-06		calculation based on Coutalides 1998	1	1.09	(2,1,2,1,1,3,3)
transport				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	3.00E-01	tkm	calculation based on Coutalides 1998	1	2.01	(2,1,2,1,1,3,5)
electricity (process)			Original publication gives only the power consumption for solid gypsum board production (without cardboard). Assumption made that input is the same for gypsum plaster board.	electricity	supply mix	No	DE	electricity, medium voltage, at grid	3.62E-02	kWh	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,2)
electricity (drying)			Original publication gives only the energy consumption for solid gypsum board production (without cardboard). Assumption made that input correlates to stucco input.	electricity	supply mix	No	DE	electricity, medium voltage, at grid	5.75E-02	kWh	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,2)
Heat (drying)				oil	heating systems	No	RER	light fuel oil, burned in industrial furnace 1MW, non-modulating	1.36E+00	MJ	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,1)
infrastructure			half the infrastructure for wooden board production assumed	wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	1.67E-11	unit	estimation	1	4.25	(5,5,5,5,5,9)
		→ Waste heat	Same amount as electricity used	air	low population density			Heat, waste	3.37E-01	MJ	calculation based on Coutalides 1998	1	1.09	(2,1,2,1,1,3,13)
		→ Dust	No data available. Since gypsum dust is biologically resorbable, no relevant impacts are expected --> neglected											
		→ Production waste	Recycled to the process. Original publication gives only the amount of waste water and it's use for solid gypsum board production (without cardboard). Assumption made that the situation is the same for gypsum plaster board.						2.50E-02	kg	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,6)
		→ gypsum plaster board		construction materials	coverings	No	CH	gypsum plaster board, at plant	1.00E+00	kg				

Fig. 5.6 Flows for “gypsum plaster board, at plant” and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
stucco	→ gypsum fibre board, at plant		calculation from other inputs incl. the part of water bound in the product.	construction materials	binder	No	CH	stucco, at plant	7.10E-01	kg	calculation based on Coutalides 1998 and Starzner & Wurmer-Weiss (2000)	1	1.09	(2,1,2,1,1,3,3)
waste paper fibres				paper & cardboard	pulps	No	RER	waste paper, sorted, for further treatment	1.50E-01	kg	Starzner & Wurmer-Weiss (2000)	1	1.09	(2,1,2,1,1,3,3)
water			1/3 ground, 2/3 tap water. Original publication gives only the water consumption for solid gypsum board production (without fibres). Assumption made that input correlates to stucco input.	resource	in water			Water, unspecified natural origin	1.59E-04	m3	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,4)
water				water supply	production	No	RER	tap water, at user	3.18E-01	kg	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,4)
siloxanes / stearates			for hydrophobing	chemicals	inorganics	No	RER	silicone product, at plant	1.29E-04	kg	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,3)
surfactants				washing agents	tensides	No	RER	alkylbenzene sulfonate, linear, petrochemical, at plant	9.68E-06	kg	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,3)
starch				agricultural production	plant production	No	DE	potato starch, at plant	2.90E-03	kg	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,3)
proteins			not inventoried						9.68E-06		calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,3)
transport				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	3.00E-01	tkm	calculation based on Coutalides 1998	1	2.06	(2,1,2,1,3,3,5)
electricity (process)			Original publication gives only the power consumption for solid gypsum board production (without fibres). Assumption made that input is the same for gypsum fibre board.	electricity	supply mix	No	DE	electricity, medium voltage, at grid	3.62E-02	kWh	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,2)
electricity (drying)			Original publication gives only the energy consumption for solid gypsum board production (without fibres). Assumption made that input correlates to stucco input.	electricity	supply mix	No	DE	electricity, medium voltage, at grid	5.03E-02	kWh	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,2)
Heat (drying)				oil	heating systems	No	RER	light fuel oil, burned in industrial furnace 1MW, non-modulating	1.19E+00	MJ	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,1)
infrastructure			half the infrastructure for wooden board production assumed	wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	1.67E-11	unit	estimation	1	4.25	(5,5,5,5,5,9)
		→ Waste heat	Same amount as electricity used	air	low population density			Heat, waste	3.11E-01	MJ	calculation based on Coutalides 1998	1	1.09	(2,1,2,1,1,3,13)
		→ Dust	No data available. Since gypsum dust is biologically resorbable, no relevant impacts are expected --> neglected											
		→ Production waste	Recycled to the process. Original publication gives only the amount of waste water and it's use for solid gypsum board production (without cardboard). Assumption made that the situation is the same for gypsum plaster board.						2.50E-02	kg	calculation based on Coutalides 1998	1	1.23	(2,1,2,1,3,3,6)
		→ gypsum plaster board		construction materials	coverings	No	CH	gypsum fibre board, at plant	1.00E+00	kg				

Fig. 5.7 Flows for “gypsum fibre board, at plant” and their representation in the ecoinvent database

6 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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Part IX

Fibre Cement Products and Processes

Data v2.0 (2007)

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Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

This part deals with three different types of fibre cement products and its packing process. The products are roof slates, corrugated slabs and facing tiles.

The production of fibre cement products mainly requires the raw materials fibre, cement, limestone meal and water.

The main part of the energy consumption in the production process for all three products is electrical energy. The cumulated energy demand is dominated by the energy for the cement production. Also the emissions mainly stem from the cement production process.

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1 Introduction

Fibre cement is a composite material made out of cement and fibres for reinforcement (Starzner & Wurmer-Weiss (2000)). In former times asbestos was used. Since 1991 asbestos containing materials are not allowed to be produced, sold and used anymore in Switzerland. Therefore today only cement products reinforced with synthetic fibres are on the market.

The products described in this report are “fibre cement roof slate”, “fibre cement corrugated slab” and “fibre cement facing tile”. These are the most used fibre cement products in the building industry. Other products are tubes, plant vessels, and pre-forms.

Fibre cement products are waterproof, frost resistant and humidity regulative. Generally facing tiles and roof slates are coated on the exposed side. Uncoated fibre cement slabs have only a very small market share and are not recommended for application on facades.

2 Reserves and Resources

Swiss Portland cement is proportionally the most important raw material. It is mainly made of clinker which uses mainly limestone and marly clay as raw materials. Its manufacture is described in the corresponding report (Part 2, Cement Products and Processes).

3 Use

The most important uses of fibre cement products in the building sector are (Starzner & Wurmer-Weiss (2000)):

- Facing tile:
 - cladding for external walls ventilated at rear and infills
 - siding for pillars
 - framework constructions, bottom of roof, parapet cladding, etc.
- Roof slates: Roof covering tiles instead of clay roof tiles
- Corrugated slab: large-size fibre cement slabs with wave profile for roof covering

4 Characterisation

Some technical specifications for the studied products are taken from the SIA building materials declaration¹ and the ecological building material information system called ECOBIS (Starzner & Wurmer-Weiss (2000)), „ökologisches Baustoffinformationssystem“, which includes environmental- and health-relevant information concerning the different building material categories for all life cycle stages (raw material, production and processing, usage and after treatment) from “cradle to grave”. It is an information system for people from the building sector and should support the choice of building products based on an ecological view. Tab. 4.1 shows some of the offered information.

¹ <http://www.sia.ch/cf/oekoinfoliste.cfm?ID=2943702&Lang=d&Mfg=72&Grp=0&Produkt=>, access date: Mai 2003

Tab. 4.1 Technical specification of the studied products (SIA)¹, Starzner & Wurmer-Weiss (2000) and Weibel & Stritz (1995)

	Fibre cement roof slates	Fibre cement corrugated slab	Fibre cement facing tile
Bulk density ρ [kg/m ³]	1800	1700	1900
Thermal conductivity λ [W/mK]	about 0.5	about 0.5	about 0.5
Weight per area [kg/m ²]	23.7 – 24.6 (used: 24.2)	17.8 – 20.1 (used: 19)	15.1 – 15.2 (used: 15.2)
Thickness [cm]	1.34	1.12	0.80
Maximum size [mm]	600 * 300	No Information	2'500/2'800/3'100 * 1'250/1'500

5 System Characterisation

Fibre cement products are composed of the materials shown in Tab. 5.1. The mixture of the materials before hardening is called mush.

Tab. 5.1 Composition of the studied products ²

[%] (w/w)	Fibre cement roof slates	Fibre cement corrugated slab	Fibre cement facing tile
Process fibres: new pulp fibres	1.1	1.8	1
Process fibres: recycled pulp fibres	1.7	1.8	1.5
Additive: limestone meal	13.9	11.1	3.7
Binding agent: portland cement	69.8	72.4	78.6
Reinforcing fibres: polyvinyl alcohol	2.6	1.8	3.7
Water: constitutional water	10.9	11.1	11.5
Total	100	100	100

A more detailed description from the ecological building material information system is reported in the following.

“The additives (in fibre cement) are used as filling material and to improve the material binding. All types of fibres are added to guarantee for high enough tensile strength and need chemical resistance to the alkaline environment of the cement. It is also important to use fibres which are not absorbable by the lung and therefore not carcinogenic. Such fibres are e.g. polyvinyl alcohol fibres which have a diameter between 3 μm and 12 to 14 μm , polyacrylnitrile fibres with a diameter between 3 μm and 14 to 18 μm , and polyethylene fibres with a length of 200 to 400 μm and a diameter between 6 μm and 3 μm . For corrugated slabs also milled fibre cement products (recycling material) can be used.

The cement particles must adhere to the fibres within the mush. For fibres with very smooth surfaces, adhesion is poor compared to the asbestos fibres with their fine and frayed structure. For this reason pulp fibres are added. They promote adhesion between the fibres and the cement and help to avoid the flush of cement particles with the surplus water when exuded.

Water as raw material is needed to hydrate the cement and assure the workability of the mush.

² <http://www.sia.ch/cf/oekoinfofoliste.cfm?ID=2943702&Lang=d&Mfg=72&Grp=0&Produkt=>, access date: Mai 2003

Air within the hardened fibre cement slab gives the freezing water expansion space thus helping to avoid destruction by frost.

For weathering and UV-radiation protection a colouring surface coating on the visible side is usually added. To adjust the diffusion resistance to the resistance of the front side, the backside is sealed.

One producer states that in most cases the the visible side is hot-coated using all-acrylate coating. It is based on inorganic colouring pigments (solvents: 2 to 4%). Some solvent and heavy metal emissions from this coating process are taken into account.

The colourless **backside sealing** is composed of 36% polyvinylidene chloride butyl acrylate, 60% water and 4% paraffin wax. For this study such sealing has not be taken into account as no data are available and the amount is very small.” (Starzner & Wurmer-Weiss (2000), translation by the author)

Fig. 5.1 shows the production process of fibre cement slabs.

The following more detailed explanation is taken from Starzner & Wurmer-Weiss (2000). The production occurs with a generally automated winding mode. The raw materials are mixed with water to obtain a homogeneous paste. Rotating sieving cylinders dip in this fibre cement mush, which drain to the inside. Thus the sieve surface gets covered with a layer of fleece of about 1 mm which is transported with a conveyor belt. Suction nozzles on the conveyor belt (fleece belt) dewater the layer.

The produced endless mat is wound up on a format-cylinder until the demanded thickness is reached. With this process the fibres of the neighbour layers are interconnecting.

After cutting up the mat cylinder, the mat is detached from the format-cylinder and by punching and pressing automatically formed to small format slates or plain and corrugated slabs. From the still soft raw slabs also ventilating tubes and special shapes are formed.

After some hours the strength of the products is high enough for them to be taken out of the form. Followed by a natural post-curing process in a store room or a vapour curing process, usually a surface coating process finishes the production process.

5.1 Raw Materials

The amounts of raw materials used for the different fibre cement products are given in the SIA building materials declaration³ and refer only to the company ”Eternit AG” with branches in Niederurnen and Payerne in Switzerland. The data in the declaration describe the composition of the different products, without taking into account the cutting waste. As the waste is internally reused it is not taken into account.

³ <http://www.sia.ch/cf/oekoinfoliste.cfm?ID=2943702&Lang=d&Mfg=72&Grp=0&Produkt=>, access date: Mai 2003

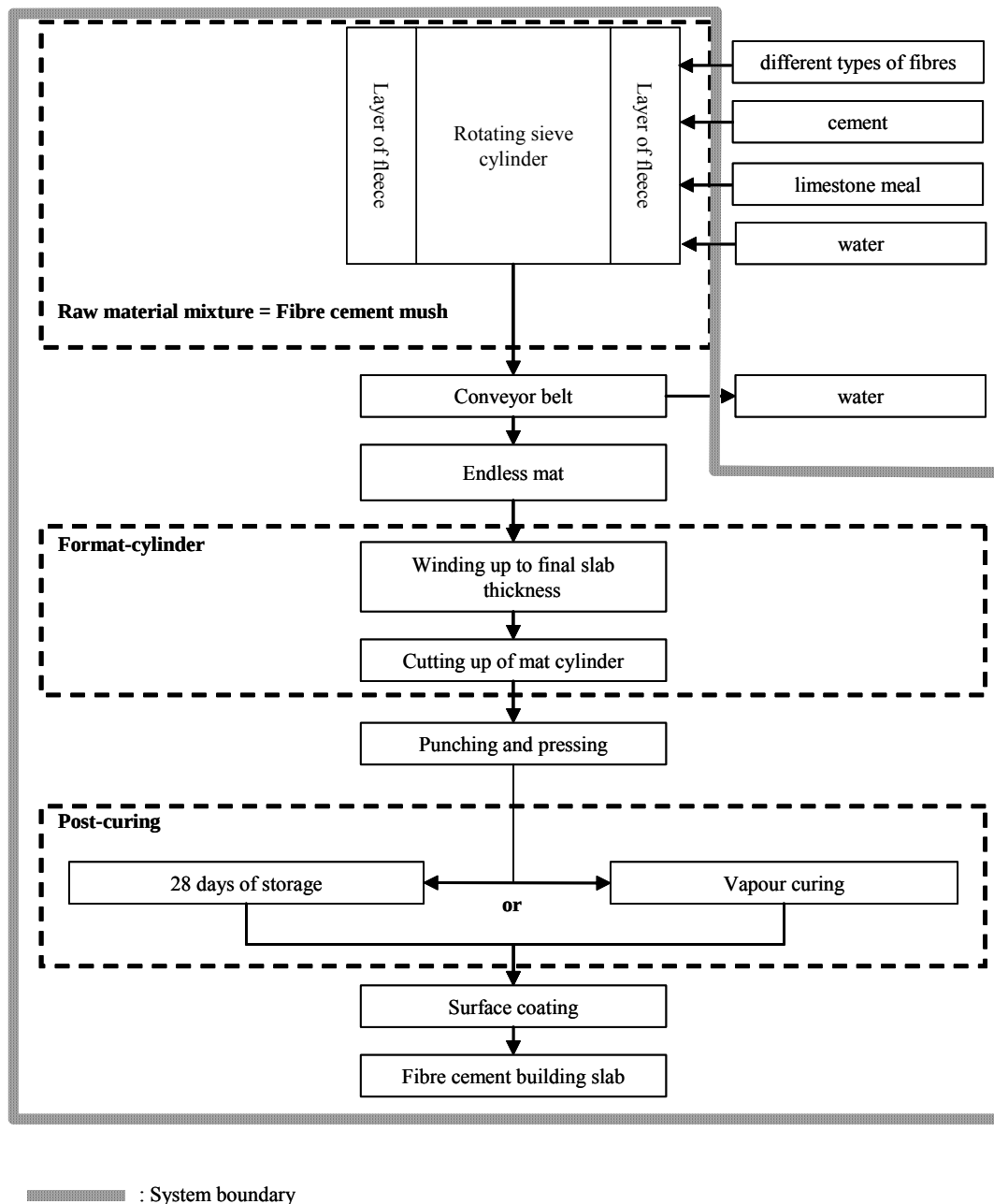


Fig. 5.1 Production process for fibre cement building slabs (Starzner & Wurmer-Weiss (2000))

5.2 Energy Consumption

The energy requirements listed in the SIA building materials declaration refer to the total amount energy consumed (including the upstream processes) for the three products fibre cement roof slates, fibre cement corrugated slabs and fibre cement facing tiles. Kasser & Pöll (1998) state that about 50 to 70% of the “grey energy” (more or less the CED) derives from the upstream processes. As no exact amount is available the worst case for the production process (without upstream processes) of fibre cement products is assumed here. A rough check with the CED for the clinker production shows that direct energy consumption even could be higher. As no better data are available for this study this means that 50% of the energy consumption indicated in the SIA building materials declaration is allocated to the upstream processes and 50% to the production process.

The total amount of energy used for the production process is based on the description in the previous section. The share of the different energy carriers concerning these two products (fibre cement roof slate and fibre cement corrugated slab) are based on (Weibel & Stritz (1995)). As no information concerning the shares of the energy inputs for “fibre cement facing tile” is available, the average of the two other products is used.

The thermal energy used for forming and drying fibre cement products at Eternit AG in Switzerland in the year 1991 is produced in three different furnaces with different sizes and using different fuels (Weibel & Stritz (1995)).

Tab. 5.2 shows the share of electric and thermal energy consumption.

Tab. 5.2 Share of electrical and thermal energy consumption (based on (Weibel & Stritz (1995))

	fibre cement roof slates [%]	fibre cement corrugated slab [%]	fibre cement facing tile [%]
CED of electricity consumption	82.6	91.4	87
Light fuel oil in furnace 1500 kW	7.1	8.1	7.6
Light fuel oil in furnace 400 kW	0.4	0.4	0.4
Propane gas in furnace 230 kW	9.8	0	4.9
Total	100	100	100

An important part of the energy (10%) is used for the neutralisation of the alkaline wastewater with the combustion gases (Weibel & Stritz (1995)).

The heating of the building and the energy consumption of administrative related actions are not included.

5.3 Transports

The transport distances are taken from (Weibel & Stritz (1995)) and presented in Tab. 5.3.

Tab. 5.3 Transport distances (Weibel & Stritz (1995))

Transport distances [km]					
	Additive	Reinforcing fibre	Polyvinyl alcohol	Pulp fibre	Cement
Transport lorry 28t	4				
Transport by freight train	132	860	36	400	90
Transport freight ship		14000			

These distances are based on one Swiss company and the year 1994. It is assumed that they have not changed significantly and that they are approximately valid for all three fibre cement products, included in this study.

5.4 Air Emissions

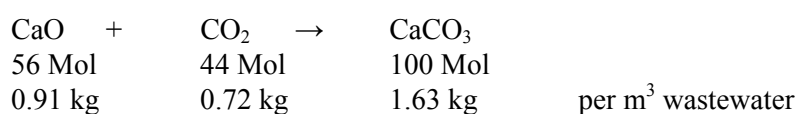
Usually the particulates arising from the production process are collected in filters and reused in the process (Starzner & Wurmer-Weiss (2000)). Due to a lack of data no dust emissions are taken into account in this study.

Most other emissions to air stem from the combustion of fuel. These emissions are taken into account in the corresponding modules.

5.5 Wastewater

The wastewater from the production process is usually very alkaline because of a high amount of calcium oxide (quicklime) from the cement, and has to be neutralised before being released to the public sewage system (Weibel & Stritz (1995)). The alkaline wastewater with pH 13.5 is corrected to pH 8.5. To neutralise the wastewater, an unspecified amount of acid is added, then CO₂ emissions from combustion are scrubbed through the wastewater. Since no data concerning the type and amount of acid is available it is not included in the inventory.

The CO₂ binds with the calcium oxide to give lime (CaCO₃) as shown in the following stoichiometric equation:



The CO₂ emissions from combustion included in the upstream processes must be corrected to take into account CO₂ binding. The amount of CO₂ bound in the wastewater is 0.72 kg/m³ (Weibel & Stritz (1995)) and is included as a negative emission to compensate the emission inventoried in the energy production module. The amount is very small because the amount of wastewater is very small (0.068 kg/kg fibre cement roof slate).

The produced amount of CaCO₃ is included in the output module “disposal of decarbonising waste”.

The total amount of solvent emissions indicated in the SIA building materials declaration⁴ is assumed to be 90% NMVOC and 10% halogenated hydrocarbons.

5.6 Land-use

No data for the fibre cement plant of Eternit AG in Switzerland are available. Therefore the module “rockwool plant” is used as proxy.

6 Life Cycle Inventories

Ecoinvent database meta information (tables), flows and their representation in the ecoinvent database are shown in (figures):

Fibre cement roof slate, at plant	Tab. 6.1 and Fig. 6.1
Fibre cement corrugated slab, at plant	Tab. 6.2 and Fig. 6.2
Fibre cement facing tile, at plant	Tab. 6.3 and Fig. 6.3
Packing fibre cement products	Tab. 6.4 and Fig. 6.4

⁴ <http://www.sia.ch/cf/oekoinfoliste.cfm?ID=2943702&Lang=d&Mfg=72&Grp=0&Produkt=>, access date: Mai 2003

Tab. 6.1 Ecoinvent meta information for the product "fibre cement roof slate, at plant"

Name	fibre cement roof slate, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce fibre cement products (raw material provision, raw material mixing, forming, post curing process coating process, and storage), transports to plant processes, and infrastructure. No acid for neutralisation of the wastewater and no requirements for administration are included.
Amount	1
Local Name	Faserzementdachschiefer, ab Werk
Synonyms	tile
General Comment to reference function	The amounts of raw materials are calculated based on the composition of fibre cement roof slate. Air emission data are lacking but emissions from the boilers are imported via the relevant modules. A certain amount of CO ₂ produced by the boiler is bound in the waste water. The amount of this entrained CO ₂ is inventoried with a negative value because it is an avoided emission. The total yearly is assumed to be 20'000 tonnes. The lifespan of the plant and the conveyor belt are assumed to be 50 years and 20 years, respectively.
Start Date	1991
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from one company in Switzerland (Eternit AG), for some exchanges RER- and OCE-modules have been used as proxy
Technology text	The raw materials are mixed with water to a homogeneous paste. Rotating sieving cylinders dip in this fibre cement mush which drain to the inside. Therefore the sieve surface gets covered with a layer of fleece of about 1mm which is transported with a conveyor belt. The produced endless mat is winded up on a format-cylinder until the demanded thickness is reached. With that process the fibres of the neighbour layers are interconnecting. After cutting up the mat cylinder the mat is detached from the format-cylinder and by punching and pressing automatically formed to small format slates, even and corrugated slabs.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Literature and "ecological building material declaration" (a sort of environmental report)
Extrapolations	See geography
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
process fibres: pulp fibres	Production of "fibre cement roof slate" (Part 1)		raw material, losses are neglected as internally reused.	paper & cardboard	pulps	No	RER	sulphate pulp, unbleached, at plant	1.10E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
process fibres: waste paper			raw material, losses are neglected as internally reused.	paper & cardboard	pulps	No	CH	waste paper, mixed, from public collection, for further treatment	1.70E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
additive: limestone meal			raw material, losses are neglected as internally reused.	construction materials	others	No	CH	limestone, milled, packed, at plant	1.39E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
fixing agent: cement			raw material, losses are neglected as internally reused.	construction materials	binder	No	CH	portland calcaereous cement, at plant	6.98E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
reinforcing fibre: polyvinyl alcohol			raw material, losses are neglected as internally reused.	plastics	monomers	No	RER	vinyl acetate, at plant	2.60E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
water: constitutional water			raw material, losses are neglected as internally reused.	water supply	production	No	RER	tap water, at user	1.09E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
total electricity consumption for complete process				electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.69E-01	kWh	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
energy consumption: light fuel oil in industrial furnace 1MW				oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	2.42E-01	MJ	SIA building product declaration	1	1.12	
energy consumption: light fuel oil in industrial furnace 100kW				oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.47E-02	MJ	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
energy consumption: gas in industrial furnace				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	3.33E-01	MJ	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
tap water			raw material	water supply	production	No	RER	tap water, at user	3.20E+00	kg	Weibel & Stritz (1995)	1	1.12	(2,1,3,1,1,1);
acid for neutralisation of waste water			no data available;								Weibel & Stritz (1995)			
dispersive acrylate			used for coating	paintings	production	No	RER	acrylic varnish, 87.5% in H2O, at plant	1.77E-02	kg	Weibel & Stritz (1995)	1	1.12	(2,1,3,1,1,1);
packing				construction materials	others	No	CH	packing, fibre cement products	1.00E+00	kg	estimation	1	1.40	(3,5,1,1,3,5);
infrastructure plant			assumption: same infrastructure for 1kg fibre cement product is inquired as for 1kg rock wool product	insulation materials	production	Yes	CH	rock wool plant	4.44E-10	unit	estimation	1	3.50	(5,1,1,1,4,5);
conveyor belt			estimation of length: 100m, estimation of lifespan 20years. Yearly production amount: 20'000t	construction processes	machinery	Yes	RER	conveyor belt, at plant	2.50E-07	m	estimation	1	3.45	(5,1,1,1,4,1);
transportation of limestone meal by lorry				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	5.56E-04	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of limestone meal by train				transport systems	train	No	CH	transport, freight, rail	1.83E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);

Fig. 6.1 Flows for "fibre cement roof slate, at plant" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
transportation of polyvinyl alcohol by train	Production of "fibre cement roof slate" (Part 2)			transport systems	train	No	CH	transport, freight, rail	9.36E-04	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);	
transportation of reinforcing fibre by train				transport systems	train	No	RER	transport, freight, rail	2.24E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);	
transportation of pulp fibres by train				transport systems	train	No	RER	transport, freight, rail	4.40E-03	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);	
transportation of cement by train				transport systems	train	No	CH	transport, freight, rail	6.28E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);	
transportation of reinforcing fibres by freight ship				transport systems	ship	No	OCE	transport, transoceanic freight ship	3.64E-01	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);	
lubricant oil				chemicals	organics	No	RER	lubricating oil, at plant	1.10E-04	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);	
		carbon dioxide reacted in waste water	Calcium oxide in waste water reacts with carbon dioxide from exhaust gas to calcium carbonate (limestone). It must be subtracted (that's why negative) from CO2 emissions from combustion which are included in the modules for boilers. 0.72 kg of CO2 per m3 waste water. Amount of waste water: 0.00154 m3/kg	air	unspecified			Carbon dioxide, fossil	-1.10E-03	kg	calculation	1	1.38	(4,5,1,1,1,5);	
		calcium carbonate to disposal	Calcium carbonate from reaction in waste water (compare above)	waste management	residual material landfill	No	CH	disposal, decarbonising waste, 30% water, to residual material landfill	2.50E-03	kg	calculation	1	1.22	(2,1,1,1,1,5);	
		waste water internally clarified by precipitating calcium carbonate	unpolluted as clarified	waste management	wastewater treatment	No	CH	treatment, sewage, unpolluted, to wastewater treatment, class 3	1.54E-03	m3	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);	
		waste water in sewage treatment plant	concrete production effluent used as proxy	waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	6.80E-06	m3	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);	
		paint in municipal waste incineration		waste management	hazardous waste incineration	No	CH	disposal, emulsion paint remains, 0% water, to hazardous waste incineration	3.41E-03	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);	
		lubricating oil to disposal		waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	1.10E-04	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);	
		90% of total solvent: NMVOC to air	share estimated	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	8.10E-05	kg	SIA building product declaration	1	1.52	(2,1,3,1,1,1);	
		10% of solvent: chlorinated halogenated hydrocarbons to air	share estimated	air	unspecified			Hydrocarbons, chlorinated	9.00E-06	kg	SIA building product declaration	1	1.52	(2,1,3,1,1,1);	
		fibre cement roof slate		construction materials	coverings	No	CH	fibre cement roof slate, at plant	1.00E+00	kg					

Fig. 6.1 Flows for "fibre cement roof slate, at plant" and their representation in the ecoinvent database (cont)

Tab. 6.2 Ecoinvent meta information for the product "fibre cement corrugated slab, at plant"

Name	fibre cement corrugated slab, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce fibre cement products (raw material provision, raw material mixing, forming, post curing process coating process, and storage), transports to plant processes, and infrastructure. No acid for neutralisation of the wastewater and no requirements for administration are included.
Amount	1
Local Name	Faserzementwellplatte, ab Werk
Synonyms	roofing//sheet//well//dach//tile
General Comment to reference function	The amounts of raw materials are calculated based on the composition of fibre cement roof slate. Air emission data are lacking but emissions from the boilers are imported via the relevant modules. A certain amount of CO ₂ produced by the boiler is bound in the waste water. The amount of this entrained CO ₂ is inventoried with a negative value because it is an avoided emission. The total yearly is assumed to be 20'000 tonnes. The lifespan of the plant and the conveyor belt are assumed to be 50 years and 20 years, respectively.
Start Date	1991
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from one company in Switzerland (Eternit AG), for some exchanges RER- and OCE-modules have been used as proxy
Technology text	The raw materials are mixed with water to a homogeneous paste. Rotating sieving cylinders dip in this fibre cement mush which drain to the inside. Therefore the sieve surface gets covered with a layer of fleece of about 1mm which is transported with a conveyor belt. The produced endless mat is winded up on a format-cylinder until the demanded thickness is reached. With that process the fibres of the neighbour layers are interconnecting. After cutting up the mat cylinder the mat is detached from the format-cylinder and by punching and pressing automatically formed to small format slates, even and corrugated slabs.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Literature and "ecological building material declaration" (a sort of environmental report)
Extrapolations	See geography
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
process fibres: pulp fibres	Production of "fibre cement corrugated slab" (Part 1)		raw material, losses are neglected as internally reused.	paper & cardboard	pulps	No	RER	sulphate pulp, unbleached, at plant	1.80E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
process fibres: waste paper			raw material, losses are neglected as internally reused.	paper & cardboard	pulps	No	CH	waste paper, mixed, from public collection, for further treatment	1.80E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
additive: limestone meal			raw material, losses are neglected as internally reused.	construction materials	others	No	CH	limestone, milled, packed, at plant	1.11E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
fixing agent: cement			raw material, losses are neglected as internally reused.	construction materials	binder	No	CH	portland calcaereous cement, at plant	7.24E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
reinforcing fibre: polyvinyl alcohol			raw material, losses are neglected as internally reused.	plastics	monomers	No	RER	vinyl acetate, at plant	1.80E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
tap water			raw material, losses are neglected as internally reused.	water supply	production	No	RER	tap water, at user	1.11E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
total electricity consumption for complete process				electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.67E-01	kWh	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
energy consumption: light fuel oil in industrial furnace 1MW				oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	2.48E-01	MJ	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
energy consumption: light fuel oil in industrial furnace 100kW				oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.35E-02	MJ	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
tap water			raw material	water supply	production	No	RER	tap water, at user	2.30E+00	kg	Weibel & Stritz (1995)	1	1.12	(2,1,3,1,1,1);
dispersive acrylate			used for coating	paintings	production	No	RER	acrylic varnish, 87.5% in H2O, at plant	5.20E-03	kg	Weibel & Stritz (1995)	1	1.12	(2,1,3,1,1,1);
packing				construction materials	others	No	CH	packing, fibre cement products	1.00E+00	kg	estimation	1	1.40	(3,5,1,1,3,5);
infrastructure plant			assumption: same infrastructure for 1kg fibre cement product is inquired as for 1kg rock wool product	insulation materials	production	Yes	CH	rock wool plant	4.44E-10	unit	estimation	1	3.50	(5,1,1,1,4,5);
conveyor belt			estimation of length: 100m, estimation of lifespan 20years.	construction processes	machinery	Yes	RER	conveyor belt, at plant	2.50E-07	m	estimation	1	3.45	(5,1,1,1,4,1);
transportation of limestone meal by lorry				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	4.44E-04	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of limestone meal by train				transport systems	train	No	CH	transport, freight, rail	1.47E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);

Fig. 6.2 Flows for "fibre cement corrugated slab, at plant" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
transportation of polyvinyl alcohol by train	Production of "fibre cement corrugated slab" (Part 2)			transport systems	train	No	CH	transport, freight, rail	6.48E-04	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of reinforcing fibre by train				transport systems	train	No	RER	transport, freight, rail	1.62E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of pulp fibres by train				transport systems	train	No	RER	transport, freight, rail	7.20E-03	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of cement by train				transport systems	train	No	CH	transport, freight, rail	6.52E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of reinforcing fibres by freight ship				transport systems	ship	No	OCE	transport, transoceanic freight ship	2.52E-01	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
lubricant oil				chemicals	organics	No	RER	lubricating oil, at plant	7.79E-04	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
		carbon dioxide reacted in waste water	Calcium oxide in waste water reacts with carbon dioxide from exhaust gas to calcium carbonate (limestone). It must be subtracted (that's why negative) from CO2 emissions from combustion which are included in the modules for boilers. 0.72 kg of CO2 per m3 waste water. Amount of waste water: 0.00115 m3/kg	air	unspecified			Carbon dioxide, fossil	-8.28E-04	kg	calculation	1	1.38	(4,5,1,1,1,5);
		calcium carbonate to disposal	Calcium carbonate from decarbonising in waste water (compare above). The modul is a proxy as the waste is carbonate-rich waste obtained during decarbonisation of water and not waste water.	waste management	residual material landfill	No	CH	disposal, decarbonising waste, 30% water, to residual material landfill	1.96E-03	kg	calculation	1	1.22	(2,1,1,1,1,5);
		waste water internally clarified by precipitating calcium carbonate	unpolluted as clarified	waste management	wastewater treatment	No	CH	treatment, sewage, unpolluted, to wastewater treatment, class 3	1.15E-03	m3	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
		waste water in sewage treatment plant	concrete production effluent used as proxy	waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	1.10E-05	m3	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
		paint in sewage treatment plant		waste management	hazardous waste incineration	No	CH	disposal, emulsion paint remains, 0% water, to hazardous waste incineration	5.08E-04	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
		lubricating oil to disposal		waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	7.79E-04	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
		solvent: NMVOC to air		air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	2.52E-04	kg	SIA building product declaration	1	1.52	(2,1,3,1,1,1);
		solvent: chlorinated halogenated hydrocarbons to air		air	unspecified			Hydrocarbons, chlorinated	2.80E-05	kg	SIA building product declaration	1	1.52	(2,1,3,1,1,1);
		fibre cement corrugated slab		construction materials	coverings	No	CH	fibre cement corrugated slab, at plant	1.00E+00	kg				

Fig. 6.2 Flows for "fibre cement corrugated slab, at plant" and their representation in the ecoinvent database (cont)

Tab. 6.3 Ecoinvent meta information for the product “fibre cement facing tile, at plant”

Name	fibre cement facing tile, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce fibre cement products (raw material provision, raw material mixing, forming, post curing process coating process, and storage), transports to plant processes, and infrastructure. No acid for neutralisation of the wastewater and no requirements for administration are included.
Amount	1
Local Name	Faserzementfassadenplatte, ab Werk
Synonyms	cladding
General Comment to reference function	The amounts of raw materials are calculated based on the composition of fibre cement roof slate. Air emission data are lacking but emissions from the boilers are imported via the relevant modules. A certain amount of CO ₂ produced by the boiler is bound in the waste water. The amount of this entrained CO ₂ is inventoried with a negative value because it is an avoided emission. The total yearly is assumed to be 20'000 tonnes. The lifespan of the plant and the conveyor belt are assumed to be 50 years and 20 years, respectively.
Start Date	1991
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from one company in Switzerland (Eternit AG), for some exchanges RER- and OCE-modules have been used as proxy
Technology text	The raw materials are mixed with water to a homogeneous paste. Rotating sieving cylinders dip in this fibre cement mush which drain to the inside. Therefore the sieve surface gets covered with a layer of fleece of about 1mm which is transported with a conveyor belt. The produced endless mat is winded up on a format-cylinder until the demanded thickness is reached. With that process the fibres of the neighbour layers are interconnecting. After cutting up the mat cylinder the mat is detached from the format-cylinder and by punching and pressing automatically formed to small format slates, even and corrugated slabs.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Literature and "ecological building material declaration" (a sort of environmental report)
Extrapolations	See geography
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
process fibres: pulp fibres	Production of "fibre cement facing tile" (Part 1)		raw material, losses are neglected as internally reused.	paper & cardboard	pulps	No	RER	sulphate pulp, unbleached, at plant	1.00E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
process fibres: waste paper			raw material, losses are neglected as internally reused.	paper & cardboard	pulps	No	CH	waste paper, mixed, from public collection, for further treatment	1.50E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
additive: limestone meal			raw material, losses are neglected as internally reused.	construction materials	others	No	CH	limestone, milled, packed, at plant	3.70E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
fixing agent: cement			raw material, losses are neglected as internally reused.	construction materials	binder	No	CH	portland calcareous cement, at plant	7.86E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
reinforcing fibre: polyvinyl alcohol			raw material, losses are neglected as internally reused.	plastics	monomers	No	RER	vinyl acetate, at plant	3.70E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
water: constitutional water			raw material, losses are neglected as internally reused.	water supply	production	No	RER	tap water, at user	1.15E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
total electricity consumption for complete process				electricity	supply mix	No	CH	electricity, medium voltage, at grid	3.46E-01	kWh	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
energy consumption: light fuel oil in industrial furnace 1MW				oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	3.17E-01	MJ	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
energy consumption: light fuel oil in industrial furnace 100kW				oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.82E-02	MJ	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
energy consumption: gas in industrial furnace				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	2.03E-01	MJ	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
tap water			raw material	water supply	production	No	RER	tap water, at user	3.00E+00	kg	estimation	1	1.12	(2,1,3,1,1,1);
dispersive acrylate			used for coating, assumed that same amount as for "fibre cement roof slate"	paintings	production	No	RER	acrylic varnish, 87.5% in H2O, at plant	1.77E-02	kg	Weibel & Stritz (1995)	1	1.12	(2,1,3,1,1,1);
packing				construction materials	others	No	CH	packing, fibre cement products	1.00E+00	kg	estimation	1	1.40	(3,5,1,1,3,5);
infrastructure plant			assumption: same infrastructure for 1kg fibre cement product is inquired as for 1kg rock wool product	insulation materials	production	Yes	CH	rock wool plant	4.44E-10	unit	estimation	1	3.50	(5,1,1,1,4,5);
conveyor belt			estimation of length: 100m, estimation of lifespan 20years.	construction processes	machinery	Yes	RER	conveyor belt, at plant	2.50E-07	m	estimation	1	3.45	(5,1,1,1,4,1);
transportation of limestone meal by lorry				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	1.48E-04	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of limestone meal by train				transport systems	train	No	CH	transport, freight, rail	4.88E-03	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of polyvinyl alcohol by train				transport systems	train	No	CH	transport, freight, rail	1.33E-03	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of reinforcing fibre by train				transport systems	train	No	RER	transport, freight, rail	3.33E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of pulp fibres by train				transport systems	train	No	RER	transport, freight, rail	4.00E-03	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);

Fig. 6.3 Flows for "fibre cement facing tile, at plant" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
transportation of cement by train	Production of "fibre cement facing tile" (Part 2)			transport systems	train	No	CH	transport, freight, rail	7.07E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);	
transportation of reinforcing fibres by freight ship				transport systems	ship	No	OCE	transport, transoceanic freight ship	5.18E-01	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);	
lubricant oil				chemicals	organics	No	RER	lubricating oil, at plant	1.00E-04	kg	estimation	1	1.24	(2,1,3,1,1,5);	
		carbon dioxide reacted in waste water	Calcium oxide in waste water reacts with carbon dioxide from exhaust gas to calcium carbonate (limestone). It must be subtracted (that's why negative) from CO2 emissions from combustion which are included in the modules for boilers. 0.72 kg of CO2 per m3 waste water. Amount of waste water: 0.0015 m3/kg	air	unspecified			Carbon dioxide, fossil	-1.08E-03	kg	calculation	1	1.38	(4,5,1,1,1,5);	
		calcium carbonate to disposal	Calcium carbonate from decarbonising in waste water (compare above). The modul is a proxy as the waste is carbonate-rich waste obtained during decarbonisation of water and not waste water.	waste management	residual material landfill	No	CH	disposal, decarbonising waste, 30% water, to residual material landfill	2.45E-03	kg	calculation	1	1.22	(2,1,1,1,1,5);	
		waste water internally clarified by precipitating calcium carbonate	unpolluted as clarified	waste management	wastewater treatment	No	CH	treatment, sewage, unpolluted, to wastewater treatment, class 3	1.50E-03	m3	estimation	1	1.24	(2,1,3,1,1,5);	
		waste water in sewage treatment plant	concrete production effluent used as proxy	waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	1.00E-05	m3	estimation	1	1.24	(2,1,3,1,1,5);	
		paint in sewage treatment plant		waste management	hazardous waste incineration	No	CH	disposal, emulsion paint remains, 0% water, to hazardous waste incineration	3.41E-03	kg	estimation	1	1.24	(2,1,3,1,1,5);	
		lubricating oil to disposal		waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	1.00E-04	kg	estimation	1	1.24	(2,1,3,1,1,5);	
		solvent: NMVOC to air		air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	3.60E-04	kg	SIA building product declaration	1	1.52	(2,1,3,1,1,1);	
		solvent: chlorinated halogenated hydrocarbons to air		air	unspecified			Hydrocarbons, chlorinated	4.00E-05	kg	SIA building product declaration	1	1.52	(2,1,3,1,1,1);	
		trace element to air: chrome, cobalt, nickel and zinc	from coating; assumed amount of chromium III emission from total: 10%	air	unspecified			Chromium	1.39E-06	kg	SIA building product declaration	1	5.02	(2,1,3,1,1,1);	
			from coating; assumed amount of cobalt emission from total: 30%	air	unspecified			Cobalt	4.17E-06	kg	SIA building product declaration	1	5.02	(2,1,3,1,1,1);	
			from coating; assumed amount of nickel emission from total: 30%	air	unspecified			Nickel	4.17E-06	kg	SIA building product declaration	1	5.02	(2,1,3,1,1,1);	
			from coating; assumed amount of zinc emission from total: 30%	air	unspecified			Zinc	4.17E-06	kg	SIA building product declaration	1	5.02	(2,1,3,1,1,1);	
		fibre cement facing tile		construction materials	coverings	No	CH	fibre cement facing tile, at plant	1.00E+00	kg					

Fig. 6.3 Flows for "fibre cement facing tile, at plant" and their representation in the ecoinvent database (cont)

Tab. 6.4 Ecoinvent meta information for the product “packing fibre cement products”

Name	packing fibre cement products
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the process of packing, transportation, and loading of the bags on paletts. The machines, the waste treatment after use by the end-user of the fibre cement products and the transports of the packing materials are included.
Amount	1
Local Name	Verpacken, Faserzementprodukte
Synonyms	
General Comment to reference function	The production amount used for normalization is 20'000t/a. This module is assumed to be valid also for plants up to 100'000t/a. The lifespan of the infrastructure and the land occupation are assumed to be 50 years.
Start Date	2000
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from a company producing lime products in Switzerland are adjusted
Technology text	Assumed that packaging is the same as for lime products
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Assumption
Extrapolations	See geography
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
infrastructure packing and loading of fibre cement products	packing and loading of fibre cement products (including energy consumption) (Part 1)		assumption: same machines as for "packing of lime products": 30t, estimated lifespan: 25years; total amount of packed material (2001): 20'000 t/a.	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	6.00E-05	kg	estimation based on observation	1	3.23	(5,na,na,na,na,na);
conveyor belt			assumption: same length of conveyor belt as for "packing of lime products": 10m	construction processes	machinery	Yes	RER	conveyor belt, at plant	2.50E-08	m	estimation based on observation	1	3.23	(5,na,na,na,na,na);
electricity consumption for packing and loading			assumption: same amount as used for module "packing lime products"	electricity	supply mix	No	CH	electricity, medium voltage, at grid	3.09E-03	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
pallets, at plant			assumption: same amount as used for module "packing lime products", estimated total weight of load on one pallet: 1 t, weight of pallet: 22 kg, number of applications: 10	wooden materials	processing	No	RER	EUR-flat pallet	1.00E-04	unit	estimation	1	1.44	(4,5,1,1,3,5);
steel band for packing (steel)				metals	extraction	No	RER	steel, low-alloyed, at plant	1.10E-03	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
steel band for packing (rolling)				metals	processing	No	RER	section bar rolling, steel	1.10E-03	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
paperbag, at plant			assumption: same amount as used for the module "packing lime products"	paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	7.87E-03	kg	estimation	1	1.24	(1,4,1,2,1,5);
plastic foil, at plant			assumption: amount used for the module "packing lime products" has been adapted to the loading weight of fibre cement products (1'000kg), circumference of pallet: 4m; high: 1m, 2 layer of plastic with a weight of 0.0123 kg/m2	plastics	processing	No	RER	packaging film, LDPE, at plant	9.84E-05	kg	estimation	1	1.21	(4,na,na,na,na,na);
transports of pallets, paper bags and plastic bags to plant by train			assumption: same amount as used for the module "packing lime products", standard distance: transport by train with distance of: 200km	transport systems	train	No	CH	transport, freight, rail	6.00E-03	tkm	standard distance	1	2.23	(5,na,na,na,na,na);
transports of pallets, paper bags and plastic bags to plant by lorry			assumption: same amount as used for the module "packing lime products", standard distance: transport by lorry 28t with distance of: 50km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	1.50E-03	tkm	standard distance	1	2.23	(5,na,na,na,na,na);
transports of paper bags and plastic bags to municipal incineration			assumption: same amount as used for the module "packing lime products", assumption: transports all by lorry 16t and total of transportdistance: 10km	transport systems	road	No	RER	transport, lorry 3.5-16t, fleet average	7.99E-05	tkm	data of KFN for the year 2001	1	1.50	(5,na,na,na,na,na);

Fig. 6.4 Flows for "packing fibre cement products" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
packing and loading of fibre cement products (including energy consumption) (Part 2)		→ waste heat	assumption: same amount as used for the module "packing lime products", same amount as electricity used	air	unspecified			Heat, waste	1.11E-02	MJ		1	1.05	(1,2,1,1,1,1);
		→ disposal pallets		waste management	municipal incineration	No	CH	disposal, wood untreated, 20% water, to municipal incineration	2.93E-03	kg		1	1.24	(1,4,1,2,1,5);
		→ paperbag, in municipal waste incineration	assumption: same amount as used for the module "packing lime products"	waste management	municipal incineration	No	CH	disposal, paper, 11.2% water, to municipal incineration	7.87E-03	kg	data of KFN for the year 2001	1	1.24	(1,4,1,2,1,5);
		→ plastic foil, in municipal waste incineration	assumption: same amount as used for the module "packing lime products"	waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	1.23E-04	kg	data of KFN for the year 2001	1	1.21	(4,na,na,na,na,na);
		→ packing and loading of 1 kg fibre cement product	process of packing 1 kg of fibre cement product	construction materials	others	No	CH	packing, fibre cement products	1.00E+00	kg				

Fig. 6.4 Flows for “packing fibre cement products” and their representation in the ecoinvent database (cont)

In general this dataset has been copied from the lime packing dataset. Some adjustments were made to account for different densities.

7 Data Quality

The uncertainties are estimated using the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

Most data for fibre cement roof slate and fibre cement corrugated slab production are directly taken from Weibel & Stritz (1995). These data are based on an inventory of one Swiss company for year 1991 (with some additional data for 1994). They give very little background information to their inventories. Thus a check whether the adaptation to the ecoinvent database modules is made correctly is not possible. The total amount of energy consumption and the raw materials for the different fibre cement products are adjusted based on the SIA building materials declaration⁵. The year of publication of this information is 2002. No data are available on the composition of the wastewater. Data for fibre cement facing tile are mostly based on assumptions.

In general both used reports do not show details on the sub-processes included in the data. Therefore a combination of the two above-mentioned reports may not be fully consistent. It was not possible to get up-to-date data from the only Swiss producer Eternit AG.

For all these reasons the data for all different types of fibre cement products are of poor overall quality and should not be used to cross compare these different materials or for direct comparison to other materials in the same function. In a larger context (e.g., use for LCA of buildings) and if the importance of these products is not too high for the application, the data may be used.

8 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

9 References

- | | |
|--------------------------------|--|
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⁵ <http://www.sia.ch/cf/oekoinfoliste.cfm?ID=2943702&Lang=d&Mfg=72&Grp=0&Produkt=>, access date: Mai 2003

Part X

Sand-Lime Brick

Data v2.0 (2007)

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Kellenberger D., Althaus H.-J., Jungbluth N., Künniger T., Lehmann M. and Thalmann P. (2007) Life Cycle Inventories of Building Products. Final report ecoinvent Data v2.0 No. 7. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.org.

Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

This part deals with the production of sand-lime brick. The raw materials are silica sand, quicklime, additives and water. The production steps are mainly mixing of the raw materials, forming and hardening. The emissions taken into account are waste water and solid waste.

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1 Introduction

Sand-lime bricks are made from limestone and sand (Starzner & Wurmer-Weiss (2000)). After intensive mixing with water it is consolidated. Forming and hardening under steam pressure completes the process.

Sand-lime lightweight bricks are insulating bricks with a low apparent density ($\rho = 700$ to 900 kg/m^3).

2 Reserves and Resources

The two resources milled limestone and sand are dealt with in the corresponding reports (Part I, Gravel and Sand Products and Processes and Part VII: Lime Products and Processes).

3 Use

The most important use of sand-lime bricks in the building sector are (Starzner & Wurmer-Weiss (2000)):

- Masonry
 - supporting walls
 - parting wall
 - casing
- fair-faced brickwork
- Facing/Casing

4 Characterisation

Some technical specifications to the studied products are taken from Starzner & Wurmer-Weiss (2000) and Cordes et al. (1996) and shown in Tab. 4.1

Tab. 4.1 Technical specification of some sand-lime brick products

	Brick ⁽¹⁾	Fair faced brick ⁽¹⁾	Facing brick ⁽¹⁾	Brick used in this study ⁽²⁾
Bulk density ρ [kg/m ³]	510 – 2'200	810 – 2'000		1'530
compression strength class [-]	4 - 60	12 - 60	20 - 60	-
thermal conductivity λ [W/mK]	0.5 – 1.3 (ρ = 1'000 – 2'200 kg/m ³)			
water vapour diffusion resistant coefficient [μ]	5 - 10	5 - 25		-
Water sorption coefficient [kg/m ² h _{0.5}]	2 - 6			
Norm sizes [cm]	-	-	-	24 * 11.5 * 7.1cm

⁽¹⁾ Starzner & Wurmer-Weiss (2000)

⁽²⁾ Cordes et al. (1996)

5 System Characterisation

5.1 Raw Materials

The raw materials used for the production of sand-lime bricks are shown in Tab. 5.1.

Tab. 5.1 Composition of sand-lime brick (Starzner & Wurmer-Weiss (2000) and Cordes et al. (1996))

Material	Amount [%] ⁽¹⁾	Amount used in study [kg] ⁽²⁾
Sand (natural silica sand)	85 - 92	0.94750
Lime (milled quick lime)	5 - 8	0.08552
Additives	2 - 3	0.03328
Water (in m ³ per kg sand-lime brick)	0.000225 m ³	0.000225 m ³

⁽¹⁾ Starzner & Wurmer-Weiss (2000)

⁽²⁾ Cordes et al. (1996)

The additives usually used are milled limestone and milled quartz. Rarely also basalt, grey wacke or gravel but no chemical additives are used. There is no information about the usage of additional additives. For colouring of the bricks, especially for fair faced bricks and facings, iron and chromium oxide pigments are common.

5.2 Production Process

The data in Cordes et al. (1996) are based on 14 sand-lime brick companies in Germany. The published data are average data.

The production process is divided in the five following steps (Fig. 5.1):

1. Limestone and sand from local mines are stored in the company's silos. The raw materials are mixed in a share of one part limestone and twelve parts sand. The raw mixture is transported by conveyor belt into a reaction vessel.
2. Inside the reaction vessel the quick lime is slaked to hydrated lime ($\text{Ca}(\text{OH})_2$). This process takes usually about 4 hours. If necessary, a pre-mixer adds water to the mixture to bring it up to pressing humidity.
3. By full automatic working presses the raw moulded blanks are formed with a pressure of usually 15 N/mm² (maximum 25 N/mm²). The raw moulded blanks are then stored on a hardening wagon.
4. The hardening of the raw moulded blanks takes place in the pressure vessel with a saturated steam at a pressure of 12 to 16 bar, a temperature of 160 to 200°C, and for a time period of 4 to 8 hours. The previous values are depending on the amounts of the components and the strength to be reached. At the hardening process the hot steam etches silicic acid on the surface of the sand granules. The silicic acid compounds form together with the adhesive hydrated lime crystalline material which epitaxy on the sand granules and stick them together in a solid way.
5. After hardening and cooling down, the sand-lime bricks are ready for use.

5.3 Energy Consumption

According to Starzner & Wurmer-Weiss (2000) the primary energy consumption for the production of sand-lime brick including limestone production changes with the mixing formula between 738 and 907 MJ/t. In average it is 845 MJ/t. The shares for the different production steps are shown in Tab. 5.1.

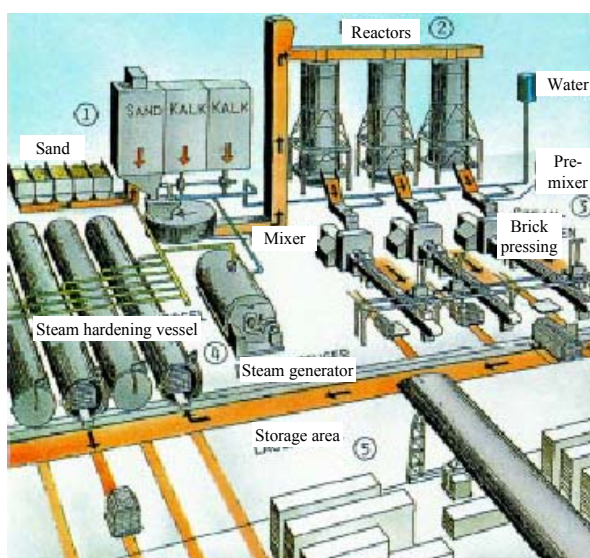


Fig. 5.1 Steps of sand-lime brick production (Cordes et al. (1996))

Tab. 5.2 Share of energy consumption for the sand-lime brick production process (Starzner & Wurmer-Weiss (2000))

	Energy consumption	
	[MJ/kg]	[%]
Limestone and additives production	0.372	44
Autoclaving (steam hardening)	0.355	42
Other processes (mixing, internal transports)	0.050	6
Transport to the user	0.068	8
Total	0.845	100

For the present modelling, the sub-process “limestone production” does not affect the direct energy demand for the production process of sand-lime bricks (energy used in the unit-process) as it is already included in the relevant upstream processes; besides, the “transports to the user” is not part of the considered system (the product is described “at plant”). Thus the total amount of primary energy used for the sub-processes “autoclaving” and “other processes” (0.405 MJ/kg) is about equal to the total amount of energy consumed in the factory (0.404 MJ/kg), as described by (Cordes et al. (1996)) (Tab. 5.3).

Tab. 5.3 Fuel consumption (Cordes et al. (1996))

	[%] of total fuel consumption	[MJ/kg]
Light fuel oil	56	0.20676
Heavy fuel oil	4	0.01477
Natural gas	40	0.14769
Electricity		0.03464
Total		0.40386

The main fuels used in Germany are natural gas, heavy and light fuel oil. The share is based on the distribution in year 1994 (Cordes et al. (1996)). As no more precise and actual data are available, the

following shares have been taken into account in this study: 56% light fuel oil, 4% heavy fuel oil and 40% natural gas.

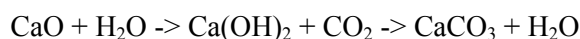
5.4 Transports

The transport distances of the raw materials to the manufacturer are estimated based on the information that the mining takes place nearby. For sand a distance of 5 km per lorry 28t and for the transportation of quicklime a distance of 100 km, 50% by train and 50% by lorry is taken into account (Weibel & Stritz (1995)).

5.5 Air Emissions

The emissions stem mainly from combustion of fuels. Since they are included in the corresponding modules for boilers, no additional emissions have been taken into account.

During the hardening process quicklime is reacting with water to calcium hydroxide. Calcium hydroxide is reacting with carbon dioxide to calcium carbonate. Therefore the amount of CO₂ from air, which is bound, is taken into account, based on stoichiometric formula:



Thus the amount of absorbed CO₂ is 0.067 kg per kg sand-lime brick.

5.6 Wastewater

The data from “Kalksandstein-Fakten zur Ökobilanz” (Cordes et al. (1996)) concerning the wastewater released by the plant are listed in Tab. 5.4.

Tab. 5.4 Wastewater emissions (Cordes et al. (1996))

Amount of wastewater	0.000083 m ³ / kg product
COD, Chemical Oxygen Demand	114 mg/l = 0.0095 kg/ kg product
BOD ₅ , Biological Oxygen Demand	76 mg/l
TOC, Total Oxygen Demand	44.6 mg/l
DOC, Dissolved Organic Carbon	44.6 mg/l

Values for TOC and BOD₅ are calculated using the standard procedure approximating the organic matter with C₆H₁₂O₆ (Frischknecht et al. (2007)).

This study uses the module “treatment of concrete production effluent to wastewater treatment” as proxy because the wastewater of the sand-lime brick production has a similar COD amount as concrete (0.0122 kg/m³) and is expected to be similarly composed.

5.7 Solid Waste

Most waste is re-used in the process. The residual industrial waste is 0.0005 m³ (Cordes et al. (1996)).

5.8 Infrastructure

The average yearly production output of one company “Xella Baustoffe GmbH”¹ in Germany is about 120'000 tons. The amount comes from the information on the total yearly production volume, which is converted to weight.

As no specific data on the land-use are available, the infrastructure module for clay products has been used as proxy. The difference of the density of clay brick to sand-lime brick is about the same as the difference between the yearly production volumes of the German plant producing sand-lime brick and the Swiss plant producing clay brick. This justifies the direct use of the infrastructure module for the clay brick as proxy for the sand-lime brick.

5.9 System Boundary

The system boundary for this study is displayed in Fig. 5.2

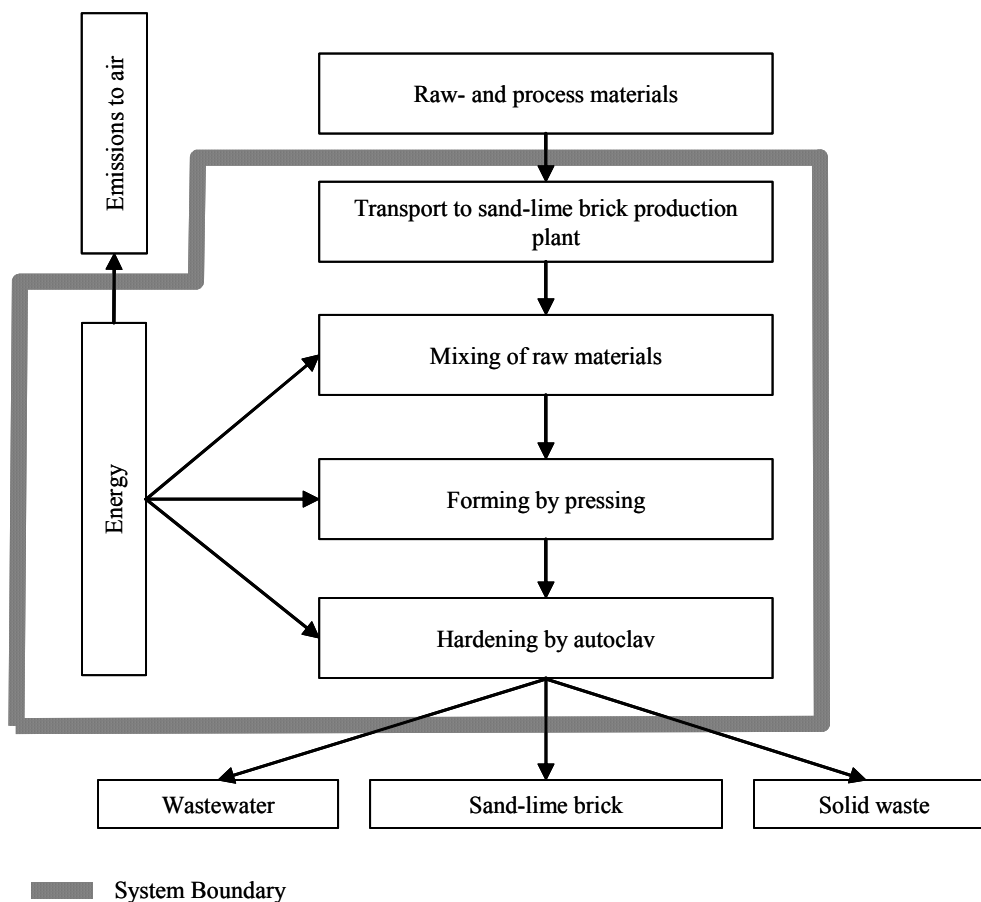


Fig. 5.2 Production process of sand-lime brick (Starzner & Wurmer-Weiss (2000); Cordes et al. (1996))

¹ http://ytong.xella.de/html/de/467.php?area_code=1, access 22.05.2003

6 Life Cycle Inventory

Ecoinvent database meta information for the module “sand-lime brick, at plant” is shown in Tab. 6.1, the flows and its representation in the ecoinvent database are shown in Fig. 6.1.

Tab. 6.1 Ecoinvent meta information for the product “sand-lime brick, at plant”

Name	sand-lime brick, at plant
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce sand-lime bricks (raw material provision, raw material mixing, forming by pressing, hardening by autoclave, packing, and storage), transports to plant, and infrastructure. No material/energy requirement for administration, are included.
Amount	1
Local Name	Kalksandstein, ab Werk
Synonyms	calcareous//chalky//sandstone//silica//calcium//silicate
General Comment to reference function	The air emissions are not measured. However, air emissions should all stem from combustion processes, which are accounted for via the modules for energy requirements. A certain amount of CO ₂ is bound from air by the hydrated lime during the hardening process and is therefore modelled as a resource. No exact information on the composition of the wastewater is available. Thus a proxy is inventoried. The total yearly production amount of a German company is assumed to be 120'000 tons (based on the yearly production volume). The lifespan of the plant is assumed to be 50 years.
Start Date	1993
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	The amount of raw materials, the fuel type and the energy consumption as well as the amount of waste are based on an average of 74 German companies. The transportation distances and the amount of CO ₂ which is bound are based on an average of four Swiss companies. For some exchanges, RER-modules have been used as proxy
Technology text	The raw materials are mixed with appropriate proportions. In a reaction vessel the quicklime is hydrated to hydrated lime. Pressing the raw moulded blanks give them a certain form. In an autoclave the blanks are hardened with saturated steam at a pressure of 12 to 16 bars and a temperature of 160 to 200°C during 4 to 8 hours.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 6.1 shows the system of the module “sand-lime brick, at plant”. It shows all in- and output flows, the allocation to the modules ecoinvent database V1.1, the sources (mainly Cordes et al. (1996) and Weibel & Stritz (1995)) and the standard deviation for each data.

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
silica sand	→	production of sand-lime brick				raw material	construction materials	additives	No	CH	sand, at mine	9.48E-01	kg	Cordes et al. (1996)	1	1.22	(1,3,2,3,3,2);
quicklime	→					raw material	construction materials	additives	No	CH	quicklime, milled, packed, at plant	8.55E-02	kg	Cordes et al. (1996)	1	1.22	(1,3,2,3,3,2);
additive (milled limestone and quartz)	→					raw material: milled limestone used as proxy	construction materials	others	No	CH	limestone, milled, packed, at plant	3.33E-02	kg	Cordes et al. (1996)	1	1.22	(1,3,2,3,3,2);
water	→					raw material	water supply	production	No	RER	tap water, at user	2.25E-01	kg	Cordes et al. (1996)	1	1.22	(1,3,2,3,3,2);
energy from light fuel oil	→					56% of total thermal energy consumption	oil	heating systems	No	CH	heat, light fuel oil, at industrial furnace 1MW	2.07E-01	MJ	Cordes et al. (1996)	1	1.22	(1,3,2,3,3,2);
energy from heavy fuel oil	→					4% of total thermal energy consumption	oil	heating systems	No	CH	heat, heavy fuel oil, at industrial furnace 1MW	1.48E-02	MJ	Cordes et al. (1996)	1	1.22	
energy from natural gas	→						natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace >100kW	1.48E-01	MJ	Cordes et al. (1996)	1	1.22	
electricity consumption	→						electricity	supply mix	No	CH	electricity, medium voltage, at grid	9.62E-03	kWh	Cordes et al. (1996)	1	1.22	(1,3,2,3,3,2);
transportation of sand to plant by lorry	→					distance: 5km; assumed that comparable with distance in Germany	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	4.74E-03	tkm	Weibel & Stritz (1995)	1	2.02	(1,2,3,1,1,3);
transportation of quicklime to plant by lorry	→					distance: 50km; assumed that comparable with distance in Germany	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	4.28E-03	tkm	Weibel & Stritz (1995)	1	2.02	(1,2,3,1,1,3);
transportation of quicklime to plant by train	→					distance: 50km; assumed that comparable with distance in Germany	transport systems	train	No	CH	transport, freight, rail	4.28E-03	tkm	Weibel & Stritz (1995)	1	2.02	(1,2,3,1,1,3);
infrastructure (including building, machines and land-use)	→					clay mine used as proxy; yearly production amount of sand-lime brick: 120'000t, estimated service life of plant: 50years	construction materials	additives	Yes	CH	mine, clay	1.67E-10	unit	estimation	1	3.55	(5,5,1,1,4,5);
packing	→					packing clay products used as proxy	construction materials	others	No	CH	packing, clay products	1.00E+00	kg	estimation	1	1.88	(5,5,1,1,4,5);
carbon dioxide binding by hardening process	→					During the hardening process CO2 is bound. It reduces the CO2 emission from combustion which is included in the heating modules	resource	in air			Carbon dioxide, in air	6.70E-02	kg	Weibel & Stritz (1995)	1	1.32	(4,2,3,1,1,5);
	→	waste heat			same amount as electricity consumption	air	unspecified			Heat, waste	3.46E-02	MJ	Cordes et al. (1996)	1	1.22	(1,3,2,3,3,2);	
	→	waste water			waste water from concrete mixing is used as proxy	waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	8.33E-05	m3	amount: Cordes et al. (1996)	1	1.57	(1,3,2,3,3,2);	
	→	solid waste			assumption: density 250kg/m3	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.25E-04	kg	Cordes et al. (1996)	1	1.22	(1,3,2,3,3,2);	
	→	sand-lime brick				construction materials	bricks	No	DE	sand-lime brick, at plant	1.00E+00	kg					

Fig. 6.1 Flows for “sand-lime brick, at plant” and their representation in the ecoinvent database

7 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

Process data from “Kalksandstein – Fakten zur Ökobilanz” (Cordes et al. (1996)) have an unspecified reference year. It is assumed to be between 1993 and 1995. The input and output products and processes are restricted to the most important ones. Compared to other building materials (e.g. part IV, Clay Products and Processes) the supplied data are rather poor.

For the packing of sand-lime brick the packing module for brick is used as proxy.

Some data have been taken from a different reference which does not provide a precise description of the system boundary.

For all these reasons the data for sand-lime brick are of fair overall quality.

8 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

9 References

- | | |
|--------------------------------|--|
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| Frischknecht et al. (2007) | Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Hirsch R., Hellweg S., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent Data v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.ch . |
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| Weibel & Stritz (1995) | Weibel T. and Stritz A. (1995) Ökoinventare und Wirkungsbilanzen von Baumaterialien; Grundlagen für den ökologischen Vergleich von Hochbaukonstruktionen. In: ESU-Reihe Nr. 1/95. Laboratorium für Energiesysteme, Eidg. Technische Hochschule (ETH), Zürich. |

Part XI

Flat Glass Products and Processes

Data v2.0 (2007)

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Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

This section describes the production and processing of uncoated and coated flat glass. The data mainly refer to a reference document on best available techniques in the glass manufacturing industry from IPPC (Integrated Pollution Prevention and Control) ([IPPC, 2001 #434]). The background of the mentioned data for the flat glass production is not transparent and therefore the uncertainties of the in- and output data are roughly estimated. The data for the coating process are based on an environmental report from a German company called Interpane Glasbeschichtungsgesellschaft mbH & Co and refer to the one year. This is taken into account in the uncertainty, to adapt to a wider geographical area.

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1 Introduction

The inventory of flat glass production is mostly based on the report (IPPC (2001)), which gives a clear overview of the glass manufacturing:

The Glass Industry within the European Union (EU) is extremely diverse, both in the products made and the manufacturing techniques employed. Product range from intricate hand-made lead crystal goblets to the huge volumes of float glass produced for the construction and automotive industries. Manufacturing techniques vary from the small electrically heated furnaces in the Ceramic Fibre Sector to the cross-fired regenerative furnaces in the flat glass sector, producing up to 700 tonnes per day. The wider glass industry also includes many smaller installations that fall below the 20 tonnes per day threshold. However, for some of the statistical data given in this chapter it has not been possible to separate the contribution from the smaller plants, but this is not considered significant since they account for less than 5 % of the total output.

The glass industry is essentially a commodity industry, although many ways of adding value to high volume products have been developed to ensure that the industry remains competitive. Over 80 % of the industry output is sold to other industries, and the glass industry as a whole is very dependent on the building, and the food and beverage industries. However, this general picture is not true for all of its components, as some of the smaller volume sectors produce high value technical or consumer products.

In the late 1990's, the glass industry was experiencing a period of reorganisation. In order to reduce costs and compete more effectively in a global market and to benefit from economies of scale companies are merging and the number of independent operators is thus decreasing.

The groups that dominate the production are becoming more international in their operation, and users are increasingly requiring homogeneous quality, regardless of the country where the products are used. The EU glass industry is at the forefront of technological developments and thus is likely to benefit from improved industrial performance in future years.

With the exception of Saint-Gobain, there are few major companies operating in more than two of the eight sectors specified in Tab. 1.1. For example, the Owens Corning Corporation specialises in glass fibre technology, continuous filament glass fibre and glass wool. PPG is a large international producer of flat glass and continuous filament glass fibre.

The Pilkington Group specialises mainly in flat glass activities. The major environmental challenges for the glass industry are emissions to air and energy consumption. Glass making is a high temperature, energy intensive activity, resulting in the emission of products of combustion and the high temperature oxidation of atmospheric nitrogen. Furnace emissions also contain dust, which arises mainly from the volatilisation and subsequent condensation of volatile batch materials. It is estimated that in 1997 the glass industry emissions to air consisted of: 9000 tonnes of dust; 103500 tonnes of NO_x; 91500 tonnes of SO_x; and 22 million tonnes of CO₂ (including electric generation). This amounted to around of 0.7 % of total EU emissions. Total energy consumption by the glass industry was approximately 265 PJ.

Emissions to water are relatively low. Solid waste levels are also generally very low, and many initiatives have been taken for further reducing waste generation, and for recycling in-house and post consumer waste.

In general, the raw materials for glass making are readily available, relatively harmless natural or man made substances. There are no major environmental problems associated with the provision of the raw materials.

Many of the sectors within the glass industry utilise large continuous furnaces with lifetimes up to twelve years. These furnaces represent a large capital commitment and the continuous operation of the furnace and the periodic rebuild provide a natural cycle of investment in the process. Major changes of

melting technology are most economically implemented if coincide with rebuild of furnaces. This can also be true for complex secondary pollution measures that must be correctly sized and any necessary gas conditioning implemented. However, many improvements to the operation of the furnace, including the installation of secondary techniques, are possible during the operating campaign. For smaller furnaces with more frequent rebuilds and lower capital costs, the advantages of co-ordinating environmental improvements and furnace repairs are less significant, but still environmental improvements may be more economical if co-ordinated with other investments.

The total production of the glass industry within the EU in 1996 was estimated to be 29 million tonnes (excluding ceramic fibres and frits); an indicative breakdown by sector is given below in Tab. 1.1. There has been a steady growth in the overall volume of sales over the last decade.

Tab. 1.1 Approximate sector based breakdown of production in the glass industry (IPPC (2001))

Sector	% of Total EU Production (1996)
Container Glass	60
Flat Glass	22
Continuous Filament Glass Fibre	1.8
Domestic Glass	3.6
Special Glass	5.8
Mineral Wool	6.8
Ceramic Fibre	-
Glass Frit and Enamel Frit	-

However, there have been fluctuations in each sector, which are discussed later in the report. The growth in total EU glass production and in production from the three largest sectors, between 1986 and 1996 is summarised in Fig. 1.1.

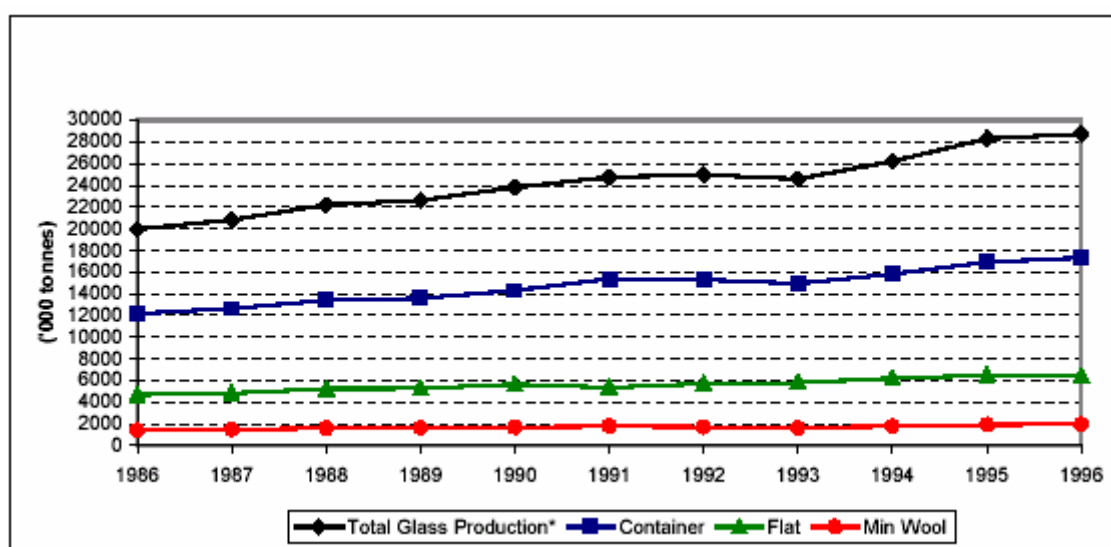


Fig. 1.1 Total glass production 1986 to 1996 (IPPC (2001)) (*excluding frits and ceramic fibres)

The outputs from the different sectors are very diverse and the links between the sectors are at times tenuous. However, the common thread linking all of the activities discussed in this document is the melting of inorganic materials to form a molten glass, or glass-like, substance which is then formed into products.

A short introduction to surface coating is translated by the authors from Richter et al. (1996): “The heat transmission through double glazing windows are caused to two third by the radiation exchange within the panes and to one third by the thermal conduction and convection of the gas between the panes. The lower the emission factor for the thermal radiation of the glass surfaces is the smaller the radiation exchange gets. The radiation exchange can be reduced significantly by a coating with low emissivity on one of the two glasses (oriented towards the glass spacing). As a result the heat-insulating value of the double insulation glazing with a glass spacing of 14 mm can be reduced from 3 to 1 W/m²K.”

2 The Flat Glass Industry Sector within the EU

The flat glass industry sector within the EU is comprehensively described in IPPC (2001):

Flat glass is the second largest sector of the glass industry in the EU representing around 22 % of the total glass production. The sector covers the production of float glass and rolled glass. In 1997 the sector produced approximately 6.9 million tonnes of glass from the 40 float tanks operating in the EU. There are five manufacturers of float glass and four rolled glass manufacturers operating in the EU. Flat glass is produced in eleven member states, the exceptions being Austria, Denmark, Greece and Ireland.

The sector directly employs 12'500 people in the manufacture of the glass, and a further 90'000 are employed in the processing suited for building and car industry. Flat glass output increased substantially between 1986 and 1997 (see Fig. 1.1), boosted in part by the entry of Sweden and Finland into the EU in 1995. However, year on year the demand can fluctuate significantly. For example 1997 showed a particularly high growth of over 6 %, but annual growth was predicted to be 2 to 3 % between 1998 and 2000.

Flat glass manufacture is a worldwide business that is dominated by five major groups; in order of worldwide sales they are Asahi (Japan), Pilkington (UK), Saint-Gobain (France), PPG Industries (USA), and Guardian Industries (USA). Pilkington and Saint-Gobain dominate EU production; together they are estimated to cover over 70 % of the worldwide market. Until recently all five major groups had factories in the EU. PPG has sold its European float glass plants (two in Italy and two in France) to the Belgian company Glaverbel, which is largely owned by Asahi. The operations of EU float glass producers in 1997 are summarised in Tab. 2.1, showing the locations of float tanks in the EU in 1997

Four new float tanks were constructed in the EU between 1998 and 2000, in Germany, UK, Italy and Spain. All four tanks will be in the 450 to 600 tonnes per day range.

Tab. 2.1 Locations of float tanks in the EU in 1997 (IPPC (2001))

Company	Number of Tanks	Locations
Saint-Gobain	13	Germany (4), France (3), Belgium (2), Spain (2), Italy (1), Portugal (1).
Pilkington	12	Germany (4), UK (3), Italy (3), Finland (1), Sweden (1)
Glaverbel	9	Belgium (4), France (2), Italy (2), Netherlands (1)
Guardian	5	Luxembourg (2), Spain (2), Germany (1)
Euroglas	1	France (1)

The geographical distribution of the sector and the range of installation sizes are shown in Tab. 2.2 and Tab. 2.3

Tab. 2.2 Number of float tanks in EU Member States in 1997 (IPPC (2001))

Member State	Number of Float Tanks	% Distribution of EU Production
Germany	9	20
France	6	15
Italy	6	15
Belgium	6	15
United Kingdom	3	10
Spain	4	10
Finland	1	14
Netherlands	1	
Luxembourg	2	
Portugal	1	
Sweden	1	
Austria	-	<1
Denmark	-	
Greece	-	
Ireland	-	
Total	40	6893000 tonnes in 1997

Tab. 2.3 Distribution of float tanks according to plant capacity in 1997 (IPPC (2001))

Capacity Range (tonnes/day)	<400	400 to 550	550 to 700	>700
% Capacity in each range in EU	3	31	38	28
% Capacity in each range world-wide	17	47	20	16

3 Products and Markets

3.1 Flat glass

The report of IPPC (2001)) describes the situation in the EU: There are two types of flat glass produced in the EU, rolled glass and float glass. Although other types of flat glass exist, they are not considered to be included in this sector, either because they are covered within the special glass sector or they do not meet the production criterion of 20 tonnes per day specified in Directive 96/61/EC. The majority of rolled glass is patterned or wired glass and accounts for around 5 % of the total flat glass output. Patterned glass is used for horticultural greenhouses, for decorative purposes and in applications where light is dispersed for example in the glass partitions and bathroom windows. Float glass makes up the other 95 % of total flat glass output and is used principally in the building and automotive industries.

Prior to the invention of the float glass process in 1962 by Pilkington, there were two main types of unpatterned glass, sheet glass and plate glass. The most widely used method for producing sheet glass was the Pittsburgh process, which involves drawing glass vertically from the tank. A refractory guidance device is placed in the glass at the drawing location and cooled grippers receive the glass. The glass passes through an annealing shaft about 12 m long and is then cut to shape. Prior to float glass,

plate glass was the highest quality glass available. Plate glass is produced from rolled glass or thick sheet glass by grinding and polishing the glass using rotating discs on large tables or conveyors. The twin process involves polishing the glass on both sides at once. The grinding and polishing process generates large amounts of solid waste for disposal.

The advantages of the float process (economy, product range, low waste and high quality) are such that, since its introduction in 1962 sheet glass and plate glass have gradually been replaced and are no longer produced within the EU. Some rolled glass products are still polished for special applications, and diminishing levels of sheet glass and plate glass are still produced in some parts of the world. For the purposes of this study, sheet glass and plate glass manufacture can be considered as essentially obsolete techniques and they are not considered here.

The most important markets for float glass are the building and automotive industries. The largest of these markets is the building industry which accounts for 75 % to 85 % of the output, while the majority of the remaining 15 % to 25 % is glass for the automotive industry. Some glass is simply cut to size and used directly, but the majority of flat glass production is further processed. For the automotive industry these are laminated windscreens, side and rear glazing, and sunroofs. The main processed product for the building industry is insulated glazing in the form of double or triple glazed units, often with one layer of coated glass. These glazed units account for 40 % to 50 % of the building market with the remainder being made up of silvered, coated, toughened, and laminated products which each making 10 % to 15 %.

3.2 Surface coating

In general three different types of surface coating are used (De Jong (1997)):

Vitreous enamels are applied by spraying, silk screening, and decalomania, i.e., the transfer of pictures from one medium to another. They are fused to the ware at high temperature.

Metallic coatings are produced either by applying a liquid suspension of metal onto the surface and then firing it or by evaporation of metal on the glass. These coatings are used on flat glass as means of controlling transmission and reflection of light, as electrical semiconductors in tin oxide films, as resistors in electronic circuitry, as defrosting agents and aircraft glazing.

Lacquers are used for decorative purposes, for example, in Christmas ornaments and electric light bulbs.

The information and data concerning the coating is based on the environmental report of the company Interpane in Lauenförde, Germany (Interpane (2000)). The purpose of the metallic coating used at Interpane is to optimise the glass with respect to preventing overheating, thermal insulation and light transmission. Two products are covered by the data. Unfortunately no information about the production share was given and therefore the products are treated together. The information on the products is taken from the company's internet homepage¹:

The coating is incorporated on the outer face in the space between the individual panes - where it is safe from adverse environmental effects and from mechanical damage.

¹ <http://www.interpane.net/>, access date: March 2003

4 Process Description

This chapter is directly taken from IPPC (2001) and gives an overview of the flat glass production process.

4.1 General

The basic principle of the float process is to pour the molten glass onto a bath of molten tin, and to form a ribbon with the upper and lower surfaces becoming parallel under the influence of gravity and surface tension.

The float tank (or bath) consists of a steel casing supported by a steel framework, and lined with refractory blocks which contain the molten tin. The float tank is about 55 m to 60 m long, 4 m to 10 m wide and divided into 15 to 20 bays. The tank is airtight and a slightly reducing atmosphere of a mixture of nitrogen and hydrogen is maintained by the injection. This is essential to prevent the oxidation of the tin surface, which would damage the crucial contact between the glass and the tin surfaces. Molten tin is used as the bath liquid because it is the only substance which remains liquid and without a significant vapour pressure over the required temperature range.

The molten glass flows from the furnace along a refractory lined canal, which can be heated to maintain the right temperature. At the end of the canal the glass pours onto the tin bath through a special refractory lip ("the spout") which ensures correct glass spreading. The glass flow is controlled by means of an adjustable suspended refractory shutter in the canal (the front "tweel"). Where the glass first makes contact with the tin, the temperature of the metal is about 1000°C, cooling to about 600°C at the exit of the bath. As it passes over the surface of the bath the glass develops a uniform thickness and assumes the almost perfect flatness of the molten tin.

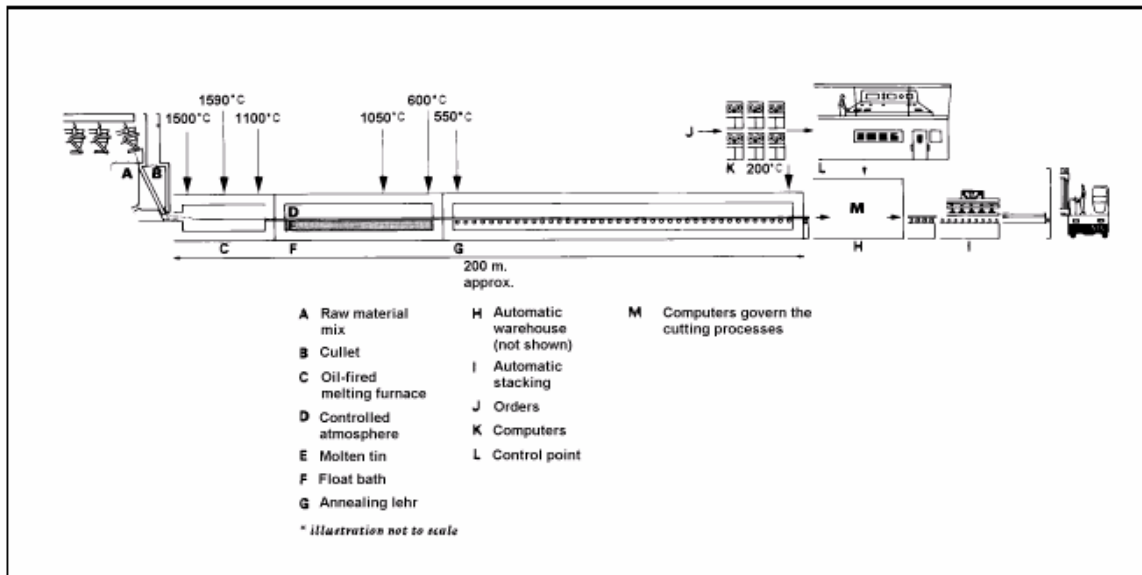


Fig. 4.1 The Float Glass Process IPPC (2001)

Inside the float tank are several pairs of water-cooled top rollers, adjustable in direction, height, penetration and angle. These rollers catch the glass sheet on both edges by cogwheels and draw it in length and width. The rate of glass flow and the rotation speeds of the rollers help to govern the thickness of

the glass, typically from 1.5 mm to 19 mm. The glass has a maximum natural thickness on the tin surface and graphite barriers can be introduced in order to produce the thicker glasses.

At the exit of the float bath the glass ribbon is taken out by lift-out rollers, and is passed through a temperature controlled tunnel, the lehr, to be annealed. At the beginning of the lehr, SO₂ is sprayed on both sides of the ribbon, providing a surface treatment to protect the glass against the contact with the rollers. The lehr is divided in sections in which there is heating and indirect or direct cooling by forced and natural convection. Glass is thus gradually cooled from 600°C to 60°C in order to reduce residual stresses, caused during the forming process, to an acceptable level. This operation needs time and space, from the pouring of glass onto the float bath to the cutting line, there is a continuous 200m ribbon of glass.

The cooled glass ribbon is cut on-line by a travelling cutter; the angle of the cutter against the line depends on the speed of the line (90° if it was not moving). The edges of the ribbon that bear roller marks are cut off and recycled to the furnace as cullet. The glass sheets are then inspected, packed and stored, either for sale or for secondary processing. On-line coatings can be applied to improve the performance of the product (e.g. low emissivity glazing). On-line coating processes are case specific and the total number of plants with on-line coating facilities is very low. A moving ribbon of glass is coated whilst hot by the impingement of silica or tin compounds onto its surface where they react to form the required film. The process generally consists of two separate coating stages, a silicon based undercoat and a separate topcoat, and e.g. fluorine doped tin oxide. Due to the nature of the chemicals used, emissions of acid gases and fine particulates can arise, which are generally treated in a dedicated abatement system.

4.2 The Melting Process

The melting process is a complex combination of chemical reactions and physical processes. Melting can be divided into several phases which all require very close control. In batch melting, these steps occur in sequence, but in continuous furnaces the melting phases occur simultaneously in different locations within the tank. The batch is fed at one end of the tank and flows through different zones in the tank and fore hearth where primary melting, fining, and conditioning occur. The refining process in a continuous furnace is by far more delicate.

4.2.1 Heating

The conventional and most common way of providing heat to melt glass is by burning fossil fuels continuously fed above a bath of batch material, and then withdrawn from the furnace in a molten condition. The temperature necessary for melting and refining the glass depends on the precise composition of batch material, but is between 1300°C and 1550°C. At these temperatures heat transfer is dominated by radioactive transmission, in particular from the furnace crown, which is heated by the flames up to 1650 °C, but also from the flames themselves. In each furnace, designed heat input is arranged to induce recirculating convective currents within the melted batch materials to ensure consistent homogeneity of the finished glass fed to the forming process. The mass of molten glass contained in the furnace is held constant, and the mean residence time is of the order of 72 hours for float glass furnaces.

4.2.2 Primary Melting

Due to the low thermal conductivity of the batch materials the melting process is initially quite slow allowing time for the numerous chemical and physical processes to occur. As the materials heat up the moisture evaporates, some of the raw materials decompose and the gases trapped in the raw materials escape. The first reactions (decarbonisation) occur around 500°C. The raw materials begin to melt between 750°C and 1200°C. First the sand begins to dissolve under the influence of the fluxing agents.

The silica from the sand combines with the sodium oxide from the soda ash and with other batch materials to form silicates. At the same time large amounts of gases escape through the decomposition of the hydrates, carbonates, nitrates and sulphates; releasing water, carbon dioxide, nitrogen and sulphur oxides. The glass melt finally becomes transparent and the melting phase is completed. The volume of the melt is about 35 - 50 % of the volume of the virgin batch materials due to the loss of gases and the elimination of interstitial spaces.

4.2.3 Fining and Homogenisation

The glass melt must be completely homogenised and free of bubbles before it can be formed into the products. This involves the complete dissolution and even distribution of all components and the elimination of all bubbles by refining.

During the melting process gas bubbles are formed mainly from carbon dioxide given off by the decomposition of the carbonate materials (principally soda ash and limestone) and to a much lesser extent from air trapped in the raw materials. These bubbles must be eliminated from the glass melt as they potentially cause defects in the finished product affecting mechanical strength and appearance. The upward movement of bubbles contributes to the physical mixing of the melt necessary to obtain a homogenous material with optimal physical properties. The bubbles rise at speeds determined by their size and the viscosity of the glass. Large bubbles rise quickly and contribute to mixing, while small bubbles move slowly, at speeds that may be small with respect to the larger scale convection currents in the furnace and are thus more difficult to eliminate. Small bubbles remaining in the finished glass are termed "seeds".

Carbon dioxide and the components of air have limited solubility in the glass melt and it is usually necessary to use chemical fining agents to effectively eliminate the small bubbles generated by the melting process. The general principle of chemical fining is to add materials which when in the melt will release gases with the appropriate solubility in the glass. Depending on the solubility of the gas in the glass melt (which is generally temperature dependant) the bubbles may increase in size and rise to the surface or be completely reabsorbed. Small bubbles have a high surface to volume ratio, which enables better exchange between the gas contained in the bubbles and the glass.

The most frequently used fining agent in the glass industry is sodium sulphate which is not taken into account in this study as no information is provided. At approximately 1450°C (1200°C if reducing agents are present) the sodium sulphate decomposes to give sodium oxide (which is incorporated into the glass), gaseous oxides of sulphur, and oxygen. The oxygen bubbles combine with or absorb other gases, particularly carbon dioxide and air, thereby increasing in size and rising to the surface. The gaseous oxides of sulphur are absorbed into the glass, or join the furnace waste gas stream.

The predominance of sodium sulphate as the fining agent is also due to its parallel action as an oxidising agent for adjusting the redox state of the colouring elements in the glass. It is also the least expensive effective fining agent for mass production of glass. Other fining agents include carbon materials and oxides of arsenic and antimony. These are more expensive and have associated potential environmental and health consequences: thus they tend to be used mainly for production of special glass. Sodium nitrate can also be used as a fining/oxidising agent particularly if a high degree of oxidation is required. Calcium sulphate and various nitrates are sometimes used for coloured flat glass.

Homogenisation can also be improved by introducing bubbles of steam, oxygen, nitrogen or more commonly air through equipment at the bottom of the tank. This encourages enhanced circulation and mixing of the glass and improves heat transfer. Some processes, for example for optical glass, may use stirring mechanisms to obtain the high degree of homogeneity required. Another technique for use in small furnaces (for special glass) is known as plaining, and involves increasing the temperature of the glass such that it becomes less viscous and the gas bubbles can rise more easily up to the surface.

The maximum crown temperatures in glass furnaces are: container glass 1600°C, flat glass 1620°C, special glass 1650°C, continuous filament 1650°C, and glass wool 1400°C.

4.2.4 Conditioning

A conditioning phase at lower temperatures follows the primary melting and fining stages. During this process, all remaining soluble bubbles are reabsorbed into the melt. At the same time, the melt cools down slowly to a working temperature between 900°C and 1350°C.

Glass does not flow through the tank in a straight way from the batch feeder to the throat where the glass reaches the working temperature. It is diverted following thermal currents. The batch pile, or the cold mixture of raw materials, is not only melted at the surface, but also from the underside by the molten glass bath. Relatively cold, bubbly glass forms below the bottom layer of batch material and sinks to the bottom of the tank. Appropriate convection currents must bring this material back to the surface, since fining occurs in tank furnaces primarily at the surface of the melt, where bubbles need to rise only a short distance to escape. If thermal currents flow too fast, they inhibit fining by bringing the glass to the conditioning zone too soon. Guiding walls or weirs can be built into the inner tank structure to create ideal glass flow paths.

4.2.5 Emissions to air

In the flat glass sector the most meaningful potential environmental emissions are emissions to air from the melting activities. The substances emitted are stemming from the melting process and presented in Chapter 5.1, Fig. 5.1. Almost all furnaces in this sector are fossil fuel fired (with natural gas or fuel oil), cross-fired regenerative furnaces.

Lead emissions for lead containing glass are very high compared with all other emissions (EPA (1998)). Since normal flat glass contains no lead, no lead emissions are inventoried

4.2.6 Emissions to water

As discussed earlier the main uses of water in this sector are for cleaning, cooling, and batch humidification. The emissions are limited to cleaning water released to surface water. The cleaning water does not present any glass industry specific pollutant that would not be common with any other industrial facility, i.e. inert solids and oil. Cooling system purges contain dissolved salts and water treatment chemicals. Surface water quality depends on the degree of drainage segregation and site cleanliness.

Excluding domestic wastewater, discharges generally only contain glass solids, potentially some oil contamination, and cooling chemicals from water system treatment. Simple abatement techniques such as settlement, screening, oil separators, and neutralising can be found.

As no data are available and as it is assumed that an efficient abatement technique is used, no emissions to water for the flat glass production process are taken into account.

4.2.7 Waste

Wherever possible, batch plant dust is returned to the raw material silos and reused in the process. Rejected batches are gradually fed back into the process by the inclusion of small amounts in subsequent batches. Typically 99 % of the glass waste from the end of the glassmaking process, trims, rejects and damaged glass is remelted.

At the end of a lifetime of the furnace the refractory structure (sometimes not regenerators) is dismantled and replaced. This material is recovered as much as practicable for reuse or sale.

In some sectors of the glass industry, chromium containing refractories are used for the construction of upper walls, crowns and regenerators. The chromium when combined with magnesia form mag-chrome bricks which are very resistant to batch carryover and combustion products at the high temperatures that exist in the regenerator chambers. The chromium used in the preparation of these materials, Cr³⁺, is essentially non-hazardous, has low solubility and presents little risk. However, at high temperatures under alkaline and oxidising conditions, small amounts of the chromium will convert to Cr⁶⁺ during the furnace campaign. Cr⁶⁺ compounds are highly soluble, toxic and carcinogenic.

As other furnace waste, every effort is made at the end of a campaign to have the materials recycled. Where this is not possible, the Cr⁶⁺ content of the used mag-chrome refractories will be determined to ensure that they are correctly classified and disposed of appropriately. The industry is gradually reducing the amount of chromium containing refractories by development and redesign.

Small tonnages of high purity chromic oxide refractories may also be used. They are generally purchased on the basis that at the end of a campaign they will be taken back by the manufacturer for recycling. In some continuous glass filament furnaces large amounts of this material are used.

Most glass raw materials are generally sold in bulk and do not give rise to packaging waste. Waste materials from packing (plastic, cardboard, wood etc) are usually reused or recycled if practicable. Other waste non-specific to the industry is disposed of by conventional means.

Based on above description no waste for the flat glass production process is taken into account.

4.3 Surface Coating

“Coated glass plays a major role in a number of applications such as heat-reflection coating systems on window panes to lower heating costs; solar protection coatings to reduce air conditioning costs in countries with high-intensity solar radiation; coated car windows to reduce thermal increases in the interior; and mirrors used both in home furniture and the automobile industry. There are several processes for applying coating on a surface. However, in order to be applied to glass surfaces coatings must meet several criteria: compatibility with glass properties; ability to form uniform films over large surfaces; ability to be produced economically, compliance with operating safety and environmental standards. Due to these restrictions, primary glass manufacturing companies today use physical vapour deposition (PVD) and chemical vapour deposition (CVD). This study will concentrate on PVD, also known as the sputtering process.”²

Metal coating (in this study mainly using bismuth, silver and nickel-chrome) is applied on float glass by cathodic sputtering in vacuum (Richter et al. (1996)). The flat glass is fed in batch quantities into the coating installation. At ambient pressure the vessel is charged and then vacuum is generated.

“The sputtering process works by passing an electrical current through ionised gas, thus bombarding the surface of a metal cathode with ions. The atoms of the desired metal are vaporised and then deposited in a thin film on the glass surface. The invention of the “planar magnetron” in 1971 increased the effectiveness of the process. This is often called a “soft coat”, because the coating is more susceptible to damage than is hard coat glass when glazed in monolithic forms. Due to its fragility, this soft-coated glass has special handling and processing requirements. Today sputtering is the most important flat-glass process in use.”²

The following description is based on the environmental report of the company Interpane in Germany Interpane (2000). It can be assumed that other companies in Western Europe are using the same principles:

² <http://www.glassonweb.com/articles/print.php?id=197>, access date: March 2003

The metals used are delivered as target materials. After the sputtering process the producer recycles the rest targets. The ancillary and operational materials are mainly argon and oxygen. Argon is used as plasma gas. This inert gas is absolutely innocuous and therefore no exhaust air decontamination installation is necessary. Further the sputtering process uses oxygen as reaction gas. It is escaping to air when not reacting to oxide with the metals. The production process needs also cooling lubricants (oils) and different purifiers. The storage and usage of these water-endangering mediums comply with German environmental laws. For the heating of the operation building a gas boiler, working in condensation mode, is used. The emissions are therefore mainly CO₂, NO_x and SO_x. The cooling water is in case of the German factory used for space heating. A regional energy supplier provides electricity. The emissions from production and cleaning activities are mainly dust emissions of the metals used for coating. These are pumped out from where they originate and separated from the outgoing air by a filter. The concentration at the work places is very low.

The only gas emissions are, except some diffuse solvent emissions, argon and oxygen used in the production process.

The drinking and process water is covered 90% from the drinking water supply and 10% from a rain water reservoir. Water is mainly used in the sputtering process, especially for the cleaning of the flat glass which occurs in a closed loop. The wastewater from cleaning is polluted by solids and the separating agent PMMA (Polymethylmethacrylate). The treatment of these is part of the closed loop process and includes an activated carbon filter, reversing osmosis and an ion exchanger. The losses are replaced by fresh water.

The waste heat from the coating process is absorbed by the cooling water and then used for space heating.

The glass waste is recycled. Fifty percent of the sputtering material is deposited on the coated glasses. The other fifty percent is deposited at the guiding plates of the sputtering chamber and arises as waste with the cleaning process. These plates are cleaned by blast cleaning. The unused targets are recycled.

Other production wastes are industrial waste, woods and waste paper, which is recycled. Waste oil and oil emulsive operating material are collected separately and disposed by a specialised company.

5 Life Cycle Inventories

Ecoinvent database meta information (tables), flows and its representation in the ecoinvent database are shown in (figures):

Flat glass, uncoated, at plant	Tab. 5.3 and Fig. 5.1
Flat glass, coated, at plant	Tab. 5.5 and Fig. 5.2
Flat glass plant	Tab. 5.6 and Fig. 5.6

The chapters start with the background information (ecoinvent database meta information) to the processes and products. This includes essential information for the understanding, which is not picked up again in the report.

5.1 Uncoated Flat Glass

The ecoinvent database meta information of “flat glass, uncoated, at plant” includes all-important and necessary information to use the data correctly. Fig. 5.1 shows all in- and output flows of the product “flat glass, uncoated, at plant”, the allocation to the modules in ecoinvent database V1.1, the sources (mainly IPPC (2001), Frischknecht et al. (1996), Flumroc (2002) and EPA (1998)) and the standard deviation for each datum.

Calculation of the input raw material based on the composition of glass

No information on the amounts of the different input materials was found. Therefore they are calculated based on the composition of the product (Tab. 5.1). This composition is taken from IPPC (2001).

Tab. 5.1 Composition of flat glass (IPPC (2001))

Components of flat glass	percentage (IPPC, 2001)	weight
	[w/w]	[kg/kg Glass]
CaO	8.6	0.0860
MgO	4.1	0.0410
Al ₂ O ₃	0.7	0.0070
K ₂ O	0.3	0.0030
SiO ₂	72.6	0.7260
Na ₂ O	13.6	0.1360
Total	99.9	0.9990

The choice of input raw materials is based on the Environmental Review 1999 of Pilkington³ and Ullmann's Encyclopaedia of Industrial Chemistry (De Jong (1997)): Limestone (CaCO₃), dolomite (MgCO₃), soda ash / salt cake (Na₂CO₃), sand (SiO₂) and feldspar ((K,Na)₂O*Al₂O₃*6SiO₂).

The following procedure leads to the calculation of the weight of the different raw materials:

- For all raw materials the mol-weight is calculated.
- Since all Al₂O₃-input comes from feldspar and Al₂O₃ is not reacting chemically the weight of Al₂O₃-input from feldspar has to be the same as the weight of Al₂O₃ in the glass. Any losses which may occur are not taken into account.
- With the mol-weight and the weight used for 1 kg of flat glass the amount of mol of Al₂O₃ is calculated.
- The same is done for K₂O.
- As either K₂O and Na₂O are components of the feldspar, the sum of their mol amounts must be the same as the amount of mol of Al₂O₃.
- With the mol-weight and the number of mol the weights of K₂O and Na₂O for 1kg of flat glass are calculated.
- The amount of mol of SiO₂ is 6 times the amount of mol of Al₂O₃. With the mol-weight the absolute weight of SiO₂-input from feldspar is calculated.
- CaCO₃ and MgCO₃ decompose in the heating process to CaO and CO₂ and to MgO and CO₂, respectively. Knowing the weight fraction of CaO and MgO in the product, the weight of CaCO₃ and MgCO₃-input are reckoned back.
- The weight of the Na₂CO₃-input is calculated by taking the weight of its Na₂O-fraction and subtracting the Na₂O-input from feldspar.

The geogenic CO₂ emissions from the rocks are also calculated based on the composition of the materials in the product and the amount of raw materials. The results are illustrated in Tab. 5.6.

Tab. 5.2 Input raw materials for flat glass production (IPPC (2001))

³ <http://www.pilkington.com/resources/environmental+review.pdf>, access date: March 2003

Input raw materials		weight [kg / kg glass]	mol	share	[%]	share	[kg]
limestone	CaCO ₃	0.1536	1.5357	CaO in CaCO ₃	56	CO ₂ in CaCO ₃	0.0675
dolomite	MgCO ₃	0.0861	1.0250	MgO in MgCO ₃	52	CO ₂ in MgCO ₃	0.0451
feldspar	6SiO ₂	0.1482	0.4118				0
	Al ₂ O ₃	0.0070	0.0686				0
	either K ₂ O	0.0030	0.0319				0
	or Na ₂ O	0.0023	0.0367				0
sand	SiO ₂	0.5778	9.6294				0
soda ash / salt cake	Na ₂ CO ₃	0.2286	2.1568	Na ₂ O in Na ₂ CO ₃	62	CO ₂ in Na ₂ CO ₃	0.0949
Total weight		1.2066				Total CO₂	0.2075

The CO₂ emissions from combustion are calculated based on the amount of fuel burned.

Sulphur trioxide (SO₃) is a part of the composition of glass. As no Sulphur is mentioned as raw material in Ullmann's Encyclopaedia of Industrial Chemistry and in the Environmental Review 1999 of Pilkington⁴, it is added separately.

All other emission data are taken from the document on "Best Available Techniques in the Glass Manufacturing Industry" (IPPC (2001)). The data are average data considering unabated as well as abated furnaces. It is assumed that 50% of the studied furnaces are abated and 50% unabated. Some additional emissions like VOC, carbon monoxide and lead as well as particle size distribution are taken from EPA (1998). These data are based on uncontrolled processes. The mistake that may be introduced is considered acceptable, though possibly increasing the emissions from the process.

It is assumed that the amount of tin to compensate the losses is released to air.

The typical production amount of a flat glass production plant is assumed about 50'000 tons per year. The life span of the flat glass plant is assumed 50 years.

⁴ <http://www.pilkington.com/resources/environmental+review.pdf>, access date: March 2003

Tab. 5.3 Ecoinvent database meta information for the product "flat glass, uncoated, at plant"

Name	flat glass, uncoated, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the whole manufacturing process to produce flat glass (raw material provision, cullet addition, melting process, forming process (on float bath), cooling process (annealing lehr), cutting process and storage), internal processes (transport, etc.) and infrastructure. No administration and packing is included.
Amount	1
Local Name	Flachglas, unbeschichtet, ab Werk
Synonyms	plate//float//Flach
General Comment to reference function	The amount of raw materials is calculated based on the composition of flat glass. The total amount of energy is partitioned in 58% from natural gas, 38% heavy fuel oil and 5% electrical. The amount of tin yearly added to the tin bath is assumed to leave the system by air, as part of the particulates. The same with lead. Both emissions are double counted: as particulate and as emission to air. Most other emissions are a mean value of plants with and plants without abatement. It is assumed that half of water evaporates and other half is treated.
Start Date	1996
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH- and DE-modules have been used as proxy
Technology text	Basic principle is pouring molten glass onto a bath of molten tin and forming a ribbon with the upper and lower surfaces becoming parallel under the influence of gravity and surface tension. Average technique used in European flat glass production.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

General Flow information					Representation in ecoinvent								Uncertainty					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
sand (silicon oxide, SiO2)	→	uncoated flat glass production (Inputs, part 1)					construction materials	additives	No	DE	silica sand, at plant	5.78E-01	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry / p. 46	1	1.62	(1,5,5,1,1,5);	
soda ash / salt cake (sodium oxide Na2O)	→						chemicals	inorganics	No	RER	soda, powder, at plant	2.29E-01	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry / p. 46	1	1.62	(1,5,5,1,1,5);	
limestone (CaCO3)	→						construction materials	others	No	CH	limestone, milled, packed, at plant	1.54E-01	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry / p. 46	1	1.62	(1,5,5,1,1,5);	
dolomite (MgCO3)	→				limestone used as proxy		construction materials	others	No	CH	limestone, milled, packed, at plant	8.61E-02	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry / p. 46	1	1.62	(1,5,5,1,1,5);	
feldspar (6SiO2)	→				limestone used as proxy		construction materials	others	No	CH	limestone, milled, packed, at plant	1.48E-01	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry / p. 46	1	1.62	(1,5,5,1,1,5);	
feldspar (Al2O3)	→				limestone used as proxy		construction materials	others	No	CH	limestone, milled, packed, at plant	7.00E-03	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry / p. 46	1	1.62		
feldspar (K2O)	→						construction materials	others	No	CH	limestone, milled, packed, at plant	3.00E-03	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry / p. 46	1	1.62		
feldspar (Na2O)	→				limestone used as proxy		construction materials	others	No	CH	limestone, milled, packed, at plant	2.28E-03	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry / p. 46	1	1.62	(1,5,5,1,1,5);	
minor materials	→				colour modifiers, fining and incidental impurities from raw materials; trace substances. No data available.										IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry / p. 46			
hydrogen	→				Assumption for the leakage of the gas mixture from the bath: about 1% per hour. Total volume is assumed to be 3'000 m3. Total anual release is 200'000 m3/year. Assumption that 1% of leaking gas is hydrogen, with density 0.09 kg/m3. Standard deviation has been estimated.		chemicals	inorganics	No	RER	hydrogen, liquid, at plant	3.60E-06	kg	estimation and calculation	1	5.00	(,,,,); Estimation of standard deviation	
nitrogen	→				assumption that about 1% gas leakage per hour (total volume assumed to be 3'000m3): 200'000 m3/year, assumption that 99% of leaking gas is nitrogen. Density: 1.25 kg/m3. Standard deviation has been estimated		chemicals	inorganics	No	RER	nitrogen, liquid, at plant	4.95E-03	kg	estimation and calculation	1	5.00	(,,,,); Estimation of standard deviation	

Fig. 5.1 Flows for “flat glass, uncoated, at plant” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty			
Input		Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
transport of raw materials to plant	→	uncoated flat glass production (Inputs part 2)		standard distance: transport by lorry 32t with distance of: 50km, for gravel/sand (proxy for all raw materials) assumption 57% of total energy consumption (heat of reaction to form glass, heat required to raise glass temp. and heat content of gas released during melting). Assumption that plant connected to high pressure pipeline.	transport systems	road	No	RER	transport, lorry >16t, fleet average	6.03E-02	tkm	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry	1	2.09	(4,5,n.a,n.a,n.a,n.a);
total energy consumption: natural gas	→			assumption 38% of total energy consumption (heat of reaction to form glass, heat required to raise glass temp. and heat content of gas released during melting).	natural gas	fuels	No	RER	natural gas, high pressure, at consumer	4.56E+00	MJ	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry / p. 73 / p. 86 / p. 252	1	1.32	(3,5,1,1,n.a.,5);
total energy consumption: heavy fuel oil	→			assumption 38% of total energy consumption (heat of reaction to form glass, heat required to raise glass temp. and heat content of gas released during melting).	oil	fuels	No	RER	heavy fuel oil, at regional storage	7.38E-02	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry / p. 73 / p. 86 / p. 252	1	1.32	(3,5,1,1,n.a.,5);
total energy consumption: electricity	→			5% of total energy consumption (heat of reaction to form glass, heat required to raise glass temp. and heat content of gas released during melting).	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.11E-01	kWh	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry / p. 73 / p. 86 / p. 252	1	1.32	(3,5,1,1,n.a.,5);
water	→				resource	in water			Water, cooling, unspecified natural origin	7.00E-04	m3	Frischknecht et al. 1996	1	2.10	(2,5,3,n.a.,n.a.,n.a.);
tin	→			2 tons tin/year, float glass production amount 600 - 800 t/day; with 600 t/day and 364 production days: 218'400 t/y	metals	extraction	No	RER	tin, at regional storage	9.16E-06	kg	personal communication Mr. Pioch, Pilkington Gladbeck	1	12.00	(,,,,); Estimation of standard deviation
steel spare parts	→			2 - 3 tons steel/year, float glass production amount 600 - 800 t/day; calculation with 3 t/y steel, 600 t/day and 364 production days	metals	extraction	No	RER	steel, converter, unalloyed, at plant	1.37E-05	kg	personal communication Mr. Pioch, Pilkington Gladbeck	1	1.26	(3,4,1,3,1,5);
fireproof brick of furnace	→			estimation: area of fireproof bricks: 20m2; assumption of thickness of bricks: 0.3m, interval of replacement of all bricks: yearly, density 2'600kg/m3	construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	3.12E-04	kg	estimation and calculation	1	10.00	(,,,,); Estimation of standard deviation
fireproof brick of float bath	→			estimation: Length of bath: 400m2, ratio length to width: 3:1, length: 34.6m, width: 11.5m glass mass filling level: 1.5m; area of fireproof bricks: 140m2; assumption of thickness of bricks: 0.3m, interval of replacement of all bricks: yearly, density: 2'600kg/m3	construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	7.53E-04	kg	estimation and calculation	1	10.00	(,,,,); Estimation of standard deviation
infrastructure (building, land-use, melting furnace etc.) [unite]	→			assumed total yearly production: 50'000 t. Total production in 50 years lifetime: 2'500'000'000 kg. "flat glass plant" is based on a production volume of 83'000t/a.	glass	construction	Yes	RER	flat glass plant	2.41E-10	unit		1	3.14	(4,5,1,3,1,5);

Fig. 5.1 Flows for “flat glass, uncoated, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
uncoated flat glass production (Output part 1)		geogenic carbon dioxide	geogenic CO ₂ ; included from calculation of interrelationship between input raw material and composition of product.	air	unspecified			Carbon dioxide, fossil	2.08E-01	kg	Calculation	1	1.38	(4,5,1,1,n.a.,5);
		Carbon dioxide for combustion	CO ₂ from combustion; included from calculation of emission from burning of used amount of natural gas and heavy fuel oil, with unabated furnace	air	unspecified			Carbon dioxide, fossil	4.85E-01	kg	fuel consumption: IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry, CO ₂ -calculation based on fuel consumption	1	1.38	(4,5,1,1,n.a.,5);
		Oxides of Nitrogen (as NO ₂)	assumption: 50% of considered plants with abatement and 50% without abatement	air	unspecified			Nitrogen oxides	3.28E-03	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry, p. 84	1	2.26	(,,,,); Calculation of standard deviation based on min. and max. values
		oxides of sulphur (as SO ₂)	assumption: 50% of considered plants with abatement and 50% without abatement	air	unspecified			Sulfur dioxide	4.04E-03	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry, p. 84	1	2.63	(,,,,); Calculation of standard deviation based on min. and max. values
		particulate matter, < 2.5um, including lead, excluding tin	assumption: 50% of considered plants with abatement and 50% without abatement. Of total particles, 80% < 2.5um	air	unspecified			Particulates, < 2.5 um	1.75E-04	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry, p. 84, EPA (1998), Glass Manufacturing, p11.15-9	1	2.75	(,,,,); Calculation of standard deviation based on min. and max. values
		particulate matter, < 2.5um, only tin	tin in particulates, additional as the only information available is the yearly amount fed into the system; it is assumed that tin leaves the system by air	air	unspecified			Particulates, < 2.5 um	9.16E-06	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry, p. 84, EPA (1998), Glass Manufacturing, p11.15-9	1	52.42	(,,,,); Calculation of standard deviation based on min. and max. values
		particulate matter, > 2.5um, < 10um	assumption: 50% of considered plants with abatement and 50% without abatement, Of total particles 10% between 2.5um and 10um	air	unspecified			Particulates, > 2.5 um, and < 10um	2.30E-05	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry, p. 84, EPA (1998), Glass Manufacturing, p11.15-9	1	2.61	(,,,,); Calculation of standard deviation based on min. and max. values
		particulate matter, > 10um	assumption: 50% of considered plants with abatement and 50% without abatement Of total particles 10% > 10um	air	unspecified			Particulates, > 10 um	2.30E-05	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry, p. 84, EPA (1998), Glass Manufacturing, p11.15-9	1	2.61	(,,,,); Calculation of standard deviation based on min. and max. values

Fig. 5.1 Flows for “flat glass, uncoated, at plant” and their representation in the ecoinvent database (cont)

Attention: The CO₂ emission in the data v2.0 is wrong. Erroneously 0.256 kg instead of the total of 0.693 kg is inventoried in the database!

General Flow information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
uncoated flat glass production (Output part 2)	→	fluorides (HF)	assumption: 50% of considered plants with abatement and 50% without abatement	air	unspecified			Hydrogen fluoride	2.10E-05	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry, p. 84	1	3.33	(,,,,); Calculation of standard deviation based on min. and max. values
	→	chlorides (HCl)	assumption: 50% of considered plants with abatement and 50% without abatement	air	unspecified			Hydrogen chloride	9.25E-05	kg	IPPC (2001), Best Available Techniques in the Glass Manufacturing Industry, p. 84	1	2.38	(,,,,); Calculation of standard deviation based on min. and max. values
	→	VOC	No information on Methan. Therefore NMVOC is chosen as Proxy	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	5.00E-05	kg	EPA (1998), Glass Manufacturing, p. 11.15-7	1	2.00	(,,,,); Calculation of standard deviation based on min. and max. values
	→	carbon monoxide	uncontrolled emission	air	unspecified			Carbon monoxide, fossil	5.00E-05	kg	EPA (1998), Glass Manufacturing, p. 11.15-7	1	5.00	(,,,,); Calculation of standard deviation based on min. and max. values
	→	tin	data over all glass production processes, assumption that the total amount of tin entering the system is leaving again the system as air emission	air	unspecified			Tin	9.13E-06	kg	estimation	1	12.00	(,,,,); Estimation of standard deviation
	→	municipal solid waste	it is assumed that the amount of 50'000 kg/y is about the same as for the rockwool production (about same yearly production amount)	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.10E-03	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);
	→	waste water to treatment	water is recycled or treated using standard techniques. It is assumed that half of water evaporates and other half is treated	waste management	wastewater treatment	No	CH	treatment, sewage, from residence, to wastewater treatment, class 2	3.50E-04	m3	EPA (1998), Glass Manufacturing, p. 11.15-7	1	5.00	(,,,,); Calculation of standard deviation based on min. and max. values
	→	flat glass, uncoated, unpacked, at plant		glass	construction	No	RER	flat glass, uncoated, at plant	1.00E+00	kg				

Fig. 5.1 Flows for “flat glass, uncoated, at plant” and their representation in the ecoinvent database (cont)

5.2 Coated Flat Glass

The basis for the data input is the environmental report of a German factory called “Interpane glass coating company”. The data represent the situation in the year 2000 and can be assumed to be typical for the average western European production.

The total amount of sputtered glass produced in the “Interpane glass coating company” in year 2000 was 5'630'000 m². With an estimated average thickness of 4 mm and an average density of 2'500 kg/m³ the production amount was equal to about 56.3 Mio kg for the year 2000.

In the case of “Interpane glass coating company” the production of flat glass and the coating process have not the same location. The transport distances for the flat glass used as raw material for the sputtering process are assumed to be 600 km by train and 100 km by lorry.

The raw materials for sputtering are bismuth, silver and nickel chromium. The total amount of these materials is represented by 50% nickel and 50% chromium, since no modules for silver and bismuth exist in the ecoinvent database.

The Interpane report includes information on emissions from fuel combustion. These emissions are not inventoried since they are included in the energy conversion datasets.

The ecoinvent database meta information of “flat glass, coated, at plant” includes all-important and necessary information to use the data correctly. Fig. 5.2 shows all in- and output flows of the product “flat glass, coated, at plant”, the allocation to the modules in ecoinvent database V1.1, the sources (Interpane (2000)) and the standard deviation for each datum.

5.3 Flat Glass Plant

The life span of the portrayed German plant is estimated to be 50 years and the yearly production amount is 83'000 tons. It is assumed that all in- and output data would be valid for plants with a yearly production of about 50'000 to 100'000 tons.

The information reported in Tab. 5.4 is supplied by a German company, which wishes to remain anonymous.

Tab. 5.4 Portray of German plant

Plant in Germany		Comments / Assumptions
Area of the site	m ²	
total	128100	
build	73200	
not build	54900	Vegetation
roads	21400	
Size and type of building	m ²	
production	28300	Steel construction hall
storage	20000	Steel construction hall
administration	3500	Multi-storey building
Location of company (e.g. industrial area, trade area, city, rural area)	No information	
Number of employees	1400	

The equipment within the plant used for producing flat glass has been roughly estimated with 100 tons. The life span of the machines is assumed to be 20 years.

It is assumed that the machines are mostly made out of steel and this material is reused to 100% at the end of life. Further it is also assumed that the tin in the bath is reused to 100% at the end of life. As no data concerning filter replacement is available and no data concerning inert materials after dismantling are available, they are not taken into account.

The ecoinvent database meta information of “flat glass plant” includes all-important and necessary information to use the data correctly. Fig. 5.6 shows all in- and output flows of the product “flat glass plant”, the allocation to the modules in ecoinvent database V1.1, the sources (mostly anonymous data supplier and estimations) and the standard deviation for each datum.

Tab. 5.5 Ecoinvent database meta information for the product “flat glass, coated, at plant”

Name	flat glass, coated, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes all measured In- and Output materials and energy carriers reported for a German glass coating plant during its operation (coating process, internal transports, packing and administration).
Amount	1
Local Name	Flachglas, beschichtet, ab Werk
Synonyms	sputtering//bedampfen//plate//sputtern//Kathodenzerstäubung// Vergütungsschicht//Wärmeschutzschicht
General Comment to reference function	All data are based on the environmental report of the company "Interpane" (DE) for the year 2000. The Targets used for sputtering are bismuth, silver and nickel-chromium. Only nickel-chromium is modelled for missing silver and bismuth modules in the database.
Start Date	2000
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from one company in Germany is extrapolated to RER. For some exchanges CH-modules are used as proxy
Technology text	Metal coating (mainly Bismuth, Silver and Nickel-Chrome) are applied on float glass by cathodic sputtering in vacuum. The installation is fed in batch quantities. At ambient pressure the vessel is charged and then vacuum is generated. The sputtering process works by passing an electrical current through ionised gas, thus bombarding the surface of a metal cathode with ions. The atoms of the desired metal are vaporised and then deposited in a thin film on the surface of glass
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Environmental report 2000 of the company Interpane
Extrapolations	See geography
Uncertainty Adjustments	none

General Flow information					Representation in ecoinvent							Uncertainty				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
flat glass, uncoated, unpacked, at plant	→	flat glass coating process (input part 1)			assumption average thickness of glass: 4mm, average density: 2'500 kg/m3, yearly production amount: 5'630'000m2 * 0.004m * 2'500kg/m3 = 56.3 Mio. kg	glass	construction	No	RER	flat glass, uncoated, at plant	1.02E+00	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
tansportation flat glass from one plant to the coating plant per train	→				assumed average transport distance: 600km per train and 100 per lorry	transport systems	train	No	RER	transport, freight, rail	6.12E-01	tkm	Interpane (2000)	1	2.09	(1,5,1,1,1,5);
transportation flat glass from one plant to the coating plant per lorry	→				assumed average transport distance: 600km per train and 100 per lorry	transport systems	road	No	RER	transport, lorry >16t, fleet average	1.02E-01	tkm	Interpane (2000)	1	2.09	(1,5,1,1,1,5);
infrastructure, coating plant	→				plant of flat glass production has been used as proxy	glass	construction	Yes	RER	flat glass plant	3.55E-10	unit	Interpane (2000)	1	10.00	(,,,,); Estimation of standard deviation
water, tab	→				90% drinking water, 10% rain water	water supply	production	No	RER	tap water, at user	1.00E-04	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
water, rain	→				10% rain water	resource	in water			Water, unspecified natural origin	1.11E-05	m3	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
electricity consumption	→					electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	6.79E-02	kWh	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
natural gas, for heating	→					natural gas	heating systems	No	RER	natural gas, burned in boiler condensing modulating >100kW	5.29E-05	MJ	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
diesel used for industrial machines (lorry, fork lift truck, etc.)	→				Total amount of diesel used for vehicles within the plant: 30'000 liter = 30 m3; Density: 840 kg/m3 -> 25'200 kg; net calorific value: 42.8 MJ/kg -> 1'078'560 MJ; for 56'300'000 kg -> 0.019157 MJ/kg, "diesel, burned in building machines" used as proxy	construction processes	machinery	No	GLO	diesel, burned in building machine	1.92E-02	MJ	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
petrol used for transporition (lorry, fork lift truck, etc.)	→				Total amount of petrol per year: 7'000l; petrol consumption per vkm in RER: 0.09l/vkm, "transport, passenger car" used as proxy, load: 1.59 passenger per car	transport systems	road	No	RER	transport, passenger car	2.20E-03	pkm	Interpane (2000)	1	2.09	(1,5,1,1,1,5);

Fig. 5.2 Flows for "flat glass, coated, at plant" and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty					
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
bismuth	→	flat glass coating process (Input part 2)			Total amount of bismuth, silver and nickel-chromium: 6'721 t, no modul to allocate								Interpane (2000)				
silver	→				Total amount of bismuth, silver and nickel-chromium: 6'721 t, no module to allocate								Interpane (2000)				
nickel-chromium (part nickel)	→				Total amount of bismuth, silver and nickel-chromium: 6'721 kg, 50% of total amount allocated to nickel	metals	extraction	No	GLO	nickel, 99.5%, at plant	5.97E-05	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);	
nickel-chromium (part chromium)	→				Total amount of bismuth, silver and nickel-chromium: 6'721 kg, 50% of total amount allocated to chromium	metals	extraction	No	RER	chromium, at regional storage	5.97E-05	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);	
argon gaseous	→				density of gaseous argon: 1.78 kg/m3	chemicals	inorganics	No	RER	argon, liquid, at plant	1.33E-05	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);	
argon compressed	→				density of liquid argon: 1400 kg/m3 = 1.4 kg/l	chemicals	inorganics	No	RER	argon, liquid, at plant	2.06E-04	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);	
oxygen	→				density of liquide oxygen: 1'150 kg/m3 = 1.15 kg/l	chemicals	inorganics	No	RER	oxygen, liquid, at plant	1.43E-03	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);	
helium	→				Argon used as Proxy for helium, density of liquide helium: 0.147 kg/l	chemicals	inorganics	No	RER	argon, liquid, at plant	3.13E-07	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);	
polymethacrylates	→					plastics	polymers	No	RER	polymethyl methacrylate, beads, at plant	3.20E-05	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);	
cleaning material	→					chemicals	organics	No	GLO	solvents, organic, unspecified, at plant	2.66E-06	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);	
lubricants	→					chemicals	organics	No	RER	lubricating oil, at plant	2.02E-04	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);	
propane	→					oil	fuels	No	RER	propane/ butane, at refinery	2.02E-04	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);	
acetylene	→					chemicals	organics	No	CH	acetylene, at regional storehouse	2.02E-04	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);	
nitrogen	→					density: 0.808 kg/l	chemicals	inorganics	No	RER	nitrogen, liquid, at plant	2.87E-06	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
welding gases	→					argon used as Proxy, density of liquide argon: 1400 kg/m3 = 1.4 kg/l	chemicals	inorganics	No	RER	argon, liquid, at plant	2.26E-04	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
paper	→						paper & cardboard	graphic paper	No	RER	paper, woodfree, uncoated, at regional storage	1.81E-04	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
packing equipment (mainly polyethylen)	→					processing	plastics	processing	No	RER	packaging film, LDPE, at plant	2.02E-04	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);

Fig. 5.3 Flows for “flat glass, coated, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
flat glass coating process (Output part 1)		waste heat	Same amount as electricity used	air	unspecified			Heat, waste	2.45E-01	MJ	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
		waste water	glass production effluent used as Proxy for coating process	waste management	wastewater treatment	No	CH	treatment, glass production effluent, to wastewater treatment, class 2	5.11E-05	m3	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
		carbon monoxide	Emissions from combustion of petrol, diesel and natural gas. They are included in the datasets chosen to represent the combustion of these fuels.						2.27E-05	kg	Interpane (2000)	1	5.10	(1,5,1,1,1,5);
		carbon dioxide							4.21E-02	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
		nitrogen oxides							1.15E-04	kg	Interpane (2000)	1	1.62	(1,5,1,1,1,5);
		sulfur dioxide							1.79E-04	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
		particulates							6.13E-05	kg	Interpane (2000)	1	3.09	(1,5,1,1,1,5);
		methane							9.75E-05	kg	Interpane (2000)	1	1.62	(1,5,1,1,1,5);
		Non-methane hydrocarbons							1.78E-05	kg	Interpane (2000)	1	1.62	(1,5,1,1,1,5);

Fig. 5.4 Flows for “flat glass, coated, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent							Uncertainty					
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
flat glass coating process (Output part 2)	→	scrap glass				waste management	inert material landfill	No	CH	disposal, glass, 0% water, to inert material landfill	1.96E-02	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
	→	oil saturated waste	Density of oil: 0.86 kg/l			waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	1.44E-05	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
	→	machine oil	Density of oil: 0.86 kg/l			waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	8.40E-06	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
	→	paper and paperboard waste	assumption: 50% of total: paper			waste management	municipal incineration	No	CH	disposal, paper, 11.2% water, to municipal incineration	7.48E-05	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
	→	paper and paperboard waste	assumption: 50% of total: paperboard			waste management	inert material landfill	No	CH	disposal, packaging cardboard, 19.6% water, to inert material landfill	7.48E-05	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
	→	scrap iron	is recycled								9.09E-05	kg	Interpane (2000)			
	→	mixed municipal solid waste				waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	2.09E-04	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
	→	wood	module "disposal, building" used as proxy			waste management	building demolition	No	CH	disposal, building, waste wood, untreated, to final disposal	1.04E-04	kg	Interpane (2000)	1	1.30	(1,5,1,1,1,5);
	→	glass, coated and packed	assumption average thickness of glass: 4mm, average density: 2'500 kg/m3, yearly production amount: 5'630'000m2 * 0.004m * 2'500kg/m3 = 56.3 Mio. kg			glass	construction	No	RER	flat glass, coated, at plant	1.00E+00	kg	Interpane (2000)			

Fig. 5.5 Flows for “flat glass, coated, at plant” and their representation in the ecoinvent database (cont)

Tab. 5.6 Ecoinvent database meta information for the product "flat glass plant"

Name	flat glass, coated, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes all measured In- and Output materials and energy carriers reported for a German glass coating plant during its operation (coating process, internal transports, packing and administration).
Amount	1
Local Name	Flachglas, beschichtet, ab Werk
Synonyms	sputtering//bedampfen//plate//sputtern//Kathodenzerstäubung// Vergütungsschicht//Wärmeschutzschicht
General Comment to reference function	All data are based on the environmental report of the company "Interpane" (DE) for the year 2000. The Targets used for sputtering are bismuth, silver and nickel-chromium. Only nickel-chromium is modelled for missing silver and bismuth modules in the database.
Start Date	2000
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from one company in Germany is extrapolated to RER. For some exchanges CH-modules are used as proxy
Technology text	Metal coating (mainly Bismuth, Silver and Nickel-Chrome) are applied on float glass by cathodic sputtering in vacuum. The installation is fed in batch quantities. At ambient pressure the vessel is charged and then vacuum is generated. The sputtering process works by passing an electrical current through ionised gas, thus bombarding the surface of a metal cathode with ions. The atoms of the desired metal are vaporised and then deposited in a thin film on the surface of glass
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Environmental report 2000 of the company Interpane
Extrapolations	See geography
Uncertainty Adjustments	none

General Flow information					Representation in ecoinvent							Uncertainty						
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
buildings for production	→	flat glass plant				area: 28'300m2	construction processes	buildings	Yes	CH	building, hall, steel construction	2.83E+04	m2	data supplier wants to stay anonymous	1	3.14	(1,5,1,3,3,5);	
buildings for storage	→					area: 20'000m2	construction processes	buildings	Yes	CH	building, hall, steel construction	2.00E+04	m2	data supplier wants to stay anonymous	1	3.14	(1,5,1,3,3,5);	
buildings for administration	→					area: 3'500m2, height 6 floors, each with a hight of 4m	construction processes	buildings	Yes	RER	building, multi-storey	8.40E+04	m3	data supplier wants to stay anonymous	1	3.14	(1,5,1,3,3,5);	
land use area built with buildings (occupation)	→					estimated service life of plant: 50 years	resource	land			Occupation, industrial area, built up	3.66E+06	m2a	data supplier wants to stay anonymous	1	1.67	(1,5,1,3,3,5);	
land use area built with roads (occupation)	→					estimated service life of plant: 50 years	resource	land			Occupation, traffic area, road embankment	1.07E+06	m2a	data supplier wants to stay anonymous	1	1.67	(1,5,1,3,3,5);	
land use area not built-up (occupation)	→					estimated service life of plant: 50 years	resource	land			Occupation, industrial area, vegetation	2.75E+06	m2a	data supplier wants to stay anonymous	1	1.67	(1,5,1,3,3,5);	
land use area built with buildings (transformation to)	→						resource	land			Transformation, to industrial area, built up	7.32E+04	m2	data supplier wants to stay anonymous	1	2.14	(1,5,1,3,3,5);	
land use area built with roads (transformation to)	→						resource	land			Transformation, to traffic area, road embankment	2.14E+04	m2	data supplier wants to stay anonymous	1	2.14	(1,5,1,3,3,5);	
land use area not built-up (transformation to)	→						resource	land			Transformation, to industrial area, vegetation	5.49E+04	m2	data supplier wants to stay anonymous	1	2.14	(1,5,1,3,3,5);	
land use (transformation from)	→						resource	land			Transformation, from unknown	1.28E+05	m2	data supplier wants to stay anonymous	1	2.14	(1,5,1,3,3,5);	
tin in bath	→					initial load	metals	extraction	No	RER	tin, at regional storage	2.00E+05	kg	personal communication Mr. Pioch, Pilkington Gladbeck	1	1.26	(3,4,1,3,1,5);	
equipment (furnace (exkl. fireproof brick), float bath, annealing lehr, warehouse, stacking)	→					very rough non-qualified estimation: 100t machine with a life time of 20 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	3.00E+05	kg	estimation	1	5.00	(,,,,); Estimation of standard deviation	
	→		steel to recycling										4.00E+04	kg	assumption			
	→		tin to recycling										2.00E+05	kg	assumption			
	→	plant, flat glass				all information for a production of about 50'000 to 100'000t of different glass products in a plant in germany	glass	construction	Yes	RER	flat glass plant	1.00E+00	unit					

Fig. 5.6 Flows for “flat glass plant” and their representation in the ecoinvent database

6 Data quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed information compare the different flow charts and the ecoinvent database meta information.

Most of the emissions from the production of flat glass are based on the document “Best Available Techniques in the Glass Manufacturing Industry” (IPPC (2001)). The standard deviation for a lognormal distribution has either been calculated based on the indicated minimum and maximum value or just roughly estimated. The calculation contains a division of the maximum value by the mean value. The same procedure is used for the emissions taken from EPA (1998).

Most of the data concerning the sputtering or coating of flat glass are based on the environmental report of only one German factory. This is taken into account in the uncertainty, to adopt to a wider geographical area.

7 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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- | | |
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Part XII

Glass Products and Processing for Solar Collectors

Data v2.0 (2007)

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Summary

This section describes the production and processing of different glass products used for the construction of solar collectors. Low-iron and borosilicate glasses are used for solar collectors due to their specific advantages which are a high solar transmission and a high resistance, respectively. Antireflection etching or coatings are applied to the glass in order to improve the efficiency of the solar collectors. Glass can be structured in order to make it milky or to change the appearance. Toughening is used in order to improve the resistance against heat and damages. Collector glass tubes are a special product used for solar collectors. The unit process raw data have been provided by manufacturers or assessed with literature data.

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1 Introduction

This section describes the production and processing of different glass products necessary for the construction of solar collectors. The following products and processes are investigated in this chapter:

- Glass products
 - Low-iron glass
 - Borosilicate glass
 - Structured glass
- Processing of glass
 - Tempering
 - Antireflection coatings
- Production of solar glass tube collectors
 - Glass tubes
 - Glass tube solar collectors

2 Reserves and Resources of Material

Silica sand and soda are the basic materials for manufacturing of glass. Coating of solar glasses uses only small amounts of materials (different metal oxides). Besides thermal energy (mainly from fossil fuels) is necessary for the glass melting process.

3 Characterisation and Use of the Products

The products investigated in this section are used for the construction of solar collectors. Special glasses for solar collectors have to fulfil certain requirements. They should allow a good transmission of solar radiation, they have to be heat resistant and they must be strong enough e.g. in winter if there is a lot of snow lying on the glass. Therefore special types of basic glass are used and the products are treated in specific ways to improve the characteristics of these glasses. Some of the products and processes investigated here can also be used for other types of applications than the solar collectors, i.e. for photovoltaic systems.

Most of today's flat glass production is used in the building industry. The market for special solar glasses is very small. The share for solar glass is estimated to be less than 1% of the total glass production.

For the production of solar collectors low-iron (or water white crystal) glass is used. There it covers the collector plate. Low-iron glass is practically colourless and allows a better transmission of solar radiation (Starzner & Wurmer-Weiß 2000).

Evacuated tubes for solar collectors, pipelines, light bulbs, photo chromic glasses, sealed-beam headlights, laboratory ware, and bake ware are examples of borosilicate glass products (Corning 2002). It has a high resistance to temperature change (necessary when heated by the sun) and chemical corrosion. Borosilicate glass is not as convenient to fabricate as either normal lime or lead glass and has higher cost than lime glass (Corning 2002).

A light textured (structured) surface or stippled pattern of solar glass provides light diffusion. It is used to change the appearance of the absorber and to make the absorber metal sheets behind it invisible. The texturing is also used for a better surface for adhesion of solar cells. It can be applied to one or both sides of the glass.

Anti-reflection, commonly abbreviated to AR, is an effect used to improve the performance of solar collectors. There are different possible processes to reduce the reflection of the flat glass surface. The effect can be achieved by making a micro structure on the flat glass surface. For that purpose it is etched, e.g. with hydrofluoric acid (HF). A thin film can be made by making a porous layer with fluorosilicic acid. Or it can be achieved by destructive interference. Such a thin film AR-coating can either be produced by sputtering technology or a sol-gel coating (Glässer 2000).

The sputtering technology is used today for several 100 million m² of flat glass coating per year in the world. This is mainly for insulation glasses. It is the most important technology for glass coatings (Glässer 2000).

Glass evacuated tubes are the key component of a solar glass tube collector. The example for an evacuated tube in Fig. 3.1 consists of two glass tubes (but there are also absorbers with only one tube on the market). The outer tube is made of extremely strong transparent borosilicate glass, that is able to resist impact from hail up to 25mm in diameter. The inner tube is also made of borosilicate glass, but coated with a special selective coating which features a good solar absorption and minimal heat reflection properties. The air is withdrawn (evacuated) from the space between the two glass tubes to form a vacuum, which eliminates conductive and convective heat loss (Focus Solar 2002).

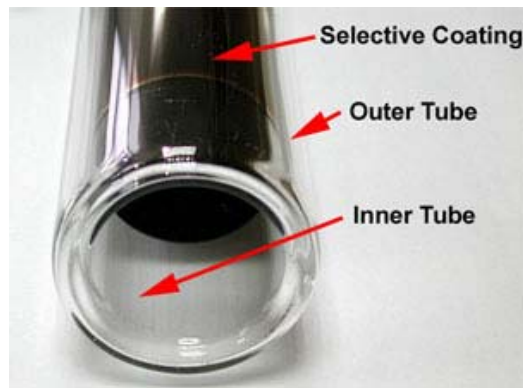


Fig. 3.1 Glass evacuated tube for solar collectors

Evacuated tube solar collectors provide good results on cloudy days. Wind and low temperatures also have less of an effect on the function of evacuated tubes when compared to flat plate solar collectors due to the insulating properties of the vacuum. The efficiency of a solar collector is dependent upon a number of factors. Tab. 3.1 shows basic specifications for evacuated tubes.

Tab. 3.1 Example for basic specifications of an evacuated collector tube

Length	1500mm
Outer tube diameter	47mm
Inner tube diameter	37mm
Glass thickness	1.6mm
Material	Borosilicate Glass 3.3
Absorptive Coating	Graded Al-N/Al
Absorptance	>92% (AM1.5)
Emittance	<8% (80°C)
Vacuum	<5x10 ⁻³ Pa
Stagnation Temperature	>200°C

4 Life Cycle Inventories

4.1 Production of Low-Iron Glass

Sand with a low-iron content (< 150ppm) is used for the production of this glass. The production process itself is very similar to the production of normal window glass.¹ The basic material low-iron silica sand and low-iron lime stone are not found very common. Thus some German manufacturers have to import this material (Starzner & Wurmer-Weiß 2000).

Low-iron glass is a very special product and there are only a few companies in the world (e.g. in DE, the US and Australia) that produce this glass. The normal flat glass production facility is used to produce a batch of low-iron glass. Therefore the raw products are changed and low-iron glass is produced for a few weeks in the same plant that is normally used for window glass.²

Due to the limited number of production plants a higher transport distance of the sand (500km by lorry) and the finished product (plus 6000km by ship) is assumed for this type of glass (Tab. 4.3). Other impacts are the same as for manufacturing normal float glass.

Tab. 4.1 Unit process raw data and data quality indicators for the low-iron glass

	Name	Location	Infrastruct	Unit	solar glass, low-iron, at regional storage	Uncertal Standard Deviation	GeneralComment
	Location				RER		
	InfrastructureProcess				0		
	Unit				kg		
product	solar glass, low-iron, at regional storage	RER	0	kg	1.00E+0		
technosphere	flat glass, uncoated, at plant	RER	0	kg	1.00E+0	1 1.54 (3,na,1,5,4,na); Assumption for a comparable product	
	transport, transoceanic freight ship	OCE	0	tkm	6.00E+0	1 2.01 (2,3,1,1,1,na); Additional transport from US, 6000km	
	transport, lorry 32t	RER	0	tkm	3.75E-1	1 2.01 (2,3,1,1,1,na); Additional transport of sand (75%), 500km	

4.2 Production of Borosilicate Glass

Borosilicate glass is any silicate glass having at least 5% of boric oxide in its composition. Tab. 4.2 shows the composition of borosilicate glass. This glass contains less sand and soda and therefore more feldspar and boron trioxide than normal lime-sodium glass (Starzner & Wurmer-Weiß 2000).

Tab. 4.2 Composition of borosilicate glass (Starzner & Wurmer-Weiß 2000).

Borosilicate Glass	Material	Formula	Compound in Glass
70 - 80 %	Silica sand		SiO ₂
4 - 8 %	Soda	Na ₂ CO ₃	Na ₂ O
	(Potash	K ₂ CO ₃	K ₂ O)
	Limestone	CaCO ₃	CaO
	Dolomite	MgCO ₃	MgO
2 - 7 %	Feldspar	NaAlSi ₃ O ₈	Al ₂ O ₃
7 - 13 %	Boron trioxide		B ₂ O ₃

Kommentar [hja1]: Zahlen jeweils tiefstellen

The production process for borosilicate glass is similar to this of normal float or casting glass. Quantitative data for the production process were not available. It is recommended to use the data for "flat glass, uncoated, at plant" to approximate the environmental impacts due to the production of flat

¹ Personal communication, Tobias Plessing, Saint-Gobain, DE, 5.2002.

² Personal communication, Andreas Wagner, Wagner-Solartechnik, DE, 6.2002.

borosilicate glasses. The production of borosilicate glass tubes from this type of glass is investigated in chapter 4.6 on page 15.

4.3 Production of Structured Glass

Structured glass can be produced in the same way as casting (or rolled) glass. The structuring is not a separate process, but a modification in the casting glass process. Casting glass is formed by a continuous double-roll process. Molten glass mass at about 1000°C is squeezed between water cooled steel rollers to produce a ribbon with controlled thickness and surface pattern (IPPC 2000). Instead of using a flat roller after the casting a structured roller is used. The further processing of structured glass is the same as for casting glass. The glass is cooled and then cut to its final format. A schematic representation of the rolled glass process is shown in Fig. 4.1.

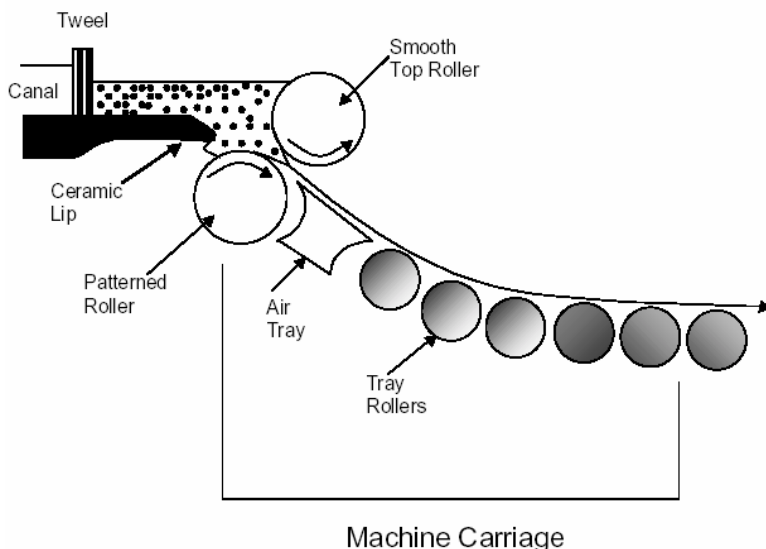


Fig. 4.1 The process for the production of casting glass (IPPC 2000) which is also used for structured glass

Unit process data were not available. The composition of raw materials is very similar to normal flat glass. Environmental impacts can be considered to be a little bit lower than these for flat (floated) glass because the expenditures for the floating process are higher (e.g. use of a zinc bath, gas atmosphere).³ The energy use for casting glass is lower than for float glass (Starzner & Wurmer-Weiß 2000), but most of the energy use is due to the melting process which is about the same for both products (IPPC 2000). It is recommended to use the unit process raw data for “flat glass, uncoated, at plant” to approximate the environmental impacts due to the use of structured glasses.

4.4 Tempering or Toughening

Tempering or toughening is a special process of solidification of a glass sheet in order to make it more resistant to breakages. This is especially important for solar collectors which face high thermal tensions. The process may be physical (thermal) or chemical. In the physical process, the glass sheet is

³ Personal communication with Mr. Braun, Saint-Gobain Glas, FR, 6.2002.

heated to a temperature just below its softening point (about 640°C) and then immediately cooled by special jets of cold-air. This hardens the surface of the glass immediately and gives the inside more time to cool. Thus a tension is build up in the glass. The result is a sheet of glass which is two or three times stronger than un-tempered glass and which, upon breakage, shatters into tiny pieces with blunt edges (the most common application is automotive glass).

The chemical process is based on the so-called ion-stuffing technique. Different chemical elements in the glass structure possess different ionic radii and therefore different densities. If glass containing sodium is cooled slowly in a salt bath of molten potassium, the sodium ions will migrate from the glass to the salt, while the potassium ions will move to the surface of the glass. There they create also a tension in the glass, due to their wider radius, and therefore stronger surface layer (of more than 0.1 mm). Glass sheets which have been chemically tempered are stronger than those which have not undergone any tempering process (glassOnline 2002).

The inventory for tempering describes the thermal process. It includes the transport to the tempering plant, the energy use for heating and cooling and losses, but not the throughput of raw glass. The process is energy intensive using gas or electricity mainly for the heating, but also for cooling. Unit process data especially for the energy use were not available. The estimation of unit process raw data and data quality indicators for tempering of glass in Tab. 4.3 is based on the assumption that the hardening uses about 20% of the energy used for the initial melting (5.5 to 8.0 MJ/kg according to (IPPC 2000)). About 10-15% of the incoming glass is normally not used due to the wastes from cutting to the final size. The production loss before the tempering is estimated to be 12%. This loss is accounted here with the glass production and its transport to the factory for tempering. Furthermore a recycling of these wastes in the glass production process can be assumed. Thus they are not regarded as wastes. Data for the infrastructure were not available. It has to be noted that the demand of flat glass has to be considered in inventory additionally to the process of tempering. The inventory is based solely on assumptions and thus the data quality is poor.

Tab. 4.3 Unit process raw data and data quality indicators for the tempering of glass

	Name	Location	Infrastructure	Unit	tempering, flat glass	Uncertainty Standard Deviation 95%	GeneralComment
product	Location				RER		
technosphere	InfrastructureProcess				0		
	Unit				kg		
	tempering, flat glass	RER	0	kg	1.00E+0		
	natural gas, burned in industrial furnace low-NOx >100kW	RER	0	MJ	1.33E+0	1 1.50	(5,na,1,1,1,na); Estimation 20% of melting energy
	flat glass, uncoated, at plant	RER	0	kg	1.20E-1	1 1.12	(3,3,1,1,1,na); Estimation 12% glass losses
	transport, lorry 32t	RER	0	tkm	1.12E-1	1 2.09	(4,5,na,na,na,na); Standard distance 100km for glass throughput including losses (1.2kg)
	transport, freight, rail	RER	0	tkm	2.24E-1	1 2.09	(4,5,na,na,na,na); Standard distance 200km for glass throughput including losses (1.2kg)

4.5 Anti-reflection for Solar Glass

4.5.1 Etching with hydrogen fluoride (HF)

Etching with hydrogen fluoride is used in order to produce milk glass used i.e. bathrooms. One problem of glass anti-reflected by etching is the ecological impact caused by the etching agent hydrofluoric acid (Glässer 2000). Detailed and reliable quantitative information for this process were not available.

4.5.2 Etching with fluosilicic acid (H₂(SiF₆))

The manufacturer Sunarc, DK uses an acid-process for its AR-solar glass. This glass is for example used by Wagner-Solartechnik, DE. The antireflection microstructure is impressed on the solar glass by an automatic etching procedure. Cleaning, etching and clarification are the necessary process stages. A

part of the glass surface is taken of and made porous by etching with fluoro silic acid (25%). Than the glass is dried and packed. It is possible to produce a film layer on structured and smooth float glass as well as on tempered or un-tempered glasses. The production capacity of this company is 175.000 m² per shift and year or over 500.000 m² per year running a 3-shift (sunarc 2002; Wagner 2002).

This producer has provided data for its production plant.⁴ The unit process raw data are shown in Tab. 4.4. The closed loop process at this company produces almost no wastes or effluents. The exhausts from the treatment chambers go through a so-called air-scrubber (air/water cleaner). The waste material from the scrubber is collected with the water from rinsing of the glasses between the different operations of the treatment and from the final cleaning. This liquid is then passed through a waste water treatment system, resulting in clean water, to be used again in process. The dry matters from the filter press consist of non-aggressive and non-toxic materials, mainly silica-, sodium-, fluoride- and calcium- carbonates and oxides. They are disposed with the normal household wastes. Fluoride is emitted with the effluents in a maximum concentration of 50mg/l. It is estimated that it can not be reduced in the wastewater treatment plant. The treatment of the used water is estimated with a generic module. The maximum concentration of fluoride in the ambient air is 5mg/m³. This emission is not considered. Standard distances have been assumed for the transport.

Tab. 4.4 Unit process raw data and data quality indicators for the AR-glass etching with fluorosilicic acid

	Name	Location	Infrastructure	Unit	anti-reflex-coating, etching, solar glass	Uncertainty vTune Standard Deviation	GeneralComment
	Location InfrastructureProcess Unit				DK 0 m2		
product	anti-reflex-coating, etching, solar glass	DK	0	m2	1.00E+0		
technosphere	natural gas, burned in industrial furnace low-NOx >100kW	RER	0	MJ	7.26E+0	1 1.09	(2,1,1,1,1,3); Data provided by manufacturer including also room heating
	electricity, low voltage, at grid	DK	0	kWh	1.20E+0	1 1.09	(2,1,1,1,1,3); Data provided by manufacturer, all electricity uses in the plant
	glass etching plant	DK	1	unit	2.00E-7	1 1.09	(2,1,1,1,1,3); Data provided by manufacturer
	tap water, at user	RER	0	kg	1.00E+1	1 1.09	(2,1,1,1,1,3); Data provided by manufacturer
	fluosilicic acid, 22% in H2O, at plant	RER	0	kg	1.33E-1	1 1.09	(2,1,1,1,1,3); Data provided by manufacturer
	lime, hydrated, packed, at plant	CH	0	kg	2.00E-2	1 1.09	(2,1,1,1,1,3); Data provided by manufacturer
	treatment, sewage, unpolluted, from residence, to wastewater treatment, class 2	CH	0	m3	1.00E-2	1 1.09	(2,1,1,1,1,3); Data provided by manufacturer, fluoride emission separate considered
	disposal, refinery sludge, 89.5% water, to sanitary landfill	CH	0	kg	2.50E-2	1 1.09	(2,1,1,1,1,3); Data provided by manufacturer
	transport, lorry 32t	RER	0	tkm	1.53E-2	1 2.09	(4,5,na,na,na,na); Standard distance for chemicals 100km
emission water, river	transport, freight, rail	RER	0	tkm	3.07E-2	1 2.09	(4,5,na,na,na,na); Standard distance chemicals 200km
	Fluoride	-	-	kg	5.00E-4	1 1.51	(2,1,1,1,1,3); Data provided by manufacturer, estimation no reduction in wastewater treatment

Infrastructure data were available for the land use and the ground area of buildings. The unit process raw data are normalized for an annual production capacity of 100'000 m². The unit process raw data for the infrastructure are shown in Tab. 4.5.

⁴ Personal communication with Göran Olsson, Sunarc Technology A/S, Herfølge, DK, 8.2002.

Tab. 4.5 Unit process raw data and data quality indicators for the glass etching plant

	Name	Location	Infrastructure	Unit	glass etching plant DK 1 unit	Uncert Stand ardDe viation	GeneralComment
	Location InfrastructureProcess Unit						
product	glass etching plant	DK	1	unit	1.00E+0		
ressource, land	Occupation, industrial area, built up	-	-	m2a	3.00E+5	1 1.50 (1,2,1,3,1,n.a.);	Standard 50a occupation
	Occupation, industrial area, vegetation	-	-	m2a	2.40E+5	1 1.50 (1,2,1,3,1,n.a.);	Standard 50a occupation
	Transformation, from unknown	-	-	m2	6.00E+3	1 2.00 (1,2,1,3,1,n.a.);	Questionnaire
	Transformation, to industrial area, vegetation	-	-	m2	4.80E+3	1 2.00 (1,2,1,3,1,n.a.);	Questionnaire
	Transformation, to industrial area, built up	-	-	m2	1.20E+3	1 2.00 (1,2,1,3,1,n.a.);	Questionnaire
technosphere	building, hall	CH	1	m2	1.20E+3	1 3.00 (1,2,1,3,1,n.a.);	Questionnaire

4.5.3 Sol-Gel technique

A recent development for a AR-coating is generated through surface roughening by means of a deposited layer (nano-particular precipitation). A silicon dioxide layer in which tin dioxide (SnO_2) particles are precipitated is deposited using the sol-gel technique. The particles incorporated in the layer create the rough surface. Therefore colloidal tin sol dissolved in water and, as a catalyst for the necessary hydrolysis stage, nitric acid (HNO_3) is added to a mixture of methyl triethoxy dihydrosilicon (MTEOS) and tetraethoxy dihydrosilicon (TEOS) (Glässer 2000).

The coating process itself is performed using the dipping technique or the spray technique, with the subsequent conversion of the hydrolysed gel film into an inorganic layer by means of firing at temperatures up to 500°C . Fig. 4.2 shows the stages of sol-gel coating process by means of the dipping technique (Glässer 2000).

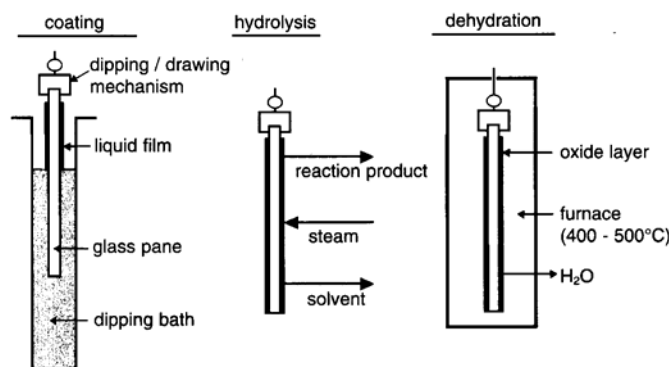


Fig. 4.2 Stages of sol-gel coating process by means of the dipping technique (Glässer 2000:176)

The course of production of sol-gel coating is shown in Fig. 4.3 (Glässer 2000:176). Flabeg (2002) produces the anti-reflex coating by a sol-gel coating with a following drying and hardening step. Low-iron float and cast glasses are coated by this company.

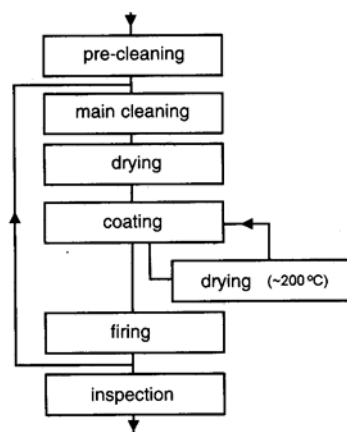


Fig. 4.3 Course of production of sol-gel coating (Glässer 2000:176)

Unit process raw data for the sol-gel coating process were not available. It is not possible to give an estimation or approximation for this process.

4.5.4 Sputtering

The sputtering process is based on a self-maintained gas discharge, known as the plasma, which is ignited at low pressure and then interacts with the coating material, known as the target. In this process the coating material is eroded from the target surface and condensed as a layer on the flat glass pane (Glässer 2000).

There are several specific processes used for sputtering. Fig. 4.4 shows a diagram of a modern in-line flat glass sputtering plant with magnetron cathodes. Flat glass panes pass continuously through a number of in-line vacuum chambers, on a horizontal roller conveyor belt. After running the pane into the lock-in chamber, the chamber is evacuated down to approximately 10^{-1} mbar. From the transfer chamber the panes pass continuously in line the pumping and coating chambers. Different types of layers are possible (Glässer 2000).

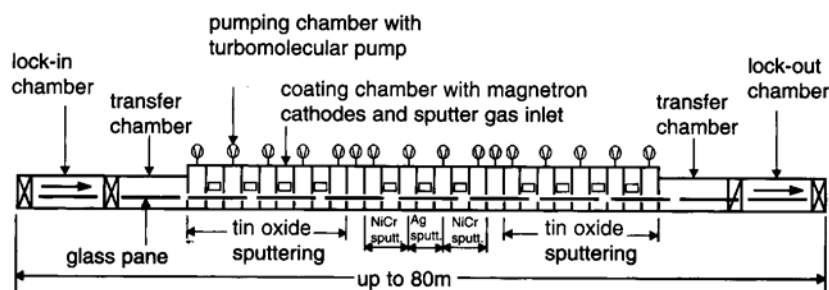


Fig. 4.4 Diagram of a modern in-line flat glass sputtering plant with magnetron cathodes (Glässer 2000:119)

Cleaning and drying are necessary before the sputtering process. Bismuth, silver and nickel are used as coating targets during sputtering process for producing insulation glass. Argon and oxygen are used for the process. The coating process for insulation glass and sputtered anti-reflex coatings for solar glass is very similar. They will mainly differ by the target materials used. Data specific for anti-reflex or mirror coatings were not available while data for the production of insulation glass are published (Interpane 2001). The anti-reflex coating made by sputtering can be assumed to have about the same

impacts as coating made for insulation glass with the sputtering technology. It is recommended to use the inventory for “flat glass, coated, at plant” as an approximation for glass with a sputtered anti-reflex coating.

4.6 Glass Tubes for Solar Collectors

The life cycle inventory for solar collector tubes is split up into three stages according to the information provided by a producer. These stages are the production of glass tubes, the manufacturing of xenon-filled collector tubes and the infrastructure for the production plant.

4.6.1 Borosilicate glass tube production

Detailed data for material and energy uses, emissions and wastes have been provided by a manufacturer⁵ of tube glasses. The unit process raw data for the production of glass tubes are shown in Tab. 4.6. The intermediate product is a tube made out of borosilicate glass, but there are also lime-sodium glasses used for solar collectors. The amount of boron used for the borosilicate glass is between 10% and 14%. The use of feldspar (5%) and boric acid (10%) has been estimated with the composition shown in Tab. 4.2. A surplus use of raw materials due to production wastes has been considered with a slight increase for the use of soda. Glass wastes are used as filling materials in landfills, but it is planned to recycle them within the process.

Normally the product output of glass production represents approximately 70% of the raw material input. The remainder largely consists of emissions to air 10-20%, made up mainly of CO₂ from the decomposition of carbonates; and scrap glass (cullet) around 10–20% arising from edge trimming, product changes and breakages. The cullet is usually continuously recycled to the furnace and so the product actually represents closer to 85% of the raw material input (IPPC 2000). A CO₂ emission of 11% has been estimated for carbonate decomposition and the surplus material use has been added to the figures given in the questionnaire. Further data, e.g. for wastes, have been provided in a personal communication by the manufacturer.

⁵ Personal communication with an anonymous manufacturer 6.2002.

Tab. 4.6 Unit process raw data and data quality indicators for the production of borosilicate glass tubes

	Name	Location	Infrastruct	Unit	glass tube, borosilicate, at plant	Uncertain Standard Deviation 95%	GeneralComment
	Location InfrastructureProcess Unit				DE 0 kg		
product	glass tube, borosilicate, at plant	DE	0	kg	1.00E+0		
technosphere	tap water, at user	RER	0	kg	3.26E-1	1 1.22	(2,2,1,3,1,5); Questionnaire
	electricity, medium voltage, at grid	DE	0	kWh	1.13E+0	1 1.22	(2,2,1,3,1,5); Questionnaire
	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	0	MJ	3.19E+0	1 1.22	(2,2,1,3,1,5); Questionnaire
	natural gas, burned in industrial furnace low-NOx >100kW	RER	0	MJ	1.33E+1	1 1.22	(2,2,1,3,1,5); Questionnaire
	silica sand, at plant	CH	0	kg	7.03E-1	1 1.22	(2,2,1,3,1,5); Questionnaire
	soda, powder, at plant	RER	0	kg	1.58E-1	1 1.22	(2,2,1,3,1,5); Reduction for feldspar + boron plus production wastes
	aluminium oxide, at plant	RER	0	kg	5.00E-2	1 1.24	(3,2,1,3,1,5); Estimation for 5% feldspar
	boric acid, anhydrous, powder, at plant	RER	0	kg	1.00E-1	1 1.24	(3,2,1,3,1,5); Estimation for 10% boron trioxide
	transport, lorry 32t	RER	0	tkm	5.13E-2	1 2.05	(2,2,1,3,1,5); Real distances for materials and wastes reported in a questionnaire
	treatment, glass production effluent, to wastewater treatment, class 2	CH	0	m3	5.43E-4	1 1.22	(2,2,1,3,1,5); Questionnaire
	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	0	kg	2.43E-3	1 1.22	(2,2,1,3,1,5); Questionnaire, unspecific wastes
	disposal, glass, 0% water, to inert material landfill	CH	0	kg	9.57E-4	1 1.22	(2,2,1,3,1,5); Glass used as sealing material
	disposal, mineral wool, 0% water, to inert material landfill	CH	0	kg	7.28E-4	1 1.31	(2,2,1,3,3,5); Questionnaire, asbestos
	disposal, building, waste wood, untreated, to final disposal	CH	0	kg	7.39E-4	1 1.22	(2,2,1,3,1,5); Questionnaire, wood wastes
	disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	0	kg	3.26E-3	1 1.22	(2,2,1,3,1,5); Questionnaire, concentrated solvents
	glass tube plant	DE	1	unit	1.00E-0	1 3.05	(2,2,1,3,1,5); Questionnaire
emission air, high population density	Heat, waste	-	-	MJ	4.07E+0	1 1.22	(2,2,1,3,1,5); Questionnaire
	Carbon dioxide, fossil	-	-	kg	1.10E-2	1 1.53	(4,2,1,3,1,5); Estimation for decomposition of carbonates
	Nitrogen oxides	-	-	kg	6.52E-3	1 1.47	(2,2,1,3,1,5); Questionnaire
	Particulates, < 2.5 um	-	-	kg	1.50E-4	1 3.05	(2,2,1,3,1,5); Waste gas purification

4.6.2 Solar tube collector production

The unit process raw data for the production of collector tubes are shown in Tab. 4.7. In this process stages a collector tube with a silver mirror is produced from the simple glass tube. These data have been provided by one manufacturer for a specific product which is not directly comparable with the one shown in Fig. 3.1. Collector tubes from other manufacturers might be produced in a different way.

This solar collector glass tube is available with a xenon or a high vacuum insulation. The amount of xenon has been calculated roughly with the volume of a glass tube. A circular silver mirror is incorporated inside the jacket tube (ICR: Internal Circular Reflector). It redirects all of the solar radiation falling onto the sectional surface of the jacket tube to the absorber tube. A radiation-selective absorber coating is applied to the inner glass tube by means of a magnetron sputtering process. This aluminium layer converts solar radiation into heat with a high efficiency while at the same time ensuring a low level of waste thermal radiation. The specific amount of used silver has not been provided. The consumption of glass tubes has been increased by the amount of generated glass wastes (12t/a). A barium getter is not necessary for the product investigated here. The effluents from the manufacturing process contain fluorides (1mg/l), tin (1-2mg/l, 240m³/a) and silver (0.05-0.1mg/l, 540m³/a). They are treated in a municipal wastewater plant. The infrastructure for producing glass tubes has already been considered for the basic glass tube.

Tab. 4.7 Unit process raw data and data quality indicators for manufacturing collector tubes with a silver mirror

	Name	Location	Infrastructure	Unit	solar collector glass tube, with silver mirror, at plant	Uncertainty	StandardDeviation5%	GeneralComment
	Location InfrastructureProcess Unit				DE 0 kg			
product	solar collector glass tube, with silver mirror, at plant	DE	0	kg	1.00E+0			
technosphere	tap water, at user	RER	0	kg	3.12E+0	1	1.22 (2,2,1,3,1.5);	Questionnaire
	electricity, medium voltage, at grid	DE	0	kWh	1.12E+0	1	1.22 (2,2,1,3,1.5);	Questionnaire
	natural gas, burned in industrial furnace low-NOx >100kW	RER	0	MJ	1.16E+0	1	1.22 (2,2,1,3,1.5);	Gasburner with open flame, no gas treatment.
	chemicals organic, at plant	GLO	0	kg	8.00E-4	1	1.57 (2,2,1,3,4.5);	Other chemicals
	aluminium, production mix, at plant	RER	0	kg	2.80E-3	1	1.31 (2,2,1,3,3.5);	Coating agents
	chromium steel 18/8, at plant	RER	0	kg	2.60E-2	1	1.22 (2,2,1,3,1.5);	Steel parts of the tube.
	xenon, gaseous, at regional storage	CH	0	kg	3.89E-3	1	1.57 (5,2,1,3,1.5);	Xenon filling of the tube, 1bar, own calculation
	hydrochloric acid, 30% in H2O, at plant	RER	0	kg	8.00E-3	1	1.22 (2,2,1,3,1.5);	Questionnaire
	EUR-flat pallet	RER	0	unit	3.37E-5	1	1.22 (2,2,1,3,1.5);	Wood pallet packaging
	glass tube, borosilicate, at plant	DE	0	kg	1.05E+0	1	1.22 (2,2,1,3,1.5);	Glass wastes added.
	transport, lorry 32t	RER	0	tkm	1.16E-1	1	2.05 (2,2,1,3,1.5);	Real distances for materials and waste reported in a questionnaire, 100 km for other materials
	transport, freight, rail	RER	0	tkm	6.33E-1	1	2.09 (4.5.na.na.na.na);	Standard distance 600 km for other materials
	treatment, tube collector production effluent, to wastewater treatment, class 2	CH	0	m3	3.12E-3	1	1.22 (2,2,1,3,1.5);	Questionnaire
	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	0	kg	2.00E-3	1	1.22 (2,2,1,3,1.5);	Questionnaire
	disposal, glass, 0% water, to inert material landfill	CH	0	kg	4.80E-2	1	1.22 (2,2,1,3,1.5);	Glass used as sealing material
	disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	0	kg	1.60E-2	1	1.22 (2,2,1,3,1.5);	Concentrated solvents
emission air, high population density	Heat, waste	-	-	MJ	4.03E+0	1	1.47 (2,2,1,3,1.5);	Calculation

4.6.3 Glass tube production plant

Infrastructure data were available for the land use and the ground area of buildings. The unit process raw data are normalized for an annual production capacity of 100'000 tones. The unit process raw data for the infrastructure are shown in Tab. 4.8.

Tab. 4.8 Unit process raw data and data quality indicators for the glass tube production plant

	Name	Location	Infrastr	Unit	glass tube plant	Uncertainty	StandardDeviation	GeneralComment
	Location InfrastructureProcess Unit				DE 1 unit			
product	glass tube plant	DE	1	unit	1.00E+0			
ressource, land	Occupation, industrial area, built up	-	-	m2a	4.41E+6	1	1.50 (1,2,1,3,1,n.a.);	Standard 50a occupation
	Occupation, industrial area, vegetation	-	-	m2a	3.31E+6	1	1.50 (1,2,1,3,1,n.a.);	Standard 50a occupation
	Transformation, from unknown	-	-	m2	1.54E+5	1	2.00 (1,2,1,3,1,n.a.);	Questionnaire
	Transformation, to industrial area, vegetation	-	-	m2	6.61E+4	1	2.00 (1,2,1,3,1,n.a.);	Questionnaire
	Transformation, to industrial area, built up	-	-	m2	8.82E+4	1	2.00 (1,2,1,3,1,n.a.);	Questionnaire
technosphere	building, hall	CH	1	m2	8.82E+4	1	3.00 (1,2,1,3,1,n.a.);	Questionnaire

5 Data Quality Considerations

The simplified approach with a pedigree matrix has been used for calculating the standard deviation of the different products.

The data for the production process for collector tubes are partly estimated and broken down from data for the production of several products. Thus the data are relatively unsure and they could not be verified. More precise data for this manufacturer can be expected after a few years of experiences with this product.

6 Cumulative Results

Results of the cumulative inventory can be downloaded from the database (www.ecoinvent.ch).

7 Conclusions

The unit process raw data investigated in this chapter are meant to be used in life cycle assessment (LCA) studies for solar collectors. Compared to the total production of these facilities the glass processing processes are normally not very important. It was difficult to find specific data for solar glasses. But it is likely that these products do not show increased impacts in comparison to normal glass products investigated for the database, besides increased transports and a slightly different composition of raw materials.

Abbreviations

AR Anti-reflection

ICR Internal Circular Reflector

Appendices: EcoSpold Meta Information

Part XII: Glass Products and Processing for Solar Collectors

ReferenceFunction	401 Name	solar glass, low-iron, at regional storage	tempering, flat glass	anti-reflex-coating, etching, solar glass	glass etching plant	glass tube, borosilicate, at plant	solar collector glass tube, with silver mirror, at plant	glass tube plant
Geography	662 Location	RER	RER	DK	DK	DE	DE	DE
ReferenceFunction	493 InfrastructureProcess	0	0	0	1	0	0	1
DataSetInformation	403 Unit	kg	kg	m2	unit	kg	kg	unit
	201 Type	1	1	1	1	1	1	1
	202 Version	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	203 energyValues	0	0	0	0	0	0	0
	205 LanguageCode	en	en	en	en	en	en	en
	206 LocalLanguageCode	de	de	de	de	de	de	de
DataEntryBy	302 Person	41	41	41	41	41	41	41
	304 QualityNetwork	1	1	1	1	1	1	1
ReferenceFunction	400 DataSeRelatesToProduct	1	1	1	1	1	1	1
	402 IncludedProcesses	Gate to gate inventory for the process including transport to RER	Gate to gate inventory for the process not including infrastructure and emissions.	Gate to gate inventory for the process including cleaning, etching and clarification and drying. Losses of glass are not included.	Infrastructure of a glass etching plant with data for land use and area of buildings. No data for machinery. Annual capacity normalized to 100000 m2/a.	Gate to gate inventory for the production of glass tubes.	Gate to gate inventory for the manufacturing of collector tubes from glass tubes including a sputter coating with a selective coating and a mirror layer. Infrastructure considered for the pre-product.	Infrastructure of a tube glass plant with data for land use and area of buildings. No data for machinery. Annual capacity normalized to 100000/a.
	404 Amount	1	1	1	1	1	1	1
	490 LocalName	Solarglas, eisenarm, ab Regionallager	härten, Flachglas	Antireflexbeschichten, Ätzen, Solarglas	Glasbeschichtungswerk	Glasrohr, Bor-Silikat, ab Werk	Kollektorrohr, mit Silberspiegel, ab Werk	Rohrglaswerk
	491 Synonyms		toughening				glass evacuated tube // magnetron sputtering // Internal Circular Reflector // Vakuumröhrenkollektor	
	492 GeneralComment	Assumption with inventory data for normal glass. This glass differs from normal glass mainly due to the type of raw materials, which is considered with a higher transport distance. Additional transports for importing it to Europe are included in the inventory.	Basic assumption of unit process raw data for the tempering of glass. Tempering or toughening is a special process of solidification of a glass sheet in order to make it particularly resistant to breakages. The process may be physical (thermal) or chemical. Here a thermal tempering is assumed. The result is a sheet of glass which is two or three times stronger than untempered glass.	The filter cake contains silicone-sodium-fluoride and calcium carbonates or oxides. They can be disposed with the normal household wastes. Standard distances have been assumed for the transport. The exhausts from the treatment chambers go through a so-called air-scrubber (air/water cleaner). The waste material from the scrubber is collected with the water from rinsing of the glasses between the different operations of the treatment and from the final cleaning. This liquid is then passed through a waste water treatment system, resulting in clean water, to be used again in process. The dry matters from the filter press consist of non-aggressive and non-toxic materials, mainly silica-, sodium-, fluoride- and calcium- carbonates and oxides	Infrastructure of a glass etching plant.	Production of borosilicate glass tubes.	Production of solar collector tubes.	Infrastructure of a tube glass plant.
	494 InfrastructureIncluded	1	1	1	1	1	1	1
	495 Category	glass	glass	glass	glass	glass	glass	glass
	496 SubCategory	construction	construction	construction	construction	construction	construction	construction
	497 LocalCategory	Glas	Glas	Glas	Glas	Glas	Glas	Glas
	498 LocalSubCategory	Bauglas	Bauglas	Bauglas	Bauglas	Bauglas	Bauglas	Bauglas
	499 Formula							
	501 StatisticalClassification							
	502 CASNumber							
	601 StartDate	2000	2000	2002	2002	2000	2002	2000
	602 EndDate	2000	2000	2002	2002	2001	2002	2001
	603 DataValidForEntirePeriod	1	1	1	1	1	1	1
	611 OtherPeriodText	Date of estimation.	Time of estimation	Time of information	Time of information	Production data for one business year.	Data for a few month of production.	Production data.
Geography	663 Text	One production plant in US. No information for RER.	Assumption for RER.	Production plant in DK. Some pre-products and waste treatment services estimated with European and Swiss data.	Production plant in DK. Building background data from CH.	Production plant in DE. Some pre-products and waste treatment services estimated with European and Swiss data.	Production plant in DE. Some pre-products and waste treatment services estimated with European and Swiss data.	Production plant in DE. Building background data from CH.
Technology	692 Text	Estimation of low-iron glass production with technology used for normal flat glass.	The glass sheet is heated to a temperature just below its softening point (about 640°C) and then immediately cooled by special jets of cold-air. These harden the surface of the glass, giving the inside more time to cool. This allows the external layer to crystallize into a wider lattice while the inside solidifies with greater compression than in the crystal lattice.	The antireflection microstructure is impressed on the solar glass by an automatic etching procedure. Etching is made with fluorosilic acid (H2SiF6). It is possible to produce a film layer on structured and smooth float glass as well as on tempered or un-tempered glasses.	Infrastructure of a plant for the anti reflex etching process.	Tube glass manufacturing. Plant data for joint production of borosilicate (20%) and lime-natron-glass (80%) tubes.	Sputtering and coating of glass.	Tube glass manufacturing.
Representativeness	722 Percent			100				
	724 ProductionVolume	not known	not known	One company. Capacity 175000m² per shift * year or 400000- 500000m² per year	100000 m2/a coated glass.	2300 t/a.	250 t/a.	63000 t/a tube glass.
	725 SamplingProcedure	Assumption	Assumption.	Personal communication	Questionnaire.	Questionnaire.	Questionnaire.	Questionnaire.
	726 Extrapolations	Assessment with European electricity mix.			none	Input of raw materials is calculated with data for borosilicate glass. Other data are from a plant with a production mix of borosilicate (20%) and lime-natron-glass (80%).	Partly based on estimations.	none
	727 UncertaintyAdjustments	none	none	none	none	none	none	none
DataGenerator	751 Person	41	41	41	41	41	41	41
AndPublication	752 DatePublishedIn	2	2	2	2	2	2	2
	757 ReferenceToPublishedSource	7	7	7	7	7	7	7
	758 Copyright	1	1	1	1	1	1	1
	759 AccessRestrictedTo	0	0	0	0	0	0	0
	766 CompanyCode							

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Part XIII

Glass Fibre Reinforced Plastic Products

Data v2.0 (2007)

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Summary

This section describes the production and processing of glass fibre reinforced plastic (GRP) products used for the construction of solar collectors. The inventory for the glass fibre is based on a literature study for the technologies used in the European glass industry. The energy use is the main environmental aspect. Glass fibres can be used in manifold ways. Therefore resins and different chemicals are necessary. Two types of processing, hand lay-up moulding with poly ester resin and injection moulding with polyamide, have been investigated based on rough assumptions given for a single manufacturer. The unit process raw data are meant to be used as background information. For other types of moulded products the background data for glass fibre can be used to assess the environmental impacts if the amount of other chemicals is known.

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1 Introduction

There are a range of different products known as glass fibre reinforced plastic. They are used for different applications. This section describes the production and processing of glass fibre reinforced plastic (GRP) products used for the construction of solar collectors.

2 Reserves and Resources of Material

Silica sand and soda are the basic materials for manufacturing of glass. Different organic chemicals are used for the production of the final product. These compounds are produced mainly from fossil resources like crude oil and natural gas. There are five producers of fibre glass in the EU, Ahlstrom, Bayer, Owens Corning, PPG Industries, and Vetrotex (IPPC 2000).

3 Characterisation of Product

Glass fibre reinforced plastics are manufactured from different basic materials. These materials and its producers are shown in Tab. 3.1. The material has the following features: light (up to 30% lighter than steel with the same strength), insulating, corrosion-resistant, shock- and heat-resistant.

Tab. 3.1 Materials and its producers for glass fibre reinforced plastics.

Material	Producers and location
Glass roving	Vetrotex/St. Gobain, GLO Owens Corning, US Skloplast, SK Camelyaf, TR
Resins	DSM/Compiègne, FR Cray-Valley/Drocourt, FR Reichhold, FR Lonza, IT
Extender	Omya, GLO Martinswerke, Bergheim, DE Alpha, Hürth, Nähe Köln, DE
Additives	Byk-Chemie, DE
Peroxides	Degussa, Peroxid-Chemie, DE Akzo-Chemie, DE

4 Use of Products

Reinforced plastics are an important class of engineering materials. They consist of one or more discontinuous phases embedded in a continuous-phase polymer matrix (Ehrenstein & Kabelka 1997). Textile and reinforcement glass strands are used in technical fabrics and in plastic parts manufactured in small or large series with multiple sizes and shapes. It is used to replace steel, wood, cardboard, etc. in high performance applications: electricity (optical cable or central strength members) and electronics (printed circuit boards), energy (windmills), building and infrastructure (building façade, bridges, road pavement). Also, consumer goods (casing for household appliances and computers) and transportation (car body panels, truck cabins) are made with GRP.

The following glass strands are used, e.g.: alkali-resistant glass (AR), glass/thermoplastic commingled and glass mat (non-woven glass fabric). These are used respectively in the reinforcement of cements and mortars, for the manufacture of high performance composite parts (automotive panels or structural parts) and waterproofing, anti-corrosion and fire protection applications.

About 1.8% of the total glass production in the EU is glass fibre products. This amounts to 475 000 tonnes of fibre from the 26 furnaces operating at the 12 sites in the EU (Saint-Gobain 2002b). The worldwide glass reinforcement market is about 2.5 million tons in the year 2000. This figure (and the ones below) does not include either the glass mats or AR-glass markets. From 1994 to 2000, this market showed an average growth of 5.7% per year. These statistics do not include either wet processes using mainly chopped strands or processes using glass mats. The major manufacturer Saint-Gobain Vetrotex produces about 550 000 tons of glass reinforcements annually (Saint-Gobain 2002c).

On a worldwide scale, glass reinforcement applications are very numerous (Fig. 4.1). Construction and public works is the largest field of application (31%) followed by transportation (26%). Electrical and electronics represent a worldwide average of 15% while it is about 30% in Asia due to the powerful electronics industry (printed circuit boards - PCB).

Marine, recreational and consumer goods represent a world average of 13% with a 20% share in North America. Industrial and agricultural equipment is 10% of the total, but that application is larger in Europe (approximately 15%) because of the high use of composites in tanks, silos, pipes, etc.

Two types of matrix can be classified: Thermoplastics are capable of being separately hardened and softened by increases and decreases, respectively, in temperature. Duroplastics are changed into a substantially infusible and insoluble material when cured by the application of heat, or through chemical means (Ehrenstein & Kabelka 1997).

Composite materials made with thermosetting resins showed a respectable increase of 4 to 5% from 1994, which was lower than average growth for composites taken as a whole (excluding PCB). The development of thermosetting composites was more or less the same in all areas of application, from traditional, semi-industrial processes (small series) such as open-mould processes (contact moulding and spray-up) to more industrial processes such as filament winding and pultrusion. For the period under consideration moulding was exceptional in the automotive segment for North America, and in the electricity and plumbing segments for Europe. In the United States, pultrusion underwent a burst of growth, especially in civil engineering. Filament winding was highly dynamic in Europe and especially in the Middle East in the petroleum and offshore industry.

Composite materials made with thermoplastics showed a $\pm 11\%$ increase at least, twice as much as thermosets for the same period. The high growth can be explained by a number of advantages associated with thermoplastics: good mechanical performance, favourable dielectric constants (electrical applications), and good recyclability. Thermoplastics can also be processed by injection moulding under pressure, allowing short production cycles that are suited to industrial mass production and complex shapes (also a high degree of functional integration).

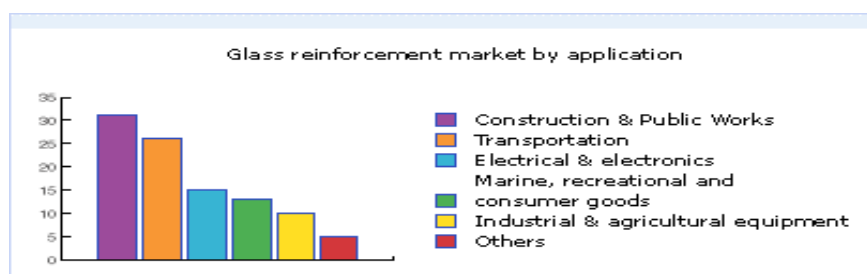


Fig. 4.1 Glass reinforcement market by application (Saint-Gobain 2002c).

5 System Characterisation

This section provides information about product characteristics, production processes and moulding processes. The production of the raw material can be roughly distinguished into production of glass strands and further converting to reinforcement products. Finally these products are moulded and then hardened in the final form with the help of heat or hardening resins. But there is no unique manufacturing path for the different glass fibre products.

The following section gives some information for the components of continuous filaments glass fibres for plastic reinforcement provided by a manufacturer (Saint-Gobain 2002c). Glass fibres are prepared from a mixture of so called E glass in the form of continuous strands and a size with a binder. The CAS number of glass fibre is 65997-17-3 (corresponding to the oxides used for production).

5.1 Glass Strands

The main type of glass used for glass fibre reinforced plastics investigated here is named E glass. This is a glass with a very low alkaline content. The composition (expressed in oxides) is within the percentages shown in Tab. 5.1. Owens Corning, an important producer in the USA uses 35% recycled glass in the manufacture of fibre glass.

Tab. 5.1 Composition of E Glass for glass fibres.

Compound	Content in E Glass	Content in E Glass
SiO ₂	52-62%	53 – 60%
Alkaline Oxides (Na ₂ O, K ₂ O)	< 2%	< 2%
Alkaline earth oxides (CaO, MgO...)	16-30%	20 – 24%
B ₂ O ₃	0-10%	5 – 10%
Al ₂ O ₃	11-16%	11 – 16%
TiO ₂	0-2%	0 – 3%
Fe ₂ O ₃	0-1%	< 1%
F ₂	0-2%	0 – 2%
Source	(IPPC 2000)	(Saint-Gobain 2002c)

The manufacture of glass is carried out in a special furnace at around 1550°C (E glass) using finely ground raw materials (Fig. 5.1). The glass leaving the furnace at a very high temperature is used to feed bushings (blocks pierced with hundreds of holes) of platinum alloy.

The majority of products are based on E glass. Other glasses are also used, for example R glass (high mechanical performance), D glass (high dielectric performance), AR glass (alkali and corrosion resistant glass) (Saint-Gobain 2002c).

The basic strand is made by forming. Forming is achieved by drawing the molten glass flowing from the holes of the bushing at high speed. This forms between 50 and several thousand filaments. These filaments are defined by diameters: from 5 to 24 microns (1µm = 1/1000 mm).

The individual filaments of reinforcements receive a coating of size at the forming stage, immediately after they leave the bushing. The size, made up of organic products dispersed in water, is designed to give the glass strand certain characteristics necessary for final processing. Each size is specially designed for a moulding or compounding process and for a different matrix type. It usually contains a silane type chemical "coupling agent" which contributes to enhancing the mechanical properties of composites and particularly their resistance to ageing. The products are either manufactured immediately after leaving the bushing (direct roving, Unifilo continuous filament mat, chopped

strands) or made up of basic strands subjected to finishing operations (cabling for yarns, assembly for rovings, chopping of chopped strands, chopping and forming of chopped strand mats).

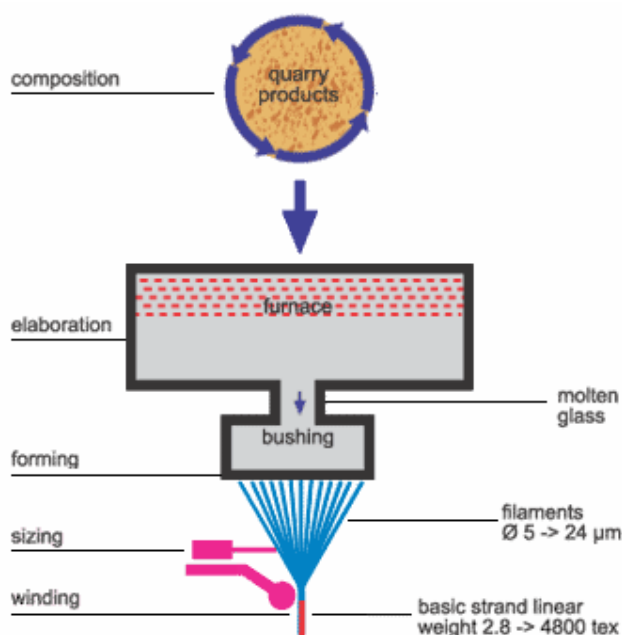


Fig. 5.1 Glass strand fabrication (Saint-Gobain 2002a)

There are many different types of materials commonly called glass fibres on the market. Tab. 5.2 shows a comparison of different types of fibres. They are made from different raw materials and they can be used in different types of applications.

Tab. 5.2 Comparison of different types of fibres used for reinforced plastics (IPPC 2000).

Fibre type	Description	Key advantages	Typical application
E grade glass fibre	Standard fibre glass	High modulus, fire retardant	Facings, tooling, roofing products, General – FRP (fibre reinforced plastic)
AR glass fibre (CEMFIL®)	Alkali resistant fibre glass	Alkali, acid resistant	Cement, Concrete, Infrastructure, Corrosion
Carbon fibre	Polyacrylonitrile (PAN) derived carbon	Ultra high modulus, high strength	High-performance FRP and advanced composites, Infrastructure, Recreational, Aerospace
Aramid fibre (KEVLAR®) (TWARON®)	Aromatic polyamide - liquid crystal polymer	Ultra high modulus, high strength, toughness	High-performance FRP and advanced composites, Marine, Recreation, Sailcloth
High tenacity (HT) polyester fibre	High strength polyester	Toughness, strength	Roofing, Automotive, Coated Fabrics
Regular tenacity (RT) polyester fibre	Regular strength polyester	Toughness	Industrial, Packaging, Laminates
Miscellaneous	Rayon, coloured yarn, polyethylene	Various	Marker yarns, Biodegradable, Aesthetics
Fibre glass forms	Description	Key advantages	Typical application
Roving	Heavy weight, untwisted	Lower cost	FRP, heavy mesh fabrics
Yarn	Light-medium weight, twisted or compacted	Light weight	Tapes, light or flexible fabrics

5.2 Size

Size is a mixture of chemicals applied to the glass strands in a maximum amount of 0.5% - more generally less than 0.2%. Most of this mixture is made up of basically non reactive high molecular weight polymers. In some cases, sizes are prepared from polymers with reactive sites or containing reactive monomers. Most of the reactive sites are polymerised during the manufacturing process. However a very small reactivity may remain.

A second type of ingredient present in almost all sizes is a member of the organo-silane family. These products account for less than 0.05% of the final weight of sized E glass. But, they are used in a higher proportion during the processing and then they are washed of in following processing stage.¹ These products are included in lists of products requiring 'hazardous product' labelling in a pure state for example in Europe R23/25 toxic if swallowed or inhaled, R21 'harmful in contact with the skin', R36 'irritant for the eyes'. The manufacturer considers this risk as negligible as, although listed as dangerous products, the concentration is extremely low and they are polymerised during the production of E glass fibres.

Other products can be used in sizes. Usually the content is extremely low (below 0.1% of total weight) and as a general rule such products are not on the dangerous product lists or, as these have reacted, any possible risk has been reduced.

5.3 Binders

Binders for mats are high molecular weight polymers deposited in quantities below 10% and polymerised on chopped or continuous glass strand mats. The binder content on the reinforcements, is typically in the range of 0.5% to 1.5% by weight (Saint-Gobain 2002c).

5.4 Reinforcement Products

There are big differences, e.g. in energy use, wastes, furnace, diameter etc. for different types of fibre glass products. The basic glass composition is the same for all products, but different chemicals are added. The production of glass fibres is not comparable to that of glass wool.²

Glass filaments are used alone or with carbon, aramid, polypropylene or polyester fibres to manufacture reinforcement products for different markets. Glass mats or strand mats can directly be used for open moulds. Resins are used in this case for hardening the final product (IPPC 2000; Saint-Gobain 2002b).

5.5 Moulding Processes

Moulding is the general term for the production process for products out of glass fibre reinforced plastics. The glass mats or glass rovings are brought into the final form. Afterwards these are hardened with the help of heat or by the use of resins and other chemicals. Glass reinforcements can be used according to many different moulding processes described in the following sections. The final composite properties are influenced significantly by the processing conditions. The fabrication processes for thermosetting matrices can be classified broadly as wet forming processes and processes involving premixes or prepreps (Ehrenstein & Kabelka 1997).

¹ Personal communication M. Overcash, 10.2002.

² Personal communication Mr. Guillermin, Vetrotex, FR, 6.2002.

Resins, extender, additives and peroxides are used in the moulding processes. The production of resins (phenol, epoxy and polyester-resins) used for GRP is described in separate parts of this database. Peroxides are used to harden the resins. The production of extender, additives and peroxides are not investigated in detail. There are many different processes and materials which can be used. Some of them are explained below.

5.5.1 Hand Lay-up Moulding

Hand lay-up moulding may be used for the production of parts of any dimensions, for example, technical parts of a few tens of m² area, as well as swimming pools of 150 m² developed area. This method is generally limited to the manufacture of parts with relatively simple shapes, requiring only one face to have a smooth appearance (the other face being rough from the moulding operation). It is recommended for small and medium volumes, for which the investment in moulds and equipment should not be very high. The contact moulding method consists of applying successively onto a mould surface (Fig. 5.2):

- A release agent
- A gel coat
- A layer of liquid (e.g. thermosetting) resin and of medium reactivity
- A layer of reinforcement (glass, aramid, carbon,...) in the form of chopped strand mat or woven roving

and to impregnate the reinforcement by hand with the aid of a roller or a brush. This operation is repeated as many times as there are layers of reinforcement, in order to obtain the desired thickness of the structure. For the manufacturing, resins are mixed with hardeners. About 2% of 50% activated methyl ethyl ketone peroxide and 0.25 % of 6 % solution of cobalt naphtanate are mixed with orthophthalic polyester resin.

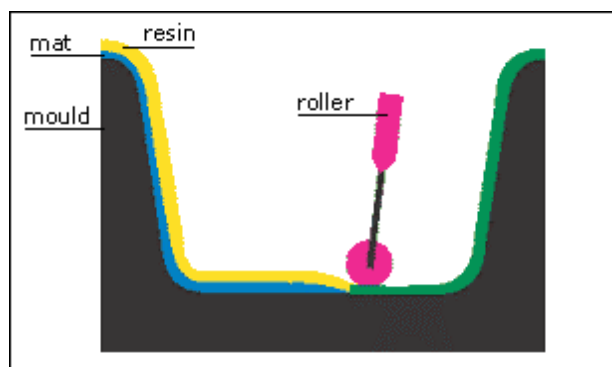


Fig. 5.2 Hand lay-up moulding.

5.5.2 Injection Moulding

Duroplastics and thermoplastics can be used for injection moulding. In reaction injection moulding (RIM), two chemically reactive liquid streams are injected into a mould where polymerisation and formation of the part occur. In Fig. 5.3, i.e. long fibre superconcentrate pellets and companion pellets are fed into a gravimetric blender which accurately proportions the components prior to feeding the injection screw. This process is also known as RTM technology (resin transfer moulding with a dry reinforcement material placed in a mould and the resin injected) or vacuum moulding. Key processing points (Ehrenstein & Kabelka 1997; Saint-Gobain 2002b):

- To prevent fibre breakage from happening during the plastification process, a screw with low compression ratios and gradual transitions should be used.
- Mould temperature is best between 40°C and 60°C.

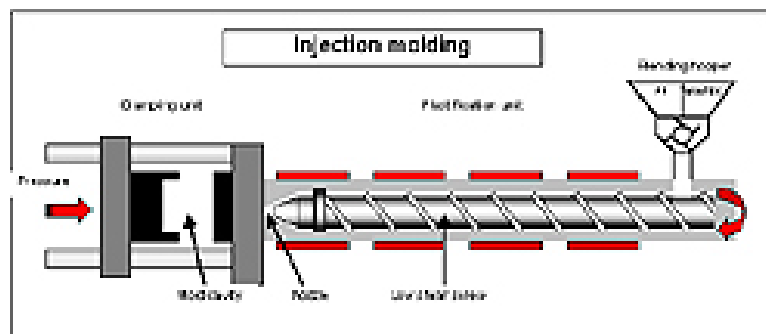


Fig. 5.3 Injection moulding

5.5.3 Other moulding processes

There are different other moulding processes which can not be described in detail here. Key processing points are described by (Saint-Gobain 2002b) for the following processes:

- Vacuum Moulding
- Filament winding
- Pultrusion
- Pulextrusion
- Thermoforming Stamping
- Extrusion Compression Moulding
- Co-moulding
- Diaphragm forming
- Panel Lamination

6 Life Cycle Inventories

There are two products used for solar collectors: Frames of the collector are made with orthophthalic polyester resin with a fibreglass content of 60% in a hand lay-up moulding. Polyamide with a fibre content 30% (injection moulding) is used for the edges of the profiles.³ Other types of moulding processes are not assessed in this report. Information about the environmental impacts might be derived from the description shown in chapter 5.5.

6.1 Glass Fibre

The main environmental issue associated with glass filament production is that it is a high temperature, energy intensive process. This results in the emission of products of combustion, and the oxidation of atmospheric nitrogen. Furnace emissions also contain dust (arising from the volatilisation and subsequent condensation of volatile batch materials) and traces of chlorides, and metals present as impurities in the raw materials. Due to the nature of the fibre-making process varying levels of fluorides are sometimes used in the batch, which can give rise to emissions of hydrogen fluoride. Melting of glass accounts for over 75% of the total energy requirements of glass manufacture. The main energy sources for glass making are natural gas, fuel oil and electricity. Continuous filament glass fibre is produced using recuperative or oxy-fuel fired furnaces (Elvebakken 2002).

The energy use for the production of glass fibre has been investigated by (Windsperger 2002) to be 12MJ/kg. This figure has been derived from a very rough calculation for a range of different products in Denmark and seems to be quite low. A study for tubes from Austria (Elvebakken 2002) has also investigated the use of GRP materials. Data from manufacturers were not available thus only an estimation could be used for this material.

The use of glass fibre reinforced plastic in comparison to other materials for the construction of automobile parts has been investigated by a manufacturer. Inventory data from this study (provided by a glass fibre manufacturer) have not been published (IPPC 2000).

Emissions to the water environment are relatively low. Water is used mainly for cleaning and cooling and can be readily recycled or treated using standard techniques. Specific issues of organic contamination can arise from continuous filament glass fibre processes (IPPC 2000).

The unit process raw data for fibre glass are shown in Tab. 6.2. These are based on the information provided in (IPPC 2000:Table 3.12, p. 87). This publication provides mainly ranges for different in- and outputs to the process. This is based on investigations for the European industry. Mean figures and standard deviation have been derived from the range of values according to the formulas:

$$Mean = \sqrt{Minimum * Maximum}$$

$$STD = \frac{Maximum}{\sqrt{Minimum * Maximum}}$$

The discharge of effluent is estimated with data for the treatment of glass manufacturing effluents (Doka 2007; IPPC 2001, Tab. 6.1).

³ Personal communication Mr. Köbbermann-Rengers, Buderus, DE, 5.2002.

Tab. 6.1 Emission levels of water pollutants in the effluents of the glass industry IPPC 2001

Name Location InfrastructureProcess Unit	Category	SubCategory	Unit	Emissions with effluents from glass manufacturing
				RER 0 1
Ammonium, ion	water	river	mg	1.00E+1
Arsenic, ion	water	river	mg	3.00E-1
Cadmium, ion	water	river	mg	5.00E-2
Chromium, ion	water	river	mg	5.00E-1
COD, Chemical Oxygen Demand	water	river	mg	1.15E+2
Copper, ion	water	river	mg	5.00E-1
Lead	water	river	mg	5.00E-1
Nickel, ion	water	river	mg	5.00E-1
Sulfate	water	river	mg	1.00E+3
Suspended solids, unspecified	water	river	mg	3.00E+1
t-Butyl methyl ether	water	river	mg	5.00E-1
Zinc, ion	water	river	mg	5.00E-1

The estimation for different types of particle emissions is based on the shares given by (Berdowski et al. 2002), but using the emission factors in (IPPC 2000). The major output mass flow is the product, which may be from 55% to 80% of raw material input. The molten glass represents around 80-85% of the furnace raw material input. Most of the loss is made up of gaseous emissions particularly CO₂ from the decomposition of carbonates (IPPC 2000). There is a significant loss of platinum which is used for the brushing appliance which could not be quantified (see footnote 1).

Tab. 6.2 Unit process raw data and data quality indicators for the continuous filament glass fibre

	Name	Location	Infrastructure	Unit	glass fibre, at plant	Uncertainty mVtVne	Standard Deviation 95%	GeneralComment
	Location InfrastructureProcess Unit				RER 0 kg			
product	glass fibre, at plant	RER	0	kg	1.00E+0			
technosphere	tap water, at user	RER	0	kg	8.94E+0	1	2.24	IPPC 2000
	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	1.61E+0	1	1.12	(1,4,1,1,1,2); Forming, utilities and processes
	natural gas, burned in industrial furnace low-NOx >100kW	RER	0	MJ	1.60E+1	1	1.43	Melting
	aluminium oxide, at plant	RER	0	kg	2.14E-2	1	7.14	For other raw materials
	boric acid, anhydrous, powder, at plant	RER	0	kg	1.88E-1	1	1.33	Estimation for colemanite CaB3O4(OH)3·H2O, Hydrated Calcium Borate Hydroxide
	chemicals organic, at plant	GLO	0	kg	3.16E-3	1	3.16	Other binders
	clay, at mine	CH	0	kg	4.64E-1	1	1.17	IPPC 2000
	silica sand, at plant	DE	0	kg	3.70E-1	1	1.23	IPPC 2000
	fluorspar, 97%, at plant	GLO	0	kg	4.47E-3	1	4.47	IPPC 2000
	limestone, milled, loose, at plant	CH	0	kg	3.51E-1	1	1.17	Calcium carbonate
	lubricating oil, at plant	RER	0	kg	2.24E-3	1	2.24	Binder
	nylon 6, at plant	RER	0	kg	2.83E-2	1	1.41	For polymers
	silicone product, at plant	RER	0	kg	1.41E-3	1	1.41	Silane
	transport, lorry 32t	RER	0	tkm	1.02E-1	1	2.09	(4,5,na,na,na,na); Standard distance sand and lime 50km, other materials 100km
	transport, freight, rail	RER	0	tkm	1.49E-1	1	2.09	(4,5,na,na,na,na); Standard distance materials 600km
	treatment, glass production effluent, to wastewater treatment, class 2	CH	0	m3	3.00E-3	1	2.00	Sewage
	disposal, mineral wool, 0% water, to inert material landfill	CH	0	kg	1.22E-1	1	2.04	Fibre glas production waste
	disposal, used mineral oil, 10% water, to hazardous waste incineration	CH	0	kg	2.24E-3	1	7.14	For other raw materials
	disposal, paint, 0% water, to municipal incineration	CH	0	kg	3.61E-3	1	3.61	Binder wastes
	flat glass plant	RER	1	unit	4.00E-10	1	3.05	(2,2,1,3,1,5); Estimation
emission air, high population density	Heat, waste	-	-	MJ	5.80E+0	1	1.12	(1,4,1,1,1,2); Calculation
	Antimony	-	-	kg	3.50E-5	1	1.52	(1,4,na,3,1,na); Emission limit in Belgium
	Arsenic	-	-	kg	2.10E-5	1	1.52	(1,4,na,3,1,na); Emission limit in Belgium
	Cadmium	-	-	kg	7.00E-6	1	1.52	(1,4,na,3,1,na); Emission limit in Belgium
	Carbon dioxide, fossil	-	-	kg	1.55E-1	1	1.29	From raw material decomposition
	Hydrogen chloride	-	-	kg	6.00E-5	1	2.00	IPPC 2000
	Hydrogen fluoride	-	-	kg	3.46E-5	1	2.00	IPPC 2000
	Nitrogen oxides	-	-	kg	4.17E-3	1	1.61	IPPC 2000
	NMVOC, non-methane volatile organic compounds, unspecified origin	-	-	kg	2.24E-4	1	2.24	IPPC 2000
	Particulates, < 2.5 um	-	-	kg	4.85E-5	1	3.46	Berdowski 2002, waste gas purification
	Particulates, > 2.5 um, and < 10um	-	-	kg	1.39E-5	1	3.46	Berdowski 2002, waste gas purification
	Particulates, > 10 um	-	-	kg	6.93E-6	1	3.46	Berdowski 2002, waste gas purification
	Sulfur dioxide	-	-	kg	2.83E-3	1	2.83	IPPC 2000

6.2 Hand Lay-up Moulding

Tab. 6.3 shows the unit process raw data and data quality indicators for hand lay-up moulded glass fibre. Orthophthalic polyester resin with a fibreglass content of 60% is used for this process. The use of glass fibre reinforced plastic does not cause much environmental impacts. It involves a lot of handwork. Pressurized air might be used. About 0.25% of the styrene is emitted. There are no effluents.⁴ Losses are estimated with 5% of the product.

⁴ Personal information Mr. Gruber, Kofmel, CH, www.kofmel.ch, 6.2002.

Tab. 6.3 Unit process raw data and data quality indicators for hand lay-up moulded glass fibre reinforced plastic

	Name	Location	Infrastructure	Unit	glass fibre reinforced plastic, polyester resin, hand lay-up, at plant	uncertainty	Standard deviation	95 %	GeneralComment
	Location	Infrastructure	Process	Unit	RER				
	Unit				0 kg				
product	glass fibre reinforced plastic, polyester resin, hand lay-up, at plant	RER	0	kg	1.00E+0				
technosphere	chemicals organic, at plant	GLO	0	kg	2.50E-2	1	1.24	(4,4,1,3,1,na);	Estimation for extender, additives and peroxides
	glass fibre, at plant	RER	0	kg	6.32E-1	1	1.24	(4,4,1,3,1,na);	Basic product, 95% use efficiency
	polyester resin, unsaturated, at plant	RER	0	kg	4.21E-1	1	1.24	(4,4,1,3,1,na);	Resin, 95% use efficiency
	disposal, mineral wool, 0% water, to inert material landfill	CH	0	kg	5.26E-2	1	1.24	(4,4,1,3,1,na);	Estimation for losses
	transport, lorry 32t	RER	0	tkm	1.08E-1	1	2.09	(4,5,na,na,na,na);	Standard distance materials 100km
	transport, freight, rail	RER	0	tkm	2.16E-1	1	2.09	(4,5,na,na,na,na);	Standard distance materials 200km
emission air, high population density	Hydrocarbons, aromatic	-	-	kg	6.25E-5	1	2.06	(4,4,1,3,1,na);	0.25% emission of styrene

6.3 Injection Moulding

Tab. 6.4 shows the unit process raw data and data quality indicators for injection moulded glass fibre. Polyamide resin with a fibre content 30% is used for this type of application. Its use has been approximated with nylon. Emission are not known. Heating is not very important for this process.

Tab. 6.4 Unit process raw data and data quality indicators for injection moulded glass fibre reinforced plastic

	Name	Location	Infrastructure	Unit	glass fibre reinforced plastic, polyamide, injection moulding, at plant	uncertainty	Standard deviation	95 %	GeneralComment
	Location	Infrastructure	Process	Unit	RER				
	Unit				0 kg				
product	glass fibre reinforced plastic, polyamide, injection moulding, at plant	RER	0	kg	1.00E+0				
technosphere	chemicals organic, at plant	GLO	0	kg	2.50E-2	1	1.24	(4,4,1,3,1,na);	Estimation for extender, additives and peroxides
	nylon 66, glass-filled, at plant	RER	0	kg	1.05E+0	1	1.32	(4,4,1,3,3,na);	95% use efficiency, polyamide resin input approximated with nylon
	injection moulding	RER	0	kg	1.00E+0	1	1.21	(4,na,1,1,1,na);	Estimation for process
	disposal, mineral wool, 0% water, to inert material landfill	CH	0	kg	3.68E-1	1	1.24	(4,4,1,3,1,na);	Estimation for losses
	transport, lorry 32t	RER	0	tkm	1.08E-1	1	2.09	(4,5,na,na,na,na);	Standard distance materials 100km
	transport, freight, rail	RER	0	tkm	2.16E-1	1	2.09	(4,5,na,na,na,na);	Standard distance materials 200km

6.4 Production Plant

Furnaces for producing glass fibres operate continuously for 8 to 12 years, after which time there are rebuilt with either partial or total replacement of the structure depending on its condition (IPPC 2000). Unit process raw data for a production plant of glass fibre were not available. They are estimated with the plant for glass tube production.

The hand lay-up moulding will take place normally at small scale manufacturing places while for injection moulding larger plants are necessary. The infrastructure shall be investigated in connection with the other production stages involved.

6.5 Disposal of Glass Fibre Reinforced Plastics

After use a part of the glass fibres can be reused. Therefore these are pulverized and than they can be used as a filler in a share of 20-30% for new composites. The electricity use for milling is estimated to be 8.3Wh/kg.⁵ This might be considered in life cycle inventories for such products which investigate also the disposal route.

⁵ Personal communication with Caroline Setterwall, Vattenfall, SE on 6.6.2002.

7 Data Quality Considerations

The data quality has been assessed for most of the entries for glass fibre with a calculation based on minimum and maximum figures because information about mean values were not available. These figures show for some inputs and outputs a great possible variation between different production plants. As the data are based on an investigation for all European plants the data quality can be assumed to be good.

The inventory for moulded products is based on assumptions for two different types of processing. A lot of other moulding technologies and product mixtures are possible. Thus the specific technology has to be taken into account if a glass fibre reinforced plastic product is investigated in detail or compared with another type of material.

8 Cumulative Results and Interpretation

Results of the cumulative inventory can be downloaded from the database (www.ecoinvent.ch).

9 Conclusions

The inventory for the glass fibre is based on a literature study for the technologies used in the European glass industry. The energy use is the main environmental aspect. Glass fibres can be used in manifold ways. Therefore resins and different chemicals are necessary. Two types of processing, hand lay-up moulding with polyester resin and injection moulding with polyamide, have been investigated based on rough assumptions given for a single manufacturer. The unit process raw data are meant to be used as background information. For other types of moulded products the background data for glass fibre can be used to assess the environmental impacts if the amount of other chemicals and the process requirements are known.

Abbreviations

AR	alkali-resistant
E glass	glass type with a very low alkaline content
FRP	fibre reinforced plastic
GRP	glass fibre reinforced plastic
PCB	printed circuit boards
SMC	sheet moulding compounds
BMC	bulk moulding compound

Glossary

Matrix	The matrix is the continuous-phase or the material compound. Two types of matrix can be classified: Thermoplastics are capable of being separately hardened and softened by increases and decreases, respectively, in temperature. Duroplastics are changed into a substantially infusible and insoluble material when cured by the application of heat, or through chemical means
Thermosetting	Thermosettings, a type of duroplastic, are changed into a substantially infusible and insoluble material when cured by the application of heat. The fabrication processes for thermosetting matrices can be classified broadly as wet forming processes and processes involving premixes or prepregs.
Duroplastic	Duroplastics are changed into a substantially infusible and insoluble material when cured by the application of heat, or through chemical means.
Thermoplastic	Thermoplastics are capable of being separately hardened and softened by increases and decreases, respectively, in temperature.

Appendices: EcoSpold Meta Information

ReferenceFunction	401 Name	glass fibre, at plant	glass fibre reinforced plastic, polyester resin, hand lay-up, at plant	glass fibre reinforced plastic, polyamide, injection moulding, at plant
Geography	662 Location	RER	plant	RER
ReferenceFunction	493 InfrastructureProcess	0	0	0
ReferenceFunction	403 Unit	kg	kg	kg
DataSetInformation	201 Type	1	1	1
	202 Version	1.0	1.0	1.0
	203 energyValues	0	0	0
	205 LanguageCode	en	en	en
	206 LocalLanguageCode	de	de	de
DataEntryBy	302 Person	41	41	41
	304 QualityNetwork	1	1	1
ReferenceFunction	400 Data_Set_Relates_To_Product	1	1	1
	402 IncludedProcesses	Gate to gate inventory for the production of glass fibre.	Gate to gate inventory for the hand lay-up moulding of glass fibre with polyester resin.	Gate to gate inventory for the injection moulding of glass fibre with polyamide resin including material inputs, process and infrastructure.
	404 Amount	1	1	1
	490 LocalName	Glasfaser, ab Werk	Glasfaserverstärkter Kunststoff, Polyesterharz, handverarbeitet, ab Werk	Glasfaserverstärkter Kunststoff, Polyamid, Spritzguss, ab Werk
	491 Synonyms	fibreglass // fiberglass // fibrous glass filter media	GFK // GRP	GFK // GRP
	492 GeneralComment	Inventory based on a state of the art report for the European glass manufacturing industry. Data had to be estimated from ranges given for different parameters.	Based on assumptions for material uses. No data for direct energy use.	Based on assumptions for material uses. Data for processing and infrastructure are assumed with generic inventory.
	494 InfrastructureIncluded	1	1	1
	495 Category	glass	plastics	plastics
	496 SubCategory	construction	others	others
	497 LocalCategory	Glas	Kunststoffe	Kunststoffe
	498 LocalSubCategory	Bauglas	Andere	Andere
	499 Formula			
	501 StatisticalClassification			
	502 CASNumber	65997-17-3		
TimePeriod	601 StartDate	2000	2000	2000
	602 EndDate	2000	2000	2000
	603 DataValidForEntirePeriod	1	1	1
	611 OtherPeriodText	Time of publication	Time of estimation	Time of estimation
Geography	663 Text	26 furnaces operating at 12 sites in Europe	Information from one manufacturer.	Information from one manufacturer.
Technology	692 Text	Recuperative or oxy-fuel fired furnaces.	Hand lay-up moulding of glass fibre.	Injection moulding of glass fibre.
Representativeness	722 Percent	100		
	724 ProductionVolume	475000 tonnes.	Not known.	Not known.
	725 SamplingProcedure	Literature	Personal communication	Personal communication
	726 Extrapolations	From ranges to mean values.	To European average.	To European average. Background data for polyamide resin were not available. Its use has been approximated with melamine formaldehyde resin.
DataGenerator	727 UncertaintyAdjustments	none	none	none
AndPublication	751 Person	41	41	41
	756 DataPublishedIn	2	2	2
	757 ReferenceToPublishedSource	7	7	7
	758 Copyright	1	1	1
	759 AccessRestrictedTo	0	0	0
	760 CompanyCode			
	761 CountryCode			
	762 PageNumbers	Glass Fibre	Glass Fibre	Glass Fibre

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Part XIV

Insulation Products and Processes

Data v2.0 (2007)

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Summary

This part deals with the production of insulation materials used in the building sector. These are glass wool, foam glass, rock wool (and basalt from mine as raw material), foamglas, polystyrene foam, cellulose fibre and urea formaldehyde foam.

Since the application of rock wool and polystyrene foam slab is very much different from the application of cellulose fibres, the blowing in of the fibres is included in the inventory. The inventories of the insulation materials in ecoinvent represent products of comparable function. Nevertheless the data are not meant for direct comparison of the different materials, rather in a broader context and if the importance of these products for an inventory is not too high.

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1 Introduction

This part deals with the production of insulation materials used in the building sector and basalt as resource for rock wool. The following materials are analysed:

- Glass wool
- Rock wool and basalt
- Foamglas
- Polystyrene foam (EPS / XPS)
- Cellulose fibre (inclusive blowing in)
- Urea formaldehyde foam

Since data for polyurethane foams (hard and soft) are adopted from APME data, they are included in the report on plastics (Hischier (2007))

1.1 Glass wool

“Glass fibres for insulation are fibres of vitreous structure produced from silicate melts by mechanical or pneumatic processes. For insulation, non-textile fibres are used exclusively. Such fibres are also called glass wool, slag wool, rock wool, or mineral wool.” (Fitzer et al. (1997))

1.2 Rock wool

With adaptations to the Swiss situation the following introduction for mineral wool and specially rock wool is based on (EPA (1998)): Mineral wool often is defined as any fibrous glassy substance made from minerals (typically natural rock materials such as basalt or diabase) or mineral products such as slag and glass (Emissions Factor Documentation for Glass Manufacturing (EPA (1998))).

In general it is possible to use slags as primary material instead of natural rock for the production of rock wool. These are for example iron blast furnace slag, copper, lead, and phosphate slags. Generally rock wool products have one of the four following forms:

- "blowing" wool or "pouring" wool, which is injected into structures of buildings;
- batts, which may be covered by paper or foil as vapour sealant shaped to fit between structural members of buildings;
- industrial and commercial products such as high-density fibre felts and blankets, which are used for insulating boilers, ovens, pipes, refrigerators, and other process equipment;
- bulk fibre, which is used as a raw material in manufacturing other products, such as ceiling tile, wall board, spray-on insulation, cement, and mortar.

The module “rock wool, at plant” includes the melting process, defibration and polymerization. The raw material “basalt, at plant” is included in this study. It is a widespread rock of volcanic origin with a broad range of applications, mainly in the building industry. Information on basalt is mainly provided by Lagaly et al. (1997), inventory data is taken from Anonymous (1997) and Köhl & Frei (1994).

1.3 Foamglas

FOAMGLAS® is a thermal insulation product manufactured and distributed throughout the world by Pittsburgh Corning Europe (PCE) und Pittsburgh Corning Corporation. FOAMGLAS® is produced in the form of slabs and pre-cut shapes. It is chiefly used as an insulation material throughout the building industry in addition to being employed for a range of specialized industrial applications. The main application areas in buildings are: flat roofs, underground walls, floors and terraces/car park decks. For

the European market, FOAMGLAS® is produced at three production sites in Tessenderlo (BE), Schmiedefeld (D) and Klasterec (CZ). In 2005, the production site in Tessenderlo produced approximately 320'000 m³ of FOAMGLAS® products of varying densities. This corresponds to 39'200'000 kg FOAMGLAS® products per year when calculating with an average density of 122.5 kg/ m³. Of this amount, approximately 70'000 m³ were delivered to Switzerland, including Lichtenstein, and 7'700 m³ were delivered to Austria.

In terms of quantity, the main product is the FOAMGLAS® board, T4 WDS, which is used for roof insulation (e.g. compact roof systems). Because of its industry prevalence, the T4 WDS production process can be regarded as being representative for all FOAMGLAS® products. 95% of the European consumption of this FOAMGLAS® board is produced in Tessenderlo. A negligible 5% is produced at the site in Germany. The market share of FOAMGLAS® products on the foam glass market for flat roof insulation in Europe is nearly 100%. Electricity use for the production volume delivered to Switzerland, France, the Netherlands and Austria, is covered by renewable energy certificates.

1.4 Polystyrene Foam

This report describes the production process of expanded and extruded polystyrene (or in short EPS and XPS). EPS and XPS find their applications mainly in the construction industry, either for flooring, roofing or walling. Extruded polystyrene shows resistance to moisture and has reliable strength and durability. The raw materials (production of ethylene and benzene) come up for the major part of the energy demand (90%). CO₂ (50% w/w), HFC-134a (25% w/w) and HFC-152a (25% w/w) are taken as blowing agents in the extruder. Emissions resulting from the production of XPS are Ethane (R134a and HFC-152a) and Styrene.

The data used in this report originates mainly from the information provided by the ECOBIS database, Germany (Starzner S. and Wurmer-Weiss P. 2000). Information on material input and the blowing agents are retrieved from Energie- und Umweltzentrum Allgäu (2002) and IPCC (2007).

2 Reserves, Resources and Raw Materials

2.1 Glass Wool

“Glass wool (wadding, fibres) are produced from mixtures of minerals and industrial chemicals that produce the required final composition. In addition to quartz sand, the minerals used include limestone, dolomite, phonolite, feldspar, nepheline-syenite, kernite, colemanite, ulexite, and heavy spar (barite). The industrial chemicals include sodium carbonate, sodium sulphate, and barium carbonate. The use of recycling glass has gained considerable importance.” (Fitzer et al. (1997))

2.2 Rock Wool / Basalt

“Basalt is the most common volcanic rock, occupying large areas of the Earth's crust. The basalts are the most important effusive rocks, and their formation can be observed even now (e.g., numerous volcanoes in Asia, the United States, Europe, and at the mid-ocean ridges). The most important types are the tholeiite basalts and the alkali basalts (feldspathoids, alkali pyroxene, alkali amphibole, and olivine). Medium to coarse grained basalts are known as dolerites, while olivine basalts (which contain feldspathoids as well as plagioclases) are known as basanites, and anchimetamorphic basalts of the Permian/Carboniferous era are known as melaphyres, diabases, and greenstones. The largest single basalt mass in Europe is in the Vogelsberg region of Germany. Other large deposits are in Hegau, the

Eifel, Westerwald, and the Rhön mountains. Another deposit extends from the Oberpfalz (Bavaria), through the Erzgebirge, and as far as the Lausitz region” (Lagaly et al. (1997)).

The most important natural resources used for producing rock wool are dolomite, basalt, and limestone. The fuel used for the melting process is coke. An example of the transport distances of the base materials and fuel are given in Tab. 2.1. The production of the greatest Swiss rock wool factory Flumroc, is used as reference in this study.

Tab. 2.1 Base materials, fuel and transport distances (production of rock wool by Flumroc AG (Flumroc (2002)))

Raw Material	Origin	Distance [km]
dolomite	surrounding	15
basalt	Germany	500
limestone	local rock	20
coke	France and Italy	500

A general comment on resources of primary materials in Switzerland has been made by the Swiss Geotechnical Commission: The resources used by the non-metallic minerals industry can be regarded as quarryable if its mining is possible with actual technology, the usage realizable with supportable financial and temporal expenditure and if the utilization does not collide with other interests. This last point causes a lot of problems in Switzerland as in the intensely populated landscape (Middle Lands) all different kinds of utilization collide. These are for example water protection, settlement area etc.. This circumstance reduces the real quarryable area fundamentally. This scenario is valid for all groups of resources in Switzerland.

For larger countries as Germany, France or Italy the mentioned scenario can also be applicable but in a longer time period.

2.3 Foamglas

FOAMGLAS® insulation materials are made from recycled, mechanically cleaned floatglass (68%) and feldspar (25%) in raw material quality. Small amounts of technical grade quality salt cake, soda ash, iron oxide, manganese oxide, sodium nitrate and carbon black are added to the raw material mix. Electricity is used for the melting process and raw mix production; natural gas is used for the production process.

Information on the reserves and resources can be found in part I on gravel products and part VII on lime products.

2.4 Polystyrene Foam

Polystyrene is a carbon based synthetic polymer, produced from Ethylbenzene. Thus the discussion is mainly revolving around oil (for benzene) and natural gas (for ethylene) resources. Depending on the process, coal is also considered as a raw material. Information on these raw materials can be found in ecoinvent report No. 5 on energy (see Dones et al. (2007)). The production of polystyrene is discussed in Hischier (2007).

2.5 Cellulose Fibre

Cellulose fibre for insulation material is made from waste paper. Thus resource availability is not critical.

2.6 Urea Formaldehyde Foam

Urea formaldehyde (UF) foam is made from UF resin. The production of the resin and the resources used therefore are discussed in Althaus et al. (2007).

3 Characterisation and Use

3.1 Glass Wool

3.1.1 Physical Properties

“Fibrous insulating materials of man-made mineral fibres are a highly disperse system of finely divided fibrous glass with a specific surface of $0.1 - 0.2 \text{ m}^2/\text{g}$. The fibres are oriented predominantly in the horizontal direction. For this reason, the material is isotropic in two dimensions but not in the third. It is an open-pore system with little resistance to flow. This is the basis for its thermal and acoustic properties.

The fibre diameters vary in a broad spectrum. The logarithm of the fibre diameter approximately follows a normal distribution. The minimum fibre diameter is process dependent and is between 0.18 and 0.30 mm. However, the properties of the individual fibres are less important than those of the final insulating material, i.e., the fibres and binder as a whole.

The most important property of insulation is the thermal conductivity. The thermal conductivity of a particular fibrous material is determined by its bulk density ρ_B and its average fibre diameter. The mechanical properties are primarily determined by bulk density and binder content; in any case, however, fibrous insulating materials cannot be used as load-bearing structural unit.” (Fitzer et al. (1997))

The glass wool mat inventoried in this study is 80 to 200 mm thick and delivered in rolls. The thermal conductivity is 0.040 W/mK . The density is 40 kg/m^3 (Richter et al. (1995)).

3.1.2 Chemical Properties

“Mineral wool insulation is inert to organic solvents, even at elevated temperatures. Extractable are mineral oils, emulsifiers, silicones, and low molecular mass components of binders.

Water and aqueous solutions attack the glass surface, the attack magnified by the large fibre surface. Attack by water also loosens the fibre–resin bond.” (Fitzer et al. (1997))

3.1.3 Use

“Mineral-fibre insulating materials are finished products for use in protection from heat, cold, fire, and noise. The major end use area is high-rise construction. Considerable quantities are used in industrial and machine construction for thermal insulation. Fibrous insulating materials are also important as noise absorbers in buildings and elsewhere.” (Fitzer et al. (1997))

3.2 Rock wool / Basalt

3.2.1 Characterisation

Basalt contains $<52\%$ SiO_2 and is the volcanic equivalent of gabbroid rocks. Basalt has a dense, compact, sometimes granular or porphyric structure, and is dark to black in color. The main constituents of the matrix are plagioclase (anorthite 50 – 75 %) and pyroxene (mainly augite). Auxiliary minerals include olivine, magnetite, ilmenite, alkali feldspar, alkali pyroxene, alkali amphibole, feldspathoids, and glass. The physical properties required depend on the use intended for the basalt. Typical properties are high compression strength (2500 – 4000 kg/cm^2 , with a maximum of 5800 kg/cm^2) and high transverse strength (150 – 250 kg/cm^2 , DIN 52 100). To use basalt it requires a high homogeneity, and should be resistant to frost, acids, and waste gases (e.g., flue gas). The density of basalt varies between 3.00 and 3.15 g/cm^3 , the bulk density between 2.95 and 3.00 g/cm^3 . The porosity varies between 20 and 25 vol % (water absorption 0.1 - 0.3 wt %). The specific heat capacity for temperatures in the range zero to 770 °C is ca. 1.09 $\text{J g}^{-1} \text{K}^{-1}$; the thermal conductivity (0 – 100 °C) ca. 0.028 $\text{W cm}^{-1} \text{K}^{-1}$; the linear expansion coefficient is $9 \times 10^{-6} \text{K}^{-1}$. Anhydrous basalt starts melting at ca. 1280 °C, while basalt containing 1 % water starts melting at only ca. 1100 °C. Basalt wool is stable up to 900 °C. Basalt is affected by weathering very soon after it is mined, especially by the heat of strong sunlight in the presence of moisture. Characteristic white spots are formed, and the rock crumbles. Sunburnt basalt is unsuitable for many applications (Lagaly et al. (1997)).

Basaltic glass in form of fibres is known from a lot of basaltic volcanoes (Kündig et al. (1997)). Very fine basaltic fibres are formed when hot (1'100°C) and mobile lava is pressed through an already solidified but perforated crust. This defibration process is triggered by small gas-air explosions in the stream of lava. The chemical composition of the glass fibres used in the rock wool production is similar to the composition of naturally cooled volcano basalt and contains the minerals listed in Tab. 3.1:

Tab. 3.1 Chemical composition of basalt (Kündig et al. (1997))

Mineral	Chemical formula
olivine	$(\text{MgFe})_2\text{SiO}_4$
plagioclase feldspar	$(\text{CaAl}_2\text{SiO}_8)$
augite	$(\text{Ca}(\text{MgFe})\text{SiO}_6)$
magnetite	$(\text{FeFe}_2\text{O}_4)$

The composition of basalt in weight percent of the oxides is shown in Tab. 3.2:

Tab. 3.2 Chemical composition of basalt in weight percentage (Kündig et al. (1997))

Oxide	Weight percent
SiO ₂	50
TiO ₂	2
Al ₂ O ₃	15.9
FeO _{total}	11
MnO	0.2
MgO	8
CaO	10
Na ₂ O	3
K ₂ O	0.5
P ₂ O ₅	0.3
Total	100

Basaltic glass threads have the ideal characteristic in terms of elasticity, corrosion and heat resistance, based on their chemical composition (Si, Al, Mg, Fe, Ca, Na, O) and diameter around 0.01 mm.

Main advantages of rock wool are that it does not burn, rot, absorb humidity and odour.

Based on (Flumroc (2002)) rock wool contains:

- A minimum of 96% artificially produced mineral fibres with non-crystalline structure which is extracted from natural basalt and dolomite as well as recycled rock wool products.
- Modified phenolic resin varying in quantity between 0.5 and 3.0 weight percent depending on the specific product characteristics.
- About 0.2 percent mineral oil to increase the hydrophobic characteristics of the final product and to decrease the amount of dust.

3.2.2 Use

Some basalts have high strength, and are therefore used in the building industry, e.g., as hard core, paving stones and as raw material for building materials. In another important application, molten basalt is converted into refractory fibres as mineral wool for insulation and fire protection (Lagaly et al. (1997)).

Flumroc AG is the leading company for rock wool production in Switzerland. It produces about 40% of the rock-wool products (mostly rock-wool mats) in the Swiss market.

The different products of Flumroc are generally applied for thermal insulation, sound insulation and/or fire protection of buildings (Flumroc (2002)).

The following products with different properties, production volumes and different technical characteristics are marketed by Flumroc and are included in the module „rock wool, at plant“:

Insulation board 1:	Thermal and sound insulation in cavity masonry, or as infilling in external timber walls.
Insulation board 2:	Thermal and sound insulation in ventilated façades with timber subframes, in sheet metal liner trays on industrial façades, and as sound insulation in partition walls.
DUO	Thermal and sound insulation in ventilated façades and in fair-faced cavity masonry.
ECCO	Thermal and sound insulation of ceiling end faces and high pitched roofs. As fire-protective cladding of combustible structural components.

NOVA	Thermal and sound insulation in steep-pitched roofs, insulation over rafters.
PRIMA	Thermal and sound insulation of flat roofs with a protective layer.

The form of the product can vary from boards to mats, stuffing wool and fine granulates. Tab. 3.3 presents main characteristics of Flumroc's different products and their production shares in year 2001.

Tab. 3.3 Most important products of Flumroc (CH) with their production volume in the year 2001 and main technical data

Product	Insulation board 1	Insulation board 3	DUO	ECCO	NOVA	PRIMA	Diverse
Production volume [%]	43.79	10.55	6.19	3.44	10.33	14.46	11.24
Mean apparent density ρ_a [kg/m ³]	32	60	50	75	90	120	60.7
Thermal conductivity λ_D [W/(mK)]	0.036	0.034	0.035	0.036	0.035	0.038	n.A.
Specific thermal capacity c [J/kgK]	830						n.A.
Coefficient of diffusion resistance μ (approx)	2						n.A.
Fire coding ¹	6q.3						n.A.
Maximum temperature in use [°C]	250						n.A.

3.3 Foamglas

Physical properties:

The product inventoried in this study is an insulation slab 40 mm to 200 mm in thickness, a thermal conductivity λ_R of 0.040 W/mK and a density of 110 kg/m³.

Chemical properties:

FOAMGLAS[®] consists of glass (>99% by weight) and small amounts of hydrogen sulfide and carbon dioxide. The product is resistant to rodents, vermin and insects, as well as being rot proof, resistant to ageing, resistant towards acids (except hydrofluoric acid) and chemical substances, especially organic solvents.

3.4 Polystyrene Foam Slab

Background information on polystyrene and its production is given in Hischier (2007)

The thermal conductivity of polystyrene foam is typically between 0.035 and 0.04 W/mK, the density is around 30 kg/m³.

Polystyrene XPS foam insulation has a combination of properties that makes it a strong and efficient insulating material. It shows high resistance to moisture, reliable strength and durability — even without a facer or cover board. Polystyrene XPS with a heat conductivity of 0.035 to 0.040 W/mK is used

¹ <http://bsronline.vkf.ch/D/Info5.asp>, access date 01.07.2003

as an insulating material for commercial, industrial and residential structures. Tab. 3.4 shows information on the material properties of extruded polystyrene (Starzner S. and Wurmer-Weiss P. 2000).

According to a publication issued by Climate Action Network Europe, about half of Polystyrene XPS is currently foamed with CO₂ as a blowing agent (CAN Europe (2004)). Canadian manufacturers of these foam products on the other hand argue that CO₂ “is not a technically or market viable option. The thermal insulation properties of CO₂-blown foam is said to be significantly compromised when compared to halocarbon-blown foam” (Environment Canada (2005)). In their 2007 Special report on “Safeguarding the Ozone Layer and the Global Climate System”, IPCC (2007) writes on this issue: “As a sole blowing agent or in conjunction with co-blowing agents (ethanol, hydrocarbons, acetone, isopropanol, water), CO₂ is principally used in XPS (extruded polystyrene) in regions which have already chosen to phase out HCFCs, and where the market considers the loss in thermal efficiency to be offset by the other inherent properties of XPS boards (moisture resistance, mechanical strength, compressive strength, dimensional stability and resistance to freeze-thaw deterioration). Higher densities or lower conversion may partially or wholly offset the low cost of CO₂ itself in some XPS applications. Nevertheless, the use of CO₂-based technology for XPS is likely to continue to grow as long as the cost of alternative HFC solutions remains high.

For extruded polystyrene (XPS) manufacturing processes, IPCC mentions HFC-134a and HFC-152a as the most common blowing agents in use, apart from CO₂.

Extruded polystyrene products only absorb 0.3% (w/w) moisture. With its closed-cell structure, XPS foam products have a natural “skin” surface which is not conducive to moisture absorbency. The closed-cell structure of XPS imparts long-term strength and durability. Products are available in a range of compressive strengths to suit varied application needs².

Tab. 3.4 Material properties of extruded polystyrene

Technical information	
Density (kg/m ³)	25 - 45
Heat conductivity (W/mK)	0.035 - 0.040 (CO ₂ -foamed, depending on density of the extruded polystyrene slab)
Melting point (°C)	>100 (deformation at temperatures over 85°C)
Water absorption (%)	0.1 – 0.5 (very low)
Pressure resistance (N/mm ²)	0.15 – 0.7 (medium)
Resistance	Resistant to leachate and non-oxidising acids, resistant to vermin, rotting, fungal decay; not resistant to solvents and UV-radiation. Has to be protected against rodents.

Polystyrene is a thermoplastic material which means that it is capable of being melted and re-inserted into the manufacture of new XPS insulation. XPS production creates virtually no “scrap” or waste material because all internal waste is recovered, ground up, re-pelletized, and fed back into the production system.

Use

Polystyrene XPS solutions are designed to cover a wide range of end use applications. Primarily as part of panels or other laminated solutions, they are both thermal and structural solutions. The following ranges of applications are some of the key areas of use of extruded polystyrene boards (KnaufInsulation (2005)):

² <http://www.xpsa.com/tech/physical.html>; accessed June 2005

- General panels (plasterboard, chipboard, cement boards etc.)
- Close tolerance panels (doors, plastisol faced etc.)
- Refrigerated vehicles.

The annual polystyrene XPS consumption in Switzerland is about 1.1 mio m³ (Hungerbühler (2004)). In Germany, the extruded polystyrene panels had a share of 3.2 % of all insulating materials in 1998 (Starzner S. and Wurmer-Weiss P. (2000)). 70% of the market is controlled by two companies in Europe, namely Dow France and BASF (Germany). Further big players on this market are Isobox Henry (UK), SIRAP GEMA (GER-ITA) and Polyglass (ES).

Like many construction materials, XPS is combustible. XPS insulation should be covered with a thermal barrier or otherwise installed in accordance with applicable building codes (XPSA (2005)).

Polystyrene as such is not poisonous. Though test results show that after production, ethylbenzene and styrene can fumigate from polystyrene products. These low emissions decrease rapidly within a few days after production (Starzner S. and Wurmer-Weiss P. (2000)). According to the Portland Cement Association and its recently published Technology Brief News article, attempts to find any emissions from polystyrene foams used in walls failed. No emissions could be detected by the most sensitive instruments. Extensive research on the polystyrenes concluded that when they were forced to burn, their emissions were “no more toxic” than the emissions of burning wood (PCA (2005)).

Other sources however note that there are low styrene gas emissions during the use-phase (eza (2004), Starzner S. and Wurmer-Weiss P. (2000)). If polystyrene XPS catches fire, carbon, carbon dioxide (CO₂), carbon monoxide, styrene and flame-retardant residue are mainly released. Hard sheets of extruded polystyrene foam may not be exposed to temperatures above 85 °C for long periods (Poliglas Dämmsysteme (1998)).

3.5 Cellulose Fibre

Cellulose fibre is used as thermal insulation in buildings. For this application, the fibres are blown into cavities e.g. between two planks of a wall to be insulated. Cellulose fibres in a typical application have a density of about 50 kg/m³ and a thermal conductivity of 0.04 W/mK.

Cellulose fibres are treated with borax and boric acid to enhance fire retarding properties.

3.6 Urea Formaldehyde Foam

UF foam is either available in prefabricated slabs or it is produced in situ to insulate cavities. The density of both foams is between 10 and 30 kg/m³ and the thermal conductivity is around 0.035 W/mK.

UF foam is mainly used as insulation material in buildings, as filler in mining and as soil conditioner (water and nutrition retention). Because UF foam can emit formaldehyde over a long time, it was banned in Canada in some States of the USA in the 1980ies³.

³ http://www.cmhc-schl.gc.ca/en/burema/gesein/abhose/abhose_ce06.cfm, access date 16.03.2004

4 System Characterisation

4.1 Glass Wool

4.1.1 General

The data for glass wool are based on a study from EMPA which is based on data from Isover SA in Switzerland. It is based on the situation by year 1991 (Tab. 4.1). Unfortunately no current data were available.

Hutter et al. (1999) state, that the amount of waste glass used as raw material can be as high as 70%. For this a high purity of the waste glass is needed which is usually not the case in practice. The amount used in this study is about 65% and refers to the year 1991 (Richter et al. (1995)).

4.1.2 Melting

“Two types of melting units are in use for the production of silicate melt: glass melting tank furnaces and cupola furnaces.

The glass melting tank furnaces are similar in construction to those used in other areas of glass manufacture. Preferred are cross-fired recuperative furnaces and U-shaped flame furnaces, but all-electric melters are also used if energy costs are high. Maximum performance is 300 t/d.

Cupola furnaces are used to produce molten rock. They require coarse feed and yield a less uniform melt. The traditional fuel is coke but oil- or gas-fired units are also used. Furnace productivity rarely exceeds 150 t/d.” (Fitzer et al. (1997)).

For this study the “cupola furnace” type has been used. The melting temperature is about 1’400°C.

4.1.3 Fibre Production

From the numerous processes and process variants for fiberizing silicate melt, the Owens process is inventoried in this study. Fitzer et al. (1997) describe this process as follows:

”In the Owens blowing process, the melt, in the form of a thick primary strand, emerges from holes at the bottom of a platinum crucible (the bushing) and is subsequently entrained by a downward-directed stream of gas from the blow-jet. The blowing medium may be steam or compressed air. The primary strand is drawn to form fibres with an average diameter of 3 – 6 mm. The length of the fibres is about 3 – 10 cm. The product contains slugs. Glass and rock melts are processed in this way.” (Fitzer et al. (1997))

Tab. 4.1 Input of raw materials for 1 kg of typical glass wool mat in this study (Richter et al. (1995))

Raw materials	Richter et al. (1995)
Silica sand	1.10E-01 kg
Sodium carbonate	8.36E-02 kg
Limestone	9.64E-03 kg
Dolomite	2.93E-03 kg
Kernite	5.76E-02 kg
Felspar (nepheline)	1.36E-01 kg
Fluorspar	2.20E-02 kg
Waste glass	6.42E-01 kg
Glass powder (not taken into account as assumed to be internally reused production waste)	1.55E-01 kg
Total raw materials	1.06E+00 kg
Binders	
Phenole, Formaldehyde and urea	1.60E-01 kg
Energy	
Electricity	2.32E+00 kWh
Natural gas	2.03E+00 kWh

4.1.4 Product Forming

“The fibres are formed into a continuous sheet by pneumatic processes, while an aqueous solution of binder is sprayed onto the freely falling individual fibres. The binders are slightly pre-condensed phenol – formaldehyde resins. They are often modified by aminoplastic resins (urea, melamine, melamine ether, dicyanodiamide) to suppress the tendency of phenolics to exothermally autoxidize. In the industry these aminoplastic resins are called anti-punking agents.

Fig. 4.1 shows schematically the overall manufacturing process. The binder resins are cured at temperatures around 180 – 250 °C in a continuous flow dryer heated by circulated air. For shaped articles or pipe insulation, for example, special ovens that operate on a similar principle are used. On the other hand, presses with purely convective heat transfer are rarely used since they require long cycle times.

Mineral-fibre insulating materials can be cut to any desired dimension with knives, saws, shears, and milling cutters.” (Fitzer et al. (1997))

The binder used in this study is a mixture of phenol, formaldehyde and urea resins. The total amount of binder based on (Hutter et al. (1999)) is between 0.5 and 9 w-% and based on Richter et al. (1995) even 16 w-% from the weight of the glass wool product. Although the difference can not be explained due to too little information, a share of 16 w-% for the binders is taken into account because most other inventoried data are based on that study (Richter et al. (1995)).

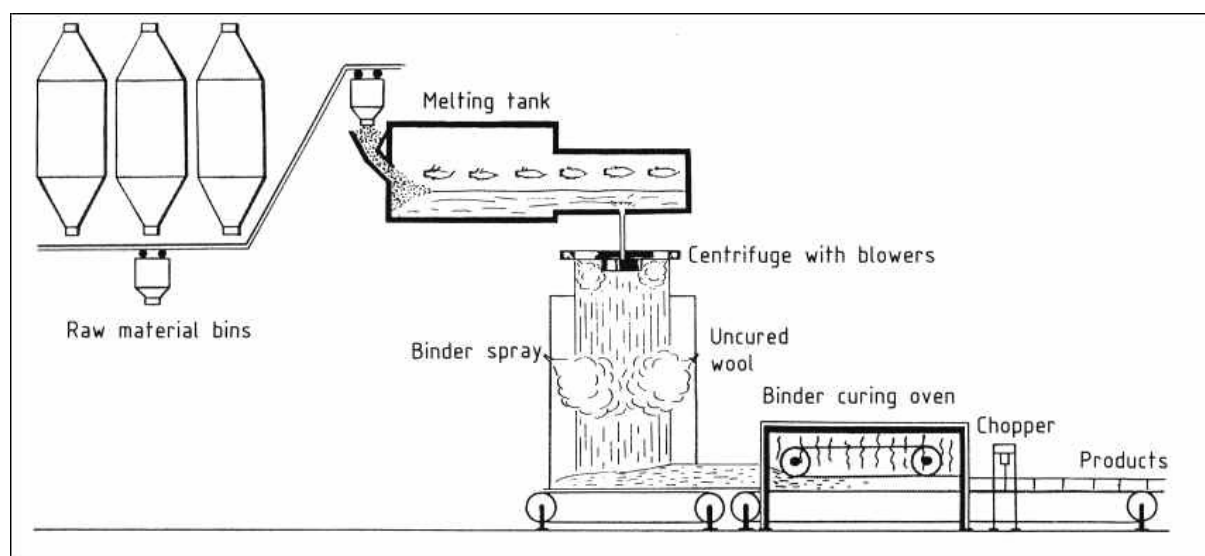


Fig. 4.1 Production process of mineral-fibre insulation (Rotary process is used to form fibres) (Fitzer et al. (1997))

4.1.5 Life Cycle Inventories

Glass wool mat, at plant

The ecoinvent database meta information of “glass wool mat, at plant” (Tab. 4.3) includes all important and necessary information to use the data correctly. Fig. 4.2 shows all in- and output flows of the product “glass wool mat, at plant”, the allocation to the modules in ecoinvent, the sources (mainly Richter et al. (1995)) and the standard deviation for each datum.

Since no specific infrastructure information is available, the “rock wool plant” is used. It includes land-use, buildings and machines.

4.1.6 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

A comparison in Tab. 4.2 of the most important data used in this study (Richter et al. (1995)) with the data of a German study (Hutter et al. (1999)) shows large differences.

The total amount of raw materials is about the same in both studies (1.06 kg per kg product). In contrast the amount of waste glass differs strongly. The Swiss study reckons with about 65% and the German study with about 30%. The data from the Swiss study is clearly related to the product glass wool mat. The German study does not explicit explain to which product it relates. It is assumed that this might be the reason for the different amount of used binder.

The electricity consumption for the Swiss data is higher than for the German data because Isover SA uses an electric heated melting furnace while in the German study the melting process is modelled with a natural gas heated melting furnace. The nevertheless higher amount of natural gas in the Swiss study could be attributed to the fact that waste glass is melted by heat from natural gas. Isover SA could not supply actual data and could not explain the difference to the German study.

There is no further information on the organic wet waste and the wet waste from filtering. For each waste a proxy is used in the study (shown in Fig. 4.2). Organic wet waste is inventoried as “disposal, hazardous waste, 25% water, to hazardous waste incineration” and wet waste from filtering as “dis-

posal, average incineration residue, 0% water, to residual material landfill". The corresponding uncertainties have been raised by estimation.

Tab. 4.2 Comparison of raw materials, binders and energy per kg glass wool product between a German (Hutter et al. (1999) and a Swiss study (Richter et al. (1995))

Raw materials	Hutter et al. (1999)	Richter et al. (1995)
Silica sand	3.20E-01 kg	1.10E-01 kg
Sodium carbonate (Soda)	1.25E-01 kg	
Natrium carbonate		8.36E-02 kg
Limestone	1.90E-02 kg	9.64E-03 kg
Dolomite	7.50E-02 kg	2.93E-03 kg
Borax (hydrated sodium borate)	8.20E-02 kg	
Kernite		5.76E-02 kg
Felspar (nepheline)	1.32E-01 kg	1.36E-01 kg
Sodium sulphate	4.00E-03 kg	
Fluorspar		2.20E-02 kg
Waste glass	3.10E-01 kg	6.42E-01 kg
Glass powder (not taken into account as assumed to be internally reused production waste)		1.55E-01 kg
Total raw materials	1.07E+00 kg	1.06E+00 kg
Binders		
Phenole, Formaldehyde and urea	7.00E-02 kg	1.60E-01 kg
Energy		
Electricity	1.40E+00 kWh	2.32E+00 kWh
Natural gas	1.05E+00 kWh	2.03E+00 kWh

The overall quality of the dataset “glass wool mat, at plant” is poor because data is relatively old (collection around 1993 to 1995; no exact information available in Richter et al. (1995)) and there are quite big inexplicable differences compared to the data in the German study. The Swiss market is not only supplied with glass wool from Isover SA but also from glass wool produced in foreign countries. As the German study shows big differences in the data compared to the Swiss company, it is expected that the companies delivering the Swiss market have quite different technology standards.

Tab. 4.3 ecoinvent meta information for the product “glass wool mat, at plant”

Name	glass wool mat, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Included processes: melting, fibre forming & collecting, hardening & curing and internal processes (workshop, etc.). Additionally transportation of raw materials and energy carrier for furnace, packing and infrastructure are included. For the heat needed, energy modules are used and the needed electricity is from Swiss grid.
Amount	1
Local Name	Glaswollematte, ab Werk
Synonyms	
General Comment to reference function	This module can be used for all different kind of glass wool mats. The density of the glass wool mat used as basis for the study is 40 kg/m ³ .
Start Date	1993
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are only from one company in Switzerland (Isover SA). For some exchanges DE-, RER- and GLO-modules are used as proxy
Technology text	The company worked on a very high technical level but the data refer to the situation before 1995. The energy for the melting process is mainly electricity and from natural gas. The amount of waste glass used as raw material is about 65%.
Representativeness [%]	
Production Volume	
Sampling Procedure	literature and environmental report
Extrapolations	see geography and technology
Uncertainty Adjustments	none

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
silica sand	→	glass wool mat, at plant (part 1)				raw material	construction materials	additives	No	DE	silica sand, at plant	1.10E-01	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
sodium carbonate (Soda)	→					raw material	chemicals	inorganics	No	RER	soda, powder, at plant	8.36E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
dolomite	→					raw material	construction materials	others	No	RER	dolomite, at plant	2.93E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
kernite	→					raw material, borax (hydrated sodium borate) has been used as proxy	chemicals	inorganics	No	US	sodium borates, at plant	5.76E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
feldspar (nepheline)	→					raw material	construction materials	others	No	RER	feldspar, at plant	1.36E-01	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
limestone	→					raw material	construction materials	others	No	CH	limestone, milled, packed, at plant	9.64E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
fluorspar	→					raw material	chemicals	inorganics	No	GLO	fluorspar, 97%, at plant	2.20E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
waste glass	→					raw material	glass	packaging	No	RER	glass, cullets, sorted, at sorting plant	7.97E-01	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
manganese oxide	→					raw material, amount of manganese in manganese oxide: 80%	metals	extraction	No	RER	manganese, at regional storage	2.62E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
sodium nitrate	→					raw material, amount of N in NaNO3	agricultural means of production	mineral fertiliser	No	RER	calcium nitrate, as N, at regional storehouse	5.76E-04	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
fixing agent: phenol	→					raw material	chemicals	organics	No	RER	phenol, at plant	2.94E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
fixing agent: formaldehyd	→					raw material	chemicals	organics	No	RER	formaldehyde, production mix, at plant	9.21E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
fixing agent: urea	→					raw material	chemicals	organics	No	RER	urea, as N, at regional storehouse	3.86E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
caustic soda hydrated	→					raw material	chemicals	inorganics	No	RER	sodium hydroxide, 50% in H2O, production mix, at plant	3.50E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
sulphuric acid	→					raw material	chemicals	inorganics	No	RER	sulphuric acid, liquid, at plant	2.06E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
hydrosilicon	→					raw material	chemicals	inorganics	No	RER	silicone product, at plant	2.94E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
ammonia	→					raw material	chemicals	inorganics	No	CH	ammonia, liquid, at regional storehouse	2.47E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
ammonium sulphate	→					raw material	agricultural means of production	mineral fertiliser	No	RER	ammonium sulphate, as N, at regional storehouse	8.03E-04	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
Mulrex (an oil-based coolant additive)	→					raw material, no chemical composition found; module "lubricating oil, at plant" used as proxy	chemicals	organics	No	RER	lubricating oil, at plant	1.11E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
Basonyl Red	→					raw material, basic dye for the printing links and specialty colorants industry, chlorine free, chemical nature: Xanthene; for of lack of alternative the module "acrylic dispersion, 65%, in H2O, at plant" is used as proxy.	paintings	production	No	RER	acrylic dispersion, 65% in H2O, at plant	5.70E-06	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
water	→					for whole production process, assumed that the whole water consumption is tap water	water supply	production	No	CH	tap water, at user	5.74E+00	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);

Fig. 4.2 Flows for "glass wool mat, at plant" and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
propane and natural gas	glass wool mat, at plant (part 2)				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	7.33E+00 MJ		Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
diesel oil in building machines					construction processes	machinery	No	GLO	diesel, burned in building machine	5.40E-02 MJ		Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
electricity consumption					electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.32E+00 kWh		Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
fireproof stones					construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	2.29E-03 kg		Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
molybdenum, material of electrode					metals	extraction	No	RER	molybdenum, at regional storage	1.25E-05 kg		Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
road transports of raw materials to plant					transport systems	road	No	RER	transport, lorry >16t, fleet average	5.34E-02 tkm		Richter et al. (1995)	1	2.07	(2,4,2,1,1,5);
train transports of raw materials to plant					transport systems	train	No	RER	transport, freight, rail	3.82E-01 tkm		Richter et al. (1995)	1	2.07	(2,4,2,1,1,5);
ship transports of raw materials to plant					transport systems	ship	No	RER	operation, barge	5.29E-02 tkm		Richter et al. (1995)	1	2.07	(2,4,2,1,1,5);
cardboard, at plant					convenience product	paper & cardboard	No	RER	corrugated board, mixed fibre, single wall, at plant	1.14E-02 kg		Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
packaging film, LDPE					convenience product	plastics	No	RER	packaging film, LDPE, at plant	1.32E-02 kg		Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
bonding agent					convenience product	chemicals	No	RER	rosin size, in paper production, at plant	3.44E-04 kg		Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
varnish/colour					convenience product, for lack of alternative the module "acrylic varnish, 87.5% in H2O, at plant" is used as proxy.	paintings	No	RER	acrylic varnish, 87.5% in H2O, at plant	3.50E-03 kg		Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
silicone					convenience product	chemicals	No	RER	silicone product, at plant	2.87E-04 kg		Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
ink					convenience product, for lack of better alternative the module "chemicals, organic, at plant" is used as proxy.	chemicals	No	GLO	chemicals organic, at plant	5.74E-05 kg		Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
solvent					convenience product, for lack of alternative the module "solvents, organic, unspecified, at plant" has been used as proxy.	chemicals	No	GLO	solvents, organic, unspecified, at plant	1.15E-05 kg		Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
road transports of different convenience products to plant					it is assumed that all transports are by lorry	transport systems	No	RER	transport, lorry >16t, fleet average	1.97E-03 tkm		Richter et al. (1995)	1	2.07	(2,4,2,1,1,5);
transports of paper bags and plastic bags to municipal incineration						waste management	No	CH	transport, municipal waste collection, lorry 21t	7.83E-04 tkm		Richter et al. (1995)	1	2.07	(2,4,2,1,1,5);
glass wool plant					"rock wool plant" used as proxy; normalized amount from rock wool production process	insulation materials	Yes	CH	rock wool plant	4.43E-10 unit		estimation and Flumroc (2002)	1	3.06	(2,4,2,1,1,5);

Fig. 4.2 Flows for "glass wool mat, at plant" and their representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
glass wool mat, at plant (part 3)	→		municipal solid waste			waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	8.49E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
	→		wet waste (organic)		For lack of further information the module "disposal, used mineral oil, 10% water, to hazardous waste incineration" is used as proxy.	waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	2.81E-03	kg	Richter et al. (1995)	1	5.00	estimation
	→		wet waste from filtering		For lack of further information the module "disposal, average incineration residue, 0% water, to residual material landfill" is used as proxy.	waste management	residual material landfill	No	CH	disposal, average incineration residue, 0% water, to residual material landfill	1.39E-02	kg	Richter et al. (1995)	1	5.00	estimation
	→		cardboard, in municipal waste incineration		internally used cardboard (a part of the totally used cardboard)	waste management	municipal incineration	No	CH	disposal, packaging cardboard, 19.6% water, to municipal incineration	1.06E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
	→		plastic foil, in municipal waste incineration		internally used plastic foil (a part of the totally used foil)	waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	6.77E-04	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
	→		waste heat from electricity consumption			air	low population density			Heat, waste	8.35E+00	MJ	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
	→		phenole		data from 1999	air	unspecified			Phenol	2.60E-04	kg	environmental report 1999 of Isover SA, Switzerland	1	2.02	(1,4,1,1,1,3);
	→		formaldehyde		data from 1999	air	unspecified			Formaldehyde	4.60E-04	kg	environmental report 1999 of Isover SA, Switzerland	1	2.02	(1,4,1,1,1,3);
	→		ammonia		data from 1999	air	unspecified			Ammonia	1.46E-03	kg	environmental report 1999 of Isover SA, Switzerland	1	2.02	(1,4,1,1,1,3);
	→		fluorine		data from 1999	air	unspecified			Fluorine	9.00E-05	kg	environmental report 1999 of Isover SA, Switzerland	1	2.02	(1,4,1,1,1,3);
	→		waste water to wastewater treatment			waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 3	1.52E-03	m3	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
	→		packed glass wool mat, at plant			insulation materials	production	No	CH	glass wool mat, at plant	1.00E+00	kg	Richter et al. (1995)			

Fig. 4.2 Flows for "glass wool mat, at plant" and their representation in the ecoinvent database (cont.)

4.2 Basalt

4.2.1 General

In this study the inventory of one German quarry is considered, using its environmental report (Anonymous (1997)). The quarry exploits mainly diabas, a variant of basalt besides regular basalt. Exploitation of diabas and basalt can be considered comparable because these two hard rocks have similar properties. Furthermore it is assumed that the different materials produced need similar exploitation efforts, hence justifying the allocation of the environmental burdens by mass. No exact data for the output in 1997 was available. Based on personal communication from the manager of the company in concern, a yearly output of 250'000 t products was assumed, of which a third is supplied to rock wool production.

4.2.2 Process Description

Basalt is mined by quarrying. The first operation is to remove the overburden (i.e., the soil, clay, and bulk rock overlying the deposit). Various techniques and types of earth moving equipment are used for this operation.

The quarry is assumed to be covered and rehabilitated after ten years of operation.

The next stage is generally to drill the bedrock. Rotary and percussive drills are used to drill holes. These are then filled with controlled amounts of explosive. After blasting, the rock is loaded into dump trucks, generally by front-end shovels or hydraulic excavators, and transported to the crushing and screening plant.

In the crushing and screening plant, the larger lumps of rock are broken in a primary jaw crusher. Once obtained the desired size reduction, the rock is washed and classified using shaking screens with different meshes then it is stored. After washing, the muddy water is clarified in a waste water treatment plant. The products are partly sold to regional construction industry and partly to national rock wool producers.

4.2.3 Energy Consumption

The process consumes energy for mining (drill, excavator, trucks), crushing (conveyor, crusher, screens) and rehabilitation (trucks).

4.2.4 Transports

Transport occurs onsite from quarry to crusher by lorries and by conveyor belts during the crushing/washing process.

4.2.5 Emissions

Direct emissions to air and water reported in Anonymous (1997) originate mainly from combustion processes. These are already covered by the modules used for the energy carriers. Therefore these releases are omitted in the module for basalt production. Dust emissions for processing processes (exclusive mining operation) are filtered with modern technology and lie under 1 mg dust per m³ outgoing air.

4.2.6 Infrastructure

The infrastructure of the facilities is modelled using the infrastructure module for the production of limestone. The infrastructure of the mobile equipment is included in the module “diesel, burned in building machine” chosen to describe the use of fuel.

4.2.7 Assignment to ecoinvent datasets

The assignment of the data from the environmental report of a German quarry to the ecoinvent datasets is illustrated in Tab. 4.4.

Tab. 4.4 Transformation of data taken from the environmental report of one German quarry (Anonymous (1997)) to the inventory in ecoinvent.

Environmental Report (Anonymous (1997))			Inventoried in ecoinvent as		
Annual inputs			Input rates per kg of product		
Motor, gear and hydraulic oil ¹⁾	t	13.46	5.38E-05	kg	lubricating oil, at plant
Lacquer	t	0.75	3.00E-06	kg	alkyd paint, white, 60% in solvent, at plant
Paper	t	0.25	1.00E-06	kg	kraft paper, unbleached, at plant
Chemicals	t	0.09	3.60E-07	kg	chemicals organic, at plant
Technical oil and lubricants	t	0.04	1.60E-07	kg	lubricating oil, at plant
Light fuel oil	kWh	21373	3.13E-04 ²⁾	MJ	light fuel oil, burned in industrial furnace 1MW, non-modulating ⁶⁾
Diesel	kWh	2879800	4.14E-02 ³⁾	MJ	diesel, burned in building machine ⁶⁾
Petrol	kWh	1500	6.78E-06 ³⁾	km	operation, passenger car ⁶⁾
Electricity	kWh	1552954	6.21E-03	kWh	electricity, medium voltage, production UCTE, at grid ⁶⁾
Water	m ³	2100	8.40E-06	m ³	Water, lake
Annual outputs					
Waste paper	t	0.13	5.20E-07	kg	disposal, paper, 11.2% water, to municipal incineration
Compost	t	0.59	2.36E-06		(not inventoried, burden free)
Waste oil (motor and gear)	t	12.38	4.95E-05	kg	(recycled, burden free)
Waste oil (hydraulic)	t	1.08	4.32E-06	kg	(recycled, burden free)
Waste, contaminated with oil	t	1.22	4.88E-06	kg	disposal, paint remains, 0% water, to hazardous waste incineration
Miscellaneous waste	t	1.45	5.80E-06	kg	disposal, municipal solid waste, 22.9% water, to municipal incineration
Total waste	t	30.74	1.23E-04 ⁵⁾	kg	
Difference total waste to sum of reported wastes	t	13.89	5.56E-05 ⁵⁾	kg	disposal, municipal solid waste, 22.9% water, to municipal incineration
CO ₂ (	t	4.02	1.61E-05	kg	- ⁶⁾
SO ₂	kg	0.05	2.00E-10	kg	- ⁶⁾
NO _x	kg	6.43	2.57E-08	kg	- ⁶⁾
CO	kg	0.73	2.92E-09	kg	- ⁶⁾
Dust (not transparent which processes are included)	t	0.18	7.20E-07	kg	Inventoried as differential fractions PM<2.5, PM>2.5-<10 and PM>10 ⁷⁾
Effluents	m ³	24	9.60E-08	m ³	treatment, sewage, from residence, to wastewater treatment, class 2
Annual products					
Total hardstone	t	250000 ⁸⁾			
Basalt only	t	83333 ⁸⁾	1	kg	Basalt, at mine

¹⁾ Calculated input from reported waste oil (motor & gear and hydraulic); losses are not taken into account

²⁾ Transformation to volume with reported 10 kWh/l and back to energy with 0.86 kg/l and 42.6 MJ/kg from (Frischknecht et al. (2007)).

³⁾ Transformation to volume with reported 10 kWh/l and back to energy with 0.84 kg/l and 42.8 MJ/kg from (Frischknecht et al. (2007)).

⁴⁾ Transformation to volume with reported 10 kWh/l and back to energy with 0.75 kg/l and 42.5 MJ/kg and 2.82 MJ/km from (Frischknecht et al. (2007)).

⁵⁾ Reported wastes do not sum up to the reported total waste of 30.74 t/a. Difference between total waste and the sum of the reported wastes is inventoried as municipal waste.

⁶⁾ The modules chosen for the fuel combustion include the relevant emissions. Therefore the emissions reported in the environmental report, which are assumed to originate from combustion, are omitted.

⁷⁾ Particulate fractions for lime production are chosen from CEPMEIP-Database (Berdowski et al. (1995), SNAP 240400, lime production, low emission level: PM<2.5 = 10%; PM>2.5-<10 = 40%; PM>10 = 50%)

⁸⁾ Estimation based on personal communication with the manager of the German quarry of concern.

4.2.8 Life Cycle Inventory

The ecoinvent database meta information of “basalt, at plant” (Tab. 4.5) includes all important and necessary information to use the data correctly. Fig. 4.3 shows all in- and output flows of the product “basalt, at plant”, the allocation to the modules in ecoinvent, the sources (mainly Anonymous (1997) and personal communication) and the standard deviation for each datum.

4.2.9 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

Different sub-processes were considered of minor importance and therefore neglected, namely the transport of the material requirements from plants to the basalt quarry. The data in the environmental report used were inconsistent in many places.

The main uncertainty remains in the production volume, which features a big fluctuation from year to year and could not be determined exactly for the year considered.

The source of the data of dust and its size categories are of poor quality because it couldn't be found out which processes are included.

For all these reasons the data for basalt are of fair overall quality and should not be used for comparisons among different construction materials. In a larger context and if the importance of these products for a LCI study is not too high, the data can be used as such.

Tab. 4.5 ecoinvent meta information for the product “basalt, at plant”

Name	basalt, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	The module includes a mining step and a subsequent beneficiation step comprising crushing, washing, and classification.
Amount	1
Local Name	Basalt, ab Abbau
Synonyms	
General Comment to reference function	The module is mainly designed for the use as raw-material for rock wool.
Start Date	2003
End Date	2003
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data for one German quarry used as proxy for whole European situation
Technology text	Basalt is exploited in quarries.
Representativeness [%]	
Production Volume	nA
Sampling Procedure	Literature
Extrapolations	see Geography and Technology
Uncertainty Adjustments	-

General Flow information					Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
basalt (resource)	→	mining of basalt (part 1)	total amount of mined basalt 1997: 250'000 t; the losses by dust emissions have been neglected.	resource	in ground			Basalt, in ground	1.00E+00	kg	estimation	1	1.26	(3,4,2,1,3,nA);	
land occupation, mine [m2a]	→		total area: 200'000 m2 (20 ha), from this 400 m2 are quarry facilities and 199'600 m2 mining area; assumption: After 10 years the used land is recultivated	resource	land			Occupation, mineral extraction site	7.98E-03	m2a	Anonymous (1997)	1	1.64	(4,4,2,1,3,3);	
land transformation mine [m2]	→		total area: 200'000 m2 (20 ha), from this 400 m2 are quarry facilities and 199'600 m2 mining area	resource	land			Transformation, to mineral extraction site	7.98E-04	m2	Anonymous (1997)	1	2.11	(4,4,2,1,3,3);	
land transformation (from forest to mine)	→		total area: 200'000 m2 (20 ha), from this 400 m2 are quarry facilities and 199'600 m2 mining area	resource	land			Transformation, from forest	7.98E-04	m2	Anonymous (1997)	1	2.11	(4,4,2,1,3,3);	
land transformation mine [m2]	→		total area: 200'000 m2 (20 ha), from this 400 m2 are quarry facilities and 199'600 m2 mining area	resource	land			Transformation, from mineral extraction site	7.98E-04	m2	Anonymous (1997)	1	2.11	(4,4,2,1,3,3);	
land transformation (from mine to unknown)	→		total area: 200'000 m2 (20 ha), from this 400 m2 are quarry facilities and 199'600 m2 mining area	resource	land			Transformation, to unknown	7.98E-04	m2	Anonymous (1997)	1	2.11		
Infrastructure (mine) [unite]	→			construction materials	additives	Yes	CH	mine, limestone	8.00E-11	unit	Anonymous (1997)	1	3.28		
blasting [kg explosive]	→		amount of blasting copied from module "limestone, at mine"	construction processes	civil engineering	No	RER	blasting	7.73E-05	kg	data of KFN for the year 2001	1	1.32	(4,nA,3,1,3,nA);	
total machines used in mine for drilling, mining and transports	→		includes infrastructure and emissions	construction processes	machinery	No	GLO	diesel, burned in building machine	4.14E-02	MJ	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	

Fig. 4.3 Flows for “basalt, at plant” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent 2000							Uncertainty					
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
petrol in cars	→	mining of basalt (part 2)			includes emissions	transport systems	road	No	CH	operation, passenger car	6.78E-06	km	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
combustion light fuel oil	→				used for heating of different buildings	oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	3.13E-04	MJ	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
electricity	→					electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	6.21E-03	kWh	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
Water	→					resource	in water			Water, lake	8.40E-07	m3	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
oil for motor, gear and hydraulics	→					chemicals	organics	No	RER	lubricating oil, at plant	5.38E-05	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
Lacquer	→					paintings	production	No	RER	alkyd paint, white, 60% in solvent, at plant	3.00E-06	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
Chemicals	→					chemicals	organics	No	GLO	chemicals organic, at plant	3.60E-07	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
Oil and lubricants	→					chemicals	organics	No	RER	lubricating oil, at plant	1.60E-07	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
			→	Waste heat		Same amount as electricity consumption	air	high population density			Heat, waste	2.24E-02	MJ		1	1.24	(1,4,2,1,3,3);
			→	Dust to air [kg]		Assumption: 10%wt of total	air	low population density			Particulates, < 2.5 um	1.60E-03	kg	Anonymous (1997)	1	3.11	(4,4,2,1,3,3);
			→	Dust to air [kg]		Assumption: 40%wt of total	air	low population			Particulates, > 2.5 um, and <	6.40E-03	kg	Anonymous (1997)	1	2.33	(4,5,2,3,4,3);
			→	Dust to air [kg]		Assumption: 50%wt of total	air	low population			Particulates, > 10 um	8.00E-03	kg	Anonymous (1997)	1	1.88	(4,5,2,3,4,3);
			→	waste paper			waste	municipal	No	CH	disposal, paper, 11.2% water, to	5.20E-07	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);
			→	waste,			waste	hazardous	No	CH	disposal, paint remains, 0%	4.88E-06	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);
			→	remaining waste			waste	municipal	No	CH	disposal, municipal solid waste, 22.9% water, to municipal	6.14E-05	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);
			→	effluents			waste management	incineration	No	CH	treatment, sewage, from residence, to wastewater	9.60E-08	m3	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);
			→	basalt, at mine		amount 1997: 250000 t	construction materials	additives	No	RER	basalt, at mine	1.00E+00	kg		1	1.00	

Fig. 4.3 Flows for “basalt, at plant” and their representation in the ecoinvent database

4.3 Rock Wool

4.3.1 Process Description

The following summary of rock wool processing has been taken mainly from the homepage of the company Flumroc in Switzerland.⁴

The stone is liquefied in a smelting furnace at approximately 1500°C, spun into fibres, impregnated to provide water repellence then hardened again in a special furnace. The stone wool is then cut to size in boards or mats on the sawing line. The finished insulation products are packaged automatically and conveyed directly for transport to customers or the warehouse. Since the raw materials are smelted in a cupola furnace, coke is the most suitable energy source. Tab. 4.4 shows schematically the production process.

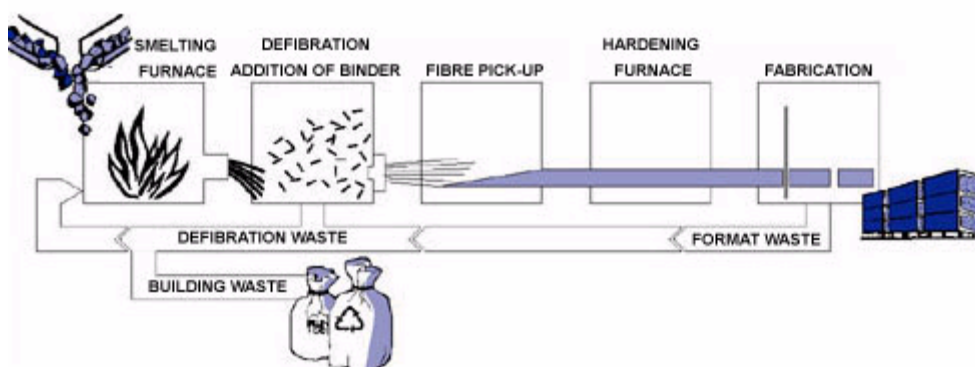


Fig. 4.4 Rock wool production process by Flumroc in Switzerland

A more detailed description of the production process of rock wool can be found in (EPA (1998)):

The first step in the process involves melting the mineral feed. The raw material (rock and sometimes slag) is loaded into a cupola melting furnace in alternating layers with coke at weight ratios of about 5 to 6 parts mineral to 1 part coke. As the coke is ignited and burned, the mineral charge is heated to the molten state at a temperature of 1300° to 1650°C. Combustion air is supplied through tuyeres located near the bottom of the furnace. Process modifications at some plants include oxygen enrichment and the use of natural gas auxiliary burners to reduce coke consumption.

The molten mineral charge exits the bottom of the cupola in a water-cooled trough and falls onto a fiberization device. Two variations of fiberization methods exist. The Powell process uses groups of rotors revolving at a high rate of speed to form the fibres. Molten material is distributed in a thin film on the surfaces of the rotors and then is thrown off by centrifugal force. As the material leaves the surface, small globules develop and form long, fibrous tails as they travel horizontally. Air or steam may be blown around the rotors to assist in fiberizing the material. A second fiberization method, the Downey process, uses a spinning concave rotor with air or steam attenuation. Molten material is distributed over the surface of the rotor, from which it flows up and over the edge and is captured by a high-velocity stream of air or steam.

During the spinning process, not all globules developed are converted into fibre (defibration waste). The non fiberized globules that remain are referred to as “shot”. In raw mineral wool, as much as half of the mass of the product may consist of shot. Shot is usually separated from the wool by gravity im-

⁴ <http://www.flumroc.ch/txtengl/frame.htm>, access date: August 2002

mediately following fiberization and then remelted. The produced amount of shot is always about the same and therefore treated as in an internal closed loop. Thus it is not part of the “final” product.

Depending on the desired product, various chemical agents may be applied to the newly formed fibre immediately following the rotor. In almost all cases, oil is applied to suppress dust and, to some degree, anneal the fibre. This oil can be either a proprietary product or a medium weight fuel or lubricating oil. If the fibre is intended for use as loose wool or bulk products, no further chemical treatment is necessary. If the mineral wool product is required to have structural rigidity, as in batts and industrial felt, a binding agent is applied with or in place of the oil treatment. This binder is typically a phenol-formaldehyde resin that requires curing at elevated temperatures. Both the oil and the binder are applied by atomizing the liquids and spraying the agents to coat the airborne fibre.

After formation and chemical treatment, the fibre is collected in a blow chamber. Resin- and/or oil-coated fibres are drawn down on a wire mesh conveyor by fans located beneath the collector. The speed of the conveyor is set so that a wool blanket of desired thickness can be obtained.

Mineral wool containing the binding agent is carried by conveyor to a curing oven, where the wool blanket is compressed to the appropriate density and the binder is baked. Hot air at a temperature of 150° to 320°C is forced through the blanket until the binder has set. Curing time and temperature depend on the type of binder used and the mass rate through the oven. A cooling section follows the oven, where blowers force air at ambient temperatures through the wool blanket.

To make batts and industrial felt products, the cooled wool blanket is cut longitudinally and transversely to the desired size. Some insulation products are then covered with a vapour barrier of aluminium foil or asphalt-coated kraft paper on one side and untreated paper on the other side. The cutters, vapour barrier applicators, and conveyors are sometimes referred to collectively as a bat machine. Those products that do not require a vapour barrier, such as industrial felt and some residential insulation batts, can be packed for shipment immediately after cutting.

Loose wool products consist primarily of blowing wool and bulk fibre. For these products, no binding agent is applied, and the curing oven is eliminated. For granulated wool products, the fibre blanket leaving the blow chamber is fed to a shredder and pelletizer. The pelletizer forms small, 1-inch-diameter pellets and separates shot from the wool (shot is reused by feeding the furnace). A bagging operation completes the processes. For other loose wool products, fibre can be transported directly from the blow chamber to a baler or bagger for packaging.

Fig. 4.5 shows a detailed flow scheme for the manufacture of rock wool:

The data in ecoinvent are not separated by the mentioned sub-processes. In co-operation with Flumroc the processes packing and administration are included in a separate module called “rock wool, packed, at plant”.

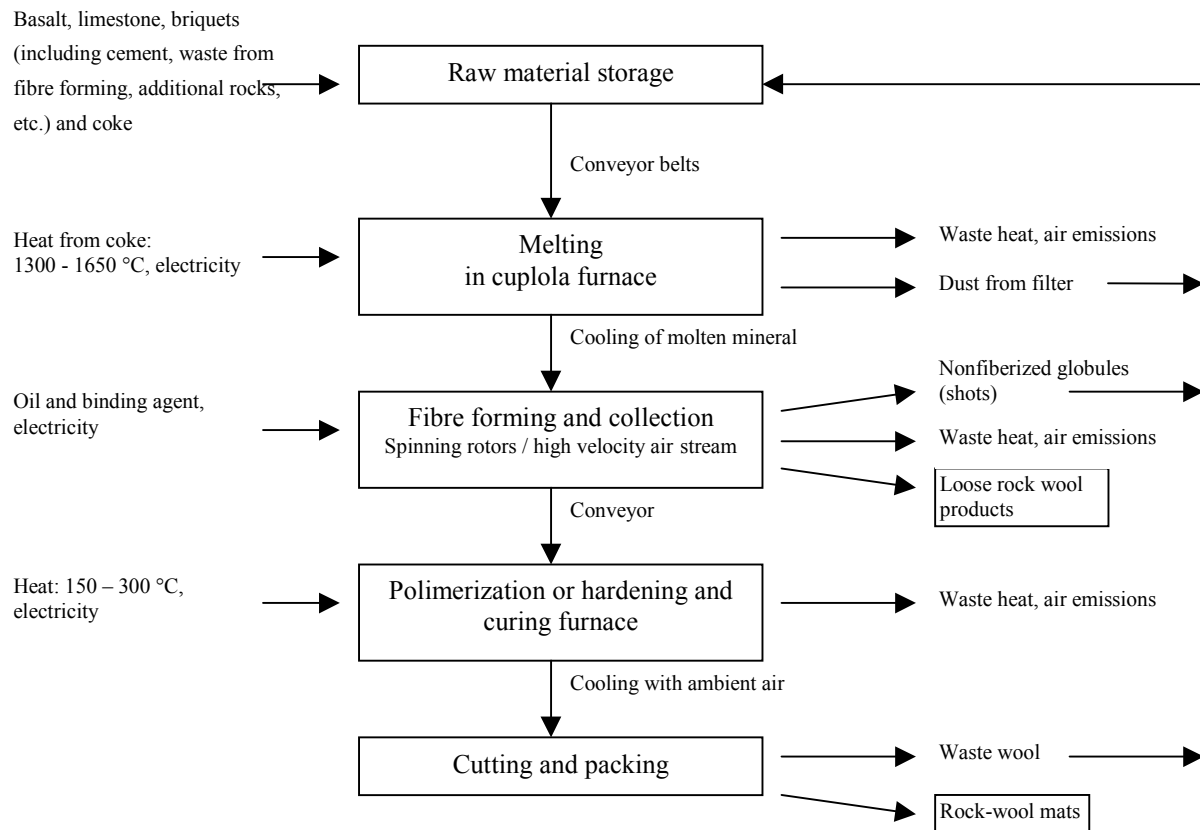


Fig. 4.5 Flow scheme of rock wool production process

4.3.2 Life Cycle Inventories

Rock wool, at plant

The dataset “rock wool, at plant” does not include any data for administration and solid waste. It is basis for the module “rock wool, packed, at plant” which includes in addition administration and solid waste. The infrastructure including its land-use, the buildings and the machines is described in the module “rock wool plant”.

The yearly output of Flumroc for the year 2001 in Switzerland is 45'159'771 kg. This includes the main products as mats and boards (44'833'837 kg) and fine granulated rock wool (325'934 kg).

The emissions of the melting furnace are reduced by carbon monoxide post combustion with recuperation of energy. Many emissions such as ammoniac, formaldehyde, volatile organic compounds, carbon monoxide and phenol are strongly reduced.

According to personal communication⁵ the formaldehyde emissions are significantly reduced in the year 2002. This is achieved by improving the production process and the inaccuracy of the measurement. This fact is not taken into account as all other data are based on the year 2001. At a next revision this must be considered.

⁵ personal communication M. Mebold, Flumroc AG, Mai 2003

The effect of the monoxide post combustion is the conversion of carbon monoxide to carbon dioxide what raises the emission of carbon dioxide by about 25%. The nitrogen oxide emissions are also raised by the transformation of ammonia.

The allocation of the dust to the different categories of particulate sizes is made based on the “Emission Factor Documentation for Mineral Wool Manufacturing (EPA (1998)). The energy balance is taken from the environmental report of Flumroc (Flumroc (2002)) and shown in Tab. 4.6, a rough mass balance is shown in Tab. 4.7.

Tab. 4.6 Energy balance for Flumroc for the year 2001 (Flumroc (2002))

Energetic production input (Group)	Type	[KWh/a]	[kg/a]	gross calorific value [MJ/kg] (*)
Fossil fuels	Coke	67879310	8340120	29.3
	Heating oil	14827243	1250072	42.7
	Diesel	537014	45275	41.9
	Propane gas	2011965	156269	46.4
Regenerative fuel	Offal timber	273611	50000	19.7
Electric energy		14775900		
Total (without electric energy)		84789143		
Total (without electric energy and heat losses to river “Seez”)		82525143		
Energetic production output (Group)				
Energy recuperation	Waste heat to internal room heating	40000000		
	Waste heat to cupola	1032000		
Water emissions	Waste heat cooling water in Seez	2264000		
Air emissions	Waste heat cupola	4539000		
	Waste heat cooling tank	16480000		
	Waste heat collecting tank	13998000		
	Waste heat hardening furnace	7470000		
	Waste heat cooling zone	2680000		
	Waste heat from hot iron / slag	2671000		
	Waste heat from basalt smelting	3325000		
	Waste heat product	596000		
	Waste heat spinning wheel cooling water	5522000		
	Waste heat packing	2021000		
	Waste heat starting up and stopping	1357000		
	Convection / solar radiation	474000		
	Waste heat from deacidification CO ₂ (Dolomite)	1318000		
	Waste heat from drying raw material	2223000		
Difference	Not identified losses	13559143		
Total		84789143		

(*) energy content used in environmental report of Flumroc for 2001 (Flumroc (2002))

Tab. 4.7 Rough mass balance for rock wool production by Flumroc for the year 2001 (Flumroc (2002))

Input	[kg/a]
Mineral resources	
Dolomite	2670562
Basalt	23865848
Additives for briquette	15969200
Cement for briquette	5384300
Fixing agents	
Phenol resin	831743
Formaldehyde resin	2317745
Urea resin	518700
Ammonia	325260
Ammonium hydrogen carbonate	56600
Waterproofing oil	134704
Calcium hydroxide	42000
Total input	52116662
Output	
Main product	
Rock wool	45159771
By-product (not assessed)	
Raw iron	1109560
Assumed emission from raw material	
A part of CO ₂ and a part of water	5847331
Total output	52116662
Difference between In- and Output	0

This table summarises all base materials necessary for the production of rock wool (Input) and makes a comparison with the produced amount of rock wool and by product (Output). The emission of CO₂ indicated in the database (0.806 kg CO₂) is higher since the CO₂ amount arising from combustion (90% carbon content in coke: $0.9 \times 8'340'120 \text{ kg (coke)} / 45'159'771 \text{ kg (rock wool)} = 0.17 \text{ kg C}$, corresponding to about 0.6 kg CO₂).

The ecoinvent database meta information of “rock wool, at plant” (Tab. 4.8) includes all important and necessary information to use the data correctly. Fig. 4.6 shows all in- and output flows of the product “rock wool, at plant”, the allocation to the modules in ecoinvent, the sources (mainly Flumroc (2002) and personal communication) and the standard deviation for each datum.

Tab. 4.8 ecoinvent meta information for the product "rock wool, at plant"

Name	rock wool, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Included processes: melting, fibre forming & collecting, hardening & curing furnace, and internal processes (workshop, etc.). Transport of raw materials and energy carrier for furnace are also included. Not included are administration, packing and infrastructure.
Amount	1
Local Name	Steinwolle, ab Werk
Synonyms	
General Comment to reference function	This product is not available on the market as it is not packed. It has been defined for internal use in this database as input for the packaging module. The mass balance does not come out as some in- and outputs are included in the module "rock wool, packed, at plant". This module shall not be used to describe rock wool in other inventories.
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are only from one company in Switzerland (Flumroc AG). For some exchanges RER-modules have been used as proxy
Technology text	The company works on a technically very high level. The fuel for the melting process is coke. The furnace is a cupola melting furnace. The emissions are abated by post-combustion with energy recuperation. The partition of the total dust emission to the categories of particulate sizes is based on an uncontrolled mineral wool cupola (EPA 1998) which only partly represents this case.
Representativeness [%]	40
Production Volume	112500t/a
Sampling Procedure	environmental report
Extrapolations	see geography and technology
Uncertainty Adjustments	Assumption: the production of Flumroc of 45'000 t per year is representative for the total consumed rock wool products in Switzerland.

General Flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
dolomite	rock wool: cupola melting furnace (part 1)		total: 2'670'562kg/a	construction materials	others	No	CH	limestone, milled, packed, at plant	5.91E-02	kg	Flumroc (2002)	1	1.07	(1,3,1,1,1,1);	
basalt			total: 23'865'848kg/a	construction materials	additives	No	RER	basalt, at mine	5.28E-01	kg	Flumroc (2002)	1	1.07	(1,3,1,1,1,1);	
cement for briquet			total: 5'384'300kg/a	construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	1.19E-01	kg	Flumroc (2002)	1	1.07	(1,3,1,1,1,1);	
additives for briquet: limestone powder (local rock)			60% of total additives (9'581'520kg/a)	construction materials	others	No	CH	limestone, milled, packed, at plant	2.12E-01	kg	Flumroc (2002)	1	1.12	(3,3,1,1,1,1);	
additives for briquet: bauxite powder			40% of total additives (6'387'680kg/a)	metals	extraction	No	GLO	bauxite, at mine	1.41E-01	kg	Flumroc (2002)	1	1.12	(3,3,1,1,1,1);	
additives for briquet: Rock wool powder from cuttings			not balanced because it is internally recycled. Preparation (e.g. milling) is not considered						1.41E-01	kg	Flumroc (2002)				
recycling rock wool material from consumers									1.45E-02	kg	Flumroc (2002)				
coke for melting			input for heating (melting process), total: 8'340'120kg/a; Flumroc calculated with a gross calorific value of 29.3 MJ/kg	hard coal	fuels	No	RER	hard coal coke, at plant	5.41E+00	MJ	Flumroc (2002)	1	1.07	(1,3,1,2,1,1);	
heating oil for recuperator, filter heating and furnace			input for recuperator, filter heating and furnace, total: 1'250'072 kg/a	oil	fuels	No	CH	light fuel oil, at regional storage	2.77E-02	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);	
diesel oil in building machines			assumption: 50% (268'507kWh/a) of total diesel consumption used for different internal transports; "diesel, burned in building machine" used as proxy	construction processes	machinery	No	GLO	diesel, burned in building machine	2.14E-02	MJ	Flumroc (2002)	1	1.22	(1,3,1,2,3,1);	
diesel oil in fork lift truck			assumption: 50% (268'507kWh/a) of total diesel consumption used in fork lift truck; "diesel, burned in building machine" used as proxy	construction processes	machinery	No	GLO	diesel, burned in building machine	2.14E-02	MJ	Flumroc (2002)	1	1.22	(1,3,1,2,3,1);	
waste wood for heating			wood used for inflaming the cupola melting furnace, not balanced						1.11E-03	kg	Flumroc (2002)				
electricity consumption (exkl. electricity consumption for packing)			total usage for cupola melting furnace and all other functions within the entire plant (14'775'900 kWh/a)	electricity	supply mix	No	CH	electricity, medium voltage, at grid	3.24E-01	kWh	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);	
fireproof stones			scone brick , total yearly amount: 2.232m3, density 2.3 t/m3 -> 5.1336 t/a	construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	1.11E-04	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);	
fireproof stones			chamotte grain, total yearly amount: 10'000kg	construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	2.21E-04	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);	
fireproof stones			burnt clay, total yearly amount: 12.24m3, density 1.7 t/m3 -> 20.808 t/a	construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	4.61E-04	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);	

Fig. 4.6 Flows for "rock wool, at plant" and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
train transports of coke to furnace	→	rock wool: cupola melting furnace (part 2)				coke: 50% from France (410 km) and 50% from Italy (535 km); average distance: 500 km	transport systems	train	No	RER	transport, freight, rail	2.71E+00	tkm	additional written information from Flumroc for the year 2000	1	2.00	(1,3,1,2,1,1);
road transports of raw materials to furnace	→				transport of raw material with lorry 40 t, average distance for basalt: 50 km, dolomite: 15 km, additives and fixing agents: 20 km, cement: 20 km	transport systems	road	No	CH	transport, lorry >28t, fleet average	3.96E-02	tkm	additional written information from Flumroc for the year 2000	1	2.05	(1,3,1,1,3,1);	
train transports of raw materials to furnace	→				transport of raw material with freight train, average distance for basalt: 170 km	transport systems	train	No	RER	transport, freight, rail	8.98E-02	tkm	additional written information from Flumroc for the year 2000	1	2.00	(1,3,1,2,1,1);	
ship transports of raw materials to furnace	→				transport of raw material with ship, average distance for basalt: 420 km	transport systems	ship	No	RER	transport, barge	2.22E-01	tkm	additional written information from Flumroc for the year 2000	1	2.00	(1,3,1,2,1,1);	
				Waste heat to air from all sources excluding electricity uses		total heat emission (data of environmental report): 82'525'143 kWh/year	air	low population density			Heat, waste	6.58E+00	MJ	Flumroc (2002)	1	1.07	(1,3,1,1,1,1);
				Waste heat to water from all sources excluding electricity uses		total heat emission (data of environmental report): 2'264'000 kWh/year	air	low population density			Heat, waste	1.80E-01	MJ	Flumroc (2002)	1	1.07	(1,3,1,1,1,1);
				waste heat from electricity consumption		waste heat from electricity consumption	air	low population density			Heat, waste	1.17E+00	MJ	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);
				dust to air [kg], > 10 um		particulates for production of rock wool, > 10 um; share for uncontrolled mineral wool cupolas has been used: 6% of total dust: 513 kg/a (total of particulates from environmental report and share from EPA-report)	air	low population density			Particulates, > 10 um	1.14E-05	kg	Flumroc (2002) and EPA (1998) (Mineral Wool Manufacturing)	1	1.78	(1,3,1,1,4,1);
				dust to air [kg], < 2.5 um		particulates for production of rock wool, < 2.5 um; share for uncontrolled mineral wool cupolas has been used: 90% of total dust: 7'695 kg/a (total of particulates from environmental report and share from EPA-report)	air	low population density			Particulates, < 2.5 um	1.70E-04	kg	Flumroc (2002) and EPA (1998) (Mineral Wool Manufacturing)	1	3.23	(1,3,1,1,4,1);
				dust to air [kg], > 2.5 um, and < 10um		particulates for production of rock wool, > 2.5 um, and < 10um; share for uncontrolled mineral wool cupolas has been used: 4% of total dust: 342 kg/a (total of particulates from environmental report and share from EPA-report)	air	low population density			Particulates, > 2.5 um, and < 10um	7.57E-06	kg	Flumroc (2002) and EPA (1998) (Mineral Wool Manufacturing)	1	2.24	(1,3,1,1,4,1);

Fig. 4.6 Flows for "rock wool, at plant" and their representation in the ecoinvent database (cont)

General Flow information					Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
rock wool: cupola melting furnace (part 3)	→	ammonia			Total measured emission from melting process only, as there are no ammonia emissions from combustion: 50250 kg/a	air	low population density			Ammonia	1.11E-03	kg	Flumroc (2002)	1	1.50	(1,3,1,1,1,1);
	→	lead			total measured emissions, only from Cupola furnace: 2.4kg/a	air	low population density			Lead	5.31E-08	kg	Flumroc (2002)	1	5.00	(1,3,1,1,1,1);
	→	cadmium			total measured emissions, only from Cupola furnace: 0.4kg/a	air	low population density			Cadmium	8.86E-09	kg	Flumroc (2002)	1	5.00	(1,3,1,1,1,1);
	→	hydrochloric acid			total measured emissions: 804.4kg/a	air	low population density			Hydrogen chloride	1.78E-05	kg	Flumroc (2002)	1	1.50	(1,3,1,1,1,1);
	→	chromium			total measured emissions, only from Cupola furnace: 0.2kg/a	air	low population density			Chromium	4.43E-09	kg	Flumroc (2002)	1	5.00	(1,3,1,1,1,1);
	→	hydrogen fluoride			total measured emissions:113.4kg/a	air	low population density			Hydrogen fluoride	2.51E-06	kg	Flumroc (2002)	1	1.50	(1,3,1,1,1,1);
	→	formaldehyde			total measured emissions after carbon monoxide post-combustion: 471 kg/a	air	low population density			Formaldehyde	1.04E-05	kg	Flumroc (2002)	1	2.05	(1,3,1,1,3,1);
	→	formaldehyde			total measured emissions from different processes: 7817 kg/a	air	low population density			Formaldehyde	1.73E-04	kg	Flumroc (2002)	1	2.05	(1,3,1,1,3,1);
	→	VOC			total measured emissions from different processes: 18'146kg/a minus formaldehyde 8'288kg/a and phenole 4'641kg/a; it is assumed that no methane is emitted	air	low population density			NMVOC , non-methane volatile organic compounds, unspecified origin	1.16E-04	kg	Flumroc (2002)	1	2.24	(1,3,1,1,4,1);

Fig. 4.6 Flows for "rock wool, at plant" and their representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
rock wool: cupola melting furnace (part 4)	→	carbon monoxide			total measured emissions from combustion: 0 kg/a, other processes: 4'920kg/a	air	low population density			Carbon monoxide, fossil	1.09E-04	kg	Flumroc (2002)	1	5.00	(1,3,1,1,1,1);
	→	carbon dioxide			total calculated emissions after carbon monoxide post-combustion: 27'521'152 kg/a	air	low population density			Carbon dioxide, fossil	6.09E-01	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);
	→	carbon dioxide			total measured emissions from process: 8'859'673 kg/a	air	low population density			Carbon dioxide, fossil	1.96E-01	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);
	→	copper			total measured emissions, only from Cupola furnace: 0.4kg/a	air	low population density			Copper	8.86E-09	kg	Flumroc (2002)	1	5.00	(1,3,1,1,1,1);
	→	phenol			total (production of rock wool) measured emissions from different processes: 4'641kg/a	air	low population density			Phenol	1.03E-04	kg	Flumroc (2002)	1	1.50	(1,3,1,1,1,1);
	→	sulfur dioxide			total measured emissions mostly from Cupola furnace: 117'251kg/a (this corresponds to 0.7% S in the coke)	air	low population density			Sulfur dioxide	2.60E-03	kg	Flumroc (2002)	1	1.07	(1,3,1,1,1,1);
	→	nitrogen oxide			total (production of rock wool) measured emissions after carbon monoxide post-combustion: 33'540 kg/a	air	low population density			Nitrogen oxides	7.43E-04	kg	Flumroc (2002)	1	1.50	(1,3,1,1,1,1);
	→	nitrogen oxide			total (production of rock wool) measured emissions from other processes: 21'334 kg/a	air	low population density			Nitrogen oxides	4.72E-04	kg	Flumroc (2002)	1	1.50	(1,3,1,1,1,1);
	→	zinc			total (production of rock wool) measured emission from Cupola furnace: 0.9kg/a	air	low population density			Zinc	1.99E-08	kg	Flumroc (2002)	1	5.00	(1,3,1,1,1,1);
	→	steel, to recycling			1'109'560kg/a; not balanced						2.46E-02	kg	Flumroc (2002)			

Fig. 4.6 Flows for "rock wool, at plant" and their representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent						Uncertainty information						
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
fixing agent: phenol	→	rock wool: fiber forming, collection, hardening and cutting (part 5)				total amount: 831743kg/a	chemicals	organics	No	RER	phenol, at plant	1.84E-02	kg	Flumroc (2002)	1	1.07	(1,3,1,2,1,1);
train transports of phenol to plant	→				transport of phenol with freight train, distance: 161km	transport systems	train	No	RER	transport, freight, rail	2.97E-03	tkm	additional written information from Flumroc for the year 2000	1	2.00	(1,3,1,2,1,1);	
fixing agent: formaldehyd	→				total amount: 2'317'745kg/a	chemicals	organics	No	RER	formaldehyde, production mix, at plant	5.13E-02	kg	Flumroc (2002)	1	1.07	(1,3,1,2,1,1);	
train transports of formaldehyd to plant	→				transport of formaldehyd with freight train, average distance: 506km	transport systems	train	No	RER	transport, freight, rail	2.60E-02	tkm	additional written information from Flumroc for the year 2000	1	2.00	(1,3,1,2,1,1);	
fixing agent: urea	→				total amount: 518'700kg/a	chemicals	organics	No	RER	urea, as N, at regional storehouse	1.15E-02	kg	Flumroc (2002)	1	1.07	(1,3,1,2,1,1);	
train transports of urea to plant	→				transport of urea with freight train, average distance: 584km	transport systems	train	No	RER	transport, freight, rail	6.71E-03	tkm	additional written information from Flumroc for the year 2000	1	2.00	(1,3,1,2,1,1);	
fixing agent: ammonia	→				total amount: 325'260kg/a	chemicals	inorganics	No	RER	ammonia, liquid, at regional storehouse	7.20E-03	kg	Flumroc (2002)	1	1.07	(1,3,1,2,1,1);	
train transports of ammonia to plant	→				transport of waterproofing oil with freight train, average distance: 342km	transport systems	train	No	RER	transport, freight, rail	2.46E-03	tkm	additional written information from Flumroc for the year 2000	1	2.00	(1,3,1,2,1,1);	
fixing agent: ammonium hydrogen carbonate	→				total amount: 56'600kg/a	chemicals	inorganics	No	RER	ammonium bicarbonate, at plant	1.25E-03	kg	Flumroc (2002)	1	1.07	(1,3,1,2,1,1);	
fixing agent: waterproofing oil	→				total amount: 134'704kg/a	chemicals	organics	No	RER	lubricating oil, at plant	2.98E-03	kg	Flumroc (2002)	1	1.07	(1,3,1,2,1,1);	

Fig. 4.6 Flows for "rock wool, at plant" and their representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
train transports of waterproofing oil to plant	→	rock wool: fiber forming, collection, hardening and cutting (part 6)				transport of waterproofing oil with lorry, average distance: 487km	transport systems	road	No	CH	transport, lorry >28t, fleet average	1.45E-03	tkm	additional written information from Flumroc for the year 2000	1	2.00	(1,3,1,1,1,1);
fixing agent: calcium hydroxide	→					total amount: 42'000kg/a	construction materials	binder	No	CH	lime, hydrated, packed, at plant	9.30E-04	kg	Flumroc (2002)	1	1.07	(1,3,1,1,1,1);
train transports of calcium hydroxide to plant	→					assumption: transport of calcium hydroxide with lorry, average distance:15km	transport systems	road	No	CH	transport, lorry >28t, fleet average	1.40E-05	tkm	additional written information from Flumroc for the year 2000 and personal communication Flumroc, Mrs. Mebold, 26.09.02	1	2.28	(5,3,1,1,3,1);
process water	→					total yearly amount: 537'438'720 kg/a total amount: 38'290kg/a, 100% primary aluminium; reason: it can be assumed that after usage the aluminium foil will be disposed with the rock wool in an inert material landfill and therefore primary aluminium has to be produced.	resource	in water			Water, well, in ground	1.19E-02	m3	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);
material: aluminium foil for coating	→						metals	extraction	No	RER	aluminium, primary, at plant	8.48E-04	kg	Flumroc (2002)	1	1.22	(1,3,1,2,3,1);
process: coating with aluminium foil	→					total amount: 38'290kg/a	metals	processing	No	RER	sheet rolling, aluminium	8.48E-04	kg	Flumroc (2002)	1	1.07	(1,3,1,2,1,1);
lamination with paper	→					total amount: 12'050kg/a	paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	2.67E-04	kg	Flumroc (2002)	1	1.22	(1,3,1,2,3,1);
diverse material	→					mostly paint for colorung some rock wool mats: 196'800kg/a	paintings	production	No	RER	acrylic dispersion, 65% in H2O, at plant	4.36E-03	kg	Flumroc (2002)	1	2.01	(1,3,1,2,5,1);
Infrastructure rock wool plant	→					amount: 1 unit	insulation materials	production	Yes	CH	rock wool plant	4.43E-10	unit	Flumroc (2002)	1	3.00	(1,3,1,1,1,1);
						770 kg/a, production waste to recycling						1.71E-05	kg	Flumroc (2002)	1		(,,,,);
	→	alu foil, to recycling															
	→	waste water to wastewater treatment			one part of process water (16'664 t/y) to wastewater treatment; rest of water evaporates	waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 3	3.69E-04	m3	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);	
	→	loose rock wool products, at plant			total produced amount of rock wool products per year: 45'159'771 kg, assumed production time: 50a	insulation materials	production	No	CH	rock wool, at plant	1.00E+00	kg	Flumroc (2002)				

Fig. 4.6 Flows for "rock wool, at plant" and their representation in the ecoinvent database (cont.)

Rock wool, packed, at plant

The ecoinvent database meta information of “rock wool, packed, at plant” (Tab. 4.9) includes all important and necessary information to use the data correctly. Fig. 4.7 shows all in- and output flows of the product “rock wool, packed, at plant”, the allocation to the modules in ecoinvent, the sources (mainly Flumroc (2002) and personal communication) and the standard deviation for each datum.

The inventories for packing and administration have been added to the module “rock wool, at plant”. For the amount of the different packaging materials (packaging film, pallet, etc.) a loading weight per pallet of 250 kg was assumed. The area of a pallet is 1m² (120×80cm)⁶, a load height of 2.5m is assumed. This gives a loading volume of 2.5m³. The average density of the packed rock wool product is assumed to be 100 kg/m³. Based on volume and the density the amount of the different packaging materials is calculated.

The machines and packing materials (pallets, cardboard, packaging film) and the waste treatment services necessary for packing are included in this module.

Tab. 4.9 ecoinvent meta information for the product “rock wool, packed, at plant”

Name	rock wool, packed, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes mechanical packing and the administration of the rock wool factory.
Amount	1
Local Name	Steinwolle, verpackt, ab Werk
Synonyms	mineralisch//Isolation//Dämm//matte//platte//mineral//insulation//mat//board
General Comment to reference function	With functional unit kg, this module can be used for all different rock wool products. Only the density of rock wool based on this module was fixed to calculate the part of the pallet use: packaging volume: 2.4m ³ , load weight: 250 kg -> 100 kg/m ³ . The mass balance does not come out as some in- and outputs are included in the module "rock wool, at plant".
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are only from one company in Switzerland (Flumroc AG). For some exchanges RER-modules have been used as proxy
Technology text	The company Flumroc AG works on a technically high level. The packing and loading process is automated.
Representativeness [%]	40
Production Volume	112500t/a
Sampling Procedure	environmental report
Extrapolations	see geography and technology
Uncertainty Adjustments	Assumption: the production of Flumroc of 45'000 t per year is representative for the total consumed rock wool products in Switzerland.

⁶ <http://www.fz-lm.de/labore/pwllabor/MARKTSP/PALSYSPalsys1.htm#EUR-Flachpalette>

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
rock wool, loose, at plant	→	packaging and administration of rock wool mat (part 1)				total produced amount of rock wool products per year: 45'159771 kg	insulation materials	production	No	CH	rock wool, at plant	1.00E+00	kg	Flumroc (2002)	1	1.07	(1,3,1,1,1,1);
infrastructure packing and loading of rock wool products	→					weight of packing- and loading-machines: assumption 2t (assumed lifetime: 20 years)	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	2.21E-06	kg	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	3.23	(5,3,1,2,1,1);
pallets	→					average loading weight: 250kg	wooden materials	processing	No	RER	EUR-flat pallet	4.00E-03	unit	Flumroc (2002)	1	1.22	(1,3,1,2,3,1);
cardboard, at plant	→					for packing of special products: 15'000kg/a	paper & cardboard	cardboard & corrugated board	No	RER	corrugated board, mixed fibre, single wall, at plant	3.32E-04	kg	Flumroc (2002)	1	1.22	(1,3,1,2,3,1);
packaging film, LDPE	→					amount: 500g per pallet with an average loading weight of 250 kg	plastics	processing	No	RER	packaging film, LDPE, at plant	2.00E-03	kg	Flumroc (2002)	1	1.07	(1,3,1,2,1,1);
propan gaz liquid	→					for heating of shrinking foil	natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	1.60E-01	MJ	Flumroc (2002)	1	1.22	(1,3,1,2,3,1);
energy consumption for packing and loading	→					based on packing of lime products	electricity	supply mix	No	CH	electricity, medium voltage, at grid	3.09E-03	kWh	data of KFN for the year 2001	1	1.51	(1,2,1,1,4,1);
transports of pallets, paper bags and plastic bags to plant by lorry	→					Distances: 0.0008kg packaging film (40%): 275km, 0.0012 kg packaging film (60%): 50km, 0.09 kg pallets (85%): 50km, 0.09 kg pallets (15%): 320km, assumption: 0.05kg cardboard: 130km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	6.85E-04	tkm	Written information M. Mebold, Flumroc, 02.09.02	1	2.06	(3,3,1,1,3,1);
administration: paper consumption	→					2730kg/a (includes the production and packaging processes)	paper & cardboard	graphic paper	No	RER	paper, woodcontaining, LWC, at plant	6.05E-05	kg	Flumroc (2002)	1	1.07	(1,3,1,2,1,1);
administration: print material	→					76400kg/a (includes the production and packaging processes)	paper & cardboard	graphic paper	No	RER	paper, woodcontaining, LWC, at plant	1.69E-03	kg	Flumroc (2002)	1	1.07	(1,3,1,2,1,1);
workshop: purifier	→				from workshop: 720kg/a, assumption: 10% tenside (most used tenside: lineare alkylbenzene sulfonates) (includes the production and packaging processes)	chemicals	organics	No	RER	alkylbenzene, linear, at plant	1.59E-06	kg	Flumroc (2002)	1	1.22	(1,3,1,2,3,1);	
tap water	→				total yearly amount: 9'317'204 kg/a (includes the production and packaging processes)	water supply	production	No	RER	tap water, at user	2.06E-01	kg	Flumroc (2002)	1	1.22	(1,3,1,2,3,1);	

Fig. 4.7 Flows for “rock wool, packed, at plant” and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
packaging and administration of rock wool mat (part 2)		municipal solid waste	55'460kg/a (includes waste from production and packing processes)	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.23E-03	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);
		waste from solvent use	760kg/a (includes waste from production and packing processes)	waste management	hazardous waste incineration	No	CH	disposal, solvents mixture, 16.5% water, to hazardous waste incineration	1.68E-05	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);
		waste oil and emulsions	8'920kg/a (includes waste from production and packing processes)	waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	1.98E-04	kg	Flumroc (2002)	1	1.07	(1,3,1,1,1,1);
		waste paper direct	assumption to recycling						1.17E-04	kg	Flumroc (2002)	1	1.07	(1,3,1,1,1,1);
		waste paper indirect	print material to costumer; assumption of final destination: waste paper to recycling						1.62E-03	kg	Flumroc (2002)	1	1.07	(1,3,1,1,1,1);
		waste water to wastewater treatment	water (9'317'204 t/y) to wastewater treatment	waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 3	2.06E-04	m3	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);
		pallets, to recycling	not balanced, because it is recycled								Flumroc (2002)	1		(,,,,);
		cardboard, in municipal waste incineration	same amount as used: 15'000kg/a	waste management	municipal incineration	No	CH	disposal, packaging cardboard, 19.6% water, to municipal incineration	3.32E-04	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);
		plastic foil, in municipal waste incineration	amount which is used minus amount which will be recycled	waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	6.20E-03	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);
		plastic foil, to recycling	4'465kg/a						9.89E-05	kg	Flumroc (2002)	1		(,,,,);
		transports of paper bags and plastic bags to municipal incineration	standard distance: 10km	waste management	others	No	CH	transport, municipal waste collection, lorry 21t	7.98E-06	tkm	Flumroc (2002)	1	1.12	(3,3,1,1,1,1);
		rock wool mat, loose, off-cut	Off-cut waste which is remelted: not balanced								Flumroc (2002)	1		(,,,,);
		packed rock wool products, at plant	total produced and packed amount of rock wool products per year: 45'159'771 kg	insulation materials	production	No	CH	rock wool, packed, at plant	1.00E+00	kg	Flumroc (2002)			

Fig. 4.7 Flows for “rock wool, packed, at plant” and their representation in the ecoinvent database

Rock wool plant

The data are all based on personal communication and plans from Flumroc AG. The production volume is about 45'000 tons per year. It is assumed that the production site will have a life span of 50 years. This includes the land occupation and the building infrastructure. The life span of the machines is estimated to be 20 years. The life span of the conveyor belt is estimated to be 10 years as it is assumed that the belt is worn faster.

The infrastructure of all „packing and loading“ modules of building materials include only the specific machines and no buildings. The reason is to avoid double counting since in most cases the packing and loading process takes place in the same building as the production process.

The ecoinvent database meta information of “rock wool plant” (Tab. 4.10) includes all important and necessary information to use the data correctly. Fig. 4.8 shows all in- and output flows of the product “rock wool plant”, the allocation to the modules in ecoinvent, the sources (mainly Flumroc (2002) and personal communication) and the standard deviation for each datum.

Tab. 4.10 ecoinvent meta information for the product “rock wool plant”

Name	rock wool plant
Location	CH
Infrastructure Process	1
Unit	unit
Data Set Version	2.0
Included Processes	Includes the area of the infrastructure (buildings, roads, etc.), the buildings and the machines used for the rock wool production
Amount	1
Local Name	Steinwollewerk
Synonyms	Fabrik//Betrieb//company//firm//mineralisch//Isolation//Dämm//matte//platte//mineral//insulation//mat//board
General Comment to reference function	The production volume of the plant is 45'000 t of rock wool products per year
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are only from one company in Switzerland (Flumroc AG). For some exchanges RER-modules have been used as proxy
Technology text	The data for the infrastructure correspond to high level of technology. The furnace is a cupola melting furnace. The emissions are abated by post-combustion with energy recuperation. As no description of the used machines exists, a total weight for these machines has been estimated and the module "machine, heavy, unspecified" used.
Representativeness [%]	40
Production Volume	112500t/a
Sampling Procedure	personal communication
Extrapolations	see Geography
Uncertainty Adjustments	Assumption: the production of Flumroc of 45'000 t per year is representative for the total consumed rock wool products in Switzerland.

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
land occupation for built area	→	infrastructure rock wool plant				86'250m2, assumption time occupation: 50 years	resource	land			Occupation, industrial area, built up	4.31E+06	m2a	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	1.50	(1,2,1,1,1,1);
land transformation to built area	→					total area with roof: 24'000m2	resource	land			Transformation, to industrial area, built up	2.40E+04	m2	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	2.00	(1,2,1,1,1,1);
land transformation to roads	→					total area with roads: 22'700m2	resource	land			Transformation, to traffic area, road network	2.27E+04	m2	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	2.00	(1,2,1,1,1,1);
land transformation to green area	→					total green area: 39'550m2	resource	land			Transformation, to industrial area, vegetation	3.96E+04	m2	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	2.00	(1,2,1,1,1,1);
land transformation from unknown	→						resource	land			Transformation, from unknown	8.63E+04	m2	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	2.00	(1,2,1,1,1,1);
buildings multi storage	→					65% of total area build: 15'600 m2 , assumption average height: 12 m	construction processes	buildings	Yes	RER	building, multi-storey	1.87E+05	m3	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	3.06	(3,3,1,2,3,1);
buildings hall	→					35% of total area built: 8'400 m2	construction processes	buildings	Yes	CH	building, hall	8.40E+03	m2	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	3.06	(3,3,1,1,3,1);
Infrastructure cupola melting furnace	→					infrastructure (machines) for the melting process; total weight without fireproof stones: assumed 30 t; life-span assumption: 20 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	7.50E+04	kg	Written information M. Mebold, Flumroc, 02.09.02	1	3.05	(1,3,1,2,3,1);
Infrastructure rest	→					rest of infrastructure (machines) for the whole process: weight: 1'500t (mostly steel); life-span assumption: 20 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	3.75E+06	kg	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	3.05	(1,3,1,2,3,1);
infrastructure conveyor belt	→					length 4000 m, width 1.5 m, life-span 20 years; corresponding module in ecoinvent has a width of 3 m and life-span of 10 years	construction processes	machinery	Yes	RER	conveyor belt, at plant	1.00E+04	m	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	3.05	(1,3,1,2,3,1);
			→	rock wool plant		total produced amount per unit (50 years): 2'257'988'550 kg	insulation materials	production	Yes	CH	rock wool plant	1.00E+00	unit				

Fig. 4.8 Flows for “rock wool plant” and their representation in the ecoinvent database

4.3.3 Comparison to other Data

Differences to other sources of data have been found. The following table shows the differences of the data according to four sources („Rock wool“ in Germany, the SIA declaration⁷, (Mötzl et al. (2000)) and the data of this study) and tries to explain the reasons for the differences:

Tab. 4.11 Comparison between different existing data and the data under study

Indicator	Data from company „Rock wool“ Germany ⁸	Data from Eyerer & Reinhardt (2000)	This study, calculations and information by Flumroc	Mötzl et al. 2000, Ökologie der Dämmstoffe
Primary energy	13 – 20 MJ/kg output ¹⁾	18.66 MJ / kg ²⁾	16.7 MJ /kg output (unit process) ³⁾	22.12 MJ /kg output ⁴⁾
Dust and gaseous emissions in the production process (cupola melting furnace)	0.01 kg / kg output ⁵⁾		0.004 kg / kg output (unit process) ⁵⁾	
CO ₂ Emissions	0.02 kg / kg output ⁶⁾		0.8 kg / kg output ⁷⁾	
¹⁾ Includes the energy for mining, transport of the raw materials and for processing. The included energy carriers and their shares are not given. ²⁾ Based on an insulating board with a density of 160 kg/m ³ . Data of one major company in Germany. The system boundary is not given (not clear whether mining is included or not) ³⁾ Includes all energy carriers, the energy for the preliminary stages of the materials and its transports ⁴⁾ Only information provided: Includes renewable and non renewable primary energy of one producer in Austria ⁵⁾ No further information available ⁶⁾ Assumption: without post-combustion of carbon monoxide; Beside coke the energy carrier of the different companies producing rock wool have often gas instead of heating oil and there are still some companies which melt with an electric furnace ⁷⁾ Includes post-combustion of carbon monoxide				

In general the system boundaries of other studies are not transparent and therefore a comparison is difficult. Nevertheless all data concerning the primary energy vary between 13 and 22 MJ/kg. This shows that there is a quite substantial variation within the standards of the different production companies and/or within the system boundaries considered in different studies.

The data in the study by Richter et al. (1995) are based on the production of Flumroc AG in the year 1992. Comparing those data with the data in this study show following differences. An interpretation is added at the end of this section:

- The consumption of raw material decreased from 1992 to 2002 by about 15%.
- The consumption of binder (phenol, formaldehyde and urea) increased from 1992 to 2002 by about 20%.
- The emissions decreased from 1992 to 2002.
- The amount of coke has stayed stable within these years.

The data in Richter et al. (1995) represent one special product of one producer with a density of 60 kg/m³. The data of the present study represents the total sum of the products of one producer (Flumroc AG) in Switzerland. A large part of their production includes products with a density higher than 60

⁷ <http://www.sia.ch/d/praxis/bauprodukte/index.cfm?Lang=d&ID=1095711>

⁸ <http://www.rockwool.de/graphics/RW-CSE/CSE/pdf/Schwerpunkte/oeko-mappe.pdf>

kg/m³. These products have usually a higher stability which is guaranteed withby a higher quota of binder.

The amount of raw material has decreased by optimising the production process and the emissions have been reduced by implementing carbon monoxide post combustion.

4.3.4 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

In general the quality of the dataset “Rock wool, at plant” is high although the share of the market in Switzerland is only 40%. It is assumed that the companies delivering to the Swiss market are located nearby and have a comparable high technology standard. The company Flumroc has been collecting data for several years and publishes them in their environmental reports at regular intervals.

4.3.5 Conclusion concerning Rock Wool Data

The advantages of the separation of the production process (module “rock wool, at plant”) from the packing process (“rock wool, packed, at plant”) are very small. This because some in- and outputs as for example “purifier”, “water” or “municipal solid waste” can’t be attributed correctly to either the production or the packing process.

4.4 Foamglas

4.4.1 System characterisation

Product

The inventory data for FOAMGLAS[®] is based on production values for 2005 and represents the product T4 WDS which is one of a number of different FOAMGLAS[®] products manufactured in Tessenlo, Belgium with a density of 110 kg/m³. In terms of quantity, T4 WDS is the most important FOAMGLAS[®] product and its manufacturing process is representative for FOAMGLAS[®] products. 95% of T4 WDS sold on the European market is produced in Tessenlo. A negligible 5% is produced at the site in Germany.

Due to the market monopoly of FOAMGLAS[®] products in Europe, the technology employed at the production site in Belgium is the de facto, current technology for producing FOAMGLAS[®] in Europe. An overview of the raw materials used to produce FOAMGLAS[®] is given in Tab. 4.12.

Tab. 4.12 Input of raw materials and energy input for 1kg of a FOAMGLAS[®] product

Raw materials	
Recycled glass	8.14E-01 kg
Feldspar	2.58E-01 kg
Iron oxide	2.77E-02 kg
Sodium nitrate	2.46E-03 kg
Salt cake	5.38E-03 kg
Carbon black	6.15E-03 kg
Energy	
Natural gas	1.16E+01 MJ
Electricity	1.83E+00 kwh

System boundary and process description

The system boundary considered for this inventory is shown in Fig. 4.9.

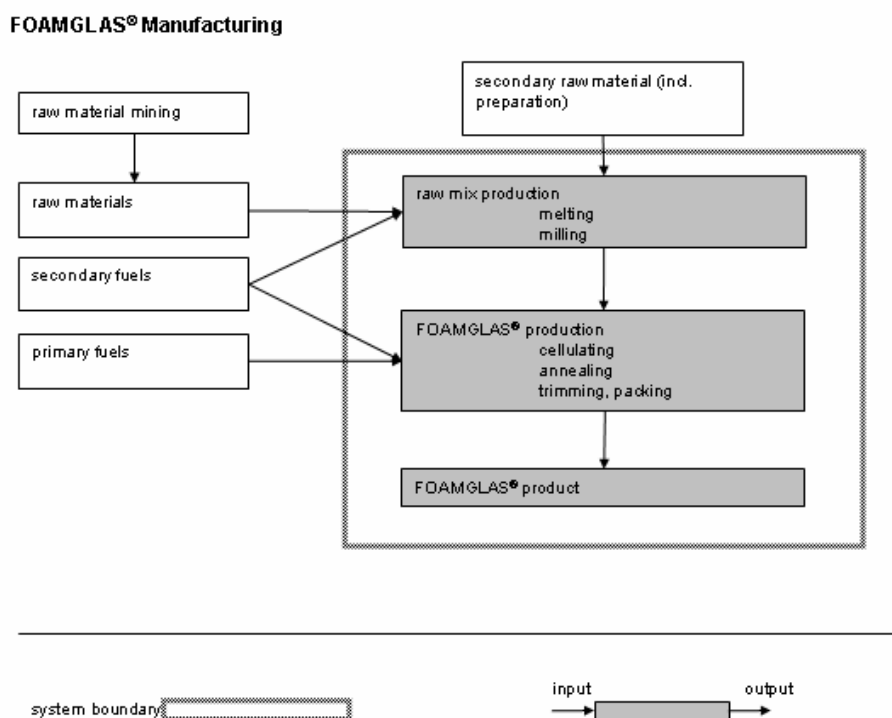


Fig. 4.9 System boundaries considered of the FOAMGLAS® production process

Over the past several years, the use of sand has been completely, if gradually, superseded by the use of recycled glass. Today, more than half of the recycled glass employed is directly injected in the ball mill (milling) without melting, allowing for a large reduction in energy input.

Details of the present manufacturing process are schematically shown in Fig. 4.10. The first step of the FOAMGLAS® production process consists of the raw mix production, a raw material conditioning process which includes the process-storage of raw material in a silo, weigh batching of raw material, melting of raw material in an oven and milling in a ball mill. Raw materials such as: recycled glass, feldspar, iron oxide, salt cake, soda ash, manganese oxide, sodium carbonate and sodium nitrate are added to the melting furnace and heated to approximately 1250 °C. The melting furnace is an electronic furnace equipped with glass melting molybdenum electrodes. The molten glass flows from the melting oven into the continuously fed ball mill where the glass is rapidly ground to colloidal fineness (approximately 10 micron or finer) with the help of corundum cylinders as a grind medium. Another batch of crushed recycled glass is added to the ball mills without melting.

During the second step - the virtual production process - the milled raw material mix undergoes a foaming process in steel moulds coated with clay and aluminium hydroxide at 850°C in the cellulating gas furnace (> 1000 kW) and is subsequently cooled down in the annealing furnace. Finally the cold foam glass cuboid is cut into pieces, packed and stored.

Production of FOAMGLAS®

(Plant Tessenderlo, Belgium)

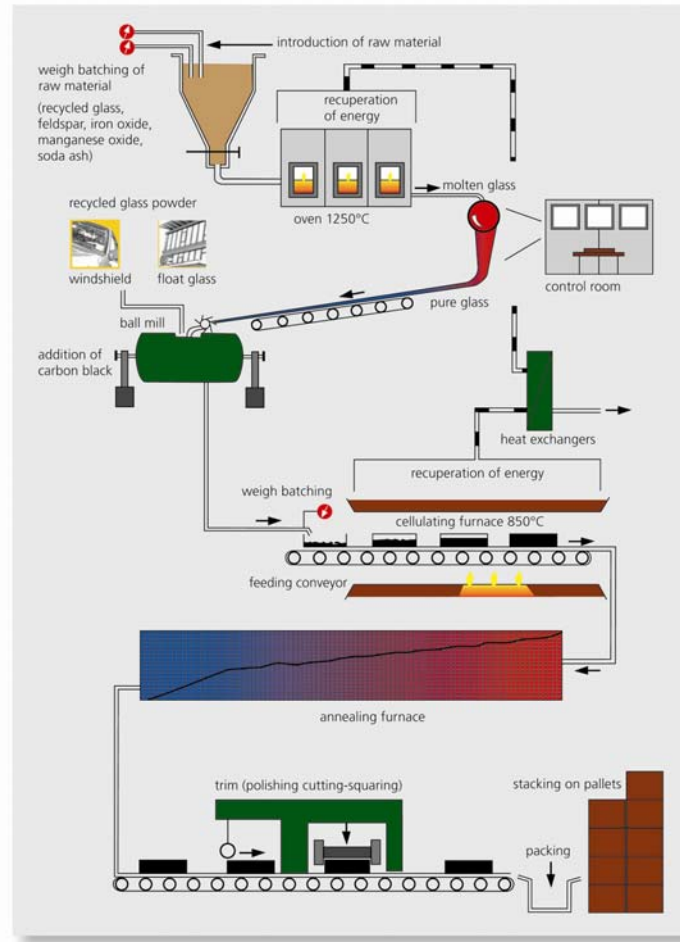


Fig. 4.10: Production process of FOAMGLAS® insulation material

Energy consumption

Electricity and natural gas are the fuels used in the production of FOAMGLAS®. 63% of the electricity is consumed by the melting oven (compare Tab. 4.13). About 18.5 % is used for the grinding or milling process in the ball mills and some 18.5% is used for supplementary systems and processes such as trimming, packing and for storage. Natural gas is used solely for the cellulating and annealing furnace.

Tab. 4.13: Electricity use for the production of 650 kg FOAMGLAS® T4WDS

Production step		Electricity consumption
Raw mix production	Oven (melting)	613 kwh
	Ball mill (milling)	180 kwh
FOAMGLAS® production	Supplementary systems and trimming, packing, storage	180 kwh
		Natural gas consumption
FOAMGLAS® production	Cellulating and annealing furnace, packing	208 Nm ³

Heat emissions from the melting oven and annealing furnace are reused in the production process. The electricity used for FOAMGLAS® production is provided via both the local electricity grid and via renewable energy certificates which consist of 2.2% wind energy and 97.8% hydro power. These certificates are provided and controlled by the independent Renewable Energy Certificate System⁹ (RECS) and sold in Tessenderlo by Fa. Electrabel¹⁰. The production facility in Tessenderlo currently has certificates for production for Switzerland, Austria, France and the Netherlands. Certificates which are reserved for products delivered to Switzerland and Austria are valid until at least 2009 and cover the use of electric power up to 20 000 000 kWh, corresponding to a production quantity of 90'000 m³ FOAMGLAS® products with an average density of 122.5 kg/m³. In 2005, approximately 70'000 m³ FOAMGLAS® products were delivered to Switzerland and 7700 m³ to Austria¹¹.

At the production facility in Tessenderlo, production lines are split up as shown in Fig. 4.11 in accordance with the grinding process in the ball mill. The FOAMGLAS® production-line 11 produces solely for Switzerland and Austria which makes it easy to attribute the quantity of used, certified energy to respective products and countries.

⁹ <http://www.recs.org>

¹⁰ Renewable Energy Certificates are RECS (Renewable Energy Certificate System)-certified and delivered by Fa. Electrabel, Belgium, Contact: Mr. Minnen 0032 (0) 25192782

¹¹ Personal communication, Email, Hans Strauven, Pittsburgh Corning Europe, 1. Feb. 2006

FOAMGLAS® Production-lines

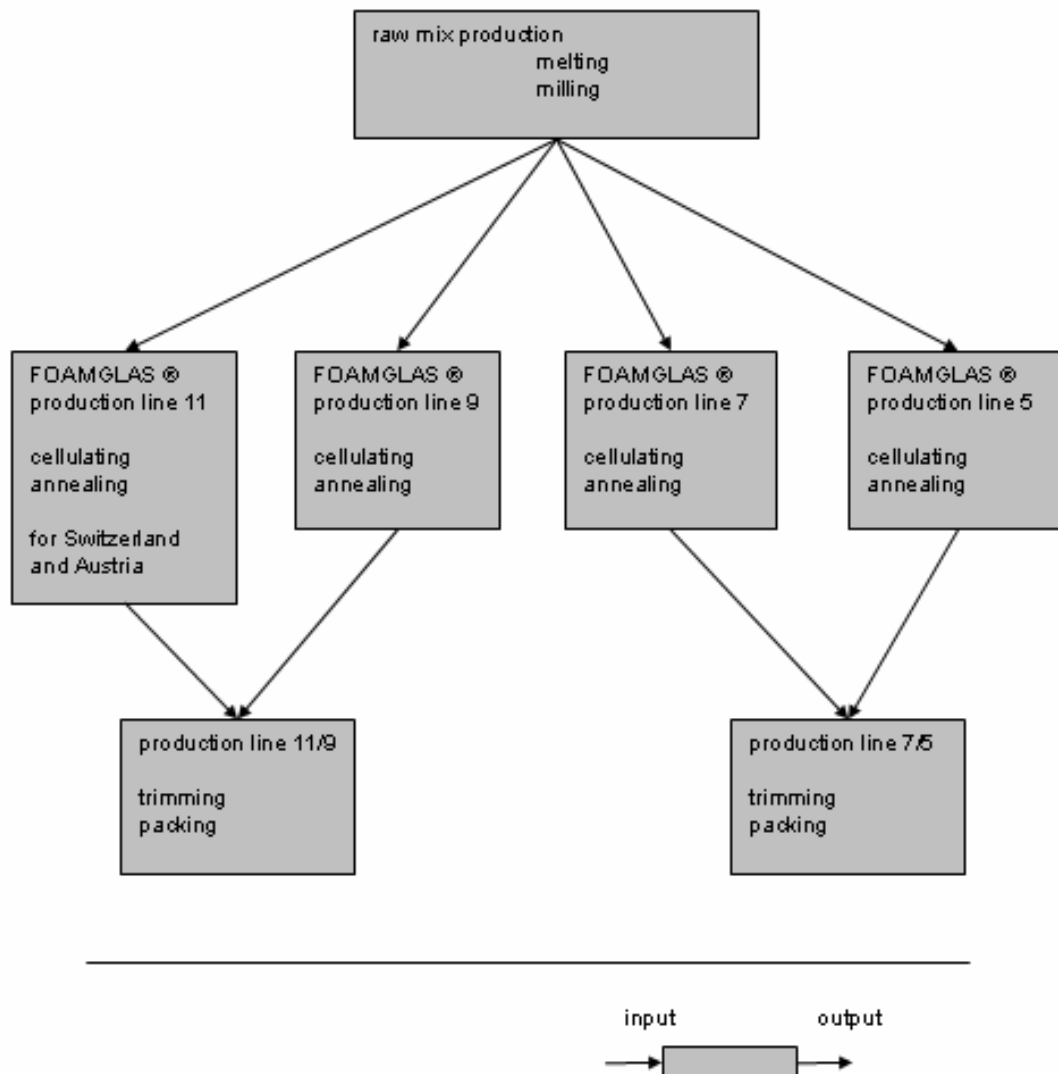


Fig. 4.11: Common and separated production steps for all countries at the production facility in Tessenderlo

Transports

Most raw materials are produced by and purchased from suppliers in Europe. As a result, the transport of raw materials is chiefly done by 40 t lorries. A negligible amount is transported by train and ship, e.g. carbon black which comes from overseas. Transport distances for road transports are estimated and calculated by the use of a route planning system¹². An overview of the transport distances is given in Tab. 4.14.

¹² www.reiseplanung.de

Tab. 4.14: Overview of transport distances

Raw materials	Approximate transport distance with lorries
Recycled glass	100 km
Feldspar	350 km
Iron oxide	25 km
Sodium nitrate	75 km
Salt cake	1000 km
Carbon black	550 km

Parts: wear-and-tear

In the FOAMGLAS[®] production process, a number of parts are subject to abrasion or corrosion and must, therefore, be periodically replaced. Parts which need to be commonly replaced include: molybdenum electrodes which are used for the melting process, corundum cylinders which serve as cylinder material for the milling process and the chromium steel moulds which are used for the foaming process. Abrasion of saw blades (97% Fe, 3% Cr) in the trimming process is included in the inventory but is of minor significance. The wear-and-tear of conveyors is too minimal to be of note¹³.

Packing

The packing facility consists of conveyor belts, a foil machine and a shrink tunnel run on electric power. FOAMGLAS[®] T4 WDS are delivered to construction sites as shrink wrapped, PE-foil (LDPE), 0.135 m³ parcels (measuring 0.5 m x 0.45 m x 0.6 m) with a corrugated cardboard cover element (0.45 m*0.6 m). The cardboard quality is a simple corrugated cardboard made from recycled fibres. The approximate net weight of a parcel is 14.85 kg. A delivery pallet commonly includes 12 parcels with a net weight of 178.2 kg which are covered with a PE-foil sheet on the top and are wrapped with PE-foil (LDPE). For deliveries, pure softwood, 10 kg, one-way pallets are used instead of 22 kg EUR-flat pallets. Every fourth pallet comes back to the factory in good shape and can be reused, which results in a reuse frequency of 0.25 or a multiplication of the pallet use by 0.75. Tab. 4.15 shows the packing materials used for one ready-for-delivery pallet.

Tab. 4.15: Packing materials used for one ready for delivery pallet of FOAMGLAS[®] T4 WDS measured by weight

Packing Material	kg per parcel	packing film per parcel	Thickness	Total per pallet
Shrink foil for parcel	0.133 kg	1.8 m ²	80E-6 m	1.596 kg
Cardboard	0.1015 kg	0.45m * 0.6m		1.218 kg
PE-foil sheet			100E-6 m	0.196 kg
Shrink foil for pallet			100E-6 m	0.786 kg
10 kg pure softwood one way pallet				7.5 kg

Electrical heating is used for shrink wrapping of FOAMGLAS[®] parcels. For the pallet packing elastic PE-foil is applied without heating.

¹³ Personal communication, Pittsburgh Corning Europe, Hans Strauven, 2006

Water and air use

Processed air is used at every production step for supplementary systems. A mix of tap water and ground-water-derived industrial water pumped at plant are used in the oven, the ball mill and the cellululating furnace. Tab. 4.16 shows the air and water use for the production of 650 kg FOAMGLAS®.

Tab. 4.16: Air and water use for the production of 650 kg FOAMGLAS® T4WDS

Water and air use	
Process air	2.75 Nm ³
Tap water	0.024 Nm ³
Industrial water	0.154 Nm ³

Land-use

The area used for the FOAMGLAS® production in Tessenderlo is approximately 0.17 km². Of that, 62'926 m² is vegetation, 44'197 m² is traffic area and 63'939 m² is industrial space with buildings. Land occupation is calculated by dividing the total area (0.17 km²) by the total amount of FOAMGLAS® products per year on the occupied area (39'200'000 kg/a). Land transformation is calculated by dividing the total area by the total life time production assuming a production of 39'200'000 kg/a over an estimated period of 50 years. Production halls occupy an area of 35'826 m² and the cubature of the multi-storey buildings is estimated to be about 5'400 m³.

Emissions to air, water and soil

Particulate matter (PM) is produced at different production steps: in the melting process, in the ball mill, in the cellululating furnace and during trimming. In the ball mill and the cellululating furnace no particulate matter emissions occur due to the re-circulation of unfiltered air in a closed loop system. The melting oven and trim station are equipped with sinter plate fabric filters from Herding¹⁴ for emission control which perform better than 1 mg/m³ air. Sinter plate fabric filters from Herding perform better than F9 Filters according to information from Herding. F9 Filters have a filtration efficiency for particulates > 2.5 µm higher than 99% in compliance with EN 779. Therefore pollutants of concern in the FOAMGLAS® production are particulate matter less than 2.5 micrometers and all emissions to air are assumed to be smaller than 2.5 µm. The particulates > 2.5 µm retained by the filter from the melting oven go into to the brick production. The particulates retained by the filter of the trimming process which are mostly bigger than 20 µm are reused as raw material in the production process.

Tab. 4.17 shows data on maximum particulate matter emissions calculated from filter efficiency, retained amount of particulates which go to the brick factory or go back to the oven and the production volume in Tessenderlo. For the calculated particle size distribution in Tab. 4.17 granulomorphometric analysis data from the oven filter was used.

Gaseous emissions to air occur either by the processing of raw materials from the melting of raw materials, cellululation and trimming or by emissions from combustion processes (i.e. heating and burning of fuels). Combustion emissions are the primary gaseous pollutants. Calculated emissions from processing raw materials are small and carbon dioxide emissions are in the order of 1-2% by weight. Emissions from processing raw materials are indicated in the flow chart in the annex and do not include the emissions from burning processes as they are included there.

Waste water is treated at the communal waste water treatment facility with the exception of the elimination of aluminiumoxide which is filtered out at the production facility.

¹⁴ Herding Filtertechnik, <http://www.herding.ch>, 2006

Tab. 4.17: Calculated particulate matter

Particulate matter (PM)	Oven		Trim	
	kg		kg	
PM >10	9.34E-02	output to brick factory	2.29E-01	input to oven
PM10-PM2.5	6.24E-02	output to brick factory	1.53E-01	input to oven
<PM2.5 (fine dust)	1.47E-6	emissions to air	4.42E-05	emissions to air
Totale particulate matter (TPM)	1.56E-01		3.82E-01	

In the production process, solid waste accrues at different production steps. General waste such as steel, other metals, electronics, plastics, batteries and paper is gathered in different containers in the factory and recycled. Damaged moulds go back to the steel supplier, where they are repaired or recycled.

4.4.2 Foamglas in ecoinvent

The ecoinvent database meta information (Tab. 4.18, Tab. 4.19) includes all important and necessary information to use the data correctly. Fig. 4.12 to Fig. 4.15 show all in- and output flows of the foam-glass products, the allocation to the modules in ecoinvent, the sources and the standard deviation for each datum.

Foam glass, packed, at plant (RER)

The dataset "foam glass, packed, at plant" includes data for administration, packing, solid waste, emissions from processing of raw material and the infrastructure, including its land-use, the buildings and the machines used.

Foam glass, packed, at plant ch and austria

The ecoinvent module Austria (A) and Switzerland (CH) takes into account the use of renewable energy certificates which are attributed to the production volume of these countries. The certificates are reserved for the aforementioned countries and are not credited to the general module foam glass, packed, at plant (BE). The additional transport distances to the centre of distribution in Switzerland and Austria are taken into account as following. From BE-3980 Tessenderlo to Austria AT-4050 Traun 877 km was brought to account and 587 km to Switzerland CH-4702 Oensingen.

Tab. 4.18 ecoinvent meta information for the foamglas production

Name	foam glass, at plant	foam glass, at regional storage
Location	RER	CH
Infrastructure Process	0	0
Unit	kg	kg
Data Set Version	2	2
Included Processes	Included processes: melting, forming to a pipe, cooling down, smashing, crushing, mixing with carbon powder. Foaming by heating in a mould, cooling, cutting to slabs and packing. Internal processes (workshop, administration etc.), transport of raw materials and energy carrier for furnace, packing and infrastructure are also included.	Included processes: melting, forming to a pipe, cooling down, smashing, crushing, mixing with carbon powder. Foaming by heating in a mould, cooling, cutting to slabs and packing. Internal processes (workshop, administration etc.), transport of raw materials and energy carrier for furnace, packing and infrastructure are also included. Renewable energy certificates are accounted and transport to Switzerland, respectively Lichtenstein is included.
Amount	1	1
Local Name	Schaumglas, ab Werk	Schaumglas, ab Regionallager
Synonyms	T4 WDS	T4 WDS
General Comment to reference function	Refers to foam glass of a density of 110 kg/m ³ and a thermal conductivity of 0.040 W/mK. Foam glass products sold to Switzerland and Lichtenstein are produced with renewable energy certificates from RECS (www.recs.org).	Refers to foam glass of a density of 110 kg/m ³ and a thermal conductivity of 0.040 W/mK. Foam glass products sold to Switzerland and Lichtenstein are produced with renewable energy certificates from RECS (www.recs.org).
Start Date	2005	2005
End Date	2005	2005
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data are from one company in Belgium which covers nearly 100% of the European foam glass market. For some exchanges RER-, CH- and GLO-modules are used as proxy. No energy certificates are accounted	Data are from one company in Belgium which covers nearly 100% of the European foam glass market. For some exchanges RER-, CH- and GLO-modules are used as proxy. No energy certificates are accounted
Technology text	All data refer to the year 2005. Today foam glass is made of 68% recycled glass and only recycled glass is used instead of silica sand. Recycled glass is to some extent directly injected to the cellulating furnace without melting, resulting in less energy use. The energy for the melting process is electricity. Natural gas is used for the cellulating process. All emissions from processing raw materials and heating are taken into account.	All data refer to the year 2005. Today foam glass is made of 68% recycled glass and only recycled glass is used instead of silica sand. Recycled glass is to some extent directly injected to the cellulating furnace without melting, resulting in less energy use. The energy for the melting process is electricity. Natural gas is used for the cellulating process. All emissions from processing raw materials and heating are taken into account.
Representativeness [%]	100	100
Production Volume	A volume of 30625 tons per year is produced	A volume of 30625 tons per year is produced
Sampling Procedure	survey	survey
Extrapolations	none	none
Uncertainty Adjustments		

Tab. 4.19 ecoinvent meta information for the foamglas production

Name	foam glass, at regional storage	foam glass plant
Location	AT	BE
Infrastructure Process	0	1
Unit	kg	unit
Data Set Version	2	2
Included Processes	Included processes: melting, forming to a pipe, cooling down, smashing, crushing, mixing with carbon powder. Foaming by heating in a mould, cooling, cutting to slabs and packing. Internal processes (workshop, administration etc.), transport of raw materials and energy carrier for furnace, packing and infrastructure are also included. Renewable energy certificates are accounted and transport to Austria is included.	Includes the buildings, and machines in Belgium
Amount	1	1
Local Name	Schaumglas, ab Regionallager	Schaumglaswerk
Synonyms	T4 WDS	
General Comment to reference function	Refers to foam glass of a density of 110 kg/m ³ and a thermal conductivity of 0.040 W/mK. Foam glass products sold to Switzerland and Lichtenstein are produced with renewable energy certificates from RECS (www.recs.org).	
Start Date	2005	2005
End Date	2005	2005
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data are from one company in Belgium which covers nearly 100% of the European foam glass market. For some exchanges RER-, CH- and GLO-modules are used as proxy. No energy certificates are accounted	Data are from one company in Belgium which covers nearly 100% of the European foam glass market. For some exchanges RER-, CH- and GLO-modules are used as proxy. No energy certificates are accounted
Technology text	All data refer to the year 2005. Today foam glass is made of 68% recycled glass and only recycled glass is used instead of silica sand. Recycled glass is to some extent directly injected to the cellulating furnace without melting, resulting in less energy use. The energy for the melting process is electricity. Natural gas is used for the cellulating process. All emissions from processing raw materials and heating are taken into account.	All data refer to the year 2005.
Representativeness [%]	100	100
Production Volume	A volume of 30625 tons per year is produced	A volume of 30625 tons per year is produced
Sampling Procedure	survey	survey
Extrapolations	none	none
Uncertainty Adjustments		

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
recycled floatglass	foam glass, at plant: Inputs part 1		raw material to oven (37.6%) and ball mill (62.4%), recycled glass: mechanically cleaned window and windshield glass	glass	packaging	No	RER	glass, cullets, sorted, at sorting plant	8.14E-01	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
FOAM GLAS scrap			processed raw material to oven, not taken into account as assumed to be burden free						3.85E-01	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
feldspar			unprocessed raw material to oven	construction materials	others	No	RER	feldspar, at plant	2.58E-01	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
soda ash			raw material to oven, from total 50% unprocessed and 50% processed raw material, "soda powder" is used as proxy for both qualities	chemicals	inorganics	No	RER	soda, powder, at plant	4.62E-02	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
iron oxide			processed raw material to oven, "iron ore at beneficiation" used as proxy	metals	extraction	No	GLO	iron ore, 65% Fe, at beneficiation	2.77E-02	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
manganese oxide			processed raw material to oven	metals	extraction	No	GLO	manganese concentrate, at beneficiation	2.62E-02	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
carbon black			processed raw material to ball mill	chemicals	inorganics	No	GLO	carbon black, at plant	6.15E-03	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
salt cake			processed raw material to oven	chemicals	inorganics	No	RER	sodium sulphate, from natural sources, at plant	5.38E-03	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
sodium nitrate			processed raw material to oven, potassium nitrate used as proxy. Calculation of N content according to stoichiometry (KNO ₃ =101.10 g/mol; N=14.007 g/mol)	agricultural means of production	mineral fertiliser	No	RER	potassium nitrate, as N, at regional storehouse	3.41E-04	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
aluminiumhydroxide			processed raw material to cellulating furnace,	chemicals	inorganics	No	RER	aluminium hydroxide, at plant	6.40E-03	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
coating material			processed raw material to cellulating furnace	construction materials	additives	No	CH	clay, at mine	3.49E-04	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
process air			use of electricity for the compressor is considered in electricity input						4.23E-03	Nm3	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);

Fig. 4.12 Flows for "foamglas, at plant" (RER) and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
tap water	foam glass, at plant: Inputs part 2		tap water for all processes	water supply	production	No	RER	tap water, at user	3.69E-02	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
industrial water			groundwater for all processes, pumped at the production facility	resource	in water			Water, well, in ground	2.37E-04	m3	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
packaging film PE			packaging film to trim for shrink wrapping	plastics	processing	No	RER	packaging film, LDPE, at plant	1.46E-02	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
packaging cardboards			cardboard for packing	paper & cardboard	cardboard & corrugated board	No	RER	corrugated board, recycling fibre, double wall, at plant	7.31E-03	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
packaging paletts (softwood)			pure softwood paletts for packing, "EUR-flat pallet" used as proxy	wooden materials	processing	No	RER	EUR-flat pallet	1.60E-03	unit	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
natural gas for cellulating and annealing			natural gas for cellulating and annealing furnace	natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	1.16E+01	MJ	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
electricity			electricity for oven, ball mill and other machines and processes	electricity	supply mix	No	BE	electricity, medium voltage, at grid	1.50E+00	kWh	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
diesel for fork lifts			diesel used for operating fork lifts	construction processes	machinery	No	GLO	diesel, burned in building machine	9.62E-02	MJ	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
heating oil			oil for heating administration buildings	oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	4.23E-02	MJ	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
molybdenum electrodes			molybdenum electrodes as wear parts for the oven, production process of the electrodes is not included	metals	extraction	No	RER	molybdenum, at regional storage	7.69E-05	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
corundum cylinders			corundum (sintered) cylinders as wear parts for the ball mill, "sinter iron" used as proxy	metals	extraction	No	GLO	sinter, iron, at plant	4.00E-04	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
moulds			chromium moulds as wear parts for the cellulation and annealing furnace	metals	extraction	No	RER	chromium steel 18/8, at plant	1.15E-03	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
saw blades			chromium blades as wear parts for the trimming machine	metals	extraction	No	RER	chromium steel 18/8, at plant	1.17E-04	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
raw material transport by barge			only carbon black is transported by barge over a distance of 1150km	transport systems	ship	No	OCE	transport, transoceanic freight ship	7.08E-03	tkm	Thalmann P, 2006	1	2.00	(1,3,1,1,1,1);
raw material transport by lorry			transportation of all raw materials, packing materials and wear parts to the factory, assumed that all transport by a lorry 40t	transport systems	road	No	CH	transport, lorry >28t, fleet average	1.94E-01	tkm	Thalmann P, 2006	1	2.00	(1,2,1,1,1,1);
infrastructure			includes machines, buildings, land occupation and land transformation, total foamglass production (all products) over 50 years: 1'960'000 tons	insulation materials	production	Yes	BE	foam glass plant	5.10E-10	unit	Thalmann P, 2006	1	3.05	(1,2,1,1,1,5);

Fig. 4.12 Flows for "foamglas, at plant" (RER) and their representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
foam glass, at plant: Outputs	→ FOAM GLAS scrap	production waste to brick factory not taken into account as it is reused							1.57E-01	kg	Thalmann P, 2006	1	1.21	(1,2,1,1,1,5);
	→ packaging film PE			waste management	municipal incineration	No	CH	disposal, PE sealing sheet, 4% water, to municipal incineration	1.46E-02	kg		1	1.07	(1,3,1,1,1,1);
	→ packaging cardboards			waste management	municipal incineration	No	CH	disposal, packaging cardboard, 19.6% water, to municipal incineration	7.31E-03	kg		1	1.07	(1,3,1,1,1,1);
	→ packaging paletts (softwood)	1 palett = 22 kg (atro)		waste management	municipal incineration	No	CH	disposal, wood untreated, 20% water, to municipal incineration	4.21E-02	kg		1	1.07	(1,3,1,1,1,1);
	→ carbon dioxide process emissions	process emission (no combustion): carbon dioxide from oxidation of raw material, 89.5% is emitted by the ball mill, 5.25% by the cellulating furnace and 5.25% by the trim process		air	low population density			Carbon dioxide, fossil	2.94E-02	kg	Thalmann P, 2006	1	1.12	(3,3,1,1,1,1);
	→ sulfur dioxide process emissions	process emissions: sulfur dioxide emissions from oven		air	low population density			Sulfur dioxide	2.31E-04	kg	Thalmann P, 2006	1	1.12	(3,3,1,1,1,1);
	→ nitrogen oxide emissions	nitrogen emissions from oven		air	low population density			Nitrogen oxides	1.00E-03	kg	Thalmann P, 2006	1	1.52	(3,3,1,1,1,1);
	→ Filter dust to recycling	in average about 99.25% of the total weight of the emitted particulates is bigger than 2.5 µm and is retained by dust filters and used in brick production							5.38E-01	kg	Thalmann P, 2006	1		
	→ dust emissions	in average about 0.75% of the total weight of the emitted particulates is smaller than 2.5 µm		air	low population density			Particulates, < 2.5 µm	4.57E-05	kg	Thalmann P, 2006	1	3.05	(1,2,1,1,1,5);
	→ waste water treated	waste water from all processes		waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 3	2.08E-04	m3	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
	→ waste heat	waste heat from electricity consumption		air	low population density			Heat, waste	5.39E+00	MJ	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
	→ FOAMGLASS			insulation materials	production	No	RER	foam glass, at plant	1.00E+00	kg	Thalmann P, 2006			

Fig. 4.12 Flows for "foamglas, at plant" (RER) and their representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
foam glass	foam glass, at regional storage (CH): Inputs			insulation materials	production	No	RER	foam glass, at plant	1.00E+00	kg	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
subtraction of total electricity consumption to replace it by green energy from certificates				electricity	supply mix	No	BE	electricity, medium voltage, at grid	-1.50E+00	kWh	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
green energy certificate wind power		2.2% wind power		wind power	power plants	No	RER	electricity, at wind power plant	3.36E-02	kWh	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
green energy certificate hydro power		97.8% hydro power		hydro power	power plants	No	CH	electricity, hydropower, at power plant	1.49E+00	kWh	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
transmission network medium voltage			transformation, transport and distribution of electricity	electricity	distribution	Yes	CH	transmission network, electricity, medium voltage	4.85E-08	km	calculation based on ecoinvent DB v1.2	1	3.10	(3,1,1,3,3,5); based on consumption statistics
transmission network high voltage				electricity	distribution	Yes	CH	transmission network, electricity, high voltage	1.28E-08	km	calculation based on ecoinvent DB v1.2	1	3.10	(3,1,1,3,3,5); based on consumption statistics
transmission network long distance				electricity	distribution	Yes	UCTE	transmission network, long-distance	4.79E-10	km	calculation based on ecoinvent DB v1.2	1	3.38	(5,5,4,5,1,5); estimates
SF6				chemicals	inorganics	No	RER	sulphur hexafluoride, liquid, at plant	1.38E-07	kg	calculation based on ecoinvent DB v1.2	1	1.07	(2,1,1,3,1,na); based on emissions

Fig. 4.13 Flows for "foamglas, at regional storage" (CH) and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
transport to CH foam glass, at regional storage (CH): Outputs → Heat, waste → Heat, waste → Dinitrogen monoxide → Ozone → Sulfur hexafluoride → FOAMGLASS in Switzerland		Heat, waste		air	unspecified			Heat, waste	6.06E-02	MJ	calculation based on ecoinvent DB v1.2	1	1.32	(4,1,3,3,1,5); estimation based on losses
		Heat, waste		soil	unspecified			Heat, waste	1.75E-02	MJ	calculation based on ecoinvent DB v1.2	1	1.32	(4,1,3,3,1,5); estimation based on losses
		Dinitrogen monoxide		air	unspecified			Dinitrogen monoxide	7.56E-06	kg	calculation based on ecoinvent DB v1.2	1	4.60	standard deviation based on variation reported in literature
		Ozone		air	unspecified			Ozone	6.80E-06	kg	calculation based on ecoinvent DB v1.2	1	5.00	standard deviation based on variation reported in literature
		Sulfur hexafluoride		air	unspecified			Sulfur hexafluoride	1.38E-07	kg	calculation based on ecoinvent DB v1.2	1	1.51	(2,1,1,3,1,na); national statistics
		FOAMGLASS in Switzerland	transport from BE-3980 Tessenderlo to CH-4702 Oensingen, distance 587 km	transport systems	road	No	CH	transport, lorry >28t, fleet average	5.87E-01	tkm	Thalmann P, 2006	1	2.00	(1,2,1,1,1,1);
				insulation materials	production	No	CH	foam glass, at regional storage	1.00E+00	kg	Thalmann P, 2006			

Fig. 4.13 Flows for "foamglas, at regional storage" (CH) and their representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
foam glass	foam glass, at regional storage (AT): Inputs			insulation materials	production	No	RER	foam glass, at plant	1.00E+00	kg	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
subtraction of total electricity consumption to replace it by green energy from certificates				electricity	supply mix	No	BE	electricity, medium voltage, at grid	-1.50E+00	kWh	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
green energy certificate wind power		2.2% wind power		wind power	power plants	No	RER	electricity, at wind power plant	3.36E-02	kWh	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
green energy certificate hydro power		97.8% hydro power		hydro power	power plants	No	CH	electricity, hydropower, at power plant	1.49E+00	kWh	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
transmission network medium voltage			transformation, transport and distribution of electricity	electricity	distribution	Yes	CH	transmission network, electricity, medium voltage	4.85E-08	km	calculation based on ecoinvent DB v1.2	1	3.10	(3,1,1,3,3,5); based on consumption statistics
transmission network high voltage				electricity	distribution	Yes	CH	transmission network, electricity, high voltage	1.28E-08	km	calculation based on ecoinvent DB v1.2	1	3.10	(3,1,1,3,3,5); based on consumption statistics
transmission network long distance				electricity	distribution	Yes	UCTE	transmission network, long-distance	4.79E-10	km	calculation based on ecoinvent DB v1.2	1	3.38	(5,5,4,5,1,5); estimates
SF6				chemicals	inorganics	No	RER	sulphur hexafluoride, liquid, at plant	1.38E-07	kg	calculation based on ecoinvent DB v1.2	1	1.07	(2,1,1,3,1,na); based on emissions

Fig. 4.14 Flows for "foamglas, at regional storage" (AT) and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
transport to A foam glass, at regional storage (AT): Outputs Heat, waste Heat, waste Dinitrogen monoxide Ozone Sulfur hexafluoride FOAMGLASS in Austria		Heat, waste		air	unspecified			Heat, waste	6.06E-02	MJ	calculation based on ecoinvent DB v1.2	1	1.32	(4,1,3,3,1,5); estimation based on losses
		Heat, waste		soil	unspecified			Heat, waste	1.75E-02	MJ	calculation based on ecoinvent DB v1.2	1	1.32	(4,1,3,3,1,5); estimation based on losses
		Dinitrogen monoxide		air	unspecified			Dinitrogen monoxide	7.56E-06	kg	calculation based on ecoinvent DB v1.2	1	4.60	standard deviation based on variation reported in literature
		Ozone		air	unspecified			Ozone	6.80E-06	kg	calculation based on ecoinvent DB v1.2	1	5.00	standard deviation based on variation reported in literature
		Sulfur hexafluoride		air	unspecified			Sulfur hexafluoride	1.38E-07	kg	calculation based on ecoinvent DB v1.2	1	1.51	(2,1,1,3,1,na); national statistics
		FOAMGLASS in Austria	transport from BE-3980 Tessenderlo to the AT-4050 Traun, distance 877 km	transport systems	road	No	CH	transport, lorry >28t, fleet average	8.77E-01	tkm	Thalmann P, 2006	1	2.00	(1,2,1,1,1,1);
				insulation materials	production	No	AT	foam glass, at regional storage	1.00E+00	kg	Thalmann P, 2006			

Fig. 4.14 Flows for "foamglas, at regional storage" (AT) and their representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
machines	→	foam glass plant			construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	3.00E+06	kg	Thalmann P, 2006	1	3.28	(3,2,1,1,4,5);
conveyor belts installed	→				construction processes	machinery	Yes	RER	conveyor belt, at plant	1.00E+03	m	Thalmann P, 2006	1	3.28	(2,2,1,1,4,5);
production halls	→				construction processes	buildings	Yes	CH	building, hall	3.58E+04	m2	Thalmann P, 2006	1	3.28	(3,2,1,1,4,5);
administration buildings	→				construction processes	buildings	Yes	RER	building, multi-storey	5.40E+03	m3	Thalmann P, 2006	1	3.28	(3,2,1,1,4,5);
fabrication area	→				resource	land			Transformation, from unknown	1.71E+05	m2	Thalmann P, 2006	1	2.29	(3,2,1,1,4,5);
industrial area with buildings	→				resource	land			Transformation, to industrial area, built up	6.39E+04	m2	Thalmann P, 2006	1	2.29	(3,2,1,1,4,5);
vegetation area	→				resource	land			Transformation, to industrial area, vegetation	6.29E+04	m2	Thalmann P, 2006	1	2.29	(3,2,1,1,4,5);
traffic area	→				resource	land			Transformation, to traffic area, road network	4.42E+04	m2	Thalmann P, 2006	1	2.29	(3,2,1,1,4,5);
occupation area	→				resource	land			Occupation, industrial area, built up	1.71E+05	m2a	Thalmann P, 2006	1	1.84	(3,2,1,1,4,5);
production plant Tessenderlo	→														
		→	Foamglass production plant	Total foamglass production (all products) over 50 years: 1'960'000 tons	insulation materials	production	Yes	BE	foam glass plant	1.00E+00	unit	Thalmann P, 2006			

Fig. 4.15 Flows for "foamglas plant" and their representation in the ecoinvent database

4.4.3 Data quality

Data ascertainments were conducted by collaborators of Pittsburgh Corning Europe in Tessenderlo in cooperation with BAU- UND UMWELTCHEMIE Beratungen + Messungen AG. An accompanying external critical review according to ISO 14040:1997 §7.3.1 was carried out by Empa (H.J. Althaus). Empa had full access to the data considered in this study which are required for data validation and for a future integration in the ecoinvent database.

The presented data collection on raw material is based chiefly on measurements (e.g. raw materials and packing materials) at the production facility in Tessenderlo. Energy consumption is measured by gas and electricity meters and is cross-checked with theoretical calculations. Data on the wear-and-tear of parts and material quality were obtained from delivery orders and from their yearly consumption across the entire production volume. Transport distances include transport from the supplier to the production facility. All data are average values for the overall production of this product per year.

Modelling principles according to ecoinvent-report 1 are used. All data, assumptions, units, energy and emission considerations and naming rules are implemented as proposed in this report. Transport distances are calculated according the chapter Transport services in ecoinvent-report No. 1 Frischknecht et al. (2007).

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta-information.

In general, the data quality of the dataset "foam glass, packed, at plant" is high due to the monopoly situation of the product FOAMGLAS® in Europe. The company Pittsburgh Corning Europe has been collecting data for the last three years and has been given full access to all data for this inventory update.

4.5 Polystyrene Foam Slab (EPS)

4.5.1 Process Description

Polystyrene foam slabs are produced by thermoforming of expandable polystyrene (EPS). This process is described in Hischier (2007).

Basically the expandable polystyrene is heated and formed. The pentane evaporates in this process and builds the foam.

4.5.2 Polystyrene Foam Slab (EPS) in ecoinvent

The production of polystyrene foam slabs is modelled by combining the modules "polystyrene, expandable, at plant" and "foaming, expanding" from Hischier (2007).

Since in Hischier (2007) 1 kg "foaming, expanding" relates to the input of 1 kg EPS and 4.15% of the input is disposed as waste, 1.0415 kg of EPS has to be inventoried to produce 1 kg polystyrene slab. The recycling or disposal of the 0.0415 kg polystyrene waste is inventoried in the "forming, expanding" module.

The ecoinvent meta-information for the polystyrene foam slab production is given in Tab. 4.20 and the inputs and output flows for "polystyrene foam slab, at plant" are shown in Fig. 4.16.

Data Quality

Since only one material and one processing modules of ecoinvent are combined, the values are certain.

The data quality for the polystyrene foam is thus the same as the data quality for EPS and for the thermoforming process (see Hischier (2007)).

Tab. 4.20 ecoinvent meta information for the polystyrene foam slab production

Name	polystyrene foam slab, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes production and thermoforming of EPS
Amount	1
Local Name	Polystyrolplatte expandiert, ab Werk
Synonyms	
General Comment to reference function	Combination of material and processing module. EPS foam slab has a density of 30 kg/m ³ and a thermal conductivity of 0.035-0.04 W/mK.
Start Date	2003
End Date	2003
Data Valid For Entire Period	1
Other Period Text	
Geography text	
Technology text	European average EPS production; thermoforming from 2 Factories in Switzerland
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	none
Extrapolations	none
Uncertainty Adjustments	none

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
EPS	polystyrene foam slab, at plant	EPS foam	foaming relates to 1 kg output. Thus input is higher to account for waste (which is disposed in the thermoforming module)	plastics	polymers	No	RER	polystyrene, expandable, at plant	1.04E+00	kg	Hischier 2003	1	1.00	uncertainties in the combined modules -> no uncertainty
thermoforming EPS				plastics	processing	No	RER	foaming, expanding	1.00E+00	kg	Hischier 2003	1	1.00	uncertainties in the combined modules -> no uncertainty
				insulation materials	production	No	RER	polystyrene foam slab, at plant	1.00E+00	kg				

Fig. 4.16 Flows for "polystyrene foam slab, at plant" and their representation in the ecoinvent database

4.6 Polystyrene extruded (XPS)

4.6.1 Process Description

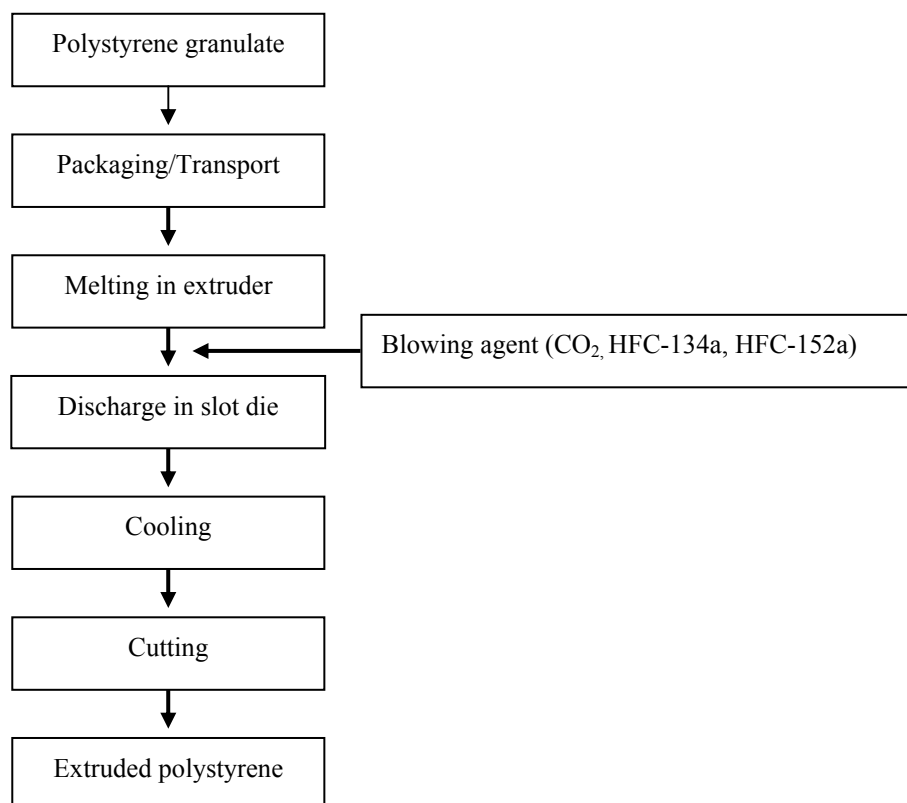
Raw Materials

XPS consists to 91% of polystyrene. The main raw material used for the production of polystyrene is benzene and ethylene. These two substances are mixed with a catalyst to produce styrene, which again is polymerised to Polystyrene (eza (2004)). Benzene and ethylene are described in the ecoinvent report No. 11 Part II on Plastics (Hischier et al. (2007)). About 7% of the XPS is blowing agent and the remaining 2% are flame retardants (bromine compounds) and other additives.

Production Process

Extruded polystyrene foam is made from solid polystyrene crystals (granulates). The polystyrene granulates, along with special additives and a blowing agent (nowadays often CO₂, but also HFC-134a and HFC-152a), are fed into an extruder. Within the extruder the mixture is combined and melted, under controlled conditions of high temperature and pressure, into a viscous plastic fluid. The hot, thick liquid is then forced in a continuous process through a die. As it emerges from the die it expands, is shaped, cooled, and trimmed to dimension. This continuous extrusion process results in a foam product with a uniform closed-cell structure, a smooth continuous skin, and consistent product qualities¹⁵ Fig. 4.17 shows the process chain of polystyrene XPS (Starzner S. and Wurmer-Weiss P. (2000)).

Fig. 4.17 Production process of extruded polystyrene



¹⁵ <http://www.diversifoam.com/xeps.htm>; access: 07.08.2007

Emissions

Similar to the process of plastic pipe extrusion, the production of XPS creates a certain amount of heat waste. This value is calculated via the electricity input for the extrusion of a plastic pipe (Käufer (1994)).

For this study, the styrene emission values of XPS extrusion are estimated at 0.1‰ of the polystyrene input. No direct CO₂-emissions during the production phase are accounted for, as most of the CO₂ used as blowing agent is captured in the foam after the extrusion process. This dataset “Polystyrene, extruded” explicitly uses 50% (w/w) CO₂ as a blowing agent¹⁶ and equally 25% (w/w) of the hydro-fluorocarbons type HFC-134a and HFC-152a (assumption). This results in 17.5% (w/w) R134a and 17.5% HFC-152a losses (or emissions) during production (see Tab. 4.21), with current production losses being 15-25% for Europe and 10-25% for North America (IPCC 2007, p.28). HFCs will most likely completely vanish in European XPS production within the next 10 to 20 years but they will survive longer in North America (Environment Canada (2005)).

Tab. 4.21 Blowing agents and emission data during XPS production (source: IPCC 2007 and assumptions)

	Weight (%)	Formula	Amount
Blowing agent(s)*			
CO ₂	50% of total input for blowing agents	$= 0.5 \cdot 0.07 \cdot 1.0$	0.035 kg
HFC-134a	25% of total input for blowing agents	$= 0.25 \cdot 0.07 \cdot 1.0$	0.0175 kg
HFC-152a	25% of total input for blowing agents	$= 0.25 \cdot 0.07 \cdot 1.0$	0.0175 kg
			0.07 kg (total)
Emissions			
R134a	17.5% losses of blowing agent during production	$= 0.175 \cdot 0.0175$	0.00306 kg
HFC-152a	17.5% losses of blowing agent during production	$= 0.175 \cdot 0.0175$	0.00306 kg

* All the blowing agents have a 7%-share (=0.07 kg) of total inputs to XPS (=1 kg for this dataset)

Energy Consumption

The production step “melting of polystyrene pearls in extruder” consumes 0.53 kWh of electrical energy per kg (value taken from Käufer (1994) for the extrusion of plastic pipes).

Transports

One explicit transport step is required for this process, when the polystyrene granulates, the flame retardants plus the rock salts are moved from the manufacturing company to the place of extrusion. For this dataset, the standard distances of ecoinvent (100/50 km by 32 t lorry and 200/100 km by train in Europe) are applied.

4.6.2 Polystyrene, extruded (XPS) in ecoinvent

The meta information of “polystyrene, extruded (XPS), at plant” (Tab. 4.22) includes all important and necessary information to use the data correctly.

¹⁶ Supported by information of http://www.climnet.org/pubs/FoamsFACTSHEET_March2004.pdf

Fig. 4.18 shows the system of the dataset “polystyrene, extruded (XPS), at plant”. It shows all in- and output flows, the allocation to the datasets in the ecoinvent database database V2.0, the sources and the standard deviation of all in- and output flows.

The production steps include the extrusion of expandable polystyrene into polystyrene XPS. The discharge process through a slot die is part of the extrusion. The dataset relates to the production of XPS polystyrene using CO₂, HFC-134a and HFC-152a as blowing agents.

Tab. 4.22 ecoinvent database meta information of the product “polystyrene, extruded (XPS), at plant”

Name	polystyrene, extruded (XPS), at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	The dataset describes the production of extruded polystyrene. Included processes are the melting of polystyrene pearls in the extruder, the discharge through a slot die, as well as the cooling with water.
Amount	1
Local Name	Polystyrol, extrudiert (XPS), ab Werk
Synonyms	XPS
General Comment to reference function	This dataset can be used in the construction sector. The cutting of the extruded polystyrene blocks is not included in the process. Blowing agent shares are 50% (w/w) CO ₂ , 25% (w/w) HFC-134a and 25% (w/w) HFC-152a. Process emissions are 17.5% (w/w) for both HFC-134a and HFC-152a.
Start Date	1994
End Date	2006
Data Valid For Entire Period	1
Other Period Text	Technology in 1994 was still using CFC or HCFC as blowing agent. In 2005, already more than 50% of companies in Europe are using CO ₂ as blowing agent. Though mainly in the USA, HFC-134a and HFC-152a as a blowing agent are still common.
Geography text	Data for leading polystyrene XPS producers in Germany and the USA. Applicable for Europe and North America.
Technology text	Extrusion process of expandable polystyrene including discharge via slot die. CO ₂ and HCF-134a/HFC-152a as blowing agents.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature, databank
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that Germany is representative for Central Europe is considered in the uncertainties.

4. System Characterisation

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Polystyrene	Polystyrene XPS production		9.1%(w/w)	plastics	polymers	No	RER	polystyrene, expandable, at plant	9.10E-01	kg	Starzner & Wurmer-Weiss (2000)	1	124	(2,4,13,15,3)
Flame retardant			1%(w/w) bromated inorganic chemical	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	1.00E-02	kg		1	124	(2,4,13,15,3)
Fluorspar				chemicals	inorganics	No	GLO	fluorspar, 97% at plant	5.00E-03	kg	Energie- & Umweltzentrum Allgäu (2002)	1	126	(3,4,13,15,3)
Rock salt				chemicals	inorganics	No	RER	sodium chloride, powder, at plant	5.00E-03	kg		1	126	(3,4,13,15,3)
Transport lorry 32 t				transport systems	road	No	RER	transport, lorry >16t, fleet average	9.43E-02	tkm	ecoinvent standard distances	1	2.05	(2,3,11,15,5)
Transport train				transport systems	train	No	RER	transport, freight, rail	1.89E-01	tkm		1	2.05	(2,3,11,15,5)
Electricity			For melting of polystyrene perls in extruder and discharge through slot die Assumption: 50% of blowing agent is CO ₂ , so half of 7% (w/w)	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	5.30E-01	kWh	Käufer (1994)	1	123	(2,3,2,3,15,2)
Blowing agent			Assumption: 25% of blowing agent is HFC-134a, so 1.75% (w/w)	chemicals	inorganics	No	RER	carbon dioxide liquid, at plant	3.50E-02	kg	Starzner & Wurmer-Weiss (2000)	1	122	(2,3,12,15,4)
Blowing agent			Assumption: 25% of blowing agent is HFC-134a, so 1.75% (w/w)	chemicals	organics	No	RER	refrigerant R134a, at plant	1.75E-02	kg	IPCC (2007)	1	114	(2,3,12,14,4)
Blowing agent			Assumption: 25% of blowing agent is HFC-152a, so 1.75% (w/w)	chemicals	organics	No	US	1,1-difluoroethane, HFC-152a, at plant	1.75E-02	kg	IPCC (2007)	1	114	(2,3,12,14,4)
		R134a emission	Emission to air: 17.5% losses during production	air	low population density			Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	3.06E-03	kg	IPCC (2007)	1	114	(2,3,12,14,4)
		HFC-152a emission	Emission to air: 17.5% losses during production	air	low population density			Ethane, 1,1-difluoro-, HFC-152a	3.06E-03	kg	IPCC (2007)	1	114	(2,3,12,14,4)
		Styrene (gas)	Assumption that styrene emissions for XPS are 0.01% of input	air	unspecified			Styrene	9.10E-05	kg	Own calculations and EPSASA (2002)	1	159	(3,4,13,15,31)
		Heat waste		air	unspecified			Heat, waste	1.91E+00	MJ	Calculation	1	126	(3,4,13,15,13)
Cooling			value for extrusion of plastic pipe	resource	in water			Water, cooling, unspecified natural origin	1.97E-02	m ³	estimation	1	126	(3,4,13,15,4)
Infrastructure			explosive production plant as proxy	construction materials	others	Yes	CH	explosive production plant	6.67E-09	unit	estimation	1	2.33	(5,5,11,5,11)
		Polystyrene, extruded (XPS), at plant		insulation materials	production	No	RER	polystyrene, extruded (XPS), at plant	1.00E+00	kg				

Fig. 4.18 Flows for “polystyrene, extruded (XPS), at plant” and their representation in the ecoinvent database

Data quality

The uncertainties are evaluated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent meta information.

Most data of this process are based on the Starzner S. and Wurmer-Weiss P. (2000), though important input was also retrieved from other sources. Information on the emissions related to the extrusion process is based on IPCC (2007), however, some assumptions had to be made (share of different blowing agents during production process).

4.7 Cellulose Fibre (inclusive blowing in)

4.7.1 Process Description

Waste paper is shredded and mixed with boric acid and borax. The mixture is milled in a hammer mill where the fibres in the paper are separated and impregnated with the boron. The fibres are separated from dust in a cyclone and packed in paper bags (Richter et al. (1995)).

Because the application of cellulose fibres as insulation material for buildings requires some electricity for blowing, this process has to be inventoried to allow for a comparison of cellulose fibres with other insulation materials such as EPS foam, rock wool or glass wool.

Infrastructure

Beside some storage capacity in the factory the main equipment is the hammer mill. It is assumed that infrastructure needed for producing 1 kg cellulose fibres insulation is about the same as for 1 kg wood pellets. Thus the wood pellet manufacturing plant is chosen as proxy for the cellulose fibre production plant.

Emissions

Dust is the only relevant direct emission from handling and milling of the paper.¹⁷

Waste

The solid wastes from the process include the dust from separating of the dust from the fibres in the cyclone and the packaging material.

Richter et al. (1995)) indicate a waste water stream in the process flow chart but does not mention any waste water in the inventory data. It is plausible that some of the water input remains in the product and in the solid waste but some waste water is nevertheless expected. The composition of the waste-water is expected to include boron and organic matter. However, no measurements are available.

4.7.2 Cellulose Fibre Production and Application in ecoinvent

The production process is based on the data given in Richter et al. (1995). For the lack of data on air emissions and wastewater they must be neglected.

The energy demand for the application of cellulose fibre as building insulation is taken from Eyerer & Reinhardt (2000)). This source indicates only the primary energy demand. As exclusively electricity is used for this process, the electricity consumption is calculated using the conversion factor primary energy to electricity of 1/3.

The ecoinvent meta-information for the cellulose fibre production is given in Tab. 4.23 and the input and output flows for "cellulose fibre, inclusive blowing in, at plant" are shown in Fig. 4.19.

¹⁷ Weibel & Stritz (1995) interpreted the amount of paper dust from the cyclone given in Richter et al. (1995) as air emission instead of waste.

Data Quality

The inventory is based on one source only and no direct emissions are known. On the other hand, the process is rather uncomplicated and the shortcomings of the inventory are not believed to be of major relevance. Thus the overall quality of the data is considered sufficient.

Tab. 4.23 ecoinvent meta information for the cellulose fibre production and application

Name	cellulose fibre, inclusive blowing in, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the input energy and material to the production processes, transports of the materials and the available process emissions. Also included is the packaging and the energy for the application of the fibres as insulation materials in buildings.
Amount	1
Local Name	Zellulosefasern, inklusive Einblasen, ab Werk
Synonyms	
General Comment to reference function	Cellulose fibre in building applications has a density of 50 kg/m ³ and a thermal conductivity of 0.04 W/mK when applied. No emissions are inventoried besides the heat waste from electricity use for lack of data.
Start Date	1995
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	production data from one major producer in Switzerland
Technology text	mechanical separation of fibres from waste paper
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	literature
Extrapolations	none
Uncertainty Adjustments	none

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Waste paper	cellulose fibre, inclusive blowing in, at plant			paper & cardboard	pulps	No	CH	waste paper, mixed, from public collection, for further treatment	8.01E-01	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,3)
boric acid				chemicals	inorganics	No	RER	boric acid, anhydrous, powder, at plant	8.87E-02	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
borax				chemicals	inorganics	No	RER	Borax, anhydrous, powder, at plant	1.09E-01	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
paper bags			for packaging	paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	3.70E-02	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
pallets			for packaging	wooden materials	processing	No	RER	EUR-flat pallet	1.21E-03	unit	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
plastic			for packaging	plastics	processing	No	RER	packaging film, LDPE, at plant	3.25E-04	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
wire			for packaging	metals	extraction	No	RER	reinforcing steel, at plant	1.81E-03	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
water				water supply	production	No	RER	tap water, at user	2.38E-02	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
Electricity			for production	electricity	supply mix	No	CH	electricity, low voltage, at grid	1.02E-01	kWh	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,2)
Electricity			0.1 MJ primary energy for application	electricity	supply mix	No	CH	electricity, low voltage, at grid	1.85E-04	kWh	calculated based on Eyerer et al. 2000	1	1.22	(1,3,2,2,1,5,2)
natural gas				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	6.41E-02	MJ	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,1)
Dielsel for internal transports				construction processes	machinery	No	GLO	diesel, burned in building machine	1.12E-02	MJ	Richter et al. 1995	1	2.05	(1,3,2,1,1,5,5)
Transport rail			600 km for chemicals, 100 km for paper and pallets	transport systems	train	No	RER	transport, freight, rail	2.05E-01	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
Transport lorry			everything 50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	5.32E-02	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
plant			wood pellet plant as proxy (1E8 kg output in 50 years)	wood energy	fuels	Yes	RER	wood pellet manufacturing, infrastructure	1.00E-08	unit	estimated	1	3.23	(5,n.A.,n.A.,n.A.,n.A.,n.A.,9)
		cellulose fibre (incl. blowing)		insulation materials	production	No	CH	cellulose fibre, inclusive blowing in, at plant	1.00E+00	kg	Richter et al. 1995			
		Waste heat into air		air	unspecified			Heat, waste	3.66E-01	MJ	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,13)
		Paper dust to incineration	dust from cyclone	waste management	municipal incineration	No	CH	disposal, paper, 11.2% water, to municipal incineration	1.12E-02	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,6)
		packaging material to incineration		waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.04E-02	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,6)
		pallets to incineration		waste management	municipal incineration	No	CH	disposal, wood untreated, 20% water, to municipal incineration	2.80E-03	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,6)
		pallets to recycling	not inventoried						1.08E-03	unit	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,6)

Fig. 4.19 Flows for "cellulose fibre, inclusive blowing in, at plant" and their representation in the ecoinvent database

4.8 Urea Formaldehyde Foam

4.8.1 Process description

“A surfactant solution containing a hardener (e.g., phosphoric acid) is foamed with compressed air and blended with an aqueous 30 – 40 % solution of urea–formaldehyde resin. The resin is distributed throughout the surfactant foam, and cross-linking is initiated by the acidic hardener. A light, whipped-cream-like foam is formed which can be shaped easily and used to fill any cavity. The foam hardens rapidly to a water-insoluble, rot-resistant, foamed plastic without creating expansion pressure. A hardening time of 15 – 40 s is generally selected, which can be controlled within certain limits by changing external variables (temperature, hardener concentration).” (Weber (1997))

Problems caused by formaldehyde release can be avoided largely by using more urea or improved resin formulations.

4.8.2 Infrastructure

A special compressor is needed for the in situ production. According to information given on a producers web page¹⁸, its weight is estimated to 50 kg at a production rate of 1 m³/15 min.

The industrial production infrastructure is approximated by the module for a generic chemical production plant. It is assumed that the output of the plant is twice the output of a comparable plant for organic chemicals in order to account for the relatively simple processes involved in the UF-foam production.

4.8.3 Emissions

A direct formaldehyde emission is estimated on the basis of emissions in the production of wooden boards with urea-formaldehyde binders. It is assumed that the industrial prefabrication process is better controlled than the in situ process and thus leads to less emissions.

4.8.4 Waste

2% of the produced slabs is inventoried as waste. This accounts for cut off in the slab production. 5% of the in situ foam is inventoried as waste to account for material loss by cleaning of the tools.

4.8.5 Urea Formaldehyde Foam Production in ecoinvent

For lack of data the material inputs, the relevant emissions and the process energy are estimated. For details see Tab. 4.24, Fig. 4.20 and Fig. 4.21.

Data Quality

The inventory is based on theoretical process description and calculations. The direct formaldehyde emission is estimated. Since this emission might be of high relevance, the overall quality of the data is considered poor, even though the process is rather uncomplicated.

¹⁸ <http://www.schaum-chemie.de/produkte/isoschaum.php>; access date: 15.03.2004

Tab. 4.24 ecoinvent meta information for the UF foam production

Name	urea formaldehyde foam, in situ foaming, at plant	urea formaldehyde foam slab, hard, at plant
Location	CH	CH
Infrastructure Process	0	0
Unit	kg	kg
Data Set Version	2.0	2.0
Included Processes	Includes raw materials and in situ production of foam. Transports to application site are not included.	Includes raw materials and production of foam.
Amount	1	1
Local Name	Ortschaum, Harnstoffformaldehyd, ab Werk	Hartschaumplatte, Harnstoffformaldehyd, ab Werk
Synonyms		
General Comment to reference function	Foam has a density of 10 - 30 kg /m3 and a thermal conductivity of 0.035 W/mK. Data is based on crude assumptions.	Foam has a density of 10 - 30 kg /m3 and a thermal conductivity of 0.035 W/mK. Data is based on crude assumptions.
Start Date	1997	1997
End Date	2003	2003
Data Valid For Entire Period	1	1
Other Period Text		
Geography text		
Technology text	On site production of foam from two components using a compressor and a portable injection device.	Production of slabs in large manufacturing plant using stationary machinery.
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	none	literature
Extrapolations	none	none
Uncertainty Adjustments	none	none

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
UF resin	urea formaldehyde foam, in situ foaming, at plant			paintings	production	No	RER	urea formaldehyde resin, at plant	9.56E-01	kg	estimated	1	2.56	(5,5,5,5,5,3)
water				water supply	production	No	CH	tap water, at user	4.10E-01	kg	estimated	1	2.56	(5,5,5,5,5,4)
surfactant				washing agents	tensides	No	RER	alkylbenzene sulfonate, linear, petrochemical, at plant	5.25E-02	kg	estimated	1	2.56	(5,5,5,5,5,3)
phosphoric acid			hardener	chemicals	inorganics	No	RER	phosphoric acid, industrial grade, 85% in H ₂ O, at plant	5.25E-02	kg	estimated	1	2.56	(5,5,5,5,5,3)
compressed air														
electricity			for compressor: Power consumption 1 kW; 1 m ³ (=20 kg) foam / 15 min	electricity	supply mix	No	CH	electricity, low voltage, at grid	1.25E-02	kWh	estimated, based on http://www.schaum-chemie.de (access 15.3.2004)	1	2.56	(5,5,5,5,5,2)
Transport rail			600 km for chemicals	transport systems	train	No	RER	transport, freight, rail	6.36E-01	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
Transport lorry			50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	5.30E-02	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
Compressor			Assumption: compressor weight 50 kg, life time foam production 100'000 kg	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	5.00E-04	kg	estimated, based on http://www.schaum-chemie.de (access 15.3.2004)	1	4.25	(5,5,5,5,5,9)
		→ waste heat	only from electricity consumption. Heat of reaction is not inventoried.	air	unspecified			Heat, waste	4.50E-02	MJ	estimated	1	2.56	(5,5,5,5,5,13)
		→ formaldehyd	1‰ of UF resin input	air	high population density			Formaldehyde	9.56E-04	kg	estimated	1	3.22	(5,5,5,5,5,23)
		→ water	drying; not inventoried						4.29E-01	kg	calculated			
		→ Waste	5% of produced foam	waste management	municipal incineration	No	CH	disposal, polyurethane, 0.2% water, to municipal incineration	5.00E-02	kg	estimated	1	2.56	(5,5,5,5,5,6)
		→ UF foam, in situ		insulation materials	production	No	CH	urea formaldehyde foam, in situ foaming, at plant	1.00E+00	kg				

Fig. 4.20 Flows for "urea formaldehyde foam, in situ foaming, at plant" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
UF resin	urea formaldehyde foam slab, hard, at plant			paintings	production	No	RER	urea formaldehyde resin, at plant	9.28E-01	kg	estimated	1	2.56	(5,5,5,5,5,3)
water				water supply	production	No	CH	tap water, at user	3.98E-01	kg	estimated	1	2.56	(5,5,5,5,5,4)
surfactant				washing agents	tensides	No	RER	alkylbenzene sulfonate, linear, petrochemical, at plant	5.10E-02	kg	estimated	1	2.56	(5,5,5,5,5,3)
phosphoric acid			hardener	chemicals	inorganics	No	RER	phosphoric acid, industrial grade, 85% in H ₂ O, at plant	5.10E-02	kg	estimated	1	2.56	(5,5,5,5,5,3)
compressed air														
electricity			Power consumption 1 kW --> 1 m3 (=20 kg) foam / 5 min	electricity	supply mix	No	CH	electricity, low voltage, at grid	4.17E-03	kWh	estimated, based on http://www.schaum-chemie.de (access 15.3.2004)	1	2.56	(5,5,5,5,5,2)
Transport rail			600 km for chemicals	transport systems	train	No	RER	transport, freight, rail	6.18E-01	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
Transport lorry			50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	5.15E-02	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
plant			Proxy: chemical production plant	chemicals	organics	Yes	RER	chemical plant, organics	2.00E-10	unit	estimated	1	4.25	(5,5,5,5,5,9)
		→ waste heat	only from electricity consumption. Heat of reaction is not inventoried.	air	unspecified			Heat, waste	1.50E-02	MJ	estimated	1	2.56	(5,5,5,5,5,13)
		→ formaldehyd	0.1% of UF resin input	air	high population density			Formaldehyde	9.28E-05	kg	estimated	1	3.22	(5,5,5,5,5,23)
		→ water	drying; not inventoried						4.17E-01	kg	calculated			
		→ Waste	2% of produced foam	waste management	municipal incineration	No	CH	disposal, polyurethane, 0.2% water, to municipal incineration	2.00E-02	kg	estimated	1	2.56	(5,5,5,5,5,6)
		→ UF foam slab		insulation materials	production	No	CH	urea formaldehyde foam slab, hard, at plant	1.00E+00	kg				

Fig. 4.21 Flows for “urea formaldehyde foam slab, hard, at plant” and their representation in the ecoinvent database

5 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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Part XV

Elastomere Tube Insulation

Data v2.0 (2007)

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Summary

This part deals with elastomere tube insulation and an appropriate production plant which includes the necessary infrastructure. The main application of this product is the insulation of warm water pipes of solar collectors. The main raw material is synthetic rubber. The data are mainly provided by one German company.

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1 Introduction

Tube insulation materials can be used for different technical applications. Within this project their use as insulation material for warm water pipes of solar collectors is of interest. But they might also be used for other types of technical processes for the transport of hot or cold fluids. The life cycle inventory investigates the production from cradle to gate.

2 Reserves and Resources of Material

The type of flexible, closed cell insulation materials investigated here, is an elastomere produced from synthetic rubber (ethylene propylene diene monomer or acrylonitrile-butadiene-styrene) and some additional chemicals. Thus the basic resource for this product is crude oil. The total production volume is not known.

3 Characterisation and Use of Tube Insulation Materials

There are different types of tube insulation materials on the market. They can be distinguished by the temperature range of application and their resistance against UV-radiation, fire and chemicals. Tube insulation materials can be used for insulation of pipe work and valves with flows of cold or hot liquids or gases. Tab. 3.1 shows a summary of different products that can be found on the market.¹

Tab. 3.1 Description of elastomere tube insulation materials from different companies

Brandname	Material, Company	Application
Armaflex HT	Elastomere, 70-80kg/m ³ , 0,040 - 0,045 W/(m·K) heat conductivity, Armacell	flexible insulation for high temperature up to +150°C, used for solar collectors
Armaflex SH	Elastomere, 0,040 W/(m·K) Armacell	insulation for sanitary and home heating use
Armaflex AF	Elastomere, 45-100kg/m ³ , 0,030 - 0,040 W/(m·K) heat conductivity, Armacell	insulation for climatization and cooling
Armaflex NH	Elastomere, 45-100kg/m ³ , 0,040 - 0,045 W/(m·K) heat conductivity, Armacell	halogen-free insulation for special applications, e.g. public buildings
Aeroflex	Closed cell elastomere, 40-60kg/m ³ , Aeroflex	insulation for warm and cold water pipes

¹ Information on www.armacell.com and www.aeroflex.ch.

4 System Characterisation

The production of tube insulation materials is investigated from gate to gate for one producer with production places in Münster and Zeulenroda, Germany. About 95% of tube insulation materials used in Switzerland are produced there. The rest is imported from Poland (Frischknecht 1999:84). Different types of products are produced (cf. Tab. 3.1), but differences in their production process could not be considered for inventory. Thus an average product is modelled.

The production process is described by Armacell GmbH (2001). Fig. 4.1 shows a flow chart for this process which can roughly be divided into pre-production and finishing process. Synthetic rubber is delivered in bulk to the plant. In a first stage different raw materials are kneaded. The rubber mass is rolled out into the size of the latter basic rubber furs which are brought to an intermediate storage.

For a part of these furs the finishing process (after step 6 in Fig. 4.1) takes place in Zeulenroda. The bulk is further processed in Münster. These rubber basic furs are warmed up again in the next rolling mill and some necessary raw materials are added. The mass is extruded to the form of the final product. Then the tubes are brought to the batch-ovens or to the conti-ovens for vulcanisation. During this process they simultaneously blow out to their final size. The product is confectionated, labelled and packed in cardboard. Most (82%) of the production wastes are recycled within the same production process.

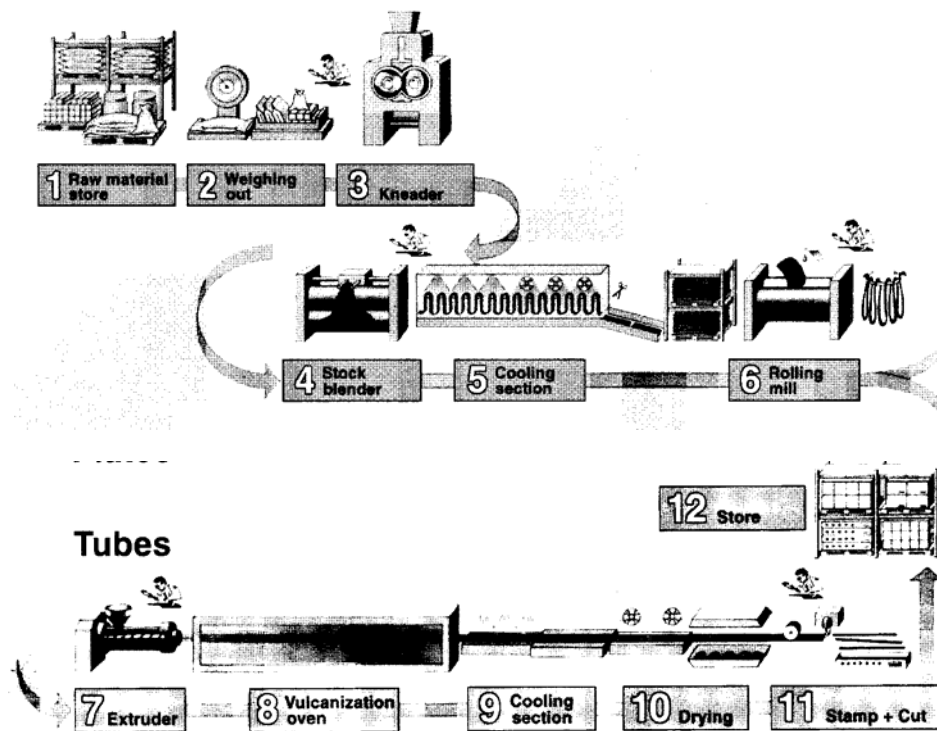


Fig. 4.1 Production process for tube insulation material.

5 Life Cycle Inventory

Different types of insulation materials do not differ much in their production process. The main difference is the use of PVC as a pre-product and the type of rubber used. Ethylene propylene diene monomer (EPDM) is used together with PVC while acrylonitrile-butadiene-styrene (NBR) can be processed without halogenes. Aeroflex products are produced without PVC nor halogenes. Good data were available for the production at (Armacell GmbH 2001). Some data have been provided by Aeroflex for a production plant in Thailand (column „Aeroflex 2001“ in Tab. 5.4).²

A generic inventory for these materials is elaborated here. The functional unit is 1kg of product. The specific weight of the material is in the range of 48-96kg/m³ for EPDM products and 40-128kg/m³ for NBR products. It is estimated with 75kg/m³. This might differ considerable between different products. The life cycle inventory for the production of tube insulation materials is shown in Tab. 5.4.

A life cycle inventory for the tube insulation material Armaflex® has been prepared by (Frischknecht 1999, column „Armacell Münster, 1996“ in Tab. 5.4). It is based on an environmental report for 1996 by the producer. This report has been updated for the year 1999 in the meantime (Armacell GmbH 2001, see “Armacell Münster, 1999 in Tab. 5.4). A part of the second production step (after step 6 in Fig. 4.1) is done in Zeulenroda. For that site environmental data could be found on the homepage (Armacell GmbH 2002, column “Armacell Zeulenroda, 1998 in Tab. 5.4).

These two reports are the basis for most of the unit process raw data. The total figures for one year are added up for the two sites and than they are divided to the summed output of elastomere tube insulation materials.

The two environmental reports do not show the data for the total production. In Tab. 5.1 it is calculated from figures for the total use of gas and electricity and data for the specific energy use per m³ of produced insulation material (Armacell GmbH 2001). According to this assumption 9500 tonnes of insulation material have been produced in 1999 in Münster. (Frischknecht 1999) has calculated a much higher output (12400t), but did not consider that a part of the rubber basic furs is processed further in Zeulenroda. In a personal communication the amount in 1999 has been given with 7000t. This figure is used here for the produced final product. For Zeulenroda the produced amount has been estimated from the input of rubber basic furs (2490t) minus the amount of wastes (402t).

Tab. 5.1 Calculation of the total production volume for the production plant in Münster

	1994	1995	1996	1997	1998	1999
Natural gas m3	2011000	1017996	1145190	1128132	1215658	1122475
Electricity kWh	13466224	12467112	11441065	10949300	11333450	11205646
Specific energy use kWh/m3	383	239	222	218	197	180
Production m3	89854	96532	105271	104132	121810	127212
Production t	6739	7240	7895	7810	9136	9541

From these figures the following production flows are estimated. Thus the higher production volume in Tab. 5.1 can be explained as the volume including the rubber basic furs.

² Personal communication with W. Merath, 7.2002.

Tab. 5.2 Assessment of production volumes in Münster and Zeulenroda

Plant	Amount processed in step 1-6 of Fig. 4.1	Amount produced in step 7-12 of Fig. 4.1
	t	t
Münster	9500	7000
Zeulenroda	0	2070
Total production	(intermediate product)	9070

About 15 different chemicals are used for the production of Aeroflex. The input of materials is derived from (Frischknecht 1999). It is assumed that synthetic rubber is mixed with 7.5% polyvinylchloride and 6.5% 1,1-azobisformamide as agent for the extrusion. 7% of the raw materials are lost during production. Ambient air is used for foaming at Aeroflex. Cooling water is used in closed-loop thus only losses have to be brought into the system. The amount of packaging cardboard has been estimated with data from Aeroflex.

The street distance from Münster to Zeulenroda is 480km. For the calculation of the transport of rubber basic furs it has to be taken into account that the specific weight is very low. The transport distance for the rubber basic furs has to be multiplied by 12 because of the low weight.³ Products from Thailand are brought by ship to Basel. Then they are transported by lorry. These imports are not considered here.

The inventory for the manufacturing plant is shown in Tab. 5.3. It is based on an environmental report (Armacell GmbH 2001) and on data provided in a questionnaire (see footnote 2). In Münster the production plant is situated within the city. Until 1960 the area has been used for agriculture. The plant in Zeulenroda is in an industrial area near to the city (Armacell GmbH 2002). Data for the equipment were not available. The annual production is 5000 tonnes and the life time 50 years.

Tab. 5.3 Unit process raw data and data quality indicators for the production plant for tube insulation. References and assumptions for the inventory are presented in the text. Annual production is 5000 tonnes. Life time 50a

	Name	Location	Infrastruc	Unit	tube insulation plant	Uncertai Standar dDevlati	GeneralComment	Armacell Münster, 1999	Armacell Zeulenrod a, 1998	Aeroflex
	Location				DE			DE	DE	TH
	InfrastructureProcess				1					
	Unit				unit			unit	unit	unit
output	tube insulation plant	DE	1	unit	1.00E+0					
technosphere	building, hall	CH	1	m2	2.39E+4	1	3.02 (3,3,1,1,1,na); Questionnaires	7.50E+4	6.20E+3	1.22E+4
resource, land	Occupation, industrial area, built up	-	-	m2a	2.39E+4	1	1.52 (3,3,1,1,1,na); 50a occupation	7.50E+4	6.20E+3	1.22E+4
	Occupation, industrial area, vegetation	-	-	m2a	5.19E+4	1	1.52 (3,3,1,1,1,na); 50a occupation	1.53E+5	1.24E+4	3.78E+4
	Transformation, from arable	-	-	m2	1.51E+3	1	2.02 (3,3,1,1,1,na); Calculation	4.56E+3	3.72E+2	1.00E+3
	Transformation, to industrial area	-	-	m2	1.51E+3	1	2.02 (3,3,1,1,1,na); Questionnaires	4.56E+3	3.72E+2	1.00E+3
Annual production								7E+6	2E+6	1E+7

³ We assume for this calculation a specific weight of 75kg/m³, a hollow space of 70% and a freight volume of 50m³ for a 40t lorry. Thus 1125kg of tube insulation material can be transported with a 40t lorry. The life cycle inventory for the lorry is calculated with an occupancy of 50% and a freight of 13.5t. Thus the transport distance in tkm has to be multiplied by 12 for this product.

Tab. 5.4 Unit process raw data and data quality indicators for the production of tube insulation. References and assumptions for the inventory are presented in the text.

	Name	Location	Infrastructure	Unit	tube insulation, elastomere, at plant	Uncertainty na	StandardDeviation95%	GeneralComment	Armace	Armace	Armace	Aeroflex
									ll Münster, 1999	ll Zeulenroda, 1998	ll Münster, 1996	ll 2001
	Location InfrastructureProcess Unit				DE 0 kg				DE 0 a	DE 0 a	DE 0 kg	TH 0 kg
output	tube insulation, elastomere, at plant	DE	0	kg	1.00E+0			(5,1,1,1,1,3); SDG for output: 1.5; Estimation from aggregated figures	7.00E+06	2.09E+06	1.00E+0	1.00E+0
technosphere	natural gas, high pressure, at consumer	DE	0	MJ	4.57E+0	1	1.51	(1,2,1,1,1,3); Environmental report, 36.3MJ/Nm3	4.07E+07	8.00E+05	3.40E+0	-
	electricity, medium voltage, at grid	DE	0	kWh	1.43E+0	1	1.51	(1,2,1,1,1,3); Environmental report	1.12E+07	1.75E+06	9.17E-1	2.00E+0
	tube insulation plant	DE	1	unit	4.00E-9	1	1.51	(1,2,1,1,1,3); Environmental report	-	-	-	-
	tap water, at user	RER	0	kg	1.34E+0	1	1.51	(1,2,1,1,1,3); Environmental report	1.16E+07	5.56E+05	1.30E+0	2.50E-1
	chemicals organic, at plant	GLO	0	kg	6.50E-2	1	2.30	(3,2,1,1,5,5); Assumption 6.5% of extrusion agent 1,1-azobisformamide	n.a.	n.a.	n.a.	n.a.
	synthetic rubber, at plant	RER	0	kg	9.30E-1	1	1.58	(3,2,1,1,1,5); Therotical assumptions, Bulk transport	n.a.	n.a.	9.30E-1	9.00E-1
	polyvinylchloride, at regional storage	RER	0	kg	7.50E-2	1	1.64	(3,2,1,1,3,5); Therotical assumptions	n.a.	n.a.	7.50E-2	-
	polyurethane, rigid foam, at plant	RER	0	kg	5.51E-2	1	1.52	(1,3,1,1,1,3); Environmental report	n.a.	1.15E+05	n.a.	n.a.
	corrugated board, mixed fibre, single wall, at plant	RER	0	kg	6.43E-3	1	1.52	(2,2,1,3,1,3); Questionnaire, product packaging	n.a.	n.a.	n.a.	6.43E-3
	transport, lorry 32t	RER	0	tkm	1.79E+0	1	2.33	(4,5,na,na,na,na); Standard distance 200km plus transport of furs to Zeulenroda	n.a.	1.43E+07	n.a.	n.a.
	transport, freight, rail	RER	0	tkm	1.06E-1	1	2.33	(4,5,na,na,na,na); Standard distances	n.a.	n.a.	n.a.	n.a.
	disposal, plastics, mixture, 15.3% water, to sanitary landfill	CH	0	kg	7.48E-2	1	1.51	(1,2,1,1,1,3); Environmental report	3.64E+05	3.16E+05	7.00E-2	n.a.
	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	0	kg	1.28E-3	1	1.51	(1,2,1,1,1,3); Environmental report	0.00E+00	1.16E+04	7.00E-2	n.a.
emission air, high population density	treatment, sewage, from residence, to wastewater treatment, class 2	CH	0	m3	1.34E-3	1	1.51	(1,2,1,1,1,3); Environmental report, Sanitary and cooling water	1.16E+04	5.56E+02	1.30E-03	2.50E-04
	Carbon monoxide, fossil	-	-	kg	2.42E-4	1	5.27	(1,2,1,1,1,3); Environmental report	3.60E+02	1.84E+03	3.10E-5	
	Carbon dioxide, fossil	-	-	kg	2.25E-1	1	1.51	(1,2,1,1,1,3); Environmental report, Calculated for Zeulenroda	2.00E+06	4.43E+04	1.70E-1	
	Nitrogen oxides	-	-	kg	2.93E-4	1	1.78	(1,2,1,1,1,3); Environmental report	1.79E+03	8.71E+02	1.50E-4	
	Particulates, > 10 um	-	-	kg	5.04E-5	1	2.24	Aerosoles measured as dust, estimation >10um		4.58E+02		
	Sulfur dioxide	-	-	kg	1.21E-2	1	1.78	(1,2,1,1,1,3); Environmental report	1.10E+05	2.29E+02	9.10E-7	
	Heat, waste	-	-	MJ	5.13E+0	1	1.51	(1,2,1,1,1,3); Calculated				
source									Armace GmbH 2001	Armace GmbH 2002	Frischknecht 1999	Personal communication

6 Data Quality Considerations

Tab. 5.4 shows the data quality indicators for the inventory of tube insulation materials. The simplified approach with a pedigree matrix has been used for calculating the standard deviation. Large uncertainties exist for data for transport distances of materials. The estimation for the product output is very uncertain. Its standard deviation has been multiplied with all entries to the inventory.

Tab. 5.3 shows the data quality indicators for the inventory of a tube insulation plant. The simplified approach with a pedigree matrix has been used for calculating the standard deviation. The data for land use are relative sure, but they had to be extrapolated to the general production in Germany.

The inventory of tube insulation materials can be considered as a reliable basis for the assessment of this material in connection with technical applications where the insulation is used in a system for the transport of hot or cold liquids. It is not meant to be used for the direct comparison of different insulation materials. For that, detailed information about product properties and type of application are necessary.

7 Cumulative Results and Interpretation

Results of the cumulative inventory can be downloaded from the database (www.ecoinvent.ch).

Abbreviations

See Tab. 3.1.

Appendicies: EcoSpold Meta Information

ReferenceFunction	401	Name	tube insulation, elastomere, at plant	tube insulation plant
Geography	662	Location	DE	DE
ReferenceFunction	493	InfrastructureProcess	0	1
ReferenceFunction	403	Unit	kg	unit
DataSetInformation	201	Type	1	1
	202	Version	1.0	1.0
	203	energyValues	0	0
	205	LanguageCode	en	en
	206	LocalLanguageCode	de	de
DataEntryBy	302	Person	41	41
	304	QualityNetwork	1	1
ReferenceFunction	400	DataSetRelatesToProduct	1	1
	402	IncludedProcesses	The production of tube insulation materials is investigated from gate to gate for one producer with production places in Münster and Zeulenroda, Germany.	Infrastructure for two producers with production places in Münster and Zeulenroda, Germany and in Thailand.
	404	Amount	1	1
	490	LocalName	Rohrsolierung, Elastomere, ab Werk	Rohrsolierungwerk
	491	Synonyms	armaflex//aeroflex//PUR//PVC	
	492	GeneralComment	Tube insulation materials can be used for different technical applications. Within this project their use as insulation material for warm water pipes of solar collectors is of interest. The inventory of tube insulation materials can be considered as a reliable data base for the assessment of this material in connection with technical applications. It is not meant to be used for the comparison of different insulation materials. Therefore detailed information about product properties and type of application would be necessary. Transport distances should be multiplied by 14 to account for low specific weight of 75kg/m3.	Infrastructure for factory including land use not including machinery. Annual production 5000 tonnes tube insulation material. Transformation based on general assumption with a lifetime of 50a.
	494	InfrastructureIncluded	1	1
	495	Category	insulation materials	insulation materials
	496	SubCategory	production	production
	497	LocalCategory	Wärmedämmstoffe	Wärmedämmstoffe
	498	LocalSubCategory	Herstellung	Herstellung
	499	Formula		
	501	StatisticalClassification		
	502	CASNumber		
TimePeriod	601	StartDate	1997	1997
	602	EndDate	1999	1999
	603	DataValidForEntirePeriod	1	1
	611	OtherPeriodText	Year of environmental reporting. Two different production places.	Year of environmental reporting. Two different production places.
Geography	663	Text	Production in Germany	Data for production plants in Germany and Thailand
Technology	692	Text	The production process can roughly be divided into pre-production and finishing process. Synthetic rubber is delivered in bulk to the plant. In a first stage different raw materials are kneaded. The rubber mass is rolled out in size of the latter rubber basic furs which are brought to an intermediate storage. For a part of these furs the finishing process takes place in Zeulenroda. The bulk is further processed in Münster. These rubber basic furs are warmed up again in the next rolling mill and some necessary raw materials are added. The mass is extruded to the form of the final product. Than the tubes are brought to the batch-ovens or to the conti-ovens for vulcanisation. During this process they simultaneously blow out to their final size. The product is cut to size, labelled and packed in cardboard. Most (82%) of the production wastes are recycled within the same production process.	Production plant for elastomere tube insulations.
Representativeness	722	Percent		
	724	ProductionVolume	12000t in two factories, 95% of CH market	20000t in three factories
	725	SamplingProcedure	Environmental report.	Environmental report and questionnaire.
	726	Extrapolations	Data for one producer are used for the total production. Packaging estimated with data from TH.	none
	727	UncertaintyAdjustments	none	none
DataGenerator	751	Person	41	41
AndPublication	756	DataPublishedIn	2	2
	757	ReferenceToPublishedSource	7	7
	758	Copyright	1	1
	759	AccessRestrictedTo	0	0
	760	CompanyCode		
	761	CountryCode		
	762	PageNumbers	Elastomere Tube Insulation	Elastomere Tube Insulation

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Part XVI

Cork

Data v2.0 (2007)

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Kellenberger D., Althaus H.-J., Jungbluth N., Künniger T., Lehmann M. and Thalmann P. (2007) Life Cycle Inventories of Building Products. Final report ecoinvent Data v2.0 No. 7. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.org.

Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

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Summary

The production of cork and cork slabs are described in this part.

The production of cork is modelled in a multi-output process together with wood. A hypothetical “average” cork slab is defined and inventoried.

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1 Introduction

This part deals with the production of raw cork and cork slabs. Because of the different possible applications of the slabs, a mixed production of different cork floorings and cork underlays is represented in this study.

2 Reserves and Resources of Materials

This information is based on De Oliveira (1991).

Cork forests exist mainly around the Mediterranean Sea and in Portugal. World-wide cork forests cover 2.2 million hectares of which 30% are in Portugal, 21% in Algeria, 20% in Spain, 16% in Morocco, 5% in France and 4% each in Italy and Tunisia. The cork forests in Portugal have been growing in the last century from around 400'000 ha in 1915 to about 660'000 ha in the 1990es.

Although Portugal has only 30% of the cork forests the country accounts for more than half of the world's cork production.

Cork oaks are stripped the first time when they are about 25 years old. The so-called "virgin cork" from the first stripping is of poor quality and thus of minor value. Afterwards the trees are stripped in intervals of about ten years until the oak is 150 to 200 years old.

The world average productivity, based on extraction every tenth year, is around 150 kg of cork per ha. However, figures between 2'000 and 2'500 kg are reported from Portugal. This large variability has different reasons: on one hand the forests in Portugal are denser than the less intensively used forests in other countries; on the other hand growth conditions in Portugal seem to be more favourable than elsewhere.

3 Characterisation and Use of the Materials

This information is based on De Oliveira (1991)).

The properties of cork derive from its honeycomb structure with 30 to 42 million cells per cubic centimetre. The almost 90% gaseous matter contained in the tissue of cork leads to a density of 0.12 to 0.2 kg/m³. Besides this lightness, the high elasticity, the impermeability, the low thermal and electrical conductivity and the high resistance to wearing are the most important technical characteristics of cork.

The uses of cork are manifold. The first that comes to mind is the use as cork stoppers. They are made from massive cork and require the best quality of cork. Other important applications are insulation and flooring materials. They are based on agglomerates of which two categories exist: pure expanded agglomerates and compound agglomerates. While the pure expanded agglomerates are agglutinated with the cork oak's own resin with heat treatment under pressure, the compound agglomerates are agglutinated with products such as asphalt, rubber, casein or melamine, urea-formaldehyde and phenolic resins.

4 System Characterisation

4.1 Process description

This chapter is mainly based on Althaus & Richter (2001). The growing of the trees is modelled in accordance with Werner et al. (2003).

The harvesting of cork is done manually using special knives and axes. The cork oaks are felled after 150 to 200 years using chain saws. The wood is of rather poor quality and usually not used as massive construction wood. Per m³ of wood, 422 kg of cork is harvested. The third main product of cork forests, the acorn, is mainly used as swine food. This product is neglected in this study.

The harvested cork is stored in staples in the forests for about two years for stabilisation before it is transported to the manufacturing of stoppers or agglomerates.

The cork for the agglomerated slab is trituated and sieved. The dust and clay¹ from this process are used as fuel in the furnace that generates the process heat. The cork granules are pressed to a slab together with the agglomeration resin. This can be done in a continuous or in a batch process. Finally the slab is sanded and stored for stabilisation. The dust from sanding is burned in the furnace, delivering together with the dust and clay the entire heat energy needed for the production. The sandpaper is disposed in landfills.

Different resins can be used depending on the application and the desired properties of the product.

4.2 Infrastructure

The infrastructure needed for agglomerated cork slab production is basically the same as for organically bound wooden boards (c.f. Werner et al. (2007)).

4.3 Emissions / waste

4.3.1 Air

Beside the emissions from heat generation there might be some dust emissions from the handling of the raw material and some formaldehyde emissions of the resins used for agglomeration.

4.3.2 Water

There is no information on water emissions in the production of cork slabs.

4.3.3 Solid waste

The cork and resin wastes are used as fuel in the furnace producing the process heat. The only waste generated in the manufacturing process that has to be landfilled is the sandpaper.

4.4 Cork production in ecoinvent

The ecoinvent meta-information for the cork production is given in Tab. 4.1 and the in- and output flows for "cork, harvesting / thinning / final cutting, under bark" and "cork slab, at plant" are given in Fig. 4.1 and Fig. 4.2 respectively.

¹ soil with a low heating value

The production of raw cork is inventoried adapting the module for hardwood production. The amount of cork produced per m^3 of wood production is calculated using the land occupation data for the growth of wood and cork. The land occupation for growing 1 m^3 wood (under bark) in Werner et al. (2007) is $1688 \text{ m}^2\text{a}$ while Althaus & Richter (2001)) report $4 \text{ m}^2\text{a}$ per kg of raw cork. Thus the value of $422 \text{ kg cork per m}^3 \text{ wood}$ is calculated.

For the production plant, the module for "wooden board manufacturing plant, organic bonded boards" is used as proxy. Since the land occupation of the industrial complex where the cork slabs are fabricated is given in Althaus & Richter (2001)) as $0.387 \text{ m}^2\text{a/m}^3$ but the production facility for organically bound wooden boards occupy $0.867 \text{ m}^2\text{a/m}^3$ (at a life time production of $30'000'000 \text{ m}^3 \rightarrow$ infrastructure demand $3.33 \cdot 10^{-08}/\text{m}^3$), only 45% of the plant that would be needed for wooden board fabrication is inventoried for cork slab production.

The allocation to the products cork and wood in the harvesting process is done for the most flows according to the overall revenue. Average prices of 2000 to 2003 from the Portuguese forest direction² were used. The factors chosen can be found in Fig. 4.1. The transport of the workers harvesting the cork is allocated to the cork only. To fulfil the mass, energy and carbon balance despite the economic allocation, an allocation correction factor for the wood is introduced (see Werner et al. (2007)). The correction factor is calculated by subtracting the economic allocation factor from the allocation factor by weight (atro) and multiplying the result with the input of wood standing in forest. The atro weight of oak wood is assumed 650 kg/m^3 , that of cork 150 kg/m^3 .

4.4.1 Data quality

Data uncertainty is derived using the pedigree matrix as described in (Frischknecht et al. (2007)).

Data of the cork oak production are based on the production of beech in Germany. For the cork itself, an additional output is defined. Because the similarity of the processes involved, the data except land use can be considered representative. Therefore the cork output is adjusted to the land use in the data. Thus the land use for the cork produced is representative while the land use for the co-produced wood is not.

All data of cork slab production beside formaldehyde emission (which is estimated) are from a fairly detailed and comprehensible LCA study conducted by EMPA Dübendorf in collaboration with a major producer. The application of the data in a generic module for different types of cork products minders the overall quality of the data from this source. Thus the overall quality of the data can be described as mediocre.

² <http://cryptomeria.dgf.min-agricultura.pt/enestatisticas.asp?prod=Cortica&Tipo=MMM&C1=&D1=&C2=Quarter&C3=Type&Unidade=1> accessed 24.04.03

Tab. 4.1 ecoinvent meta information for the cork production processes

Name	cork, harvesting / thinning / final cutting, under bark	cork slab, at plant
Location	RER	RER
Infrastructure Process	0	0
Unit	kg	kg
Data Set Version	2.0	2.0
Included Processes	Includes manual harvesting of the cork (every 10 years) and motor-manual processes for thinning and final cutting of the trees. Also included are the transport of the workers to the forests for harvesting cork and the transport of the products to the nearest forest road.	Includes the material and energy inputs to the production processes, transports of the material inputs and the available process emissions.
Amount	1	1
Local Name	Kork, Ernte / Durchforstung / Endnutzung, Festkubikmeter	Korkplatte, ab Werk
Synonyms		
General Comment to reference function	The volume of wood produced refers to the wood not including the bark. The multi-output process "cork, harvesting / thinning / final cutting, under bark delivers the two coproducts "wood, cork oak, under bark, u=70%, at forest road" and "raw cork, at forest road". The allocation is based on the overall revenue of wood and cork production.	This cork product is used e.g. as underlay for floating floorings or as insulation material.
Start Date	1996	2001
End Date	2003	2001
Data Valid For Entire Period	1	1
Other Period Text	Cork yield data from 1993	
Geography text	Data for Germany (forestry processes) and Portugal used for Europe	Production data for one major producer in Portugal used for Europe
Technology text	The harvesting of wood considers the motor-manual processes for "Buche, II. Ertragsklasse, starke Durchforstung"	Mix of continuous fabrication and fabrication in moulds and splitting (batch process)
Representativeness [%]		
Production Volume	Unknown	Unknown
Sampling Procedure	Literature	Literature

General Flow information							Representation in Ecoinvent 2000								
Input		Process Name		Output	Allocation	source allocation	Remarks	Cate gory	Sub category	Infra structure	Loca tion	Modul name in Ecoinvent 2000	Mean value	Unit	Source mean value
Cork oak in forest	→	cork, harvesting / thinning / final cutting, under bark													
Stand establishment / tending / site development beech	→						including bark (12%)	wooden materials	extraction	No	RER	hardwood, standing, under bark, in forest	1.00E+00	m3	Frühwald et al. 1996
Diesel burned in building machine	→						under bark	wooden materials	extraction	No	RER	hardwood, stand establishment / tending / site development, under bark	1.00E+00	m3	Schweinle 2000; Werner 2002
Chainsaw carburator without catalytic converter	→							construction processes	machinery	No	GLO	diesel, burned in building machine	4.54E+01	MJ	Schweinle 2000; Werner 2002
Transport	→							construction processes	machinery	No	RER	power sawing, without catalytic converter	1.48E-01	h	Schweinle 2000; Werner 2002
Hardwood, allocation correction	→				100% to cork		workers and equipment	transport systems	road	No	RER	transport, passenger car	5.00E-02	pkm	estimated
Hardwood, allocation correction	→				100% to wood		adds the amount of CO2 uptake, ressource consumption and embodied energy that is lacking due to economic allocation and the fact, that cork is actually bark.	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	9.66E-01	m3	calculated
					100% to cork			wooden materials	extraction	No	RER	hardwood, allocation correction, 2	-4.72E-01	m3	calculated
				wood cork oak, at forest road (u=70%)	Weighted average (year 2000-2003): 54.14 CHF/m3; revenue: 54.14 CHF; revenue process: 1788 CHF; alloc. factor: 0.03	prices from http://cryptomeria.dgf.min-agricultura.pt (accessed 24.04.03)	under bark	wooden materials	extraction	No	RER	wood, cork oak, under bark, u=70%, at forest road	1.00E+00	m3	Source for moisture content: Frühwald et al. 1996; yield: Schweinle 2000, Schweinle 2001; BFS/BUWAL 2000
			Raw cork, at forest road	Weighted average (year 2000-2003): 4.11 CHF/kg; revenue: 1734 CHF; revenue process: 1788 CHF; alloc. factor: 0.97	prices from http://cryptomeria.dgf.min-agricultura.pt (accessed 24.04.03)	u= 5%, Density: 150 kg/m3. Raw cork includes 18.4% clay. The fact that the cork is harvested up to 175 years before the wood is not taken into account in calculating the allocation factor (no discounting)	wooden materials	extraction	No	RER	raw cork, at forest road	4.22E+02	kg	Source for moisture content: Frühwald et al. 1996; yield: Ceuterick 1993	

Fig. 4.1 Flows for "cork, harvesting / thinning / final cutting, under bark" and their representation in the ecoinvent database

Part XVI: Cork

General Flow information						Representation in Ecoinvent 2000								Uncertainty information		
Input	Process Name	Output	Allocation	source allocation	Remarks	Cate gory	Sub category	Infra struc ture	Loca tion	Modul name in Ecoinvent 2000	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Raw cork	cork slab, at plant				Cork content is 0.893 kg. Rest of input is lost as "dust and clay" and burned for energy production	wooden materials	extraction	No	RER	raw cork, at forest road	1.06E+00	kg	Althaus & Richter 2001	1	1.13	(1,3,1,3,1,4,3)
melamine formaldehyde resin, at plant					total glue input is 140 g. Melamine is assumed to be 40%	paintings	production	No	RER	melamine formaldehyde resin, at plant	5.60E-02	kg	Althaus & Richter 2001	1	1.13	(1,3,1,3,1,4,3)
Urea-formaldehyde 100%					total glue input is 140 g. UF is assumed to be 40%	paintings	production	No	RER	urea formaldehyde resin, at plant	5.60E-02	kg	Althaus & Richter 2001	1	1.13	(1,3,1,3,1,4,3)
Phenole-formaldehyde resin 100%					total glue input is 140 g. PF is assumed to be 20%	paintings	production	No	RER	phenolic resin, at plant	2.80E-02	kg	Althaus & Richter 2001	1	1.13	(1,3,1,3,1,4,3)
Residual wood chips from process burned in 300kW furnace					4 to 10 MJ of fuel is needed depending on the product.	wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 300kW	6.00E+00	MJ	Althaus & Richter 2001	1	1.13	(1,3,1,3,1,4,1)
Electricity medium voltage - at grid UCTE					energy consumption (including lighting, administration...) = 0.4 to 1.5 kWh	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.00E+00	kWh	Althaus & Richter 2001	1	1.13	(1,3,1,3,1,4,2)
Transport lorry					Chemicals: 50 km, cork: 320 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	3.45E-01	tkm	Althaus & Richter 2001	1	2.02	(1,3,1,3,1,4,5)
plant					amount calculated to fit land use in source	wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	1.00E-10	unit	area from Althaus & Richter 2001	1	3.02	(1,3,1,3,1,4,9)
		cork slab			Density 150 kg/m3	insulation materials	production	No	RER	cork slab, at plant	1.00E+00	kg				
		Waste heat into air				air	unspecified			Heat, waste	3.60E+00	MJ	calculated	1	1.13	(1,3,1,3,1,4,13)
	Formaldehyde into air			0.1% of agglomeration resin input assumed	air	unspecified			Formaldehyde	1.40E-04	kg	calculated	1	2.06	(4,3,1,3,1,4,23)	
	shreds and dust			0.6 - 0.7 kg: used as fuel --> not inventoried						6.50E-01	kg	Althaus & Richter 2001	1	1.13	(1,3,1,3,1,4,6)	
	sandpaper to landfill				waste management	sanitary landfill	No	CH	disposal, paper, 11.2% water, to sanitary landfill	3.71E-03	kg	Althaus & Richter 2001	1	1.14	(2,3,1,3,1,4,6)	

Fig. 4.2 Flows for "cork slab, at plant" and their representation in the ecoinvent database

5 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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Part XVII

Mortar and Plaster Products and Processes

Data v2.0 (2007)

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Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

This part deals with the production process of different types of mortar and plasters and plaster mixing and with the production process of anhydrite floor and adhesive mortar

The main raw materials for plasters and mortar are binders as acrylate, stucco, anhydrite or cement and aggregates (mainly normal sand or silica sand). In most cases the products are inventoried as a dry bulked and packed mixture of the raw materials. In certain cases (organic cover coat and clay plaster) the product is sold in a moist ready for use condition and thus inventoried.

The plaster mixing process includes only the energy (electricity) for the operation and the equipment.

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1 Introduction

A general description of mortar is given in (Locher & Kropp (1997)): “Mortar is distinguished from concrete by the maximum size of its aggregates. In general, mortar contains only sand with a maximum grain size of 2 – 4 mm. Other binding materials such as hydraulic lime can be used in addition to cement.”

Because of the small maximum size of aggregates, mortars contain higher water to cement ratios than are normally encountered in concrete making. Therefore they exhibit lower compressive strength and volume stability than concrete. These disadvantages can be offset by the use of water-reducing and expansion-producing additives.

“Mortars for masonries are mixtures of sand, binders, water and if needed, additives. Their function is to join the bricks and to guarantee a compensation of possibly unequal geometry between the brick layers.

Plaster is a mixture of one or several binding agents, aggregates with the majority of the grains between 0.25 and 4 mm, water and if needed additives. Plasters are used as coatings on walls and ceilings.

The final quality of the plasters and mortars is reached after hardening on the construction.” (translated by the author from Starzner & Wurmer-Weiss (2000))

2 Reserves and Resources

The resources of the different raw materials as cement, sand, hydraulic and hydrated lime, stucco, anhydrite, etc. are dealt with in the corresponding parts of this report.

3 Use

Cement mortar can be used indoor as well as outdoor. It is suitable for walling, plastering and mending of parts. It is also used to make socles and walls underground or for plastering areas which have to be highly shock-resistant.¹

In concrete constructions, cement mortars are used for connecting pre-cast elements, as injection mortars in ducts of post-tensioned structures with bonded tendons, and for the repair of old concrete surfaces with areas of deterioration (Locher & Kropp (1997))

The composition and strength requirements for masonry mortars are specified in national standards. Lightweight aggregate mortars with reduced thermal conductivity have been developed to improve the thermal insulation properties of a masonry structure. Polymer-modified cement-based mortars with fine sand as the aggregate enable bed joints to have a thickness of only ca. 3 mm (thin-layer mortar). Centrally mixed mortars may contain retarding additives to allow placement of the mortar up to 48 h after mixing.

¹ http://www.quick-mix.de/sortiment/sortiment_profi/287.html, access date: 29.05.03

A variety of polymer-modified mortars designed for repair of concrete structures with deteriorated surface zones is available. Basic requirements of mortar for such applications are the ability to form strong bonds with old concrete and a low shrinkage strain.

A summary of the uses of mortars and plasters is given in Tab. 3.1.

Tab. 3.1 Field of application / specification of studied mortar and plasters (Kasser & Pöll (1998), ²)

Type of mortar or plaster	Field of application / specification
Cement mortar	Normal mortar for masonry constructions
Lime mortar	Base and finishing coat for indoor and outdoor use
Light mortar	Mortar with expanded clay for lightweight masonries
Base plaster	Base and finishing coat, dado, for indoor and outdoor use
Organic cover coat	Finishing coat
Mineral cover coat	Finishing coat for indoor use
Thermal plaster	Plaster for outdoor use
Clay plaster	Ready mixed plaster
Acrylic filler	Knifing filler, for flattening unevenness on base coats, concrete or wood, for indoor and outdoor use
Anhydrite floor	Anhydrite floor can be used in the whole indoor area including damp rooms and heated floor pavement. It is not suitable for damp rooms with gradient and floor drain. Further it is not suitable for areas with medium and high load requirements.
Adhesive mortar	Adhesive mortar is used for laying of tiles, plates, small mosaics, lightweight and high resistance foam slab for indoor and outdoor use. It glues to concrete, plaster, floor pavement, anhydrite floor, masonry, gypsum plaster board, lightweight and autoclaved aerated concrete.

4 Characterisation

Cement mortar is classified depending on the mixing ratio of cement and sand. The studied cement mortar belongs to group III, which means that the ratio of the ingredients is one part by volume of cement and four part of volume of sand with a “natural” humidity (Wendehorst & Vollenschaar (1998)). The technical specifications reported in Tab. 4.1 are based on a description from a German company¹ and (Starzner & Wurmer-Weiss (2000)).

Tab. 4.1 Technical specification of cement mortar

Compressive strength (built-in)	> 10	N/mm ²
Bulk density (built-in)	1'500 – 2'000	kg/m ³
Thermal conductivity (built-in)	1.4	W/(mK)
Grain size	< 1	mm
Water requirement	6 litres par 40 kg	
Processing time	~ 2	hours

Some characteristics of cement mortar after application are: ³

- Good bonding;
- Weathering and frost resistance after hardening;
- High shock resistance.

Some characteristics of anhydrite floor after application are:

Tab. 4.2 Technical specification of anhydrite floor (Starzner and Wurmer-Weiss 2000)

Bulk density (built-in)	450 – 1'200	kg/m ³
Thermal conductivity (built-in)	1.2	W/(mK)
Ratio water to grain	0.15 to 0.23	

5 System Characterisation

5.1 Raw materials

The raw materials used for the production of cement mortar are shown in Tab. 5.1. Other fixing agents can be used instead of cement such as air hardening lime or hydraulic lime. Additives used instead of sand are other natural minerals and light weight aggregates (e.g. exfoliated mica, exfoliated perlite, swelling clay, etc.) (Starzner & Wurmer-Weiss (2000)). The additives are responsible for the quality of the mortar. For this study only cement as fixing agent and no additives other than sand are taken into account.

Tab. 5.1 Composition of cement mortar (Wendehorst & Vollenschaar (1998))

Material	% (v/v)	Bulk density [kg/m ³] (*)	weight [kg]
Slightly humid sand (natural silica sand)	0.8	1'600	0.8
Cement (Portland)	0.2	1'700	0.2
Total	1		1

(*):moisture content about 5%

Plasters (except clay plaster, acrylic filler and organic cover coat) are dry bulked and offered in packed form. The minimum amount of water (chemical bound) needed for mixing the mortar or plaster for a ready to use product is illustrated in Tab. 5.2. This amount of water is calculated stoichiometrically based on the water binding raw materials (cement, hydraulic lime, and stucco).

5.2 Production Process

The production process is in general the same for all plaster and mortars and includes the following process steps (Starzner & Wurmer-Weiss (2000)):

1. Transportation of raw materials to production place;

³ http://www.quick-mix.de/sortiment/sortiment_profi/287.html, access date: 29.05.03

2. Proportioning;
3. Mixing;
4. Conveying;
5. Packing and storing.

5.3 Energy Consumption

The energy requirements are in general the same for all plaster and mortars and include 0.1 MJ direct electricity for mixing one kg cement mortar (Kasser & Pöll (1998)). Not included in the modules for plaster and mortars is the energy of 0.2 MJ per kg sand for drying sand (Kasser & Pöll (1998)), since it is already accounted for in the module “silica sand, at plant”. Also not included is the energy for packing since it is already included in the module “packing, cement products”.

5.4 Transport Requirements

For the cement mortar production the transport distances from the production sites of raw materials to the manufacturer are the same as for the production of sand-lime bricks because the raw materials (sand, quicklime) are similar. Since mining takes place nearby, sand is assumed to be transported 5 km by 28 t lorries. For the transportation of cement a distance of 100 km, 50% by train and 50% by lorry is taken into account (compare part 10, Sand-Lime Brick). For all other plaster and mortars an average transport distance of 20 km for all raw materials from its productions sites to the ready mixing site (either construction site or plant) is assumed.

5.5 Emissions

No direct air and water emissions and no solid waste are taken into account because no data are available and because they are presumably marginal (Starzner & Wurmer-Weiss (2000)).

5.6 Infrastructure

For all modules no additional buildings and land-use have been taken into account as it is assumed that the mixing process takes place in an existing building of the sand/cement industry or on the construction site.

For the mixing process a mixing machine is assumed to weight in total 20 tons; the length of the conveyor belts is assumed 100 m. The service life of these machines is assumed 20 years. The yearly production volume cement mortar is based on the Austrian company “Wietersdorfer und Peggauer Zementwerke GmbH”,⁴ producing about 150'000 tonnes per year.

⁴ http://www.wup.at/umwelt/UE_W2_99.pdf, access date: 30.05.03

Tab. 5.2 Raw materials of the studied mortar and plasters including the minimum amount of water needed for ready-mixed product (Kasser & Pöll (1998))

Type of mortar or plaster	Raw materials (weight kg product)	Minimum amount of water needed for mixing [kg]
Cement mortar	Sand (0.800 kg) and cement, (0.200 kg)	0.080
Lime mortar	Sand (0.290 kg), cement (0.560 kg), lime hydraulic (0.105 kg), lime hydrated (0.043 kg) and additives (0.002 kg)	0.037
Light mortar	Sand (0.187 kg), cement (0.342 kg), lime hydrated (0.026 kg), expanded clay (0.444 kg) and additives (0.001 kg)	0.137
Base plaster	Sand (0.772 kg), cement (0.205 kg), lime hydrated (0.023 kg) and additives (0.0002 kg)	0.082
Organic cover coat	Sand (0.888 kg), organic binder (0.110 kg), solvent (0.001 kg) and additives (0.001 kg)	Not needed
Mineral cover coat	Sand (0.244 kg), stucco (0.743 kg), lime hydrated (0.013 kg) and additives (0.00008 kg)	0.138
Thermal plaster	Sand (0.285 kg), cement (0.523 kg), lime hydraulic (0.107 kg), lime hydrated (0.033 kg), expanded polystyrene (0.048 kg) and additives (0.004 kg)	0.246
Clay plaster	Sand (0.550 kg), clay, (0.250 kg) and water (0.200 kg)	Included in mixture
Acrylic filler	Sand (0.690 kg), acrylic binder, (0.150 kg), water (0.100 kg), pigment (0.020 kg) and ancillary products (0.040 kg)	Included in mixture
Anhydrite floor	Sand (0.732 kg), binding agent (0.260 kg), different reagents (0.006 kg) and an additive (0.002 kg)	0.048
Adhesive mortar	Limestone powder (0.473 kg), cement (0.106 kg), water (0.047 kg), acrylate dispersion (0.310 kg) and diverse other products (0.064 kg)	Included in mixture

6 Life Cycle Inventories

Ecoinvent database meta information (tables), flows and their representation in the ecoinvent database are shown in (figures):

Cement mortar, at plant	Tab. 6.1 and Fig. 6.1
Lime mortar, at plant	Tab. 6.2 and Fig. 6.2
Light mortar, at plant	Tab. 6.3 and Fig. 6.3
Base plaster, at plant	Tab. 6.4 and Fig. 6.4
Cover coat, organic, at plant	Tab. 6.5 and Fig. 6.5
Cover coat, mineral, at plant	Tab. 6.6 and Fig. 6.6
Thermal plaster, at plant	Tab. 6.7 and Fig. 6.7
Clay plaster, at plant	Tab. 6.8 and Fig. 6.8
Acrylic filler, at plant	Tab. 6.9 and Fig. 6.9
Plaster mixing	Tab. 6.10 and Fig. 6.10
Anhydrite floor, at plant	Tab. 6.11 and Fig. 6.11
Adhesive mortar, at plant	Tab. 6.12 and Fig. 6.12

6.1 Cement Mortar

The ecoinvent database meta information of “cement mortar, at plant” (Tab. 6.1) includes all important and necessary information to use the data correctly.

Tab. 6.1 Ecoinvent meta information for the product “cement mortar, at plant”

Name	cement mortar, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce cement mortar (raw material provision, raw material mixing, packing, and storage), transports to plant, and infrastructure. No requirements for administration are included. No additional buildings and land-use have been taken into account as it is assumed that the mixing process takes place in an existing building of the sand/cement industry or on the construction site
Amount	1
Local Name	Zementmörtel, ab Werk
Synonyms	compo
General Comment to reference function	The raw materials are mixed in a certain ratio, transported to the packing installation and packed. No direct process emissions are taken into account as they play a subordinate role compared to the emissions of the raw material production (e.g. drying of silica sand) and because no data are available. For inventorying the infrastructure, a yearly production amount of 150'000 tons of an Austrian producer is assumed. The lifespan of the machines is assumed to be 20 years.
Start Date	1994
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	The amount of raw materials is based on the definition of the group of mortar III (cement mortar). The transportation distances are based on an average of four Swiss companies. For the infrastructure RER-modules are used as proxy
Technology text	Assumed average current technology. The raw materials are mixed in a certain ratio, transported to the packing installation and packed.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature and environmental report
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.1 shows the module “cement mortar, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly Locher & Kropp (1997) and Weibel & Stritz (1995)) and the standard deviation for each data.

6.2 Lime Mortar

The ecoinvent database meta information of “lime mortar, at plant” (Tab. 6.2) includes all important and necessary information to use the data correctly.

Tab. 6.2 Ecoinvent meta information for the product “lime mortar, at plant”

Name	lime mortar, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce lime mortar (raw material provision, raw material mixing, packing, and storage), transports to plant, and infrastructure. No requirements for administration are included. No additional buildings and land-use have been taken into account as it is assumed that the mixing process takes place in an existing building of the sand/cement industry or on the construction site
Amount	1
Local Name	Kalkmörtel, ab Werk
Synonyms	Kalkzementputz
General Comment to reference function	The raw materials per kg powder blend are mixed in following ratio: sand: 0.029kg, cement: 0.56kg lime hydraulic: 0.105kg, lime hydrated: 0.043kg, additives: 0.002kg. After mixing it is internally transported to the packing installation and packed. For the use 0.261 kg water has to be added (not included in this module). No direct process emissions are taken into account as they play a subordinate role compared to the emissions of the raw material production (e.g. drying of silica sand) and because no data are available. For inventorying the infrastructure, a yearly production amount of 150'000 tons is assumed. The lifespan of the machines is assumed to be 20 years.
Start Date	1995
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	The amount of raw materials is based on estimations. The transportation distance is an estimation. For the infrastructure RER-, for the additives and silica sand DE- and for the organic chemicals GLO-modules are used as proxy
Technology text	Assumed average current technology. The raw materials are mixed in a certain ratio, transported to the packing installation and packed.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	
Uncertainty Adjustments	none

Fig. 6.2 shows the module “lime mortar, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly Kasser & Pöll (1998) and estimations) and the standard deviation for each data.

Fig. 6.2 Flows for “lime mortar, at plant” and their representation in the ecoinvent database

6.3 Light Mortar

The ecoinvent database meta information of “light mortar, at plant” (Tab. 6.3) includes all important and necessary information to use the data correctly.

Tab. 6.3 Ecoinvent meta information for the product “light mortar, at plant”

Name	light mortar, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce light mortar (raw material provision, raw material mixing, packing, and storage), transports to plant, and infrastructure. No requirements for administration are included. No additional buildings and land-use have been taken into account as it is assumed that the mixing process takes place in an existing building of the sand/cement industry or on the construction site
Amount	1
Local Name	Leichtmörtel, ab Werk
Synonyms	Blähton//Zuschlag//Lecca
General Comment to reference function	The raw materials per kg powder blend are mixed in following ratio: sand: 0.187kg, cement: 0.205kg, lime hydrated: 0.026kg, expanded clay: 0.444kg, additives: 0.001kg. After mixing it is internally transported to the packing installation and packed. For the use 0.137kg water has to be added (not included in this module). No direct process emissions are taken into account as they play a subordinate role compared to the emissions of the raw material production (e.g. drying of silica sand) and because no data are available. For inventorying the infrastructure, a yearly production amount of 150'000 tons is assumed. The lifespan of the machines is assumed to be 20 years.
Start Date	1995
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	The amount of raw materials is based on estimations. The transportation distance is an estimation. For the infrastructure RER-, for the expanded clay and the silica sand DE- and for the organic chemicals GLO-modules are used as proxy
Technology text	Assumed average current technology. The raw materials are mixed in a certain ratio, transported to the packing installation and packed.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	
Uncertainty Adjustments	none

Fig. 6.3 shows the module “light mortar, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly Kasser & Pöll (1998) and estimations) and the standard deviation for each data.

Fig. 6.3 Flows for “light mortar, at plant” and their representation in the ecoinvent database

6.4 Base Plaster

The ecoinvent database meta information of “base plaster, at plant” (Tab. 6.4) includes all important and necessary information to use the data correctly.

Tab. 6.4 Ecoinvent meta information for the product “base plaster, at plant”

Name	base plaster, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce base plaster (raw material provision, raw material mixing, packing, and storage), transports to plant, and infrastructure. No requirements for administration are included. No additional buildings and land-use have been taken into account as it is assumed that the mixing process takes place in an existing building of the sand/cement industry or on the construction site
Amount	1
Local Name	Grundputz, ab Werk
Synonyms	Zementputz
General Comment to reference function	The raw materials are mixed in following ratio: sand: 0.772kg, cement: 0.205kg, lime hydrated: 0.023kg, additives: 0.0002kg. After mixing it is internally transported to the packing installation and packed. For the use 0.082 kg water has to be added (not included in this module). No direct process emissions are taken into account as they play a subordinate role compared to the emissions of the raw material production (e.g. drying of silica sand) and because no data are available. For inventorying the infrastructure, a yearly production amount of 150'000 tons is assumed. The life-span of the machines is assumed to be 20 years.
Start Date	1995
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	The amount of raw materials is based on estimations. The transportation distance is an estimation. For the infrastructure RER-, for the silica sand DE- and for the organic chemicals GLO-modules are used as proxy
Technology text	Assumed average current technology. The raw materials are mixed in a certain ratio, transported to the packing installation and packed.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	
Uncertainty Adjustments	none

Fig. 6.4 shows the module “base plaster, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly Kasser & Pöll (1998) and estimations) and the standard deviation for each data.

6.5 Organic Cover Coat

The ecoinvent database meta information of “cover coat, organic, at plant” (Tab. 6.5) includes all important and necessary information to use the data correctly.

Tab. 6.5 Ecoinvent meta information for the product “cover coat, organic, at plant”

Name	cover coat, organic, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce organic cover coat (raw material provision, raw material mixing, packing, and storage), transports to plant, and infrastructure. No requirements for administration are included. No additional buildings and land-use have been taken into account as it is assumed that the mixing process takes place in an existing building of the sand/cement industry or on the construction site
Amount	1
Local Name	Deckputz kunststoffmodifiziert, ab Werk
Synonyms	Kunstharzputz
General Comment to reference function	The raw materials per kg ready mix product are mixed in following ratio: sand: 0.888kg, organic binder: 0.110kg, solvents: 0.001kg, additives: 0.001kg. After mixing it is internally transported to the packing installation and packed. No direct process emissions are taken into account as they play a subordinate role compared to the emissions of the raw material production (e.g. drying of silica sand) and because no data are available. For inventorying the infrastructure, a yearly production amount of 150'000 tons is assumed. The lifespan of the machines is assumed to be 20 years.
Start Date	1995
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	The amount of raw materials is based on estimations. The transportation distance is an estimation. For the infrastructure, acrylic binder and white spirit RER-, for the additives DE- and for the organic chemicals GLO-modules are used as proxy
Technology text	Assumed average current technology. The raw materials are mixed in a certain ratio, transported to the packing installation and packed.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	
Uncertainty Adjustments	none

Fig. 6.5 shows the module “cover coat, organic, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly Kasser & Pöll (1998) and estimations) and the standard deviation for each data.

6.6 Mineral Cover Coat

The ecoinvent database meta information of “cover coat, mineral, at plant” (Tab. 6.6) includes all important and necessary information to use the data correctly.

Tab. 6.6 Ecoinvent meta information for the product “cover coat, mineral, at plant”

Name	cover coat, mineral, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce mineral cover coat (raw material provision, raw material mixing, packing, and storage), transports to plant, and infrastructure. No requirements for administration are included. No additional buildings and land-use have been taken into account as it is assumed that the mixing process takes place in an existing building of the sand/cement industry or on the construction site
Amount	1
Local Name	Deckputz mineralisch, ab Werk
Synonyms	Weissputz//Gipsputz
General Comment to reference function	The raw materials per kg powder blend are mixed in following ratio: sand: 0.244kg, plaster of Paris: 0.743kg, lime hydrated: 0.013kg, additives: 0.00008kg. After mixing it is internally transported to the packing installation and packed. For the use 0.138kg water has to be added (not included in this module). No direct process emissions are taken into account as they play a subordinate role compared to the emissions of the raw material production (e.g. drying of silica sand) and because no data are available. For inventorying the infrastructure, a yearly production amount of 150'000 tons is assumed. The lifespan of the machines is assumed to be 20 years.
Start Date	1995
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	The amount of raw materials is based on estimations. The transportation distance is an estimation. For the infrastructure and silica sand RER-, for the silica sand DE- and for the organic chemicals GLO-modules are used as proxy
Technology text	Assumed average current technology. The raw materials are mixed in a certain ratio, transported to the packing installation and packed.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	
Uncertainty Adjustments	none

Fig. 6.6 shows the module “cover coat, mineral, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly Kasser & Pöll (1998) and estimations) and the standard deviation for each data.

6.7 Thermal Plaster

The ecoinvent database meta information of “thermal plaster, at plant” (Tab. 6.7) includes all important and necessary information to use the data correctly.

Tab. 6.7 Ecoinvent meta information for the product “thermal plaster, at plant”

Name	thermal plaster, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce thermal plaster (raw material provision, raw material mixing, packing, and storage), transports to plant, and infrastructure. No requirements for administration are included. No additional buildings and land-use have been taken into account as it is assumed that the mixing process takes place in an existing building of the sand/cement industry or on the construction site
Amount	1
Local Name	Wärmedämmputz, ab Werk
Synonyms	EPS
General Comment to reference unction	The raw materials per kg powder blend are mixed in following ratio: sand: 0.285kg, cement: 0.523kg lime hydraulic: 0.107kg, lime hydrated: 0.033kg, expanded polystyrene: 0.048kg, additives: 0.004kg. After mixing it is internally transported to the packing installation and packed. For the use 0.246 kg water has to be added (not included in this module). No direct process emissions are taken into account as they play a subordinate role compared to the emissions of the raw material production (e.g. drying of silica sand) and because no data are available. For inventorying the infrastructure, a yearly production amount of 150'000 tons is assumed. The lifespan of the machines is assumed to be 20 years.
Start Date	1995
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	The amount of raw materials is based on estimations. The transportation distance is an estimation. For the infrastructure and polystyrene foam slab RER-, for the silica sand DE- and for the organic chemicals GLO-modules are used as proxy
Technology text	Assumed average current technology. The raw materials are mixed in a certain ratio, transported to the packing installation and packed.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	
Uncertainty Adjustments	none

Fig. 6.7 shows the module “thermal plaster, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly Kasser & Pöll (1998) and estimations) and the standard deviation for each data.

6.8 Clay Plaster

The ecoinvent database meta information of “clay plaster, at plant” (Tab. 6.8) includes all important and necessary information to use the data correctly.

Tab. 6.8 Ecoinvent meta information for the product “clay plaster, at plant”

Name	clay plaster, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce clay plaster (raw material provision, raw material mixing, packing, and storage), transports to plant, and infrastructure. No requirements for administration are included. No additional buildings and land-use have been taken into account as it is assumed that the mixing process takes place in an existing building of the sand/cement industry or on the construction site
Amount	1
Local Name	Lehmputz, ab Werk
Synonyms	Lehmmörtel
General Comment to reference function	The raw materials per kg ready mix product are mixed in following ratio: sand: 0.55kg, clay: 0.25kg, water: 0.2kg. After mixing it is internally transported to the packing installation and packed. Can be used without adding water. No direct process emissions are taken into account as they play a subordinate role compared to the emissions of the raw material production (e.g. drying of silica sand) and because no data are available. For inventorying the infrastructure, a yearly production amount of 150'000 tons is assumed. The lifespan of the machines is assumed to be 20 years.
Start Date	1995
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	The amount of raw materials is based on estimations. The transportation distance is an estimation. For the infrastructure RER- and for the silica sand DE-modules are used as proxy
Technology text	Assumed average current technology. The raw materials are mixed in a certain ratio, transported to the packing installation and packed.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	
Uncertainty Adjustments	none

Fig. 6.8 shows the module “clay plaster, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly Kasser & Pöll (1998) and estimations) and the standard deviation for each data.

General Flow information						Representation in ecoinvent 2000								Uncertainty information			
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
dry silica sand	→	Production of clay plaster (for direct use)				raw material, amount assumed	construction materials	additives	No	DE	silica sand, at plant	5.50E-01 kg		estimation	1	5.00	estimation
clay	→					raw material	construction materials	additives	No	CH	clay, at mine	2.50E-01 kg		estimation by Kasser & Pöll (1998)	1	5.00	estimation
water	→					raw material, amount assumed	water supply	production	No	CH	tap water, at user	2.00E-01 kg		estimation	1	5.00	estimation
electricity consumption	→					for mixing, the source does not specify whether the data describes the electricity consumption or the primary energy: it is assumed that it's the electricity consumption	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.78E-02 kWh		estimation by Kasser & Pöll (1998)	1	5.00	estimation
transportation of raw materials to plant by lorry	→					assumed average distance: 20km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	1.60E-02 tkm		estimation	1	5.00	estimation
infrastructure: mixing equipment	→					estimated weight of equipment: 20t, yearly production amount of lime mortar: 150'000t, estimated service life of machine: 20years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	6.67E-06 kg		estimation	1	5.00	estimation
infrastructure: conveyor belt	→					estimated length of conveyor belt: 100m, yearly production amount of organic cover coat: 150'000t, estimated service life of conveyor belt: 20years	construction processes	machinery	Yes	RER	conveyor belt, at plant	3.33E-08 m		estimation	1	5.00	estimation
packing	→					not taken into account as packaging material unknown									1		
	→			waste heat		same amount as electricity consumption	air	unspecified			Heat, waste	1.00E-01 MJ		after electricity requirements	1	5.00	estimation
	→			clay plaster, at plant		dried product	mortar and plaster	production	No	CH	clay plaster, at plant	1.00E+00 kg					

Fig. 6.8 Flows for “clay plaster, at plant” and their representation in the ecoinvent database

6.9 Acrylic filler

Acrylic fillers are the most commonly used plasters or mortars based on organic binders. An aqueous solution of acrylic resin is used as binder and the fillers are for example chalk, lime meal, talcum or quartz. Small amounts of pigments, anti foaming agents, softeners, etc. are usually added. Commonly the share of binder is between 7 and 13%.

Acrylic fillers are mixed at plant and delivered either ready to use or as powders that only need additional water before application.

The exemplary mixture inventoried in this study is given in Starzner & Wurmer-Weiss (2000).

Tab. 6.9 gives the meta information and Fig. 6.9 shows the data inventoried for the acrylic filler.

Tab. 6.9 Ecoinvent meta information for the product “acrylic filler, at plant”

Name	acrylic filler, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Transport of raw materials and production of filler. The requirements of raw materials, the electricity and the infrastructure are inventoried. Packaging is neglected.
Amount	1
Local Name	Acrylatspachtelmasse, ab Werk
Synonyms	
General Comment to reference function	Acrylates are part of a large family of diverse polyesters and acrylic fillers can contain varying amounts of different filler materials. This dataset stands for one specific acrylic filler ready to apply (including water) and should not be used for other acrylic fillers if their contribution to the overall result of an assessment might be relevant. The data quality is not sufficient to allow for e.g. a comparative assessment of different fillers.
Start Date	
End Date	2000
Data Valid For Entire Period	2003
Other Period Text	1
Geography text	
Technology text	
Representativeness [%]	The dataset models a mixing process with its electricity consumption.
Production Volume	
Sampling Procedure	Unknown
Extrapolations	Raw material consumption for a "typical" acrylic filler is chosen. Energy data are taken from a similar process.
Uncertainty Adjustments	None

General Flow information					Representation in ecoinvent								Uncertainty information				
Input		Process Name		Output	Remarks	Cate gory	Sub category	Infra struc ture	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Acrylic binder	→	acrylic filler, at plant			acrylic binder in ecoinvent refers to the acrylate only (without water)	paintings	production	No	RER	acrylic binder, 34% in H2O, at plant	1.50E-01	kg	Starzner & Wurmer-Weiss (2000)	1	1.68	(4,5,5,2,1,5,3)	
Filler	→					construction materials	additives	No	DE	silica sand, at plant	6.90E-01	kg	Starzner & Wurmer-Weiss (2000)	1	1.68	(4,5,5,2,1,5,3)	
Water	→					water supply	production	No	CH	tap water, at user	1.00E-01	kg	Starzner & Wurmer-Weiss (2000)	1	1.68	(4,5,5,2,1,5,3)	
Pigment	→					chemicals	inorganics	No	RER	titanium dioxide, production mix, at plant	2.00E-02	kg	Starzner & Wurmer-Weiss (2000)	1	1.68	(4,5,5,2,1,5,3)	
ancillary materials	→					chemicals	organics	No	GLO	chemicals organic, at plant	4.00E-02	kg	Starzner & Wurmer-Weiss (2000)	1	1.68	(4,5,5,2,1,5,3)	
Transport rail	→					standard distance for raw material transport	transport systems	train	No	RER	transport, freight, rail	1.26E-01	tkm	estimate	1	3.22	(5,5,5,5,5,5,5)
Transport lorry	→					standard distance for raw material transport	transport systems	road	No	RER	transport, lorry >16t, fleet average	5.55E-02	tkm	estimate	1	3.22	(5,5,5,5,5,5,5)
Electricity [kWh/kg]	→					concrete as proxy	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.83E-03	kWh	estimate	1	2.56	(5,5,5,5,5,5,2)
Infrastructure (plant)	→					concrete as proxy	construction materials	others	Yes	CH	concrete mixing plant	1.92E-10	unit	estimate	1	4.25	(5,5,5,5,5,5,9)
	→			waste heat			air	unspecified			Heat, waste	6.59E-03	MJ	calculated	1	2.56	(5,5,5,5,5,5,14)
	→		production waste		no info --> neglected												
	→		Acrylic filler			mortar and plaster	production	No	RER	acrylic filler, at plant	1.00E+00	kg					

Fig. 6.9 Flows for “acrylic filler, at plant” and their representation in the ecoinvent database

6.10 Plaster Mixing

This dataset includes only the electricity and infrastructure needed for the mixing of ready-mixed plaster. The functional unit is one kilogram ready-mixed plaster. Therefore the raw materials as for example cement, hydraulic and/or hydrated lime and water must be added.

The ecoinvent database meta information of “plaster mixing” (Tab. 6.10) includes all important and necessary information to use the data correctly.

Tab. 6.10 Ecoinvent meta information for the product “plaster mixing”

Name	plaster mixing
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes an estimated amount of electricity and infrastructure for the mixing of the plaster/mortar with water at the site of use. No other requirements are included.
Amount	1
Local Name	Putzmischen
Synonyms	
General Comment to reference function	Only the mixing process is taken into account; no mortar and water are included
Start Date	2001
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	
Technology text	Assumed average current technology
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	
Uncertainty Adjustments	none

Fig. 6.10 shows the module “plaster mixing”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly estimation) and the standard deviation for each data.

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
electricity consumption	→	mixing of 1 kg of plaster					electricity	supply mix	No	CH	electricity, medium voltage, at grid	8.00E-02	kWh	estimation	1	10.00	estimation
equipment for mixing	→			estimated life span: 10 years, estimated yearly amount of plaster mixed 100'000 kg, estimated weight of machine: 200 kg	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	2.00E-04	kg	estimation	1	10.00	estimation		
				waste heat	same amount as electricity consumption	air	unspecified			Heat, waste	2.88E-01	MJ	after electricity requirements	1	10.00	estimation	
				mixing of 1 kg of plaster		mortar and plaster	production	No	CH	plaster mixing	1.00E+00	kg					

Fig. 6.10 Flows for “plaster mixing” and their representation in the ecoinvent database

6.11 Anhydrite Floor

The ecoinvent database meta information of “anhydrite floor, at plant” (Tab. 6.10) includes all important and necessary information to use the data correctly.

Tab. 6.11 Ecoinvent meta information for the product “anhydrite floor, at plant”

Name	anhydrite floor, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce anhydrite floor (raw material provision, raw material mixing, packing, and storage), transports to plant, and infrastructure. No requirements for administration are included. No additional buildings and land-use are taken into account as it is assumed that the mixing process takes place in an existing building of the sand/cement industry or on the construction site
Amount	1
Local Name	Anhydritfliessmörtel, ab Werk
Synonyms	Anhydritestrich, Calciumsulfat-Fliessestrich
General Comment to reference function	The raw materials for ready mix product are mixed. After mixing it is internally transported to the packing installation and packed. For the use 0.048 kg water has to be added per kg dry product (not included in this dataset). No direct process emissions are taken into account as they play a subordinate role compared to the emissions of the raw material production (e.g. drying of silica sand) and because no data is available. For inventorying the infrastructure, a yearly production amount of 150'000 tons is assumed. The lifespan of the machines is assumed to be 20 years.
Start Date	2000
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	For the infrastructure RER- and for the silica sand DE-modules are used as proxy
Technology text	Assumed average current technology. The raw materials are mixed in an assumed average ratio, transported to the packing installation and packed.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	
Uncertainty Adjustments	none

Fig. 6.10 shows the module “anhydrite floor, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly (Vogel, Pestalozzi et al. 2003) and estimations) and the standard deviation for each data.

General Flow information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
binding agent	anhydrite floor, at plant		binding agent: assumption: anhydrate mix	construction materials	binder	No	CH	anhydrite, at plant	2.60E-01	kg	Vogel et al. (2003)	1	1.50	estimation
Activator			assumption: potassium sulphate	agricultural means of production	mineral fertiliser	No	RER	potassium chloride, as K ₂ O, at regional storehouse	1.50E-03	kg	Vogel et al. (2003) and estimation	1	5.00	estimation
Retarder			assumption: citric acid (acetic acid used as proxy)	chemicals	organics	No	RER	acetic acid, 98% in H ₂ O, at plant	1.50E-03	kg		1	5.00	estimation
fluxing agent			assumption: melamine sulfonate	paintings	production	No	RER	melamine formaldehyde resin, at plant	1.50E-03	kg		1	5.00	estimation
Stabilizer			assumption: polyethylen oxide	plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	1.50E-03	kg		1	5.00	estimation
additive			additive: assumption: limestone powder	construction materials	additives	No	CH	limestone, milled, loose, at plant	2.00E-03	kg	Vogel et al. (2003)	1	5.00	estimation
aggregate			aggregate 1: assumption: siliceous substances	construction materials	additives	No	DE	silica sand, at plant	7.32E-01	kg	Vogel et al. (2003) and estimation	1	1.50	estimation
electricity consumption				electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.78E-02	kWh	copied from cement mortar production	1	5.00	estimation
transportation of raw materials to plant by lorry			assumed average distance: 20km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	2.00E-02	tkm		1	5.00	estimation
infrastructure: mixing equipment			estimated weight of equipment: 20t, yearly production amount of anhydrite floor: 150'000t, estimated service life of machine: 20years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	6.67E-06	kg		1	5.00	estimation
infrastructure: conveyor belt			estimated length of conveyor belt: 100m, yearly production amount of anhydrite floor: 150'000t, estimated service life of conveyor belt: 20years	construction processes	machinery	Yes	RER	conveyor belt, at plant	3.33E-08	m		1	5.00	estimation
packing			"packing cement products" used as proxy	construction materials	others	No	CH	packing, cement	1.00E+00	kg	estimation	1	5.00	estimation
		→ waste heat	same amount as electricity consumption	air	unspecified			Heat, waste	1.00E-01	MJ	calculated	1	5.00	estimation
		→ anhydrite floor		mortar and plaster	production	No	CH	anhydrite floor, at plant	1.00E+00	kg				

Fig. 6.11 Flows for “anhydrite floor, at plant” and their representation in the ecoinvent database

6.12 Adhesive Mortar

The ecoinvent database meta information of “adhesive mortar, at plant” (Tab. 6.10) includes all important and necessary information to use the data correctly.

Tab. 6.12 Ecoinvent meta information for the product “adhesive mortar, at plant”

Name	adhesive mortar, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce adhesive mortar (raw material provision, raw material mixing, packing, and storage), transports to plant, and infrastructure. No requirements for administration are included. No additional buildings and land-use are taken into account as it is assumed that the mixing process takes place in an existing building of the sand/cement industry or on the construction site
Amount	1
Local Name	Klebemörtel, ab Werk
Synonyms	
General Comment to reference function	The raw materials for the ready mix product are mixed. After mixing it is internally transported to the packing installation and packed. Water is already contained in the product and does not need to be added for the application. No direct process emissions are taken into account as they play a subordinate role compared to the emissions of the raw material production (e.g. drying of silica sand) and because no data is available. For inventorying the infrastructure, a yearly production amount of 150'000 tons is assumed. The lifespan of the machines is assumed to be 20 years.
Start Date	2000
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	For the infrastructure RER- and for the silica sand DE-datasets are used as proxy
Technology text	Assumed average current technology. The raw materials are mixed in an assumed average ratio, transported to the packing installation and packed.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	
Uncertainty Adjustments	none

Fig. 6.10 shows the module “adhesive mortar, at plant”. It shows all in- and output flows, the allocation to the modules in ecoinvent, the sources (mainly ⁵) and the standard deviation for each data.

⁵ ecoinvent-report No. 7
http://www.baustoffchemie.de/richtrezepturen/Flex-Fliesenkleber_2003.html, access date 24.05.2005

General Flow information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
limestone powder	adhesive mortar, at plant		component 1	construction materials	additives	No	CH	limestone, milled, loose, at plant	4.73E-01	kg	http://www.baustoffchemie.de/richtrezepturen/Fliesenkleber_2003.html	1	1.62	(1,1,5,3,3,5);
cement			component 1	construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	1.06E-01	kg		1	1.62	(1,1,5,3,3,5);
epoxy			component 1	plastics	monomers	No	RER	epoxy resin, liquid, disaggregated data, at plant	1.50E-02	kg		1	1.62	(1,1,5,3,3,5);
rapid hardening or aluminous cement			component 1	construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	1.00E-02	kg		1	1.62	(1,1,5,3,3,5);
cellulose ether			component 1	chemicals	organics	No	GLO	chemicals organic, at plant	2.00E-03	kg		1	1.62	(1,1,5,3,3,5);
antifoaming agent			component 1	chemicals	organics	No	GLO	chemicals organic, at plant	3.00E-03	kg		1	1.62	(1,1,5,3,3,5);
lithium carbonate as accelerator			component 1	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	7.00E-03	kg		1	1.62	(1,1,5,3,3,5);
citric acid as restrainer			component 1	chemicals	organics	No	GLO	chemicals organic, at plant	3.00E-03	kg		1	1.62	(1,1,5,3,3,5);
acrylate dispersion			component 2	paintings	production	No	RER	acrylic dispersion, 65% in H2O, at plant	3.10E-01	kg		1	1.62	(1,1,5,3,3,5);
polypropylene glycol alcy l phenyl ethers			component 2	chemicals	organics	No	RER	propylene glycol, liquid, at plant	1.20E-02	kg		1	1.62	(1,1,5,3,3,5);
polyamines addition compound			component 2	chemicals	organics	No	GLO	chemicals organic, at plant	1.20E-02	kg	copied from cement mortar production	1	1.62	(1,1,5,3,3,5);
water			component 2 for mixing, the source does not specify weither the data describes the electricity consumption or the primary energy: it is assumed that it's the electricity consumption	water supply	production	No	CH	tap water, at user	4.70E-02	kg		1	1.62	(1,1,5,3,3,5);
electricity consumption				electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.78E-02	kWh		1	5.00	estimation
transportation of raw materials to plant by lorry			assumed avarage distance: 20km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	1.23E-02	tkm		1	5.00	estimation
infrastructure: mixing equipment			estimated weight of equipment: 20t, yearly production amount of anhydrite floor: 150'000t, estimated service life of machine: 20years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	6.67E-06	kg		1	5.00	estimation
infrastructure: conveyor belt			estimated length of conveyor belt: 100m, yearly production amount of anhydrite floor: 150'000t, estimated service life of conveyor belt: 20years	construction processes	machinery	Yes	RER	conveyor belt, at plant	3.33E-08	m		1	5.00	estimation
packing			"packing cement products" used as proxy	construction materials	others	No	CH	packing, cement	1.00E+00	kg	estimation	1	5.00	estimation
		→ waste heat	same amount as electricity consumption	air	unspecified			Heat, waste	1.00E-01	MJ	calculated	1	5.00	estimation
		→ adhesive mortar		mortar and plaster	production	No	CH	adhesive mortar, at plant	1.00E+00	kg				

Fig. 6.12 Flows for “adhesive mortar, at plant” and their representation in the ecoinvent database

7 Data quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

The data on raw material consumption for all studied mortars and plasters except cement mortar and acrylic filler are taken from Kasser & Pöll (1998). The authors of this study refer to own estimations of the amount of direct raw materials. No further information on characterisation and use of the products is offered in that reference.

The transport distances are taken directly from the module “sand-lime brick, at plant”. It is assumed that the transport distances of the raw material quicklime are the same as for cement (in the module “cement mortar”).

The data for the heating energy to dry sand and for the electricity consumption for mixing are based on Kasser & Pöll (1998), who do not describe the system boundary. Therefore the uncertainty concerning these inputs is relatively high.

No direct emissions from the processing are taken into account as no data are available.

The packing module for cement that is used as proxy for the packing of all different dry bulk plasters and mortar can be considered to be sufficiently appropriate.

For all the above reasons the data for cement mortar are of fair overall quality and the data for all other mortars and plasters are of poor overall quality.

These modules should not be used for direct comparison among each others or with other materials. In a broader context and if the importance of these products for an inventory is not too high, the data can be used.

8 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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- | | |
|--|---|
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Part XVIII

Additional Products and Processes

Data v2.0 (2007)

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Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

This part deals with different machines and processes, different types of buildings, the EUR-flat pallet and explosives including the blasting process. Most data (except operation of building machines) are rough estimations and therefore it's not advisable to use them if they are of high importance for a specific inventory.

The machines inventoried in this study are a hydraulic digger and a general building machine. For these infrastructure modules a corresponding operation module is included. The functional unit for the excavation with a hydraulic digger is one m³ of dislocated material. The functional unit for unspecified work with a building machine is one MJ of diesel consumed. Additionally the excavation with a skid-steer loader based on the general building machine is taken into account.

For inventorying unspecified machines with low importance within a certain context a general module "industrial machine, heavy, unspecified, at plant" is offered. The functional unit is one kilogram.

The process of rock crushing is inventoried on the basis of a certain type of rock crusher. The functional unit is one kilogram of crushed rock. No dust emission from rock itself is taken into account.

The products power saw are offered for two types of two-stroke cycle engine power chain saws ones with and ones without catalytic converter. Besides the converter the same material composition has been used. To each type the operation phase is inventoried. These sawing processes are not comparable as the measured emissions are based on quite different types of power saws.

The module conveyor belt is very rough inventoried and the functional unit is one meter.

Three types of buildings are included in this study: multi-storey building, a steel and a wooden construction hall. In case no information about the construction type is available also a unspecified hall based on a assumed share of 30% wood and 70% steel construction hall is offered. Both building hall modules are theoretical modules because the data are based on a theoretical building with a assumed geometry and service-life of 50 years.

The data for the explosive module are based on a specific type of explosive called "Tovex". The appropriate blasting process includes emissions which are based on an assumed and possible chemical reaction based on the composition.

The data concerning the auxiliary module "EUR-flat pallet" include only the material inputs wood (two types) and nails but no energy and waste.

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1 Introduction

This part deals with different machines and processes, different types of buildings, the EUR-flat Pallet and explosives including the blasting process.

2 Machines and Processes

2.1 Introduction

This chapter deals with building machines (skid-steer loader, hydraulic digger and building machine in general), mining equipment (rock crusher and crushing), logging equipment (motor saw and sawing, with and without catalytic converter) and transportation equipment (conveyor belt)

We thank Dr. Bernard Couson from the company Electrowatt Infra AG in Zurich, Switzerland for his help with emissions of building machines.

2.2 Skid-Steer Loader, Hydraulic Digger, Building Machine

2.2.1 Use and Application

Out of a great variety of building machines two different kind of crawler-type vehicles for earthwork are chosen. The emissions and energy consumption of these two types are based on an average excavation capacity. For all the other building machines with diesel engine a general module called “building machine” is defined. For this type the emissions and the energy consumption are based on the diesel consumption.

All these modules are approximate but good enough to serve as “machine requirements” for other modules. However they should not be used for direct comparison of different types of building machines.

Tab. 2.1 Types of building machines in ecoinvent after (Frischknecht et al. (1996))

Building machine	Name of module	Average weight	Average excavation capacity	Average service-life
	Skid-steer loader	10 t	110 m ³ /h (*)	10'000 h
	Hydraulic digger	15 t	100 m ³ /h (*)	10'000 h
	Building machine	10 t	17.4 kg diesel / h (*)	10'000 h

(*): Based on BUWAL (2000b)

2.2.2 System Characterisation of Manufacturing

The material and energy uses for the machines are copied from Frischknecht et al. (1996):

Lorry and cars are made to about 70 % of steel. The weight of steel in building machines can be presumably extrapolated to 90% of the total. However this study assumes conservatively 100% of steel.

The direct energy consumption for the manufacturing of a car is about 20 MJ of primary energy per kg of car weight. The same amount is used here for all different building machines. It is assumed that half of this energy is provided by natural gas and half by electricity.

For transportation of the steel a distance of 600 km by freight train and 50 km by 40t lorry are assumed.

2.2.3 System Characterisation of Operation

System Boundary

The modules describing the operation of the considered three types of building machines include the machine itself, energy requirements and direct emissions. However the infrastructure of the machine producing factories and the maintenance of the machines are not considered.

Fuel and Lubricating Oil Consumption and Emissions

The emissions in (Frischknecht et al. (1996)) have been reworked based on the “Off-road Database” (BUWAL (2000b)) and current information from the project leader¹. The emissions can be calculated from this database for certain machines as well as for groups of vehicles for a certain year or in certain time progression. The emission factors and the coefficient of correction for the main emissions (HC, CO, PM and NO_x) are based on the information reported in Appendix 9. The emission factors for the other emission species are taken from (BUWAL (2000b)). The assumptions made to calculate the emissions are shown in Tab. 2.2:

¹ Internal communication 26.02.1999, between Mr. A. Mayer, TTM, 5443 Niederrohrdorf and Mr. B. Couson, Electrowatt Infra AG, 8037 Zürich and personal communication 29.04.2003, Mr. B. Couson, Electrowatt Infra AG, 8037 Zürich

Tab. 2.2 Characteristic data of different types of building machines (BUWAL (2000b))

	Skid-steer loader		Hydraulic digger		Building machine	
Type in "Off-road Database"	"Raupenlader" = traxcavator		"Hydraulik-Bagger" = Hydraulic digger		Average over different types of building machines excluding road building machines	
Function of machine	Raising		Excavation, digging		All different functions as rollers, hydraulic equipment, twist driller, etc.	
Average mechanical power per single machine (with 100% work-load)	110	kW	100	kW	120	kW
Average mechanical energy per single machine (with 50% work-load)	55	kWh/h	50	kWh/h	60	kWh/h
Average mechanical energy per load or time	0.5	kWh/m ³	0.5	kWh/m ³	60	kWh/h
Average specific time need (*)	9.09	h / 1000m ³	10	h / 1000m ³		
Diesel consumption used for all types:	290 g Diesel / kWh					
Diesel consumption rate (1990)	145	g / m ³	145	g / m ³	17.4	kg / h
Coefficient of correction for diesel consumption from 1990 to 2000	0.9		0.9		0.9	
Diesel consumption per unit (2000) used in this study	0.013	kg / m ³	0.013	kg / m ³	0.023	kg / MJ
Lubricating oil consumption (**)	0.0025	kg / m ³	0.003	kg / m ³	0.0005	kg / MJ

(*): Quotient of the average mechanical energy per single machine (with 50% work-load) and average mechanical energy per load

(**): Frischknecht et al. (1996)

2.2.4 Modules in Ecoinvent

Building Machine in Ecoinvent

The ecoinvent database meta information for "building machine" and "diesel, burned in building machine" (Tab. 2.3) include all important and necessary information to use the data correctly.

Tab. 2.3 Ecoinvent database meta information for the product “building machine” and the process “diesel, burned in building machine”

Name	Building machine	Diesel, burned in building machine
Location	RER	GLO
Infrastructure Process	1	0
Unit	unit	MJ
Data Set Version	2.0	2.0
Included Processes	Includes the most important materials, the transportation of the parts to the assembly plant and of the building machine to the end-user. Also included is the requirement of electricity and heating energy. No disposal is included as it is assumed that all materials used in the machine are 100% recycled.	Includes the inputs “building machine” for infrastructure, lubricating oil and fuel consumption, and some measured air emissions as output.
Amount	1	1
Local Name	Baumaschine	Diesel, in Baumaschine
Synonyms	construction//engine	construction//engine
General Comment to reference function	The module describes a fictitious machine made 100% of steel. The energy consumption for the production process is assumed to be the same as the energy consumption for the production of a car. The transportation distances for delivering the parts to the assembly plant are estimated. This module should not be used if its relative importance would be high in a certain environmental inventory. Average service life: 10'000 hours.	This module is based on the data included in the report Frischknecht et al. (1996). The diesel consumption and the emissions are updated using the Swiss “Off-road database” and applied to year 2000. This module should not be used if its relative importance would be high in a certain environmental inventory.
Start Date	1996	1996
End Date	2001	2001
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Certain elementary flows or intermediate product flows are extrapolated from Swiss conditions.	Certain elementary flows or intermediate product flows are extrapolated from RER and CH conditions.
Technology text	Average current technology for one typical machine representing the category.	Average current technology for one typical machine representing the category.
Representativeness [%]		
Production Volume	Unknown	Unknown
Sampling Procedure	Most data from the LCA study Frischknecht et al. (1996).	Some data from the LCA study Frischknecht et al. (1996) and emissions from the Swiss “Off-road” database.
Extrapolations	See geography	See geography
Uncertainty Adjustments	None	None

Fig. 2.1 and Fig. 2.2 show all in- and output flows of the infrastructure “building machine“ and the process “diesel, burned in building machine”, the allocation to the modules in ecoinvent, the sources (mainly Frischknecht et al. (1996) and BUWAL (2000b)) and the standard deviation for each value.

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
steel low-alloyed	→	building machine					metals	extraction	No	RER	steel, low-alloyed, at plant	3.00E+03 kg		Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);
steel unalloyed	→					metals	extraction	No	RER	reinforcing steel, at plant	7.00E+03 kg		Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);	
electricity consumption for production of building machine	→					electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	9.17E+03 kWh		Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);	
natural gas used within production of building machine	→					natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	1.00E+05 MJ		Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);	
transport lorry 40t for transportation of steel for building machine	→					transport systems	road	No	CH	transport, lorry >28t, fleet average	1.00E+03 tkm		Frischknecht et al. (1996)	1	2.05	(1,5,2,3,1,3);	
transport train for transportation of steel for building machine	→					transport systems	train	No	RER	transport, freight, rail	2.00E+03 tkm		Frischknecht et al. (1996)	1	2.05	(1,5,2,3,1,3);	
transportation of building machine to end-user by lorry	→					distance assumption for Switzerland: 50km	transport systems	road	No	CH	transport, lorry 28t	3.50E-01 tkm		assumption EMPA	1	2.43	(4,5,5,3,5);
						Is not taken into account as it's assumed that all parts will be reused to 100%						7.00E+00 kg					
							air	unspecified			Heat, waste	3.30E+04 MJ		Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);
							construction processes	machinery	Yes	RER	building machine	1.00E+00 unit					
															</		

Fig. 2.1 Flows for “building machine” and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
infrastructure building machine	diesel, burned in building machine (part 1)		average life-span: 10'000h, average diesel used for excavation: 17.4 kg diesel per hour; net calorific value: 42.8 MJ/kg	construction processes	machinery	Yes	RER	building machine	1.34E-07	unit	Frischknecht et al. (1996)	1	3.05	(1,5,2,3,1,3);
lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	5.14E-04	kg	Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);
diesel consumption				oil	fuels	No	RER	diesel, at regional storage	2.34E-02	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.22	(1,5,1,3,1,3);
		emission to air: NOx		air	unspecified			Nitrogen oxides	1.03E-03	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		emission to air: CO		air	unspecified			Carbon monoxide, fossil	2.66E-04	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	5.06	(1,5,1,3,1,3);
		emission to air: PM<2.5um		air	unspecified			Particulates, < 2.5 um	1.05E-04	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	3.05	(1,5,1,3,1,3);
		emission to air: CO2		air	unspecified			Carbon dioxide, fossil	7.30E-02	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.22	(1,5,1,3,1,3);
		emission to air: SO2		air	unspecified			Sulfur dioxide	2.36E-05	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.22	(1,5,1,3,1,3);
		emission to air: CH4		air	unspecified			Methane, fossil	3.75E-06	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		emission to air: NMVOC	From the total indicated the PAH modules and benzo(a)pyrene are deducted	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	1.21E-04	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		emission to air: N2O		air	unspecified			Dinitrogen monoxide	2.80E-06	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		emission to air: NH3		air	unspecified			Ammonia	4.67E-07	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		emission to air: Dioxine		air	unspecified			Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin	1.40E-15	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		emission to air: Benz(a)anthracene (C18H12)	PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	1.87E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	3.05	(1,5,1,3,1,3);
		emission to air: Benzo(b)fluoranthracene	PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	1.17E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	3.05	(1,5,1,3,1,3);
		emission to air: Dibenzo(a,h)anthracene	PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	1.89E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	3.05	(1,5,1,3,1,3);

Fig. 2.2 Flows for “diesel, burned in building machine” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
diesel, burned in building machine (part 2)	→	emission to air: Benzo(a)pyrene				air	unspecified			Benzo(a)pyrene	7.01E-10	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	3.05	(1,5,1,3,1,3);
	→	emission to air: Chrysene			PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	4.67E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	3.05	(1,5,1,3,1,3);
	→	emission to air: Fluoranthracene			PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	1.05E-08	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	3.05	(1,5,1,3,1,3);
	→	emission to air: Phenanthrene			PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	5.84E-08	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	3.05	(1,5,1,3,1,3);
	→	emission to air: Cd				air	unspecified			Cadmium	2.34E-10	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	5.06	(1,5,1,3,1,3);
	→	emission to air: Cu				air	unspecified			Copper	3.97E-08	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	5.06	(1,5,1,3,1,3);
	→	emission to air: Cr				air	unspecified			Chromium	1.17E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	5.06	(1,5,1,3,1,3);
	→	emission to air: Ni				air	unspecified			Nickel	1.64E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	5.06	(1,5,1,3,1,3);
	→	emission to air: Se				air	unspecified			Selenium	2.34E-10	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	5.06	(1,5,1,3,1,3);
	→	emission to air: Zn				air	unspecified			Zinc	2.34E-08	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	5.06	(1,5,1,3,1,3);
	→	waste heat				air	unspecified			Heat, waste	1.06E+00	MJ	Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);
	→	waste oil			as no data available, total of oil is disposed; no losses are assumed	waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	5.14E-04	kg	Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);
	→	diesel in building machine				construction processes	machinery	No	GLO	diesel, burned in building machine	1.00E+00	MJ				

Fig. 2.2 Flows for “diesel, burned in building machine” and their representation in the ecoinvent database (cont.)

Hydraulic digger in Ecoinvent

The ecoinvent database meta information for “hydraulic digger” and “excavation, hydraulic digger” (Tab. 2.4) include all important and necessary information to use the data correctly.

Tab. 2.4 Ecoinvent database meta information for the product “hydraulic digger” and the process “excavation, hydraulic digger”

Name	Hydraulic digger	Excavation, hydraulic digger
Location	RER	RER
Infrastructure Process	1	0
Unit	unit	m3
Data Set Version	2.0	2.0
Included Processes	Includes the most important materials, the transportation of the parts to the assembly plant and of the hydraulic digger to the end-user. Also included is the requirement of electricity and heating energy. No disposal is included as it is assumed that all materials used in the machine are 100% recycled.	Includes the inputs "hydraulic digger" for infrastructure, lubricating oil and fuel consumption, and some measured air emissions as output.
Amount	1	1
Local Name	Hydraulikbagger	Aushub Hydraulikbagger
Synonyms	excavator//engine	excavator//engine
General Comment to reference function	The module describes a fictitious machine made 100% of steel. The energy consumption for the production process is assumed to be the same as the energy consumption for the production of a car. The transportation distances for delivering the parts to the assembly plant are estimated. This module should not be used if its relative importance would be high in a certain environmental inventory. Average service life: 10'000 hours.	This module is based on the data included in the report "Ökoinventare von Energiesysteme" (Frischknecht et al. (1996)). The diesel consumption and the emissions are updated using the Swiss "Off-road database" and applied to year 2000 (BUWAL (2000b)). This module should not be used if its relative importance would be high in a certain environmental inventory.
Start Date	1996	1996
End Date	2001	2001
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Certain elementary flows or intermediate product flows are extrapolated from Swiss conditions.	Certain elementary flows or intermediate product flows are extrapolated from Swiss conditions.
Technology text	Average current technology for one typical machine representing the category.	Average current technology for one typical machine representing the category.
Representativeness [%]		
Production Volume	Unknown	Unknown
Sampling Procedure	Most data from the LCA study Frischknecht et al. (1996) (Frischknecht et al. (1996)).	Some data from the LCA study Frischknecht et al. (1996) and emissions from the Swiss "Off-road" database (BUWAL (2000b)).
Extrapolations	See geography	See geography
Uncertainty Adjustments	None	None

Fig. 2.3 and Fig. 2.4 show all in- and output flows of the infrastructure “hydraulic digger“, and the process “excavation, hydraulic digger”, the allocation to the modules in ecoinvent, the sources (mainly Frischknecht et al. (1996) and BUWAL (2000b)) and the standard deviation for each value.

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
steel low-alloyed	→	hydraulic digger					metals	extraction	No	RER	steel, low-alloyed, at plant	4.50E+03 kg		Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);
steel unalloyed	→					metals	extraction	No	RER	reinforcing steel, at plant	1.05E+04 kg		Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);	
electricity consumption for production of building machine	→					electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.39E+04 kWh		Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);	
natural gas used within production of building machine	→					natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	1.50E+05 MJ		Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);	
transport lorry 40t for transportation of steel for building machine	→					transport systems	road	No	CH	transport, lorry >28t, fleet average	1.50E+03 tkm		Frischknecht et al. (1996)	1	2.05	(1,5,2,3,1,3);	
transport train for transportation of steel for building machine	→					transport systems	train	No	RER	transport, freight, rail	3.00E+03 tkm		Frischknecht et al. (1996)	1	2.05	(1,5,2,3,1,3);	
transportation of hydraulic digger to end-user by lorry	→					distance assumption for Switzerland: 50km	transport systems	road	No	CH	transport, lorry 28t	3.50E-01 tkm		assumption EMPA	1	2.43	(4,5,5,5,3,5);
			→	disposal of hydraulic digger		Is not taken into account as it's assumed that all parts will be reused to 100%						7.00E+00 kg					
			→	waste heat			air	unspecified			Heat, waste	5.00E+04 MJ		Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);
			→	hydraulic digger			construction processes	civil engineering	Yes	RER	hydraulic digger	1.00E+00 unit					

Fig. 2.3 Flows for “hydraulic digger” and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
infrastructure hydraulic digger	excavation hydraulic digger (part 1)		average life-span: 10'000h, average excavation capacity: 100 m ³ /h	construction processes	civil engineering	Yes	RER	hydraulic digger	1.00E-06	unit	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.22	(1,5,2,3,1,3);
lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	2.50E-03	kg	Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);
diesel				oil	fuels	No	RER	diesel, at regional storage	1.31E-01	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.22	(1,5,1,3,1,3);
		→ emission to air: NOx		air	unspecified			Nitrogen oxides	5.76E-03	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		→ emission to air: CO		air	unspecified			Carbon monoxide, fossil	1.49E-03	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		→ emission to air: PM<2.5um		air	unspecified			Particulates, < 2.5 um	5.85E-04	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		→ emission to air: CO2		air	unspecified			Carbon dioxide, fossil	4.08E-01	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		→ emission to air: SO2		air	unspecified			Sulfur dioxide	1.32E-04	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		→ emission to air: CH4		air	unspecified			Methane, fossil	2.09E-05	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		→ emission to air: NMVOC	From the total indicated the PAH modules and benzo(a)pyrene are deducted	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	6.76E-04	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		→ emission to air: N2O		air	unspecified			Dinitrogen monoxide	1.57E-05	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		→ emission to air: NH3		air	unspecified			Ammonia	2.61E-06	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		→ emission to air: Dioxine		air	unspecified			Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin	7.84E-15	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		→ emission to air: Benz(a)anthracene (C18H12)	PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	1.04E-08	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		→ emission to air: Benzo(b)fluoranthracene	PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	6.53E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
		→ emission to air: Dibenzo(a,h)anthracene	PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	1.06E-08	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);

Fig. 2.4 Flows for “excavation, hydraulic digger” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
excavation hydraulic digger (part 2)	→		emission to air: Benzo(a)pyrene			air	unspecified			Benzo(a)pyrene	3.92E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Chrysene		PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	2.61E-08	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Fluoranthracene		PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	5.87E-08	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Phenanthrene		PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	3.26E-07	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Cd			air	unspecified			Cadmium	1.31E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Cu			air	unspecified			Copper	2.22E-07	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Cr			air	unspecified			Chromium	6.53E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Ni			air	unspecified			Nickel	9.14E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Se			air	unspecified			Selenium	1.31E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Zn			air	unspecified			Zinc	1.31E-07	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		waste heat			air	unspecified			Heat, waste	5.20E+00	MJ	Frischknecht et al. (1996)	1	1.57	(1,5,2,3,1,3);
	→		waste oil		as no data available, total of oil is disposed; no losses are assumed	waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	2.50E-03	kg	Frischknecht et al. (1996)	1	1.57	(1,5,2,3,1,3);
	→		excavation, hydraulic digger			construction processes	civil engineering	No	RER	excavation, hydraulic digger	1.00E+00	m3				

Fig. 2.4 Flows for “excavation, hydraulic digger” and their representation in the ecoinvent database (cont.)

Skid-Steer Loader in Ecoinvent

The machine “skid-steer loader” is not included in this study because no data are available. Therefore the infrastructure “building machine” has been used as proxy.

The ecoinvent database meta information for “skid-steer loader” (Tab. 2.5) includes all important and necessary information to use the data correctly.

Tab. 2.5 Ecoinvent database meta information for the product skid-steer loader”

Name	Excavation, skid-steer loader
Location	RER
Infrastructure Process	0
Unit	m3
Data Set Version	2.0
Included Processes	Includes the inputs "building machine" for infrastructure, lubricating oil and fuel consumption, and some measured air emissions as output.
Amount	1
Local Name	Aushub Frontladerraupe
Synonyms	loading//shovel//loader//front-end//engine
General Comment to reference function	This module is based on the data included in the report Frischknecht et al. (1996). The diesel consumption and the emissions are updated using the Swiss "Off-road database" and applied to year 2000. This module should not be used if its relative importance would be high in a certain environmental inventory.
Start Date	1996
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Certain elementary flows or intermediate product flows are extrapolated from Swiss conditions.
Technology text	Average current technology for one typical machine representing the category.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Some data from the LCA study Frischknecht et al. (1996) and emissions from the Swiss "Off-road" database.
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 2.5 shows all in- and output flows of the process “excavation, skid-steer loader”, the allocation to the modules in ecoinvent database, the sources (mainly Frischknecht et al. (1996) and BUWAL (2000b)) and the standard deviation for each data.

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
infrastructure skid steer loader	→	excavation skid steer loader (part 1)				"building machine" used as proxy, average life-span: 10'000h, average excavation capacity: 110 m3/h	construction processes	machinery	Yes	RER	building machine	9.09E-07	unit	Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3)
lubricating oil	→						chemicals	organics	No	RER	lubricating oil, at plant	3.00E-03	kg	Frischknecht et al. (1996)	1	1.22	(1,5,2,3,1,3);
diesel	→						oil	fuels	No	RER	diesel, at regional storage	1.30E-01	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.22	(1,5,1,3,1,3);
			→	emission to air: NOx			air	unspecified			Nitrogen oxides	5.76E-03	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
			→	emission to air: CO			air	unspecified			Carbon monoxide, fossil	1.48E-03	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
			→	emission to air: PM<2.5um			air	unspecified			Particulates, < 2.5 um	5.85E-04	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
			→	emission to air: CO2			air	unspecified			Carbon dioxide, fossil	4.08E-01	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
			→	emission to air: SO2			air	unspecified			Sulfur dioxide	1.32E-04	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
			→	emission to air: CH4			air	unspecified			Methane, fossil	2.09E-05	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
			→	emission to air: NMVOC			From the total indicated the PAH modules and benzo(a)pyrene are deducted	air	unspecified		NMVOC, non-methane volatile organic compounds, unspecified origin	6.76E-04	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
			→	emission to air: N2O				air	unspecified		Dinitrogen monoxide	1.57E-05	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
			→	emission to air: NH3				air	unspecified		Ammonia	2.61E-06	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
			→	emission to air: Dioxine				air	unspecified		Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin	7.84E-15	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
			→	emission to air: Benz(a)anthracene (C18H12)			PAH module used as proxy	air	unspecified		PAH, polycyclic aromatic hydrocarbons	1.04E-08	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
			→	emission to air: Benzo(b)fluoranthracene			PAH module used as proxy	air	unspecified		PAH, polycyclic aromatic hydrocarbons	6.52E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
			→	emission to air: Dibenzo(a,h)anthracene			PAH module used as proxy	air	unspecified		PAH, polycyclic aromatic hydrocarbons	1.06E-08	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);

Fig. 2.5 Flows for “excavation, skid-steer loader” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
excavation skid steer loader (part 2)	→		emission to air: Benzo(a)pyrene			air	unspecified			Benzo(a)pyrene	3.91E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Chrysene		PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	2.61E-08	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Fluoranthracene		PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	5.87E-08	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Phenanthrene		PAH module used as proxy	air	unspecified			PAH, polycyclic aromatic hydrocarbons	3.26E-07	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Cd			air	unspecified			Cadmium	1.30E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Cu			air	unspecified			Copper	2.22E-07	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Cr			air	unspecified			Chromium	6.52E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Ni			air	unspecified			Nickel	9.13E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Se			air	unspecified			Selenium	1.30E-09	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		emission to air: Zn			air	unspecified			Zinc	1.30E-07	kg	BUWAL (2000), Handbuch: Offroad-Datenbank	1	1.56	(1,5,1,3,1,3);
	→		waste heat			air	unspecified			Heat, waste	6.00E+00	MJ	Frischknecht et al. (1996)	1	1.57	(1,5,2,3,1,3);
	→		waste oil		as no data available, total of oil is disposed; no losses are assumed	waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	3.00E-03	kg	Frischknecht et al. (1996)	1	1.57	(1,5,2,3,1,3);
	→		excavation, skid steer loader			construction processes	civil engineering	No	RER	excavation, skid-steer loader	1.00E+00	m3				

Fig. 2.5 Flows for “excavation, skid-steer loader” and their representation in the ecoinvent database (cont.)

2.2.5 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

The data of the modules for infrastructure include a rough estimation of the production energy/material requirements. The overall quality is poor and these modules should not or only with great care be used if they were important contributors in a specific case.

The data for the operation phase include detailed fuel consumption and emissions from an up-to-date and comprehensive study (BUWAL (2000b)) and therefore the overall quality is good.

2.3 Heavy Industrial Machine

2.3.1 Use, Application and Characterisation

The materials used for the construction of a heavy industrial machine are based on the rock crusher Nordberg HP400 SX. It is assumed that this material composition can represent all different kinds of heavy industrial machines. Therefore from this specific rock crusher an undefined heavy industrial machine has been derived. Ninety-six percent of the mass of the materials in the crusher was inventoried, with the remaining 4% of the components in the rock crusher making up miscellaneous metal items (such as nuts and bolts). The materials contained in the rock crusher, along with their contribution to the total mass, are provided in Tab. 2.6. The manufacturing and assembly of the crusher are not included in the module.

Tab. 2.6 HP400 SX rock crusher materials (Landfield & Karra (2000))

Material	Weight [kg]	% of total weight
Steel	20684	90.5
Iron	1733	7.6
Bronze	338	1.5
Epoxy resin	80	0.35
Aluminium	17	0.07
Brass	0.64	0.003

2.3.2 System Boundary

The module “industrial machine, heavy, unspecified, at plant” includes the materials listed in Tab. 2.6, the transportation of the materials to the assembly plant and after end-of-life to the secondary metal producer. It is assumed that all materials are reused and therefore no burdens for disposal are accounted for. No spares and wears are included as they are part of the operation phase.

2.3.3 Heavy Industrial Machine in Ecoinvent

The ecoinvent database meta information of “industrial machine, heavy, unspecified, at plant” (Tab. 2.7) includes all important and necessary information to use the data correctly.

The functional unit is 1 kg of machine!

Tab. 2.7 Ecoinvent database meta information for the product “industrial machine, heavy, unspecified, at plant”

Name	industrial machine, heavy, unspecified, at plant
Location	RER
Infrastructure Process	1
Unit	kg
Data Set Version	2.0
Included Processes	Includes the most important materials and the transportation of the new parts to the assembly plant. No disposal is included as it is assumed that all materials are recycled. No energy for assembling is included.
Amount	1
Local Name	Industriemaschine, schwer, unspezifisch, ab Werk
Synonyms	
General Comment to reference function	This infrastructure is based on the materialization of a rock crusher. The transportation distances are estimated and the transportation requirements added. This module should not be used if its relative importance would be high in a certain environmental inventory. Service life declared by the producer: 25 years. The functional unit is one kilogram.
Start Date	1999
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Certain elementary flows or intermediate product flows are extrapolated from Swiss conditions.
Technology text	Rock crusher type Nordberg HP400 SX produced in the USA in 1999, used as basis
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	most data from LCA study
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 2.6 shows all in- and output flows of the product “industrial machine, heavy, unspecified, at plant”, their allocation to the modules in ecoinvent, the sources (mainly Landfield & Karra (2000)) and the standard deviation for each value.

General Flow information					Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
low-alloyed steel	→	heavy industrial machine			Amount of material based on composition of rock crusher Nordberg HP400 SX	metals	extraction	No	RER	steel, low-alloyed, at plant	9.05E-01	kg	Landfield & Karra (2000)	1	1.38	(1,5,1,3,3,5);
processing of steel	→				Amount of steel processed based on composition of rock crusher Nordberg HP400 SX	metals	processing	No	RER	section bar rolling, steel	9.05E-01	kg	Landfield & Karra (2000)	1	1.38	(1,5,1,3,3,5);
iron	→				Amount of material based on composition of rock crusher Nordberg HP400 SX	metals	extraction	No	RER	cast iron, at plant	7.58E-02	kg	Landfield & Karra (2000)	1	1.38	(1,5,1,3,3,5);
bronze	→				Amount of material based on composition of rock crusher Nordberg HP400 SX	metals	extraction	No	CH	bronze, at plant	1.48E-02	kg	Landfield & Karra (2000)	1	1.38	(1,5,1,3,3,5);
epoxy resin	→				Amount of material based on composition of rock crusher Nordberg HP400 SX	plastics	monomers	No	RER	epoxy resin, liquid, at plant	3.50E-03	kg	Landfield & Karra (2000)	1	1.38	(1,5,1,3,3,5);
aluminium	→				Amount of material based on composition of rock crusher Nordberg HP400 SX	metals	extraction	No	RER	aluminium, production mix, wrought alloy, at plant	7.44E-04	kg	Landfield & Karra (2000)	1	1.38	(1,5,1,3,3,5);
brass	→				Amount of material based on composition of rock crusher Nordberg HP400 SX	metals	extraction	No	CH	brass, at plant	2.80E-05	kg	Landfield & Karra (2000)	1	1.38	(1,5,1,3,3,5);
transportation of diverse parts to assembly plant	→				Transportation of diverse material to assembly plant, distance assumption: 50km	transport systems	road	No	RER	transport, lorry >16t, fleet average	5.00E-02	tkm	estimation	1	2.32	(5,5,1,1,1,5);
					Is not taken into account as it is assumed that the steel compenents will be reused to 100%						1.00E+00	kg				
			disposal of metal components at end-of-life industrial machine, heavy, unspecified, at plant			construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.00E+00	kg				

Fig. 2.6 Flows for “industrial machine, heavy, unspecified, at plant” and their representation in the ecoinvent database

2.3.4 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

All data for the unspecified heavy industrial machine is based on the materialisation of a rock crusher Nordberg HP400 SX (Landfield & Karra (2000)). This main uncertainty is taken into account by raising the value concerning the indicator “technology” in the uncertainty matrix. Thus the data are of insufficient quality for such a generic module. As pointed out also in the previous chapters for building machines, the module of “rock crusher” should not be used in case its relevance in a specific context could be high.

2.4 Rock Crushing

Consistently with the module for its infrastructure, the rock crusher used as basis for this study is the Nordberg HP400 SX (Landfield & Karra (2000)). At the time of data collection (1999), the Nordberg HP400 SX is one of Nordberg’s newest models of rock crushers and has only been in use for about 4–5 months, so these data are based on specifications and limited use. Fig. 2.7 shows a picture of the rock crusher. Its operational characteristics, taken from a life cycle assessment (Landfield & Karra (2000)) are shown in Tab. 2.8.



Fig. 2.7 Nordberg NW400-IOC portable crushing plant incorporates an HP400 SX cone together with 1067 mm rear feed and rear discharge conveyors on a triple-axle chassis. ²

² <http://rocktoroad.com/miller.html>, access date 24.04.2002

Tab. 2.8 Characteristic of the Nordberg HP400 SX (Landfield & Karra (2000))

Indicator	Value / Description	Unit
Type of rock crushed	Glacier round rock	
Crusher discharge (final product)	3.2	cm
Measured crusher throughput capacity	454	t/h
Crusher production tie	5000	h/year
Total annual throughput	2270000	t/year
Measured Electricity consumption	5850000	MJ/year
Spare parts ("spares")		
Liner	703	kg/part
Wear parts ("wears")		
Mantle	1089	kg/part
Bowl liner	1075	kg/part
Torch ring	6.8	kg/part
Total spare and wear parts per year	15898	kg/year
Lubricating oil	1136	l/year
Assumed lifetime	25	years

System Boundary

The system boundaries for describing the operation of the Nordberg rock crusher include the use phase of the machine, its end-of-life phase, and various transportation requirements.

Use phase: Use phase modelling includes oil and lubricant consumption, and parts replacement requirements of the crusher. The Nordberg HP400 SX rock crusher data were measured and provided by a Nordberg HP400 SX user.

The electricity consumption is not included in the module to allow for the use of specific measured consumption data. If no situation specific electricity consumption is available the measured data for a Nordberg HP400 SX (0.000716 kWh per kg crushed material) can be used.

End-of-life phase: It is assumed that all metal components of the rock crusher are recycled. This mass just leaves the system without further burdens or benefits for the module except for the transport to the scrap metal dealer.

Emissions: Particulates

In the LCA study Landfield & Karra (2000) particulate matter from the rock crushing plant itself is accounted for, and EPA AP-42 (EPA (1998)) provided the emission factor: 1.2 g of particulate matter (PM10) emitted per short ton of crushed rock in tertiary rock processing and crushing operations. This module for rock crushing does not include any particulate emissions as they are already included in the rock processing modules.

Rock Crushing in ecoinvent

The ecoinvent database meta information for "crushing rock" (Tab. 2.9) includes all important and necessary information to use the data correctly.

Tab. 2.9 Ecoinvent database meta information for the product “crushing, rock”

Name	crushing, rock
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the machine, spares and wears, lubricating oil, and the transportation of the new parts to the assembly plant as well as transportation of the crusher to the operational sites. Electricity consumption and dust emissions are not included.
Amount	1
Local Name	Brechen, Gestein
Synonyms	Breaker
General Comment to reference function	Most data are taken from an LCA report and are based on a specific type of crusher (Nordberg HP 400 SX). The transportation distances are estimated and the transport requirements added. This module may not be used if having a high importance in a certain context. Crusher throughput capacity declared by the producer: 454 tons per hour
Start Date	1999
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Certain elementary flows or intermediate product flows are extrapolated from Swiss conditions.
Technology text	Rock crusher type Nordberg HP400 SX produced in the USA in 1999, used as basis
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Most data from LCA study
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 2.8 shows all in- and output flows of the process rock crushing, their allocation to modules in ecoinvent, the sources (mainly Landfield & Karra (2000)) and the standard deviation for each value.

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
electricity	→	crushing of rock				No data for electricity consumption included. Should be added separately. If no specific data is available 7.16°E-4 kWh for a yearly throughput capacity of 2.27 E9 kg/a can be used.								Landfield & Karra (2000)			
iron (spares and wears)	→					yearly amount of spares and wears: 15.898 tons, yearly throughput capacity: 2.27 E9 kg/a	metals	extraction	No	RER	cast iron, at plant	7.00E-06	kg	Landfield & Karra (2000)	1	1.38	(1,5,1,3,3,5);
lubricating oil	→					yearly throughput capacity: 2.27 E9 kg/a, density: 900 kg/m3	chemicals	organics	No	RER	lubricating oil, at plant	4.50E-07	kg	Landfield & Karra (2000)	1	1.38	(1,5,1,3,3,5);
industrial machine, heavy, unspecified, at plant	→					weight of machine: 25'000kg; crusher throughput capacity: -> 56'750'000 t per lifespan (25 years)	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	4.41E-07	kg		1	3.32	(5,5,1,1,1,5);
transportation of diverse parts to assembly plant	→					Transportation of diverse material to assembly plant, distance assumption: 50km	transport systems	road	No	RER	transport, lorry >16t, fleet average	3.73E-07	tkm		1	2.32	(5,5,1,1,1,5);
transportation of crusher to range of application	→					distance assumption: 100km	transport systems	road	No	RER	transport, lorry >16t, fleet average	4.45E-08	tkm		1	2.32	(5,5,1,1,1,5);
			→	particulates PM10		0.0000012 kg/kg crushed rock. Assumption: total of Particulates, > 2.5 µm, and < 10µm. It is not included in this module as it is already taken into account in the corresponding rock processing modules.								Landfield & Karra (2000)			
			→	rock crushing		crushing of 1 kg of rock	construction processes	machinery	No	RER	crushing, rock	1.00E+00	kg				

Fig. 2.8 Flows for “crushing, rock” and their representation in the ecoinvent database

Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

All data for the process of crushing rock are based on (Landfield & Karra (2000)). For the infrastructure the module “industrial machine, heavy, unspecified, at plant” is used. This module is based on the information of the Nordberg HP400 SX crusher. Thus data are of good quality for this type of crusher. It is assumed that the difference to other types of rock crusher is very small and therefore the data of the generic module “crushing, rock” is also of good quality.

2.5 Power Saw and Power Sawing

Introduction

We thank Mr. H. Pichler from the Stihl AG for his help with the material composition of a motor saw.

Chainsaws are used for cutting, disbranching and crosscutting trees. A two-stroke cycle engine powers chainsaws. Apart from typical emissions from combustion motors, chainsaws emit high amounts of chain lubricant oil. During sawing, this lubricant oil is ejected onto the chainsaw and finally drops to the soil. Mainly biologically degradable lubricant oil (rape oil (*brassica napus oleifera*)) is used.

An average chainsaw weights about 7 kg. Based on personal communication³ the composition of the motor saw is 30% low-alloyed steel (2.1 kg), 25% aluminium diecast (1.75 kg), 35% plastics (2.45 kg) and 10% magnesium diecast (1.05 kg). The weight of the chain is 0.4 kg and of the running rail (blade) is 1.2 Kg (Frischknecht et al. (1996)). Chain and running rail consist of high alloyed steel. Service life of a chainsaw is around 2'500 service hours. The chain (0.4 kg) is replaced every 65 service hours; the running rail (1.2 kg) is replaced every 200 hours.⁴ Lubricant oil used for the chain is rape oil and because of lack of a specific module it is inventoried as “Soya oil”. Soya oil is mainly produced in South-America. In contrast rape oil is mainly locally produced. Some additional negative transports are included in the power sawing modules to compensate for these transports.

Data on fuel consumption differs considerably for various sources. BUWAL (1990) assumes 0.8 l/h whereas EMPA-measurements indicate 2.85 litres per operating hour for a certain not further defined type of chainsaw without catalytic converter and 1.68 litre per operating hour for another not further defined type of chainsaws equipped with a catalytic converter.

In order to refer to a consistent data set, emission data and fuel consumption are taken from the EMPA-report (Walter (1993)).

Along with ecoinvent standards, steel is assumed to be transported over 600 km by train and 50 km by lorry. Lubricant oil is transported 200 km by train and 50 km by lorry. All other materials involved are transported 50 km to the final user by lorry.

System Boundary

The modules “power saw, with catalytic converter” and “power saw, without catalytic converter” include the different materials described in the previous section and the transportation of the materials to the assembly plant and from the assembly plant to the end-user. Natural gas and electricity are used in

³ Mr. H. Pichler, Stihl AG, 21.09.2003

⁴ Personal communication, Stihl Vertriebs AG, Mönchaltorf

the production process (Frischknecht et al. (1996)). It is assumed that all materials are reused and therefore no burdens for waste disposal are added. The module does not include any spares and wears as these are accounted for in the operation phase.

Power saw without catalytic converter and the according sawing process in Ecoinvent

The ecoinvent database meta information for “power saw, without catalytic converter” and “power sawing, without catalytic converter” (Tab. 2.10) includes all important and necessary information to use the data correctly.

Fig. 2.9 shows all in- and output flows of the product “power saw, without catalytic converter”, the allocation to the modules in ecoinvent database, the sources (mainly Frischknecht et al. (1996)) and the standard deviation for each value.

Tab. 2.10 Ecoinvent database meta information for the power saw and sawing modules without catalytic converter

Name	power saw, without catalytic converter	power sawing, without catalytic converter
Location	RER	RER
Infrastructure Process	1	0
Unit	unit	h
Data Set Version	2.0	2.0
Included Processes	Includes the most important materials, the transportation of the parts to the assembly plant and of the power saw to the end-user. Also included is a certain amount of electricity and heating energy. No disposal is included.	Includes the Inputs: power saw (infrastructure), fuel use, lubricating oil requirements, and some measured air emissions (including emissions from combustion of the fuel).
Amount	1	1
Local Name	Motorsäge, ohne Katalysator	Motorsägen, ohne Katalysator
Synonyms	chain//motor//Kettensäge	chain//motor//Kettensäge
General Comment to reference function	The composition is based on one type of power saw (produced in Stihl, Germany) and the transportation distances are partly estimated. This module should not be used if its relative importance would be high in a certain environmental inventory. Service life capacity, declared by producer: 2500 hours	All measured emissions are taken from an old (1993) EMPA report. The type of power saw probably does not correspond to the type described in the infrastructure, but this discrepancy can be accepted. This module should not be used if its relative high importance would be high in a certain environmental inventory.
Start Date	1995	1993
End Date	2001	2001
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	The power saw studied is produced in Germany and used as proxy for RER. For some elementary flows proxies (other geographical origin) have been used.	The type of machine whose air emissions were measured is unknown. Certain elementary flows or intermediate product flows are extrapolated from other geographical areas than indicated.
Technology text	An average chainsaw weights about 7 kg. It is assumed that about 5.4 kg are unalloyed steel. The remaining 1.6 kg consists of the chain (0.4 kg) and the running rail (blade). Chain and running rail are made of highly alloyed steel. Service life of a chainsaw is around 2'500 service hours. The chain is replaced every 65 service hours; the running rail is replaced every 200 hours.	The module represents an example for an average technology used in Europe in the late 1990s
Representativeness		
Production Volume	Unknown	Unknown
Sampling Procedure	Most data obtained by personal communication	Most data from one technical report
Extrapolations	See geography	See geography
Uncertainty adjustments	None	None

General Flow information					Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
Steel low-alloyed	→	power saw, without catalytic converter			total weight of an avarage saw: 7kg; 30% steel low-alloyed	metals	extraction	No	RER	steel, low-alloyed, at plant	2.10E+00 kg		E-mail from Stihl (DE) on 21.09.2001	1	1.40	(3,5,1,2,3,5);
Processing steel	→				30% of total weight for steel parts	metals	processing	No	RER	section bar rolling, steel	2.10E+00 kg		E-mail from Stihl (DE) on 21.09.2001	1	1.40	(3,5,1,2,3,5);
aluminium diecast	→				25% of total weight for aluminium parts	metals	extraction	No	RER	aluminium, production mix, cast alloy, at plant	1.75E+00 kg		E-mail from Stihl (DE) on 21.09.2001	1	1.40	(3,5,1,2,3,5);
magnesium diecast	→				10% of total weight for magnesium parts	metals	extraction	No	RER	magnesium-alloy, AZ91, diecasting, at plant	7.00E-01 kg		E-mail from Stihl (DE) on 21.09.2001	1	1.40	(3,5,1,2,3,5);
plastic granulate	→				35% of total weight for plastics	plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	2.45E+00 kg		E-mail from Stihl (DE) on 21.09.2001	1	1.40	(3,5,1,2,3,5);
forming of plastic granulate to parts	→				same weight as plastic granulate used	plastics	processing	No	RER	injection moulding	2.45E+00 kg		E-mail from Stihl (DE) on 21.09.2001	1	1.40	(3,5,1,2,3,5);
Transportation of steel parts to assembly plant by lorry	→				Transportation of steel to assembly plant, distance: 600km	transport systems	road	No	CH	transport, lorry >28t, fleet average	1.26E+00 tkm		Frischknecht et al. (1996)	1	2.32	(5,5,1,3,1,5);
Transportation of steel parts to assembly plant by train	→				Transportation of steel to assembly plant, distance: 50km	transport systems	train	No	RER	transport, freight, rail	1.05E-01 tkm		Frischknecht et al. (1996)	1	2.43	(4,5,5,5,3,5);
Transportation of non-steel parts to assembly plant by lorry	→				Transportation of diverse parts to assembly plant, assumption of distance: 300km	transport systems	road	No	CH	transport, lorry >28t, fleet average	1.47E-01 tkm		assumption	1	2.37	(5,5,1,1,3,5);
Transportation of power saw to end-user by lorry	→				distance assumption for Switzerland: 50km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	3.50E-01 tkm		Frischknecht et al. (1996)	1	2.43	(4,5,5,5,3,5);
Natural gas burnt in industrial furnace	→				assumed for a heating process	natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	2.04E-01 MJ		Frischknecht et al. (1996)	1	1.74	(4,5,5,5,3,5);
Electricity	→				assumed for production process	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.87E-02 kWh		Frischknecht et al. (1996)	1	1.74	(4,5,5,5,3,5);
			→	waste heat from electricity consumption			air	unspecified			Heat, waste	6.73E-02 MJ		Frischknecht et al. (1996)	1	1.74
		→	disposal of power saw		Is not taken into account since it is assumed that all parts will be reused to 100%						7.00E+00 kg					
		→	power saw, without catalytic converter		total weight of an avarage saw: 7kg	construction processes	machinery	Yes	RER	power saw, without catalytic converter	1.00E+00 unit					

Fig. 2.9 Flows for “power saw, without catalytic converter” and their representation in the ecoinvent database

Fig. 2.10 shows all in- and output flows of the product “power sawing, without catalytic converter”, the allocation to the modules in ecoinvent database, the sources (mainly personal communication with people of Stihl AG, EMPA Report No 148’849 and Frischknecht et al. (1996)) and the standard deviation for each value.

General Flow information					Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
Fuel (two stroke blend)	→	power sawing, carburetor without catalytic converter (part 1)				oil	fuels	No	RER	petrol, two-stroke blend, at regional storage	2.13E+00 kg		Source EMPA Report No 148'849, Th. Walter Abt. 137	1	1.32	(1,5,3,3,1,5);
Lubricating oil	→					chemicals	organics	No	RER	soya oil, at plant	5.40E-01 kg		personal communication, Mr. H. Pichler, Stihl AG, 12.06.03	1	1.68	(5,5,1,3,3,5);
reduction "transport lorry 32t" from "soya oil"-module	→				Reduction of total transportation amount from "soya oil"-module (1.46 tkm/kg soya oil) and addition of estimated 200 km for inner european transports by lorry	transport systems	road	No	RER	transport, lorry >16t, fleet average	-6.80E-01 tkm		Stihl (1994)	1	2.37	(5,5,1,3,3,5);
removing "transport, barge" from "soya oil"-module	→				Reduction of total transportation amount from "soya oil"-module (0.806 tkm/kg soya oil)	transport systems	ship	No	RER	transport, barge	-4.35E-01 tkm		Stihl (1994)	1	1.00	(,,,,); calculated
removing "transport, transpceanic freight ship" from "soya oil"-module	→				Reduction of total transportation amount from "soya oil"-module (12.1 tkm/kg soya oil)	transport systems	ship	No	OCE	transport, transoceanic freight ship	-6.53E+00 tkm		Stihl (1994)	1	1.00	(,,,,); calculated
removing "transport, freight, rail" from "soya oil"-module	→				Reduction of total transportation amount from "soya oil"-module (0.351 tkm/kg soya oil)	transport systems	train	No	RER	transport, freight, rail	-1.90E-01 tkm		Stihl (1994)	1	1.00	(,,,,); calculated
Transportation of lubricating oil to assembly plant by train	→				Transportation of lubricating oil to assembly plant by train, distance: 200km	transport systems	train	No	RER	transport, freight, rail	1.08E-01 tkm		Frischknecht et al. (1996)	1	2.43	(4,5,5,5,3,5);
Transportation of lubricating oil to end-user by lorry	→				distance assumption for Switzerland: 50km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	2.70E-02 tkm		Frischknecht et al. (1996)	1	2.43	(4,5,5,5,3,5);
chain	→					metals	extraction	No	RER	chromium steel 18/8, at plant	6.15E-03 kg		Stihl (1994)	1	3.10	(1,5,3,3,1,5);
blade	→					metals	extraction	No	RER	chromium steel 18/8, at plant	6.00E-03 kg		Stihl (1994)	1	3.10	(1,5,3,3,1,5);
power saw	→					construction processes	machinery	Yes	RER	power saw, without catalytic converter	4.00E-04 unit			1	3.10	(1,5,3,3,1,5);

Fig. 2.10 Flows for “power sawing, without catalytic converter” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
power sawing, carburettor without catalytic converter (part 2)	→	Oils (biol. degr..) into soil	assumption: 10% of the used amount of lubricating oil	soil	forestry			Oils, biogenic	1.00E-01	kg		1	1.63	(1,5,3,3,1,5);
	→	CH4 Methane		air	low population density			Methane, fossil	1.12E-02	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	1.63	(1,5,3,3,1,5);
	→	CO Carbonmonoxide		air	low population density			Carbon monoxide, fossil	1.33E+00	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	5.12	(1,5,3,3,1,5);
	→	CO2 Carbondioxide		air	low population density			Carbon dioxide, fossil	2.80E+00	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	1.32	(1,5,3,3,1,5);
	→	NMVOC (Non-Methane VOC)		air	low population density			NMVOC, non-methane volatile organic compounds, unspecified origin	5.49E-01	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	2.11	(1,5,3,3,1,5);
	→	NOx Nitrogenoxide as NO2		air	low population density			Nitrogen oxides	3.00E-03	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	1.63	(1,5,3,3,1,5);
	→	Pb lead		air	low population density			Lead	1.48E-04	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	5.12	(1,5,3,3,1,5);
	→	SO2 Sulfurdioxide		air	low population density			Sulfur dioxide	1.37E-03	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	2.11	(1,5,3,3,1,5);
	→	Dust	no data available						-			1		(.....);
	→	power saw carburettor without catalytic converter		construction processes	machinery	No	RER	power sawing, without catalytic converter	1.00E+00	h				

Fig. 2.10 Flows for “power sawing, without catalytic converter” and their representation in the ecoinvent database (cont.)

Power saw with catalytic converter and the according sawing process in Ecoinvent

The ecoinvent database meta information for “power saw, with catalytic converter” and “power sawing, with catalytic converter” (Tab. 2.11) includes all important and necessary information to use the data correctly.

Tab. 2.11 Ecoinvent database meta information for the power saw and sawing modules with catalytic converter

Name	power saw, with catalytic converter	power sawing, with catalytic converter
Location	RER	RER
Infrastructure Process	1	0
Unit	unit	h
Data Set Version	2.0	2.0
Included Processes	Based on the module for power saw without catalytic converter, adding some ceramics, platinum and rhodium to describe the catalytic converter.	Includes the Inputs: power saw (infrastructure), fuel use, lubricating oil requirements, and some measured air emissions (including emissions from combustion of the fuel). The saw is equipped with a catalytic converter.
Amount	1	1
Local Name	Motorsäge, mit Katalysator	Motorsägen, mit Katalysator
Synonyms	chain//motor//Kettensäge	chain//motor//Kettensäge
General Comment to reference function	The composition is based on one type of power saw (produced by Stihl, Germany) and the transportation distances are partly estimated. This module should not be used if its relative importance would be high in a certain environmental inventory. Service life capacity, declared by producer: 2500 hours	All measured emissions are taken from an old (1993) EMPA report. The type of power saw probably does not correspond to the type described in the infrastructure, but this discrepancy can be accepted. This module should not be used if its relative high importance would be high in a certain environmental inventory.
Start Date	1995	1993
End Date	2001	2001
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	The power saw studied is produced in Germany and used as proxy for RER. For some elementary flows proxies (other geographical origin) have been used.	The type of machine whose air emissions were measured is unknown. Certain elementary flows or intermediate product flows are extrapolated from other geographical areas than indicated.
Technology text	An average chainsaw weights about 7 kg. It is assumed that about 5.4 kg are unalloyed steel. The remaining 1.6 kg consists of the chain (0.4 kg) and the running rail (blade). Chain and running rail are made of highly alloyed steel. Service life of a chainsaw is around 2'500 service hours. The chain is replaced every 65 service hours; the running rail is replaced every 200 hours.	The module represents an example for an average technology used in Europe in the late 1990s
Representativeness		
Production Volume	Unknown	Unknown
Sampling Procedure	Most data obtained by personal communication. Data for the catalytic converter were extrapolated from data for car equipment.	Most data from one technical report
Extrapolations	See geography	See geography
Uncertainty adjustments	None	None

Fig. 2.11 shows all in- and output flows of the product “power saw, with catalytic converter”, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each value.

General Flow information					Representation in ecoinvent							Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
power saw, without catalytic converter	power saw, with catalytic converter			construction processes	machinery	Yes	RER	power saw, without catalytic converter	1.00E+00	unit		1	3.00	(1,1,1,1,1,1);
Platinum			raw estimation: 10% of catalytic converter from car/van (0.0016 kg)	metals	extraction	No	RER	platinum, at regional storage	1.60E-04	kg		1	1.93	(5,5,5,2,3,5);
Palladium			raw estimation: 10% of catalytic converter from car/van (0.0003 kg)	metals	extraction	No	RER	palladium, at regional storage	3.00E-05	kg		1	1.93	(5,5,5,2,3,5);
Ceramic monolithic carrier			no data											
		power saw, with catalytic converter		construction processes	machinery	Yes	RER	power saw, with catalytic converter	1.00E+00	unit				

Fig. 2.11 Flows for “power saw, with catalytic converter” and their representation in the ecoinvent database

Fig. 2.12 shows all in- and output flows of the product “power sawing, with catalytic converter”, the allocation to the modules in ecoinvent database, the sources (mainly personal communication with people of Stihl AG and Frischknecht et al. (1996)) and the standard deviation for each value.

General Flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
Fuel (two stroke blend)	power sawing, injection with catalytic converter (part 1)				oil	fuels	No	RER	petrol, two-stroke blend, at regional storage	1.25E+00	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	1.32	(1,5,3,3,1,5);
Lubricating oil				Soya oil used as proxy. Transports taken out by including them negativ	chemicals	organics	No	RER	soya oil, at plant	5.40E-01	kg	personal communication, Mr. H. Pichler, Stihl AG, 12.06.03	1	1.68	(5,5,1,3,3,5);
reduction "transport lorry 32t" from "soya oil"-module				Reduction of total transportation amount from "soya oil"-module (1.46 tkm/kg soya oil) and addition of estimated 200 km for inner european transports by lorry	transport systems	road	No	RER	transport, lorry >16t, fleet average	-6.80E-01	tkm	Stihl (1994)	1	2.37	(5,5,1,3,3,5);
removing "transport, barge" from "soya oil"-module				Reduction of total transportation amount from "soya oil"-module (0.806 tkm/kg soya oil)	transport systems	ship	No	RER	transport, barge	-4.35E-01	tkm	Stihl (1994)	1	1.00	(.....); calculated
removing "transport, transoceanic freight ship" from "soya oil"-module				Reduction of total transportation amount from "soya oil"-module (12.1 tkm/kg soya oil)	transport systems	ship	No	OCE	transport, transoceanic freight ship	-6.53E+00	tkm	Stihl (1994)	1	1.00	(.....); calculated
removing "transport, freight, rail" from "soya oil"-module				Reduction of total transportation amount from "soya oil"-module (0.351 tkm/kg soya oil)	transport systems	train	No	RER	transport, freight, rail	-1.90E-01	tkm	Stihl (1994)	1	1.00	(.....); calculated
Transportation of lubricating oil to assembly plant by train				Transportation of lubricating oil to assembly plant, distance: 200km	transport systems	train	No	RER	transport, freight, rail	1.08E-01	tkm	Frischknecht et al. (1996)	1	2.43	(4,5,5,5,3,5);
Transportation of lubricating oil to end-user by lorry				distance assumption for Switzerland: 50km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	2.70E-02	tkm	Frischknecht et al. (1996)	1	2.43	(4,5,5,5,3,5);
chain					metals	extraction	No	RER	chromium steel 18/8, at plant	6.15E-03	kg	Stihl (1994)	1	3.10	(1,5,3,3,1,5);
blade					metals	extraction	No	RER	chromium steel 18/8, at plant	6.00E-03	kg	Stihl (1994)	1	3.10	(1,5,3,3,1,5);
power saw					construction processes	machinery	Yes	RER	power saw, with catalytic converter	4.00E-04	unit		1	3.10	(1,5,3,3,1,5);

Fig. 2.12 Flows for “power sawing, with catalytic converter” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
power sawing, injection with catalytic converter (part 2)	→	Oils (biol. degr..) into soil			assumption: 10% of the used amount of lubricating oil	soil	forestry			Oils, biogenic	2.80E-01	kg		1	1.63	(1,5,3,3,1,5);
	→	CH4 Methane				air	low population density			Methane, fossil	1.60E-03	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	1.63	(1,5,3,3,1,5);
	→	CO Carbonmonoxide				air	low population density			Carbon monoxide, fossil	1.60E-01	kg	Consumption pro m3 Holz aus Wegener et al., 1994	1	5.12	(1,5,3,3,1,5);
	→	CO2 Carbondioxide				air	low population density			Carbon dioxide, fossil	3.47E+00	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	1.32	(1,5,3,3,1,5);
	→	NMVOC (Non-Methane VOC)				air	low population density			NMVOC, non-methane volatile organic compounds, unspecified origin	7.84E-02	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	2.11	(1,5,3,3,1,5);
	→	NOx Nitrogenoxide as NO2				air	low population density			Nitrogen oxides	1.80E-02	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	1.63	(1,5,3,3,1,5);
	→	Pb lead				air	low population density			Lead	8.74E-05	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	5.12	(1,5,3,3,1,5);
	→	SO2 Sulfurdioxide				air	low population density			Sulfur dioxide	8.06E-04	kg	Source EMPA Report No 148'849, Th. Walter Abt. 137	1	2.11	(1,5,3,3,1,5);
	→	Dust			no data available					-				1		(,...);
	→	power saw injection with catalytic converter				construction processes	machinery	No	RER	power sawing, with catalytic converter	1.00E+00	h				

Fig. 2.12 Flows for “power sawing, with catalytic converter” and their representation in the ecoinvent database (cont.)

Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

The data of the infrastructure module “power saw, without catalytic converter” include the main materials of a certain power saw, the transportation and energy requirements for the production. The module “power saw, with catalytic converter” is based on the power saw without catalytic converter and adding an estimation of materials for the converter. The overall quality is poor and these modules should not or only with great care be used if they are important contributors in specific cases.

The data of the operation phase include the fuel consumption and some emissions. The developments of the power saw production industry are fast and the emission data available are 10 years old. The number of emitted species considered is quite small compared to other combustion processes. Thus the data are of insufficient quality even for such general modules. It is recommended to use the module only if it plays an unimportant role in an inventory.

2.6 Conveyor belt

System Boundary

Conveyor belts are used for transportation of great amounts of heavy loads as earthwork, coal and other mining ores.

The conveyor belt used as basis for the ecoinvent module has the characteristics shown in Tab. 2.12 (Beitz & Dubbel (2001)).

Tab. 2.12 Characteristic data of conveyor belt used as basis

Characteristic data per km length	Value	Unit
Stream of conveyed material	3900	kg/s
Hauling speed	5.2	m/s
Width of belt	3	m
Thickness of belt	0.005	m
Volume of belt	15	m ³
Supporting structure (steel)	300000	kg
Rollers	200000	kg

The belt is made out of rubber with an assumed density of 910 kg/m³.

Conveyor belt in ecoinvent

The ecoinvent database meta information for “conveyor belt, at plant” (Tab. 2.13) includes all important and necessary information to use the data correctly.

Tab. 2.13 Ecoinvent database meta information for the product “conveyor belt, at plant”

Name	conveyor belt, at plant
Location	RER
Infrastructure Process	1
Unit	m
Data Set Version	2.0
Included Processes	Includes the most important materials, the transportation of the new parts to the assembly plant and of all dismantled parts at the end of life either to the secondary metal producer or to the disposal site. No disposal of metal is included as it is recycled. Disposal of concrete from foundation and conveyor rubber band is included. No energy for assembling is included.
Amount	1
Local Name	Förderband, ab Werk
Synonyms	conveying//band//Förderanlage
General Comment to reference function	Most data are taken from a description of one type of conveyor belt and completed by assuming transportation distances for components. This module should not be used if its relative importance would be high in a certain environmental inventory.
Start Date	2000
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Certain elementary flows or intermediate product flows are extrapolated from Swiss conditions.
Technology text	The module represents an example for an average technology used in Europe in the late 1990s
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Most data from literature
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 2.13 shows all in- and output flows of the product “conveyor belt, at plant”, the allocation to the modules in ecoinvent database, the sources (mainly Beitz & Dubbel (2001)) and the standard deviation for each value.

Data Quality

The uncertainties are estimated using the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the Ecoinvent database meta information.

Most data are unqualified estimations and therefore the module “conveyor belt” is of poor overall quality. It is advisable not to use the module if conveyor belts are of high importance for a specific inventory.

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
low-alloyed steel	→	conveyor belt			assumption from literature: 300 t/km for channels, 200t/km for rollers	metals	extraction	No	RER	steel, low-alloyed, at plant	5.00E+02 kg		E-mail A. Primas vom 13.08.2002	1	1.30	(1,5,1,3,1,5);	
processing of steel	→				same weight as steel used	metals	processing	No	RER	section bar rolling, steel	5.00E+02 kg			1	1.30	(1,5,1,3,1,5);	
band, part rubber	→				assumption from literature: total volume: 15m3/km; density of rubber 910 kg/m3	plastics	polymers	No	RER	synthetic rubber, at plant	1.37E+01 kg		Beitz & Dubbel (2001)	1	1.32	(3,5,1,3,1,5);	
band, part steel cable, steel	→				assumption from literature: volume: 25% of total volume: steel part: 3.75m3/km; density of steel: 7900 kg/m3	metals	extraction	No	RER	steel, low-alloyed, at plant	2.96E+01 kg		Beitz & Dubbel (2001)	1	1.32	(3,5,1,3,1,5);	
band, part steel cable, wire drawing	→				same amount as steel used for cable	metals	processing	No	RER	wire drawing, steel	2.96E+01 kg		Beitz & Dubbel (2001)	1	1.32	(3,5,1,3,1,5);	
fundament concrete	→				calculated from plan: volume of fundament: 0.01m3	construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	1.00E-02 m3		Beitz & Dubbel (2001)	1	1.32	(3,5,1,3,1,5);	
driving gear	→				not balanced because no data were available												
Transportation of diverse parts to assembly plant	→				Transportation of diverse material to assembly plant, distance assumption for Switzerland: 50km	transport systems	road	No	RER	transport, lorry >16t, fleet average	2.83E+01 tkm		assumption	1	2.32	(5,5,1,3,1,5);	
Transportation of crap metal from crushing plant to secondary metal producer	→				transportation of dismantled metal parts at end of life, distance assumption for Switzerland: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	2.50E+01 tkm		assumption	1	2.32	(5,5,1,3,1,5);	
Transportation of the belt to waste disposal	→				transportation of dismantled metal parts at end of life, distance assumption for Switzerland: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	2.16E+00 tkm		assumption	1	2.32	(5,5,1,3,1,5);	
	→			disposal of metal components at end-of-life		Is not taken into account as it is assumed that the steel compenens will be reused to 100%						5.00E+02 kg					
	→			disposal of reinforced concrete		density of reinforced concrete: 2'300 kg/m3	waste management	building demolition	No	CH	disposal, building, reinforced concrete, to final disposal	2.30E+01 kg			1	1.32	(3,5,1,3,1,5);
	→			disposal of steel part of band			waste management	municipal incineration	No	CH	disposal, steel, 0% water, to municipal incineration	2.96E+01 kg			1	1.32	(3,5,1,3,1,5);
	→		disposal of rubber part of band			waste management	municipal incineration	No	CH	disposal, rubber, unspecified, 0% water, to municipal incineration	1.37E+01 kg			1	1.32	(3,5,1,3,1,5);	
	→		conveyor belt			construction processes	machinery	Yes	RER	conveyor belt, at plant	1.00E+00 m						

Fig. 2.13 Flows for “conveyor belt, at plant” and their representation in the ecoinvent database

3 Buildings

3.1 Acknowledgement

Thanks to Markus Meili, Gerold Janzi and René Steiger from EMPA Dübendorf for their help with assumptions concerning the material use in the different buildings.

3.2 Multi-Storey Building

3.2.1 System Characterisation

Mauch et al. (1995) describe an average material composition of a multi-storey apartment building. The average is derived from two buildings: one multifamily residential building in Germany built in 1972 (about 60% of total Swiss buildings) and one two-family residential building in Switzerland built in 1927 (about 40% of Swiss buildings). The construction weight is on average 0.5 t per m³ building volume above ground.

Further information from Mauch et al. (1995) is the primary energy used for the building, operation and demolition phase. The wastes arising at the end of life are categorised based on the destination: landfill, incineration, and recycling.

Tab. 3.1 Characteristic material composition of a multi-storey building (Mauch et al. (1995))

Material-Input	[%] (v/v)	Modelled as	Construction phase [kg/m ³]	Refurbishment [kg/m ³]	Total [kg/m ³]
Reinforced Concrete ⁵	31	Concrete (97 %)	231.86	0.00	231.86
		Reinforcement steel (3 %)	7.17	0.00	7.17
Brick	48	Brick	177.66	0.00	177.66
Mortar	5	Mortar	30.84	30.84	61.69
Glass	0.2	Glass	1.54	1.54	3.08
Wood	10	Wood	15.42	15.42	30.84
Metals	2	Steel (1%)	16.96	0.00	16.96
		Aluminium (0.5%)	8.48	0.00	8.48
		Copper (0.5%)	8.48	0.00	8.48
Insulation	3	Rock-wool	0.65	0.65	1.30
Plastics	0.4	PVC	0.46	0.46	0.93
Others	0.4	PE	0.46	0.46	0.93
Total	100		500	95	600

The amount of waste arising after demolition and the allocation to the “end” destination is shown in Tab. 3.2.

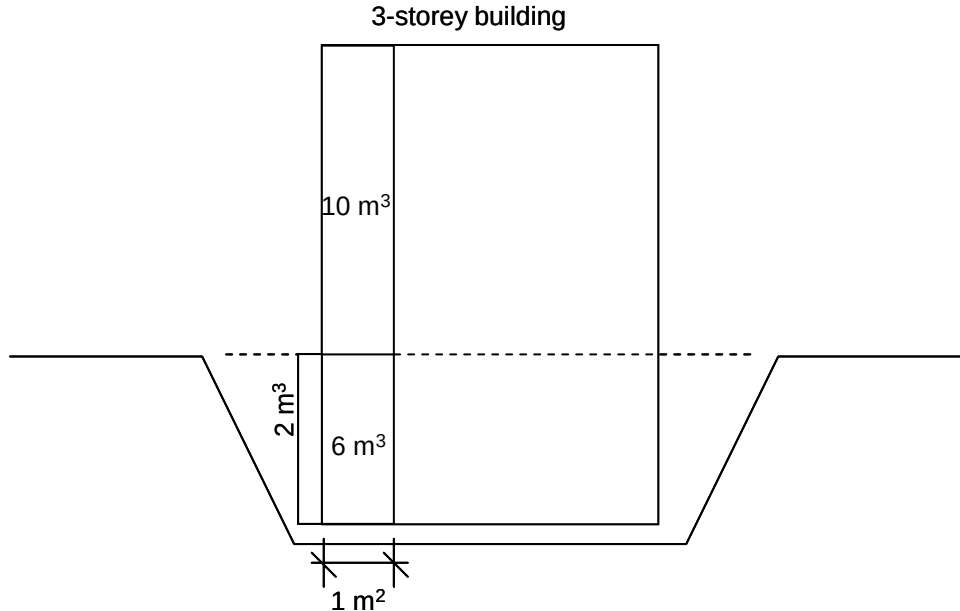
The estimation/calculation of the primary energy demand is partitioned in construction (excavation), refurbishment and demolition phase.

⁵ The values for reinforcing steel and concrete are wrong in the ecoinvent v2.0 Database. The value for concrete is used for steel and vice versa.

Tab. 3.2 End of life waste composition and destination of a multi-storey building (Mauch et al. (1995))

Disposal building...	[kg/m ³ material]
<i>... to final disposal:</i>	
Concrete	155
Reinforcement steel	5
Brick	240
Mortar	50
Glass	2
Rock-wool	30
Wood	100
PVC	4
PE	4
Total	590
<i>... to sorting plant</i>	
Copper	2.5
Steel	5
Total	7.5
<i>... to recycling</i>	
Aluminium	2.5
Total	2.5
Total Material-Output	607.5

The main usage of diesel powered building machines happens during the excavation and demolition phases of a building. The estimation of energy is based on the following calculation:



Based on the above sketch it has been assumed that 1 m³ building volume above ground needs 0.8 m³ of excavation volume. A skid-steer loader has an energy consumption of 0.13 kg diesel per m³ excavation volume (compare chapter 2.2). With reference to 1 m³ building this is 0.104 kg diesel. As the en-

ergy content (lower heating value) of diesel is 42.8 MJ/kg, the energy for the excavation of 1 m³ building is 4.45 MJ. This value has been rounded up to 5 MJ/m³.

The demolition has been assumed to be about double of the excavation energy and therefore about 10 MJ/m³. Thus the total energy consumption in form of “diesel in building machine” is 15 MJ/m³.

The electricity needed for construction, refurbishment and demolition is based on personal communication and can be set at about 0.30 kWh/ m³.

The service-life is assumed to be 80 years.

3.2 Building hall

3.2.1 System Characterisation

The data are based on a theoretical building with geometry described in Tab. 3.1. The service-life is assumed to be 50 years.

Tab. 3.1 Geometry of building used as basis in this study

Geometry	value	Unit
length	50	m
width	30	m
height	7	m
floor area (or ceiling)	1500	m ²
external area	1120	m ²
daylight opening (10% of external area)	110	m ²

3.2.2 Hall, Steel Construction

Based on Tab. 3.1 the materials used for the steel construction hall have been calculated. Tab. 3.2 shows the calculation and the total amount of the different materials included in the module “building, hall, steel construction”, where the functional unit is the m².

Tab. 3.2 Material for the reference steel construction hall

Building component	Considered sizes	value	Unit	(*)
wall	masonry thickness	0.16	m	
	brick volume	160	m ³	
	brick density	1000	kg/m ³	
	total weight masonry (256'000 kg / 1'500 m ²)	107	kg/m ²	
	Cement mortar: Amount bricks (rounded up)	20000	unit	
	Length of mortar joint	10000	m	
	Thickness of joint	0.01	m	
	Thickness of wall	0.15	m	
	Weight of cement mortar (density: 2'100 kg/m ³)	31500	kg	
	Weight of cement mortar	21	kg/m ²	

Tab. 3.2 Material for the reference steel construction hall (cont)

Building component	Considered sizes	value	Unit	(*)
insulation	insulation thickness	0.1	m	
	insulation volume (ceiling and wall)	262	m ³	
	insulation density (mineral wool)	50	kg/m ³	
	insulation density (Polystyrol)	20	kg/m ³	
	total weight insulation mineral wool (50 vol.-%)	6550	kg	
	total weight insulation Polystyrol (50 vol.-%)	2620	kg	
	Assumption half of weight mineral wool: glass wool (6'550/2 kg / 1'500 m ²)	2.2	kg/m ²	20
	Assumption other half of weight mineral wool: rock wool (6'550/2 kg / 1'500 m ²)	2.2	kg/m ²	20
	weight Polystyrol (6'550/2 kg / 1'500 m ²)	1.8	kg/m ²	20
foundation	reinforced concrete foundation thickness	0.3	m	
	reinforced concrete volume	450	m ³	
	total volume concrete (450 m ³ / 1'500 m ²)	0.3	m ³ /m ²	
steel pillars and beams	steel for supporting structure total estimation by EMPA (weight per m ² floor area)	40	kg/m ²	
	steel for supporting structure (75% of 40 kg/m ² weight per m ² floor area)	30	kg/m ²	
reinforced concrete pillars and beams	reinforced concrete (instead of pure steel) for supporting structure (25% of 40 kg/m ² = 10 kg/m ² steel) floor area: 1'500 m ² -> 15 t (3 % (w/w)), density steel: 7'850 -> 2m3 (used for estimation of volume of reinforced concrete pillars and beams (instead of steel))	2	m ³	
	Reinforced concrete for supporting structure; (assumed that about double volume of a pure steel structure: 4 m ³ , density 2'500 kg/m ³ = 10'000 kg)	7	kg/m ²	
	Steel fraction	3	weight-%	
	Concrete fraction	97	weight-%	
	total weight steel	0.2	kg/m ²	
	total weight concrete	6.8	kg/m ²	
window glass	double glazing (10% of 1'120 m ²)	110	m ²	
	glass weight (thickness: 2*4 mm, density 2600 kg/m ³ = 2'300 kg)	1.5	kg/m ²	20
window frame	Aluminium: Assumption of a specific window: 1.3 * 1.65 m, 44.61 kg glass per window and 30 kg frame per window; --> weight of frame per m ² window: 14 kg / m ² glass	1	kg/m ²	20
facing	fibre cement board (thickness: 8mm, area: 1'120 m ² , density: 1'800 kg/m ³)	11	kg/m ²	20
roof	Steel sheet with trapezoidal corrugations for flat roofs (as trapezoidal: assumption of total steel area: 2'000 m ² , thickness: 1.5 mm, density: 7'850 kg/m ³): 1.6 kg/m ² ; 10% added due to screws and joints.	1.8	kg/m ²	30

(*): Assumed service-life if shorter than 50 years.

The energy demand for the construction and refurbishment is assumed to be one third of the consumption for a multi-storey building.

3.2.3 Hall, Wood Construction

Based on Tab. 3.1 the material used for the wood construction hall has been calculated. Tab. 3.3 shows the calculation and the total amount of the different materials included in the module “building, hall, wood construction”.

Tab. 3.3 Materials for the reference wooden construction hall

Building component	Considered sizes	value	Unit	(*)
primary beams	glued laminated timber beams , cross sectional area, profile 1.5 m * 0.2 m = 0.3 m ²	0.3	m ²	
	total length of beams (2 * 50 m)	100	m	
	total volume of beams	30	m ³	
	total volume of primary beams (30 m ³ / 1'500 m ²)	0.02	m ³ /m ²	
secondary beams	glued laminated timber beams , cross sectional area, profile 0.8 m * 0.2 m = 0.16 m ²	0.16	m ²	
	total length of beams (9 * 30 m)	270	m	
	total volume of beams	43.2	m ³	
	total volume of secondary beams (43.2 m ³ / 1'500 m ²)	0.03	m ³ /m ²	
tertiary beams	structural timber beams (softwood) , profile 0.15 m * 0.3 m = 0.045 m ²	0.045	m ²	
	total length of beams (29 * 50 m)	1450	m	
	total volume of beams	65	m ³	
	total volume of tertiary beams (65 m ³ / 1'500 m ²)	0.04	m ³ /m ²	
Wall: framework construction	structural timber (softwood) , profile 0.14 * 0.08 m	0.0112	m ²	
	total length of timber (270 * 7 m + 640 m)	2530	m	
	total volume of timber	25	m ³	
	total volume of framework construction (25 m ³ / 1'500 m ²)	0.02	m ³ /m ²	
	structural timber board (softwood), area	1120	m ²	
	thickness of board	0.02	m	
	total volume of structural timber	23	m ³	
	total volume of structural timber (23 m³ / 1'500 m²)	0.015	m ³ /m ²	20
Wall: panelling (inside)	particle board, indoor use, area	1120	m ²	
	thickness of board	0.02	m	
	total volume of particle board	23	m ³	
	total volume of particle board (23 m ³ / 1'500 m ²)	0.015	m ³ /m ²	30
Insulation	insulation thickness	0.1	m	
	insulation volume (ceiling and wall)	262	m ³	
	insulation density (mineral wool)	50	kg/m ³	
	insulation density (Polystyrol)	20	kg/m ³	
	total weight insulation mineral wool (50% of volume)	6550	kg	
	total weight insulation Polystyrol (50% of volume)	2620	kg	

Tab. 3.3 Materials for the reference wooden construction hall

Building component	Considered sizes	value	Unit	(*)
Insulation	Assumption half of weight mineral wool: glass wool (6'550/2 kg / 1'500 m ²)	2.2	kg/m ²	20
	Assumption other half of weight mineral wool: rock wool (6'550/2 kg / 1'500 m ²)	2.2	kg/m ²	20
	weight Polystyrol (6'550/2 kg / 1'500 m ²)	1.8	kg/m ²	20
foundation	reinforced concrete foundation thickness	0.3	m	
	reinforced concrete volume	450	m ³	
	total volume concrete (450 m ³ / 1'500 m ²)	0.3	m ³ /m ²	
pillars	structural timber beams (softwood), profile 0.4 * 0.4 m	0.16	m ²	
	total length of pillars (8 * 7 m)	56	m	
	total volume of pillars	9	m ³	
	total volume of pillars per area (9 m ³ / 1'500 m ²)	0.006	m ³ /m ²	
window glass	double glazing (10% of 1'120 m ²)	110	m ²	
	glass weight (thickness: 2 * 4 mm, density 2600 kg/m ³ = 2'300 kg)	1.5	kg/m ²	20
window frame	Aluminium: Assumption of a specific window: 1.3 * 1.65 m, 44.61 kg glass per window and 30 kg frame per window; --> weight of frame per m ² window: 14 kg / m ² glass	1	kg/m ²	20
roof	steel sheet with trapezoidal corrugations for flat roofs (as trapezoidal: area assumption: 2'000 m ² , thickness: 1.5 mm, density: 7'850 kg/m ³)	1.6	kg/m ²	30
steel parts	Total weight of wood parts in hall, used for the calculation of the weight of the steel parts (average density: 500 kg/m ³ , total volume: 0.146 m ³ /m ²)	73	kg/m ²	
	steel parts (estimation 15% of total weight of wood)	10	kg/m ²	
	area of steel parts used for galvanizing coating (density: 7'850 kg/ m ³ , average thickness: 0.002 m)	0.64	m ² /m ²	

(*): Assumed service-life if shorter than 50 years.

The energy demand for the construction and refurbishment is assumed to be a third of the consumption for a multi-storey building.

3.2.4 Hall, Unspecified

The construction of an unspecified hall is assumed to be composed of 30% wood and 70% steel construction, which is about the current share for Swiss halls.⁶

⁶ Personal communication, 17.06.2002, G. Janzi, EMPA Dübendorf, Switzerland

3.3 Life Cycle Inventories for each unit Process

The following sections start with background information included in the specific ecoinvent database meta information. This includes essential information for the understanding, which is not picked up again in the report.

3.3.1 Multi-Storey Building in Ecoinvent

The ecoinvent database meta information of “building, multi-storey” (Tab. 3.4) includes all important and necessary information to use the data correctly.

Tab. 3.4 Ecoinvent database meta information for “building, multi-storey”

Name	building, multi-storey
Location	RER
Infrastructure Process	1
Unit	m3
Data Set Version	2.0
Included Processes	Includes the most important materials used and their disposal, the transportation of the parts to the building site and to the final disposal at the end of life. Also included is the requirement of electricity for construction, maintenance and demolition. Operation is not included.
Amount	1
Local Name	Gebäude, mehrstöckig
Synonyms	multi-story//house//Haus
General Comment to reference function	The combination of two concrete buildings, one built in 1927 and the other in 1972, makes this module a sort of average of existing multi-storey buildings. The life of the building is assumed to be 80 years. This module should not be used if its relative importance would be high in a certain environmental inventory.
Start Date	1995
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Certain elementary flows or intermediate product flows are RER, UCPTE or GLO modules
Technology text	The two concrete multi-storey buildings used to make this module were built in 1927 and 1972, but they can be assumed to reflect a sort of average of existing buildings as they are weighted.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Most data obtained by personal communication
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 3.1 shows all in- and output flows of the product “building, multi-storey”, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each value.

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
reinforced concrete, steel fraction	→	building, multi-storey (part 1)					metals	extraction	No	RER	reinforcing steel, at plant	7.17E+00	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
reinforced concrete, concrete fraction	→						construction materials	concrete	No	CH	concrete, normal, at plant	9.27E-02	m3	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
brick	→						construction materials	bricks	No	RER	brick, at plant	1.78E+02	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
mortar	→						mortar and plaster	production	No	CH	cement mortar, at plant	6.17E+01	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
glass	→						glass	construction	No	RER	flat glass, coated, at plant	3.08E+00	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
wood	→						wooden materials	extraction	No	RER	sawn timber, softwood, planed, kiln dried, at plant	6.17E-02	m3	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
steel	→						metals	extraction	No	RER	reinforcing steel, at plant	1.70E+01	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
aluminium	→						metals	extraction	No	RER	aluminium, production mix, at plant	8.48E+00	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
copper	→						metals	extraction	No	RER	copper, at regional storage	8.48E+00	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
mineral wool	→						insulation materials	production	No	CH	rock wool, packed, at plant	1.30E+00	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
pvc raw material	→						plastics	polymers	No	RER	polyvinylchloride, at regional storage	9.25E-01	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
pvc extrusion to foil	→						plastics	processing	No	RER	extrusion, plastic film	9.25E-01	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);

Fig. 3.1 Flows for “building, multi-storey” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
PE	building, multi-storey (part 2)			plastics	processing	No	RER	packaging film, LDPE, at plant	#BEZUG!	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
transportation from factory gate to building ground construction, maintenance and demolition with building machines			avarage distance assumption within Switzerland: 20km	transport systems	road	No	RER	transport, lorry >16t, fleet average	1.10E+01	tkm	estimation and calculation	1	2.55	(5,5,2,3,4,5);
electricity use for construction, maintenance and demolition			mainly construction (5 MJ/m3) and demolition (10 MJ/m3)	construction processes	machinery	No	GLO	diesel, burned in building machine	1.50E+01	MJ	estimation	1	1.63	(2,5,2,3,4,5);
				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.00E-01	kWh	personal communication	1	1.63	(2,5,2,3,4,5);
		reinforced concrete to final disposal		waste management	building demolition	No	CH	disposal, building, reinforced concrete, to final disposal	2.39E+02	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
		diverse steel parts to sorting plant		waste management	building demolition	No	CH	disposal, building, bulk iron (excluding reinforcement), to sorting plant	1.70E+01	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
		brick to final disposal		waste management	building demolition	No	CH	disposal, building, brick, to final disposal	1.78E+02	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
		mortar to final disposal		waste management	building demolition	No	CH	disposal, building, cement (in concrete) and mortar, to final disposal	6.17E+01	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
		glass to final disposal		waste management	building demolition	No	CH	disposal, building, glass sheet, to final disposal	3.08E+00	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
		aluminium to recycling							8.48E+00	kg	Mauch et al. (1996)			
		rock wool mat to final disposal		waste management	building demolition	No	CH	disposal, building, mineral wool, to final disposal	1.30E+00	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
		wood to final disposal	with u=20%: density 545 kg/m3)	waste management	building demolition	No	CH	disposal, building, waste wood, untreated, to final disposal	3.08E+01	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
		pvc to final disposal		waste management	building demolition	No	CH	disposal, building, polyvinylchloride products, to final disposal	9.25E-01	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
		pe to final disposal		waste management	building demolition	No	CH	disposal, building, PE sealing sheet, to final disposal	9.25E-01	kg	Mauch et al. (1996)	1	1.63	(2,5,2,3,4,5);
		copper to recycling							8.48E+00	kg	Mauch et al. (1996)			
		building, multi-storey		construction processes	buildings	Yes	RER	building, multi-storey	1.00E+00	m3				

Fig. 3.1 Flows for “building, multi-storey” and their representation in the ecoinvent database (cont.)

3.3.2 Steel Construction Building Hall in Ecoinvent

The ecoinvent database meta information of “building hall, steel construction” (Tab. 3.5) includes all important and necessary information to use the data correctly.

Tab. 3.5 Ecoinvent database meta information for “building hall, steel construction”

Name	building hall, steel construction
Location	CH
Infrastructure Process	1
Unit	m2
Data Set Version	2.0
Included Processes	Includes the most important materials used and their disposal, the transportation of the parts to the building site and to the final disposal at the end of life. Also included is the requirement of electricity for construction, maintenance and demolition. Operation is not included.
Amount	1
Local Name	Gebäude, Halle, Stahlkonstruktion
Synonyms	
General Comment to reference function	The module is based on a fictitious building hall with size: length 50 m, width 30 m, height 7 m, daylight opening 110 m2. The service life is assumed to be 50 years. This module should not be used if its relative importance would be high in a certain environmental inventory.
Start Date	2000
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Certain elementary flows or intermediate product flows are RER or GLO modules
Technology text	Example representing current metal industrial structures in Switzerland
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Most data obtained by personal communication
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 3.2 shows all in- and output flows of the product “building, hall, steel construction”, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each value.

General Flow information					Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
wall, masonry	building, steel construction with masonry and insulation (part 1)				construction materials	bricks	No	RER	brick, at plant	1.07E+02	kg	estimation, based on fictitious building	1	2.15	(4,5,1,1,5,5);
insulation glass wool				rock wool used as proxy, assumed service life: 20 years; hence, three replacements are envisaged.	insulation materials	production	No	CH	rock wool, packed, at plant	6.60E+00	kg	estimation, based on fictitious building	1	2.15	(4,5,1,1,5,5);
insulation mineral wool				assumed service life: 20 years; hence, three replacements are envisaged.	insulation materials	production	No	CH	rock wool, packed, at plant	6.60E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
insulation polystyrol				assumed service life: 20 years; hence, three replacements are envisaged.	insulation materials	production	No	RER	polystyrene foam slab, at plant	5.40E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
mortar					mortar and plaster	production	No	CH	cement mortar, at plant	2.10E+01	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
foundation, concrete					construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	3.00E-01	m3	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
steel pillars and beams					metals	extraction	No	RER	reinforcing steel, at plant	3.00E+01	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
reinforced concrete pillars and beams, steel part					metals	extraction	No	RER	reinforcing steel, at plant	2.00E-01	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
reinforced concrete pillars and beams, concrete part					construction materials	concrete	No	CH	concrete, normal, at plant	2.72E-03	m3	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
window, double glazing				assumed service life: 20 years; hence, three replacements are envisaged.	glass	construction	No	RER	flat glass, uncoated, at plant	4.50E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
aluminium window frame				assumed service life: 20 years; hence, three replacements are envisaged.	metals	extraction	No	RER	aluminium, production mix, at plant	3.00E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
facing, fiber cement board				assumed service life: 20 years; hence, three replacements are envisaged.	construction materials	coverings	No	CH	fibre cement corrugated slab, at plant	3.30E+01	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
roof, steel sheet				assumed service life: 30 years; hence, two replacements are envisaged. Module "reinforcing steel" used as proxy	metals	extraction	No	RER	reinforcing steel, at plant	3.20E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
roof, rolling steel sheet				assumed service life: 30 years; hence, two replacements are envisaged.	metals	processing	No	RER	sheet rolling, steel	3.20E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
roof, steel sheet, zink coating				assumed service life: 30 years; hence, two replacements are envisaged.	metals	processing	No	RER	zinc coating, coils	5.33E+00	m2	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);

Fig. 3.2 Flows for “building, hall, steel construction” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
transportation of all materials from factory gate to building ground	→	building, steel construction with masonry and insulation (part 2)			average distance for all materials: 100km	transport systems	road	No	RER	transport, lorry >16t, fleet average	9.77E+01	tkm	estimation and calculation	1	2.98	(5,5,1,1,5,5);
construction, maintenance and demolition with building machines	→				based on diesel consumption of multi-storey building	construction processes	machinery	No	GLO	diesel, burned in building machine	4.17E+00	MJ	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
electricity use for construction, maintenance and demolition	→				based on electricity consumption of multi-storey building	electricity	supply mix	No	CH	electricity, medium voltage, at grid	8.33E-02	kWh	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
			reinforced concrete to final disposal		Reinforcing steel and concrete are summed up	waste management	building demolition	No	CH	disposal, building, reinforced concrete, to final disposal	7.57E+02	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
			steel pillars and beams and roof steel to recycling			waste management	building demolition	No	CH	disposal, building, bulk iron (excluding reinforcement), to sorting plant	3.32E+01	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
			fibre cement shingle in landfill		assumption based on information of G. Doka: fibre cement can approximately be replaced by cement fibre slab	waste management	building demolition	No	CH	disposal, building, cement-fibre slab, to final disposal	3.30E+01	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
			brick in landfill			waste management	building demolition	No	CH	disposal, building, brick, to final disposal	1.07E+02	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
			mortar to final disposal			waste management	building demolition	No	CH	disposal, building, cement (in concrete) and mortar, to final disposal	2.10E+01	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
			glass in landfill			waste management	building demolition	No	CH	disposal, building, glass sheet, to final disposal	4.50E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
			aluminium to recycling								3.00E+00	kg	estimation by EMPA, based on fictitious building			
			polystyrol to municipal incineration			waste management	building demolition	No	CH	disposal, building, polystyrene isolation, flame-retardant, to final disposal	5.40E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
			glass wool mat to final disposal			waste management	building demolition	No	CH	disposal, building, mineral wool, to final disposal	6.60E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
			rock wool mat to final disposal			waste management	building demolition	No	CH	disposal, building, mineral wool, to final disposal	6.60E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
			building, hall, steel construction			construction processes	buildings	Yes	CH	building, hall, steel construction	1.00E+00	m2				

Fig. 3.2 Flows for “building, hall, steel construction” and their representation in the ecoinvent database (cont.)

3.3.3 Wood Construction Building Hall in Ecoinvent

The ecoinvent database meta information of “building hall, wood construction” (Tab. 3.6) includes all important and necessary information to use the data correctly.

Tab. 3.6 Ecoinvent database meta information for “building hall, wood construction”

Name	building hall, wood construction
Location	CH
Infrastructure Process	1
Unit	m2
Data Set Version	2.0
Included Processes	Includes the most important materials used and their disposal, the transportation of the parts to the building site and to the final disposal at the end of life. Also included is the requirement of electricity for construction, maintenance and demolition. Operation is not included.
Amount	1
Local Name	Gebäude, Halle, Holzkonstruktion
Synonyms	
General Comment to reference function	The module is based on a fictitious building hall with size: length 50 m, width 30 m, height 7 m, daylight opening 110 m2. The service life is assumed to be 50 years. This module should not be used if its relative importance would be high in a certain environmental inventory.
Start Date	2000
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Certain elementary flows or intermediate product flows are RER or GLO modules
Technology text	Example representing current wooden industrial structures in Switzerland
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Most data obtained by personal communication
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 3.3 shows all in- and output flows of the product “building, hall, wood construction”, the allocation to the modules in ecoinvent database, the sources (mainly Mauch et al. (1995)) and the standard deviation for each value.

General Flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
primary beams, glued laminated timber beams	building, wood construction with timber board and insulation (part 1)			wood density	wooden materials	extraction	No	RER	glued laminated timber, indoor use, at plant	2.00E-02	m3	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
secondary beams, glued laminated timber beams					wooden materials	extraction	No	RER	glued laminated timber, indoor use, at plant	3.00E-02	m3	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
tertiary beams, structural timber beams					wooden materials	extraction	No	RER	sawn timber, softwood, raw, kiln dried, u=20%, at plant	4.00E-02	m3	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
wall: framework construction					wooden materials	extraction	No	RER	sawn timber, softwood, raw, kiln dried, u=20%, at plant	2.00E-02	m3	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
wall: facing (outside)				assumed service life: 20 years	wooden materials	extraction	No	RER	sawn timber, softwood, raw, kiln dried, u=20%, at plant	4.50E-02	m3	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
wall: facing (inside)				assumed service life: 30 years	wooden materials	extraction	No	RER	particle board, indoor use, at plant	3.00E-02	m3	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
insulation glass wool				rock wool used as proxy, assumed service life: 20 years	insulation materials	production	No	CH	rock wool, packed, at plant	6.60E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
insulation mineral wool				assumed service life: 20 years	insulation materials	production	No	CH	rock wool, packed, at plant	6.60E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
insulation polystyrol				assumed service life: 20 years	insulation materials	production	No	RER	polystyrene foam slab, at plant	5.40E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
foundation, concrete					construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	3.00E-01	m3	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
wooden pillars					wooden materials	extraction	No	RER	sawn timber, softwood, planed, kiln dried, at plant	6.00E-03	m3	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
window, double glazing				assumed service life: 20 years	glass	construction	No	RER	flat glass, uncoated, at plant	4.50E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
aluminium window frame				assumed service life: 20 years	metals	extraction	No	RER	aluminium, production mix, at plant	3.00E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
roof, steel sheet				assumed service life: 30 years, module "reinforcing steel" used as proxy.	metals	extraction	No	RER	reinforcing steel, at plant	3.20E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
roof, rolling of steel sheet				assumed service life: 30 years	metals	processing	No	RER	sheet rolling, steel	3.20E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
roof, steel sheet, zink coating				assumed service life: 30 years	metals	processing	No	RER	zinc coating, coils	5.33E+00	m2	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);

Fig. 3.3 Flows for “building, hall, wood construction” and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
steel parts	building, wood construction with timber board and insulation (part 2)			metals	extraction	No	RER	reinforcing steel, at plant	1.00E+01	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
rolling of steel parts				metals	processing	No	RER	sheet rolling, steel	1.00E+01	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
steel parts, zink coating				metals	processing	No	RER	zinc coating, coils	6.37E-01	m2	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
transportation from factory gate to building ground			avarage distance assumption for Switzerland: 105km	transport systems	road	No	RER	transport, lorry >16t, fleet average	9.75E+01	tkm	estimation and calculation	1	2.80	(4,5,1,1,5,5);
construction, maintenance and deconstruction with building machine			based on diesel consumption of multi-storey building	construction processes	machinery	No	GLO	diesel, burned in building machine	4.17E+00	MJ	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
electricity use for construction, maintenance and deconstruction			based on electricity consumption of multi-storey building	electricity	supply mix	No	CH	electricity, medium voltage, at grid	8.33E-02	kWh	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
		glued laminated timber to final disposal		waste management	building demolition	No	CH	disposal, building, fibre board, to final disposal	3.70E+01	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
		structural timber to final disposal		waste management	building demolition	No	CH	disposal, building, waste wood, untreated, to final disposal	7.99E+01	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
		particle board to final disposal		waste management	building demolition	No	CH	disposal, building, fibre board, to final disposal	2.22E+01	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
		roof steel to sorting plant		waste management	building demolition	No	CH	disposal, building, bulk iron (excluding reinforcement), to sorting plant	1.32E+01	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
		reinforced concrete to final disposal		waste management	building demolition	No	CH	disposal, building, reinforced concrete, to final disposal	7.50E+02	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
		glass in landfill		waste management	building demolition	No	CH	disposal, building, glass sheet, to final disposal	4.50E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
		aluminium to recycling							3.00E+00	kg	estimation by EMPA, based on fictitious building	1	1.05	(,...);
		polystyrol to municipal incineration		waste management	building demolition	No	CH	disposal, building, polystyrene isolation, flame-retardant, to final disposal	5.40E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
		glass wool mat to final disposal		waste management	building demolition	No	CH	disposal, building, mineral wool, to final disposal	6.60E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
		rock wool mat to final disposal		waste management	building demolition	No	CH	disposal, building, mineral wool, to final disposal	6.60E+00	kg	estimation by EMPA, based on fictitious building	1	2.15	(4,5,1,1,5,5);
		building, hall, wood construction		construction processes	buildings	Yes	CH	building, hall, wood construction	1.00E+00	m2				

Fig. 3.3 Flows for “building, hall, wood construction” and their representation in the ecoinvent database (cont.)

3.3.4 Building Hall in Ecoinvent

The ecoinvent database meta information of “building, hall” (Tab. 3.7) includes all important and necessary information to use the data correctly.

Tab. 3.7 Ecoinvent database meta information for “building, hall”

Name	building, hall
Location	CH
Infrastructure Process	1
Unit	m2
Data Set Version	2.0
Included Processes	Includes a combination of the wooden building hall (30%) and the steel building hall (70%) to represent the estimated current Swiss share.
Amount	1
Local Name	Gebäude, Halle
Synonyms	
General Comment to reference function	This module should not be used if its relative importance would be high in a certain environmental inventory.
Start Date	2000
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	
Technology text	Estimated share of current wooden and steel halls in Switzerland.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Most data obtained by personal communication
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 3.4 shows all in- and output flows of the product “building hall”, the allocation to the modules in ecoinvent database, the sources and the standard deviation for each value.

3.4 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed information compare the different flow charts and the ecoinvent database meta information.

Most data on the steel and wood construction building hall and the share of these for the general module building hall are estimations and therefore the modules are of poor overall quality. It is advisable not to use any of these modules if they are of high importance for a specific inventory.

The data for the multi-storey building are based on two specific buildings, one in Switzerland and one in Germany. The average for Western Europe based on only this information is of poor overall quality. It is advisable not to use the modules “building, multi-storey” if it is of high importance for a specific inventory.

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
building, hall, wood construction	→	undefined building hall				estimated share: 30% wood construction, 70% steel construction	construction processes	buildings	Yes	CH	building, hall, wood construction	3.00E-01	m2	Estimation EMPA	1	3.14	(4,5,1,1,1,5);
building, hall, steel construction	→					estimated share: 30% wood construction, 70% steel construction	construction processes	buildings	Yes	CH	building, hall, steel construction	7.00E-01	m2	Estimation EMPA	1	3.14	(4,5,1,1,1,5);
					building, hall		construction processes	buildings	Yes	CH	building, hall	1.00E+00	m2				

Fig. 3.4 Flows for “building hall” and their representation in the ecoinvent database

4 Explosives and Blasting

4.1 Acknowledgement

Thanks to O. Ringgenberg from Gasser AG in Lungern and Mr. M. Delaloye from SSE (Societe Suisse des Explosives), Switzerland for his useful help.

4.2 Introduction

The area of action of explosives can be divided in two parts: Underground and quarrying.

Explosive can be separated in the following categories (Grisiger et al. (2000)):

- Homogeneous explosives: Blasting oil, nitro glycol, nitropenta, trinitrophenol, trinitro-toluol (TNT) and hexogen.
- Mixture of explosives: Black blasting powder, gelatinous permitted explosives, powder permitted explosives, slurries, water gel explosives and emulsion explosives.
- Detonator: Mercury fulminate, lead azide.

The data for explosives mentioned in Frischknecht et al. (1996)) refer to an explosive named „Gamsit“. This is a gelatinous explosive, which contains nitro-glycerine and nitro glycol (Grisiger et al. (2000)). In Switzerland „Gamsit“ is no longer produced but there is still a certain amount being imported (Ringgenberg (2002)). The SUVA declares it use generates bad working hygiene, so that „Gamsit“ will no longer be used in 10 to 20 years.

In general it is fact that the market of explosives changes very quickly. Nowadays the explosives „Tovex“ and „Amolit/Anfolit“ are used for surface mining with great spreading in Switzerland Ringgenberg (2002). In a few years it can be expected that explosives on the base of emulsions will dominate the market.

The only explosive for which data are available is „Tovex“. Therefore it represents the module „explosives“. The limited relevance of explosives for the environmental inventory of building processes and products justifies this simplification.

The percentage of the Swiss market covered by the mixture of the explosives „Tovex“ and „Amolit/Anfolit“ was not found out.

4.3 Characterisation

The most common explosive nowadays is a mixture of „Tovex“ (slurry explosive) in the foot charge and „Amolit“ or „Anfolit“ in the leg charge of a borehole. The foot charge is loaded always up to 3 to 5 meters from bottom. The rest (leg charge) is filled with „Amolit“ or „Anfolit“.

Tab. 4.1 shows the composition and some characteristics of „Tovex“. Further characterisation as calculated and measured parameters (e.g. temperature of explosion, volume of the gas, density, speed of detonation, etc.) are published in (Grisiger et al. (2000)).

4.4 Production Process

The production process is described based on a personal communication (Ringgenberg (2002)):

In a two-storey production hall of an area of about 20 to 30 meter the raw materials are stored in the second floor. By free fall these materials arrive in the mixer on the first floor. The mixer has a diameter of about 3m. It is made out of chromium steel. Its charge is about 1'500 kg and each mixing procedure takes about 15 minutes. After having been well mixed, the mass goes in a hot basin to keep it sleek. After that it will be put in sausages-shape by a cartridging machine and then packed in card-

board boxes. By conveyor belt the boxes are stapled on pallets and then stored. The area of the storage halls are assumed to be 2'000 m².

Tab. 4.1 Composition of the explosive "Tovex" ⁷

Components	Formula	Share [%]	Chosen share [%]
Methylamine nitrate	CH ₆ N ₂ O ₃	23 - 34	23
Ammonium nitrate	NH ₄ NO ₃	24 - 43	24
Sodium nitrate	NaNO ₃	11 - 30	17.1
Calcium nitrate	Ca(NO ₃) ₂	0 - 20	16
Guar flour	(C ₁₂ O ₁₁ H ₂₂) _n	0.9 – 1.1	0.9
Aluminium powder	Al	0 - 11	8
Water	H ₂ O	8 - 13	8
Glass		2 - 4	3
Total			100
Parameter	Amount	Unit	
Density	0.8 – 1.35	g/cm ³	
Oxygen demand	0 – 1	%	
Energy	680 – 1200	kcal/kg	
Gas	640 – 950	Litre/kg	
Temperature	1900 – 2900	°C	
Pressure	10 – 25	kbar	

4.5 Blasting

As no information on the emissions produced by the blasting of Tovex was found, a plausible chemical reaction has been drawn up and the masses calculated. Following Tab. 4.2 shows the inputs and some possible outputs.

Tab. 4.2 Possible Input materials and corresponding emissions from blasting with Tovex

Input	Mol	Weight [g]	Output	Mol	Weight [g]
CH ₆ N ₂ O ₃	0.245	23	NO _x	1.139	33.1
NH ₄ NO ₃	0.300	24	NH ₃	0.347	5.9
2 NaNO ₃	0.201	17.1	O ₂	0.010	0.3
Ca(NO ₃) ₂	0.098	16	CO	0.276	7.7
(C ₁₂ O ₁₁ H ₂₂) _n	0.002	0.9	H ₂ O	1.287	23.2
Al	0.296	8	Na ₂ O	0.101	6.2
H ₂ O	0.444	8	CaO	0.098	5.5
Total		97	Al ₂ O ₃	0.148	15.1
			Total		97

⁷ Written information handed over by Mr. O. Ringgenberg from Gasser AG, 02.09.2002 at EMPA

The chemical reaction is stoichiometric correct but some possible emissions as CO₂ and carbon hydride are not taken into account. Dust emissions are also not taken into account on one hand because no data are available and on the other hand it is assumed that the amount is negligible compared to the dust from breaking rock. No detonator has been taken account because no data are available.

4.6 Explosive “Tovex” in Ecoinvent

The ecoinvent database meta information for the “explosives, Tovex, at plant” (Tab. 4.3) includes all important and necessary information to use the data correctly.

Tab. 4.3 Ecoinvent database meta information for “Explosives, Tovex, at plant”

Name	explosives, tovox, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce Tovex (raw material provision, mixing process, packing, internal processes (transport, etc.) and infrastructure.
Amount	1
Local Name	Sprengstoff, Tovex, ab Werk
Synonyms	blasting//agent
General Comment to reference function	As almost no data are available most information is assumed. Only the information on the raw material is handed over in written form. This module should not be used if its relative importance would be high in a certain environmental inventory.
Start Date	1997
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER- and DE-modules have been used as proxy
Technology text	The basic materials are poured by gravity in a mixer. When filled with 1'500kg the mixing procedure starts and lasts about 15min. To keep it warm it's filled in a hot basin from where it is put in sausages by a cart ridging machine and then packed in card board boxes. A conveyor belt transports the boxes and puts it on pallets for storing
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Share of basic material by personal communication. All other In- and Output modules are estimations
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 4.1 shows all in- and output flows of the product “explosives, tovox, at plant”, the allocation to the modules in ecoinvent database, the sources (mainly personal communication and estimations) and the standard deviation for each value.

General Flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
methylamine nitrate	→ production of Tovex (part 1)			23 - 34% of 1 kg endproduct, assumption: 23%; "ammonium nitrate" used as proxy, 33% of total weight is N	agricultural means of production	mineral fertiliser	No	RER	ammonium nitrate, as N, at regional storehouse	7.67E-02	kg	personal communication Mr. O. Ringgenberg, Gasser AG, 02.09.2002	1	1.68	(4,5,5,1,1,5);
ammonium nitrate				24 - 43% of 1 kg endproduct, assumption: 24%, 2.86 kg ammonium nitrate for 1 kg "ammonium nitrate as N"	agricultural means of production	mineral fertiliser	No	RER	ammonium nitrate, as N, at regional storehouse	8.39E-02	kg	personal communication Mr. O. Ringgenberg, Gasser AG, 02.09.2002	1	1.68	(4,5,5,1,1,5);
sodium nitrate				11 - 30% of 1 kg endproduct, assumption: 17.1%; "calcium nitrate" used as proxy, 17% of total weight is "calcium nitrate, as N"	agricultural means of production	mineral fertiliser	No	RER	calcium nitrate, as N, at regional storehouse	2.92E-02	kg	personal communication Mr. O. Ringgenberg, Gasser AG, 02.09.2002	1	1.68	(4,5,5,1,1,5);
calcium nitrate				0 - 20% of 1 kg endproduct, assumption: 16%, 17% of total weight is "calcium nitrate, as N"	agricultural means of production	mineral fertiliser	No	RER	calcium nitrate, as N, at regional storehouse	2.74E-02	kg	personal communication Mr. O. Ringgenberg, Gasser AG, 02.09.2002	1	1.68	(4,5,5,1,1,5);
guar flour				0.9 - 1.1% of 1 kg endproduct, assumption: 0.9%; "potato starch" used as proxy	agricultural production	plant production	No	DE	potato starch, at plant	9.00E-03	kg	personal communication Mr. O. Ringgenberg, Gasser AG, 02.09.2002	1	1.68	(4,5,5,1,1,5);
aluminium powder				0 - 11% of 1 kg endproduct, assumption: 8%	metals	extraction	No	RER	aluminium, production mix, at plant	8.00E-02	kg	personal communication Mr. O. Ringgenberg, Gasser AG, 02.09.2002	1	1.68	
water					water supply	production	No	RER	tap water, at user	8.00E-02	kg	personal communication Mr. O. Ringgenberg, Gasser AG, 02.09.2002	1	1.68	
glass				2 - 4% of 1 kg endproduct, assumption: 3%	glass	construction	No	RER	flat glass, uncoated, at plant	3.00E-02	kg	personal communication Mr. O. Ringgenberg, Gasser AG, 02.09.2002	1	1.68	(4,5,5,1,1,5);

Fig. 4.1 Flows for "explosives, tovex, at plant" and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
road transports of raw materials to plant	production of Tovex (part 2)		assumptions: 50% of transport of raw material with lorry 40 t, average distance: 100 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	5.00E-02	tkm	assumption	1	2.32	(5,5,1,1,1,5);
train transports of raw materials to plant			assumption 50% of transport of raw material with freight train (assumption), average distance: 100km (assumption)	transport systems	train	No	CH	transport, freight, rail	5.00E-02	tkm	assumption	1	2.32	(5,5,1,1,1,5);
heating oil for buildings			SIA 380/1: 260MJ/m2/a (mix of heating energy for production and administration buildings), area: 1'800m2, assumed yearly production amount: 3'000t (total factory outlet; 2'100t of other products)	oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.56E-01	MJ	assumption	1	1.68	(4,5,1,1,4,5);
explosive production plant			assumed total yearly production amount of factory (sum of TOVEX and other products): 3'000t, assumed service-life: 50 years	construction materials	others	Yes	CH	explosive production plant	6.67E-09	unit	assumption	1	3.99	(5,5,1,1,5,5);
diesel oil in fork lift truck			internal transports, based on diesel consumption of rock-wool production	construction processes	machinery	No	GLO	diesel, burned in building machine	2.14E-02	MJ	assumption	1	2.15	(4,5,1,1,5,5);
electricity consumption			mostly for heating of the warming basin, assumption of total consumption: 500MJ/m2/a (based on SIA 380/4, Food Warehouse), area: 1'800m2, assumed yearly production amount: 3'000t	electricity	supply mix	No	CH	electricity, medium voltage, at grid	8.33E-02	kWh	assumption	1	1.68	(4,5,1,1,4,5);
		waste heat of electricity consumption	waste heat of electricity consumption	air	unspecified			Heat, waste	3.00E-01	MJ	assumption	1	1.68	(4,5,1,1,4,5);
		Particulate matter	total of Particulates, < 2.5 um from Tovex production; no data and assumed that neglectable						-		assumption			
		further emissions	no data						-		assumption			

Fig. 4.1 Flows for “explosives, tovex, at plant” and their representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent							Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
Infrastructure packing and loading of Tovax	production of Tovex (part 3: packing)		weight of packing- and loading-machines: assumption 10t; assumed total amount of produced explosive per year: 3'000t; assumed service-life: 20 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.67E-04	kg	assumption	1	3.32	(5,5,1,1,1,5);
electricity consumption			electricity consumption for packing and loading machines is based on the consumption for packing and loading of lime products: total consumption: 3kWh/t	electricity	supply mix	No	CH	electricity, medium voltage, at grid	3.00E-03	kWh	assumption	1	2.15	(4,5,1,1,5,5);
paletts, at plant			assumed loading weight: 500kg, cycle of reuse: 5 times	wooden materials	processing	No	RER	EUR-flat pallet	4.00E-04	unit	assumption	1	1.68	(4,5,1,1,4,5);
cardboard, at plant			cardboard for packing of diverse products: assumption: 1kg per pallet, assumed load per pallet: 500kg	paper & cardboard	cardboard & corrugated board	No	RER	corrugated board, mixed fibre, single wall, at plant	2.00E-03	kg	assumption	1	1.62	(5,5,1,1,1,5);
transports of paletts and cardboard to plant by lorry			assumption: transport of pallet and cardboard with lorry 28t, average distance: 100km (assumption)	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	2.40E-04	tkm	assumption	1	2.32	(5,5,1,1,1,5);
administration: paper consumption			based on "rockwool"-module; assumption: 20'000kg/a, 900t of explosives is produced and packed yearly	paper & cardboard	graphic paper	No	RER	paper, woodcontaining, LWC, at plant	2.22E-02	kg	assumption	1	2.33	(5,5,1,1,5,5);
water from administration and other diverse utilizations			based on "rockwool"-module; assumption: 5'000'000 kg/a, 900t of explosives is produced and packed yearly	water supply	production	No	RER	tap water, at user	5.56E+00	kg	assumption	1	2.33	(5,5,1,1,5,5);
		waste heat	waste heat of electricity consumption for packing	air	unspecified			Heat, waste	1.08E-02	MJ	assumption	1	2.15	(4,5,1,1,5,5);
		municipal solid waste	Assumption: 1/4 of packing material (cardboard and paper) after delivery and unpacking of the explosive	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	6.06E-03	kg	assumption	1	1.62	(5,5,1,1,1,5);
		waste paper	rest of paper: assumption to recycling								assumption			
		waste water	same amount as water used	waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 3	5.56E-03	m3	assumption	1	1.62	(5,5,1,1,1,5);
		paletts to municipal incineration	not balanced, as it is recycled								assumption			
		cardboard, in municipal waste incineration	assumption: 3/4 of cardboard used	waste management	municipal incineration	No	CH	disposal, packaging cardboard, 19.6% water, to municipal incineration	1.50E-03	kg	assumption	1	1.62	(5,5,1,1,1,5);
		Transports of paper bags and plastic bags to municipal incineration	standard distance: 10km	waste management	others	No	CH	transport, municipal waste collection, lorry 21t	7.56E-02	tkm	assumption	1	2.32	(5,5,1,1,1,5);
		explosives, tovex, at plant	yearly production amount: 900'000 kg/a	chemicals	inorganics	No	CH	explosives, tovex, at plant	1.00E+00	kg	personal communication, Mr. Delaloye, 29.04.2003			

Fig. 4.1 Flows for "explosives, tovex, at plant" and their representation in the ecoinvent database (cont.)

4.7 Explosive Production Plant in Ecoinvent

The ecoinvent database meta information of an “explosive production plant” (Tab. 4.4) includes all important and necessary information to use the data correctly.

Tab. 4.4 Ecoinvent database meta information for a explosive production plant

Name	explosive production plant
Location	CH
Infrastructure Process	1
Unit	unit
Data Set Version	2.0
Included Processes	Includes the infrastructure (buildings multi-storey and halls and equipment) and the land-use (built and not built areas)
Amount	1
Local Name	Sprengstoffwerk
Synonyms	blasting//agent//
General Comment to reference function	All data are estimations based on personal communication and aerial photographs. This module should not be used if its relative importance would be high in a certain environmental inventory.
Start Date	2000
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER- and DE-modules have been used as proxy
Technology text	Assumption that data valid for average Swiss plant
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	All other In- and Output modules are estimations
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 4.2 shows all in- and output flows of the product “explosive production plant”, the allocation to the modules in ecoinvent database, the sources (mainly assumptions) and the standard deviation for each value.

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
land occupation for built area	→	explosive production plant				assumption 2'600 m2, assumption for time occupation: 50 years	resource	land			Occupation, industrial area, built up	1.30E+05	m2a	assumption	1	1.67	(4,5,1,3,1,5);
land occupation for road area	→					assumption total area with roads: 200 m2; assumption for time occupation: 50 years	resource	land			Occupation, traffic area, road network	1.00E+04	m2a	assumption	1	1.67	(4,5,1,3,1,5);
land occupation for green area	→					assumption total green area: 200 m2; assumption for time occupation: 50 years	resource	land			Occupation, industrial area, vegetation	1.00E+04	m2a	assumption	1	1.67	(4,5,1,3,1,5);
land transformation to built area	→					assumption total area under roof: 2'600m2	resource	land			Transformation, to industrial area, built up	2.60E+03	m2	assumption	1	2.14	(4,5,1,3,1,5);
land transformation to roads	→					assumption total area with roads: 200m2	resource	land			Transformation, to traffic area, road network	2.00E+02	m2	assumption	1	2.14	(4,5,1,3,1,5);
land transformation to green area	→					assumption total green area: 200m2	resource	land			Transformation, to industrial area, vegetation	2.00E+02	m2	assumption	1	2.14	(4,5,1,3,1,5);
land transformation from unknown	→					assumption total area: 3'000m2	resource	land			Transformation, from unknown	3.00E+03	m2	assumption	1	2.14	(4,5,1,3,1,5);
buildings multi storage	→					two storey hall: assumption concrete structure, assumption height: 12 m	construction processes	buildings	Yes	RER	building, multi-storey	7.20E+03	m3	assumption	1	3.14	(4,5,1,3,1,5);
buildings hall	→					Assumption: 2000 m2	construction processes	buildings	Yes	CH	building, hall	2.00E+03	m2	assumption	1	3.14	(4,5,1,3,1,5);
infrastructure machines	→					weight assumption for the infrastructure for the whole process (storing, mixing, putting in cartridges, packing, etc.): 500 t (mostly steel) (Mixer is of chromium steel); life-span assumption: 25 years.	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.00E+06	kg	assumption	1	3.14	(4,5,1,3,1,5);
infrastructure conveyor belt	→					assumption length 500 m, width: 1 m, life-span 15 years; the corresponding module in Ecoinvent has a width of 3 m	construction processes	machinery	Yes	RER	conveyor belt, at plant	2.50E+02	m	assumption	1	3.14	(4,5,1,3,1,5);
						not balanced because it is entirely reused									assumption		
					Assumed total produced amount per year and factory (sum explosives and other products): 5'000t	construction materials	others	Yes	CH	explosive production plant	1.00E+00	unit					

Fig. 4.2 Flows for “explosive production plant” and their representation in the ecoinvent database

4.8 Blasting Process in Ecoinvent

The ecoinvent database meta information of the process “blasting” (Tab. 4.5) includes all important and necessary information to use the data correctly.

Tab. 4.5 Ecoinvent database meta information for blasting

Name	blasting
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the raw material (explosive Tovex) and the emissions (calculated stoichiometric)
Amount	1
Local Name	Sprengen
Synonyms	blasting//agent//Tovex
General Comment to reference function	All emissions are calculated based on the composition of the raw materials and completed by the report BUWAL (2000a). No emissions of the blasted rock is included. This module should not be used if its relative importance would be high in a certain environmental inventory.
Start Date	1990
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	The explosive Tovex is a CH-module and is used as proxy for RER
Technology text	Blasting of 100% Tovex
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Emissions are based on stoichiometric calculations and all other In- and Out-put modules are estimations
Extrapolations	See geography
Uncertainty Adjustments	None

Fig. 4.3 shows all in- and output flows of the product “blasting”, the allocation to the modules in ecoinvent database, the sources (mainly calculations and BUWAL (2000a)) and the standard deviation for each value.

4.9 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

Most data are estimations, many chemicals are described with proxies, dust emissions are missing and therefore all explosive modules are of poor overall quality. It is advisable not to use the modules if they are of high importance for a specific inventory.

General Flow information					Representation in ecoinvent							Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
explosiv	blasting			chemicals	inorganics	No	CH	explosives, tovox, at plant	1.00E+00 kg			1	2.12	(5,5,5,3,4,5);
detonator			for detonation a detonator is necessary; no data, no module											
electricity consumption			for detonation electricity is necessary; no data	electricity	supply mix	No	CH	electricity, medium voltage, at grid	-	kWh				
		Nitrogen oxide to air	calculated data	air	unspecified			Nitrogen oxides	3.31E-01 kg	Calculation		1	1.68	(4,5,5,1,1,5);
		ammonia to air	calculated data	air	unspecified			Ammonia	5.90E-02 kg	Calculation		1	1.92	(4,5,5,1,1,5);
		Oxygen to air							3.00E-03 kg	Calculation		1	1.92	(4,5,5,1,1,5);
		Carbon monoxide to air	calculated data	air	unspecified			Carbon monoxide, fossil	7.70E-02 kg	Calculation		1	1.68	(4,5,5,1,1,5);
		water to air	calculated data	air	unspecified			Water	2.32E-01 kg	Calculation		1	1.92	(4,5,5,1,1,5);
		sodium oxide to air	no exchange in ecoinvent						6.20E-02 kg	Calculation		1	1.92	(4,5,5,1,1,5);
		Calcium oxide to air	no exchange in ecoinvent						5.50E-02 kg	Calculation		1	1.92	(4,5,5,1,1,5);
		Aluminum oxide to air	aluminum used as proxy	air	unspecified			Aluminum	1.51E-01 kg	Calculation		1	1.92	(4,5,5,1,1,5);
		NMVOC to air	data from 1990 and for another kind of explosive called "explosives civil"	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	6.00E-02 kg	BUWAL (2000) Emissionsfaktoren für stationäre Quellen		1	2.34	(2,5,3,1,5,5);
		dust to air [kg]	data TSP from 1990 and for another kind of explosive called "explosives civil"; Particulates, < 2.5 um are assumed 90% of TSP	air	unspecified			Particulates, < 2.5 um	5.40E-03 kg	BUWAL (2000) Emissionsfaktoren für stationäre Quellen		1	3.78	(2,5,3,1,5,5);
		dust to air [kg]	data TSP from 1990 and for another kind of explosive called "explosives civil"; Particulates, > 2.5 um, and < 10 um are assumed 5% of TSP	air	unspecified			Particulates, > 2.5 um, and < 10um	3.00E-04 kg	BUWAL (2000) Emissionsfaktoren für stationäre Quellen		1	2.77	(2,5,3,1,5,5);
		dust to air [kg]	data TSP from 1990 and for another kind of explosive called "explosives civil"; Particulates, > 10 um are assumed 5% of TSP	air	unspecified			Particulates, > 10 um	3.00E-04 kg	BUWAL (2000) Emissionsfaktoren für stationäre Quellen		1	2.34	(2,5,3,1,5,5);
		blasting		construction processes	civil engineering	No	RER	blasting	1.00E+00 kg					

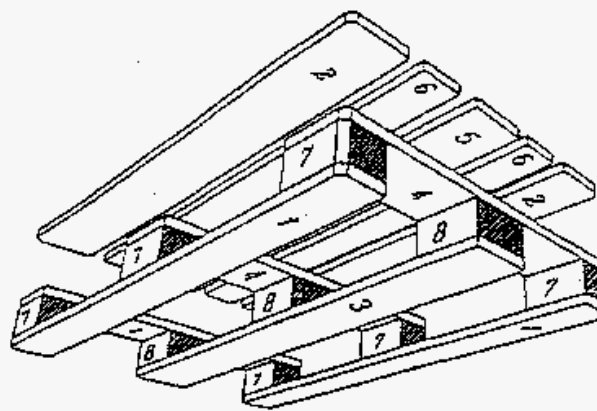
Fig. 4.3 Flows for "blasting" and their representation in the ecoinvent database

5 EUR-flat pallet

The European Pallet Association (EPAL) defines the geometry of a standard EUR-flat pallet as described in Fig. 5.1.

Part	Component ¹⁾	Number of components	Dimensions at 22 % moisture content		
			Länge	Breite	Dicke
1	Bottom deck lead board	2	1 200 ± 3	100 ± 3	22 + 2/0
2	Top deck lead board	2	1 200 ± 3	145 +5/-3	22 + 2/0
3	Central bottom deck board	1	1 200 ± 3	145 +5/-3	22 + 2/0
4	Stringer board	3	800 ± 3	145 + 5/-3	22 + 2/0
5	Central top deck board	1	1 200 ±	145 + 5/-3	22 + 2/0
6	Intermediate top deck board	2	1 200 ± 3	100 ± 3	22 + 2/0
7	outer skid block	6	145 +5/-3	100 ± 3	78 + 1/0
8	Centre skid block	3	145 +5/-3	145 + 5/-3	78 + 1/0

¹⁾ See figure below



800 mm x 1200 mm pallet component parts

Fig. 5.1 Definition of a EUR-flat pallet ⁸

The quality label EPAL prescribes 78 nails and their arrangement to join the various parts. The weight of the nails is estimated based on information from Swiss Nails⁹, which reports an average number of 400 nails per kg weight. According to (Ulitzsch (2001)) the EUR-flat pallets are not treated with wood preservatives and other chemicals. EPAL quality label prescribes a wood humidity of about 22%. The corner and middle bearer are made out of glued pressed particle blocks. The average weight of a EUR-flat pallet is about 22 kg¹⁰. All these pieces of information have been taken into account in the module “EUR-flat pallet”.

The ecoinvent database meta information of “EUR-flat pallet” (Tab. 5.1) includes all important and necessary information to use the data correctly.

⁸ <http://www.epal-pallets.org/engl/framee.htm>

⁹ <http://www.opd.ch/page.php?id=1218>

¹⁰ E-Mail from T. Ulitzsch, Gütegemeinschaft Paletten, www.gpal.de

Tab. 5.1 Ecoinvent database meta information for the product “EUR-flat pallet”

Name	EUR-flat pallet
Location	RER
Infrastructure Process	0
Unit	unit
Data Set Version	2.0
Included Processes	Includes only the materials and not the process of construction. The examined system is from gate to gate and as in most cases the pallets have a long life-span the waste treatment is not included. It must be included in the packing module
Amount	1
Local Name	EUR-Flachpalette
Synonyms	stillage//Palett
General Comment to reference function	Module includes only materials (no production process) and should only be used for the packing and transportation of products.
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	
Technology text	Standard composition of materials
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Internet search
Extrapolations	
Uncertainty Adjustments	None

Fig. 5.2 shows all in- and output flows of the product “EUR-flat pallet”, the allocation to the modules in ecoinvent database, the sources (mainly the European Pallet Association) and the standard deviation for each data.

5.1 Data Quality

Most data concerning the EUR-Flat pallet are based on the description of the materials used without taking into account the assemblage energy. Therefore the data are of poor overall quality. It is advisable not to use this module if it is of high importance for a specific inventory.

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
wooden boards	production of "Euro flat-pallets"	EUR-flat pallet	volume: 0.0335m3, density: 495kg/m3; weight: 16.6kg	wooden materials	extraction	No	RER	sawn timber, softwood, raw, air dried, u=20%, at plant	3.35E-02	m3	http://www.gpal.de/1024/1024ie.htm , access Oct. 2003	1	1.05	(1,1,1,1,1,1);
pressed, glued particle wood			volume: 0.0117m3, density: 680kg/m3; weight: 8kg	wooden materials	extraction	No	RER	particle board, outdoor use, at plant	1.17E-02	m3	http://www.gpal.de/1024/1024ie.htm , access Oct. 2003	1	1.05	(1,1,1,1,1,1);
nails			78 nails per pallet; 400 pieces per kg: 0.0025kg/nail	metals	extraction	No	RER	steel, low-alloyed, at plant	1.95E-01	kg	http://www.gpal.de/1024/1024ie.htm , access Oct. 2003	1	1.22	(2,1,1,1,3,1);
				wooden materials	processing	No	RER	EUR-flat pallet	1.00E+00	unit	http://www.gpal.de/1024/1024ie.htm , access Oct. 2003			

Fig. 5.2 Flows for "EUR-flat pallet" and its representation in the ecoinvent database

6 Abbreviations

EMF	Emission factors (Emissionsfaktoren)
EMPA	Swiss Federal Laboratories for Materials Testing and Research (Eidgenössische Materialprüfungs- und Forschungsanstalt)
EPA	U.S. Environmental Protection Agency
EPAL	European Pallet Association
EUR	European
KF	Correction factors (Korrekturfaktoren)
PVC	Polyvinylchloride
PE	Polyethylene
SUVA	Swiss Laboratories for Accident Insurance (Schweizerische Unfallversicherungsanstalt)

7 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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- | | |
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[illegible]

Fig. 9.1 Emission factors coefficient of correction for Diesel engines from 1990 to 2030 based on capacity class (cont.)

(Szenario 1 gemäss TTM-Mitteilung vom 26.02.99 *)																										
		6	27.5	56	102.5	345	Mittel- Wert	Regression y=b*m^x		6	27.5	56	102.5	345		6	27.5	56	102.5	345		6	27.5	56	102.5	345
		Leistungsklassen in kW																								
		< 18	18-37	37-75	75-130	130-560	K M	m	b	Abschätzung mit der Regression						Vergleich Regression / Originalwerte						Vergleich Mittelwert / Originalwerte				
EMF_BE	1990	314	302	296	290	287	298																			
	1995	305	286	281	275	272	284																			
	2000	295	271	266	261	258	270																			
	2005	290	265	260	255	250	264																			
	2010	285	259	254	249	241	258																			
	2015	280	253	248	243	238	252																			
	2020	275	247	242	237	235	247																			
	2025	275	244	239	235	232	245																			
KF_BE	2030	275	241	236	232	229	243																			
	1990	1.00	1.00	1.00	1.00	1.00	1.00	1.000	1.000	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	
	1995	0.97	0.95	0.95	0.95	0.95	0.95	1.000	0.956	0.96	0.96	0.95	0.95	0.95	0.98	1.01	1.01	1.00	1.00	0.98	1.01	1.00	1.00	1.00	1.01	
	2000	0.94	0.90	0.90	0.90	0.90	0.91	1.000	0.912	0.91	0.91	0.91	0.91	0.89	0.97	1.02	1.01	1.01	1.00	0.97	1.01	1.01	1.01	1.01	1.01	
	2005	0.92	0.88	0.88	0.88	0.87	0.89	1.000	0.895	0.89	0.89	0.89	0.89	0.86	0.97	1.02	1.01	1.01	0.99	0.96	1.01	1.01	1.01	1.01	1.02	
	2010	0.91	0.86	0.86	0.86	0.84	0.86	1.000	0.877	0.88	0.87	0.87	0.86	0.84	0.97	1.02	1.01	1.01	0.99	0.95	1.01	1.01	1.01	1.01	1.03	
	2015	0.89	0.84	0.84	0.84	0.83	0.85	1.000	0.857	0.86	0.85	0.85	0.85	0.82	0.96	1.02	1.02	1.01	0.99	0.95	1.01	1.01	1.01	1.01	1.02	
	2020	0.88	0.82	0.82	0.82	0.82	0.83	1.000	0.837	0.84	0.83	0.83	0.83	0.81	0.96	1.02	1.02	1.02	0.99	0.95	1.01	1.01	1.01	1.01	1.01	
	2025	0.88	0.81	0.81	0.81	0.81	0.82	1.000	0.831	0.83	0.83	0.83	0.82	0.80	0.95	1.03	1.02	1.02	0.99	0.94	1.02	1.02	1.02	1.02	1.02	
2030	0.88	0.80	0.80	0.80	0.80	0.81	1.000	0.824	0.82	0.82	0.82	0.81	0.79	0.94	1.03	1.03	1.02	0.99	0.93	1.02	1.02	1.02	1.02	1.02		
1) Mittlere Leistungen pro Klasse (kW)																										
2) Berechnung gem. Offroad (A-B*P**0.1)																										
* Interne Mitteilung von Herrn A. Mayer, TTM 26.02.1999																										
EMF: Emissionsfaktoren																										
KF: Korrekturfaktoren																										

Fig. 9.1 Emission factors coefficient of correction for Diesel engines from 1990 to 2030 based on capacity class (cont.)

Part XIX

Bentonite Products and Processes

Data v2.0 (2007)

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Kellenberger D., Althaus H.-J., Jungbluth N., Künniger T., Lehmann M. and Thalmann P. (2007) Life Cycle Inventories of Building Products. Final report ecoinvent Data v2.0 No. 7. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.org.

Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

This report deals with the mining process of bentonite, the mine itself, the processing of bentonite and the recultivation of the mine. The inventory data include a production mix of acid activated bentonite (61%), alkaline activated bentonite (38%) and catalytic converters (1%).

The main part of the cumulated energy demand is due to the very thin bentonite layer (around 3 m), overlaid by a thick layer of overburden material (around 30 m) which has to be removed and carried back for recultivation.

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1 Introduction

Bentonite is an industrial mineral with a high content of swellable threelayer clay minerals, often originated by transformation of volcanic ashes and tuffs. The name originates from the first place where it was found, which is nearby Benton, Wyoming (USA). The exceptional characteristics are given by the clay mineral “montmorillonite”.

A large company in Germany called “Süd-Chemie AG” mines raw bentonite and produces alkaline and acid activated as well as rheological bentonite. This company provides the only data available.

2 Reserves and Resources

The largest occurrences of bentonite in Germany are in a geographical triangle between “Moosburg”, “Mainburg” and “Landshut” in the Bundesland Bayern (Süd-Chemie (1997)). It was formed in complex geological processes 12 million years ago. The lens-shaped, up to three-meter thick bentonite layers are covered by about 30 meter of overburden material.

Other important countries with large production of bentonite are listed in Tab. 2.1.

Tab. 2.1 Bentonite production by country in 1998 (Stockwell et al. (2000))

Most important countries	Amount [tonnes]
Greece	1'148'408
Italy	592'000
Russia	650'000
Turkey	565'708
USA	3'820'000
Germany	478'000
World total	10'170'000

3 Use

The most important uses of **alkaline** activated bentonite from “Süd-Chemie AG” are (Süd-Chemie (1997)):

- Foundry Bentonite: The high binding and sizing characteristics are used to produce casting moulds.
- Building Bentonite: The characteristics of suspensions with alkaline activated bentonite are of big importance for special underground working and tunnelling. It is used to stabilise sidewalls of foundation ditches or to replace lavish boarding. It is used for mineral waterproofing of building constructions, deposits, ponds and embankments. Other applications are as additives in the cement production, in concrete to raise the water tightness, etc..
- Wine Bentonite: Used to clarify (refining) wine.
- Paper Bentonite: Removes printing ink at the recycling process of waste paper.

The most important uses of **acid** activated bentonite from “Süd-Chemie AG” are:

- Bleaching earth: The most important product is the powdery bleaching earth used in oil mills to bleach alimentary oils.
- Paper Bentonite: It is used for the production of self-copying paper.

A more extensive list of possible applications is offered in the World Wide Web.¹

4 Characterisation

4.1 General

A very simple description of the characterisation of alkaline activated bentonite is found in the environmental report of “Süd-Chemie AG” (Süd-Chemie (1997)):

Alkaline activated bentonite is able to absorb a big quantity of water. In accordance it is swelling many times its original volume and has a jelly-like consistence.

Thus two important uses of alkaline activated bentonite are for thickening liquids or using the adhesive and binding force of dried bentonite gel.

Alkaline activated bentonite is thixotropic: If leaving the thickened liquid, which is treated with alkaline bentonite, it is going to be gel-like or even solid. By churning it returns to liquid state. This enables the production of non-dripping wall paint.

The description of acid activated bentonite is also taken from the environmental report of “Süd-Chemie AG” (Süd-Chemie (1997)):

Acid activated bentonite is also called bleaching earth. It is an absorber, which can absorb secondary substances in different liquids as fresh pressed oil or wine. The escort substances will stick at the bleaching earth, which is filtered and deposited. For bleaching one kg of sunflower oil one piled spoon of bleaching earth is required. The surface of the bleaching earth increases by a factor of five, from an area of about 60 up to about 300 m² per gram (about the size of a football ground) when activated.

4.2 Raw Materials

The main raw material used for the production of acid and alkaline activated bentonite is based on a clay mineral, digged in open pit mining.

For the alkaline activation soda is used as activator. In contact with water it is resulting in a slight alkaline and soapy solution. Other important aggregates of foundry bentonite are glance carbon and natural resins and other “renewable” raw materials (Süd-Chemie (1997)).

The activation of acid activated bentonite needs hydrochloric acid.

4.3 Production Process and System Boundary

The data used in this study include all processes taking place in the factory of “Süd-Chemie AG” in Moosburg, Germany (Süd-Chemie (1997)). The two main products acid and alkaline activated bentonite are described.

¹ <http://www.bentonit.de/>, access date: 04.06.03

4.3.1 Alkaline Activated Bentonite

For the alkaline activation raw bentonite is mixed with soda, then milled carefully and dried simultaneously. Depending on the application different shaping and mixing processes are performed before the final product is stored, packed and loaded.

The production process of alkaline activated bentonite is shown in Fig. 4.1

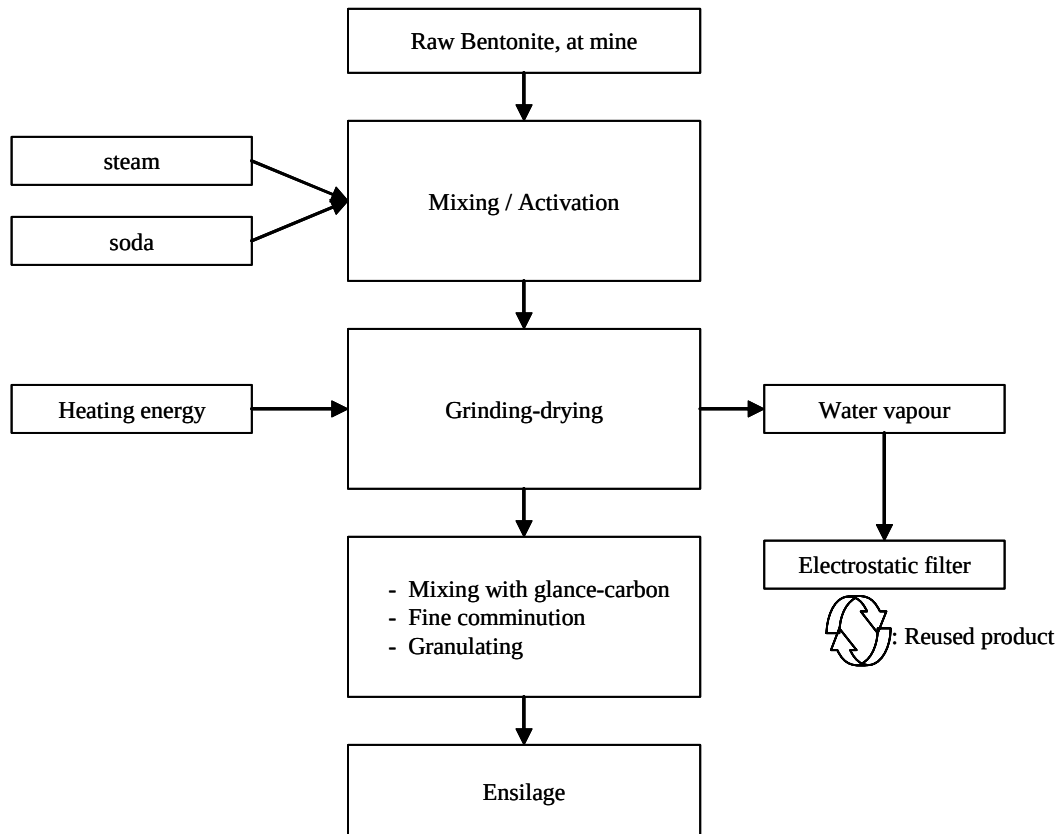


Fig. 4.1 Production process of alkaline activated bentonite (Süd-Chemie (1997))

4.3.2 Acid Activated Bentonite

The acid activated bentonite production process is more complicated. Raw bentonite is crushed, elutriated and cleared of sand and gravel, which are reused as building or filling material for gravel pits. The activation takes place in a wet-chemical process using hydrochloric acid: Bentonite gets purified in this process by leaching a big part of the enclosed aluminium, iron and trace elements as e.g. zinc, copper, lead and arsenic and exchanging them with hydronium ions (H_3O^+). In the activation process the surface is increased by the factor of five. The process is followed by an intensive washing process to remove the remaining acid. Afterwards the acid activated bentonite is dried and milled.

The valuable by-products of the acid activation process (e.g. aluminium and iron salts) are further processed to coupled products.

Fig. 4.2 shows the production process of acid activated bentonite.

4.4 Energy Consumption

The biggest amount of natural gas (70%) is used for the drying process and for heating. About 28% is used for the production of steam and about 2% for self-generation of electricity. These 2% produces about 10% of the total electricity consumption.

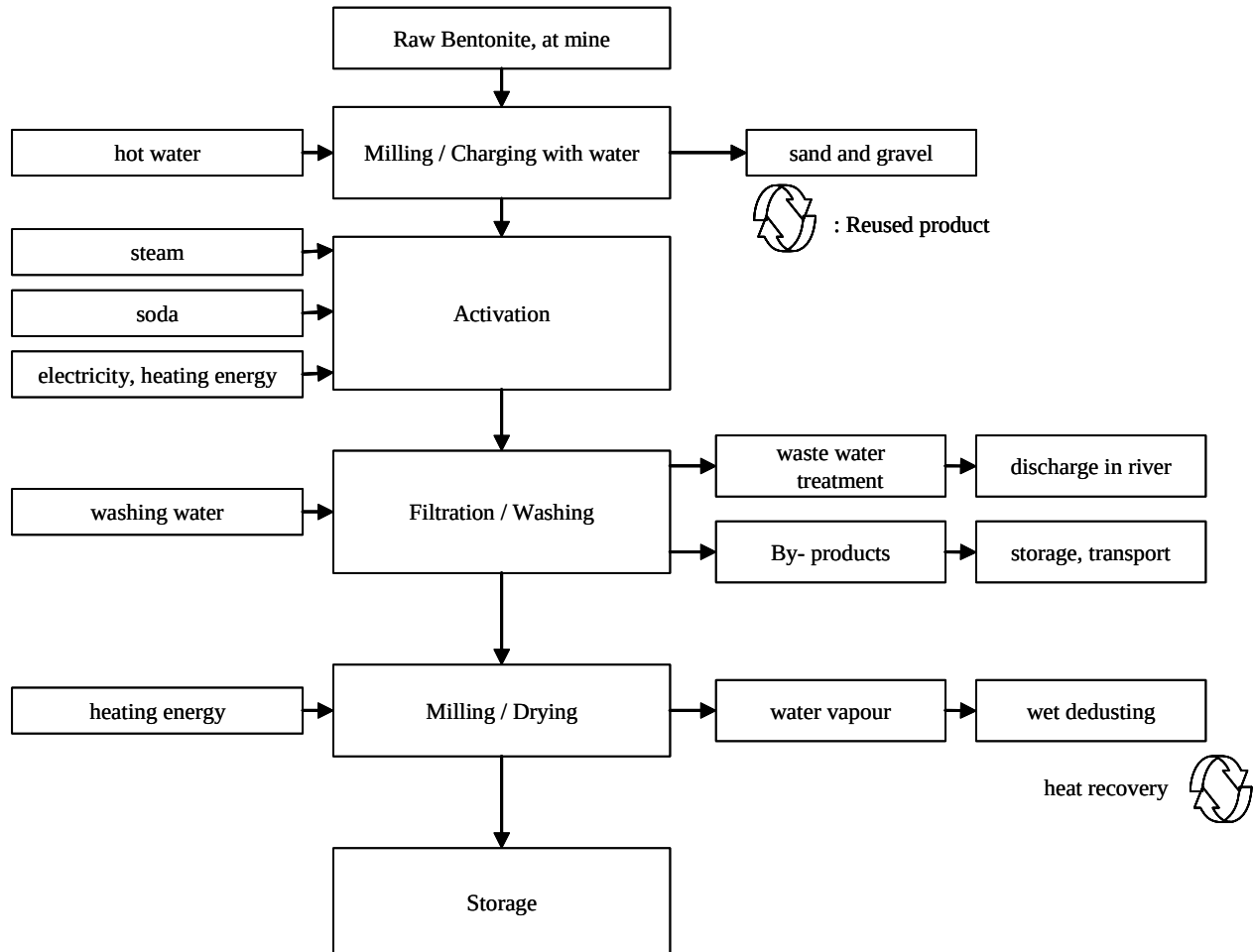


Fig. 4.2 Production process of acid activated bentonite (Süd-Chemie (1997))

4.5 Transports

The diesel is used for mining, transportation of raw bentonite to the processing plant, and mine restoration. As the raw bentonite deposit is about 30 m underground the overburden has to be removed first. When the 3 m thick layer of raw bentonite is exploited the overburden material is refilled and the area is recultivated. Therefore about 95% of the diesel consumption is used for moving the overburden and only about 5% for the real product raw bentonite. (Süd-Chemie (1997))

4.6 Wastewater

A huge amount of acid wastewater is arising from the production process of bleaching earth. The company "Süd-Chemie AG" has an internal wastewater treatment plant. One part of the cleaned wastewater is discharged into a river called "Schleiferbach" and another in a river called "Isar". The wastewater leaving the sewage treatment plant still bears a considerable amount of the heavy metals

lead, nickel, chromium, copper, cobalt, zinc and arsenic. Also problematic is the salt-fright (chloride and sulphates) still in the treated wastewater.

4.7 Solid Waste

Of the total amount of solid waste 97.5% is disposed and 2.5% reused. The disposed waste is composed of 73.5% sludge arising in flushing of the activated bentonite, 12.6% filter cake from the waste water treatment, and 9.5% mineral waste. The rest is declared as different waste and household rubbish.

The largest amount of disposed waste is composed of sand and gravel, which are used for restoration of the mine and thus can be considered burden-free in the context of the present study.

On average the filter cake from wastewater treatment is composed of 58% water, 35% not reusable bentonite and 7% metal hydroxides (mainly iron and aluminium hydroxide) which cannot be reused economically.

Either the waste is deposited in a dump for inert material, in the company's own landfill or in a landfill for household rubbish. Only a very small part of the deposited waste is put in a specially monitored landfill.

4.8 Infrastructure

No data on infrastructure is available. As an approximation, the data of the module "aluminium hydroxide plant" are adjusted as shown in Tab. 4.1.

Tab. 4.1 Estimation of the infrastructure of the bentonite plant of Süd-Chemie AG (based on Classen et al. (2007))

Infrastructure	Aluminium hydroxide plant	Bentonite plant (1999)	Unit	Assumed reduction factor (*)
Yearly production throughput	1'220'000'000	345'743'000	kg	
Transformed area	522'500	150'000	m ²	
Building, multi story	25'000	30'000	m ³	
Building, hall	5'600	30'000	m ²	
Concrete in reinforced concrete (assumption: 91 vol %)	3'020	3'000	m ³	0.3
Steel in reinforced concrete and other applications (assumption: 9 vol %)	2'090'775	696'925	kg	0.3
Heavy machines, unspecified	220'000	73'333	kg	0.3
Concrete for additional foundation	100	33	m ³	0.3
Conveyor belt	640	213	m	0.3
Chromium steel (stainless)	1'865'911	621'971	kg	0.3
Steel low alloyed	24'630	8'210	kg	0.3

(*) based on the difference in the yearly production capacity and the transformed area

The lifetime is generally 50 years for stationary equipment (e.g. buildings, concrete foundations, etc.) and 25 years for mobile equipment and is included in above data from Classen et al. (2007).

4.9 Dust Emissions

The total amount of dust emitted during the bentonite production process in 1999 is 0.16 g per kg bentonite (Süd-Chemie (1999)). The share is taken from Inventory (1999). According to it, 17.5 % of total dust is allocated smaller than 2.5 µm, 56.5 % between 10 and 20 µm and 26% larger than 10 µm.

5 Life Cycle Inventories

Ecoinvent database meta information (tables), flows and their representation in the ecoinvent database are shown in (figures):

Mine, bentonite	Tab. 5.1 and Fig. 5.1
Bentonite, at mine	Tab. 5.2 and Fig. 5.2
Recultivation, bentonite mine	Tab. 5.4 and Fig. 5.3
Bentonite, at processing	Tab. 5.5 and Fig. 5.4

5.1 Bentonite Mine

The ecoinvent database meta information of “mine, bentonite” (Tab. 5.1) includes all important and necessary information to use the data correctly.

Fig. 5.1 shows the system of the infrastructure module “mine, bentonite”. It shows all in- and output flows, the allocation to the modules in ecoinvent database V1.1, the sources (mainly Süd-Chemie (1997) and estimations) and the standard deviation for each datum.

Since this is an infrastructure module, the described land-use includes only the build part. The land-use of the mine itself is included in the module “bentonite, at mine” as it is a part of the production. Considering that the bentonite factory is practically attached to the mine and that the land exploitation develops in parallel to bentonite processing operations, the land use of the excavated area is attributed to this module.

The yearly production amount (345’743 tons) of the Süd-Chemie factory in Moosburg was used as basis for the module “mine, bentonite”.

Tab. 5.1 Ecoinvent database meta information for the product "mine, bentonite"

Name	mine, bentonite
Location	DE
Infrastructure Process	1
Unit	unit
Data Set Version	2.0
Included Processes	includes the area of the "built" infrastructure (buildings, roads, etc.) as well as the buildings, machines, and conveyor belts used for bentonite production. It does not include the area disturbed during mining operation.
Amount	1
Local Name	Mine, Bentonit
Synonyms	
General Comment to reference function	The total land occupation is 16.4 ha and consists of 6 ha buildings (3 ha multi-storey and 3 ha hall), 7.8 ha paved roads, 1.5 ha vegetation and 1.1 ha vessels and containers. The yearly production of the factory used as basis for the module "mine, bentonite" is for data concerning 1999, 345'743 tons (for data concerning 1997, 316'000 t). The lifetime taken into account is generally 50 years for stationary (e.g. buildings, concrete foundations, etc.) and 25 years for mobile equipment. The amount of equipment has been extrapolated from the equipment used at the module "aluminium hydroxide, at plant"
Start Date	1997
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For the exchanges, RER and CH modules have been used as proxy
Technology text	The data of the infrastructure corresponds to a high level of technology
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	The estimation of land use and infrastructure relates to one company in Germany (Süd-Chemie AG)
Extrapolations	See geography
Uncertainty Adjustments	none

General Flow information					Representation in ecoinvent 2000							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
land occupation for build area	Infrastructure bentonite processing plant (part 1)			total area: 15 ha, built area: 6 ha, lifetime of the plant for bentonite production is assumed to be 50 years total area: 15 ha, paved area: 7.8 ha, lifetime of the roads within the plant for bentonite production is assumed to be 50 years total area: 15 ha, green area: 1.5 ha, lifetime of the green area within the plant for bentonite production is assumed to be 50 years total area: 15 ha, area with vessels and containers: 1.1 ha, lifetime of the components within the plant for bentonite production and of their land use are assumed to be 50 years	resource	land			Occupation, industrial area, built up	3.00E+06	m2a	Süd-Chemie (1997)	1	1.50	(1,3,1,1,1,1);
land occupation for roads					resource	land			Occupation, traffic area, road network	3.90E+06	m2a	Süd-Chemie (1997)	1	1.50	(1,3,1,1,1,1);
land occupation for green area					resource	land			Occupation, industrial area, vegetation	7.50E+05	m2a	Süd-Chemie (1997)	1	1.50	(1,3,1,1,1,1);
land occupation for vessels and containers					resource	land			Occupation, industrial area	5.50E+05	m2a	Süd-Chemie (1997)	1	1.50	(1,3,1,1,1,1);
land transformation to build area					resource	land			Transformation, to industrial area, built up	6.00E+04	m2	Süd-Chemie (1997)	1	2.00	(1,3,1,1,1,1);
land transformation to roads					resource	land			Transformation, to traffic area, road network	7.80E+04	m2	Süd-Chemie (1997)	1	2.00	(1,3,1,1,1,1);
land transformation to green area					resource	land			Transformation, to industrial area, vegetation	1.50E+04	m2	Süd-Chemie (1997)	1	2.00	(1,3,1,1,1,1);
land transformation to area for vessels and containers					resource	land			Transformation, to industrial area	1.10E+04	m2	Süd-Chemie (1997)	1	2.00	(1,3,1,1,1,1);
land transformation from unknown					resource	land			Transformation, from unknown	1.64E+05	m2	calculation	1	2.00	(1,3,1,1,1,1);

Fig. 5.1 Flows for “mine, bentonite” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
buildings multi storey	→	Infrastructure bentonite processing plant (part 2)				assumed area of multi-storey buildings: 3 ha; assumed average height: 10 m	construction processes	buildings	Yes	RER	building, multi-storey	3.00E+05 m3		estimation by EMPA based on areal view	1	3.23	(5,3,1,1,1,3);
buildings hall	→				assumed area of building halls: 3 ha	construction processes	buildings	Yes	CH	building, hall, steel construction	3.00E+04 m2		estimation by EMPA based on areal view	1	3.23	(5,3,1,1,1,3);	
infrastructure (machines)	→				assumed weight of machines based on extrapolations from the infrastructure of "aluminium hydroxide plant"	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	7.33E+04 kg		estimation by EMPA	1	3.23	(5,3,1,1,1,3);	
concrete in reinforced concrete (91 vol%)	→				assumed to be one third of the infrastructure used for production of "aluminium hydroxide, plant"	construction materials	concrete	No	CH	concrete, normal, at plant	1.01E+03 m3		estimation by EMPA	1	1.51	(5,3,1,1,1,3);	
steel in reinforced concrete (9 vol%)	→				assumed to be one third of the infrastructure used for production of "aluminium hydroxide, plant"	metals	extraction	No	RER	reinforcing steel, at plant	6.97E+05 kg		estimation by EMPA	1	1.51	(5,3,1,1,1,3);	
poor concrete for additional foundation	→				assumed to be one third of the infrastructure used for production of "aluminium hydroxide, plant"	construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	3.33E+01 m3		estimation by EMPA	1	1.51	(5,3,1,1,1,3);	
conveyor belt	→				assumed to be one third of the infrastructure used for production of "aluminium hydroxide, plant"	construction processes	machinery	Yes	RER	conveyor belt, at plant	2.13E+02 m		estimation by EMPA	1	3.23	(5,3,1,1,1,3);	
chromium steel (stainless)	→				assumed to be one third of the infrastructure used for production of "aluminium hydroxide, plant"	metals	extraction	No	RER	chromium steel 18/8, at plant	6.22E+05 kg		estimation by EMPA	1	1.51	(5,3,1,1,1,3);	
steel low alloyed	→				assumed to be one third of the infrastructure used for production of "aluminium hydroxide, plant"	metals	extraction	No	RER	steel, low-alloyed, at plant	8.21E+03 kg		estimation by EMPA	1	1.51	(5,3,1,1,1,3);	
			→	bentonite processing plant		total production amount 1999 (including acid and alkaline activation and catalytic converter and other products): 345743tons	construction materials	additives	Yes	DE	mine, bentonite	1.00E+00 unit					

Fig. 5.1 Flows for "mine, bentonite" and their representation in the ecoinvent database (cont.)

5.2 Bentonite at Mine

The ecoinvent database meta information of “bentonite, at mine” (Tab. 5.2) includes all important and necessary information to use the data correctly.

Tab. 5.2 Ecoinvent database meta information for the product “bentonite, at mine”

Name	bentonite, at mine
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes excavation by digger, transportation to first grinding machine, the land-use of the mine and the recultivation
Amount	1
Local Name	Bentonit, ab Abbau
Synonyms	
General Comment to reference function	Assumed thickness of bentonite layer: 3 m. Thickness of overburden material: 30 m. Digging and transportation is made by heavy diesel machines.
Start Date	1997
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	"Operation, passenger car" has been used as proxy for the petrol consumption. For the exchanges, RER, UCTE, GLO and CH modules have been used as proxy
Technology text	The technology used in the modelled German mine is probably state-of-art (first mining company in Europe taking part at an eco-audit based on EG-eco-audit requirements)
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	environmental report of one German company (Süd-Chemie AG)
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 5.2 shows the system of the module “bentonite, at mine”. It shows all in- and output flows, the allocation to the modules in ecoinvent database V1.1, the sources (mainly Süd-Chemie (1997) and Süd-Chemie (1999)) and the standard deviation for each data.

The data for the calculation of the land occupation for the mineral extraction site are based on information from 1997, as no more recent data are available. Therefore the mining amount is not consistent with the mining amount used for the infrastructure module “mine, bentonite”. The annual amount mined in 1997 was 2.75 Mio m³ (bentonite and overburden material). From the total area of 247 ha, the storage area is 233.1 ha and the mining area 18.9 ha (Süd-Chemie (1997)). With a total thickness of the layer being excavated (33 m) it takes thus 2.2 years to exploit 13.9ha.

General Flow information					Representation in ecoinvent 2000							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
bentonite	excavation of bentonite from nature (part 1)			possible losses are neglected	resource	in ground			Clay, bentonite, in ground	1.00E+00	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
occupation mineral extraction site (mine) [m2a]				area of winning: 13.9ha, amount of winning 1997: 380'000tons; timeperiod until the area will be exhausted is about 1.7 years	resource	land			Occupation, mineral extraction site	3.66E-04	m2a	Süd-Chemie (1997)	1	1.50	(1,3,1,1,1,1);
occupation storage area (mine) [m2a]				total area 1997: 247ha, amount of winning 1997: 380'000tons; area of storage 1997: 233.1ha, timeperiod until the area will have an other utilization is assumed to be 1.7 years	resource	land			Occupation, mineral extraction site	6.13E-03	m2a	Süd-Chemie (1997)	1	1.50	(1,3,1,1,1,1);
land transformation to mineral extraction (and storage) site (mine) [m2]				total area 1997: 247ha, amount of material mined 1997: 380'000tons	resource	land			Transformation, to mineral extraction site	6.50E-03	m2	Süd-Chemie (1997)	1	2.00	(1,3,1,1,1,1);
land transformation from unknown (mine) [m2]				total area 1997: 247ha, amount of material mined 1997: 380'000tons	resource	land			Transformation, from unknown	6.50E-03	m2	Süd-Chemie (1997)	1	2.00	(1,3,1,1,1,1);
recultivation (mine) [m2]				total material recultivated per 1m2: 66'000 kg overburden material with a density of 2'000 kg/m3	construction materials	additives	No	DE	recultivation, bentonite mine	1.55E-05	m2	Süd-Chemie (1997)	1	2.00	(1,3,1,1,1,1);
diesel (mine)				diesel consumption for excavation of a 3m layer of bentonite and a 30m overburden material layer; about 50% of total diesel consumption in mine. As recultivation takes place at the same time the other 50% are used for that. Data of 1999; 50% of diesel consumption: 60'045GJ, amount of material mined in 1997: 380'000t	construction processes	machinery	No	GLO	diesel, burned in building machine	7.90E-02	MJ	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);

Fig. 5.2 Flows for "bentonite, at mine" and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent 2000							Uncertainty information						
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
heating oil, burned in boiler	→	excavation of bentonite from nature (part 2)				data of 1999; heating oil consumption: 2'811GJ, amount of material mined in 1999: 326'000t	oil	heating systems	No	RER	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	8.62E-03	MJ	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);	
petrol	→					data of 1999; fuel: proxy transport passenger car; fuel consumption: 2.82 MJ/km, amount of material mined in 1999: 326'000t, load: 1.59 persons per car.	transport systems	road	No	RER	transport, passenger car	6.13E-02	pkm	Süd-Chemie (1999)	1	2.00	(1,3,1,1,1,1);	
electricity consumption	→					data of 1999; electricity consumption: 232'661kWh, amount of material mined in 1999: 326'000t	electricity	production mix	No	DE	electricity, medium voltage, production DE, at grid	7.14E-04	kWh	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);	
antifreezer	→					data of 1999; antifreezer consumption: 1.18t, amount of material mined in 1999: 326'000t	chemicals	organics	No	RER	ethylene glycol, at plant	3.62E-06	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);	
lubricating oil	→					data of 1999; lubricating oil consumption: 26.26t, amount of material mined in 1999: 326'000t	chemicals	organics	No	RER	lubricating oil, at plant	8.06E-05	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);	
drinking water	→					data of 1999; drinking water consumption: 2'231t, amount of material mined in 1999: 326'000t	water supply	production	No	RER	tap water, at user	6.84E-03	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);	
rain water	→					data of 1997; rain water consumption: 14'700t, amount of material mined in 1997: 380'000t	resource	in water			Water, unspecified natural origin	3.87E-05	m3	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);	
	→			waste heat				air	unspecified			Heat, waste	2.57E-03	MJ	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
	→			waste water treatment			data of 1999; waste water treatment: 1'931t, amount of material mined in 1999: 326'000t	waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 3	5.92E-06	m3	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
	→			solid waste to municipal waste incineration			data of 1999; solid waste: 126t, amount of material mined in 1999: 326'000t	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	3.87E-04	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
	→		bentonite, at mine			total production amount for the year 1999: 326'000 tons	construction materials	additives	No	DE	bentonite, at mine	1.00E+00	kg					

Fig. 5.2 Flows for "bentonite, at mine" and their representation in the ecoinvent database (cont)

Tab. 5.3 gives an overview of the amounts of mined bentonite and overburden material. The basis year is 1997 because the land-use data are also based on that year.

Tab. 5.3 Composition of weight, volume and mining area of bentonite and overburden material at Süd-Chemie AG

Basis year: 1997	amount	unit
Total amount of material mobilized (bentonite incl. overburden material) (*1)	2.75 Mio	m ³ /a
Total weight of bentonite mined (*1)	380'000	t/a
Total weight of overburden material	5'000'000	t/a
Assumed density of bentonite	1'520	kg/m ³ /a
Assumed density of overburden material	2'000	kg/m ³ /a
Total volume of bentonite mined	250'000	m ³ /a
Total volume of overburden material	2'500'000	m ³ /a
Thickness of overburden material layer (*1)	30	m/a
Thickness of bentonite layer (*1)	3	m/a
Area of overburden material with a thickness of 30 m	83'333	m ² /a
Area of bentonite with a thickness of 3 m	83'333	m ² /a

(*1): based on environmental report 1997 (Süd-Chemie (1997))

The amount of diesel used for mining has been estimated by the thickness of the layers. 50% of the total diesel consumption is allocated to the mining process and 50% to the recultivation process.

5.3 Recultivation Bentonite Mine

The ecoinvent database meta information of “recultivation, bentonite mine” (Tab. 5.4) includes all important and necessary information to use the data correctly.

Fig. 5.3 shows the system of the module “recultivation, bentonite mine”. It shows all in- and output flows, the allocation to the modules in ecoinvent database V1.1, the sources (mainly Süd-Chemie (1997)) and the standard deviation for each data.

As already described in chapter 5.3, half of total diesel consumption is allocated to the recultivation process. As the module “recultivation, bentonite mine” functional unit is the m² the diesel consumption is converted based on the density of 2'000kg/m³ and the area to be recultivated of 83'333 m².

Tab. 5.4 Ecoinvent database meta information for the product “recultivation, bentonite mine”

Name	recultivation, bentonite mine
Location	DE
Infrastructure Process	0
Unit	m2
Data Set Version	2.0
Included Processes	includes only transformation from mineral extraction site to unknown and the diesel consumption for refilling of overburden material (thickness of layer 30m) and of additional material (thickness layer 3m). The additional material and its transport to the mine is not included.
Amount	1
Local Name	Rekultivierung, Bentonitmine
Synonyms	Renaturierung
General Comment to reference function	May only be used as Input for the module "bentonite, at mine". It is assumed that the total of mined material is immediately refilled (from on side to the other...). Therefore half of the total amount of diesel is allocated to this recultivation module.
Start Date	1997
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are only from one company in Germany (Süd-Chemie AG); reclamation practices may differ with the sites and the country, hence prudence should be taken to use this module for different mines.
Technology text	Recultivated soil thickness 33 m (30 m overburden material and 3 m additional material)
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	environmental report of one German company (Süd-Chemie AG)
Extrapolations	data which are only available for GLO have been used as proxy for D
Uncertainty Adjustments	none

General Flow information						Representation in ecoinvent 2000						Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
land transformation from mineral extraction (and storage) site (mine) [m2]	→	recultivation bentonite mine					resource	land			Transformation, from mineral extraction site	1.00E+00	m2	Süd-Chemie (1997)	1	2.00	(1,3,1,1,1,1);
land transformation to unknown (recultivation) [m2]	→					Considering the different practices for reclamation in Europe and the different requirements for the final status of the restored mine, "transformation, to unknown" is used as proxy.	resource	land			Transformation, to unknown	1.00E+00	m2	Süd-Chemie (1997)	1	2.00	(1,3,1,1,1,1);
diesel in machines used for recultivation	→					0.079 MJ per kg mined or recultivated material, weight of material recultivated: 66'000 kg/m2	construction processes	machinery	No	GLO	diesel, burned in building machine	5.21E+03	MJ	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
				→	recultivation bentonite mine		construction materials	additives	No	DE	recultivation, bentonite mine	1.00E+00	m2				

Fig. 5.3 Flows for "recultivation, bentonite mine" and their representation in the ecoinvent database

5.4 Bentonite at Processing

The ecoinvent database meta information of “bentonite, at processing” (Tab. 5.5) includes all important and necessary information to use the data correctly.

The company “Süd-Chemie AG” produced in the year 1999 three different products (Süd-Chemie (1999)):

1. Acid activated bentonite: 212'076 tons (61%)
2. Alkaline activated bentonite: 130'211 tons (38%)
3. Catalytic converters: 3'456 tons (1%)
4. Total production amount: 345'743 tons (year 1999)

The product of the module “bentonite, at processing” is a mixture of the three products acid and alkaline activated bentonite and catalytic converter. No allocation to single products is made.

Fig. 5.4 shows the system boundary of the module “bentonite, at processing”. It shows all in- and output flows, the allocation to the modules in ecoinvent database V1.1, the sources (mainly Süd-Chemie (1999)) and the standard deviation for each data.

Tab. 5.5 Ecoinvent database meta information for the product "bentonite, at processing"

Name	bentonite, at processing
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	"Bentonite, at processing" is a virtual product, put together from the production processes of alkaline activated bentonite (38%), acid activated bentonite (61%) and catalytic converter (1%). The production process of acid activated bentonite includes the raw materials, milling, activation, filtration/washing, milling, drying and storage. The production process of alkaline activated bentonite includes the raw materials, mixing, activation, grinding-drying, mixing/milling/granulation and storage. The total production amount for all products for 1997 is 316'000t and for 1999 is 345'743t.
Amount	1
Local Name	Bentonit, ab Verarbeitung
Synonyms	alcaline//acid//activated//bleaching//earth//Fuller//Bleich//Erde
General Comment to reference function	The emissions to water are direct emissions as they are released into the river after treatment. Emissions to air are from processes. The emissions from combustion of fuels are imported via the requirements of fossil energy.
Start Date	1997
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For the exchanges, RER, UCTE, GLO and CH modules have been used as proxy. For Isopropanol the proxy "NMVOC to water" is used. The salt in the waste water is assumed to be made of 50% sodium chloride and 50% sodium sulphate. Sodium chloride is divided into "sodium ion" and "chloride ion" and sodium sulphate into "sodium ion" and "sulphate ion". For unspecified metals to water, aluminium is used as proxy
Technology text	Mix of different production processes for different products in one German plant: 61% acid activated bentonite, 38% alkaline activated bentonite and 1% catalytic converter
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	environmental report of one German company (Süd-Chemie AG)
Extrapolations	See geography
Uncertainty Adjustments	none

General Flow information					Representation in ecoinvent 2000							Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
bentonite, at mine	production of different bentonite products (Inputs)		Share between total yearly in- and output is neglected as not significant	construction materials	additives	No	DE	bentonite, at mine	1.00E+00	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
water			Total yearly amount: 10'577kg	water supply	production	No	RER	tap water, at user	3.06E-02	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
ground water			Total yearly amount: 2'590'842'000kg	resource	in water			Water, well, in ground	7.49E-03	m3	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
natural gas in industrial furnace				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	2.53E-01	MJ	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
electricity			Total yearly amount, assumed to be measured: 156'785 GJ	electricity	production mix	No	DE	electricity, medium voltage, production DE, at grid	1.26E-01	kWh	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
hydrochloric and sulphuric acid (H2SO4)			Total yearly amount of hydrochloric- and sulphuric acid: 69'349 t; 50% is assumed to be sodium chloride. Total yearly production of acid activated bentonite: 212'076 t	chemicals	inorganics	No	RER	hydrochloric acid, 30% in H2O, at plant	1.64E-01	kg	Süd-Chemie (1999)	1	1.51	(5,3,1,1,1,1);
hydrochloric and sulphuric acid (H2SO4)				chemicals	inorganics	No	RER	sulphuric acid, liquid, at plant	1.64E-01	kg	Süd-Chemie (1999)	1	1.51	(5,3,1,1,1,1);
soda, powder, at plant			Total yearly amount of soda: 4'307 t. Total yearly production of alkaline activated bentonite: 130'211 t	chemicals	inorganics	No	RER	soda, powder, at plant	3.31E-02	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
alkaline solution			sodium hydroxide used as proxy, total yearly amount: 427 t. Total yearly production of alkaline activated bentonite: 130'211 t	chemicals	inorganics	No	RER	sodium hydroxide, 50% in H2O, production mix, at plant	3.28E-03	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
chemicals organic, at plant			Total yearly amount of various chemicals (used for all products): 31'740t	chemicals	organics	No	GLO	chemicals organic, at plant	9.18E-02	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
coal			Total yearly amount of coal; 9'745 t	hard coal	fuels	No	UCTE	hard coal mix, at regional storage	2.82E-02	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
infrastructure bentonite plant				construction materials	additives	Yes	DE	mine, bentonite	5.78E-11	unit	Süd-Chemie (1999)	1	3.00	(1,3,1,1,1,1);
diesel in building machine			Total yearly amount of diesel: 3'229 GJ	construction processes	machinery	No	GLO	diesel, burned in building machine	9.34E-03	MJ	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);

Fig. 5.4 Flows for “bentonite, at processing” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent 2000							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
production of different bentonite products (Outputs part 1)	→			waste water to external waste water treatment		as no information available on the composition of the waste water the module "treatment sewage, from residence" has been used as proxy	waste management	wastewater treatment	No	CH	treatment, sewage, from residence, to wastewater treatment, class 2	5.10E-05	m3	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
	→			waste heat			air	unspecified			Heat, waste	4.53E-01	MJ	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
	→			hydrogen chloride			air	unspecified			Hydrogen chloride	2.89E-08	kg	Süd-Chemie (1999)	1	1.50	(1,3,1,1,1,1);
	→			ammonia (NH3)			air	unspecified			Ammonia	2.89E-06	kg	Süd-Chemie (1999)	1	1.50	(1,3,1,1,1,1);
	→			Isopropanol		NMVOC used as proxy	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	3.18E-06	kg	Süd-Chemie (1999)	1	1.50	(1,3,1,1,1,1);
	→			particulates, < 2.5um		bentonite dust, 17.5% < 2.5um	air	unspecified			Particulates, < 2.5 um	2.83E-05	kg	Süd-Chemie (1999)	1	2.50	
	→			particulates, 10 20 um		bentonite dust, 56.5% between 10 and 20um	air	unspecified			Particulates, > 2.5 um, and < 10um	9.15E-05	kg	Süd-Chemie (1999)	1	2.50	
	→			particulates, >10 um		bentonite dust, 26% > 10um	air	unspecified			Particulates, > 10 um	4.21E-05	kg	Süd-Chemie (1999)	1	2.50	
	→			sulfur dioxide (SOx as SO2)			air	unspecified			Sulfur dioxide	1.97E-06	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
	→			ammonium, ion			water	river			Ammonium, ion	2.89E-07	kg	Süd-Chemie (1999)	1	1.50	(1,3,1,1,1,1);
	→			AOX, Adsorbable Organic Halogen as Cl			water	river			AOX, Adsorbable Organic Halogen as Cl	8.68E-09	kg	Süd-Chemie (1999)	1	1.50	(1,3,1,1,1,1);
	→			COD, Chemical Oxygen Demand			water	river			COD, Chemical Oxygen Demand	3.56E-05	kg	Süd-Chemie (1999)	1	1.50	(1,3,1,1,1,1);
	→			aluminum			water	river			Aluminum	6.94E-08	kg	Süd-Chemie (1999)	1	5.00	(1,3,1,1,1,1);
	→			arsenic			water	river			Arsenic, ion	3.18E-08	kg	Süd-Chemie (1999)	1	5.00	(1,3,1,1,1,1);
	→			lead			water	river			Lead	5.78E-09	kg	Süd-Chemie (1999)	1	5.00	(1,3,1,1,1,1);
	→			cadmium, ion			water	river			Cadmium, ion	5.78E-09	kg	Süd-Chemie (1999)	1	5.00	(1,3,1,1,1,1);
	→			chromium, ion			water	river			Chromium, ion	5.78E-09	kg	Süd-Chemie (1999)	1	5.00	(1,3,1,1,1,1);
	→			iron, ion			water	river			Iron, ion	3.88E-07	kg	Süd-Chemie (1999)	1	5.00	(1,3,1,1,1,1);
	→			copper, ion			water	river			Copper, ion	8.68E-09	kg	Süd-Chemie (1999)	1	5.00	(1,3,1,1,1,1);
	→			nickel, ion			water	river			Nickel, ion	1.45E-08	kg	Süd-Chemie (1999)	1	5.00	(1,3,1,1,1,1);
	→			mercury			water	river			Mercury	8.68E-09	kg	Süd-Chemie (1999)	1	5.00	(1,3,1,1,1,1);
	→			zinc, ion			water	river			Zinc, ion	1.16E-08	kg	Süd-Chemie (1999)	1	5.00	(1,3,1,1,1,1);
	→			hydrocarbons, unspecified			water	river			Hydrocarbons, unspecified	1.45E-08	kg	Süd-Chemie (1999)	1	3.00	(1,3,1,1,1,1);
	→			metals, unspecified		aluminium used as proxy	water	river			Aluminum	4.05E-07	kg	Süd-Chemie (1999)	1	5.00	(1,3,1,1,1,1);
	→			nitrate			water	river			Nitrate	2.14E-05	kg	Süd-Chemie (1999)	1	1.50	(1,3,1,1,1,1);
	→			phosphorus			water	river			Phosphorus	6.94E-05	kg	Süd-Chemie (1999)	1	1.50	(1,3,1,1,1,1);

Fig. 5.4 Flows for "bentonite, at processing" and their representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent 2000							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
production of different bentonite products (Outputs part 2)	→		salt in waste water: chloride and sulphate		assumed that 50% of total fright is sodium chloride, of which 40% is sodium ions.	water	river			Sodium, ion	1.11E-03	kg	Süd-Chemie (1999)	1	1.78	(5,3,1,1,1,1);
	→		salt in waste water: chloride and sulphate		assumed that 50% of total fright is sodium chloride, of which 60% is chloride ions.	water	river			Chloride	1.66E-03	kg	Süd-Chemie (1999)	1	1.78	(5,3,1,1,1,1);
	→		salt in waste water: chloride and sulphate		assumed that 50% of total fright is sodium sulphate, of which 32% is sodium ions.	water	river			Sodium, ion	8.86E-04	kg	Süd-Chemie (1999)	1	1.78	(5,3,1,1,1,1);
	→		salt in waste water: chloride and sulphate		assumed that 50% of total fright is sodium sulphate, of which 68% is sulphate ions.	water	river			Sulfate	1.88E-03	kg	Süd-Chemie (1999)	1	1.78	(5,3,1,1,1,1);
	→		suspended solids, unspecified			water	river			Suspended solids, unspecified	1.82E-02	kg	Süd-Chemie (1999)	1	1.50	(1,3,1,1,1,1);
	→		solid waste to disposal, monitored		1999: 29291t (97.7% - 13% (filter sludge) = 84.7%); includes mineral waste, soil- and mineral sludges, module "disposal inert waste to inert material landfill" is used as proxy	waste management	inert material landfill	No	CH	disposal, inert waste, 5% water, to inert material landfill	7.18E-02	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
	→		solid waste to disposal, specially monitored		1999: 86t (0.3%); composition not declared, module "disposal hazardous waste to underground deposit" is used as proxy	waste management	underground deposit	No	DE	disposal, hazardous waste, 0% water, to underground deposit	2.49E-04	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
	→		solid waste to recycling		1999: 589t (2%); not balanced as assumed to be burden-free						1.70E-03	kg	Süd-Chemie (1999)			
	→		filter cake from waste water treatment		1999: 29291t (13%); sludge arising in flushing of the activated bentonite, module "disposal hazardous waste to underground deposit" is used as proxy	waste management	underground deposit	No	DE	disposal, hazardous waste, 0% water, to underground deposit	1.10E-02	kg	Süd-Chemie (1999)	1	1.07	(1,3,1,1,1,1);
	→		bentonite		3690.666	construction materials	additives	No	DE	bentonite, at processing	1.00E+00	kg				

Fig. 5.4 Flows for "bentonite, at processing" and their representation in the ecoinvent database (cont.)

6 Data quality

The uncertainties are evaluated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

Most data are based on the environmental report of the German company “Süd-Chemie AG” (Süd-Chemie (1999)). The total amount of mined bentonite in Germany in 1998 was 478'000 tons. The total amount mined bentonite by “Süd-Chemie AG” in the same year was 343'520 tons. The share of production of Germany is therefore about 70%. The market share is unknown, as no data for Import and Export are available.

The data on processing are detailed and comprehensive and give a good portray of the German production.

For all these reasons the data for a general bentonite module are of good overall quality. When using the data it must be taken into account that the product “bentonite, at processing” is a mixture of three different bentonite products.

7 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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Part XX

Synthetic rubber (EPDM)

Data v2.0 (2007)

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Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

This part deals with the production of profiles from EPDM (a specific synthetic rubber) for seals of windows, doors etc.

Since there are many different synthetic rubbers with different compositions, different raw materials and different production processes, this inventory should only be used for EPDM. If it is used for another (unknown) type of rubber, a considerable uncertainty is introduced.

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1 Introduction

This part deals with the production of profiles from EPDM (a specific synthetic rubber) for seals of windows, doors etc.

Natural rubber production is completely different and there are many other synthetic rubbers made from very different input materials in very different processes. Thus these data have to be used with care if the type of rubber for an application is not exactly known.

2 Reserves and Resources of Materials

EPDM is made from ethylene, propylene and a dien. For information about reserves and resources see the information in the respective chapters.

The fillers used with the EPDM for profile production are soot and calcium carbonate (limestone). For information about reserves and resources see the information in the respective chapters in Hischer (2007) and in part VII of this report.

3 Characterisation and Use of the Materials

3.1 Definitions

Rubber

Rubber (raw rubber) is a non-cross-linked, but cross-linkable (vulcanisable) polymer with rubber-elastic properties at room temperature. Rubber-elastic behaviour is characterised by a relatively low shear modulus with rather slight temperature dependence. At higher temperature or under the influence of deforming forces, raw rubber exhibits increasing viscous flow, so that it can be moulded under suitable conditions. Raw rubber is a starting material for the production of elastomers. (Greve (1997))

Elastomer

Elastomers are polymeric materials that are cross-linked (vulcanised) up to their decomposition points. They are hard and glassy at low temperature and do not exhibit viscous flow even at high temperature. They have rubber-elastic properties, particularly from room temperature up to their decomposition point. (Greve (1997))

Vulcanisation

Vulcanisation is a process in which rubber, through a change in its chemical structure (for example, cross-linking), is converted to a condition in which the elastic properties are conferred or re-established or improved or extended over a greater range of temperatures. In some cases, the process is carried to a point where the substances become rigid. (Greve (1997))

3.2 Synthetic rubbers / elastomers

This information is based on Obrecht et al. (1997).

A variety of more than 20 synthetic rubbers (SR) exists, most of them having been developed after World War II. SR can be classified into several groups according to their chemical structure. The group designation according to internationally accepted nomenclature is given in Tab. 3.1.

Tab. 3.1 Group designation of rubber according to the chemical structure Obrecht et al. (1997)

Group designation	Chemical structure	Examples
M	rubber having a fully saturated polymer backbone of the polymethylene type	EPM, EPDM, CM, CSM, AEM, ACM, EVM
N	rubber containing nitrogen	NBR
O	rubber having oxygen atoms in the polymer backbone (polyether rubber)	CO, ECO, GPO
Q	rubber having silicon atoms in the polymer backbone (silicon rubber)	VMQ, MQ, FMVQ
R	rubber having an unsaturated carbon main chain	NR, SBR, BR, NBR, CR, IIR
T	rubber having sulfur in the polymer backbone (polythioether rubber)	OT, EOT
U	rubber of the polyurethane type containing carbon, oxygen, and nitrogen in the polymer backbone	AU, EU
Z	rubber having phosphorus and nitrogen in the polymer backbone	FZ, PZ

Based on this system the abbreviations listed in Tab. 3.2 are used for the different types of rubber according to DIN/ISO 1629. The last letter of the abbreviation refers to the chemical group to which the specific rubber belongs.

Elastomers are made of rubber, different fillers and pigments (often soot and limestone), softeners, other additives and vulcanisation agents. Fillers and pigments usually make up more than 50% of the elastomer weight.

There are scores of different elastomers with very different properties and applications. Some characteristics of EPDM elastomers are discussed in Chapter 3.2.1. For detailed information on other elastomers see Obrecht et al. (1997) or Engels et al. (1997).

3.2.1 Ethylene-propene-diene terpolymer (EPDM) rubbers / elastomers

Ethylene-propene elastomers are a family of random elastomeric copolymers with a completely saturated polymer backbone. These elastomers are therefore designated with an "M" according to the ASTM standard DIN/ISO 1629.

There are two basic types of ethylene – propene elastomers:

- Ethylene-propene copolymers (EPM)
- Ethylene-propene-diene terpolymers (EPDM)

The copolymers are generally amorphous elastomeric polymers without unsaturated bonds, thus necessitating the use of peroxide cure systems for vulcanisation. The terpolymers contain small quantities (< 12 wt %) of a nonconjugated diene. Only one of the double bonds of each diene molecule takes part in polymerisation, resulting in a saturated polymer backbone with a pendant double bond that facilitates the use of conventional sulphur curing.

The saturated carbon backbone structure accounts for the following properties:

- Outstanding ozone resistance
- Excellent weathering stability
- Excellent heat resistance (< 175 °C)
- Good insulating properties
- Low-temperature flexibility (glass transition temperature : ca. – 55 °C)
- Good chemical resistance (toward polar organic and inorganic compounds)
- Good moisture and steam resistance

EPM and EPDM have the lowest density (860 kg/m³) of any commercial elastomer. In addition, during compounding, EP(D)M can be heavily loaded with oil and fillers resulting in low compound costs.

The structure of the polymer can generally be described by three features:

- Ethylene : propene ratio and monomer sequence distribution
- Molar mass and molar mass distribution
- Content and type of diene incorporated

These structural features are directly related to the polymerisation conditions. By varying the conditions, polymers with different properties can be obtained and a large selection can be provided, suitable for various applications.

Tab. 3.2 Abbreviations for the different types of rubber according to DIN/ISO 1629 (Obrecht et al. (1997))

ACM	copolymer of ethyl acrylate or other acrylates and a small amount of a monomer that facilitates vulcanization
a A-CSM	chlorosulfonated polyethylene with a high content of alkyl groups
a AEM	copolymers of ethyl acrylate or other acrylates with ethylene
AU	polyester urethane
BIIR	brominated isobutene–isoprene copolymers (bromobutyl rubber)
BR	butadiene rubbers
CIIR	chlorinated isobutene – isoprene copolymers (chlorobutyl rubber)
CM	chlorinated polyethylene
CO	polychloromethyloxirane(epichlorohydrin rubber)
CR	chloroprene rubber
CSM	chlorosulfonated polyethylene
a E-BR	emulsion butadiene rubber
ECO	ethylene oxide (oxirane) and epichlorohydrin (chloromethyloxirane) copolymer
a ENM	copolymer of ethylene and acrylonitrile
a ENR	epoxidized natural rubber
EPDM	terpolymer of ethylene, propene, and a diene with the residual unsaturated portion of the diene in the side chain
EPM	ethylene–propylene copolymer
a E-SBR	emulsion styrene–butadiene rubber
EU	polyether urethane
a EVM	ethylene–vinyl acetate copolymer
FMQ	silicone rubber having fluorine and methyl groups on the polymer chain
FMVQ	silicone rubber having fluorine, methyl, and vinyl substituents on the polymer chain
FPM	rubber with fluoro, fluoroalkyl, or fluoroalkoxy substituents on the saturated carbon main chain
FZ	polyphosphazene rubber having fluoroalkyl or fluoroalkoxy substituents on the polymer chain
GPO	copolymer of propylene oxide and allyl glycidyl ether
HIIR	halogenated butyl rubber (halobutyl)
HNBR	hydrogenated acrylonitrile–butadiene rubber (highly saturated nitrile rubber)
IIR	isobutene–isoprene copolymers (butyl rubber)
IR	isoprene rubber, synthetic
MQ	silicone rubber having only methyl substituents on the polymer chain, such as dimethyl polysiloxane
NBR	acrylonitrile–butadiene rubber
PMQ	silicone rubber having phenyl and methyl substituents on the polymer chain
PNR	polynorbornene rubber
PZ	polyphosphazene rubber with phenoxy substituents on the polymer chain
SBR-HS	styrene–butadiene rubber with high styrene content
a S-SBR	solution styrene–butadiene rubber
a S-SIBR	solution styrene–isoprene–butadiene rubber
a TPE	thermoplastic elastomer
VMQ	silicone rubber having both methylvinyl and vinyl substituent groups on the polymer chain
XNBR	carboxylated acrylonitrile–butadiene rubber
XSBR	carboxylated styrene–butadiene rubber
YAU	thermoplastic polyester urethane
YEU	thermoplastic polyether urethane
YSBR	block copolymer of styrene and butadiene
YSIR	block copolymer of styrene and isoprene

a: not officially approved by DIN/ISO 1629.

3.3 Use of synthetic rubber / elastomer

This information is based on Obrecht et al. (1997).

In 1991 the combined annual consumption of synthetic (SR) and natural rubber (NR) was nearly 15 Million tons world-wide. Approximately two-thirds of the total rubber consumed was SR.

The areas of SR application in the Western world and their proportion of total consumption (percent) are listed in Tab. 3.3.

Tab. 3.3 Application of synthetic rubber

Application	%	Application	%
Tires	59	Construction	3
Automotive mechanical goods	10	Wire covering	2
Non-automotive mechanical goods	9	Adhesives	1
Plastic modification	6	Others	6
Footwear	4		

According to market shares and prices three different categories of SR are usually distinguished.

General-purpose or tire rubber

General-purpose rubber includes various types of rubber with a high market volume that are available at a modest price. This category consists of the R rubbers NR, SBR, BR, and IR that are used mainly in tires. The annual consumption of each of the general-purpose rubbers is more than 1 million tons.

Special rubber

Special rubber includes a series of products that are higher priced and produced in smaller volumes than general-purpose rubber. Representatives of this category are EPDM, NBR, CR, and IIR. These products are superior to general-purpose rubber in at least one property. EPDM for example features a good heat and ageing resistance, NBR good oil resistance and IIR low air permeability. Most of the special rubbers are used for a variety of technical rubber goods such as extruded profiles, seals, gas-kets, hoses, or power transmission belts. The annual consumption of each special rubber is in the range 200 000 – 650 000 tons.

Speciality rubber or high-performance elastomers

The group of high-performance elastomers (speciality rubber) consists of small-volume, high-priced products. These products are distinguished by at least two major properties. Examples are FPM with excellent oil and heat resistance, VMQ with excellent low-temperature flexibility and excellent heat resistance. As for special rubber, the most important application of high-performance elastomers is in technical rubber goods. Apart from silicon rubber with an annual consumption of more than 100 000 tons, the annual consumption of the other high-performance elastomers is generally less than 100 000 tons each.

3.3.1 Uses of EPDM rubber / elastomer

The EP(D)M grades (ca. 250 available world-wide) are constantly optimised to fit special applications. Although many manufacturers produce particular grades which are reasonable substitutes for one another, EP(D)M elastomers are not standardised as are general-purpose rubbers.

Uses: As a result of their versatility, EP(D)M polymers find use in a number of market segments, both in the traditional thermoset areas, and increasingly in thermoplastic blends and alloys. Typical applications for EP(D)M are given in Tab. 3.4.

Tab. 3.4 Applications of EPM and EPDM Obrecht et al. (1997)

Market segment	Share	Typical applications
Automotive	32 %	radiator/heater hoses, weatherstrips, window channels, sponge seals, mounts, car mats
Plastics modification	15 %	impact-modified polypropylene, thermoplastic olefins
Construction	14 %	roofing membrane, window seals
Industrial	11 %	steam and garden hoses, gaskets O-rings, mounts
Oil	10 %	oil viscosity improver
Electrical	6 %	cable jacketing, insulation, cable filler, connectors
Tire	4 %	sidewall construction, inner tubes, tire valves
Miscellaneous	8 %	footwear, carpet underlay

In the 1960s, EPDM was expected to become the new tire elastomer. But until recently, its utilisation for tire applications has been limited to some inner tubes and to blends for white side-walls. The main deficiency of EPDM is its inability to develop adhesion and to covulcanise with unsaturated polymers such as natural rubber, or polybutadiene.

In blends with NR, excellent physical properties and adhesion can be achieved when the EPDM has high molar mass and high (> 8 %) ethyldenenorbornene content. This concept is commercially being used in veneers for radial passenger tires. The veneer is resistant to weathering, since the EPDM phase is immune to ozone attack, and the NR phase domains are so small that cracks cannot easily grow. No chemical antidegradants are required in the formulation, so discoloration from staining antioxidants is avoided. Development of this technology for tire sidewalls and other tire components is underway in tire companies world-wide.

4 System Characterisation

4.1 Synthetic rubber

This chapter is mainly based on Obrecht et al. (1997).

The great variety of manufacturing processes for SR can be classified from both a chemical and from a chemical engineering point of view. Chemically, the mechanism governing the polymerisation reaction and the chemistry involved in the key reaction are important. However, technologically speaking, the process and the series of process steps involved are major issues. In Tab. 4.1 the most important synthetic rubbers are listed according to both chemical and chemical engineering criteria.

As the product modelled in this study is EPDM only this specific production is described in more detail. For the production of other synthetic rubbers see e.g. Obrecht et al. (1997).

Tab. 4.1 Production of synthetic rubber in terms of chemical mechanisms and process features Obrecht et al. (1997)

Chemical mechanism	Process features				
	Emulsion	Solution	Dispersion	Bulk	Gas phase
Radical polymerisation	E-SBR; CR; NBR; E-BR; ACM; FPM; EVM	EVM	EVM	AEM; EVM (ENM)	
Ziegler–Natta polymerisation		BR; EPM; EPDR	EPM; EPDM	EPM; EPDM; EPDM (BR)	(EPM); (EPDM)
Anionic polymerisation		BR; S-SBR; IR		Q	
Cationic polymerisation		ECO; CO	IIR	Q	
Polyaddition and polycondensation		AU; EU	EU	AU; Q	
Polymer modification		CIIR; BIIR; CM; CSM; HNBR; FZ	CM; CSM (HNBR)		

Processes under development are given in parentheses.

4.1.1 Ethylene-propene-diene terpolymer (EPDM) rubber

Monomers

The monomers ethylene and propene are normally used as polymerisation grade (purity > 99 %) and must be free of impurities such as carbon monoxide, oxygen, water, acetylene, methanol, or methylacetylene (content < 10 ppm). These impurities are potent poisons of Ziegler – Natta catalysts, which are exclusively used for EPDM polymerisation.

The ideal termonomer has to meet several important criteria. It should have two double bonds of widely different reactivity. One double bond should react rapidly in the terpolymerisation, while the second one should be relatively inert. However, this second double bond must be reactive toward the vulcanising agent. The monomer should also be volatile, so that the residual unreacted monomer can be easily removed from the polymer after polymerisation. Finally, the termonomer should be readily available and inexpensive.

Ethylenenorbornene (ENB) is the most commonly used termonomer because it reacts rapidly in the polymerisation and gives high quality, fast curing elastomers. ENB polymers cure faster than polymers containing other termonomers. This is essential for high productivity in continuous curing extruded applications.

Catalysts

Commercial EPDM polymerisation processes employ vanadium or titanium catalysts. Metallocene catalysts, although commercially used for polyolefin polymerisation are currently in the pilot plant stage for EPDM polymerisation.

Production Processes

At present, commercial EPM and EPDM polymers are produced by solution or suspension processes. Most manufacturers use a solution process. Both solution and suspension are continuous processes which typically consist of four operating sections, i.e., polymerisation, monomer and solvent recycling, product recovery, and production finishing.

Solution Process

Typically, polymerisation is carried out at 10 – 50 °C and 1 – 20 bar in a single continuous stirred-tank reactor (CSTR). The polymerisation is exothermic (DH » –2.4 MJ per kilogram of polymer). The heat

of reaction is removed by precooling the reactor feeds and/or evaporative cooling of the reaction mixture. The residence times in the reactor range from 15 – 60 min, depending on the catalyst employed, temperature, etc. Ethylene conversion is nearly complete in a single pass through the reactor. Unreacted propene is recovered and recycled to the polymerisation. Solvent (usually hexane) and unreacted diene are also recovered for reuse after purification. Immediately downstream of the reactor, the polymerisation is shortstopped with water, alcohol, or other polar liquids. The polymer solution is then mixed with water, aqueous sodium hydroxide, or dilute acid under vigorous agitation to remove catalyst residues. After addition of chemicals such as extender oils and stabilisers, the polymer solution is coagulated and stripped with steam and hot water to remove unreacted monomers and solvents. Anti-agglomerants and antifoaming agents may be added to control crumb agglomeration and facilitate the separation of gas and liquid. The rubber crumb is then passed over a screen to remove free water and subsequently dried. Either extrusion drying or hot air drying may be used. EPDM is sold in the form of crumb, bales, or pellets.

Suspension Process

Suspension polymerisation is carried out in an excess of liquid propene. The polymer precipitates out of the reaction medium as soon as it is formed. Consequently, the viscosity of the polymerisation medium is increased only little (unlike that of solution polymerisation medium). The solids content in the reaction medium can therefore be higher than in the solution process, e.g., 20-30 % (w/w) instead of 5-10 % (w/w). Monomer recovery, stripping, and finishing are carried out as described for the solution process except that the coagulation step is not required since the polymer already exists as particles. The slurry process is well suited to the production of grades with high Mooney viscosity, which can be oil extended for easier processing.

Infrastructure

The infrastructure needed for EPDM production is basically a chemical plant as needed for the production of other polymers.

4.2 Synthetic elastomer

This chapter is mainly based on Von Langenthal & Schnetger (1997).

The production of rubbery materials with industrially useful properties from solid rubber involves the incorporation of a range of additives. The latter critically affect the properties with regard to process technology and permit cross-linking or vulcanisation. To achieve a sufficiently high and uniform property level of the rubber articles produced, homogeneous distribution of all additives in the tough rubber matrix is necessary.

The raw compound is then processed further and moulded by different procedures such as calendaring, extrusion, pressing, injection moulding, or coating processes, and vulcanised through further energy input. Cross-linked structures are thus built up, which convert the rubber blank into a rubber-elastic material. Processing is generally carried out in internal mixers, less frequently in open mills.

4.2.1 Ethylene-propene-diene terpolymer (EPDM) elastomer

A possible mixture for EPDM elastomers is given in Tab. 4.2.

Tab. 4.2 Possible mixture of EPDM elastomer (Richter et al. (1996))

[%] (w/w)	low	high	mean
EPDM rubber	27.9	34.7	31.3
soot	31.8	41.7	36.8
plasticisers	17.0	13.9	15.5
filler (CaCO ₃)	15.9	1.7	8.8
peptizers	3.0	2.6	2.8
activator	1.9	1.9	1.9
auxiliary materials	2.5	3.5	3.0

The input materials are mixed batch-wise at a temperature of 120 - 180 °C for about 5 minutes. The driving power required to achieve the necessary mixing and shearing action is at least 6 kW, and usually 10 kW per litre. The heat of friction in this process is so high that cooling is necessary. The elastomer is rolled and the sheets are dried and stored. For further processing, the sheets are granulated, extruded and vulcanised. Vulcanisation takes place in a salt bath directly following the extruder. The extrudate is fed into a molten eutectic salt mixture consisting of 53 % potassium nitrate, 40 % sodium nitrite, and 7 % sodium nitrate. After vulcanisation and before unrolling, the salt partially adhering to the vulcanisate surface is removed with brushes in a water bath and reprocessed in special recovery plants to avoid wastewater contamination.

4.3 EPDM elastomer production in ecoinvent

As no input- / output information on the EPDM rubber production is available it is approximated by the production of high density polyethylene (HDPE) and polypropylene (PP). HDPE is produced by the Ziegler-Natta polymerisation of ethylene, which means that the process is very similar to the rubber production. PP is not commonly produced in solution polymerisation.

Richter et al. (1996) assumes an ethylene-share of 55 % (w/w) and neglects the diene. Obrecht et al. (1997) gives a range of 40-80 % (w/w) for the ethylene share and a diene content of up to 12%. In this study we are using 80 % (w/w) HDPE and 20 % PP for the EPDM rubber because the HDPE production fits the EPDM production better than the PP production and because no data for the polymerisation process of the diene are available. In this way, the inventory of the rubber production includes material and energy inputs and waste treatment.

For the production of the EPDM elastomer sheets the material inputs as shown in Tab. 4.2 are considered. The electricity consumption for mixing is assumed as 10 kW/litre for 5 minutes. With EPDM's density of 860 kg/m³ this equals to 0.717 kWh/kg.

The extrusion step is approximated by the process for pipe extrusion. As this module is scaled to the product output and 0.00369 kg plastic waste is generated, the polymer input per kg output has to be 1.00369 kg.

The vulcanisation of the product is modelled using the process and salt mixture described in chapter 4.2.1. It is assumed that 10 g of the salt mixture is lost per kg vulcanised product. The energy for heating the bath is assumed to be part of the waste heat from the mixing process.

The cooling water consumption is estimated to 45 kg/kg. This amount is derived by the assumption that the temperature of the cooling water is increased 10 K and that about 70% of the 0.7 kWh electrical energy is used to heat the water. The rest of the energy dissipates to air partially after being used for heating the salt bath for vulcanisation.

The only available data on direct emissions to air and water are given in Richter et al. (1996). As these data include (possibly exclusively) the emissions of the heat generation from oil, which are included in the upstream processes in the model used here, they are not inventoried.

The ecoinvent meta-information for the EPDM elastomer production is given in Tab. 4.3 and the in- and output flows for "synthetic rubber, at plant" are given in Fig. 4.1.

4.3.1 Data quality

Data uncertainty is derived using the Pedigree matrix as described in (Frischknecht et al. (2007)).

All data are based on theoretical process descriptions. They use proxies for processes for which not enough information for a proper modelling is available. Thus the overall quality of these is poor. Data are not to be used for direct comparison of synthetic rubber with alternative materials or in other context where the impact of this material is relatively high.

Tab. 4.3 Ecoinvent meta information for the EPDM elastomer production processes

Name	synthetic rubber, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Production of EPDM-rubber, production of EPDM elastomer, extrusion and vulcanisation of EPDM profiles. Also included are the transports of the raw materials to the polymerisation and elastomer production plant.
Amount	1
Local Name	Gummi EPDM, ab Werk
Synonyms	EPDM//Elastomer
General Comment to reference function	This module refers to the EPDM elastomer as it is used in technical products. Thus, according to DIN the name "rubber" (meaning only the un-vulcanised polymer without any fillers etc.) would actually be wrong. EPDM is one of many different rubbers and there are EPDM elastomers of many different compositions. The elastomer modelled in this data could typically be used as seals (for e.g. windows)
Start Date	1995
End Date	2003
Data Valid For Entire Period	1
Other Period Text	
Geography text	
Technology text	Ziegler-Natter solution polymerisation of EPDM. Internal mixing of elastomer. Salt vulcanisation after extrusion
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Literature
Extrapolations	None
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	STdv 95%	General Comment
high density polyethylene	synthetic rubber, at plant			plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	2.51E-01 kg		estimation	1	1.32	(4,2,3,2,1,5,3);
polypropylene				plastics	polymers	No	RER	polypropylene, granulate, at plant	6.28E-02 kg		estimation	1	1.32	(4,2,3,2,1,5,3);
soot				chemicals	inorganics	No	GLO	carbon black, at plant	3.69E-01 kg		estimation	1	1.32	(4,2,3,2,1,5,3);
plasticiser			mostly mineral oils	chemicals	organics	No	RER	lubricating oil, at plant	1.55E-01 kg		estimation	1	1.32	(4,2,3,2,1,5,3);
filler (CaCO3)				construction materials	others	No	CH	limestone, milled, packed, at plant	8.83E-02 kg		estimation	1	1.32	(4,2,3,2,1,5,3);
pepticiser			often pentachlorothiophenol	chemicals	organics	No	GLO	chemicals organic, at plant	2.81E-02 kg		estimation	1	1.32	(4,2,3,2,1,5,3);
activator			often zinc oxide --> beneficiated zinc ore as proxy	metals	extraction	No	GLO	zinc concentrate, at beneficiation	1.91E-02 kg		estimation	1	1.32	(4,2,3,2,1,5,3)
auxiliaries				chemicals	organics	No	GLO	chemicals organic, at plant	3.01E-02 kg		estimation	1	1.32	(4,2,3,2,1,5,3)
Electricity [kWh/kg]				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	7.19E-01 kWh		estimation	1	1.32	(4,2,3,2,1,5,3)
extrusion				plastics	processing	No	RER	extrusion, plastic pipes	1.00E+00 kg		calculated	1	1.00	(4,2,3,2,1,5,3)
KNO3				agricultural means of production	mineral fertiliser	No	RER	potassium nitrate, as N, at regional storehouse	7.34E-04 kg		estimation	1	1.32	(4,2,3,2,1,5,3)
NaNO2				agricultural means of production	mineral fertiliser	No	RER	potassium nitrate, as N, at regional storehouse	1.33E-03 kg		estimation	1	1.32	(4,2,3,2,1,5,3)
NaNO3				agricultural means of production	mineral fertiliser	No	RER	potassium nitrate, as N, at regional storehouse	1.89E-04 kg		estimation	1	1.32	(4,2,3,2,1,5,3)
cooling water				resource	in water			Water, cooling, unspecified natural origin	4.50E-02 m3		estimation	1	1.58	(4,2,3,2,1,5,3)
Transport rail			standard distance for raw material transport	transport systems	train	No	RER	transport, freight, rail	6.08E-01 tkm		estimation	1	2.09	(4,2,3,2,1,5,2)
Transport lorry			standard distance for raw material transport	transport systems	road	No	RER	transport, lorry >16t, fleet average	1.01E-01 tkm		estimation	1	2.09	(4,2,3,2,1,5,3)
Infrastructure (plant)				chemicals	organics	Yes	RER	chemical plant, organics	4.00E-10 unit		estimation	1	3.28	(4,2,3,2,1,5,3)
		→ waste heat	to water	water	unspecified			Heat, waste	1.81E+00 MJ		estimation	1	1.32	(4,2,3,2,1,5,3)
		→ waste heat	to air	air	unspecified			Heat, waste	7.77E-01 MJ		estimation	1	1.32	(4,2,3,2,1,5,3)
		→ water	cooling water						4.50E-02 m3					(5,2,3,2,1,5,4)
		→ EPDM elastomer, vulcanised		plastics	polymers	No	RER	synthetic rubber, at plant	1.00E+00 kg					

Fig. 4.1 Flows for "synthetic rubber, at plant" and their representation in the ecoinvent database

5 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

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Part XXI

Lightweight Products and its Preliminary Products and Processes

Data v2.0 (2007)

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Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

This part deals mainly with the production process of lightweight concrete blocks. Its raw materials are sand, cement and water for the concrete structure, mixed with lightweight aggregates. The lightweight aggregates which are studied are pumice from nature, polystyrene, expanded vermiculite, perlite and clay. The mining of the raw materials are included too, except clay which is offered in other part. The expanding process is a very energy intensive process. The needed energy is covered either by heavy (expanded clay) or light fuel oil. The moulded blocks are air-dried.

Additionally to the lightweight concrete blocks also autoclaved aerated concrete blocks and light clay bricks are studied. The raw materials for autoclaved aerated concrete blocks are silica sand and binding agents (milled quicklime, cement and anhydrite) which are mixed with water to a suspension. The expanding agent is aluminium powder or paste is added to the suspension and poured into a casting mould. The semisolid raw block is hardened in a autoclave and exposed to a saturated steam atmosphere. The needed energy is covered by natural gas. The raw materials for light clay bricks are clay and straw. After mixing them with water and pressed into a form they are dried by heat from natural gas.

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1 Introduction

This report describes the processes and their inventories for the production of different types of lightweight concrete blocks and light clay bricks as they are modelled in ecoinvent data v 1.1. The production of the used lightweight aggregates and the mining of its raw materials are dealt with too in this part.

Most of the relevant information to the processes is compiled in the figures depicting the input and output flows to the processes together with information about their use in the “real world” and about their representation in ecoinvent. This figure shows also the sources and the uncertainty information to the individual values.

2 Reserves, Resources and Raw Materials

2.1 Vermiculite

“Vermiculite is a natural silicate, belonging to the phyllosilicates (layer silicates), and has a mica-like foliated structure. It displays the typical properties of the montmorillonite – saponite group to a pronounced degree : the layers have a greater excess charge and a greater capacity for cation exchange than talc or pyrophyllite layers.” (Lagaly et al. (1997))

“Deposits of economic importance are found in northern Transvaal (Republic of South Africa), and in Montana and the Enoree district of South Carolina (USA). Smaller deposits occur in Kenya, Australia, India, and Japan. The vermiculite in these deposits has been formed from phlogopite and biotite by weathering (leaching of K) and hydrothermal change. The vermiculite in the Enoree district of South Carolina occurs in metamorphic rocks in contact with acidic intrusions interspersed with pegmatites. Accompanying minerals include corundum, apatite, serpentine, chlorite, talc, and mica.” (Lagaly et al. (1997))

2.2 Perlite

“Perlite (obsidian) is a natural glass that is usually black or gray, but sometimes brownish red. It has curved shrinkage cracks, and therefore breaks up into spheroidal granules. It is usually produced by rapid cooling of rhyolitic melts (rhyolite being the volcanic equivalent of granite). Being a glass, perlite contains few crystals and has a low water content. Naturally-occurring glasses containing 3 – 8 % water are known as pitchstone.” (Lagaly et al. (1997))

“The most important European deposits are found in Greece (Milos), Italy (Sardinia), the former Czechoslovakian Republic, Hungary, Bulgaria, and Iceland. Further important deposits are located in Turkey (Cumaovasi, Izmir), Canada, Australia, and the United States (New Mexico).” (Lagaly et al. (1997))

2.3 Pumice

“Pumice is a volcanic rock which occurs naturally in expanded form. It is formed from volcanic slag which by eruption under high pressure and together with the enclosed gas form a glassy liquid foam. By cooling down the foamy structure stays intact.” (translation by author from Starzner & Wurmer-Weiss (2000))

“Important commercially exploited pumice deposits occur in Germany (Koblenz-Neuwied district); in Italy (isle of Lipari, northeast to Sicily, and around Naples); in volcanic Greek islands (Dodecanese group : islands of Yali and Nisyros ; Cyclades group : Thera Island); Spain (Province of Ciudad Real,

Canada area on isle of Tenerife); Portugal (on the volcanic island of Sao Miguel in the Azores); United States (in central New Mexico in the area of Santa Fe, at Gate in the state of Oklahoma); in the Caribbean islands (Dominicana, Martinique); volcanic islands of Japan (southern parts of Hokkaido and Kyushu); New Zealand (North Island).” (Lagaly et al. (1997))

3 Use

3.1 Expanded Vermiculite

“In its most important applications, vermiculite is used in its expanded state to make use of its low density, good thermal and noise insulating properties, and high absorption capacity for liquids.

Vermiculite is utilized in the construction industry for thermal and sound insulation :

- Loose filling for hollow spaces, e.g., cavity walls
- Production of insulating plaster by bonding with gypsum or cement ; cement-bonded material is also used to prevent condensation on ceilings and walls in rooms with moist atmospheres
- Waterglass-bonded slabs for covering load-bearing constructional units made of steel or concrete, and for providing a fireproof covering
- As an additive to produce lightweight aggregate concrete, in competition with porous and pumice-stone concrete Vermiculite is usually used in expanded form” (Lagaly et al. (1997))

3.2 Expanded Perlite

Perlite usually is used in expanded form:

“The most important uses of expanded perlite (normally treated to render it hydrophobic) include filled plaster board and wall elements, insulating silicate-bonded boards for heat and flame protection, insulating bitumen-bonded boards for back facing, insulating boards and coating materials with polymer binders, mortar (bonded with cement, plaster, or silicate), polymer dispersion mortars and plasters for thermal insulation and prevention of condensation on walls and ceilings (e.g., in swimming pools, high humidity rooms). Expanded perlite, with a bulk density of 40 – 60 kg/m³, is used as an adsorptive filtering aid in precoated filters (e.g., for drinking water or medicinal products).” (Lagaly et al. (1997))

This study portrays expanded perlite which is used as insulation filling material in steep roofs, in walls (core insulation in double wall masonry) and floor constructions. (from Starzner & Wurmer-Weiss (2000))

3.3 Pumice

“Crushed and classified pumice is employed as an additive in light insulating concrete used to make constructional units. Due to its limited compression strength, this building material can only be used for two-storey buildings, or for cladding load-bearing frameworks.

Building blocks consist of pure, natural pumice granules treated to remove organic and inorganic impurities. They are mainly used to build houses.

Powdered pumice is used as an abrasive, as a polishing medium, and as an additive to pumice soap. It is also used as an absorption medium due to its large surface area (e.g., in packages carrying breakable liquid containers).

As an insulator, pumice competes with expanded perlite and expanded vermiculite.” (Lagaly et al. (1997))

3.4 Expanded Clay

“Expanded clay is mainly used as a lightweight aggregate for lightweight concrete blocks and thermal plaster. It is also used as sound and thermal bulk insulation (translated by author from Starzner & Wurmer-Weiss (2000))

3.5 Lightweight and Autoclaved Aerated Concrete Blocks

The main use of lightweight and autoclaved aerated concrete blocks are precision blocks, precision building plates, roof and floor panels, reinforced wall partition blocks, assembly beam, head and pillar units.

3.6 Light Clay and Cobwork

Light clay bricks are mainly used for framework renovation, for insulation of wooden beam ceilings and floors and for inner and exterior walls. The use as exterior walls needs a water-repellent outside plaster.

For the cobwork production, the raw materials clay, wooden chips and water are usually mixed on the construction site. After mixing it is put on hurdle work. The function of the wooden chips is to stabilize the construction and to increase the thermal insulation. The thickness of the walls can have in maximum the size of the girder.

Clay is particularly suitable as filling material in wooden constructions because clay has the quality to transport humidity of neighbour construction to the outside very quickly. This is due to the low humidity of dry clay (2 to 5%).

4 Characterisation

4.1 Vermiculite

The bulk density of crude vermiculite or vermiculite concentrate is in the range of 640-1120 kg/m³.

The following physical properties of expanded vermiculite are of interest for industrial applications (mostly taken from Lagaly et al. (1997)):

- | | |
|------------------------------------|---|
| - Density of expanded vermiculite: | 64 - 160 kg/m ³ ¹ |
| - Specific heat capacity : | 0.84 kJ kg ⁻¹ K ⁻¹ |
| - Thermal conductivity : | 0.062 – 0.065 W m ⁻¹ h ⁻¹ K ⁻¹ |

¹ <http://www.vermiculite.org/aboutvermiculite.htm>, access date: February 2004

4.2 Perlite

“The most important properties for the diverse applications of expanded perlite are its extremely low density, low thermal conductivity, low thermal expansion, good sound absorption, adequate compression strength at a bulk density of $> 100 \text{ kg/m}^3$, and resistance to acids, weathering, and microbial attack. The average properties of industrial perlite (from geographical origin Milos in Greece) after expanding and grinding to 98 % $< 40 \text{ mm}$ are”: (Lagaly et al. (1997))

- Density of expanded perlite: $60 - 300 \text{ kg/m}^3$ (Starzner & Wurmer-Weiss (2000))
- Thermal conductivity (25 °C) at a bulk density of 400 kg/m^3 : $0.04 \text{ W m}^{-1} \text{ h}^{-1} \text{ K}^{-1}$
 at a bulk density of 900 kg/m^3 : $0.05 \text{ W m}^{-1} \text{ h}^{-1} \text{ K}^{-1}$
 at a bulk density of $1'300 \text{ kg/m}^3$: $0.06 \text{ W m}^{-1} \text{ h}^{-1} \text{ K}^{-1}$
- Thermal conductivity (bulk density 900 kg/m^3) at 20 °C: $0.06 \text{ W m}^{-1} \text{ h}^{-1} \text{ K}^{-1}$
 at 400 °C: $0.11 \text{ W m}^{-1} \text{ h}^{-1} \text{ K}^{-1}$
 at 800 °C: $0.22 \text{ W m}^{-1} \text{ h}^{-1} \text{ K}^{-1}$

4.3 Pumice

The following physical properties of pumice are of interest for industrial applications (mainly taken from Lagaly et al. (1997)):

- Bulk density of pumice: $400 - 900 \text{ kg/m}^3$ (Starzner & Wurmer-Weiss (2000))
- Specific heat capacity: $1.00 \text{ kJ kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of pumice concrete: $1.005 \text{ kJ kg}^{-1} \text{ K}^{-1}$
- Thermal conductivity at 50°C: $0.274 \text{ W m}^{-1} \text{ h}^{-1} \text{ K}^{-1}$
- Thermal conductivity of pumice concrete: $0.457 \text{ W m}^{-1} \text{ h}^{-1} \text{ K}^{-1}$ at a density of ca. $1'000 \text{ kg/m}^3$

4.4 Expanded Vermiculite, Perlite and Clay

“The main raw material of expanded vermiculite (loose boulders) is mica schist, which is a volcanic rock. Another name for it is aluminium-iron-magnesium-silicate.” (translation by author from Starzner & Wurmer-Weiss (2000))

“Expanded perlite is made from raw perlite, a volcanic glass also called rhyolite or quartz-porphry glass. It is a result of volcanic activities. It is mainly made of about 75% SiO_2 and about 15% Al_2O_3 .” (translated by author from Starzner & Wurmer-Weiss (2000))

“Expanded clay is made out of clay which is rich in illite. As filling material in the case of fat clay, dust and upraised rejects from expanded clay production, are used. All ingredients (binding agent, additives, filling materials) are of natural origin and added to the mixture as solid material.” (translation by author from Starzner & Wurmer-Weiss (2000))

The following physical properties of expanded clay are of interest for industrial applications (mainly taken from Mötzl et al. (2000)):

- Bulk density of expanded clay: $260 - 500 \text{ kg/m}^3$
- Specific heat capacity: $1.00 \text{ kJ kg}^{-1} \text{ K}^{-1}$
- Thermal conductivity: $0.076 - 0.091 \text{ W m}^{-1} \text{ K}^{-1}$

4.5 Lightweight Concrete Blocks

“Lightweight concrete blocks are made from aggregates with porous structure, which is mixed with cement and water.” (translation by author from Starzner & Wurmer-Weiss (2000))

The products inventoried in this study are lightweight concrete blocks with following aggregates:

- Expanded vermiculite
- Expanded perlite
- Expanded clay
- Pumice
- Polystyrene

The following physical properties of lightweight concrete blocks are of interest for industrial applications (mainly taken from (mainly from Starzner & Wurmer-Weiss (2000)):

- Bulk density of lightweight concrete blocks: $200 - 2'000 \text{ kg/m}^3$.
- Thermal conductivity: $0.05 - 1.2 \text{ W m}^{-1} \text{ K}^{-1}$

4.6 Autoclaved Aerated Concrete Blocks

The following physical properties of autoclaved aerated concrete blocks are of interest for industrial applications (mainly taken from Starzner & Wurmer-Weiss (2000)):

- Bulk density of autoclaved aerated concrete blocks: $350 - 1'000 \text{ kg/m}^3$
- Specific heat capacity: $1.00 \text{ kJ kg}^{-1} \text{ K}^{-1}$ (Wendehorst & Vollenschaar (1998))
- Thermal conductivity: $0.12 - 0.29 \text{ W m}^{-1} \text{ K}^{-1}$ (Wendehorst & Vollenschaar (1998))

4.7 Light Clay and Cobwork

Light clay bricks are made of clay and the additive used in this study is straw. Other mineral or organic lightweight additives are possible. Clay has the effect of a binding agent. (Starzner & Wurmer-Weiss (2000))

Cobwork is a mixture of clay and an additive which in this study are wooden chips. It can also be mixed with saw dust or straw.

Two of the main physical characteristics of light clay bricks from Starzner & Wurmer-Weiss (2000)) are:

- Gross density: $600 \text{ to } 1'200 \text{ kg/m}^3$.
- Thermal conductivity: $0.17 \text{ W m}^{-1} \text{ K}^{-1}$ (for a density $\leq 600 \text{ kg/m}^3$)
 $0.47 \text{ W m}^{-1} \text{ K}^{-1}$ (for a density $\leq 1'200 \text{ kg/m}^3$)

5 Production Process

5.1 Expanded Vermiculite

The vermiculite-bearing contact zones are selectively extracted by open-cast mining. Harder rock may need to be broken up by explosives before removing it by front loaders. Softer rock is directly excavated.²

² <http://www.palabora.co.za>

“In the first stage, the rock is usually broken into < 20 mm or < 16 mm granules, and the adhering fine material is removed by washing and screening. Further size reduction separates the vermiculite from the accompanying rock.

In a process operated by Palabora Mining, the vermiculite flakes are separated from the accompanying rock, including mica, by multistage screening in a horizontal stream of air using various suspension techniques.

In industrial processes, the vermiculite flakes are expanded in 5 – 8 s at 1250 – 1500 °C and then cooled immediately to 400 °C. The hydroxyl groups of the silicate layers must not be split off by using an incorrect temperature–time profile because this results in intercrystalline rearrangement and break-down of the material into powder. The volume increase is 15 – 30 fold, depending on the raw material and method of furnace operation, and the weight loss is 10 – 12 %. The density of the expanded product is 64 – 160 kg/m³.” (Lagaly et al. (1997))

5.2 Expanded Perlite

“Perlite is always extracted by open-cast mining. It is then crushed (usually by cone crushers), and classified using vibratory screens into several grain sizes with closely specified tolerances.

The industrial value of the classified ore depends on the temperature and time necessary for expansion, relative volume increase, ratio of closed to open cells, nonexpandable fraction, and compression strength of the product.

Expansion is performed in two types of furnace:

1. tilted horizontal rotary furnaces through which the feed material passes in co-untercurrent flow to the combustion gas, or
2. vertical blast furnaces through which the expanding grains and hot gases pass in co-current flow.

The expansion results from shock heating the grains to 870 – 1090 °C (the softening range). Steam is formed inside the grains and inflates them, producing very large numbers of closed, thin-walled cells, and a 15 – 30 fold volume increase. The optimum temperature–time profile must be closely followed. Longer heating times lead to shrinkage of the heated aggregates, and higher temperatures cause shrinkage and/or melting. In some cases, the expanded perlite is ground by pendulum or roller mills. If the product is to be used as a filler or additive, it is treated to make it hydrophobic ; it may be silanized with dilute aqueous silane suspensions, or coated with dilute, aqueous silicone oil emulsions.” (Lagaly et al. (1997))

5.3 Pumice

“Pumice is mined with bucket excavators and wheel loaders and/or by blasting. The mined material is either transported by heavy load lorry or by conveyor belt to the washing and sieving location. After washing and sieving the pumice is possibly crushed and milled again before classification and storage. (Starzner & Wurmer-Weiss (2000), translated by the author)

5.4 Expanded Clay

“To produce expanded clay the raw clay is prepared by adding water, then ground, sometimes granulated and burned at reverse flow method in a rotary furnace at 1’200 °C. The organic materials which are naturally included and the added heavy fuel in the clay burn and the clay pellets expand. In the same time the surface melts and a sintered shell is produced.

Depending on the usage and after burning the clay pellets are possibly crushed and stored in bins separated in different grain size fractions.” (Starzner & Wurmer-Weiss (2000))

5.5 Lightweight Concrete Blocks

“The raw material mixture (aggregates, cement and water) of lightweight concrete blocks are poured into a mould then squeeze moulded and at the end hardened by air or by steam treatment.” (Starzner & Wurmer-Weiss (2000), translation by author). In this study the hardening process is assumed to be by air.

5.6 Autoclaved Aerated Concrete Blocks

“Silica sand is ground in a dry process in a ball triturator or prepared in a wet process by levigation. Prepared silica sand is then mixed under addition of binding agents (milled quicklime and/or cement) and under addition of water to a aqueous suspension. Shortly before the suspension is poured into the casting moulds the expanding agent (aluminium powder or paste) is added. Water slakes the lime by development of heat. Aluminium reacts with the alkaline water of the suspension, producing hydrogen as result. The hydrogen produces pores and escapes the blocks without residue. Depending on the raw density grade, from one m³ raw material about five m³ autoclaved aerated concrete blocks are produced.

A first binding process results in a semisolid raw block which is cut by machine into desired block formats. The not hardened production waste is reused in the production process as returned sludge.

For the steam curing the cut semisolid blocks are put into autoclaves (hardening vessel) and exposed to a saturated steam atmosphere of 180 to 200 °C with a pressure of about 8 to 12 bar for a time of 6 to 12 hours. After the hardening process, steam and condensate are passed on to a next autoclave or to a reservoir. The hardened autoclaved aerated concrete blocks consists mainly of calcium silicate hydrate and air voids.” (Starzner & Wurmer-Weiss (2000), translation by author).

“Arising process water is recycled by returning it to the production process. Therefore the amount of arising wastewater is very small” (Starzner & Wurmer-Weiss (2000), translation by author). Based on this description no wastewater is taken into account in this study.

“The amount of aluminium powder used for the expanding process is very small and recycling material from punchings is used.” (Starzner & Wurmer-Weiss (2000), translation by author).

Tab. 5.1 (raw materials needed for the production of autoclaved aerated concrete blocks) shows the raw materials needed for the production of autoclaved aerated concrete blocks with an assumed density of 500 kg/m³.

Tab. 5.1 Example of raw material mixtures for a block density of 500 kg/m³ (Hutter et al. (1999))

Raw materials	[kg/kg]
Sand/gravel	0.504
quicklime	0.104
portland cement	0.260
anhydrite	0.044
aluminium powder	0.0007
water	0.648

Production waste is totally reused as secondary raw material and therefore not taken into account. (Starzner & Wurmer-Weiss (2000))

“The cumulated emissions as result from the production of autoclaved aerated concrete blocks are small and stem mainly from burning of natural gas for the steam hardening process and the production process of the electricity mix. Most rest emissions come from the production of the raw materials.” (Hutter et al. (1999)). According to that fact this study does not take direct emissions into account.

The production process of sand-lime brick and autoclaved aerated concrete blocks are very similar. Slight differences are the varying saturated steam pressure in the vessel which is between 12 to 16 bar for sand-lime bricks and 6 to 12 bar for autoclaved aerated concrete blocks. Both products have the same hardening temperature of 160 to 200°C but different hardening times. For sand-lime bricks its 4 to 8 hours and for autoclaved aerated concrete blocks its 6 to 12 hours.

5.7 Light Clay and Cobwork

The raw materials clay and straw are mixed in a weight ratio of about 85% clay and 15% straw. The straw is mainly added chaffed. These materials are mixed with water and then pressed in forms. The formed bricks are then technically dried in a furnace heated by natural gas. After this drying process they are packed. (Starzner & Wurmer-Weiss (2000) and Weibel & Stritz (1995))

The raw materials clay, wooden chips and water for the cobwork are mixed on the construction site and put on the hurdle work. The cobwork is naturally dried.

6 Life Cycle Inventory

Ecoinvent database meta information (tables), flows and their representation in the ecoinvent database are shown in (figures):

Vermiculite, at mine	Tab. 6.1 and Fig. 6.1
Perlite, at mine	Tab. 6.2 and Fig. 6.2
Pumice, at mine	Tab. 6.3 and Fig. 6.3
Mine, Vermiculite	Tab. 6.4 and Fig. 6.4
Expanded vermiculite, at plant	Tab. 6.5 and Fig. 6.5
Expanded perlite, at plant	Tab. 6.6 and Fig. 6.6
Expanded clay, at plant	Tab. 6.7 and Fig. 6.7
Lightweight concrete block, expanded vermiculite, at plant	Tab. 6.8 and Fig. 6.8
Lightweight concrete block, expanded perlite, at plant	Tab. 6.9 and Fig. 6.9
Lightweight concrete block, expanded clay, at plant	Tab. 6.10 and Fig. 6.10
Lightweight concrete block, pumice, at plant	Tab. 6.11 and Fig. 6.11
Lightweight concrete block, polystyrene, at plant	Tab. 6.12 and Fig. 6.12
Autoclaved aerated concrete block, at plant	Tab. 6.13 and Fig. 6.13
Light clay brick, at plant	Tab. 6.14 and Fig. 6.14
Cobwork, at plant	Fig. 6.15 and Fig. 6.15

6.1 Vermiculite, at mine

The ecoinvent database meta information of “vermiculite, at mine” (Tab. 6.1) includes all important and necessary information to use the data correctly.

Tab. 6.1 Ecoinvent database meta information for the product “vermiculite, at mine”

Name	vermiculite, at mine
Location	ZA
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes excavation by digger, blasting, transportation within mine, the land-use for quarrying, the recultivation
Amount	1
Local Name	Vermiculit, ab Mine
Synonyms	Glimmer
General Comment to reference function	For a lack of data the dust emissions for bauxite mining are used as proxy. For the recultivation process the module "recultivation, limestone mine" has been used as proxy. To account for the phosphate mined together with the vermiculite, the resource "phosphorus, 18% in apatite, 12% in crude ore, in ground" is inventoried. Since the apatite is dumped but afterwards used by other companies it is neither considered co-product nor waste. The paraffin fueled heating system is approximated with "light fuel oil, burned in industrial furnace 1 MW, non-modulating". For the electricity mix the UCPTe-mix has been used.
Start Date	2000
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For the exchanges, RER, GLO and CH modules have been used as proxy.
Technology text	High level of technology used in one of the biggest vermiculite mines in the world, in South-Africa.
Representativeness [%]	
Production Volume	2'390'000 tons (2001)
Sampling Procedure	Data from questionnaire from one company called Parabora mining company ltd. in South-Africa.
Extrapolations	data which are only available for GLO, RER and CH have been used as proxy for ZA
Uncertainty Adjustments	none

Fig. 6.1 shows the system of the module “vermiculite, at mine”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly personal communication³, Künniger et al. (2001), Süd-Chemie (1999) and estimations) and the standard deviation for each data.

Most of the vermiculite used in Europe comes from South Africa. and therefore the data in this study are based on information from Palabora Mining Company Ltd. in South Africa.⁴

³ personal communication: Mr. Gabe van den Berg, Palabora Mining Company Limited, South Africa, August 13th 2002

“Vermiculite mining and concentration in the Republic of South Africa was started in the 1940's. The industry was established primarily to satisfy European demand and later extended to North America and Australia. Most of Palabora's vermiculite production is exported. At Palabora vermiculite ore is derived from three sources:

- The vermiculite pit which is mined as an open cast truck and loader operation. The pit is roughly 1600 m long on the north-south axis and 1400 m wide on the east-west axis. Overall depth is 50 m below average surface elevation.
- The original plant dump of plant tailings has become economically more significant due to an ever increasing demand for the finer grades of vermiculite. The dump is reclaimed using trucks and front-end loaders.
- Palabora Phosphate and Vermiculite (PP&V) mine. A deposit mined as an open cast truck and loader operation similar to the original pit.”

General Flow information					Representation in ecoinvent							Uncertainty Information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
vermiculite in ground	excavation of vermiculite from nature (part 1)			total mined amount of vermiculite ore in Palabora mine in 2001: 2'390'000 tons, total amount of waste (mainly phosphate) in 2001: 710'000 tons, no losses taken into account as no data available	resource	in ground			Vermiculite, in ground	1.00E+00	kg	After questionnaire Palabora Mining Company Ltd. (2002)	1	1.24	(1,4,1,1,1,5);
phosphate in ground				total amount of secondary product (mainly phosphate) mined in 2001: 710'000 tons, module "Phosphorus, 18% in apatite, 12% in crude ore, in ground" used as proxy	resource	in ground			Phosphorus, 18% in apatite, 12% in crude ore, in ground	3.00E-01	kg	After questionnaire Palabora Mining Company Ltd. (2002)	1	2.06	(1,4,1,1,1,5,5);
occupation mineral extraction site (mine) [m2a]				transformation = mining area, assumed occupation time: 50 years; assumed density of mining material: 2'000 kg/m3; mining depth: 50 m	resource	land			Occupation, mineral extraction site	6.50E-04	m2a	estimation	1	1.58	(1,4,1,1,1,5);
land transformation to mineral extraction site (mine) [m2]				transformation = mining area; assumed density: 2'000 kg/m3 of mining material; mining depth: 50 m	resource	land			Transformation, to mineral extraction site	1.30E-05	m2	estimation	1	2.06	(1,4,1,1,1,5);
land transformation mine from unknown				transformation = mining area; assumed density: 2'000 kg/m3 of mining material; mining depth: 50 m	resource	land			Transformation, from unknown	1.30E-05	m2	estimation	1	2.06	(1,4,1,1,1,5);
diesel (mine)				total diesel consumption for in 2001: 601'000 liter (=504'840 kg)	construction processes	machinery	No	GLO	diesel, burned in building machine	6.97E-03	MJ	After questionnaire Palabora Mining Company Ltd. (2002)	1	1.24	(1,4,1,1,1,5);
heating - drying of feed material to plant				paraffin is used as heating fuel, yearly amount 2001: 450'000 liter, heating oil EL (used as proxy; 0.86 kg/l, 42.6 kg/MJ)	oil	heating systems	No	RER	light fuel oil, burned in industrial furnace 1MW, non-modulating	5.32E-03	MJ	After questionnaire Palabora Mining Company Ltd. (2002)	1	2.06	(1,4,1,1,1,5,5);
electricity consumption				includes crusher (0.1 kWh/ton crushed material) and winnowing machine (0.0045 kWh/ton)	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.05E-04	kWh	After questionnaire Palabora Mining Company Ltd. (2002)	1	1.24	(1,4,1,1,1,5);
lubricating oil				amount copied from bentonite mining	chemicals	organics	No	RER	lubricating oil, at plant	8.06E-05	kg	Süd-Chemie (1999)	1	5.00	estimation
blasting				Type of explosive used: HEF 100 Explosive; "Tovex" used as proxy	construction processes	civil engineering	No	RER	blasting	1.66E-04	kg	After questionnaire Palabora Mining Company Ltd. (2002)	1	2.06	(1,4,1,1,1,5,5);
vermiculite mine				total production amount for the year 2001: 3'100'000 tons, assumed life time: 50 years	construction materials	additives	Yes	ZA	mine, vermiculite	6.45E-12	unit	After questionnaire Palabora Mining Company Ltd. (2002)	1	3.73	(1,4,1,1,1,5,5);

Fig. 6.1 Flows for "vermiculite, at mine" and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty Information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
excavation of vermiculite from nature (part 2)	→	waste heat	From electricity uses	air	unspecified			Heat, waste	3.76E-04	MJ	After questionnaire Palabora Mining Company Ltd. (2002)	1	1.24	(1,4,1,1,1,5);
	→	Dust to air [kg]	Assumptions: Same as Limestone	air	low population density			Particulates, < 2.5 um	8.00E-06	kg	BUWAL, 2001	1	10.00	estimation
	→	Dust to air [kg]	Assumptions: Same as Limestone	air	low population density			Particulates, > 2.5 um, and < 10um	4.00E-05	kg	BUWAL, 2001	1	10.00	estimation
	→	Dust to air [kg]	Assumptions: Same as Limestone	air	low population density			Particulates, > 10 um	1.12E-04	kg	BUWAL, 2001	1	10.00	estimation
	→	phosphate bearing waste material dumped on designated dump site	P2O5 bearing material is dumped on designated dump site but processed by another near-by mine (Foskor mine Ltd.) and therefore treated as being recycled.						3.00E-01		After questionnaire Palabora Mining Company Ltd. (2002)	1		
	→	vermiculite, at mine	total production amount for the year 2001: 2'390'000 tons	construction materials	additives	No	ZA	vermiculite, at mine	1.00E+00	kg		1		

Fig. 6.1 Flows for “vermiculite, at mine” and their representation in the ecoinvent database (cont)

6.2 Perlite, at mine

The ecoinvent database meta information of “perlite, at mine” (Tab. 6.2) includes all important and necessary information to use the data correctly.

Tab. 6.2 Ecoinvent database meta information for the product “perlite, at mine”

Name	perlite, at mine
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes excavation by digger, blasting, transportation within mine, the land-use of the mine, the recultivation and dust emissions
Amount	1
Local Name	Perlit, ab Mine
Synonyms	
General Comment to reference function	As no data are available, data from vermiculite mining are copied. Uncertainties are raised accordingly.
Start Date	2000
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For the exchanges, RER, GLO and CH modules have been used as proxy.
Technology text	Technology of vermiculite mining used.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Data adapted from module "vermiculite, at mine" based on literature
Extrapolations	data which are only available for GLO, RER and CH have been used as proxy for D
Uncertainty Adjustments	none

Fig. 6.2 shows the system of the module “perlite, at mine”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly estimations) and the standard deviation for each data.

General Flow information					Representation in ecoinvent							Uncertainty Information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
perlite	→	excavation of perlite from nature (part 1)				losses are neglected	resource	in ground			Perlite, in ground	1.00E+00	kg	estimation and questionnaire Polabora Mining Comany Ltd. (2002)	1	5.00	estimation
occupation mineral extraction site (mine) [m2a]	→					transformation = mining area, assumed occupation time: 50 years; assumed density of mining material: 2'000 kg/m3; mining depth: 50 m	resource	land			Occupation, mineral extraction site	5.00E-04	m2a	estimation and questionnaire Polabora Mining Comany Ltd. (2002)	1	5.00	estimation
land transformation to mineral extraction site (mine) [m2]	→					transformation = mining area; assumed density: 2'000 kg/m3 of mining material; mining depth: 50 m	resource	land			Transformation, to mineral extraction site	1.00E-05	m2	estimation and questionnaire Polabora Mining Comany Ltd. (2002)	1	5.00	estimation
land transformation mine from unknown	→					transformation = mining area; assumed density: 2'000 kg/m3 of mining material; mining depth: 50 m	resource	land			Transformation, from unknown	5.00E-04	m2	estimation and questionnaire Polabora Mining Comany Ltd. (2002)	1	5.00	estimation
diesel (mine)	→					Amount of diesel per kg mined material copied from vermiculite mining process	construction processes	machinery	No	GLO	diesel, burned in building machine	6.97E-03	MJ	estimation and questionnaire Polabora Mining Comany Ltd. (2002)	1	5.00	estimation
heating - drying of feed material	→					Amount of light fuel oil per kg mined material copied from vermiculite mining process	oil	heating systems	No	RER	light fuel oil, burned in industrial furnace 1MW, non-modulating	5.32E-03	MJ	estimation and questionnaire Polabora Mining Comany Ltd. (2002)	1	5.00	estimation
electricity consumption	→					Amount of electricity per kg mined material copied from vermiculite mining process	electricity	production mix	No	DE	electricity, medium voltage, production DE, at grid	1.05E-04	kWh	estimation	1	5.00	estimation
lubricating oil	→					Amount copied from bentonite mining	chemicals	organics	No	RER	lubricating oil, at plant	8.06E-05	kg	estimation and Süd-Chemie (1999)	1	5.00	estimation
explosives	→					Amount copied from vermiculite mining process	construction processes	civil engineering	No	RER	blasting	1.66E-04	kg	estimation	1	5.00	estimation
infrastructure mine	→					copied from vermiculite mining process, assumed life time: 50 years	construction materials	additives	Yes	ZA	mine, vermiculite	6.45E-12	unit	estimation	1	5.00	estimation

Fig. 6.2 Flows for “perlite, at mine” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent										Uncertainty Information		
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
excavation of perlite from nature (part 2)	→	waste heat				from electricity uses	air	unspecified			Heat, waste	3.76E-04	MJ	calculated after the electricity use	1	5.00	estimation
	→	Dust to air [kg]				Assumptions: Same as Limestone	air	low population density			Particulates, < 2.5 um	8.00E-06	kg	BUWAL, 2001	1	5.00	estimation
	→	Dust to air [kg]				Assumptions: Same as Limestone	air	low population density			Particulates, > 2.5 um, and < 10um	4.00E-05	kg	BUWAL, 2001	1	5.00	estimation
	→	Dust to air [kg]				Assumptions: Same as Limestone	air	low population density			Particulates, > 10 um	1.12E-04	kg	BUWAL, 2001	1	5.00	estimation
	→	overburden material (topsoil)				neglected											
	→	perlite, at mine					construction materials	additives	No	DE	perlite, at mine	1.00E+00	kg				

Fig. 6.2 Flows for “perlite, at mine” and their representation in the ecoinvent database (cont)

6.3 Pumice, at mine

The ecoinvent database meta information of “pumice, at mine” (Tab. 6.3) includes all important and necessary information to use the data correctly.

Tab. 6.3 Ecoinvent database meta information for the product “pumice, at mine”

Name	pumice, at mine
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes excavation by digger, transportation within mine, washing, the land-use of the mine, the recultivation and dust emissions
Amount	1
Local Name	Naturbims, ab Mine
Synonyms	
General Comment to reference function	For a lack of data the dust emissions for bauxite mining are used as proxy. Further following proxys have been used: The amount of water has been copied from the water consumption in "sand, at mine"-wasing and for heating the "light fuel oil"-heating module has been used.
Start Date	2000
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For the exchanges, RER, GLO and CH modules have been used as proxy.
Technology text	technology unknown
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	sampled from different other modules (some data copied from "limestone crushing", some from "bentonite mining", some from "recultivation, limestone mine", etc.).
Extrapolations	data which are only available for GLO, RER and CH have been used as proxy for D
Uncertainty Adjustments	none

Fig. 6.3 shows the system of the module “pumice, at mine”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly Hutter et al. (1999), estimations and ⁵) and the standard deviation for each data.

⁵ ecoinvent-report No. 7
<http://www.bgr.de/b121/ctn1903.pdf>, access data: February 2004

General Flow information					Representation in ecoinvent							Uncertainty Information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
pumice, at ground	→	excavation of pumice from nature (part 1)			yearly mining amount in Germany 2001: 124'000 tons	resource	in ground			Pumice, in ground	2.17E+00	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
occupation mineral extraction site (mine) [m2a]	→				occupied area Germany 2001: 31'000m2, assumed duration of occupation: 50 years	resource	land			Occupation, mineral extraction site	1.25E-02	m2a	http://www.bgr.de/b121/ctn1903.pdf	1	1.89	(3,5,5,3,1,5);
land transformation to mineral extraction site (mine) [m2]	→				occupied area Germany 2001: 31'000m2,	resource	land			Transformation, to mineral extraction site	2.50E-04	m2	http://www.bgr.de/b121/ctn1903.pdf	1	2.05	(4,1,1,1,1,1);
land transformation mine from unknown	→				occupied area Germany 2001: 31'000m2,	resource	land			Transformation, from unknown	2.50E-04	m2	estimation and questionnaire Polabora Mining Comany Ltd. (2002)	1	5.00	estimation
water for washing and separating the mined pumice	→				copied from module "sand, at mine", water used for washing 1 kg sand: 0.00138 m3, the amount of water applied on the total mined pumice	resource	in water			Water, well, in ground	2.99E-03	m3	http://www.bgr.de/b121/ctn1903.pdf	1	5.00	estimation
diesel, burned in building machine	→				0.9 liter diesel per m3 mined pumice is needed, average density 500kg/m3	construction processes	machinery	No	GLO	diesel, burned in building machine	1.51E-03	MJ	Hutter et al. (1999)	1	2.34	(3,5,5,3,1,5);
electricity consumption	→				for crushing process, copied from limestone crushing	electricity	production mix	No	DE	electricity, medium voltage, production DE, at grid	7.20E-04	kWh	estimation and data of KFN for the year 2001	1	5.00	estimation
lubricating oil	→				amount copied from bentonite mining	chemicals	organics	No	RER	lubricating oil, at plant	8.06E-05	kg	estimation and Süd-Chemie (1999)	1	5.00	estimation
infrastructure mine	→			only machines taken into account as it is assumed that no buildings are needed; estimation machines concerning crushing process; total weight 25t, assumed life time: 25 years, estimated yearly mining amount: 60'000'000 kg	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.67E-05	kg	estimation	1	5.00	estimation	

Fig. 6.3 Flows for "pumice, at mine" and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty Information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
excavation of pumice from nature (part 2)	→	waste heat	from electricity uses	air	unspecified			Heat, waste	2.59E-03	MJ	calculated after the electricity use	1	5.00	estimation
	→	Dust to air [kg]	Assumptions: Same as Limestone	air	low population density			Particulates, < 2.5 um	8.00E-06	kg	BUWAL, 2001	1	5.00	estimation
	→	Dust to air [kg]	Assumptions: Same as Limestone	air	low population density			Particulates, > 2.5 um, and < 10um	4.00E-05	kg	BUWAL, 2001	1	5.00	estimation
	→	Dust to air [kg]	Assumptions: Same as Limestone	air	low population density			Particulates, > 10 um	1.12E-04	kg	BUWAL, 2001	1	5.00	estimation
	→	heavy fraction not used for lightweight products	neglected; assumed to be about 1 kg						1.00E+00					
	→	overburden material (topsoil)	neglected; assumed to be about 0.17 kg						1.70E-01					
	→	pumice, at mine	yearly mining amount in Germany 2001: 124'000 tons	construction materials	additives	No	DE	pumice, at mine	1.00E+00	kg				

Fig. 6.3 Flows for “pumice, at mine” and their representation in the ecoinvent database (cont)

6.4 Mine, vermiculite

The ecoinvent database meta information of “vermiculite mining plant” (Tab. 6.4) includes all important and necessary information to use the data correctly.

Tab. 6.4 Ecoinvent database meta information for the product “vermiculite mining plant”

Name	vermiculite mining plant
Location	ZA
Infrastructure Process	1
Unit	unit
Data Set Version	2.0
Included Processes	includes land-use (occupation, transformation) for infrastructure, buildings, roads and green areas. The quarried area is not included here, but in the module describing mining activities
Amount	1
Local Name	Mine, Vermiculit
Synonyms	
General Comment to reference function	The mine studied is the Palabora mine in South Africa
Start Date	2000
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	data are only from one company in South-Africa (Palabora Mining Company Ltd.);
Technology text	infrastructure from one of the biggest mines in the world, which is in South-Africa, has been used.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Data from questionnaire from one company called Palabora mining company ltd. in South-Africa.
Extrapolations	data which are only available for CH, RER have been used as proxy for D
Uncertainty Adjustments	none

Fig. 6.4 shows the system of the module “vermiculite mining plant”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly personal communication⁶) and the standard deviation for each data.

⁶ personal communication: Mr. Gabe van den Berg, Palabora Mining Company Limited, South Africa, August 13th 2002

General Flow information					Representation in ecoinvent							Uncertainty Information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
infrastructure (machines)	→	vermiculite mine			includes 10 crusher (total 36 tons) and some additional machines (weight estimated: 14 tons), no further infrastructure taken into account, assumed life-time: 25 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	1.00E+05	kg	After questionnaire Palabora Mining Company Ltd. (2002)	1	2.06	(1,4,1,1,1,5);
infrastructure: building hall					assumed that used buildings are similar with those from limestone mine	construction processes	buildings	Yes	CH	building, hall	5.00E+02	m2	estimation	1	2.06	(1,4,1,1,1,5);
infrastructure: building multistore					assumed that similar with limestone mine and therefore amount copied	construction processes	buildings	Yes	RER	building, multi-storey	4.80E+02	m3	estimation	1	2.06	(1,4,1,1,1,5);
total land transformation from unknown	→				sum of green part, road part and railway track part	resource	land			Transformation, from unknown	5.72E+06	m2	After questionnaire Palabora Mining Company Ltd. (2002)	1	1.58	(1,4,1,1,1,5);
occupation green area [m2a]	→				Green area in 2001: 5'612'000 m2; assumed occupation time period 100 years	resource	land			Occupation, industrial area, vegetation	5.61E+08	m2a	After questionnaire Palabora Mining Company Ltd. (2002)	1	2.06	(1,4,1,1,1,5);
land transformation to green part of area [m2]	→				Green area in 2001: 5'612'000 m2	resource	land			Transformation, to industrial area, vegetation	5.61E+06	m2	After questionnaire Palabora Mining Company Ltd. (2002)	1	1.58	(1,4,1,1,1,5);
occupation road area [m2a]	→				Road area in 2001: 52'000 m2; assumed occupation timeperiod 100 years	resource	land			Occupation, traffic area, road network	5.20E+06	m2a	After questionnaire Palabora Mining Company Ltd. (2002)	1	2.06	(1,4,1,1,1,5);
land transformation to road area [m2]	→				Road area in 2001: 52'000 m2	resource	land			Transformation, to traffic area, road network	5.20E+04	m2	After questionnaire Palabora Mining Company Ltd. (2002)	1	1.58	(1,4,1,1,1,5);
occupation building area [m2a]	→				building area in 2001: 57'200 m2; assumed occupation timeperiod 100 years	resource	land			Occupation, industrial area, built up	5.72E+06	m2a	After questionnaire Palabora Mining Company Ltd. (2002)	1	2.06	(1,4,1,1,1,5);
land transformation to building area [m2]	→				building area in 2001: 57'200 m2	resource	land			Transformation, to industrial area, built up	5.72E+04	m2	After questionnaire Palabora Mining Company Ltd. (2002)	1	1.58	(1,4,1,1,1,5);
occupation railway track area [m2a]	→				Railway track area in 2001: 48'400 m2; assumed occupation timeperiod 200 years	resource	land			Occupation, traffic area, rail network	9.68E+06	m2a	After questionnaire Palabora Mining Company Ltd. (2002)	1	2.06	(1,4,1,1,1,5);
land transformation to railway track area (mine) [m2]	→				Railway track area in 2001: 48'400 m2	resource	land			Transformation, to traffic area, rail network	4.84E+04	m2	Questionnaire Palabora Mining Company Ltd. (2002)	1	2.06	(1,4,1,1,1,5);
			→	vermiculite mine			construction materials	additives	Yes	ZA	mine, vermiculite	1.00E+00	unit			

Fig. 6.4 Flows for “vermiculite mining plant” and their representation in the ecoinvent database

6.5 Expanded vermiculite, at plant

The ecoinvent database meta information of “expanded vermiculite, at plant” (Tab. 6.5) includes all important and necessary information to use the data correctly.

Tab. 6.5 Ecoinvent database meta information for the product “expanded vermiculite, at plant”

Name	expanded vermiculite, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the raw material vermiculite, the transport to the finishing plant and some internal transports, the energy carrier, some packaging materials, the infrastructure and the emissions
Amount	1
Local Name	Blähvermiculit, ab Werk
Synonyms	Blähglimmer
General Comment to reference function	Vermiculite transported by ship 13'500 km from South-Africa to Germany. It is expanded by the heat from light fuel oil in 5 – 8 s at 1250 – 1500 °C and then cooled immediately to 400 °C. Density of expanded vermiculite: 64 - 160 kg/m ³
Start Date	1995
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH, RER, GLO, OCE and ZA modules have been used as proxy.
Technology text	Basic principle is a quick heating of the vermiculite flakes to high temperature followed by a quick cooling. The volume increase is 15 – 30 fold
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.5 shows the system of the module “expanded vermiculite, at plant”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly Richter et al. (1995) and EPA (1998b)) and the standard deviation for each data.

General Flow information					Representation in ecoinvent							Uncertainty Information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
vermiculite, at mine	→	swelling of vermiculite (part 1)				construction materials	additives	No	ZA	vermiculite, at mine	1.14E+00	kg	Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);
transport of total amount of raw vermiculite with freight rail	→			transport distance: 400 km	transport systems	train	No	RER	transport, freight, rail	4.55E-01	tkm	Richter et al. (1995)	1	2.15	(4,5,3,1,1,5);	
transport of total amount of raw vermiculite with freight ship	→			transport distance: 13'500 km	transport systems	ship	No	OCE	operation, transoceanic freight ship	1.53E+01	tkm	Richter et al. (1995)	1	2.15	(4,5,3,1,1,5);	
transport of total amount of raw vermiculite with lorry	→			transport distance: 42 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	4.77E-02	tkm	Richter et al. (1995)	1	2.15	(4,5,3,1,1,5);	
infrastructure (incl. Land-use and machines)	→			clay manufacturing plant used as proxy; amount copied from module "brick, at plant"	construction materials	additives	Yes	CH	mine, clay	2.00E-10	unit	estimation	1	5.00	estimation	
water	→			assumed that tap water has been used used for high-lift truck, module "diesel, burnded in building machine" used as proxy, amount: 0.000267 l/kg	water supply	production	No	RER	tap water, at user	1.34E-02	kg	Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);	
diesel	→				construction processes	machinery	No	GLO	diesel, burned in building machine	9.61E-03	MJ	Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);	
electricity	→			UCTE supply mix used as proxy.	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.71E-02	kWh	Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);	
light fuel oil	→			the amount of light fuel oil (0.055kg/kg expanded vermiculite) is equivalent to about 2.4 MJ/kg	oil	fuels	No	RER	light fuel oil, at regional storage	5.55E-02	kg	Richter et al. (1995)	1	1.21	(4,1,1,1,1,1);	
packing material (paper)	→			One paper bag for 25 kilo product. Assumed weight of paper bag 0.05 kg	paper & cardboard	cardboard & corrugated board	No	RER	corrugated board base paper, kraftliner, at plant	2.00E-03	kg	Richter et al. (1995)	1	1.41	(4,5,3,5,1,5);	
packing material (plastics)	→				plastics	processing	No	RER	packaging film, LDPE, at plant	4.81E-04	kg	Richter et al. (1995)	1	1.41	(4,5,3,5,1,5);	

Fig. 6.5 Flows for "expanded vermiculite, at plant" and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty Information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
swelling of vermiculite (part 2)	→	waste heat				from electricity uses	air	unspecified			Heat, waste	9.77E-02 MJ		calculated after the electricity use	1	1.40	(4,5,3,1,1,5);
	→	vermiculite waste				losses during production process						5.08E-02 kg		Richter et al. (1995)			
	→	disposal of dust from filter				total amount: 1kg/t product	waste management	sanitary landfill	No	CH	disposal, inert material, 0% water, to sanitary landfill	1.00E-03 kg		EPA (1998), chapter 11.28	1	5.00	estimation
	→	carbon dioxide				measured data supplied by production plant(s)	air	unspecified			Carbon dioxide, fossil	1.75E-01 kg		Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);
	→	nitrogen oxide as NO2				measured data supplied by production plant(s)	air	unspecified			Nitrogen oxides	3.18E-05 kg		Richter et al. (1995)	1	1.69	(4,5,3,1,1,5);
	→	sulfur oxide as SO2				measured data supplied by production plant(s)	air	unspecified			Sulfur dioxide	2.25E-04 kg		Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);
	→	carbon monoxide				measured data supplied by production plant(s)	air	unspecified			Carbon monoxide, fossil	2.33E-05 kg		Richter et al. (1995)	1	5.17	(4,5,3,1,1,5);
	→	hydrocarbon				measured data supplied by production plant(s)	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	5.00E-07 kg		Richter et al. (1995)	1	3.15	(4,5,3,1,1,5);
	→	expanded vermiculite					construction materials	additives	No	CH	expanded vermiculite, at plant	1.00E+00 kg					

Fig. 6.5 Flows for “expanded vermiculite, at plant” and their representation in the ecoinvent database (cont)

In Starzner & Wurmer-Weiss (2000) the amount of cumulated energy demand (definition based on Kasser & Pöll (1998)) needed for the whole production process is 5.7 MJ/kg expanded vermiculite. The main part of this is necessary for the thermal expansion process. The published data of Richter et al. (1995) shows a much lower amount of cumulated energy demand (about 2.4 MJ per kg expanded vermiculite). The data stem from a certain company and rely to the year 1994. Unfortunately no more information is available to discuss the reason for the difference. More actual data are needed for a profound answer.

6.6 Expanded perlite, at plant

The ecoinvent database meta information of “expanded perlite, at plant” (Tab. 6.6) includes all important and necessary information to use the data correctly.

Tab. 6.6 Ecoinvent database meta information for the product “expanded perlite, at plant”

Name	expanded perlite, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the raw material, the transport to the finishing plant and some internal transports, the heating energy carrier, some packaging materials, the infrastructure and the emissions
Amount	1
Local Name	Blähperlite, ab Werk
Synonyms	
General Comment to reference function	Expanded by shock heating at a temperature of 870 – 1090 °C. It is assumed that light fuel oil is used for heating. Density of expanded perlite: 60 - 300kg/m ³
Start Date	1995
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH, RER and OCE modules have been used as proxy.
Technology text	Basic principle is a heating of the perlite grains to high temperature. The volume increase is 15 – 30 fold
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.6 shows the system of the module “expanded perlite, at plant”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly Richter et al. (1995), EPA (1998a), Weibel & Stritz (1995) and ⁷) and the standard deviation for each data.

The data of this study are mainly based on (Richter et al. (1995)) and include transports of the raw materials from mine to factory and some packaging materials.

“The cumulative energy demand (Kasser & Pöll (1998)) for the whole production process for expanded perlite is 9.3 MJ/kg. The thermal process needs most of the grey energy” (Starzner & Wurmer-Weiss (2000)), translated by the author). Therefore the energy needed is assumed to be 8.5 MJ/kg. The energy carrier is light fuel oil. (Richter et al. (1995))

⁷ http://www.agoef.de/baustoffe_daemmung/geblaechte.html, access date: February 2004

General Flow information					Representation in ecoinvent							Uncertainty Information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
perlite, at mine	→	swelling of perlite				construction materials	additives	No	DE	perlite, at mine	1.02E+00	kg	Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);
transport with freight rail	→			transport distance of one part of raw perlite to plant: 500 km	transport systems	train	No	RER	transport, freight, rail	4.50E-01	tkm	Richter et al. (1995)	1	2.15	(4,5,3,1,1,5);	
transport with freight ship	→			transport distance of one part of raw perlite to plant over 1'300 km and another part ovr 1'500 kg	transport systems	ship	No	OCE	transport, transoceanic freight ship	1.35E+00	tkm	Richter et al. (1995)	1	2.15	(4,5,3,1,1,5);	
transport with lorry	→			different transport distances for different raw materials	transport systems	road	No	RER	transport, lorry >16t, fleet average	4.87E-01	tkm	Richter et al. (1995)	1	2.15	(4,5,3,1,1,5);	
infrastructure (incl. Land-use and machines)	→			clay manufacturing plant used as proxy; amount copied from module "brick, at plant"	construction materials	additives	Yes	CH	mine, clay	2.00E-10	unit	estimation	1	5.00	estimation	
water	→			assumed that tap water is used	water supply	production	No	RER	tap water, at user	5.00E-04	kg	Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);	
electricity	→				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	6.50E-02	kWh	Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);	
light fuel oil	→				oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	8.50E+00	MJ	Starzner & Wurmer-Weiss (2000) and estimation	1	5.00	estimation	
packing material (paper)	→				paper & cardboard	cardboard & corrugated board	No	RER	corrugated board base paper, kraftliner, at plant	1.44E-03	kg	[Weibel, 1995]	1	1.41	(4,5,3,5,1,5);	
packing material (plastics)	→				plastics	processing	No	RER	packaging film, LDPE, at plant	1.15E-02	kg	Weibel & Stritz (1995)	1	1.41	(4,5,3,5,1,5);	
filterbags	→			no information on material					4.00E-05		Weibel & Stritz (1995)					
silicone for impregnating	→			about 5 w-%	chemicals	inorganics	No	RER	silicone product, at plant	5.00E-02	kg	http://www.agoef.de/baustoffe_daemmung/geblaehnte.html	1	1.68	(4,5,5,3,1,5);	
	→	waste heat		from electricity uses	air	unspecified			Heat, waste	2.34E-01	MJ	calculated after the electricity use	1	1.40	(4,5,3,1,1,5);	
	→	undefined losses		losses during production process						2.00E-01	kg					
	→	disposal of dust from filter		total amount: 0.3kg/t product	waste management	sanitary landfill	No	CH	disposal, inert material, 0% water, to sanitary landfill	3.00E-04	kg	EPA (1998), chapter 11.30	1	5.00	estimation	
	→	expanded perlite			insulation materials	production	No	CH	expanded perlite, at plant	1.00E+00	kg					

Fig. 6.6 Flows for “expanded perlite, at plant” and their representation in the ecoinvent database

6.7 Expanded clay, at plant

The ecoinvent database meta information of “expanded clay, at plant” (Tab. 6.7) includes all important and necessary information to use the data correctly.

Tab. 6.7 Ecoinvent database meta information for the product “expanded clay, at plant”

Name	expanded clay, at plant
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the raw materials, the transport to the finishing plant and some internal transports, the energy carrier, some packaging materials, the infrastructure and the emissions
Amount	1
Local Name	Blähton, ab Werk
Synonyms	
General Comment to reference function	Expanded at 1'200 °C in a rotary furnace. Heavy fuel oil is used for heating. Bulk density of expanded clay: 260 - 500kg/m ³
Start Date	1995
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH, RER and GLO modules have been used as proxy.
Technology text	The mined clay is mixed with water, ground, sometimes granulated and burned at reverse flow in a rotary furnace at 1'200 °C. The included organic materials burn and the clay pellets expand with the help of heavy fuel oil. The surface melts and produces a sintered shell.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.7 shows the system of the module “expanded clay, at plant”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly Richter et al. (1995), Hutter et al. (1999) and ⁸) and the standard deviation for each data.

⁸ ecoinvent report No. 7
http://www.agocf.de/baustoffe_daemmung/geblaechte.html, access date: February 2004

General Flow information						Representation in ecoinvent						Uncertainty Information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
clay, at mine	→	swelling of clay (part 1)					construction materials	additives	No	CH	clay, at mine	1.10E+00	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
transport with lorry within production in Germany	→			50 tkm for 1,1 t of raw material (from mine beside production plant; about 5 km)	transport systems	road	No	RER	transport, lorry >16t, fleet average	4.55E-02	tkm	Hutter et al. (1999)	1	2.34	(3,5,5,3,1,5);		
infrastructure (incl. Land-use and machines)	→			clay manufacturing plant used as proxy; amount copied from module "brick, at plant"	construction materials	additives	Yes	CH	mine, clay	2.00E-10	unit	estimation	1	5.00	estimation		
water	→			copied from vermiculite production	water supply	production	No	RER	tap water, at user	1.34E-02	kg	Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);		
heavy fuel oil as additive	→			about 5 w-%	oil	fuels	No	RER	heavy fuel oil, at regional storage	5.50E-02	kg	http://www.agoef.de/baustoffe_daemmung/geblaehnte.html	1	1.68	(4,5,5,3,1,5);		
diesel	→			copied from "expanded vermiculite", used for high-lift truck, module "diesel, burnded in building machine" used as proxy, amount: 0.000267 l/kg	construction processes	machinery	No	GLO	diesel, burned in building machine	9.61E-03	MJ	Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);		
electricity	→				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.50E-02	kWh	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);		
heavy fuel oil for burning	→			the used amount of heavy fuel oil (0.075 l/kg expanded clay) is equivalent to about 3.09 MJ/kg	oil	fuels	No	RER	heavy fuel oil, at regional storage	7.50E-02	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);		
packing material (paper)	→			copied from "expanded vermiculite", 0.04 paper bags for one kilo product. Assumed weight of paper bag 0.05 kg	paper & cardboard	cardboard & corrugated board	No	RER	corrugated board base paper, kraftliner, at plant	2.00E-03	kg	Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);		
packing material (plastics)	→			copied from "expanded vermiculite"	plastics	processing	No	RER	packaging film, LDPE, at plant	4.81E-04	kg	Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);		

Fig. 6.7 Flows for "expanded clay, at plant" and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty Information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
swelling of clay (part 2)	→	waste heat			from electricity uses	air	unspecified			Heat, waste	1.26E-01	MJ	calculated after the electricity use	1	1.64	(3,5,5,3,1,5);
	→	vermiculite waste			losses during production process						9.99E-02	kg				
	→	dust to air [kg]			total amount: 0.086kg/t product; same share as in module "heavy fuel oil, burned in industrial furnace 1MW, non-modulating, RER, [MJ]" : 70%<2.5um	air	low population density			Particulates, < 2.5 um	6.02E-05	kg	Hutter et al. (1999)	1	3.33	(3,5,5,3,1,5);
	→	dust to air [kg]			total amount: 0.086kg/t product; same share as in module "heavy fuel oil, burned in industrial furnace 1MW, non-modulating, RER, [MJ]" : 10um>10%>2.5um	air	low population density			Particulates, > 2.5 um, and < 10um	8.60E-06	kg	Hutter et al. (1999)	1	2.34	(3,5,5,3,1,5);
	→	dust to air [kg]			total amount: 0.086kg/t product; same share as in module "heavy fuel oil, burned in industrial furnace 1MW, non-modulating, RER, [MJ]" : 20%>10um	air	low population density			Particulates, > 10 um	1.72E-05	kg	Hutter et al. (1999)	1	1.89	(3,5,5,3,1,5);
	→	carbon dioxide				air	unspecified			Carbon dioxide, fossil	2.88E-01	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
	→	carbon monoxide				air	unspecified			Carbon monoxide, fossil	2.30E-04	kg	Hutter et al. (1999)	1	5.38	(3,5,5,3,1,5);
	→	nitrogen oxide as NO2				air	unspecified			Nitrogen oxides	5.44E-04	kg	Hutter et al. (1999)	1	1.89	(3,5,5,3,1,5);
	→	sulfur oxide as SO2				air	unspecified			Sulfur dioxide	1.59E-03	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
	→	methane				air	unspecified			Methane, fossil	1.26E-04	kg	Hutter et al. (1999)	1	1.89	(3,5,5,3,1,5);
	→	NMVOC				air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	6.60E-05	kg	Hutter et al. (1999)	1	1.89	(3,5,5,3,1,5);
	→	laughing gas				air	unspecified			Dinitrogen monoxide	5.00E-07	kg	Hutter et al. (1999)	1	1.89	(3,5,5,3,1,5);
	→	expanded clay				construction materials	additives	No	DE	expanded clay, at plant	1.00E+00	kg				

Fig. 6.7 Flows for "expanded clay, at plant" and their representation in the ecoinvent database (cont)

6.8 Lightweight concrete block, expanded vermiculite, at plant

The ecoinvent database meta information of “lightweight concrete block, expanded vermiculite, at plant” (Tab. 6.8) includes all important and necessary information to use the data correctly.

Tab. 6.8 Ecoinvent database meta information for the product “lightweight concrete block, expanded vermiculite, at plant”

Name	lightweight concrete block, expanded vermiculite, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the raw materials, their transport to the finishing plant, the air-drying, packing, the infrastructure and the disposal of wastewater and some solid household (e.g. packing material) waste
Amount	1
Local Name	Leichtbetonstein, Blähvermiculit, ab Werk
Synonyms	
General Comment to reference function	Hardening is assumed to be by air-drying. It is assumed that there is wastewater and solid waste, but as no exact description on the composition of the wastewater and solid waste is available a proxy for each ("treatment, concrete production effluent, to wastewater treatment, class 3" and "disposal, municipal solid waste, 22.9% water, to municipal incineration") is used. The lifespan of the plant is assumed to be 50 years.
Start Date	1995
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH, RER and GLO modules have been used as proxy.
Technology text	The raw material mixture (aggregates, cement and water) of lightweight concrete blocks are poured into a mould then squeezed and finally hardened by air (assumption for this study)
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.8 shows the system of the module “lightweight concrete block, expanded vermiculite, at plant”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly Hutter et al. (1999) and estimations) and the standard deviation for each data.

General Flow information					Representation in ecoinvent							Uncertainty Information					
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
expanded vermiculite	→	production of lightweight concrete block with expanded vermiculite			raw material, amount copied from module "lightweight concrete block, expanded clay"	construction materials	additives	No	CH	expanded vermiculite, at plant	9.00E-01	kg	estimation and Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
portland cement	→				raw material, amount copied from module "lightweight concrete block, expanded clay"	construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	1.00E-01	kg	estimation and Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
water	→				raw material, amount copied from module "lightweight concrete block, expanded clay"	water supply	production	No	RER	tap water, at user	5.00E-02	kg	estimation and Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
electricity consumption	→				for mixing, conveying, forming, copied from module "lightweight concrete block, expanded clay"	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	6.50E-03	kWh	estimation and Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
diesel for high-lift truck	→				copied from module "lightweight concrete block, expanded clay"; for internal transports, module "diesel, burned in building machine" used as proxy, amount: 0.00044 l/kg	construction processes	machinery	No	GLO	diesel, burned in building machine	1.58E-02	MJ	estimation and Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
transportation of raw materials to plant by lorry	→				distance, copied from module "lightweight concrete block, expanded clay", distance about 75 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	7.69E-02	tkm	estimation and Hutter et al. (1999)	1	5.00	estimation	
infrastructure (including building, machines and land-use)	→				copied from module "sand-lime brick"	construction materials	additives	Yes	CH	mine, clay	1.67E-10	unit	estimation	1	5.00	estimation	
packing	→				packing clay products used as proxy	construction materials	others	No	CH	packing, clay products	1.00E+00	kg	estimation	1	5.00	estimation	
	→		waste heat			from electricity uses	air	unspecified			Heat, waste	2.34E-02	MJ	calculated after the electricity use	1	1.64	(3,5,5,3,1,5);
	→		waste water			copied from sand-lime brick module; waste water from concrete mixing is used as proxy	waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	8.33E-05	m3	estimation	1	5.00	estimation
	→	solid waste			household waste (e.g. packing material) within production process, copied from sand-lime brick module: 0.5l/t, assumption density of solid waste: 250kg/m3	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.25E-04	kg	estimation	1	5.00	estimation	
	→	lightweight concrete block, expanded vermiculite				construction materials	additives	No	CH	lightweight concrete block, expanded vermiculite, at plant	1.00E+00	kg					

Fig. 6.8 Flows for "lightweight concrete block, expanded vermiculite, at plant" and their representation in the ecoinvent database

The mass balance does not work because a certain amount of water is evaporating during the air-drying process.

6.9 Lightweight concrete block, expanded perlite, at plant

The ecoinvent database meta information of “lightweight concrete block, expanded perlite, at plant” (Tab. 6.9) includes all important and necessary information to use the data correctly.

Tab. 6.9 Ecoinvent database meta information for the product “lightweight concrete block, expanded perlite, at plant”

Name	lightweight concrete block, expanded perlite, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the raw materials, their transport to the finishing plant, the air-drying, the packing, the infrastructure and the disposal of waste-water and some solid household (e.g. packing material) waste
Amount	1
Local Name	Leichtbetonstein, Blähperlite, ab Werk
Synonyms	
General Comment to reference function	Hardening is assumed to be by air-drying. It is assumed that there is wastewater and solid waste, but as no exact description on the composition of the wastewater and solid waste is available a proxy for each ("treatment, concrete production effluent, to wastewater treatment, class 3" and "disposal, municipal solid waste, 22.9% water, to municipal incineration") is used. The lifespan of the plant is assumed to be 50 years.
Start Date	1995
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH, RER and GLO modules have been used as proxy.
Technology text	The raw material mixture (aggregates, cement and water) of lightweight concrete blocks are poured into a mould then squeezed and finally hardened by air (assumption for this study)
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.9 shows the system of the module “lightweight concrete block, expanded perlite, at plant”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly Hutter et al. (1999) and estimations) and the standard deviation for each data.

General Flow information					Representation in ecoinvent							Uncertainty Information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
expanded perlite	→	production of lightweight concrete block with expanded perlite				raw material, amount copied from module "lightweight concrete block, pumice"	insulation materials	production	No	CH	expanded perlite, at plant	9.00E-01	kg	estimation and Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
portland cement	→				raw material, amount copied from module "lightweight concrete block, pumice"	construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	1.00E-01	kg	estimation and Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
water	→				raw material, amount copied from module "lightweight concrete block, pumice"	water supply	production	No	RER	tap water, at user	5.00E-02	kg	estimation and Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
electricity consumption	→				for mixing, conveying, forming, copied from module "lightweight concrete block, pumice"	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	7.53E-03	kWh	estimation and Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
diesel for high-lift truck	→				copied from module "lightweight concrete block, pumice"; for internal transports, module "diesel, burnd in building machine" used as proxy, amount: 0.00042 l/kg	construction processes	machinery	No	GLO	diesel, burned in building machine	1.51E-02	MJ	estimation and Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
transportation of raw materials to plant by lorry	→				distance, copied from module "lightweight concrete block, expanded clay", distance about 75 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	7.69E-02	tkm	estimation and Hutter et al. (1999)	1	5.00	estimation	
infrastructure (including building, machines and land-use)	→				copied from module "sand-lime brick"	construction materials	additives	Yes	CH	mine, clay	1.67E-10	unit	estimation	1	5.00	estimation	
packing	→				packing clay products used as proxy	construction materials	others	No	CH	packing, clay products	1.00E+00	kg	estimation	1	5.00	estimation	
	→			waste heat	from electricity uses	air	unspecified			Heat, waste	2.71E-02	MJ	calculated after the electricity use	1	1.64	(3,5,5,3,1,5);	
	→			waste water	copied from sand-lime brick module; waste water from concrete mixing is used as proxy	waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	8.33E-05	m3	estimation and amount: Cordes et al. (1996)	1	5.00	estimation	
	→		solid waste	household waste (e.g. packing material) within production process, copied from sand-lime brick module: 0.5l/t, assumption density of solid waste: 250kg/m3	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.25E-04	kg	estimation and amount: Cordes et al. (1996)	1	5.00	estimation		
	→		lightweight concrete block, expanded perlite			construction materials	concrete	No	CH	lightweight concrete block, expanded perlite, at plant	1.00E+00	kg					

Fig. 6.9 Flows for "lightweight concrete block, expanded perlite, at plant" and their representation in the ecoinvent database

The mass balance does not work because a certain amount of water is evaporating during the air-drying process.

6.10 Lightweight concrete block, expanded clay, at plant

The ecoinvent database meta information of “lightweight concrete block, expanded clay, at plant” (Tab. 6.10) includes all important and necessary information to use the data correctly.

Tab. 6.10 Ecoinvent database meta information for the product “lightweight concrete block, expanded clay, at plant”

Name	lightweight concrete block, expanded clay, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the raw materials, their transport to the finishing plant, the air-drying, the packing, the infrastructure and the disposal of wastewater and some solid household (e.g. packing material) waste
Amount	1
Local Name	Leichtbetonstein, Blähton, ab Werk
Synonyms	
General Comment to reference function	Hardening is assumed to be by air-drying. It is assumed that there is wastewater and solid waste, but as no exact description on the composition of the wastewater and solid waste is available a proxy for each ("treatment, concrete production effluent, to wastewater treatment, class 3" and "disposal, municipal solid waste, 22.9% water, to municipal incineration") is used. The lifespan of the plant is assumed to be 50 years.
Start Date	1995
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH, RER and GLO modules have been used as proxy.
Technology text	The raw material mixture (aggregates, cement and water) of lightweight concrete blocks are poured into a mould then squeezed and finally hardened by air (assumption for this study)
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.10 shows the system of the module “lightweight concrete block, expanded clay, at plant”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly Hutter et al. (1999) and estimations) and the standard deviation for each data.

General Flow information					Representation in ecoinvent							Uncertainty Information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
expanded clay	→	production of lightweight concrete block with expanded clay				raw material	construction materials	additives	No	DE	expanded clay, at plant	9.00E-01	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
portland cement	→					raw material	construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	1.00E-01	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
water	→					raw material	water supply	production	No	RER	tap water, at user	5.00E-02	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
electricity consumption	→					for mixing, conveying, forming	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	6.50E-03	kWh	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
diesel for high-lift truck	→					for internal transports, module "diesel, burnded in building machine" used as proxy, amount: 0.00044 l/kg	construction processes	machinery	No	GLO	diesel, burned in building machine	1.58E-02	MJ	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
transportation of raw materials to plant by lorry	→					distance, calculated from indicated energy use (123 MJ/t) and the operation energy use of "lorry 32t" (0.038 kg diesel/tkm or 1.6 MJ/tkm), transported distance about 75 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	7.69E-02	tkm	Hutter et al. (1999)	1	2.34	(3,5,5,3,1,5);
infrastructure (including building, machines and land-use)	→					copied from module "sand-lime brick"	construction materials	additives	Yes	CH	mine, clay	1.67E-10	unit	estimation	1	5.00	estimation
packing	→					packing clay products used as proxy	construction materials	others	No	CH	packing, clay products	1.00E+00	kg	estimation	1	5.00	estimation
	→			waste heat		from electricity uses	air	unspecified			Heat, waste	2.34E-02	MJ	calculated after the electricity use	1	1.64	(3,5,5,3,1,5);
	→			waste water		copied from sand-lime brick module; waste water from concrete mixing is used as proxy	waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	8.33E-05	m3	estimation and amount: Cordes et al. (1996)	1	5.00	estimation
	→		solid waste		household waste (e.g. packing material) within production process, copied from sand-lime brick module: 0.5l/t, assumption density of solid waste: 250kg/m3	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.25E-04	kg	estimation and amount: Cordes et al. (1996)	1	5.00	estimation	
	→		lightweight concrete block, expanded clay			construction materials	concrete	No	CH	lightweight concrete block, expanded clay, at plant	1.00E+00	kg					

Fig. 6.10 Flows for "lightweight concrete block, expanded clay, at plant" and their representation in the ecoinvent database

The mass balance does not work because a certain amount of water is evaporating during the air-drying process.

6.11 Lightweight concrete block, pumice, at plant

The ecoinvent database meta information of “lightweight concrete block, pumice, at plant” (Tab. 6.11) includes all important and necessary information to use the data correctly.

Tab. 6.11 Ecoinvent database meta information for the product “lightweight concrete block, pumice, at plant”

Name	lightweight concrete block, pumice, at plant
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the raw materials, their transport to the finishing plant, the additional washing, the air-drying, the packing, the infrastructure and the disposal of wastewater and some solid household (e.g. packing material) waste
Amount	1
Local Name	Leichtbetonstein, Naturbims, ab Werk
Synonyms	
General Comment to reference function	Hardening is assumed to be by air-drying. It is assumed that there is wastewater and solid waste, but as no exact description on the composition of the wastewater and solid waste is available a proxy for each ("treatment, concrete production effluent, to wastewater treatment, class 3" and "disposal, municipal solid waste, 22.9% water, to municipal incineration") is used. The lifespan of the plant is assumed to be 50 years.
Start Date	1995
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH, RER and GLO modules have been used as proxy.
Technology text	The raw material mixture (aggregates, cement and water) of lightweight concrete blocks are poured into a mould then squeezed and finally hardened by air (assumption for this study)
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.11 shows the system of the module “lightweight concrete block, pumice, at plant”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly Hutter et al. (1999) and estimations) and the standard deviation for each data.

General Flow information					Representation in ecoinvent							Uncertainty Information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
pumice	→	production of lightweight concrete block with pumice				raw material, contains still a part water which evaporates during the hardening process	construction materials	additives	No	DE	pumice, at mine	1.40E+00	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
portland cement	→					raw material	construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	2.22E-01	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
water	→					raw material	water supply	production	No	RER	tap water, at user	5.00E-02	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
electricity consumption	→					for washing of the pumice	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.12E-03	kWh	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
electricity consumption	→					for further production process	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	7.53E-03	kWh	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
diesel for high-lift truck	→					for internal transports, module "diesel, burnded in building machine" used as proxy, amount: 0.00042 l/kg	construction processes	machinery	No	GLO	diesel, burned in building machine	1.51E-02	MJ	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
transportation of raw materials to plant by lorry	→					distance of 75 km, copied from module "lightweight concrete, expanded clay"	transport systems	road	No	RER	transport, lorry >16t, fleet average	1.21E-01	tkm	Hutter et al. (1999)	1	2.34	(3,5,5,3,1,5);
infrastructure (including building, machines and land-use)	→					copied from module "sand-lime brick"	construction materials	additives	Yes	CH	mine, clay	1.67E-10	unit	estimation	1	5.00	estimation
packing	→					packing clay products used as proxy	construction materials	others	No	CH	packing, clay products	1.00E+00	kg	estimation	1	5.00	estimation
	→			waste heat		from electricity uses	air	unspecified			Heat, waste	3.83E-02	MJ	calculated after the electricity use	1	1.64	(3,5,5,3,1,5);
	→		waste water		copied from sand-lime brick module; waste water from concrete mixing is used as proxy	waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	8.33E-05	m3	estimation and amount: Cordes et al. (1996)	1	5.00	estimation	
	→		solid waste		household waste (e.g. packing material) within production process, copied from sand-lime brick module: 0.5l/t, assumption density of solid waste: 250kg/m3	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.25E-04	kg	estimation and amount: Cordes et al. (1996)	1	5.00	estimation	
	→		lightweight concrete block, pumice			construction materials	concrete	No	DE	lightweight concrete block, pumice, at plant	1.00E+00	kg					

Fig. 6.11 Flows for "lightweight concrete block, pumice, at plant" and their representation in the ecoinvent database

The mass balance does not work because a certain amount of water is evaporating during the air-drying process.

6.12 Lightweight concrete block, polystyrene, at plant

The ecoinvent database meta information of “lightweight concrete block, polystyrene, at plant” (Tab. 6.12) includes all important and necessary information to use the data correctly.

Tab. 6.12 Ecoinvent database meta information for the product “lightweight concrete block, polystyrene, at plant”

Name	lightweight concrete block, polystyrene, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the raw materials, their transport to the finishing plant, the air-drying, the packing, the infrastructure and the disposal of wastewater and some solid household (e.g. packing material) waste
Amount	1
Local Name	Leichtbetonstein, Polystyrol, ab Werk
Synonyms	
General Comment to reference function	Hardening is assumed to be by air-drying. It is assumed that there is wastewater and solid waste, but as no exact description on the composition of the wastewater and solid waste is available a proxy for each ("treatment, concrete production effluent, to wastewater treatment, class 3" and "disposal, municipal solid waste, 22.9% water, to municipal incineration") is used. The lifespan of the plant is assumed to be 50 years.
Start Date	1995
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH, RER and GLO modules have been used as proxy.
Technology text	The raw material mixture (aggregates, cement and water) of lightweight concrete blocks are poured into a mould then squeezed and finally hardened by air (assumption for this study)
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.12 shows the system of the module “lightweight concrete block, polystyrene, at plant”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly Hutter et al. (1999) and estimations) and the standard deviation for each data.

The amount of the raw materials cement and expanded polystyrene are calculated on the basis of the densities:

- Polystyrene lightweight concrete block 400 kg/m³
- Expanded polystyrene 40 kg/m³
- Dissolved and hardened cement 2'350 kg/m³

The sum of the masses from expanded polystyrene and dissolved and hardened cement is equal to the output of 1 kg polystyrene lightweight concrete block which is equal to 1/400 m³.

General Flow information					Representation in ecoinvent							Uncertainty Information					
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
polystyrene	→	production of lightweight concrete block with polystyrene			raw material, amount calculated on basis of the densities	insulation materials	production	No	RER	polystyrene foam slab, at plant	8.40E-02	kg	Hutter et al. (1999)	1	4.00	estimation	
portland cement	→				raw material, amount calculated on basis of the densities	construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	9.16E-01	kg	Hutter et al. (1999)	1	4.00	estimation	
water	→				raw material, amount copied from module "lightweight concrete block, expanded clay"	water supply	production	No	RER	tap water, at user	5.00E-02	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
electricity consumption	→				for mixing, conveying, forming, copied from module "lightweight concrete block, expanded clay"	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	6.50E-03	kWh	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
diesel for high-lift truck	→				copied from module "lightweight concrete block, expanded clay"; for internal transports, module "diesel, burned in building machine" used as proxy, amount: 0.00042 l/kg	construction processes	machinery	No	GLO	diesel, burned in building machine	1.51E-02	MJ	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
transportation of raw materials to plant by lorry	→				distance, copied from module "lightweight concrete block, expanded clay", distance about 75 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	7.69E-02	tkm	estimation and Hutter et al. (1999)	1	5.00	estimation	
infrastructure (including building, machines and land-use)	→				copied from module "sand-lime brick"	construction materials	additives	Yes	CH	mine, clay	1.67E-10	unit	estimation	1	5.00	estimation	
packing	→				packing clay products used as proxy	construction materials	others	No	CH	packing, clay products	1.00E+00	kg	estimation	1	5.00	estimation	
	→		waste heat			from electricity uses	air	unspecified			Heat, waste	2.34E-02	MJ	calculated after the electricity use	1	1.64	(3,5,5,3,1,5);
	→		waste water			copied from sand-lime brick module; waste water from concrete mixing is used as proxy	waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	8.33E-05	m3	estimation and amount: Cordes et al. (1996)	1	5.00	estimation
	→	solid waste			household waste (e.g. packing material) within production process, copied from sand-lime brick module: 0.5l/t, assumption density of solid waste: 250kg/m3	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.25E-04	kg	estimation and amount: Cordes et al. (1996)	1	5.00	estimation	
	→	lightweight concrete block, expanded perlite				construction materials	concrete	No	CH	lightweight concrete block, polystyrene, at plant	1.00E+00	kg					

Fig. 6.12 Flows for "lightweight concrete block, polystyrene, at plant" and their representation in the ecoinvent database

The mass balance does not work because a certain amount of water is evaporating during the air-drying process.

6.13 Autoclaved aerated concrete block, at plant

The ecoinvent database meta information of “autoclaved aerated concrete block, at plant” (Tab. 6.13) includes all important and necessary information to use the data correctly.

Tab. 6.13 Ecoinvent database meta information for the product “autoclaved aerated concrete block, at plant”

Name	autoclaved aerated concrete block, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the raw materials, their transport to the finishing plant, the energy for the autoclaving process, the packaging, the infrastructure and the disposal of wastewater and some solid household (e.g. packing material) waste
Amount	1
Local Name	Gasbetonstein, ab Werk
Synonyms	Bimshohlblockstein/Ytong
General Comment to reference function	Hardening is assumed to be by air-drying. It is assumed that there is wastewater and solid waste, but as no exact description on the composition of the wastewater and solid waste is available a proxy for each ("treatment, concrete production effluent, to wastewater treatment, class 3" and "disposal, municipal solid waste, 22.9% water, to municipal incineration") is used. The lifespan of the plant is assumed to be 50 years.
Start Date	1995
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH, RER and GLO modules have been used as proxy.
Technology text	The raw materials (sand/gravel, quicklime, anhydrite, cement and aluminium powder) are mixed with water and poured into a mould. The aluminium powder is reacting with the alkaline and siliceous additives and forms, by hydrogen release, pores with a diameter between 2 and 3mm.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.13 shows the system of the module “autoclaved aerated concrete block, at plant”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly Hutter et al. (1999) and estimations) and the standard deviation for each data.

General Flow information					Representation in ecoinvent							Uncertainty Information					
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
sand, gravel	→	production of autoclaved aerated concrete block			raw material	construction materials	additives	No	CH	sand, at mine	5.04E-01	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
quicklime	→				raw material	construction materials	additives	No	CH	quicklime, milled, packed, at plant	1.04E-01	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
portland cement	→				raw material	construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	2.60E-01	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
anhydrite	→				raw material	construction materials	binder	No	CH	anhydrite, at plant	4.40E-02	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
aluminium powder	→				raw material	metals	extraction	No	RER	aluminium, secondary, from new scrap, at plant	7.00E-04	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
water	→					water supply	production	No	RER	tap water, at user	6.48E-01	kg	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
natural gas	→				energy for autoclaving, amount: 0.0178m3/kg	natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace >100kW	6.46E-01	MJ	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
electricity consumption	→					electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	5.75E-02	kWh	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
diesel for high-lift truck	→				for high-lift truck, module "diesel, burned in building machine" used as proxy, amount: 0.0007 l/kg	construction processes	machinery	No	GLO	diesel, burned in building machine	2.52E-02	MJ	Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);	
transportation of raw materials to plant by lorry	→				distance, calculated from indicated energy use (61 MJ/t) and the operation energy use of "lorry 32t" (0.038 kg diesel/tkm or 1.6 MJ/tkm), transported distance about 75 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	3.81E-02	tkm	Hutter et al. (1999)	1	2.34	(3,5,5,3,1,5);	
infrastructure (including building, machines and land-use)	→					copied from module "sand-lime brick"	construction materials	additives	Yes	CH	mine, clay	1.67E-10	unit	estimation	1	5.00	estimation
packing	→					packing clay products used as proxy	construction materials	others	No	CH	packing, clay products	1.00E+00	kg	estimation	1	5.00	estimation
	→	waste heat			from electricity uses	air	unspecified			Heat, waste	2.07E-01	MJ	calculated after the electricity use	1	1.64	(3,5,5,3,1,5);	
	→	waste water			copied from sand-lime brick module; waste water from concrete mixing is used as proxy	waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	8.33E-05	m3	estimation and amount: Cordes et al. (1996)	1	5.00	estimation	
	→	solid waste			household waste (e.g. packing material) within production process, copied from sand-lime brick module: 0.5l/t, assumption density of solid waste: 250kg/m3	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.25E-04	kg	estimation and amount: Cordes et al. (1996)	1	5.00	estimation	
	→	autoclaved areated concrete block			Density: 500kg/m3	construction materials	bricks	No	CH	autoclaved aerated concrete block, at plant	1.00E+00	kg					

Fig. 6.13 Flows for “autoclaved aerated concrete block, at plant” and their representation in the ecoinvent database

The mass balance does not work because a certain amount of water is evaporating during the autoclaving process.

6.14 Light clay brick, at plant

The ecoinvent database meta information of “light clay brick, at plant” (Tab. 6.14) includes all important and necessary information to use the data correctly.

Tab. 6.14 Ecoinvent database meta information for the product “light clay brick, at plant”

Name	Light clay brick, at plant
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the raw materials clay, straw and water, their transport to the finishing plant, the drying energy, the packaging and the infrastructure. Beside the product, some solid household (e.g. packing material) waste is taken into account.
Amount	1
Local Name	Leichtlehmstein, ab Werk
Synonyms	
General Comment to reference function	technical drying by heat from liquid gas (natural gas used as proxy). It is assumed that there is wastewater but as no data is available it is not included. The lifespan of the plant is assumed to be 50 years.
Start Date	1992
End Date	1995
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH, RER and GLO modules have been used as proxy.
Technology text	The raw materials (clay and straw) are mixed with water and poured into a mould. The drying process is technical with the use of a natural gas heater
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.14 shows the system of the module “light clay brick, at plant”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly Weibel & Stritz (1995) and estimations) and the standard deviation for each data.

General Flow information						Representation in ecoinvent						Uncertainty Information						
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
clay at mine	→	production of ligh clay brick, at plant				raw material	construction materials	additives	No	CH	clay, at mine	1.01E+00	kg	Weibel & Stritz (1995)	1	1.34	(3,5,3,1,1,5);	
straw	→					raw material	agricultural production	plant production	No	CH	straw IP, at farm	1.62E-01	kg	Weibel & Stritz (1995)	1	1.34	(3,5,3,1,1,5);	
water	→					raw material	water supply	production	No	RER	tap water, at user	1.13E-01	kg	Weibel & Stritz (1995)	1	1.64	(3,5,5,3,1,5);	
electricity consumption	→					for mixing, conveying, forming	electricity	supply mix	No	DE	electricity, medium voltage, at grid	5.56E-03	kWh	Weibel & Stritz (1995)	1	1.64	(3,5,5,3,1,5);	
diesel in building machine	→					for internal transports, module "diesel, burned in building machine" used as proxy, amount: 0.00044 l/kg	construction processes	machinery	No	GLO	diesel, burned in building machine	1.58E-02	MJ	Weibel & Stritz (1995)	1	1.64	(3,5,5,3,1,5);	
heat from natural gas	→					drying by heat from natural gas used as proxy for heat from liquid gas	natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace >100kW	1.78E+00	MJ	Weibel & Stritz (1995)	1	1.64	(3,5,5,3,1,5);	
transportation of raw materials to plant by lorry	→					about 15km	transport systems	road	No	RER	transport, lorry >16t, fleet average	1.76E-02	tkm	Weibel & Stritz (1995)	1	2.34	(3,5,5,3,1,5);	
infrastructure (including building, machines and land-use)	→					copied from module "sand-lime brick"	construction materials	additives	Yes	CH	mine, clay	1.67E-10	unit	estimation		1	5.00	estimation
packing	→					packing clay products used as proxy	construction materials	others	No	CH	packing, clay products	1.00E+00	kg	estimation		1	5.00	estimation
						from electricity uses	air	unspecified			Heat, waste	2.00E-02	MJ	calculated after the electricity use		1	1.64	(3,5,5,3,1,5);
					no data													
						waste management	inert material landfill	No	CH	disposal, inert waste, 5% water, to inert material landfill	1.25E-04	kg	Weibel & Stritz (1995)	1	1.64	(3,5,5,3,1,5);		
						construction materials	bricks	No	DE	light clay brick, at plant	1.00E+00	kg	Weibel & Stritz (1995)					

Fig. 6.14 Flows for "light clay brick, at plant" and their representation in the ecoinvent database

The mass balance does not work because a certain amount of water is evaporating during the air-drying process.

6.15 Cobwork, at plant

The ecoinvent database meta information of “cobwork, at plant” (Tab. 6.15) includes all important and necessary information to use the data correctly.

Tab. 6.15 Ecoinvent database meta information for the product “cobwork, at plant”

Name	Cobwork, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	It includes the raw materials clay in ground, wooden chips and water. It includes also the digging, crumbling and mixing of clay, and the transports of the raw materials to the construction site. No packaging and no infrastructure is taken into account.
Amount	1
Local Name	Leichtlehmausfachung, ab Werk
Synonyms	
General Comment to reference function	mixing of raw materials with machines, depositing by hand and drying by air. For the transportation of the raw materials an average distance of 20 km has been assumed.
Start Date	1994
End Date	1995
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges CH, RER and GLO modules have been used as proxy.
Technology text	The raw materials (clay and wooden chips) are mixed with water and smacked by hand on mainly hurdle work.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	none

Fig. 6.15 shows the system of the module “cobwork, at plant”. It shows all in- and output flows, the allocation to the modules in the ecoinvent database, the sources (mainly estimations and Weibel & Stritz (1995)) and the standard deviation for each data.

General Flow information					Representation in ecoinvent							Uncertainty Information					
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
clay in ground	→	production of cobwork, at plant			raw material, mixed on site, amount per kg of cobwork product is copied from light clay brick	resource	in ground			Clay, bentonite, in ground	1.01E+00	kg	estimation	1	5.00	estimation	
wooden chips	→				raw material, mixed on site; copied from light clay brick with moist bulked density of wooden chips: 264.04 kg/m3	wood energy	fuels	No	RER	wood chips, mixed, from industry, u=40%, at plant	6.14E-04	m3	estimation	1	5.00	estimation	
water	→				raw material, mixed on site, copied from light clay brick	water supply	production	No	RER	tap water, at user	1.13E-01	kg	estimation	1	5.00	estimation	
electricity consumption	→				for mixing, conveying, forming	electricity	supply mix	No	DE	electricity, medium voltage, at grid	1.77E-03	kWh	Weibel & Stritz (1995)	1	1.64	(3,5,5,3,1,5);	
diesel in building machine	→				for internal transports, module "diesel, burnded in building machine" used as proxy, amount: 0.00044 l/kg	construction processes	machinery	No	GLO	diesel, burned in building machine	8.95E-03	MJ	Weibel & Stritz (1995)	1	1.64	(3,5,5,3,1,5);	
transportation of raw materials to plant by lorry	→				about 20km for about 25% of the wooden chips	transport systems	road	No	RER	transport, lorry >16t, fleet average	8.10E-04	tkm	Weibel & Stritz (1995)	1	2.34	(3,5,5,3,1,5);	
			</														

Fig. 6.15 Flows for "cobwork, at plant" and their representation in the ecoinvent database

The mass balance does not work because a certain amount of water is evaporating during the air-drying process.

7 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed information on the uncertainties for each In- and Output compare the flows and the associated meta information.

Most data of the vermiculite mining process are based on a questionnaire filled in by Palabora Mining Company Ltd. in South-Africa. It is one of the biggest vermiculite mines in the world and supplies a large part of Europe's demand. Therefore the area of validity of this module is South-Africa. For these reasons, the data for vermiculite mining are of very good overall quality, although the recultivation, the heating system the electricity mix and the dust emissions are roughly estimated.

As no information for perlite mining is available and the mining process is quite similar, most data have been assumed to be identical to vermiculite mining. This adaptation results in a higher uncertainty. Accordingly, the data are of poor overall quality.

For the mining process of pumice no data covering the entire process are available. For different exchanges, proxies with related amounts from other mining processes are used. This results in a mediocre overall quality.

The infrastructure module "mine, vermiculite" is based on data of the Palabora Mining Company Ltd. which are estimated to be of good overall quality.

Most data for expanded vermiculite, perlite and clay modules are based on a comprehending LCI study each having different sources. As there is no indication of the data quality we assume that it is of good overall quality.

Most data of the different lightweight concrete blocks including autoclaved aerated concrete block are from the same above mentioned literature and we therefore assume that also these data are of good overall quality.

8 Abbreviations

KFN	Limestone Factory Netstal (Kalkfabrik Netstal)
-----	--

9 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

10 References

- | | |
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Part XXII:

Natural Stone Plate

Data v2.0 (2007)

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Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

This part deals with three types of natural stone mined and used in Switzerland. The three types are cut natural stone, ground natural stone and polished natural stone.

The drilling for the mining process is assumed to be done with compressed air, which is produced with electricity. The sawing is assumed to be accomplished with diamond cables and the grinding with diamond studded grinding wheels. The diamond as infrastructure and its losses are not taken into account as no data is available even though they might be significant. The polishing is assumed to be fulfilled with a polishing machine with very fine grinding medium. The dust emissions are estimated based on the cutting/grinding and polishing losses. It's assumed that only 5% of the dust emissions go to air and the rest is bound in the cooling water which is disposed in the ground. No water and soil emissions are taken into account.

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1 Introduction

This report describes the processes and their inventories for the production of different natural stone plate products.

Most of the relevant information about the processes is compiled in the figures depicting the input and output flows to the processes together with information about their use and about their representation in the ecoinvent database. This figure shows also the sources and the uncertainty information valid for the individual values.

2 Reserves and Resources

In Switzerland a huge variety of different stones coming from all over the world is used. Countries beside Switzerland, which deliver the Swiss market are South America (often Brazil), India, China, Spain, France, Italy, Portugal, etc.

The main types of Swiss rock are:

- Bright ticino gneiss
- Maggia- /Calanca-gneiss
- Grisons gneiss
- Valais quartzite
- Marbles
- Granite sandstone
- Plate sandstone
- Shell limestone
- Jurassic limestone

3 Use

The most important application of Swiss natural stones in the building sector and its most used types are (according to Kündig et al. (1997)):

Tab. 3.1 Important applications of natural stone**Raw products after breaking**

Raw blocks:	Configuration, lining, block walling in gardening, road construction and water engineering
Split tile:	Tessellated- or rectangular base plates, stairs, covering in garden and road constructions
Creb- and Kerbstone:	Split and roughly adjusted stones in garden and road constructions
Building stone:	Cyclopic- and quarry stone masonry, floor paving in garden, road, building and river construction
Paving stone:	Paving stones in all types from 4 to 6, 8 to 11 up to 14 to 16 cm size in road and garden construction
Work piece:	Roughly trimmed sockets, bridge stone slabs, steles, etc.

Processed products after breaking

Plank log:	Blocks which are ideal in form and size for sawing up
Vast quantity plates:	Intermediate product for the production of floor slabs, stairs, work pieces for all industries
Slabs of different types:	Floor plates, stairs, sockets, covering plates in building, garden and road construction
Facing tile:	Cladding for external walls ventilated at rear, soffits, ledges, lintel in the building construction
Creb- and Kerbstone:	Sawn, solidificated, scorched and adjusted properly in the road and garden construction
Building stone:	Adjusted masonry, moellons, renovations, restorations in the building, garden and road constructions
Working piece outdoor:	Restoration / Monument, Door and window curbs, pillars in the building construction
Working piece indoor:	Kitchen covering, washing tables, tables, open fireplaces, pillars, ovens in the building construction
Sculpture and Gravestones:	Blanks / Rock, convenience product, plastic arts, altars, handicraft, art

4 Characterisation

The main Swiss rock types which have comparable qualities within certain boundaries are characterized in this chapter (Kündig et al. (1997)):

Granite and granite-similar rocks have, based on their main components quartz and feldspar relative high sclerometric hardness, a very low solubility and therefore a good chemical resistance. The polish on the exterior surface is stable. The inner structure is in general very dense, the porosity and the water absorption are in general small. The compression strength is mostly high and the grain size varies in wide limits.

Marble is composed of only one mineral. Calcite or dolomite form the grain structure and defines in this way important characteristics of the rock masses. The sclerometric hardness is small and there is a respective solubility. The structure is densely packed and porosity and water adsorption are small. The compression strength in freshly mined, un-weathered condition ranges between middle and high.

Limestone is, like marble, mainly composed of calcite and has similar characteristics. The rock texture is mainly compact and the porosity is small. Though there are several, partly extreme exceptions (e.g. shell marl, calcareous tuff). Except the porous and coarse porous limestone, the compression strength is middle to high. Limestone with siliceous share has very high compression strength.

Sandstone varies very strongly in composition, large structure, texture, grain size and porosity. From soft, porous sandstones to very conglobate stones everything can be found. To judge sandstone regarding technical inspections concerning the material, an overall characterisation is not possible. Therefore it is important to judge each stone independently.

5 System Characterisation

5.1 Natural Stone

There are many different ways of mining and processing of granite. Tab. 5.1 gives an incomplete overview of possible procedures. The procedures considered in the study are marked with bold frames.

Tab. 5.1 Scheme of natural stone processing (Kündig et al. (1997))

Process step	Procedure	Instruments
Working in mine, mining of boulders	Blasting	Explosives -> small pieces Gun powder -> smooth blasting
	Trench procedure - Direct separation from rock - Whole to whole drilling - Kerving - Wire saw - Oxygen lance - Water jet	Cotters, only when scissile rock Air-pressure drilling machine Chain-saw Diamond-wire Oxygen lance: rock melts resp. falls apart from thermal stresses Super-pressure jet (2000 bar)
Cutting the boulders in plates and work pieces	Splitting	Direct splitting of the boulder with plain key, only for easy splitting rock
	Reciprocating saw	Sand mill saw or diamond studded mill saw
	Circular saw	Diamond studded
	Wire saw	Forced conducted wire
Surface machining	Splitting, sand sawing, diamond sawing	
	Grinding: raw, middle and fine	Grinding wheels and heads with grind graining
	Polishing	Polishing agent, very fine grinding medium
	Burning	Acetylene or hydrogen burner

All the datasets studied in this report are mainly based on estimations based on different sources.

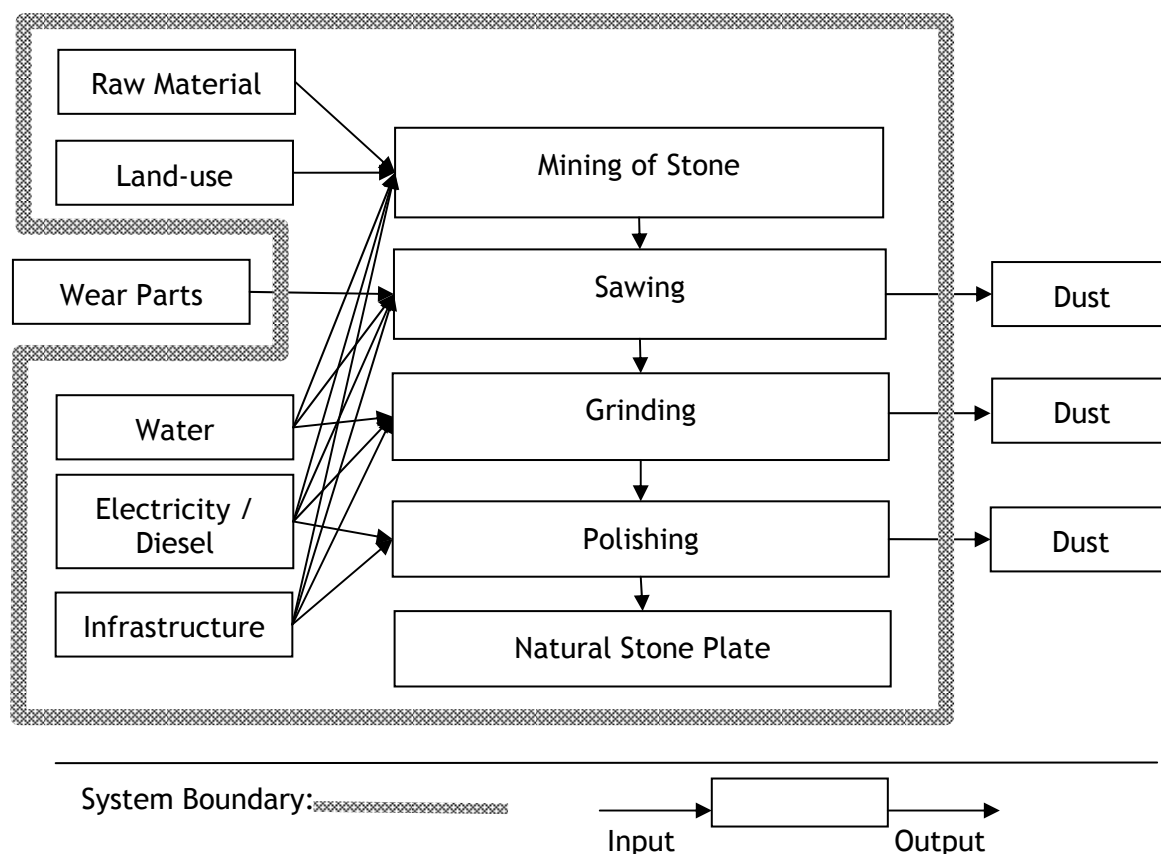


Fig. 5.1 Production process of natural stone plates

The unit for all datasets is kg. The geometry of the piece of natural stone is decisive for the different in-and outputs, for example the amount of dust. A granite plate with a specific geometry (0.5 * 0.5 * 0.03 m) is used for calculation. The weight of that plate with a density of 2'750 kg/m³ is about 20.6 kg.

5.1.1 Air Emissions (mainly Dust)

A very important air emission from stone processing is dust. It is expected in all sub-processes from mining to polishing. The amount is estimated based on the losses in the mining, sawing, grinding and polishing processes.

The total amount of dust in air and cooling water from the two first production processes (mining and sawing) is assumed to be equal to the losses from cutting. The cuts are along the surface area, on the length and on the short edge. The slitting width is assumed to be 1 mm¹. For the calculation half of each cutting losses is allocated to one plate. Thus the volume of rock which is transformed to dust by the mining and sawing process for one

¹ http://www.diamant-boart.com/pdf/cables_GE.pdf, access date: 19.04.05

plate is $0.001 * (0.03 * 0.5 * 2 + 0.5 * 0.5) = 0.00028 \text{ m}^3$. The density of granite is $2'750 \text{ kg/m}^3$. Thus the amount of dust per kg granite plate is 0.037 kg . It is assumed that about 95% of the dust is bound in the cooling water.

For the grinding process it has been assumed that about 0.5 mm on each surface of the plate is removed and transferred to dust. The amount of dust per slab is therefore: $(0.5 * 0.5 * 2 + 0.5 * 0.03 * 4) * 0.0005 = 0.00028 \text{ m}^3$ or 0.77 kg and consequently, the amount of dust per kg of plate is 0.037 kg . It is assumed that the main part of the dust (assumed that about 95%) is bound in the cooling water.

For the polishing process it has been assumed that about 0.05 mm on each side of the plate is emitted as dust. The amount of dust per slab is therefore: $(0.5 * 0.5 * 2 + 0.5 * 0.03 * 4) * 0.00005 = 0.000028 \text{ m}^3$ or 0.077 kg and consequently, the amount of dust per kg of plate is 0.0037 kg . It is assumed that no cooling water is used in this process.

The share of the dust emission categories is based on the EPA report (US Environmental Protection Agency) "Generalized Particle Size Distributions"²: which shows following result:

- 85% of the particulates are $< 10 \mu\text{m}$
- 15% $> 10 \mu\text{m}$
- 30% $< 2.5 \mu\text{m}$

Thus the share of dust between 2.5 and $10 \mu\text{m}$ is 55%.

Applied to the total amount of dust for the different processes the results are:

mining and sawing:	Particulates $> 10 \mu\text{m}$ = $0.15 * 0.5 * 0.037 \text{ kg} = 0.00028 \text{ kg}$
	Particulates $< 10 \mu\text{m}$ and $> 2.5 \mu\text{m}$ = $0.55 * 0.5 * 0.037 \text{ kg} = 0.001027 \text{ kg}$
	Particulates $< 2.5 \mu\text{m}$ = $0.3 * 0.5 * 0.037 \text{ kg} = 0.00056 \text{ kg}$
grinding:	Particulates $> 10 \mu\text{m}$ = $0.15 * 0.5 * 0.037 \text{ kg} = 0.00028 \text{ kg}$
	Particulates $< 10 \mu\text{m}$ and $> 2.5 \mu\text{m}$ = $0.55 * 0.5 * 0.037 \text{ kg} = 0.001027 \text{ kg}$
	Particulates $< 2.5 \mu\text{m}$ = $0.3 * 0.5 * 0.037 \text{ kg} = 0.00056 \text{ kg}$
polishing:	Particulates $> 10 \mu\text{m}$ = $0.15 * 0.0037 \text{ kg} = 0.00056 \text{ kg}$
	Particulates $< 10 \mu\text{m}$ and $> 2.5 \mu\text{m}$ = $0.55 * 0.0037 \text{ kg} = 0.00205 \text{ kg}$
	Particulates $< 2.5 \mu\text{m}$ = $0.3 * 0.0037 \text{ kg} = 0.00112 \text{ kg}$

Dust from burning processes is included in the respective processes.

² <http://www.epa.gov/ttn/chief/ap42/appendix/appb-2.pdf>: access date: 19.04.05

5.1.2 Energy Consumption

The energy consumption is taken from www.energie.ch³. 70% of the total electricity consumption in the stone processing industry is used for the production machines, 10% for compressed air, 10% for water pumps and 10% for different activities. The mining process is dominated by the use of compressed air. Thus the total electricity consumption of 0.8kWh per kg processed stone is assumed to be as follows:

The electricity used for the production machines (70%) subdivided in:

- 10% for mining
- 30% for sawing
- 30% for grinding

The electricity used for the production of compressed air (10%)

- 10% for mining

The electricity used for the water pumps and different application (20%)

- 10% for mining
- 10% for sawing

With above assumption the total electricity consumption is for

- Mining: $0.3 * 0.8 = 0.24$ kWh/kg
- Sawing: $0.4 * 0.8 = 0.32$ kWh/kg
- Grinding: $0.3 * 0.8 = 0.24$ kWh/kg.

For the process of polishing it is assumed that the electricity consumption is comparable with the grinding process (0.24 kWh/kg).

5.1.3 Diesel consumption and Transports

Within the mine the amount of diesel used for different transportation is 0.05 litre per kg transported stone⁴. It has been assumed that 0.03 litres are used for mining, 0.01 litres for sawing and 0.01 litres for polishing.

For the transportation from the mine to the regional storage a distance of 300km has been estimated. This estimation is based on the assumption that the mine is located in the south of Switzerland and the storage place is in the centre of Switzerland. No further transportations have been taken into account.

³ <http://www.energie.ch/themen/industrie/infel/steinwerke.htm>, access date: 19.04.05

⁴ <http://www.energie.ch/themen/industrie/infel/steinwerke.htm>, access date: 19.04.05

5.1.4 Water Consumption / Wastewater

20 litres of water is needed per minute for sawing the stone with a wire fitted with diamonds⁵. It is assumed that the area which is cut per hour is about 4 m²⁵. This results in an area of 0.067 m² per minute. As the geometry of the plate is 0.25m² and the weight 20kg the amount of water needed is: $(0.25 * 20 / 0.067) / 20 = 3.73$ litre or 0.00373m³ water per kg stone. The water is used in the mining, sawing and grinding process. No further water consumption has been taken into account

5.1.5 Infrastructure

The estimation of the used infrastructure is based on following ideas:

- Mining and sawing: A limestone mine is used as proxy as no data are available. The limestone mine used, has a lifespan of 50 years and a yearly mining amount of 100'000 tons. It includes the land-use (occupation and transformation) and the buildings but not the machines. The weight of the used machines has been estimated with 30 tons. It includes the saw, the water pump, a stone cracking machine and diverse steel parts.
- Grinding: The weight of the grinding machine is estimated with 5 tons.
- Polishing: The weight of the polishing machine is estimated with 5 tons.

5.1.6 Wear Parts

There are no wear parts taken into account as no data on their production is available. Nevertheless, their amount is calculated.

The life-span of a wire used for sawing granite is declared to be about 12m² per m wire⁶. The cut area allocated to one plate is 0.25m². Thus 48 plates (or 960 kg stone) can be cut with 1 m wire. The wire is tipped with 40 diamond pearls per m. The geometry of the pearls is shown in Fig. 5.2

⁵ http://www.diamant-boart.com/pdf/cables_GE.pdf, access date: 20.04.05

⁶ http://www.diamant-boart.com/pdf/cables_GE.pdf, access date: 20.04.05

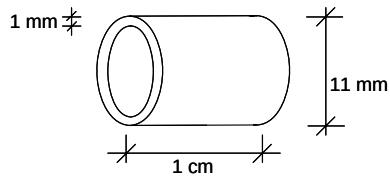


Fig. 5.2 Geometry of diamond fitted pearl

The volume of the pearl shell is $6.5 \times 10^{-7} \text{ m}^3$. It is assumed that about 30% of it is made up of synthetic diamond with a density of 3520 kg/m^3 . The weight of synthetic diamond for one pearl is therefore 0.00068 kg and thus $2.8 \times 10^{-5} \text{ kg}$ of synthetic diamond is needed per kg stone plate.

The amount of diamond needed for processing stone is not taken into account because no data for the production of synthetic diamond is available. This process needs high temperature (1500°C) and high-pressure (around 60000 atm) to produce diamond from graphite.

5.1.7 Land-use

Land-use is included in the infrastructure of “mine, limestone” which is used as proxy for the infrastructure of granite mining and processing.

6 Life Cycle Inventories

ecoinvent database meta information (tables), flows and their representation in the ecoinvent database are shown in:

Natural stone plate, cut, at regional storage	Tab. 6.1 and Fig. 6.1
Natural stone plate, grinded, at regional storage	Tab. 6.2 and Fig. 6.2
Natural stone plate, polished, at regional storage	Tab. 6.3 and Fig. 6.3

6.1 Cut Natural Stone

The meta information of “natural stone, cut, at regional storage” (Tab. 6.1) includes all important and necessary information to use the data correctly. Fig. 6.1 shows all in- and output flows of the product “natural stone, cut, at regional storage”, the allocation to the datasets in ecoinvent database v2.0, the sources and the standard deviation of all in- and output flows.

6.2 Ground Natural Stone

The meta information of “natural stone, ground, at regional storage” (Tab. 6.2) includes all important and necessary information to use the data correctly. Fig. 6.2 shows all in- and output flows of the product “natural stone, ground, at regional storage”, the allocation to the datasets in ecoinvent database v2.0, the sources and the standard deviation of all in- and output flows.

6.3 Polished Natural Stone

The meta information of “natural stone, polished, at regional storage” (Tab. 6.3) includes all important and necessary information to use the data correctly. Fig. 6.3 shows all in- and output flows of the product “natural stone, polished, at regional storage”, the allocation to the datasets in ecoinvent database v2.0, the sources and the standard deviation of all in- and output flows.

Tab. 6.1 ecoinvent meta information of the product “natural stone, cut, at regional storage”

Name	natural stone, cut, at regional storage
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the raw material granite, the mining, sawing and cooling, the transports within the mine and to the regional storage in Switzerland and the infrastructure (machines). No administration, wastewater, wear and spare parts are included. Dust is accounted for.
Amount	1
Local Name	Natursteinplatten, geschnitten, ab Regionallager
Synonyms	
General Comment to reference function	The drilling for the mining process is assumed to be done with compressed air, which is produced with electricity. The sawing is assumed to be accomplished with diamond wires. The diamond losses are not taken into account even though they might be significant as no data on diamond production is available. The dust emissions are estimated based on the losses (volume of granite with a density of 2'750 kg/m ³) by cutting a plate with the geometry 0.5 * 0.5 * 0.03 m and a weight of 20 kg. It is assumed that 95% of the dust is bound in the cooling water.
Start Date	2000
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER- and GLO-modules are used as proxy
Technology text	Typical technology for average Swiss production. Mining of boulders by direct separation from rock, whole to whole drilling, kerving wire saw, oxygen lance and water jet. Cutting by wire saw.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
Granite in ground	natural stone plate, cut, at regional storage		Assumed that Granite plate with a geometry of 0.5 * 0.5 * 0.03m and a weight of 20 kg (2750 kg/m ³)	resource	in ground			Granite, in ground	1.39E+00	kg	assumption	1	1.68	(5,5,1,1,3,5);
Electricity for rock mining			(mainly for drilling)	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.40E-01	kWh	http://www.energie.ch/themen/industrie/infel/steinwerke.htm, access data 14.04.05	1	1.40	(3,5,1,1,3,5);
Electricity for sawing of stone			assumed that diamond cables have been used (diameter: 11mm)	electricity	supply mix	No	CH	electricity, medium voltage, at grid	3.20E-01	kWh		1	1.40	(3,5,1,1,3,5);
Transports within mine			total diesel consumption for mining and processing 50l; assumption 30l for mining and cutting and 10l for grinding and 10l for polishing	construction processes	machinery	No	GLO	diesel, burned in building machine	1.52E+00	MJ		1	1.40	(3,5,1,1,3,5);
Transports to storage place in Switzerland			Assumed that mainly imported from south of Switzerland with a lorry 32t	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	3.00E-01	tkm	assumption	1	2.19	(4,5,1,1,3,5);
Infrastructure			"mine, limestone" used as proxy, estimation of mining duration (50 years) and yearly mining amount (100'000 tons)	construction materials	additives	Yes	CH	mine, limestone	2.00E-10	unit	http://www.stbg.de/Zeitung/se603/granit.htm	1	3.15	(3,5,1,2,3,5);
Water			20 litres per min.; 4m ² per hour (=0.067 m ² /min.); geometry of plate 0.25m ² , weight 20kg	resource	in water			Water, river	3.73E-03	m ³	http://www.diamant-boart.com/pdf/cables_GE.pdf	1	1.38	(2,5,1,3,3,5);
Diamond losses			calculation of amount described in report						2.80E-05	kg		1	1.38	(2,5,1,3,3,5);
		waste heat		air	unspecified			Heat, waste	2.02E+00	MJ	http://www.energie.ch/themen/industrie/infel/steinwerke.htm, access data 14.04.05	1	1.40	(3,5,1,1,3,5);
		Dust < 2.5 um	assumption 5% to air, 95% with water in ground, particle distribution: 30% < 2.5um	air	unspecified			Particulates, < 2.5 um	5.60E-04	kg		1	3.79	(2,5,3,5,5,5);
		Dust > 2.5, < 10um	assumption 5% to air, 95% with water in ground, particle distribution: 10um > 55% > 2.5um	air	unspecified			Particulates, > 2.5 um, and < 10um	1.03E-03	kg		1	2.78	(2,5,3,5,5,5);
		Dust > 10um	assumption 5% to air, 95% with water in ground, particle distribution: 15% > 10um	air	unspecified			Particulates, > 10 um	2.80E-04	kg		1	2.35	(2,5,3,5,5,5);
		waste water to ground	same amount as water input; not inventoried						3.73E-03	m ³				
		natural stone plate, cut		construction materials	others	No	CH	natural stone plate, cut, at regional storage	1.00E+00	kg				

Fig. 6.1 Flows for “natural stone, cut, at regional storage” and their representation in the ecoinvent database

Tab. 6.2 **ecoinvent meta information of the product “natural stone, ground, at regional storage”**

Name	natural stone, ground, at regional storage
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Basis is the cut natural granite plate in a storage which is ground and cooled. Beside infrastructure some internal transports are included. No administration and wear and spare parts are taken into account. Dust is accounted for.
Amount	1
Local Name	Natursteinplatten, geschliffen, ab Regionallager
Synonyms	
General Comment to reference function	The grinding is assumed to be accomplished with diamond studded grinding wheels. The diamond losses are not taken into account even though they might be significant as no data on diamond production is available. The dust emissions are estimated based on the losses (0.5 mm abrasion around the granite plate with geometry of 0.5 * 0.5 * 0.03 m and a weight of 2'750 kg/m ³). It is assumed that 95% of the dust is bound in the cooling water.
Start Date	2000
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER- and GLO-modules are used as proxy
Technology text	Typical technology for average Swiss production
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
natural stone plate, cut	natural stone plate, grounded, at regional storage		Assumed that Granite plate with a geometry of 0.5 * 0.5 * 0.03m and a weight of 20 kg (2750 kg/m ³)	construction materials	others	No	CH	natural stone plate, cut, at regional storage	1.08E+00	kg		1	1.68	(5,5,1,1,3,5);
Electricity for grinding				electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.40E-01	kWh	http://www.energie.ch/themen/industrie/infel/steinwerke.htm, access data 14.04.05	1	1.40	(3,5,1,1,3,5);
Transports within plant			total diesel consumption for mining and processing 50l; assumption 30l for mining and cutting and 10l for grinding and 10l for polishing	construction processes	machinery	No	GLO	diesel, burned in building machine	5.06E-01	MJ		1	1.40	(3,5,1,1,3,5);
Infrastructure			weight of the machines has been assumed to be 5 tons with life span of 10 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	5.00E-06	kg		1	3.36	(5,5,1,2,3,5);
Water			assumption: same amount as for cutting	resource	in water			Water, river	3.73E-03	m ³	http://www.energie.ch/themen/industrie/infel/steinwerke.htm, access data 14.04.05	1	1.38	(2,5,1,3,3,5);
		waste heat		air	unspecified			Heat, waste	8.64E-01	MJ		1	1.40	(3,5,1,1,3,5);
		Dust < 2.5 um	assumption 5% to air, 95% with water in ground, particle distribution: 30% < 2.5um	air	unspecified			Particulates, < 2.5 um	5.60E-04	kg		1	3.79	(2,5,3,5,5,5);
		Dust > 2.5, < 10um	assumption 5% to air, 95% with water in ground, particle distribution: 10um > 55% > 2.5um	air	unspecified			Particulates, > 2.5 um, and < 10um	1.03E-03	kg		1	2.78	(2,5,3,5,5,5);
		Dust > 10um	assumption 5% to air, 95% with water in ground, particle distribution: 15% > 10um	air	unspecified			Particulates, > 10 um	2.80E-04	kg		1	2.35	(2,5,3,5,5,5);
		waste water to ground	same amount as water input; not inventoried						3.73E-03	m ³				
		natural stone plate, grounded		construction materials	others	No	CH	natural stone plate, grounded, at regional storage	1.00E+00	kg				

Fig. 6.2 Flows for “natural stone, ground, at regional storage” and their representation in the ecoinvent database

Tab. 6.3 **ecoinvent meta information of the product “natural stone, polished, at regional storage”**

Name	natural stone, polished, at regional storage
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Basis is a ground natural granite plate in a storage. Beside infrastructure some internal transports are included. No administration and wear and spare parts are taken into account. Dust is accounted for.
Amount	1
Local Name	Natursteinplatten, poliert, ab Regionallager
Synonyms	
General Comment to reference function	The polishing is assumed to be done using a polishing machine and very fine grinding medium. No losses are taken into account as no data are available. The dust emissions are estimated based on the losses (0.05 mm abrasion around the granite plate with geometry of 0.5 * 0.5 * 0.03 m and a weight of 2'750 kg/m ³).
Start Date	2000
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	For some exchanges RER- and GLO-modules are used as proxy
Technology text	Typical technology for average Swiss production
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Literature
Extrapolations	See geography
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
natural stone plate, grounded	natural stone plate, polished, at regional storage		Assumed that Granite plate with a geometry of 0.5 * 0.5 * 0.03m and a weight of 20 kg (2750 kg/m ³)	construction materials	others	No	CH	natural stone plate, grounded, at regional storage	1.01E+00	kg		1	1.68	(5,5,1,1,3,5);
Electricity for polishing				electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.40E-01	kWh	http://www.energie.ch/themen/industrie/infel/steinwerke.htm, access data 14.04.05	1	1.40	(3,5,1,1,3,5);
Transports within plant			total diesel consumption for mining and processing 50l; assumption 30l for mining and cutting and 10l for grinding and 10l for polishing	construction processes	machinery	No	GLO	diesel, burned in building machine	5.06E-01	MJ		1	1.40	(3,5,1,1,3,5);
Infrastructure			weight of the machines has been assumed to be 5 tons with life span of 10 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	5.00E-06	kg		1	3.36	(5,5,1,2,3,5);
		waste heat		air	unspecified			Heat, waste	8.64E-01	MJ		1	1.40	(3,5,1,1,3,5);
		Dust < 2.5 um	assumption 100% to air as no cooling needed, particle distribution: 30% < 2.5um	air	unspecified			Particulates, < 2.5 um	1.12E-03	kg	http://www.energie.ch/themen/industrie/infel/steinwerke.htm, access data 14.04.05	1	3.79	(2,5,3,5,5,5);
		Dust > 2.5, < 10um	assumption 100% to air as no cooling needed, particle distribution: 10um > 55% > 2.5um	air	unspecified			Particulates, > 2.5 um, and < 10um	2.05E-03	kg		1	2.78	(2,5,3,5,5,5);
		Dust > 10um	assumption 100% to air as no cooling needed, particle distribution: 15% > 10um	air	unspecified			Particulates, > 10 um	5.60E-04	kg		1	2.35	(2,5,3,5,5,5);
		natural stone plate, polished		construction materials	others	No	CH	natural stone plate, polished, at regional storage	1.00E+00	kg				

Fig. 6.3 Flows for “natural stone, polished, at regional storage” and their representation in the ecoinvent database

7 Data Quality

For the following reasons the data quality for all different types of natural stone is rather poor.

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007b)). For detailed results compare the different flows and the ecoinvent database meta information.

The public data situation is very poor. A lot of assumptions were needed. Following list gives an overview of the most important ones:

- The basis of all the calculations is a granite plate with a certain geometry
- The mining is assumed to be done by drilling with compressed air and without explosives.
- The sawing is assumed to be accomplished with diamond cables. The diamond losses are not taken into account even though they might be significant as no data on diamond production is available.
- The grinding is assumed to be accomplished with diamond studded grinding wheels. The diamond losses are not taken into account even though they might be significant as no data on diamond production is available.
- The polishing is assumed to be done using a polishing machine and very fine grinding medium.
- The amount of dust which can be expected has been estimated by calculating the cutting/grinding/polishing losses for a granite plate.
- It has been assumed that the sawing and grinding processes are cooled with water. On the other hand it has been assumed that the polishing process does not need cooling.
- The dust emission to air has been assumed with 5% of the losses and therefore the amount bound in the cooling water with 95%.
- The size distribution for particulate emission is based on average data for ores and non-metallic minerals. It covers material handling and processing of ores and minerals. The particulate emissions are a result of agitating the materials by screening or transfer during size reduction and beneficiation of the materials by grinding and fine milling and by drying⁷.
- The dust in the cooling water which is deposited in the ground has not been taken into account as no environmental burdens are expected.
- No administration and wear and spare parts are taken into account.
- The infrastructure which is used in the natural stone manufacturing processes is copied from the infrastructure for mining of limestone

These datasets should not be used for direct comparison with other materials. In a broader context and if the contribution of these products to the LCIA results is not too high, the data can be used.

⁷ <http://www.epa.gov/ttn/chief/ap42/appendix/appb-2.pdf>: access date: 19.07.05

8 Literature

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Part XXIII: **Bituminous Products and Processes**

Data v2.0 (2007)

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Summary

Bituminous products consist of a carrier material that is soaked with bitumen. The top coating of the bitumen sealing is normally oxidation bitumen with additives like talcum and sand. The bitumen sealings find their application mainly in flat roofed houses and in road construction work, basically where water intrusion is undesired. The bitumen sealing types described in this report are V60, Alu80, VA4 and EP4 flam. Further datasets are established for hot and cold bitumen adhesive compounds, as well as for mastic asphalt. No chemical reactions occur during the production of bituminous products, though certain emissions to the air are caused. Bituminous products can be recycled to a certain extent, but as the built-in sealing is very often dirty and contaminated, recycling does usually not take place for economic reasons.

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1 Introduction

This report describes the processes and their inventories of the production of bituminous and bituminous based products.

Most of the relevant information to the processes is compiled in the figures depicting the input and output flows of the processes together with information on their use and on their representation in the ecoinvent database. These figures show also the sources and the uncertainty information of the individual values.

Two different product groups are dealt with in this report. Beside some bituminous products also mastic asphalt is studied. Bitumen can be found in nature in form of asphalten (discharge of asphalten to the surface of the island of Trinidad as a natural resource in form of an asphalt lake) and asphalt rock which were produced in long geological time periods by evaporation of the easier boiling fractions of the crude oil. Industrially used bitumen is the fraction of the crude oil production which remains after distillation. Asphalt as main ingredient of mastic asphalt is a mixture of bitumen and aggregates.

The bituminous products are made out of a processed mixture of crude-oil, which contains a number of substances (almost 95 % are carbon (83%) and hydrogen (10%)). Bitumen remains after benzene and oils are distilled from crude oil. Therefore bitumen is a natural substance that has been created in the rock – under exclusion of oxygen – over millions of years. Bituminous products find their application mainly in civil engineering to protect buildings and roads from water intrusion. This report describes the features of different types of bituminous sealing lanes, as well as a hot and cold adhesive bitumen compound. The data for the bituminous products are mainly derived from Starzner & Wurmer-Weiss (2000) and Vogel et al. (2003). Some assumptions and calculations have been made.

Mastic asphalt is defined by the Mastic Asphalt Council¹ as follows: Mastic asphalt is a type of asphalt composed of graded mineral matter and asphaltic cement in such proportions as to form a cohesive, impermeable mass. It's solid or semi-solid under normal temperature conditions, but sufficiently fluid when brought to a suitable temperature to be spread by means of a hand float or by mechanical means without compaction. Mastic asphalt is used as protection for a wide range of construction applications. It offers total water

¹ <http://www.masticasphaltpcouncil.co.uk/>, access date: 02.05.2005

impermeability for roofing and tanking, acts as a working surface and damp-proof membrane in flooring and is ideal for paving, offering a durable wearing surface.

2 Raw Materials

The main raw materials of bituminous products are bitumen itself, some carrier material (e.g. polyester fleece, glass fleece, glass fibre or aluminium foil), plus fillers like talcum and sand. The top coating normally consists of oxidation bitumen, sprinkled with limestone and schist meal. The availability of bitumen is closely linked to the existence of crude oil. But not all crude oil contains the same amount of bitumen. It varies from the North Sea with hardly any bitumen content in crude-oil up to an almost 30%-content in Kuwait. More information on bitumen and bitumen sealing in general can be derived from the ecoinvent report No. 8 on Chemicals (Althaus et al. (2007)). Tab. 2.1 shows the material properties of an average bitumen sealing product (Bitumen V60).

Tab. 2.1 Material composition of bitumen sealing V60 (Starzner & Wurmer-Weiss 2000)

	Weight per m²	Material
Carrier	60 g/m ²	Glass fleece
Top coatings	3330 g/m ² 1110 g/m ²	Oxidation bitumen Limestone and/or schist meal
Filler	200 g/m ²	Sand
Moisture barrier	6 g/m ²	Polypropylene
Total weight	4.7 kg/m ²	

The raw materials of mastic asphalt are²:

- asphaltic cement, which consists of bitumen asphaltite, lake asphalt, or blends of these with one another.
- polymers for the modification of the asphaltic cement.
- limestone fine aggregate which is crushed to a finely graded powder.
- coarse aggregate: either crushed stone, e.g. granite, limestone or naturally occurring siliceous material (e.g. grit).
- pigments, binders, modifiers and special aggregates.

² <http://www.masticasphaltpcouncil.co.uk/>, access date: 02.05.2005

3 Use

The main application of bituminous products is the sealing of buildings and also road construction against water intrusion. Around 70 % of flat roof buildings in Europe are estimated to be coated with bitumen sealing (Starzner & Wurmer-Weiss 2000). Tab. 3.1 shows where the different types of bituminous products are applied:

Tab. 3.1 Application of different bitumen types in civil engineering (Starzner & Wurmer-Weiss 2000)

Type of bitumen	Application field in building industry
Distillation bitumen	Asphalt road construction
High vacuum bitumen	Bitumen lacquer and bitumen coating
Oxidation bitumen	Sealing lanes, roof lanes, pipe insulation
Polymer bitumen	Road construction (asphalt facing), roof lanes

Mastic asphalt is used in the building industry in different application ranges:

- Floor pavement in domestic and industry buildings
- Sealant in connection with mastic asphalt consisting of a water proofing layer in combination with a layer of mastic asphalt on it.
- Protective coating on a separation layer on sealants from bituminous materials
- Root protection layer under grassy areas
- Covering (floor pavement) on concrete in chemical engineering installations
- Liquid impermeable sealing in installations where water endangering substances are handled
- Surfacing in road works
- Protective and conversion coating on civil engineering works

Mastic asphalt can be adapted to different strains by changing the composition of the mineral aggregates and the amount and type of enamel.

4 Characterisation of the materials

4.1 Bituminous Products

The information regarding the characteristics of bituminous products is retrieved from Starzner & Wurmer-Weiss (2000).

4.1.1 Resistance

Bituminous products have a high resistance by nature. Bitumen is highly resistant to leachates, acids and saline solutions. It is not resistant to organic solutions, benzenes or oils. There is no restriction to heat exposure. Bitumen embrittles under high UV-radiation and air exposure.

4.1.2 Long-term emissions

Bitumen is not mentioned in any water pollution class³. Though, recent studies show that certain substances added to bitumen are washed out after some time. Long-term emissions to air in indoor applications should be taken into account. There, the crucial factor is probably the temperature.

4.1.3 Fire

Bitumen is hardly combustible under normal temperatures. Regarding the chemical structure of bitumen (and in particular hot bitumen), it has to be expected that the fumes are rather toxic (sulphuric fumes are released).

4.1.4 Material recycling

Material recycling of bituminous products is generally possible and is practiced in the case of asphalt. In civil engineering, recycling of bitumen sealing is not very reasonable. Bituminous products are in most cases bonded with other construction parts, and are difficult to separate into clean recycling fractions.

4.1.5 Waste incineration

Bitumen has a heating value of approximately 40 MJ/kg, which is similar to fuel oil. It should be incinerated in plants equipped with flue gas cleaning, as the sulphur content is in the range of that of heavy fuel oil (1-4%).

³ German characterization method based on a biological test procedure and other characteristics to assess the potential of pollutants and preparations which can affect the characteristics of water in a negative way

4.1.6 Land-fill behaviour

Bituminous products are not counted as hazardous waste. Still, land-filling of bitumen products makes no sense due to the high heating value.

4.2 Mastic Asphalt

Mastic asphalt⁴ is a type of asphalt composed of suitable graded mineral matter and asphaltic cement in such proportions as to form a cohesive, impermeable mass, solid or semi-solid under normal temperature conditions, but sufficiently fluid when brought to a suitable temperature to be spread by means of a hand float or by mechanical means without compaction.

4.2.1 Resistance to water

Mastic asphalt is impervious to water.

4.2.2 Resistance to biological attack

Mastic asphalt is vermin-proof and rot-proof.

4.2.3 Fire

Because of its high mineral content, mastic asphalt is virtually incombustible. Indicative tests have been performed on samples of insulated mastic asphalt roof decks. No significant flame spread was observed and no flame penetration occurred in either test. Mastic asphalt fulfils all the external fire resistance required for a roof covering and achieved the highest rating when tested.

4.2.4 Thermal conductivity

Mastic asphalt has a thermal conductivity, (λ) value, of between 0.43 W/m K and 1.15 W/m K. 0.50 W/m K may be assumed for design purposes.

4.2.5 Thermal expansion

Mastic asphalt is thermoplastic and is capable of accommodating normal movements encountered in well-designed building structures.

4.2.6 Coefficient of cubic expansion

The coefficient of cubic expansion is $15 \times 10^{-5}/^{\circ}\text{C}$.

4.2.7 Toxicity

Mastic asphalt is non-toxic and is generally suitable for use in contact with potable water.

⁴ <http://www.masticasphaltpcouncil.co.uk/>, access date: 02.05.2005

4.2.8 Odour

Mastic asphalt is odourless after application.

4.2.9 Density

The density of mastic asphalt varies due to a number of factors such as the differing proportions of constituents utilised in its manufacture and the nature and quality of coarse aggregate incorporated. For practical and load calculation purposes, however, its density can be taken to be 2'400 kg/m³.

4.2.10 Compressive strength

When mastic asphalt is fully confined it has the same compressive strength as the containing material. When not confined, the compressive strength is dependent on a number of factors including the temperature to which it may be subjected. Advice on individual cases should be sought from a manufacturer or the Mastic Asphalt Council.

4.2.11 Durability

Asphalt roofing properly designed and laid should prove capable of lasting 50-60 years.

4.2.12 Recycling

Mastic asphalt can be recycled.

5 System Characterisation

5.1 Raw Materials

All the described bitumen sealings in this report consist of bitumen (major input in terms of mass), a fleece layer (glass fibre, polyester, glass etc.), limestone and schist meal, as well as sand and talcum. The exception is the bitumen sealing Alu80, where the aluminium foil (moisture barrier) supplies the function of the fleece. The exact material inputs of the different bitumen sealings are given in chapter 6, Fig. 6.1 to Fig. 6.4.

The cold bitumen adhesive compound is an emulsion of bitumen (> 30 % of total weight) in water (50-60%), solvent (<8 %) and emulsifier (<3%) (Starzner & Wurmer-Weiss 2000). The hot bitumen adhesive compound is only composed of bitumen as raw material, as this type of adhesive compound is heated and not dissolved.

Mastic asphalt is composed of the materials shown in Tab. 5.1.

Tab. 5.1 Raw materials for the mastic asphalt production (Vogel et al. 2003)

	kg / kg product
limestone powder	0.26
sand/grit	0.66
bitumen	0.06
natural bitumen	0.02
Total	1

5.2 Production process

The production process includes the whole manufacturing process including raw material provision and transportation, heating and mixing, storage, flush mounting, and the infrastructure. No transportation from the plant to the construction site and no requirements for administration are included.

5.3 Energy Consumption

All bitumen types (V60, Alu80, VA4, EP4 flam) require a certain amount of energy during the production phase. The energy consumption in form of electricity for all process is around 0.42 kWh for 1 m² of bitumen sheet (Starzner & Wurmer-Weiss 2000). Included in this amount is the final rolling of the bitumen sheet into a coil.

The cold bitumen adhesive compound asks for electricity input in order to emulsify the bitumen. This input is estimated to be the same as for plaster mixing (0.08 kWh per 1 m²).

In case of hot bitumen compound, the energy consumption relates to the heating of the bitumen by a gas burner up to ca. 140 °C. This is calculated via a heat capacity for bitumen of 2100 J/kgK and an efficiency factor of 20%⁵.

In the production process of mastic asphalt the bitumen to be applied is already delivered with the temperature needed for transport and working. Therefore, energy consumption is only modelled for the drying of the minerals (1.5 MJ/kg, Starzner & Wurmer-Weiss 2000) and for holding up the temperature during the transport and storage in insulated buckets. The amount of electricity is comparable with the energy needed for the production of hydraulic-setting cast plaster floors.

5.4 Transports

Various transports by lorry and railway can be linked to the delivery of bituminous products. Transports for the raw materials from the excavation to the production site are already assigned to the input materials. For these datasets, ecoinvent standard transport distances in Europe for individual material groups are applied. These are 50 km by lorry for mineral products like sand, talcum and limestone, and 100 km for plastics, bitumen, glass fibre and metals. The additional standard transport distances of mineral products by railway is set to 0 km, whereas for plastics, glass fibres, bitumen and metals it is assumed to be 200 km.

A diesel consumption of 0.0006 l/kg is assumed for all internal transports in the production process of mastic asphalt, based on the production of concrete. As the transportation system is not known, the dataset "diesel, burned in building machine" is used as proxy.

5.5 Air Emissions

The four bitumen sealings mentioned in this report show similar emissions to the air during their production phase. Emissions depend only on the percentage of bitumen in the product. The main air emissions of the processing of bitumen are VOC, volatile organic compounds. Starzner & Wurmer-Weiss 2000 indicate that per 1 kg hot processed bitumen, about 1-2 g of VOC is emitted to the air. BUWAL 2000 reports the emissions given in Tab. 5.2. However, these values refer to an old technology, using about 20% of organic solvents in the production. Today's production uses almost no solvents any more. Thus, 1.5 g NMVOC emissions per kg of bitumen input to bitumen sealing production are used in this study.

⁵ Efficiency is chosen that low because the process usually is taking place on the construction site and often is
ecoinvent report No. 7

Tab. 5.2 Emissions to air from bitumen sealing production (BUWAL 2000)

	CO	NM VOC	CH ₄	Dust
g/m ² sealing	350	40	1.3	18
kg bitumen / m ² sealing	1.2	1.2	1.2	1.2
g/kg bitumen input	292	33.3	1.08	15.0

VOC emissions from bitumen adhesive are reported to be 1 to 2 g/kg hot bitumen adhesive compound and 0 to 80 g/kg cold bitumen adhesive compound (Starzner & Wurmer-Weiss 2000). Consequently 1.5 and 40 g NMVOC per kg hot and cold bitumen adhesive compound are used in this study.

The VOC emission of the mastic asphalt production is quite small. Based on Starzner & Wurmer-Weiss 2000 it's smaller than 0.2 g/kg. For this study the amount of 0.19 g/kg mastic asphalt is used.

BUWAL 2001a indicates Benzo(a)pyrene (PaB) emission factors for different types of bitumen. Emissions due to the production of bitumen sealing, hot bitumen adhesive compound and mastic asphalt can be estimated from these values. Emission factors strongly depend on the temperature and range between 0.03 and 0.3 µg/m²*min at 150 °C, 0.21 and 0.68 µg/m²*min at 180 °C and between 2.12 and 5.25 µg/m²*min at 250 °C. Emission rates between 0.25 and 0.30 µg/m²*min at 150 °C are reported for oxidation bitumen, the most common bitumen for sealings and hot bitumen adhesive compound. Using the upper value and assuming that the bitumen is at 150 °C for 30 minutes during sealing production and adhesive compound application, PaB emissions of 9 µg/m² and the emissions per functional unit given in Tab. 5.3 are calculated.

Tab. 5.3 PaB emissions of bitumen sealing and hot bitumen adhesive production

	bitumen sealing V60	bitumen sealing Alu80	bitumen sealing VA4	bitumen sealing, polymer EP4 flame retardant	bitumen adhesive compound, hot
Weight [kg / m ²]	4.7	2.6	5.0	5.2	1.3
Emission [µg/kg]	1.91	3.46	1.80	1.73	6.92

Emission rates of 0.21 µg/m²*min at 180 °C and 2.12 µg/m²*min at 250 °C are reported in the same source for the most common bitumen for mastic asphalt, high-vacuum bitumen. Assuming that mastic asphalt is at 180 °C for 30 minutes while heating and cooling and at

poorly controlled.

250 °C for 10 minutes while lying, the PaB emission of mastic asphalt is 27.5 µg/m². Further assuming a demand of 59 kg mastic asphalt per m² (Starzner & Wurmer-Weiss 2000), the resulting emission per kg mastic asphalt is 0.466 µg/kg. This value is used in this study.

Long-term air emission caused by bitumen sealings and mastic asphalt cannot be excluded, causing potential effects mainly if applied for interior purposes. The risk however is diminished, as bitumen sealings and mastic asphalt are seldom used in heated room interiors. Since the ecoinvent datasets relate to the production of bitumen sealings only, such emissions are not taken into account.

5.6 Water / Soil Emissions

No water emissions are accounted for in the production phase. Though, it has to be considered that in the use-phase biocides like Mecoprop might leak to water and soil.

5.7 Infrastructure

For the datasets bitumen sealing V60, Alu80, VA4, EP4 flam the infrastructure is calculated similar to the ecoinvent dataset “bitumen sealing, at plant”. 10% of the infrastructure demand of bitumen sealings is accounted for the production of cold and hot bitumen adhesive compound.

As no data for the infrastructure of the mastic asphalt production is available, some estimation is necessary. Mastic asphalt is mainly produced in stationary asphalt mixing plants. The mineral nutrients are dried in a rotary drier. The heated minerals are then stored in insulated silos from where they are mixed in the required ratio. The equipment used after the production of mastic asphalt is not taken into account; for instance the transportation of the hot asphalt in heated and insulated agitating buckets to the installation site. The weight of the equipment inventoried is assumed to be 20t, the annual production amount 150'000t, and the service life of the machines 20 years. The length of the conveyor belt is assumed to be 100m.

6 Life Cycle Inventories

The meta information (tables), flows and their representation in the ecoinvent database are shown in (figures):

Bitumen sealing V60, at plant	Tab. 6.1 and Fig. 6.1
Bitumen sealing Alu80, at plant	Tab. 6.2 and Fig. 6.2
Bitumen sealing VA4, at plant	Tab. 6.3 and Fig. 6.3
Bitumen sealing, polymer EP4 flam, at plant	Tab. 6.4 and Fig. 6.4
Bitumen adhesive compound, cold, at plant	Tab. 6.5 and Fig. 6.5
Bitumen adhesive compound, hot, at plant	Tab. 6.6 and Fig. 6.6
Mastic asphalt, including laying, at plant	Tab. 6.7 and Fig. 6.7

6.1 Bitumen sealing V60

The meta information of “Bitumen sealing V60, at plant” (Tab. 6.1) includes all important and necessary information to use the data correctly. Fig. 6.1 shows all in- and output flows of the product “Bitumen sealing V60, at plant”, the allocation to the datasets in the ecoinvent database, the sources and the standard deviation of all in- and output flows.

Tab. 6.1 ecoinvent meta information of the product “bitumen sealing V60, at plant”

Name	bitumen sealing V60, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2
Included Processes	The dataset describes the production of bitumen sealing V60. Included processes are the extrusion of polypropylene foil, the fabrication of the sealing, the infrastructure, the transport by road and rail as well as the disposal of the bitumen cuttings in a municipal incineration. The production of bitumen sealing does not include any chemical processes.
Amount	1
Local Name	Bitumenbahn V60, ab Werk
Synonyms	
General Comment to reference function	The dataset is applicable in the construction building sector. Bitumen sealing is used to protect the construction against water intrusion.
Start Date	1994
End Date	2000
Data Valid For Entire Period	1
Other Period Text	The production of bitumen sealing has not changed very much over the years, as there are no chemical processes involved.
Geography text	The data of this dataset is taken from literature studies and producer information.
Technology text	Simple processes involved, a carrier material is being soaked with bitumen. The bitumen coatings are then gritted with minerals (talcum, sand).
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature, database
Extrapolations	see Geography
Uncertainty Adjustments	The used data is mainly valid for Central Europe, the dataset is applicable for the whole European region.

General flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Cate gory	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
Bitumen	Bitumen sealing V60		70% of total weight; losses included	oil	fuels	No	RER	bitumen, at refinery	7.15E-01	kg	Starzner & Wurmer-Weiss (2000)	1	1.24	(2,4,1,3,1,5,3)	
Glass fibre fleece			Proxy for glass fibre fleece, 1,3% of total weight	glass	construction	No	RER	glass fibre, at plant	1.31E-02	kg		1	1.24	(2,4,1,3,1,5,3)	
Limestone			Half of filler amount, 11.7% weight-%	construction materials	additives	No	CH	limestone, milled, loose, at plant	1.18E-01	kg		1	1.24	(2,4,1,3,1,5,3)	
Schist meal			Proxy (weight% 11.7)	construction materials	additives	No	CH	limestone, milled, loose, at plant	1.18E-01	kg		1	1.24	(2,4,1,3,1,5,3)	
Sand			Silica sand	construction materials	additives	No	CH	sand, at mine	2.17E-02	kg		1	1.24	(2,4,1,3,1,5,3)	
Talcum			Proxy for talcum	construction materials	additives	No	CH	limestone, milled, loose, at plant	2.17E-02	kg		1	1.24	(2,4,1,3,1,5,3)	
Polypropylene			For PP-welding film	plastics	polymers	No	RER	polypropylene, granulate, at plant	1.01E-03	kg		1	1.24	(2,4,1,3,1,5,3)	
Foil extrusion				plastics	processing	No	RER	extrusion, plastic film	1.01E-03	kg		1	1.24	(2,4,1,3,1,5,2)	
Infrastructure			Chemical plant, analog to ecoinvent dataset "bitumen sealing, at plant"	chemicals	organics	Yes	RER	chemical plant, organics	4.00E-10	unit	ecoinvent	1	3.06	(2,4,1,3,1,5,9)	
		PaB		air	unspecified			Benzo(a)pyrene	1.91E-09	kg	calculation, based on BUWAL 2001	1	5.00	estimation	
		VOC	Starzner & Wurmer-Weiss (2000): 1-2 g VOC per kg of bitumen input. 1.5 g NMVOC per kg chosen in this study.	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	1.07E-03	kg	Starzner & Wurmer-Weiss (2000)	1	2.06	(2,4,1,3,1,5,23)	
			Included in total electricity									1			
Rolling			For all the processing steps only	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	4.20E-01	kWh	Calculated with ecoinvent standard transport distances	1	1.22	(2,3,1,3,1,5,2)	
Electricity				transport systems	road	No	RER	transport, lorry >16t, fleet average	8.69E-02	tkm		1	2.10	(4,3,1,3,1,5,5)	
Transport road				transport systems	train	No	RER	transport, freight, rail	1.46E-01	tkm		1	2.10	(4,3,1,3,1,5,5)	
Transport rail											Calculated	1	1.35	(2,4,3,3,3,5,6)	
		Waste municipal incineration bitumen	For incineration of the bitumen; assumption 1% of total output	waste management	municipal incineration	No	CH	disposal, bitumen sheet, 1.5% water, to municipal incineration	7.16E-03	kg		1	1.35	(2,4,3,3,3,5,6)	
		Waste municipal incineration rest	Proxy for incineration of the mineral materials	waste management	municipal incineration	No	CH	disposal, glass, 0% water, to municipal incineration	2.93E-03	kg	Calculated via electricity input	1	1.35	(2,4,3,3,3,5,6)	
		Heat		air	unspecified			Heat, waste	1.51E+00	MJ		1	1.24	(2,4,1,3,1,5,13)	
		Bitumen sealing V60, at plant	Bitumen V60 weighs 4.7 kg/m2	plastics	others	No	RER	bitumen sealing V60, at plant	1.00E+00	kg					

Fig. 6.1 Flows for “Bitumen sealing V60, at plant” and their representation in the ecoinvent database

6.2 Bitumen sealing Alu80

The meta information of “Bitumen sealing Alu80, at plant” (Tab. 6.2) includes all important and necessary information to use the data correctly. Fig. 6.2 shows all in- and output flows of the product “Bitumen sealing Alu80, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

Tab. 6.2 ecoinvent meta information of the product “Bitumen sealing Alu80, at plant”

Name	Bitumen sealing Alu80, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2
Included Processes	The dataset describes the production of bitumen sealing Alu80. Included processes are the extrusion of polypropylene foil, the fabrication of the sealing, the infrastructure, the transport by road and rail as well as the disposal of the bitumen cuttings in a municipal incineration. The production of bitumen sealing does not include any chemical processes. Bitumen sealing Alu80 has as a specification a moisture barrier.
Amount	1
Local Name	Bitumenbahn Alu80, ab Werk
Synonyms	
General Comment to reference function	The dataset is applicable in the construction building sector. Bitumen sealing is used to protect the construction against water intrusion.
Start Date	1994
End Date	2000
Data Valid For Entire Period	1
Other Period Text	The production of bitumen sealing has not changed very much over the years, as there are no chemical processes involved.
Geography text	The data of this dataset is taken from literature studies and producer information.
Technology text	Simple processes involved; a carrier material is being treated with bitumen. The bitumen coatings are then gritted with minerals (talcum, sand).
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature, database
Extrapolations	see Geography
Uncertainty Adjustments	The used data is mainly valid for Central Europe, the dataset is applicable for the whole European region.

General flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Cate gory	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
Bitumen	Bitumen sealing Alu80		71,8% of total weight; losses included	oil	fuels	No	RER	bitumen, at refinery	6.65E-01	kg	Calculated from Vogel et al. (2003) and Starzner & Wurmer-Weiss (2000)	1	1.24	(2,4,1,3,1,5,3)	
Limestone			Half of filler amount	construction materials	additives	No	CH	limestone, milled, loose, at plant	1.11E-01	kg		1	1.24	(2,4,1,3,1,5,3)	
Schist meal			Proxy, half of filler weight	construction materials	additives	No	CH	limestone, milled, loose, at plant	1.11E-01	kg		1	1.24	(2,4,1,3,1,5,3)	
Sand			Silica sand	construction materials	additives	No	CH	sand, at mine	1.99E-02	kg		1	1.24	(2,4,1,3,1,5,3)	
Talcum			Proxy for talcum	construction materials	additives	No	CH	limestone, milled, loose, at plant	1.99E-02	kg		1	1.24	(2,4,1,3,1,5,3)	
Aluminium foil			Aluminium foil (80µm)	metals	extraction	No	RER	aluminium, production mix, at plant	8.39E-02	kg	Calculated	1	1.22	(2,2,1,3,1,5,3)	
Aluminium rolling			Aluminium foil; double of weight input for foil	metals	processing	No	RER	sheet rolling, aluminium	1.68E-01	kg		1	1.26	(3,4,1,3,1,5,2)	
Infrastructure			Chemical plant, analog to ecoinvent dataset "bitumen sealing, at plant"	chemicals	organics	Yes	RER	chemical plant, organics	4.00E-10	unit	ecoinvent	1	3.06	(2,4,1,3,1,5,9)	
		PaB		air	unspecified			Benzo(a)pyrene	3.46E-09	kg	calculation, based on BUWAL 2001	1	5.00	estimation	
		VOC		air	unspecified			NM VOC, non-methane volatile organic compounds, unspecified origin	9.97E-04	kg	Starzner & Wurmer-Weiss (2000)	1	2.06	(2,4,1,3,1,5,23)	
			Included in total electricity									1			
Rolling				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	4.20E-01	kWh	Calculated with ecoinvent standard transport distances	1	1.22	(2,3,1,3,1,5,2)	
Electricity				transport systems	road	No	RER	transport, lorry >16t, fleet average	8.79E-02	tkm		1	2.10	(4,3,1,3,1,5,5)	
Transport road				transport systems	train	No	RER	transport, freight, rail	1.50E-01	tkm		1	2.10	(4,3,1,3,1,5,5)	
Transport rail											Calculated	1	1.35	(2,4,3,3,3,5,6)	
		Waste municipal incineration bitumen	For incineration of the bitumen	waste management	municipal incineration	No	CH	disposal, bitumen sheet, 1.5% water, to municipal incineration	6.65E-03	kg		1	1.35	(2,4,3,3,3,5,6)	
		Waste municipal incineration aluminium	For incineration of the aluminium foil	waste management	municipal incineration	No	CH	disposal, aluminium, 0% water, to municipal incineration	8.39E-04	kg		1	1.35	(2,4,3,3,3,5,6)	
		Waste municipal incineration rest	Proxy for incineration of the mineral materials	waste management	municipal incineration	No	CH	disposal, glass, 0% water, to municipal incineration	2.61E-03	kg	Calculated via electricity input	1	1.35	(2,4,3,3,3,5,6)	
		Heat		air	unspecified			Heat, waste	1.51E+00	MJ		1	1.24	(2,4,1,3,1,5,13)	
		Bitumen sealing Alu80, at plant	Bitumen Alu80 weighs 2.6 kg / m2	plastics	others	No	RER	bitumen sealing Alu80, at plant	1.00E+00	kg					

Fig. 6.2 Flows for “Bitumen sealing Alu80, at plant” and their representation in the ecoinvent database

6.3 Bitumen sealing VA4

The meta information of “Bitumen sealing VA4, at plant” (Tab. 6.3) includes all important and necessary information to use the data correctly.

Fig. 6.3 shows all in- and output flows of the product “Bitumen sealing VA4, at plant”, the allocation to the datasets in the ecoinvent database, the sources and the standard deviation of all in- and output flows.

Tab. 6.3 ecoinvent meta information of the product “Bitumen sealing VA4, at plant”

Name	Bitumen sealing VA4, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2
Included Processes	The dataset describes the production of bitumen sealing VA4. Included processes are the extrusion of polypropylene foil, the fabrication of the sealing, the infrastructure, the transport by road and rail as well as the disposal of the bitumen cuttings in a municipal incineration. The production of bitumen sealing does not include any chemical processes. In this specific type of bitumen sealing VA4, aluminium is being sputtered onto a PE-foil.
Amount	1
Local Name	Bitumenbahn VA4, ab Werk
Synonyms	
General Comment to reference function	The dataset is applicable in the construction building sector. Bitumen sealing is used to protect the construction against water intrusion.
Start Date	1994
End Date	2000
Data Valid For Entire Period	1
Other Period Text	The production of bitumen sealing has not changed very much over the years, as there are no chemical processes involved.
Geography text	The data of this dataset is taken from literature studies and producer information.
Technology text	Simple processes involved (beside sputtering), a carrier material is being soaked with bitumen. The bitumen coatings are then gritted with minerals (talcum, sand).
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature, database
Extrapolations	see Geography
Uncertainty Adjustments	The used data is mainly valid for Central Europe, the dataset is applicable for the whole European region.

General flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Cate gory	Sub category	Infra struc ture	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
Bitumen	→	Bitumen sealing VA4, part 1				71.8% of total weight; losses included	oil	fuels	No	RER	bitumen, at refinery	7.23E-01	kg	Calculated from Vogel et al. (2003) and Starzner & Wurmer-Weiss (2000)	1	1.24	(2,4,1,3,1,5,3)
Glass fibre fleece	→					Proxy for glass fibre fleece	glass	construction	No	RER	glass fibre, at plant	1.31E-02	kg	Starzner & Wurmer-Weiss (2000)	1	1.24	(2,4,1,3,1,5,3)
Limestone	→					Half of filler weight	construction materials	additives	No	CH	limestone, milled, loose, at plant	1.08E-02	kg	Calculated from Vogel et al. (2003) and Starzner & Wurmer-Weiss (2000)	1	1.24	(2,4,1,3,1,5,3)
Schist meal	→					Proxy, half of filler weight	construction materials	additives	No	CH	limestone, milled, loose, at plant	1.08E-02	kg		1	1.24	(2,4,1,3,1,5,3)
Sand	→					Silica sand	construction materials	additives	No	CH	sand, at mine	1.20E-01	kg		1	1.24	(2,4,1,3,1,5,3)
Talcum	→					Proxy for talcum	construction materials	additives	No	CH	limestone, milled, loose, at plant	1.20E-01	kg		1	1.24	(2,4,1,3,1,5,3)
Aluminium	→					Aluminium, sputtered on PE-foil (assumed thickness 100 nm)	metals	extraction	No	RER	aluminium, production mix, at plant	5.45E-04	kg		1	1.22	(2,2,1,3,1,5,3)
Polyester-compound foil	→					Proxy PE-vlies for polyester foil in compound with Aluminium, 40 µm; density for polyester 1,1 g/cm3	plastics	polymers	No	RER	fleece, polyethylene, at plant	8.89E-03	kg	1	1.24	(2,4,1,3,1,5,3)	
Aluminium sputtering	→					Proxy for aluminium sputtering	metals	refinement	No	DE	selective coating, copper sheet, sputtering	2.00E-01	m2	Calculated	1	1.26	(3,4,1,3,1,5,2)
Copper	→					Taken off from dataset "selective coating, copper sheet, sputtering"	metals	extraction	No	RER	copper, at regional storage	-1.64E-02	kg	ecoinvent	1	1.26	(3,4,1,3,1,5,2)
Tin	→					Taken off from dataset "selective coating, copper sheet, sputtering"	metals	extraction	No	RER	tin, at regional storage	-1.72E-03	kg		1	1.26	(3,4,1,3,1,5,2)
Chromium	→					Taken off from dataset "selective coating, copper sheet, sputtering"	metals	extraction	No	RER	chromium, at regional storage	-1.48E-03	kg		1	1.26	(3,4,1,3,1,5,2)
Infrastructure	→					Chemical plant, analog to ecoinvent dataset "bitumen sealing, at plant"	chemicals	organics	Yes	RER	chemical plant, organics	4.00E-10	unit		1	3.06	(2,4,1,3,1,5,9)

Fig. 6.3 Flows for “Bitumen sealing VA4, at plant” and their representation in the ecoinvent database

General flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment	
Rolling Electricity Transport road Transport rail	Bitumen sealing VA4, part 2	→ PaB		air	unspecified			Benzo(a)pyrene	1.80E-09	kg	calculation, based on BUWAL 2001	1	5.00	estimation	
		→ VOC	Starzner & Wurmer-Weiss (2000): 1-2 g VOC per kg of bitumen input. 1.5 g NMVOC per kg chosen in this study.	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	1.08E-03	kg	Starzner & Wurmer-Weiss (2000)	1	2.06	(2,4,1,3,1,5,23)	
			Included in total electricity									1			
			For all the processing steps only	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	4.20E-01	kWh		1	1.22	(2,3,1,3,1,5,2)	
				transport systems	road	No	RER	transport, lorry >16t, fleet average	8.77E-02	tkm	Calculated with ecoinvent standard transport distances	1	2.10	(4,3,1,3,1,5,5)	
				transport systems	train	No	RER	transport, freight, rail	1.49E-01	tkm		1	2.10	(4,3,1,3,1,5,5)	
		→ Waste municipal incineration bitumen	For incineration of the bitumen	waste management	municipal incineration	No	CH	disposal, bitumen sheet, 1.5% water, to municipal incineration	7.23E-03	kg	Calculated	1	1.35	(2,4,3,3,3,5,6)	
		→ Waste municipal incineration aluminium	For incineration of the aluminium foil	waste management	municipal incineration	No	CH	disposal, aluminium, 0% water, to municipal incineration	5.45E-06	kg		1	1.35	(2,4,3,3,3,5,6)	
		→ Waste municipal incineration rest	Proxy for incineration of the mineral materials	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	2.84E-03	kg		1	1.35	(2,4,3,3,3,5,6)	
		→ Heat		air	unspecified			Heat, waste	1.51E+00	MJ	Calculated via electricity input	1	1.24	(2,4,1,3,1,5,13)	
		→ Bitumen sealing VA4, at plant	Bitumen VA4 weighs 5.0 kg/m2	plastics	others	No	RER	bitumen sealing VA4, at plant	1.00E+00	kg					

Fig. 6.3 Flows for “Bitumen sealing VA4, at plant” and their representation in the ecoinvent database (cont)

6.4 Bitumen sealing, polymer EP4 flam

The meta information of “Bitumen sealing, polymer EP4 flam, at plant” (Tab. 6.4) includes all important and necessary information to use the data correctly. Fig. 6.4 shows all in- and output flows of the product “Bitumen sealing, polymer EP4 flam, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

Tab. 6.4 ecoinvent meta information of the product “Bitumen sealing, polymer EP4 flam, at plant”

Name	Bitumen sealing, polymer EP4 flam, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2
Included Processes	The dataset describes the production of bitumen sealing EP4 flam. Included processes are the extrusion of polypropylene foil, the fabrication of the sealing, the infrastructure, the transport by road and rail as well as the disposal of the bitumen cuttings in a municipal incineration. The production of bitumen sealing does not include any chemical processes. This type of bitumen sealing is built around a polyester fleece layer.
Amount	1
Local Name	Polymerbitumenbahn EP4 flam, ab Werk
Synonyms	
General Comment to reference function	The dataset is applicable in the construction building sector. Bitumen sealing is used to protect the construction against water intrusion.
Start Date	1994
End Date	2000
Data Valid For Entire Period	1
Other Period Text	The production of bitumen sealing has not changed very much over the years, as there are no chemical processes involved.
Geography text	The data of this dataset is taken from literature studies and producer information.
Technology text	Simple processes involved, a carrier material is being soaked with bitumen. The bitumen coatings are then gritted with minerals (talcum, sand).
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature, database
Extrapolations	see Geography
Uncertainty Adjustments	The used data is mainly valid for Central Europe, the dataset is applicable for the whole European region.

General flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Cate gory	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
Bitumen	bitumen sealing, polymer EP4 flame retardant, at plant		56,4% of total weight; losses included	oil	fuels	No	RER	bitumen, at refinery	5.70E-01	kg	Calculated from Vogel et al. (2003) and Starzner & Wurmer-Weiss (2000)	1	1.24	(2,4,1,3,1,5,3)	
Polymer SBS			Proxy for styrene-butadiene-styrene polymer	plastics	polymers	No	RER	synthetic rubber, at plant	7.27E-02	kg		1	1.24	(2,4,1,3,1,5,4)	
Limestone			Half of filler amount, 13% weight-%	construction materials	additives	No	CH	limestone, milled, loose, at plant	1.31E-01	kg	Starzner & Wurmer-Weiss (2000)	1	1.24	(2,4,1,3,1,5,3)	
Schist meal			Proxy (weight-% 13)	construction materials	additives	No	CH	limestone, milled, loose, at plant	1.31E-01	kg		1	1.24	(2,4,1,3,1,5,3)	
Sand			Silica sand	construction materials	additives	No	CH	sand, at mine	3.89E-02	kg		1	1.24	(2,4,1,3,1,5,3)	
Talcum			Proxy for talcum	construction materials	additives	No	CH	limestone, milled, loose, at plant	3.89E-02	kg		1	1.24	(2,4,1,3,1,5,3)	
Polyester fleece			Proxy for polyester fleece	plastics	polymers	No	RER	fleece, polyethylene, at plant	2.73E-02	kg		1	1.24	(2,4,1,3,1,5,4)	
Infrastructure			Chemical plant, analog to ecoinvent dataset "bitumen sealing, at plant"	chemicals	organics	Yes	RER	chemical plant, organics	4.00E-10	unit	ecoinvent	1	3.06	(2,4,1,3,1,5,9)	
		PaB		air	unspecified			Benzo(a)pyrene	1.73E-09	kg	calculation, based on BUWAL 2001	1	5.00	estimation	
		VOC	Starzner & Wurmer-Weiss (2000): 1-2 g VOC per kg of bitumen input. 1.5 g NMVOC per kg chosen in this study.	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	8.54E-04	kg	Starzner & Wurmer-Weiss (2000)	1	2.06	(2,4,1,3,1,5,23)	
			Included in total electricity									1			
Rolling			For all the processing steps only	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	4.20E-01	kWh	Calculated with ecoinvent standard transport distances	1	1.22	(2,3,1,3,1,5,2)	
Electricity				transport systems	road	No	RER	transport, lorry >16t, fleet average	8.40E-02	tkm		1	2.10	(4,3,1,3,1,5,5)	
Transport road				transport systems	train	No	RER	transport, freight, rail	1.34E-01	tkm		1	2.10	(4,3,1,3,1,5,5)	
Transport rail											Calculated	1	1.35	(2,4,3,3,3,5,6)	
		Waste municipal incineration bitumen	For incineration of the bitumen	waste management	municipal incineration	No	CH	disposal, bitumen sheet, 1.5% water, to municipal incineration	5.70E-03	kg		1	1.35	(2,4,3,3,3,5,6)	
		Waste municipal incineration rest	Proxy for incineration of mineral materials, polymer and fleece	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	4.40E-03	kg		1	1.35	(2,4,3,3,3,5,6)	
		Heat		air	unspecified			Heat, waste	1.51E+00	MJ	Calculated via electricity input	1	1.24	(2,4,1,3,1,5,13)	
		bitumen sealing, polymer EP4 flame retardant, at plant	Bitumen EP4 weighs 5.2 kg/m2	plastics	others	No	RER	bitumen sealing, polymer EP4 flame retardant, at plant	1.00E+00	kg					

Fig. 6.4 Flows for “Bitumen sealing, polymer EP4 flam, at plant” and their representation in the ecoinvent database

6.5 Bitumen adhesive compound, cold

The meta information of “Bitumen adhesive compound, cold, at plant” (Tab. 6.5) includes all important and necessary information to use the data correctly. Fig. 6.5 shows all in- and output flows of the product “Bitumen adhesive compound, cold, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

Tab. 6.5 ecoinvent meta information of the product “Bitumen adhesive compound, cold, at plant”

Name	bitumen adhesive compound, cold, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2
Included Processes	The dataset describes the production of cold bitumen adhesive compound. Included sub-processes are the mixing of bitumen and solvent, the infrastructure, the transport by road and rail as well as the disposal of the compound cuttings in a hazardous waste municipal incineration.
Amount	1
Local Name	Bitumenklebemasse kalt, ab Werk
Synonyms	cold bonding compound, Kaltkleber
General Comment to reference function	The dataset is applicable in the construction building sector. Bitumen adhesive compound is used to protect roads and roofs against water intrusion.
Start Date	1994
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	The data of this dataset is extracted from literature and producer information. Some own calculations (e.g. energy input, waste heat) were made.
Technology text	The technology involved is mixing of bitumen and solvent in a cold process.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature, database
Extrapolations	see Geography
Uncertainty Adjustments	The used data is mainly valid for Central Europe, the dataset is applicable for the whole European region.

General flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment	
Bitumen	Bitumen adhesive compound cold	NMVOC	39% of total weight (1% loss included)	oil	fuels	No	RER	bitumen, at refinery	3.94E-01	kg	Starzner & Wurmer-Weiss (2000)	1	1.24	(2,4,1,3,1,5,3)	
Solvent / Emulsifier			(8+3)% of total weight	chemicals	organics	No	GLO	chemicals organic, at plant	8.08E-02	kg		1	1.24	(2,4,1,3,1,5,4)	
Water			50% of total weight	water supply	production	No	CH	water, deionised, at plant	5.05E-01	kg		1	1.24	(2,4,1,3,1,5,4)	
Infrastructure			Chemical plant, analog to ecoinvent dataset "bitumen sealing, at plant" Starzner & Wurmer-Weiss (2000): 0-80 g VOC per kg of bitumen compound. 40 g NMVOC per kg chosen in this study.	chemicals	organics	Yes	RER	chemical plant, organics	4.00E-11	unit	ecoinvent	1	3.06	(2,4,1,3,1,5,9)	
			Electricity input similar to value in ecoinvent dataset "plaster mixing"	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	4.00E-02	kg	Starzner & Wurmer-Weiss (2000)	1	2.16	(4,4,1,3,3,5,23)	
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	8.00E-02	kWh	Starzner & Wurmer-Weiss (2000)	1	1.22	(2,3,1,3,1,5,2)	
Transport road				transport systems	road	No	RER	transport, lorry >16t, fleet average	4.75E-02	tkm	Calculated with ecoinvent standard transport distances	1	2.10	(4,3,1,3,1,5,5)	
Transport rail				transport systems	train	No	RER	transport, freight, rail	1.27E-01	tkm		1	2.10	(4,3,1,3,1,5,5)	
			For incineration of the bitumen	waste management	hazardous waste incineration	No	CH	disposal, hazardous waste, 25% water, to hazardous waste incineration	9.80E-03	kg	Calculated	1	1.35	(2,4,3,3,3,5,6)	
				air	unspecified			Heat, waste	2.88E-01	MJ	Calculated via electricity input	1	1.24	(2,4,1,3,1,5,13)	
		Waste incineration bitumen	0.15 - 0.4 kg needed per m2 (Starzner & Wurmer-Weiss 2000)	plastics	others	No	RER	bitumen adhesive compound, cold, at plant	1.00E+00	kg					
		Heat													
		Bitumen adhesive compound, cold, at plant													

Fig. 6.5 Flows for “bitumen adhesive compound, cold, at plant” and their representation in the ecoinvent database

6.6 Bitumen adhesive compound, hot

The meta information of “bitumen adhesive compound, hot, at plant” (Tab. 6.6) includes all important and necessary information to use the data correctly. Fig. 6.6 shows all in- and output flows of the product “Bitumen adhesive compound, hot, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

Tab. 6.6 ecoinvent meta information of the product “bitumen adhesive compound, hot, at plant”

Name	bitumen adhesive compound, hot, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2
Included Processes	The dataset describes the production of hot bitumen adhesive compound. Included sub-processes are the heating of the bitumen compound, the infrastructure, the transport by road and rail as well as the disposal of the compound cuttings in a municipal incineration.
Amount	
Local Name	Bitumenklebemasse heiss, ab Werk
Synonyms	hot bonding compound, Heisskleber
General Comment to reference function	The dataset is applicable in the construction building sector. Bitumen adhesive compound is used to protect roads and roofs against water intrusion.
Start Date	1994
End Date	2000
Data Valid For Entire Period	
Other Period Text	
Geography text	The data of this dataset is extracted from literature and producer information. Some own calculations (e.g. energy input, waste heat) were made.
Technology text	The technology involved is basically heating up the bitumen with a natural gas burner.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature, database
Extrapolations	see Geography
Uncertainty Adjustments	The used data is mainly valid for Central Europe, the dataset is applicable for the whole European region.

General flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment	
Bitumen	Bitumen adhesive compound hot	NM VOC PaB	100% of total weight; losses included	oil	fuels	No	RER	bitumen, at refinery	1.01E+00	kg	Starzner & Wurmer-Weiss (2000)	1	1.24	(2,4,1,3,1,5,3)	
Infrastructure			Chemical plant, analog to ecoinvent dataset "bitumen sealing, at plant"	chemicals	organics	Yes	RER	chemical plant, organics	4.00E-11	unit	ecoinvent	1	3.06	(2,4,1,3,1,5,9)	
			Starzner & Wurmer-Weiss (2000): 1-2 g VOC per kg of bitumen compound. 1.5 g NMVOC per kg chosen in this study.	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	1.50E-03	kg	Starzner & Wurmer-Weiss (2000)	1	2.06	(2,4,1,3,1,5,23)	
				air	unspecified			Benzo(a)pyrene	6.92E-09	kg	calculation, based on BUWAL 2001	1	5.00	estimation	
Gas		Waste municipal incineration bitumen Bitumen adhesive compound, hot, at plant	Bitumen is heated at ca. 140°C; calculated via heat capacity (2100 J/kgK); efficiency factor 20%. Industrial furnace used as proxy.	natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	1.26E+00	MJ	Starzner & Wurmer-Weiss (2000) and Pavatex (2002)	1	1.22	(2,3,1,3,1,5,2)	
Transport road				transport systems	road	No	RER	transport, lorry >16t, fleet average	1.01E-01	tkm	Calculated with ecoinvent standard transport distances	1	2.10	(4,3,1,3,1,5,5)	
Transport rail				transport systems	train	No	RER	transport, freight, rail	2.02E-01	tkm		1	2.10	(4,3,1,3,1,5,5)	
			For incineration of the bitumen	waste management	municipal incineration	No	CH	disposal, bitumen sheet, 1.5% water, to municipal incineration	1.01E-02	kg	Calculated	1	1.35	(2,4,3,3,3,5,6)	
			1 to 1.6 kg needed per m2 (Starzner & Wurmer-Weiss 2000)	plastics	others	No	RER	bitumen adhesive compound, hot, at plant	1.00E+00	kg					

Fig. 6.6 Flows for “bitumen adhesive compound, hot, at plant” and their representation in the ecoinvent database

6.7 Mastic Asphalt

The meta information of “mastic asphalt, at plant” (Tab. 6.7) includes all important and necessary information to use the data correctly. Fig. 6.7 shows all in- and output flows of the product “mastic asphalt, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

Tab. 6.7 ecoinvent meta information of the product “mastic asphalt, at plant”

Name	mastic asphalt, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce mastic asphalt: raw material provision and transport, heating and mixing, storage, flush mounting, and the infrastructure. No transportation from the plant to the construction site and no requirements for administration are included. No additional buildings and land-use have been taken into account as it is assumed that the mixing process takes place at the construction site.
Amount	1
Local Name	Gussasphalt, ab Werk
Synonyms	
General Comment to reference function	The raw materials per kg ready mix product are mixed in following ratio: Limestone powder (0.260 kg), sand (0.66 kg), bitumen (0.060 kg) and natural bitumen (0.020 kg). The only process emissions taken into account are the VOC-emissions. Further emissions are not taken into account as they play a subordinate role compared to the emissions of the raw material production (e.g. drying of silica sand) and because no data are available. For inventorying the infrastructure, a yearly production amount of 150'000 tons is assumed. The lifespan of the machines is assumed to be 20 years.
Start Date	2000
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	For the infrastructure RER-datasets are used as proxy
Technology text	Mixing, heating and casting of raw materials according to current technology used in Switzerland. It can be assumed that the technology is basically the same all over Europe and probably also for the rest of the world.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	
Uncertainty Adjustments	none

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
limestone powder	mastic asphalt, at plant			construction materials	others	No	CH	limestone, milled, packed, at plant	2.60E-01	kg	Vogel et al. (2003)	1	5.00	estimation
sand/grit			"wet" sand is dried in the mixing machine	construction materials	additives	No	CH	sand, at mine	6.60E-01	kg		1	5.00	estimation
bitumen				oil	fuels	No	CH	bitumen, at refinery	6.00E-02	kg		1	5.00	estimation
nature bitumen			"bitumen at refinery" used as proxy for internal transports, dataset "diesel, burned in building machine" used as proxy, amount: 0.0006 l/kg	oil	fuels	No	CH	bitumen, at refinery	2.00E-02	kg		1	5.00	estimation
diesel in building machine				construction processes	machinery	No	GLO	diesel, burned in building machine	2.16E-02	MJ	copied from dataset "concrete block, at plant"	1	5.00	estimation
electricity consumption			Estimation; dataset "anhydrite, at plant" used as proxy	electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.78E-02	kWh	copied from dataset "anhydrite floor, at plant"	1	3.00	estimation
transportation of raw materials to plant by lorry			assumed distance: minerals 50 km, bitumen 100 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	5.40E-02	tkm	Calculated with ecoinvent standard transport distances	1	2.10	(4,3,1,3,1,5,5)
Transport rail			assumed distance: minerals 0 km, bitumen 200 km	transport systems	train	No	RER	transport, freight, rail	1.60E-02	tkm		1	2.10	(4,3,1,3,1,5,5)
energy mainly for drying of minerals			heated with oil heating	oil	heating systems	No	CH	heat, light fuel oil, at industrial furnace 1MW	1.50E+00	MJ	Starzner & Wurmer-Weiss (2000)	1	5.00	estimation
infrastructure: mixing equipment			estimated weight of equipment: 20t, yearly production amount of mastic asphalt: 150'000t, estimated service life of machine: 20years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	6.67E-06	kg	copied from cement mortar production	1	5.00	estimation
infrastructure: conveyor belt			estimated length of conveyor belt: 100m, yearly production amount of mastic asphalt: 150'000t, estimated service life of conveyor belt: 20years	construction processes	machinery	Yes	RER	conveyor belt, at plant	3.33E-08	m		1	5.00	estimation
		waste heat	same amount as electricity consumption	air	unspecified			Heat, waste	1.00E-01	MJ	calculated	1	5.00	estimation
		VOC emissions	in Ecobis: <0.2 g/kg; assumption: 0.19 g/kg	air	unspecified			NM VOC, non-methane volatile organic compounds, unspecified origin	1.90E-04	kg	Starzner & Wurmer-Weiss (2000)	1	5.00	estimation
		PaB emissions		air	unspecified			Benzo(a)pyrene	4.66E-10	kg	calculation based on BUWAL 2001a and Starzner & Wurmer-Weiss (2000)	1	5.00	estimation
		mastic asphalt		construction materials	others	No	CH	mastic asphalt, at plant	1.00E+00	kg				

Fig. 6.7 Flows for "mastic asphalt, at plant" and their representation in the ecoinvent database

7 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007b)). For detailed results compare the different flows and the ecoinvent database meta information.

Most of the information for the four bitumen sealing types V60, Alu80, VA4 and EP4 flam is retrieved from Starzner & Wurmer-Weiss (2000) or Vogel et al. (2003). The applied weight measures for the individual materials are quite accurate.

Some of the materials had to be approximated by similar materials (e.g. proxy for “glass fibre fleece” is “glass fibre, at plant”).

Transport distances (road, rail) are calculated with the standard distances recommended in Frischknecht et al. (2007a).

The datasets of the hot and cold adhesive compounds are of lower quality than for the bitumen sealing types.

The air emissions (NMVOC and Benzene) are determined based on own assumptions.

Transport distances by road and rail are estimated using ecoinvent standard transport distances.

The data on raw material consumption for mastic asphalt is taken from Vogel et al. (2003). A comparison with the published data in Starzner & Wurmer-Weiss (2000) shows a good accordance.

The internal transports, electricity consumption and infrastructure are derived from similar processes.

Beside VOC (including benzene) no direct emissions from the processing of mastic asphalt are taken into account as no data are available.

For all the above reasons the datasets for bituminous products and processes are of mediocre to quite good overall quality.

These datasets should not be used for direct comparison among each others or with other materials. In a broader context and if the importance of these products in an inventory is not too high, the data can be used.

8 Literature

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Part XXIV:

Titanium Zinc Plate

Data v2.0 (2007)

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Kellenberger D., Althaus H.-J., Jungbluth N., Künniger T., Lehmann M. and Thalmann P. (2007) Life Cycle Inventories of Building Products. Final report ecoinvent Data v2.0 No. 7. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.org.

Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

Titanium zinc plate is an alloy of 99.995 % refined zinc with very small fractions of copper, titanium and aluminium. Its main uses are gutters and drainpipes, roofing and façade cladding, copings, dormers, or building profiles. The post consumer recycling rate for titanium zinc is in the order of 94%. The production of titanium zinc includes two rather energy-intensive melting processes in induction furnaces. The pre-weathering process is only applied for certain products. This ecoinvent dataset describes the titanium zinc plate production without the pre-weathering step.

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1 Introduction

Titanium zinc is an alloy of 99.995 % electrolytically refined zinc with accurately defined parts of copper, titanium and aluminium. Titanium zinc plates are mainly used for gutters and drainpipes, roofing and façade cladding, copings, dormers, or building profiles.

Titanium zinc provides effective protection from graffiti and can be used for solar photovoltaic system technology (Rheinzink (2005)).

The data used for this report originates mainly from the Internet information provided by one of the leading titanium zinc producers in Europe, Rheinzink GmbH. & Co. KG, Germany (Rheinzink (2002)).

2 Reserves and Resources

More than 99% of titanium zinc is zinc. Details of zinc extraction, refining and supply are given in Classen et al. (2007).

Ilmenite supplies about 90 % of the worlds demand for titanium minerals. Rutile is another resource suitable for the extraction of titanium. World ilmenite resources total about 1 billion tons of titanium dioxide. Identified world resources of rutile (including anatase) total about 230 million tons of contained TiO_2 (U.S. Geological Survey (2003)). TiO_2 is described in Althaus et al. (2007).

3 Characterisation of the material

Titanium zinc is an alloy of electrolytically refined zinc with a 99.995 % degree of purity and additional accurately defined parts of copper, titanium and aluminium (Rheinzink (2005)).

Tab. 3.1, Tab. 3.2 and Tab. 3.3 show information on the material composition and material properties of titanium zinc.

Tab. 3.1 Basic material composition of titanium zinc (RheinZink (2005))

Basic material	% of total weight
Refined zinc	≤ 99.835
Copper	0.08 - 0.17
Titanium	0.07 - 0.12
Aluminium	≤ 0.015

Tab. 3.2 Material properties of titanium zinc (RheinZink (2005))

Density (specific weight)	7.2 g/cm ³
Melting point	418 °C
Recrystallization limit	> 300 °C
Coefficient of lengthwise expansion:	2.2 mm/m x 100K

Tab. 3.3 Material weight and thickness of titanium zinc metal sheet¹

Metal thickness range (mm)	Metal sheet, nominal size cuts (mm)	Weight range (kg/m ²)
0.7 - 1.0	200 - 1000	5.04 - 7.2

4 Use

The main applications of titanium zinc plates are gutters and drainpipes, roofing and façade cladding, copings, dormers, or building profiles. (RheinZink (2005)).

During the chemical surface treatment of the titanium zinc plate, the atmospheric oxygen reacts with zinc into zinc oxide. With the impact of water during precipitation, zinc hydroxide is formed, which again reacts with carbon monoxide (from air) to an alkaline zinc carbonate (so-called Patina). This Patina is a very dense, bonding and water-insoluble coating. As a result, titanium zinc plates do not need any maintenance or servicing during their use phase (AUB (2002)).

The crucial factor for further degradation of the zinc surface is the acidity of the precipitation. With the decrease of SO₂ concentration in the air in recent years, the zinc ion concentration in the precipitation water could be reduced noticeably. No critical concentration values have been exceeded in rivers because of titanium zinc plates. The zinc content of the precipitation water can temporarily increase the zinc content of the soil (AUB (2002)).

¹ <http://www.rheinZink.de/67.aspx>: accessed June 2005

Material recycling of titanium zinc plates and its fabrication materials happens throughout their life cycle. In the fabrication phase, all trim scrap is being melted down into new products. Cuttings remaining from construction are collected and passed on to secondary metal producers either directly or via scrap metal dealers. The energy consumption for the recycling of the titanium zinc plates only makes up 5% of the energy that goes into primary zinc plate production. Recycling rates for titanium zinc plates were in the order of 94% in 2001/2002. One third of the recycled material is reused for zinc coating of steel, approximately one third find its way into brass production, and another third goes as zinc oxide into the production of zinc compounds. Zinc compounds are used in the pharmaceutical and chemical industry, for rubber, ceramics and glass production, as well as for animal feed, cosmetics or varnishes (AUB (2002)).

Because of the existing high-tech recycling processes, there is practically no titanium zinc going for disposal. Tests analysing household waste and collected rubbish of enterprises only showed traces of zinc (AUB (2002)).

5 System Characterisation

5.1 Raw Materials

The main raw material used for the production of titanium zinc plates is refined zinc (cf. Classen et al (2007), Part VIII). Only primary material is used for the production of titanium zinc plates (AUB 2002). Beside that, less than 0.1% of copper, titanium and aluminium are needed as alloy materials.

5.2 Production Process

The data used in this study include all processes taking place in the factory of “RheinZink” in Datteln, Germany, except for the pre-weathering. The production process for titanium zinc plates consists of 8 different steps (AUB (2002)):

1. *Pre-alloying*: Refined zinc, titanium, copper and aluminium are melted in a 760 °C induction furnace.
2. *Casting*: The pre-alloy mix is cast into so-called “pre-alloyed blocks”. The resulting blocks contain more titanium and copper than the final rolled alloys.
3. *Melting*: The pre-alloyed blocks and refined zinc are melted in large induction furnaces at temperatures of 500 - 550 °C. Induced currents help to mix the melting compound.

4. *Casting*: The ready-made alloy is cooled down below its melting point in the casting machine. A hard cast strand gets formed.
5. *Rolling*: After cooling down, the cast material is rolled by 5 rolling pairs. By this rolling step, the thickness of the material is reduced by up to 50%. At the same time, the material is being cooled and lubricated with a special emulsion.
6. *Coiling*: The titanium zinc is rolled up to 20 t coils. As the material still has a temperature of around 100 °C, it is interim stored for further cooling.
7. *Stretch forming and cutting*: The material stress caused during the rolling phase is eliminated by a “stretch-bend-truing” treatment of the material.
8. *Pre-weathering*: This procedure only applies for products with pre-weathered surface. After a thorough cleaning, the coil material undergoes the etching-process. The complete etching process takes place in closed, continuously working facilities.

5.3 Working Materials

Working materials are materials that are in direct contact with the basic raw materials, but do not undergo any reaction with them. Working materials used for the production of titanium zinc are (AUB (2002)):

- Roll-oil emulsion (0.8 kg/t zinc): A high-temperature resistant mixture of synthetic ester, fat essence, and emulsifier agents, used as a cooling and lubricant agent. The roll-oil emulsion is biologically degradable.
- Etching: Process for surface treatment. The composition of the pickle is the following: Sulphuric acid (15 kg/t zinc), nitric acid (5 kg/t zinc); alkaline degreasing agent (0,5 kg/t zinc); and temporary Cr-containing protection (0,5 - 0,8 kg/t zinc).

5.4 Emissions - Measures taken to reduce environmental impacts from production process

Only emissions to air are considered for this dataset. Some zinc emissions are reported during the pre-alloying phase of the production process (see Fig. 6.1).

Air, water and soil compartments are taken care of by the following measures (AUB (2002)):

- Air: Special filter systems clean the process air below legal critical values.
- Water/soil: The cooling of the casting process runs in a closed water circulation. Waste water is produced during the etching process (which is not considered in the dataset). These waters are neutralised and run into the municipal waste water treatment plant after daily analysis.

5.5 Energy Consumption

The production steps pre-alloying and melting of titanium zinc consume about 0.27 kWh and 0.142 kWh of electrical energy per kg. This was calculated via the specific heat capacity and the heat of fusion for zinc, and compared to energy consumption information of an induction furnace for various ferrous and non-ferrous materials (ZAM (2003)). The energy efficiency is assumed to be 50% for the induction crucible furnace and 60% for the induction channel furnace (for pre-alloying, temperature 760°C and for melting, temperature 500-550°C, respectively AUB (2003)).

Tab. 5.1 Calculation of energy demand for pre-alloying and melting

	Heat capacity (c) (Formeln & Tafeln 1977)	Temperature difference (dT)	Enthalpy of fusion (dH) (Formeln & Tafeln 1977)	Theoretical energy demand for melting (Eth)	Efficiency (eta)	Effective energy demand (Eeff)
Unit	[J*kg ⁻¹ *K ⁻¹]	K	J/kg	kWh/kg		kWh/kg
Formula				$(c \cdot dT + dH) / 3.6E6$		$= Eth / eta$
Prealloy	387	740	$1.11 \cdot 10^5$ (Zinc)	0.138	50%	0.27
Titanium-zinc	387	500	$1.11 \cdot 10^5$ (Zinc)	0.084	60%	0.142

Due to a lack of actual data, the titanium zinc alloy casting process is approximated by “casting, brass”, which shows energy consumption levels of 0.0197 kWh/kg for electricity, 0.354 MJ/kg for natural gas heating and 0.283 MJ/kg for oil heating. The final rolling and coiling of the titanium zinc is approximated by “contour, brass”, which uses 1.81 kWh/kg electricity (UCTE) and 1.6 MJ/kg of natural gas heating per kg alloy (Classen et al. (2007)).

5.6 Transports and Infrastructure

Rheinzink GmbH buys the refined zinc from a company in Duisburg, Germany, and from other zinc-electrolysis enterprises. As far as the other elements titanium and aluminium are concerned, they are ordered from various countries all over the world. Copper comes as a secondary raw material in form of fractured scrap (AUB (2002)).

The infrastructure is inventoried with the proxy “casting plant” for aluminium production. This is a rather good approximation of the titanium zinc factory in Germany.

6 Life Cycle Inventory

6.1 Titanium zinc plate, without pre-weathering, at plant

The meta information of “titanium zinc plate, without pre-weathering, at plant” (Tab. 6.1) includes all important and necessary information to use the data correctly.

Fig. 6.1 shows the system of the dataset “titanium zinc plate, without pre-weathering, at plant”. It shows all in- and output flows, the allocation to the datasets in the ecoinvent database database V2.0, the sources (mainly AUB (2002) and estimations) and the standard deviation of all in- and output flows.

The information of this dataset was mainly provided by the company Rheinzink GmbH in Datteln, Germany (Rheinzink (2005) and AUB (2002)). Rheinzink has an annual production volume of titanium zinc and refined zinc of 160'000 t/a. The electricity consumption for both induction furnace models is based on physical properties and information published by the Polish company ZAM Kety. ZAM Kety produces and sells induction furnaces, induction heaters, machines for steel plants and founding, welded constructions, or systems of automatic control (ZAM (2003)).

Tab. 6.1 ecoinvent meta information of the product “titanium zinc plate, without pre-weathering, at plant”

Name	titanium zinc plate, without pre-weathering, at plant
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	The dataset describes the production of titanium zinc alloy from primary zinc and the fabrication of titanium zinc plates with all included fabrication steps. Pre-weathering of the plate is not included.
Amount	
Local Name	Titanzinkblech, nicht vorbewittert, ab Werk
Synonyms	
General Comment to reference function	The production steps pre-alloying, melting, casting, rolling, coiling, stretch forming and cutting are approximated using "casting, brass", "contour brass" and "sheet rolling, aluminium".
Start Date	1997
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from a leading titan zinc plate producer in Germany. Applicable for Europe.
Technology text	Induction furnace for all the pre-alloying and melting steps, roll stands with 5 rolling pairs for the rolling step. Coiling facilities.
Representativeness [%]	
Production Volume	160'000 t/a (Titanium zinc plus refined zinc)
Sampling Procedure	Environmental report, certificate
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that Germany is representative for Central Europe is considered in the uncertainties.

General flow information				Representation in ecoinvent								Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment	
Zinc	titanium zinc plate, without pre-weathering, at plant			metals	extraction	No	RER	zinc, primary, at regional storage	106E+00	kg	Arbeitsgemeinschaft Umwelterträgliches Bauprodukt (AUB 2002)	1	124	(14,13,15,3)	
Copper				metals	extraction	No	RER	copper, at regional storage	127E-03	kg		1	124	(14,13,15,3)	
Titanium			Primary aluminium as proxy for Titanium	metals	extraction	No	RER	aluminium, primary, at plant	101E-03	kg		1	124	(14,13,15,3)	
Aluminium				metals	extraction	No	RER	aluminium, production mix, at plant	159E-04	kg		1	124	(14,13,15,3)	
Electricity pre-alloying			Pre-alloying process	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.86E-01	kWh	ZAM Kety Company Poland (2003) Internet, plus own calculations	1	124	(14,13,15,2)	
Casting			Proxy	metals	processing	No	CH	casting, brass	106E+00	kg	ecoinvent 11	1	145	(4,5,13,3,5,3)	
Casting plant			Pre-alloying process; assumption same infrastructure and same allocation to different products as in "casting, brass"	metals	extraction	Yes	RER	aluminium casting, plant	5.15E-11	unit		1	3.06	(2,4,13,15,9)	
			Pre-alloying and melting process	air	unspecified			Heat, waste	157E+00	MJ		1	124	(2,4,13,15,13)	
			Pre-alloying process	air	unspecified			Zinc	3.18E-04	kg		1	158	(2,4,13,15,31)	
Electricity for melting			Melting process	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	149E-01	kWh	ZAM Kety Company Poland (2003) Internet, plus own calculations	1	124	(14,13,15,2)	
Casting			Proxy	metals	processing	No	CH	casting, brass	105E+00	kg	ecoinvent 11	1	145	(4,5,13,3,5,3)	
Casting plant			Proxy; assumption same infrastructure and same allocation to different products as in "casting, brass"	metals	extraction	Yes	RER	aluminium casting, plant	4.90E-11	unit		1	3.06	(2,4,13,15,9)	
Rolling			Contour brass as proxy for sheet rolling; cuttings 0.0396 kg per 1kg	metals	processing	No	RER	contour, brass	100E+00	kg		1	133	(3,3,11,3,5,3)	
Rolling			weight for brass taken off	metals	extraction	No	CH	brass, at plant	-3.96E-02	kg		1	133	(3,3,11,3,5,3)	
				metals	extraction	No	DE	titanium zinc plate, without pre-weathering, at plant	100E+00	kg					

Fig. 6.1 Flows for "titanium zinc plate, without pre-weathering, at plant" and their representation in the ecoinvent database

7 Data quality

The uncertainties are evaluated with the help of the pedigree matrix (see Frischknecht et al. (2007b)). For detailed results compare the different flows and the ecoinvent database meta information.

Most data are based on the environmental certificate issued to the German company “Rheinzink GmbH” by the Arbeitsgemeinschaft Umweltverträgliches Bauprodukt (AUB (2002)).

8 Literature

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Part XXV:

Tin plating

Data v2.0 (2007)

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Kellenberger D., Althaus H.-J., Jungbluth N., Künniger T., Lehmann M. and Thalmann P. (2007) Life Cycle Inventories of Building Products. Final report ecoinvent Data v2.0 No. 7. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.org.

Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

The tin plating process finds its main application in the manufacturing and packaging sector. Tin plated products are used to package soups, pet foods, juices, or for paint cans, food trays, electrical connectors etc. Due to a lack of data, many of the sub-processes in this dataset are approximated via the more common “zinc coating, pieces” process. Chromium steel (18/8) is one of the major materials on which tin plating is applied. During the different baths and rinsing processes of tin plating, several acids and leachates are added. Waste water and exhaust air are normally treated and filtered within the production plant.

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1 Introduction

Tin coated materials have been a packaging and manufacturing medium for more than a century. Tin plating is a multi-step process including alkaline and acid baths, rinsing processes, the so called “nickel strike” and neutralisation baths in the end.

The processes described in this report derive mainly from the galvanising company Collini Zimmermann located in Dübendorf, Switzerland. The input/output flows are mainly adapted from the ecoinvent dataset for “zinc coating, pieces”.

2 Characterisation of the material

A plating process generally serves to protect basic materials from corrosion and friction. As this coating is very thin, the properties of the basic material remain the same. The process gives a decorative finish depending on what raw material is being used. Plated tin coatings are non-toxic, ductile, resistant to corrosion, easy to coat, and have high throwing and good distribution properties. Tin finishes are also easier to solder components onto (IPPC 2006). Tin plating is preferred for cans to store food and drinks, as the more common zinc (zinc plating) affects the food quality.

Despite the fact that tin is only used as a very thin coating layer, it is a relatively expensive raw material (extracted from tin ore). For that reason tin-plating companies are very careful not to waste any tin during the plating process. Tab. 2.1 shows information on the material properties of tin (Formeln&Tafeln (1977) and Farmingdale (2004)).

Tab. 2.1 Material properties of tin (applied as coating to e.g. chromium steel)

Density (specific weight)	7.3 g/cm ³
Melting point	232 °C
Recrystallization temperature	300 °C (Farmingdale (2004))
Atomic weight	118.7 g/mol

3 Use

Tin plated metal is widely used in many applications because of its unique properties. Tin's main applications are the coating of steel for packaging of food, beverages and aerosols, etc. It is also widely applied in printed circuit boards, electrical connectors, appliance chassis, as well as for kitchen utensils (IPPC (2006)). Furthermore, tin plated metals are used as oil filters, paint cans, aerosol cans, baking pans, food trays, and alkaline battery jackets. Tin coated steel e.g. has been a packaging and manufacturing medium for more than a century (OCC (2005)).

4 System Characterisation

4.1 Raw Materials

The main raw material used for the tin plating process is tin. Still, in order to run the various process steps, many more substances are required: degreasing agents like sodium hydroxide, hydrochloric acid for the pickling, stripping and fluxing processes, as well as a nickel compound to assure the fluxing. Tin is often plated on the carrier material steel, in particular chromium steel. Chromium steel and its LCI are described in Althaus et al (2004).

4.2 Production Process

The tin plating process is very similar to the more common zinc coating process. Still, each individual company optimises the process according to its needs. Therefore, a detailed description of the sub processes is generally kept as a professional secret.

The data used in this study include all processes taking place in the factory of "Collini Zimmermann AG" in Dübendorf, Switzerland. The production process of tin-plated chromium steel consists of 10 different steps (Fig. 4.1). In between each of the steps there is a rinsing procedure with circulating water (Collini (2005)¹):

¹ Personal Communication with Mr. Joachim Buk, Process Engineer at Collini Zimmermann AG, Dübendorf, April, 2005

1. Degreasing in alkaline dipping bath: In sodium hydroxide solution with ultrasound (density of solution 1,1 g/cm³; temperature ca. 70 °C; bath volume 250 l).
To guarantee satisfactory galvanising and to enhance the performance of filtering separators, a degreasing step is used to remove traces of coolants and lubricants from the fabricated steel. This is commonly done using alkaline degreasing baths. The resulting unstable emulsions float on the surface of the bath and can be removed by gravitational separators, skimmers, micro- or ultra filtration etc. (IPPC (2001)).
2. Etching (optional step): Hydrochloric acid and water (volume share 1:1) plus some organic inhibitors at room temperature (bath volume 150 l).
To remove casting skin, roll skin, grid or scale, the items are pickled in diluted hydrochloric acid. To prevent excessive pickling of steel items, especially in pickling high tensile steels, and to protect the steel pickling vats, pickling inhibitors (e.g. hexamethylenetetramine) are added to the bath. During operation the iron content of the pickling bath increases, while the amount of free acid decreases, making it necessary to top up the bath occasionally by adding fresh acid. Pickling in plants with open pre-treatment is usually done at ambient air temperature. Gaseous emissions of the pickling agent can arise from the pickling baths, depending on concentration and temperature of the bath, and from the pickled items (IPPC (2001)).
3. Electrolytic degreasing: Cathode process, which is common for chromium steel (density of solution 1,1 g/cm³; temperature ca. 50 °C; bath volume 1000 l).
Process for removal of grease, oil, etc. from metal surfaces in preparation for electroplating. The metal is made the cathode in an electrolytic cell containing strongly alkaline (sometimes hot) solution that dissolves these coatings (YCES (2005)).
4. Pickling (Stripping): In sulphuric acid (10 Vol.-%), in order to neutralise the oxidation-film on the steel surface created during the previous process step "electrolytic degreasing".
5. Nickel-strike process: Ni-containing hydrochloric acid (density 1,1 g/cm³, up to 50°C, current density 1-5 A/dm², ca. 5 minutes)
Nickel strike electrolytes are based on nickel sulphate and sulphuric acid or on nickel chloride and hydrochloric acid (see above). Nickel strike deposits are applied for activating passive nickel surfaces and stainless steel prior to further metal plating.
6. Nickel bath: Ni-sulphate liquid (density 1,1 g/cm³, at 50 °C, 2 µm coating, 30-45 minutes)
Electroless nickel electrolytes are based on nickel sulphate and nickel chloride (Nickel content 2-10 g/l). A thin undercoating of 2 µm is applied.
7. Tin-plating in acid bath: Methane sulphonate acid bath (MSA system) with an antioxidant and some organic additives (density 1,2 g/cm³, so-called BT250 bath; BT stands for "bright tin"), is cooled down to room temperature (25 °C). Refined tin is added (current density 5 A/dm², 2-5 µm). The stability, low sludge formation, deposit properties and the structure of the deposit ensures the market acceptance of MSA.
8. Neutralisation bath with neutralization solvent, a mild alkaline leachate (Product: "Neutra Rinse": density 1,1 g/cm³, ca. 2 minutes at 40 °C)
9. Separation of unwanted salts from water by an exchanger ("Mischbetttauscher") using deionised warm water (density 1,0 g/cm³, at up to 70 °C)
10. Drying: Oven or centrifuge, ca. 5 minutes at 60 – 70 °C.

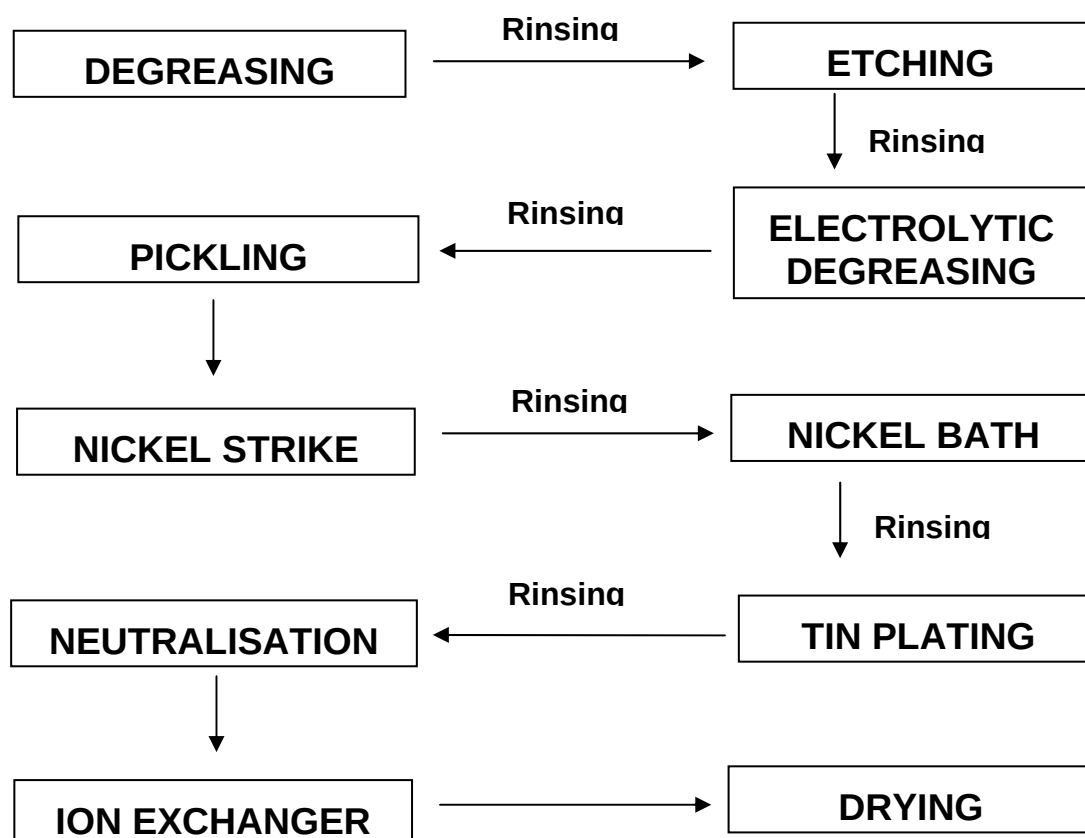


Fig. 4.1 Process description of tin plating of chromium steel (after Collini (2005); personal communication)

Each of the process steps causes its respective air and water emissions. The galvanising industry is using various abatement technologies to address these issues. (IPPC (2006)).

Waste water: The waste water of galvanising production plants consists mainly of metal-ions, toxic anions, chromate and neutral salts. Metal-ions are chemically precipitated. Toxic anions are destroyed by oxidation and reduction processes. Neutral (non-ionic) salts can only be removed by distillation, which is a very energy intensive process. The precipitation normally happens either with sodium hydroxide and lime water, in some cases also with carbonates or sulphides. The resulting precipitate is disposed as sludge.

Waste materials: The sludge accumulating during the plating process has a water content of normally more than 95%. The solid content is increased to 40% by dewatering, and disposed of as galvanic sludge.

Exhaust air: Galvanic companies are obliged to take measures to protect their employees from harmful emissions to air. This is achieved by a highly developed ventilation

technology. Dangerous emission substances are toxic gases like NO_x, HCl, HF, as well as aerosols loaded with SO_x, NaH and chromium compounds. The main sources of these emissions are the pickling baths, the degreasing baths and the electro deposition of metals.

In order to keep within the MAC (maximum allowable concentration), the following parameters have to be taken into account (IPPC (2006)):

- Size of bath
- Continuous or discontinuous operation of the bath
- Bath temperature
- Physical and chemical properties of the applied chemicals
- Categorisation of the hazardous substances
- Application of pickling additives to reduce or prevent emissions of HF, NO_x and Cr(VI)
- Purification of waste gases

4.3 Energy Consumption

The tin plating process consumes energy for heating in the individual process steps (see tables in Chapter 5). Collini Zimmermann AG is using but electricity as energy input.

4.4 Transports

Collini Zimmermann AG buys the chromium steel and tin from regional dealers and suppliers. This requires certain transportation efforts, which are approximated by 32 t lorries and rail. Standard distances are applied according to the ecoinvent methodology.

5 Life Cycle Inventories

ecoinvent meta information (tables), flows and their representation in the ecoinvent database are shown in (figures):

Tin plating, pieces

Tab. 5.1 and Fig. 5.1

Tin plated chromium steel sheet

Tab. 5.2 and Fig. 5.2

5.1 Tin plating, pieces

The meta information of “tin plating, pieces” (Tab. 5.1) includes all important and necessary information to use the data correctly. Fig. 5.1 shows the system of the dataset “tin plating, pieces”. It shows all in- and output flows, the allocation to the modules in ecoinvent database V2.0, the sources and estimations and the standard deviation of all in- and output flows. The dataset “tin plating, pieces” can be applied where mainly steel pieces are coated with tin. The main application of tin is the coating of steel, in order to protect it against corrosion. The dataset “Tin plated chromium steel sheet” (Tab. 5.2 and Fig. 5.2) provides a ready-made dataset for chromium steel sheet coated with tin.

Due to a lack of data on the tin plating process, the ecoinvent data for “zinc coating, pieces” is taken as a basis. The dataset “tin plating, pieces” was adjusted with information received by the Collini-Zimmermann Company, Dübendorf, Switzerland.

Tab. 5.1 ecoinvent database meta information of the product “tin plating, pieces”

Name	tin plating, pieces
Location	RER
Infrastructure Process	0
Unit	m2
Data Set Version	2.0
Included Processes	The dataset describes the tin plating process of chromium steel. Included processes are degreasing, electrolytic degreasing, pickling, stripping, fluxing and rinsing. The treatment of the resulting waste water is also inventoried, as well as the transportation of materials by a 32t lorry. The infrastructure is approximated with a steel rolling mill.
Amount	1
Local Name	Stückverzinnung
Synonyms	
General Comment to reference function	The dataset can be applied for the tin plating process of steel, preferably of chromium steel (18/8).
Start Date	2001
End Date	2005
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from an established galvanising company in Central Europe. Data applicable for Europe.
Technology text	Tin plating of chromium steel sheet, including processes of tin plating and sheet rolling.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature, company information
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that one galvanising company in Switzerland is representative for Central Europe is considered in the uncertainties.

General flow information				Representation in ecoinvent								Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment	
Degreasing agent	tin plating, pieces, part degreasing		solution of 10 %needed, --> 1/5 of 50% solution is inventoried	chemicals	inorganics	No	RER	sodium hydroxide, 50% in H ₂ O, production mix, at plant	6.33E-03	kg	IPPC 2001	1	2.00	calculated from max value assuming lognormal distribution	
Water				resource	in water			Water, unspecified natural origin	1.67E-04	m ³		1	2.00	calculated from max value assuming lognormal distribution	
Heat from oil			Assumption: each energy carrier is used from the same number of plants	oil	heating systems	No	RER	heat, light fuel oil, at industrial furnace 1M W	3.35E-01	M J		1	2.00	calculated from max value assuming lognormal distribution	
Heat from natural gas				natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace >100kW	3.35E-01	M J		1	2.00	calculated from max value assuming lognormal distribution	
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	9.29E-02	kWh		1	2.00	calculated from max value assuming lognormal distribution	
heat				chemicals	organics	No	RER	heat, unspecified, in chemical plant	3.35E-01	M J		1	2.00	calculated from max value assuming lognormal distribution	
		Liquor and sludge	Including discarded degreasing bath. Recycled -> not inventoried						4.50E-02	kg		1	2.00	calculated from max value assuming lognormal distribution	
		Oily sludge	disposed	waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	2.67E-03	kg		1	156	(15,1,1,3,4,1)	
Hydrochloric acid (30%)	tin plating, pieces, part pickling			chemicals	inorganics	No	RER	hydrochloric acid, 30% in H ₂ O, at plant	3.20E-01	kg	IPPC 2001	1	2.09	calculated from max value assuming lognormal distribution	
Inhibitor				hexamethylenetetramine	organics	No	GLO	chemicals organic, at plant	1.67E-03	kg		1	2.00	calculated from max value assuming lognormal distribution	
Water				resource	in water			Water, unspecified natural origin	2.92E-04	m ³		1	2.00	calculated from max value assuming lognormal distribution	
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.04E-01	kWh		1	4.00	calculated from max value assuming lognormal distribution	
		Hydrogen chloride	Assumption: same amount of waste gas per kg of product as for HCl-pickling in cold sheet rolling of steel	air	unspecified			Hydrogen chloride	1.18E-06	kg		1	7.07	calculated from max value assuming lognormal distribution	
		Hydrogen		air	unspecified			Hydrogen	5.00E-05	kg		1	2.00	calculated from max value assuming lognormal distribution	
		Waste acid and sludge	assumed as recycled --> not inventoried						3.33E-01	l		1	2.00	calculated from max value assuming lognormal distribution	
		Discarded pickle liquor	No data on composition and amount of discarded pickling liquor.						n.a.						

Fig. 5.1 Flows for "tin plating, pieces" and their representation in the ecoinvent database

General flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
Hydrochloric acid (30%)	tin plating, pieces, stripping	Spent strip liquor		chemicals	inorganics	No	RER	hydrochloric acid, 30% in H ₂ O, at plant	5.00E-02	kg	IPPC 2001	1	2.00	calculated from max value assuming lognormal distribution
Water				resource	in water			Water, unspecified natural origin	5.83E-05	m ³		1	2.00	calculated from max value assuming lognormal distribution
			assumed as recycled --> not inventoried						2.24E-02	kg		1	1.12	calculated from max value assuming lognormal distribution
Nickel compound	tin plating, pieces, part fluxing			metals	extraction	No	GLO	nickel, 99.5% at plant	1.78E-05	kg	Collini (2005)	1	1.24	(2,4,1,3,15,3)
Hydrochloric acid (30%)				chemicals	inorganics	No	RER	hydrochloric acid, 30% in H ₂ O, at plant	5.00E-02	kg	IPPC 2001	1	1.05	(3,4,2,3,15,4)
Water				resource	in water			Water, unspecified natural origin	1.67E-04	m ³		1	2.00	calculated from max value assuming lognormal distribution
Energy			Energy for heating up the flux bath is assumed to be supplied by internally recovered heat						n.a.					
	tin plating, pieces, part overall process 1	Sludge							n.a.	kg	IPPC 2001	1	2.45	calculated from max value assuming lognormal distribution
			Inventoried for the overall process	resource	in water			Water, unspecified natural origin	1.67E-04	m ³		1	2.00	calculated from max value assuming lognormal distribution
Water			Organic chemicals as proxy	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	2.50E-02	kg	IPPC 2001	1	1.26	(3,4,2,3,15,4)
Methan sulphate			Calculated by density 7.3 g/cm ³ and thickness 2µm; assumed yield 80% IWZ (2005) mentions 1-5,6 g tin per m ² for tin plated steel	metals	extraction	No	RER	tin, at regional storage	3.65E-02	kg	Collini (2005)	1	1.24	(2,4,1,3,15,3)
Tin			Assumption: 49% of the plants are using natural gas, 49% light fuel oil and 2% electricity as energy carrier for heating up the zinc bath	oil	heating systems	No	RER	heat, light fuel oil, at industrial furnace 1M W	1.25E+01	MJ	IPPC 2001	1	2.36	calculated from max value assuming lognormal distribution
Steel coils				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	1.25E+01	MJ		1	2.36	calculated from max value assuming lognormal distribution
Natural gas				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.41E-01	kWh		1	2.36	calculated from max value assuming lognormal distribution
Electricity		Extracted gas from kettle	unknown composition. Air emissions inventoried separately						7.07E+01	m ³		1	2.83	calculated from max value assuming lognormal distribution
		Methan sulphate	0.1% of input assumed	air	unspecified			NM VOC, non-methane volatile organic compounds, unspecified origin	2.50E-05	kg		1	2.08	(3,4,2,3,15,23)
		Dross	Returned to the secondary tin industry for recovery of the tin content --> not inventoried						2.04E-01	kg		1	2.45	calculated from max value assuming lognormal distribution

Fig. 5.1 Flows for "tin plating, pieces" and their representation in the ecoinvent database (cont.)

General flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Cate gory	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
Flocculating agents tin plating, pieces, part waste water treatment	→		Reference contained no data. Assumption: Same amount per kg product as for coil coating. Aluminium sulphate as proxy.	chemicals	inorganics	No	RER	aluminium sulphate, powder, at plant	1.51E-05	kg	IPPC 2001	1	1.22	(2,5,1,1,1,3,4)	
		Waste water	One part of the incoming cooling and process water leaves the system as treated effluent; it is assumed to be discharged in surface water --> not inventoried. Infrastructure for waste water treatment plant is comprised in the infrastructure module.						1.67E-03	m3		1	1.22	(2,5,1,1,1,3,4)	
		→ TSS	Assumption: same amount per m3 treated water, as for zinc coating process	water	unspecified			Suspended solids, unspecified	3.74E-06	kg		1	1.57	(2,5,1,1,1,3,32)	
		→ Iron		water	unspecified			Iron, ion	4.10E-07	kg		1	5.06	(2,5,1,1,1,3,35)	
		→ Nickel		water	unspecified			Nickel, ion	1.67E-08	kg		1	5.06	(2,5,1,1,1,3,35)	
		→ Zinc		water	unspecified			Zinc, ion	2.63E-07	kg		1	5.06	(2,5,1,1,1,3,35)	
		→ Lead		water	unspecified			Lead	2.51E-08	kg		1	5.06	(2,5,1,1,1,3,35)	
		→ Chromium		water	unspecified			Chromium, ion	1.10E-07	kg		1	5.06	(2,5,1,1,1,3,35)	
		→ Cr6+		water	unspecified			Chromium VI	1.67E-08	kg		1	5.06	(2,5,1,1,1,3,35)	
		→ COD		water	unspecified			COD, Chemical Oxygen Demand	2.20E-04	kg		1	1.57	(2,5,1,1,1,3,32)	
		→ BOD		water	unspecified			BOD5, Biological Oxygen Demand	2.20E-04	kg		1	2.50	(5,5,5,5,5,32)	
		→ DOC		water	unspecified			DOC, Dissolved Organic Carbon	4.95E-06	kg		1	2.45	(5,5,5,5,5,32)	
		→ TOC		water	unspecified			TOC, Total Organic Carbon	4.95E-06	kg		1	1.57	(2,5,1,1,1,3,32)	
		→ Hydrocarbons		water	unspecified			Hydrocarbons, unspecified	1.98E-06	kg		1	3.05	(2,5,1,1,1,3,34)	
		→ Oil and fat		water	unspecified			Oils, unspecified	1.73E-06	kg		1	1.57	(2,5,1,1,1,3,32)	
		→ Phosphorus		water	unspecified			Phosphorus	1.50E-06	kg		1	1.57	(2,5,1,1,1,3,33)	
		→ Water treatment sludge (dry material)	Waste water treatment sludge from steel forming processes is inventoried as proxy. Assumption: same amount per m3 treated water, as for coil coating process.	waste management	residual material landfill	No	CH	disposal, sludge from steel rolling, 20% water, to residual material landfill	1.51E-03	kg		1	1.22	(2,5,1,1,1,3,6)	
		→ Tin	Same amount as chromium according to IPPC 2006	water	unspecified			Tin, ion	1.10E-07	kg	IPPC 2001 / IPPC 2006	1	5.06	(2,5,1,1,1,3,35)	

Fig. 5.1 Flows for “tin plating, pieces” and their representation in the ecoinvent database (cont.)

General flow information				Representation in ecoinvent								Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment	
Lorry 32t	tin plating, pieces, part overall process 2	Water	standard distance ecoinvent: 100 / 200 (lorry / train) km for chemicals and metals	transport systems	road	No	RER	transport, lorry >16t, fleet average	148E-01	tkm	Calculation	1	2.09	(4,5,n.a.,n.a.,n.a.,5)	
Train transport				transport systems	train	No	RER	transport, freight, rail	2.95E-01	tkm		1	2.09	(4,5,n.a.,n.a.,n.a.,5)	
Soda			Data taken from zinc coating	chemicals	inorganics	No	RER	soda, powder, at plant	1.67E-03	kg	Frischknecht et al. 1996	1	2.05	(15,2,113,5)	
Ammonia				chemicals	inorganics	No	RER	ammonia, liquid, at regional storehouse	5.10E-04	kg		1	122	(15,2,113,4)	
Water (softened)				water supply	production	No	CH	water, deionised, at plant	3.33E+00	kg		1	123	(2,5,2,113,12)	
	tin plating, pieces, part overall process 3	Heat waste Dust >10 Dust 2.5-10 HCl NH3 As Cd Pb Zn Fe Chloride Ion Zinc Ion Iron Tin plating of pieces	Process and cooling water evaporated (uncontaminated -> not inventoried) or already included in the waste disposal. Assumption: ratio between treated and untreated water is the same as for coil coating process.						2.51E-03	m3	Frischknecht et al. 1996	1	123	(2,5,2,113,12)	
Infrastructure			10% of rolling mill for steel forming per kg output as rough approximation	metals	processing	Yes	RER	rolling mill	2.71E-12	unit	estimation	1	3.15	(4,5,13,3,4,9)	
			from electricity consumed	air	unspecified			Heat, waste	1.22E+00	MJ	Calculation	1	122	(15,1113,13)	
			Assumption: Emissions as for zinc coating	air	unspecified			Particulates, > 10 µm	2.58E-05	kg	Frischknecht et al. 1996	1	157	(15,2,113,25)	
				air	unspecified			Particulates, > 2.5 µm, and < 10 µm	7.73E-05	kg		1	2.05	(15,2,113,26)	
				air	unspecified			Hydrogen chloride	2.04E-04	kg		1	2.05	(15,2,113,23)	
				air	unspecified			Ammonia	7.27E-04	kg		1	2.05	(15,2,113,23)	
				air	unspecified			Arsenic	3.16E-09	kg		1	157	(15,2,113,31)	
				air	unspecified			Cadmium	2.18E-08	kg		1	157	(15,2,113,31)	
				air	unspecified			Lead	5.67E-08	kg		1	157	(15,2,113,31)	
				air	unspecified			Zinc	1.12E-05	kg		1	157	(15,2,113,31)	
				air	unspecified			Iron	2.33E-05	kg		1	157	(15,2,113,31)	
				water	unspecified			Chloride	3.13E-05	kg		1	157	(15,2,113,31)	
				water	unspecified			Zinc, ion	3.13E-05	kg		1	157	(15,2,113,31)	
				water	unspecified			Iron, ion	9.38E-05	kg		1	157	(15,2,113,31)	
			applies to 1m2 of coating layer of 2 µm thickness	metals	processing	No	RER	tin plating, pieces	1.00E+00	m2					

Fig. 5.1 Flows for “tin plating, pieces” and their representation in the ecoinvent database (cont)

5.2 Tin plated chromium steel sheet

The meta information of “tin plated chromium steel sheet” (Tab. 5.2) includes all important and necessary information to use the data correctly. Fig. 5.2 shows the system of the dataset “tin plated chromium steel sheet”. It shows all in- and output flows, the allocation to the datasets in the ecoinvent database database V2.0, the sources and estimations and the standard deviation of all in- and output flows. The dataset “tin plated chromium steel sheet” can be applied for chromium steel that is coated with a 2µm layer of tin.

Tab. 5.2 ecoinvent database meta information for the product “tin plated chromium steel sheet”

Name	tin plated chromium steel sheet
Location	RER
Infrastructure Process	0
Unit	m2
Data Set Version	2.0
Included Processes	The dataset describes the production of tin plated chromium steel sheet. Included sub processes are "sheet rolling" and "tin plating". The basic material coated is chromium steel 18/8.
Amount	1
Local Name	Verzinnertes Chromstahlblech, 2 mm, ab Werk
Synonyms	
General Comment to reference function	The dataset can be applied for tin plated chromium steel sheets, preferably of steel 18/8 and sheet thickness of around 2 mm.
Start Date	2001
End Date	2005
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from an established galvanising company in Central Europe. Data applicable for Europe.
Technology text	Tin plating of chromium steel sheet, including processes of tin plating and sheet rolling.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature mainly
Extrapolations	see Geography
Uncertainty Adjustments	None

General flow information				Representation in ecoinvent								Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment	
Chromium steel	Tin plated chromium steel sheet	Tin plated chromium steel sheet		metals	extraction	No	RER	chromium steel 18/8, at plant	156E+01	kg	Calculation	1	100	Direct modelling	
Sheet rolling Cr steel				metals	processing	No	RER	sheet rolling, chromium steel	156E+01	kg		1	100	Direct modelling	
Tin plating				metals	processing	No	RER	tin plating, pieces	100E+00	m2		1	100	Direct modelling	
			Sheet thickness assumed to be 2 mm	metals	extraction	No	RER	tin plated chromium steel sheet, 2 mm, at plant	100E+00	m2					

Fig. 5.2 Flows for “tin plated chromium steel sheet” and their representation in the ecoinvent database

6 Data quality

The uncertainties are evaluated with the help of the pedigree matrix (see Frischknecht et al. (2007b)). For detailed results compare the different flows and the ecoinvent meta information.

The data is based on the ecoinvent dataset “zinc coating, pieces” and on limited information on the tin coating process in one specific company. The dataset is of poor to medium quality. A reliable data source (IPPC (2001)) was combined with a number of assumptions.

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Part XXVI:

Glazing, Window frames, Claddings and Doors

Data v2.0 (2007)

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Kellenberger D., Althaus H.-J., Jungbluth N., Künniger T., Lehmann M. and Thalmann P. (2007) Life Cycle Inventories of Building Products. Final report ecoinvent Data v2.0 No. 7. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.org.

Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Summary

Glazing, window frames, claddings and doors are dedicated parts of building construction. This report discusses the datasets of three different types of glazing (two double, one triple), four window frames (wood, wood-metal, plastic and aluminium), one cladding (aluminium) and four doors (two outer, two inner). The respective disposal datasets are also established.

The laminated safety glass type (double glazing) requires an additional layer of a PVB plastic film. The triple glazing has inserted Krypton gas to decrease heat exchange. The laminated safety glass (2-IV) requires 3.5 times the energy input during the production phase compared to a normal double glazing. The material input of the three glazing types varies a lot.

The window frames show rather big differences regarding material input and energy input. The wood-metal window frame needs more than 3.5 times the amount of aluminium as the wooden frame, whereas the metal window frame requires almost 10 times the amount of aluminium. The wooden window frame requires almost 8 times the electricity consumption for the production process compared to the plastic frame. The metal window frame only comes up for about one fourth of the electricity consumption of the plastic frame.

The materials for the aluminium cladding production range from metals, plastics, chemicals, glass and insulation materials to wood and paintings. The dataset is derived from an existing cladding of a school-building in Switzerland.

The outer and inner door datasets in this report are based on earlier EMPA reports and assumptions. Wooden doors are the most common type of doors applied in the indoor construction building.

The report describes a number of disposal datasets for different glazing, window frames and doors. Metals like aluminium or chromium steel are assumed to be recycled and thus leave the system without burdens or credits (cut-off). Though, smaller metal pieces that are difficult to separate from other materials (e.g. glued together) are incinerated. Non-metal materials of these building products are all incinerated.

Three auxiliary datasets are established as input to the glazing or window frame production. These are the polysulphide sealing compound for the glazing, the rubber-based sealing compound and the metal adhesive compound for the aluminium window frame.

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1 Introduction

The report deals with the production and the disposal of four basic construction components, namely glazing, window frame, cladding and door.

Chapter 2 describes the raw materials that form the various types of glazing, window frames, claddings and doors. The use and basic characteristics of these products is elaborated in Chapter 3. The system characterisation in Chapter 4 includes a detailed description of the production processes, including raw materials entering the system, energy consumption, required transport activities, produced air emissions and waste water, plus the infrastructure involved. For all the construction components respective disposal datasets are generated. Metals are assumed to be recycled (mainly) and thus are modelled to leave the system without burdens or credits, whereas wood or plastic parts end up in an incineration plant.

Chapter 5 focuses on the life cycle inventories, displaying all the flow charts and meta information tables of these datasets. Chapter 6 with a few remarks on the data quality concludes this report.

2 Raw Materials

2.1 Glazing

The raw materials for the production of the double or triple glazing are mainly glass, some aluminium, plastics and argon or krypton as filling.

2.2 Window frames

Four types of window frames are discussed: wood, wood-metal (aluminium), plastic (PVC) and aluminium.

The main raw material inputs for the wooden and wood-metal window frame are wood, steel, aluminium and coatings. The main raw material inputs for the plastic window frame are PVC, steel and insulating materials. The aluminium window frame consists mainly of aluminium, steel and some plastics.

2.3 Claddings

The raw material inputs for the aluminium cladding façade are mainly steel, aluminium, different kind of plastics (EPDM, polyurethane, XPS and PVB), glass, plywood, insulation materials and krypton gas.

2.4 Doors

Four types of doors are differentiated, two outer doors and two inner doors. The raw material inputs for the doors (including door frames) are as follows:

- The outer door, wood-aluminium, consists of the materials aluminium, softwood, steel, some plastic and rubber.
- The outer door, wood-glass, is made out of aluminium, softwood, glass (laminated safety glass), steel, some plastic and rubber.
- The inner door, wood, consists of aluminium, different wood types, and acrylic varnish, steel, some plastic and rubber.
- The inner door, glass-wood, is produced out of aluminium, different wood types, glass, an acrylic varnish, steel, some plastic and rubber.

3 Use

Glazing, window frames, claddings and doors are all used in construction. The chapters 3.1 to 3.4 describe the application of these components and their situation on the market.

3.1 Glazing

The volume of the global market for flat glass in calendar year 2004 was approximately 36 million tonnes. Of this tonnage, around 70 % was consumed in windows for buildings, 10 % in glazing products for automotive applications and 20 % in furniture and other interior applications (Pilkington (2005)).

The glazing has a number of functions. It divides between indoor and outdoor temperature, lets in light and radiative energy and reduces heat loss to the outside. The higher the transmittance value and the lower the U-value of a glazing are, the better it fulfils these functions. In 1995, in the first 15 European Union member states, almost 60% of the dwellings still had single glazing.

3.2 Window frames

The window frame market in Germany showed the following shares in the year 2003: plastic 54.1%, wood 20.6%, aluminium 19.8% and aluminium-wood 5.5%. These shares are

similar in Switzerland. More than 14 million window units were sold in Germany in the year 2003 (Fensterplatz 2005). Plastic window frames hold a higher market share than wooden frames since the beginning of the 1990s. Wooden window frames have generally lost market share since 1993.

Depending on the function of the window, the frames are chosen accordingly. Wooden frames have a more natural look and create a comfortable atmosphere. All three window frame types are very durable, given that wooden frames are regularly treated to prevent weather impacts. Plastic window frames have existed on the market only for 35 years. Wooden frames show excellent heat insulation, are produced from renewable material and can be disposed of in an environmentally friendly way. Plastic window frames also have good heat insulation values, and can be recycled. Aluminium frames have almost equally good heat insulation, if the profiles are thermally separated. Aluminium is recycled, as secondary aluminium is in high demand (Fensterplatz (2005)).

Comparing the prices of the different window frames, the metal (aluminium) frame is by far the most expensive, followed by the wood frame and the plastic frame (Fensterplatz (2005)).

3.3 Claddings

Aluminium claddings grew more and more popular during the last decade. Aluminium claddings can be installed rather easily due to the low weight of the material.

Aluminium claddings imply high investment costs. However, as these types of claddings do not need any repainting or other maintenance during their use phase (apart from washing), the costs over the whole life cycle become reasonable. (Klempnerhandwerk (2005)).

3.4 Doors

Wooden room doors are the most common type of doors applied in interior construction. Still, in some houses, a glass-wood door is preferred. One reason to install this type of door is the light factor. Another reason could be, if two rooms are meant to be separated from each other, but with a see-through door.

The world-wide demand for doors was 45 billion US\$ in 1997 and 54.4 billion US\$ in 2002. This growth is expected to continue to a projected 71.9 billion US\$ in the year 2007. The demand for windows and doors will accelerate especially in Western Europe and Japan. However, market gains will be less robust than in developing areas. Global demand for vinyl, fibreglass and other plastic door products will outpace the increases for other

product types. The residential building market dominates world door demand, accounting for two thirds of all sales in 2002 (Global Information (2003)).

4 System Characterisation

4.1 Glazing

Compared to single glazing, double glazing reduces heat loss by 50% due to the insulating air space between the glass layers (Efficient Windows (2005)). Insulation glazing (IV) reduces the heat losses even more due to a low emissive coating and the low conductivity of the gas filling between the panes. Three types of glazing are characterised in this report; two insulation glazings with a U-value of $1.1 \text{ W/m}^2\text{K}$, one of them being a laminated safety glass (LSG). The third type is a triple glazing with a U-value of $0.5 \text{ W/m}^2\text{K}$. Tab. 4.1 shows some characteristics of the three glazing types. Depending on the glazing type (2-IV or 3-IV), different weights occur per functional unit (m^2 of visible glass). Using these datasets it has to be considered that not only the U-values but also light and total energy transmissions are different. If different glazing is to be compared in an LCA study of windows or total buildings, the characteristics of the actual products shall be considered to calculate heat gains and losses in the actual situation.

The visible part (daylight opening) of an installed window glazing is smaller than the total glazing fitted into a window frame. The invisible or hidden part of the glazing surface makes up about 6% (depending on the dimension of the window) of the total area (Richter et al. 1996). This fact has been taken into account for all the input materials of a glazing process.

Tab. 4.1 Characteristics of the glazing

Type of glazing	Total glass thickness	Measures (h/w)	Space between glazing (sbg)	Weight per m^2 visible glass)	Filling type
Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$	4+4 mm	1.18 m x 0.72 m	16 mm	20.1 kg	Argon (gas)
Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$, LSG	4+4 mm	1.18 m x 0.72 m	16 mm	21.1kg	Argon (gas)
Glazing, triple (3-IV), $U < 0.5 \text{ W/m}^2\text{K}$	4+4+4 mm	1.18 m x 0.72 m	12+12 mm	30.1 kg	Krypton (gas)

The amounts of input material to the three glazing datasets are mainly calculated based on the geometry of the glazing. Electricity and water demand as well as waste water generation are taken from Richter et al. 1996. The following formulae are used:

$$\text{Amount glass} = [h * w * t * \rho_{\text{glass}} / (1 - \text{wastages})] * 1.06$$

--> with height (h) = 1180 mm, width (w) = 720 mm, thickness (t) = 8 mm (all panes), density of glass (ρ_{glass}) = 2.5E-06 kg/mm³ and the scaling factor 1.06 from visible area to total area

$$\text{Amount alu} = [2 * (h - t_{\text{sealing}}) + 2 * (w - t_{\text{sealing}}) * (2 * \text{gap} + 2 * h_{\text{spacer}}) * t_{\text{spacer}} * \rho_{\text{alu}}] * 1.06$$

--> with h = 1180 mm, w = 720 mm, t_{sealing} = 4 mm, gap = 16 mm, h_{spacer} = 12 mm, t_{spacer} = 0.5 mm, density of alu (ρ_{alu}) = 2.7E-06 kg/mm³ and the scaling factor 1.06 from visible area to total area

$$\text{Amount sealing} = (2 * h + 2 * w) * \text{gap} * t_{\text{sealing}} * \rho_{\text{polysulfide}} * 1.06$$

--> with h = 1180 mm, w = 720 mm, t_{sealing} = 4 mm, gap = 16 mm, $\rho_{\text{polysulfide}}$ = 1.7E-06 kg/mm³ and the scaling factor 1.06 from visible area to total area

Fig. 4.1 shows a typical configuration of a double glazing, taken from Richter et al. 1996.

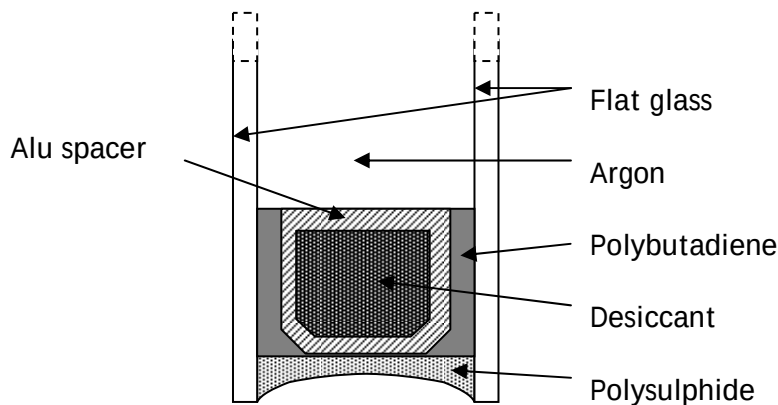


Fig. 4.1 Configuration of a typical double-glazing described in ecoinvent

The three datasets represent the supply of the raw materials (including the transport activities) and the production of the glazing. Process steps involved are section bar rolling of the aluminium parts plus the assembly of the different parts like flat glass, sealing compounds etc.

Laminated safety glass (LSG) has a transparent plastic film (PVB) between two or more pieces of glass. This film is laminated to one of the panes of glass (lamination in an autoclave). This process is done under the action of heat and pressure. It is quite difficult

to penetrate laminated safety glass, compared to normal window plane glass. Besides that, LSG reduces transmission of high frequency sound and also blocks ultraviolet radiation. LSG is also used in thermometers, greenhouse windows, shower enclosures, office partitions, etc. (ULT (2005)).

4.1.1 Energy consumption

Richter et al. 1996 reports 10.9 MJ (3.03 kWh) electricity consumption for the production of 1 m² of 2-IV glazing. This value refers to the actual area of the glazing. Since in this inventory we are referring to the visible area, this value needs to be multiplied by 1.06, resulting in the demand of 3.21 kWh electrical energy per functional unit. The lamination process in the autoclave additionally requires 9.67 kWh electricity for the LSG glass production¹. The electricity consumption is assumed to be linearly related to the number of glass panes. Thus, 4.81 kWh/m² electricity is inventoried for the 3-IV glazing production.

4.1.2 Transports

Road transports from Richter et al. 1996 (650 km for aluminium, 700 km for sealings and 90 km for zeolite) are used to inventory transports of materials to the production facility of the glazing.

4.1.3 Air emissions

No direct air emissions are inventoried for the glazing production.

4.1.4 Waste water

Per 1 m² of glazing about 0.00094 m³ of water is used for cleaning purposes (Richter et al. 1996). The same amount of waste water is generated. For the 2-IV glazing, this amount is scaled by 1.06 to account for the difference in the functional unit in this study. For the 3-IV glazing, the 2-IV glazing value is scaled by 3/2 since water use is linearly related to the number of glass panes.

4.1.5 Infrastructure

No separate infrastructure for the three glazing datasets is accounted for since the production of the glazing is done in the flat glass factory which is inventoried in the datasets for flat glass.

¹ Personal Communication with Temperit AG, June 2005

4.2 Window Frames

Data for the various window frame types represent average values from different unpublished studies carried out by Empa for different customers. The four types of window frames (wood, wood-metal, plastic, aluminium) show all similar U-values (air-to-air heat transmission value) between 1.5 and 1.6 W/m²K. The window frame inventories are calculated per 1 m² of visible window frame. Data are derived for a window of 1.6x1.3m² (including wings) with visible frame surfaces of 0.45 m² in case of the plastic, wooden and wood-metal window frames, and 0.48 m² for the aluminium window. 1 m² of visible window frame weighs 80.2 kg (wood), 83.4 kg (wood-metal), 94.5 kg (plastic) and 50.7 kg (aluminium). The main components of each window frame type consist of the following materials (Tab. 4.2):

Tab. 4.2 Main material components of the four window frames (in kg per m² of visible frame surface; cuttings not included):

Wooden frame	Wood-metal frame	Plastic frame	Aluminium frame
65.2 kg Wood	56.9 kg Wood	55.5 kg PVC	39.59 kg Aluminium (frame)
5.23 kg Paint	12.22 kg Aluminium	36.1 kg Steel	5.17 kg Polyamide
4.93 kg Steel, low-alloyed	4.88 kg Steel	1.04 kg Aluminium	4.87 kg EPDM (sealing)
2.86 kg Aluminium	4.8 kg Paint	0.798 kg Rubber	0.95 kg Steel (low- and high-alloyed)
1.12 kg Rubber	2.88 kg EPDM rubber	0.392 kg Expanded polystyrene	0.398 kg ABS
0.349 kg Nylon	1.09 kg Nylon	0.368 kg PE/PP	0.288 kg Metal adhesive
0.29 kg Zinc	0.29 kg Zinc	0.325 kg Zinc	0.0146 kg Nylon
0.256 kg Polyethylene	0.256 kg Polyethylene	0.00698 kg Copper	
0.006 kg Copper	0.006 kg Copper		

Further frame characteristics are demonstrated on the plastic window frame (Tab. 4.3).

Tab. 4.3 Characteristics of the reported plastic window frame

Characteristics Plastic window frame	Value
Circumference of fenestra	5.82 m
Visible frame surface	0.45 m ²
Fenestra (h / w)	1.6 m / 1.31 m

The production of a wooden or wood-metal window frame include the steps wood cutting, profiling, varnishing, plugging and stopping, as well as joining and fitting. These process steps require electricity and a number of chemicals (see Fig. 5.7 and Fig. 5.8).

The production of the plastic window frame requires a different set of process steps, reaching from moulding to plastic extrusion, to cutting and welding, and to section bar extrusion of aluminium. The production of an aluminium window frame finally includes the steps section bar extrusion of aluminium, section bar rolling of steel, powder coating of the aluminium sheet as well as some material sealing.

4.2.1 Energy consumption

The electrical input for the production of the *wooden* window frame is 0.711 kWh for the cutting, 12.8 kWh for the profiling, 7.53 kWh for the varnishing process, 0.903+0.085 kWh for the joining steps, and finally 35.7 kWh for the fitting (data retrieved from horizontal aggregation of various confidential studies). Apart from that, there is a 54.0 MJ energy input in form of heat (heat from waste wood incineration).

The electrical input for the production of the *wood-metal* window frame is assumed to be similar to the wooden frame.

The *plastic* window frame on the other hand has electrical inputs for the PVC-extrusion process, the welding and cutting. The energy input for the extrusion is part of the applied dataset “extrusion, plastic pipes”. For the welding and cutting steps, 13.8 kWh per m² window frame have been allocated.

An average of 1.27 kWh of medium voltage UCTE is consumed for the production of 1m² *aluminium* window frame. The rest of the processing energy input is already included in the process datasets “section bar extrusion of aluminium” for frame and fittings, “section bar rolling of steel” for fittings; and “extrusion of PEHD” (production step).

4.2.2 Transports

For the wooden window frame the transport activities include 39.25 tkm by 20-28 t lorry for 1 m² of frame. The wood-metal window frame consumes 37.05 tkm of transport (by 20-28 t lorry), where as the plastic window frame requires a total of 30.27 tkm. This is due to the assumption that PVC and EPDM materials have to be transported over 500 km from their storage to the window production site. The aluminium window frame (1 m² visible area) again asks for 4.21 tkm for the frame materials, 0.304 tkm of transport for the fittings, and 0.0556 tkm for the transport during the production phase (total of 4.57 tkm).

4.2.3 Air emissions

No air emissions are considered for production phase of the window frames. Emissions to air (VOC, hydrogen chloride etc.) that are caused during coating processes (zinc coating, powder coating) are already accounted for in the respective ecoinvent datasets that enter these window frame datasets.

4.2.4 Waste water

The manufacturing of window frames does not result in any waste water; therefore no waste water is modelled for the window frame datasets. Waste water production from materials going into these window frame parts are calculated in their individual ecoinvent datasets.

4.2.5 Infrastructure

The infrastructure for the window frame datasets is accounted for and is approximated by the ecoinvent dataset “metal working factory”. The life-span of the window frame factory is assumed to be 50 years.

4.3 Claddings

An aluminium cladded façade with integrated glazing is described in this report. The dataset is calculated for 1m² of cladding (86.7 kg/m²). The data is retrieved from a datasheet provided by the company “Metall-Bau-Technik” in Switzerland. The data characterises the cladding of an existing school-building in 2005.



Fig. 4.2 New aluminium cladding at school-house in Zurich (inventoried for ecoinvent)²

The aluminium cladding is constructed from the following parts:

- | | |
|----------------------------------|---|
| - Steel sections (primed/coated) | - Element window (aluminium) |
| - Steel sheet (zinc-plated) | - Insulation materials (mineral/glass wool) |
| - Screw holes | - Glazing (LSG) |
| - Glazing system | - Coatings/paints |
| - Connectors (Anschlüsse) | |

The cladding part facing the inside of the building is made out of steel for static reasons. The part facing the outside is manufactured from aluminium, and so are the elements. The material cuttings for steel, aluminium, plastics, insulation materials and wood are considered in this dataset (each 2%). Steel and aluminium cuttings are recycled (cut-off), plastics and wood are being disposed in a municipal incineration plant, mineral insulation is landfilled in an inert material landfill. Apart from the glass (20.71 kg of 3-IV laminated safety glass per m² of cladding), this type of cladding has high steel input (ca. 26.4 kg/m²), followed by aluminium (3.82 kg/m²).

The dataset includes the materials for profiles, coatings (powder, zinc, paint) on metals, the glazing system, fittings, windows, insulation materials, plus the energy consumption of the production processes. The transportation of the cladding elements from the production to the construction site is not accounted for; though train and road transport of materials between regional storage and production site is inventoried.

4.3.1 Energy consumption

Due to a lack of data the energy input for the aluminium crossbar-pole cladding production was assumed on the basis of the aluminium window frame. 1.27 kWh for 1m² of aluminium cladding has been estimated. The additional electricity consumption for the lamination of the 3-IV glass is referred to with 6.66 kWh.

4.3.2 Transports

The transport activities for the aluminium pole-crossbar cladding are calculated with the following assumed distances:

- in case of lorry transport (>16 t, average fleet): 100 km for metals and plastics, 50 km for wood and mineral insulation, 100 km for glazing;
- in case of rail transport: 200 km for metals and plastics, 0 km for wood, 100 km for mineral insulation and 600 km for glazing.

² Picture taken from http://www.stadt-zuerich.ch/intermet/hardau/home/projekte/pro_3.html
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No transportation from the production to the construction site has been considered.

4.3.3 Air emissions

No direct air emissions are considered for the dataset “Cladding, crossbar-pole, aluminium, at plant”.

4.3.4 Waste water

No direct waste water is modelled for the dataset “Cladding, crossbar-pole, aluminium, at plant”.

4.3.5 Infrastructure

The infrastructure for the aluminium cladding dataset is accounted for and is approximated by a metal working plant. The calculated output is 43675 t/a, which results in 4.57E-10 unit/kg product. The life-span of the window frame factory is assumed to be 50 years.

4.4 Doors

The four doors are basically characterised by the door frame, the door core, the door lock and handle, the rubber frame, and in one case the door mantle. The datasets describe the different stages of production, thus the background processes, the frame manufacturing, the frame installation, the surface treatment, and – where required - the planking of the wooden door leaf (Werner et al. 1997)

The four door datasets describe the material inputs of a door system, the energy consumption (medium voltage electricity UCTE) of the door production process as well as an average distance for the transportation of the various components.

The measurements of the door leaf are 205cm x 86cm x 4.5cm. Door lock, handle and hinges are made from chromium steel, independent whether it is an inner door or outer door.

In case of the two glass-wood doors (one inner, one outer), the assumption was made that around 62% of the door is wood and ca. 38% glass (surface-%!).

The four door types have the following weights per m² of door (including the door frame, but without mortar for installation!): Outer door wood-aluminium 38.8 kg; outer door

wood-glass 36.5 kg; inner door wood 27.6 kg; and inner door glass-wood 27.6 kg. Main components of the door types are mentioned in (Tab. 4.4)³.

Tab. 4.4 Main material components of the four door types (kg per m² of door including door frame, but without mortar for installation)

Outer door, wood-aluminium (38.8 kg/m ²)	Outer door, wood-glass (36.5 kg/m ²)	Inner door, wood (27.6 kg/m ²)	Inner door, glass-wood (27.6 kg/m ²)
24.3 kg Wood (total of particle board, fibre board and softwood)	15.1 kg Wood (total of particle board, fibre board and softwood)	24.97 kg Wood (total of particle board, fibre board and softwood)	17.82 kg Core wood (total of particle board, fibre board and softwood)
10.98 kg Steel	10.97 kg Steel	1.12 kg Varnish	8.02 kg Glass (2-IV)
2.85 kg Aluminium	8.02 kg Glass (2-IV LSG)	0.747 kg Plastics	0.798 kg Varnish
0.313 kg Rubber frame	1.82 kg Aluminium	0.478 kg Steel	Plastics 0.464 kg
0.263 kg Varnish	0.313 kg Rubber frame	0.153 kg Aluminium	0.324 kg Steel
0.115 kg Plastics	0.263 kg Varnish	0.143 kg Rubber frame	0.143 kg Rubber frame
			0.0946 kg Aluminium

The four door datasets have the following specific material inputs (the information is retrieved and calculated from Werner et al. 1997):

- Outer door, wood-aluminium:

The door frame is made of 10.5 kg of steel, whereas the installation frame requires 50 kg of cement mortar. The frame surface of 0.938 m² is being zinc-coated. Further background processes include wood and PE film manufacturing for packaging, as well as paper, EPDM rubber and some alkyd resin as varnish (see Fig. 5.15). The steel fittings weigh 0.45 kg and are also zinc-coated. Before the application of 2.7 kg of aluminium planking, the wooden door leaf (21.95 kg) is being varnished.

The production processes (including packaging) cause wood, plastic (PE) and paper disposal mainly to the municipal incineration plant and to some extent to the sanitary landfill (Werner et al. 1997).

- Outer door, wood-glass:

³ Values calculated with moisture content of 20% for wood. Since moisture content of wood input is only 10%, the values are slightly higher than the sum of the inputs.

The outer door made from 62 % wood and 38 % glass requires the same background and manufacturing processes as the wood-aluminium door mentioned above. The only difference is the reduced amount of wood fabricated (62 %), and the additional glazing. The input for the aluminium planking is therefore also reduced by 38 %.

- Inner door, wood:

For this type of door, frame and door leaf are both made out of wood. The door leaf has the measures 200cm x 80cm x 4.5cm, and 0.844 kg varnish are applied on the whole door leaf (Werner et al. 1997). The steel fittings are galvanised (zinc coating). Further (process) materials for the manufacturing of the fittings are EPDM, polyamide, solder, ammonia, HCl, Zinc, Ethylene, lubricating oil, etc. The manufacturing of the fittings create certain waste products in form of plastics, hazardous waste, waste oil, NMVOC etc. The process heat required in the raw frame manufacturing is used from directly burnt wood chips released during the construction of the wooden door. As a result, this amount of wood chips for heating can be taken off from the total wood input to this door type. The data for all these values are based on Werner et al. (1997).

The frame surface treatment of the wooden inner door includes acrylic filler, acrylic dispersion and varnish, plus some packaging, electricity and different kinds of transport.

The door leaf requires a large number of input materials and process steps. Particle board, fibre board and softwood are glued together to serve as door leaf. Different resins, ammonia, HCl, polyamide, alkyd varnish, mineral wool, lubricating oil, etc. are applied during the door leaf production processes.

Energy consumption, transport and infrastructure are described in the following Chapters.

- Inner door, glass-wood:

For this dataset, the same background data as for the wooden inner door has been applied. With the exception that instead of 100 % wood for the door leaf, 38 % of it is glass.

4.4.1 Energy consumption

The total energy requirement for the production of a wood-aluminium outer door is 5.51 kWh (Werner et al. 1997). This electricity is finally released as heat waste. Process heat from gas in the range of 0.0396 MJ is needed during manufacturing.

The wood-glass outer door also consumes 5.51 kWh of electricity during the manufacturing processes (Werner et al. 1997).

The total electricity used for the wooden inner door is almost three times as high (15.34 kWh), mainly due to the surface processing of the raw frame and the door leaf. Both door frame and leaf are made of wood and therefore need more treatment steps, which result in a higher energy consumption.

Due to missing information on the electricity consumption of the wood-glass inner door, a similar value to the wooden inner door is assumed. The wood-glass door manufacturing requires more energy for sawing, whereas the surface treatment is applied to a smaller area. This is assumed to balance out.

4.4.2 Transports

The total transport activities for the four door types are:

- Outer door, wood-aluminium: barge 6.06 tkm, 40 t lorry 6.48 tkm, 16 t lorry 1.54 tkm;
- Outer door, wood-glass: barge 6.06 tkm, 40 t lorry 9.121 tkm, 16 t lorry 2.27 tkm;
- Inner door, wood: rail 0.01125 tkm, 40 t lorry 6.28 tkm, 16 t lorry 3.67 tkm;
- Inner door, wood-glass: rail 0.01125 tkm, 40 t lorry 8.92 tkm, 16 t lorry 4.4 tkm;

The data on goods transport is retrieved from Werner et al. (1997) and calculated per m² of door.

4.4.3 Air emissions

The only air emissions reported of during door production steps are 4.88E-04 kg NMVOC and 3.67E-06 kg Ethylene dichloride (per m² of door) in case of the wooden inner door and the wood-glass inner door. The aromatic hydrocarbon and heptane emissions to air during the surface treatment processes for the two inner door types are mentioned in the flow charts, but not explicitly inventoried. The reason for that is that nowadays, the coatings causing these emissions are replaced by water based coatings.

4.4.4 Waste water

No direct waste water is reported for the four different door types.

4.4.5 Infrastructure

The infrastructure is inventoried applying the proxy “metal working factory” (ecoinvent dataset). The life-time of the factory is set to 50 years.

4.5 Disposal datasets

For most of the glazing, window frame and door datasets, related disposal datasets were generated (see Figures in Chapter 5). These disposal datasets describe the standard way of

disposing in Switzerland, which is in most cases burning in a municipal incineration plant. Bigger metal parts (e.g. steel, aluminium) used in the window frames and doors are partly recycled and thus cut off. Smaller metal parts are often glued into other materials, which means that they cannot be efficiently collected and are therefore also disposed in the municipal incineration plant. The transport distance to the incineration plant is generally assumed to be 50 km for these disposal datasets.

4.6 Auxiliary datasets

In order to establish the glazing and window frame datasets, a number of auxiliary datasets were inventoried and applied:

- Polysulphide (sealing compound)
- Sealing compound, rubber (caoutchouc)-based
- Metal adhesive compound

The following subchapters shortly describe the datasets mentioned above.

4.6.1 Polysulphide sealing compound

The production of Polysulphide sealing compound requires limestone as filling material, chemicals, a hardening paste (manganese oxide), some polyurethane (rigid foam) and energy in form of heat and electricity. Polysulphide is used as a sealing compound in the glazing production.

4.6.2 Sealing compound, rubber-based (caoutchouc)

This caoutchouc-based sealing compound dataset was established with the information from a report by a German producer (Personal communication, Teroson AG (1997)). Due to a lack of a matchingecoinvent dataset, caoutchouc is approximated by the more general dataset “chemicals, organic”. Further input material is white spirit. Transport by lorry (100 km distance) and rail (600 km distance) is inventoried, as well as the required infrastructure in form of a chemical plant. This rubber-based sealing compound is an essential part in the construction of e.g. aluminium claddings.

4.6.3 Metal adhesive compound

Metal adhesive compounds are applied in the production of aluminium window frames. Basic materials going into the production are organic chemicals, liquid epoxy resin, hydrated lime and silica sand. Transport by lorry (100 km distance) and rail (600 km distance) is inventoried, as well as the required infrastructure in form of a chemical plant. The data is based on the material safety data sheet by a German producer (Personal communication Mr. Geissler, Agomet Klebstoffe GmbH (1997)).

5 Life Cycle Inventories

ecoinvent database meta information (tables), flows and their representation in the ecoinvent database are shown in (figures):

Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$, at plant	Tab. 5.1 and Fig. 5.1
Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$, laminated safety glass, at plant	Tab. 5.2 and Fig. 5.2
Glazing, triple (3-IV), $U < 0.5 \text{ W/m}^2\text{K}$, at plant	Tab. 5.3 and Fig. 5.3
Disposal, glazing 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, in municipal incineration	Tab. 5.4 and Fig. 5.4
Disposal, glazing 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, LSG, in municipal incineration	Tab. 5.5 and Fig. 5.5
Disposal, glazing 3-IV, $U < 0.5 \text{ W/m}^2\text{K}$, in municipal incineration	Tab. 5.6 and Fig. 5.6
Window frame, wood, $U = 1.5 \text{ W/m}^2\text{K}$, at plant	Tab. 5.7 and Fig. 5.7
Window frame, wood-metal, $U = 1.6 \text{ W/m}^2\text{K}$, at plant	Tab. 5.8 and Fig. 5.8
Window frame, plastic (PVC), $U = 1.6 \text{ W/m}^2\text{K}$, at plant	Tab. 5.9 and Fig. 5.9
Window frame, aluminium, $U = 1.6 \text{ W/m}^2\text{K}$, at plant	Tab. 5.10 and Fig. 5.10
Disposal, window frame, wood, in municipal incineration	Tab. 5.11 and Fig. 5.11
Disposal, window frame, wood-metal, in municipal incineration	Tab. 5.12 and Fig. 5.12
Disposal, window frame, plastic, in municipal incineration	Tab. 5.13 and Fig. 5.13
Cladding, crossbar-pole, aluminium, at plant	Tab. 5.14 and Fig. 5.14
Door, outer, wood-aluminium, at plant	Tab. 5.15 and Fig. 5.15
Door, outer, wood-glass, at plant	Tab. 5.16 and Fig. 5.16
Door, inner, wood, at plant	Tab. 5.17 and Fig. 5.17
Door, inner, glass-wood, at plant	Tab. 5.18 and Fig. 5.18
Disposal, door, outer, wood-aluminium, in municipal incineration	Tab. 5.19 and Fig. 5.19
Disposal, door, outer, wood-glass, in municipal incineration	Tab. 5.20 and Fig. 5.20
Disposal, door, inner, wood, in municipal incineration	Tab. 5.21 and Fig. 5.21
Disposal, door, inner, glass-wood, in municipal incineration	Tab. 5.22 and Fig. 5.22
Polysulphide, sealing compound, at plant	Tab. 5.23 and Fig. 5.23
Sealing compound, rubber-based (caoutchouc)	Tab. 5.24 and Fig. 5.24
Metal adhesive compound	Tab. 5.25 and Fig. 5.25

5.1 Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$, at plant

The meta information of “Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$, at plant” (Tab. 5.1) includes all important and necessary information to use the data correctly.

Tab. 5.1 ecoinvent meta information of the product “Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$, at plant”

Name	Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$, at plant
Location	RER
Infrastructure Process	0
Unit	m^2
Data Set Version	2.0
Included Processes	Included in this dataset are the materials and production processes of 1.06 m^2 (corresponding to 1 m^2 visible area) double glazing (2-IV) with a U-value below $1.1 \text{ W/m}^2\text{K}$ including disposal of waste water and production wastes.
Amount	1
Local Name	Verglasung, 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, ab Werk
Synonyms	
General Comment to reference function	This dataset describes the processes included in the production of 1.06 m^2 (corresponding to 1 m^2 visible area) double glazing (2-IV) with a U-value below $1.1 \text{ W/m}^2\text{K}$. It can be applied for double glazing used in the building sector. 1 m^2 visible glazing area has a final weight of 20.07 kg .
Start Date	1996
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data estimated according to a European company.
Technology text	The technology represents average European production.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Company documentation, calculations based on geometry
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this dataset is representative for the whole Europe is taken into account in the uncertainties.

Fig. 5.1 shows all in- and output flows of the product “Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000								Uncertainty						
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
Transport road	→	Glazing 2-IV U<1.1W/m2K		Lorry, 28 t		transport systems	transport	road	No	CH	transport, lorry 20-28t, fleet average	6.22E-01	tkm	Calculation based on Richter 1996	1	2.05	(13,13,15,5)	
Sealing compound	→			Polysulphide		plastics	others	No	RER		polysulphide, sealing compound, at plant	4.38E-01	kg	Calculation based on geometry	1	2.05	(13,13,15,5)	
Molecular sieve	→			Zeolith		washing agents	builders	No	RER		zeolite, powder, at plant	132E+00	kg	Calculation based on geometry	1	2.10	(4,3,13,5,5,4)	
Aluminium	→					metals	extraction	No	RER		aluminium, production mix, at plant	3.03E-01	kg	Calculation based on geometry	1	122	(13,13,15,3)	
Section bar rolling, aluminium	→					metals	processing	No	RER		sheet rolling, aluminium	3.03E-01	kg	Calculation based on geometry	1	122	(13,13,15,3)	
Argon	→					chemicals	inorganics	No	RER		argon, liquid, at plant	5.80E-02	kg	Calculation based on geometry	1	122	(13,13,15,4)	
Flat glass coated	→					glass	construction	No	RER		flat glass, coated, at plant	113E+01	kg	Calculation based on geometry	1	122	(13,13,15,3)	
Flat glass uncoated	→					glass	construction	No	RER		flat glass, uncoated, at plant	113E+01	kg	Calculation based on geometry	1	122	(13,13,15,3)	
Polybutadiene	→					plastics	polymers	No	RER		polybutadiene, at plant	9.37E-06	kg	Calculation based on geometry	1	122	(13,13,15,3)	
Water	→						water supply	production	No	RER		water, completely softened, at plant	9.96E-01	kg	Richter 1996	1	122	(13,13,15,12)
Electricity low voltage	→						electricity	production mix	No	UCTE		electricity, low voltage, production UCTE, at grid	3.21E+00	kWh	Richter 1996	1	122	(2,3,13,15,2)
							air	unspecified				Heat, waste	116E+01	MJ	Calculation	1	126	(3,4,13,15,13)
							Unpolluted, in waste treatment plant	waste management	wastewater treatment	No	CH	treatment, sewage, unpolluted, to wastewater disposal, glass, 0%water, to inert material landfill	9.96E-04	m3	Richter 1996	1	122	(13,13,15,6)
							7% waste according to Richter 1996, LCA of	waste management	inert material landfill	No	CH		136E+00	kg	Calculation based on geometry	1	122	(13,13,15,6)
						Unspecific, in m2	building components	windows	No	RER	glazing, double (2-IV), U<1.1 W/m2K, at plant	100E+00	m2					

Fig. 5.1 Flows for “Glazing, double (2-IV), U<1.1 W/m²K, at plant” and their representation in the ecoinvent database

5.2 Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$, laminated safety glass, at plant

The meta information of “Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$, laminated safety glass, at plant” (Tab. 5.2) includes all important and necessary information to use the data correctly.

Tab. 5.2 ecoinvent meta information of the product “Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$, laminated safety glass, at plant”

Name	Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$, laminated safety glass, at plant
Location	RER
Infrastructure Process	0
Unit	m^2
Data Set Version	2.0
Included Processes	Included in this dataset are the materials and production processes of 1.06 m^2 (corresponding to 1 m^2 visible area) LSG double glazing (2-IV) with a U-value below $1.1 \text{ W/m}^2\text{K}$ including disposal of waste water and production wastes.
Amount	1
Local Name	Verglasung, 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, Verbundsicherheitsglas, ab Werk
Synonyms	
General Comment to reference function	This dataset describes the processes included in the production of 1.06 m^2 (corresponding to 1 m^2 visible area) double glazing (2-IV, LSG) with a U-value below $1.1 \text{ W/m}^2\text{K}$. It can be applied for laminated safety double glazing used in the building sector. 1 m^2 visible glazing area has a final weight of 21.12 kg .
Start Date	1996
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data estimated according to a European company.
Technology text	The technology represents average European production.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Company documentation, calculations based on geometry
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this single dataset is representative for the whole Europe is taken into account in the uncertainties.

Fig. 5.2 shows all in- and output flows of the product “Glazing, double (2-IV), $U < 1.1 \text{ W/m}^2\text{K}$, laminated safety glass, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000								Uncertainty					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Glass 2-IV, U<1.1	→	Glazing 2-IV LSG U<1.1W/m2K				Production LSG glazing identical with production of "normal" glazing except for lamination of PVB Film	building components	windows	No	RER	glazing, double (2-IV), U<1.1 W/m2K, at plant	100E+00	m2	-	1	2.05	(13,13,15,5)
Electricity medium for lamination	→				3000 h per year; 608 kW; 200'000 m2 per year	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	9.67E+00	kWh	Temperit AG, personal communication	1	124	(2,4,13,15,2)	
PVB	→				Polyvinyl butyral (PVB) Film: 1mm thick, density 1083 kg/m3, 7%loss in production	plastics	polymers	No	RER	polyvinylchloride, at regional storage	105E+00	kg	Calculation based on geometry	1	122	(13,13,15,3)	
Extrusion, plastic film	→					plastics	processing	No	RER	extrusion, plastic film	105E+00	kg	Own calculations and assumptions	1	122	(2,3,13,15,3)	
				Heat waste		air	unspecified				Heat, waste	3.48E+01	MJ	Calculation	1	126	(3,4,13,15,13)
				Glass 2-IV, U<1.1 LSG			Unspecific, in m2	building components	windows	No	RER	glazing, double (2-IV), U<1.1 W/m2K, laminated safety glass, at plant	100E+00	m2			

Fig. 5.2 Flows for “Glazing, double (2-IV), U<1.1 W/m²K, laminated safety glass, at plant” and their representation in the ecoinvent database

5.3 Glazing, triple (3-IV), $U < 0.5 \text{ W/m}^2\text{K}$, at plant

The meta information of “Glazing, triple (3-IV), $U < 0.5 \text{ W/m}^2\text{K}$, at plant” (Tab. 5.3) includes all important and necessary information to use the data correctly.

Tab. 5.3 ecoinvent meta information of the product “Glazing, triple (3-IV), $U < 0.5 \text{ W/m}^2\text{K}$, at plant”

Name	Glazing, triple (3-IV), $U < 0.5 \text{ W/m}^2\text{K}$, at plant
Location	RER
Infrastructure Process	0
Unit	m^2
Data Set Version	2.0
Included Processes	Included in this dataset are the materials and production processes of 1.06m^2 (corresponding to 1m^2 visible area) triple glazing (3-IV) with a U-value below $0.5 \text{ W/m}^2\text{K}$ including disposal of waste water and production wastes.
Amount	1
Local Name	Verglasung, 3-IV, $U < 0.5 \text{ W/m}^2\text{K}$, ab Werk
Synonyms	
General Comment to reference function	This dataset describes the processes included in the production of 1.06m^2 (corresponding to 1m^2 visible area) triple glazing (3-IV) with a U-value below $0.5 \text{ W/m}^2\text{K}$. It can be applied for triple glazing used in the building sector. 1 m^2 visible glazing area has a final weight of 30.12 kg.
Start Date	1996
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data estimated according to a European company.
Technology text	The technology represents European production.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Company documentation, calculations based on geometry
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this single dataset is representative for the whole Europe is taken into account in the uncertainties.

Fig. 5.3 shows all in- and output flows of the product “Glazing, triple (3-IV), $U < 0.5 \text{ W/m}^2\text{K}$, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000								Uncertainty				
Input		Process Name		Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Krypton	→	Glazing 3-IV U<0.5 W/m2K				chemicals	inorganics	No	CH	krypton, gaseous, at regional storage	183E-01	kg	Calculation based on geometry	1	122	(13,13,15,4)
Transport road	→			Lorry, 28 t	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	9.72E-01	tkm	Calculation based on Richter 1996	1	2.05	(13,13,15,5)	
Sealing compound	→			Polysulphide	plastics	others	No	RER	polysulphide, sealing compound, at plant	6.57E-01	kg	Calculation based on geometry	1	2.05	(13,13,15,5)	
Molecular sieve	→			Zeolite	washing agents	builders	No	RER	zeolite, powder, at plant	193E+00	kg	Calculation based on geometry	1	2.10	(4,3,13,5,5,4)	
Aluminium	→				metals	extraction	No	RER	aluminium, production mix, at plant	5.20E-01	kg	Calculation based on geometry	1	122	(13,13,15,3)	
Section bar rolling, aluminium	→				metals	processing	No	RER	sheet rolling, aluminium	5.20E-01	kg	Calculation based on geometry	1	122	(13,13,15,3)	
Flat glass coated	→				glass	construction	No	RER	flat glass, coated, at plant	194E+01	kg	Calculation based on geometry	1	122	(13,13,15,3)	
Flat glass uncoated	→				glass	construction	No	RER	flat glass, uncoated, at plant	9.68E+00	kg	Calculation based on geometry	1	122	(13,13,15,3)	
Polybutadiene	→				plastics	polymers	No	RER	polybutadiene, at plant	187E-05	kg	Calculation based on geometry	1	122	(13,13,15,3)	
Water	→				water supply	production	No	RER	water, completely softened, at plant	149E+00	kg	Calculation based on Richter 1996	1	122	(13,13,15,12)	
Electricity low voltage	→				electricity	production mix	No	UCTE	electricity, low voltage, production UCTE, at grid	4.81E+00	kWh	Calculation based on Richter 1996	1	122	(2,3,13,15,2)	
			→	Heat waste		air	unspecified			Heat, waste	173E+01	MJ	Calculation	1	126	(3,4,13,15,13)
			→	Waste water	Unpolluted, in waste treatment plant	waste management	wastewater treatment	No	CH	treatment, sewage, unpolluted, to wastewater disposal, glass, 0%water, to inert material landfill	149E-03	m3	Own calculations and assumptions	1	122	(13,13,15,6)
			→	Disposal production	7% waste according to Richter 1996, LCA of	waste management	inert material landfill	No	CH		2.16E+01	kg	Calculation based on geometry	1	122	(13,13,15,6)
			→	Glass 3-IV, U<0.5	Unspecific, in m2	building components	windows	No	RER	glazing, triple (3-IV), U<0.5 W/m2K, at plant	100E+00	m2				

Fig. 5.3 Flows for “Glazing, triple (3-IV), U<0.5 W/m²K, at plant” and their representation in the ecoinvent database

5.4 Disposal, glazing 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, in municipal incineration

The meta information of “Disposal, glazing 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, in municipal incineration” (Tab. 5.4) includes all important and necessary information to use the data correctly.

Tab. 5.4 ecoinvent meta information of the product “Disposal, glazing 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, in municipal incineration”

Name	Disposal, glazing 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, in municipal incineration
Location	RER
Infrastructure Process	0
Unit	m^2
Data Set Version	2.0
Included Processes	The dataset describes the disposal of 1.06 m^2 double glazing (2-IV) (not laminated safety glass)) in a municipal incineration. 80% of the glass goes for landfill, 20% to MSWI. 1.06 m^2 corresponds to 1 m^2 visible glass.
Amount	1
Local Name	Entsorgung, Verglasung 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, in Kehrrechtverbrennung
Synonyms	
General Comment to reference function	This dataset is applied for the disposal of 2-IV glazing of windows.
Start Date	1996
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data estimated according to Swiss practice.
Technology text	The technology described involves the disposal of aluminium, rubber and glass of a 2-IV glazing (not LSG) in a municipal incineration plant (glass is partly landfilled).
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Calculation from waste composition
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this dataset is representative for the whole Europe is taken into account in the uncertainties.

Fig. 5.4 shows all in- and output flows of the product “Disposal, glazing 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, in municipal incineration”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Disposal, glazing 2-IV, U<1.1 W/m2K, in MSWI	→	Disposal, aluminium and Zeolithe, in incineration	Assumption: 80%in landfill, 20%(edges) in MWI	waste management	municipal incineration	No	CH	disposal, aluminium, 0% water, to municipal incineration	162E+00	kg	Calculation based on composition	1	122	(13,13,15,6)
	→	Disposal, glass, in landfill		waste management	inert material landfill	No	CH	disposal, glass, 0%water, to inert material landfill	144E+01	kg		1	122	(13,13,15,6)
	→	Disposal, glass, in incineration		waste management	municipal incineration	No	CH	disposal, glass, 0%water, to municipal incineration	3.60E+00	kg		1	122	(13,13,15,6)
	→	Disposal, rubber, in incineration		waste management	municipal incineration	No	CH	disposal, rubber, unspecified, 0%water, to municipal incineration	4.38E-01	kg		1	122	(13,13,15,6)
	→	Disposal, glazing, in incineration		waste management	building demolition	No	RER	disposal, glazing 2-IV, U<1.1W/m2K, in municipal incineration	100E+00	m2				

Fig. 5.4 Flows for “Disposal, glazing 2-IV, U<1.1W/m²K, in municipal incineration” and their representation in the ecoinvent database

5.5 Disposal, glazing 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, LSG, in municipal incineration

The meta information of “Disposal, glazing 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, LSG, in municipal incineration” (Tab. 5.5) includes all important and necessary information to use the data correctly.

Tab. 5.5 ecoinvent meta information of the product “Disposal, glazing 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, LSG, in municipal incineration”

Name	Disposal, glazing 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, LSG, in municipal incineration
Location	RER
Infrastructure Process	0
Unit	m^2
Data Set Version	2.0
Included Processes	The dataset describes the disposal of 1.06 m^2 double glazing 2-IV (laminated safety glass) in a municipal incineration. 80% of the glass goes for landfill, 20% to MSWI. 1.06 m^2 corresponds to 1 m^2 visible glass.
Amount	1
Local Name	Entsorgung, Verglasung 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, VSG, in Kehrrechtverbrennung
Synonyms	
General Comment to reference function	This dataset is applied for the disposal of 2-IV glazing (LSG) of windows.
Start Date	1996
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data estimated according to Swiss practice.
Technology text	The technology described involves the disposal of aluminium, rubber and glass of a 2-IV glazing (LSG) in a municipal incineration plant (glass is partly landfilled).
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Calculation from waste composition
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this dataset is representative for the whole Europe is taken into account in the uncertainties.

Fig. 5.5 shows all in- and output flows of the product “Disposal, glazing 2-IV, $U < 1.1 \text{ W/m}^2\text{K}$, LSG, in municipal incineration”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Disposal, glazing 2-IV LSG, U<1.1 W/m2K, in MSWI	→	Disposal, aluminium and Zeolithe, in incineration		waste management	municipal incineration	No	CH	disposal, aluminium, 0% water, to municipal incineration	162E+00	kg	Calculation based on composition	1	122	(13,13,15,6)
	→	Disposal, glass, in landfill	Assumption: 80%in landfill, 20%(edges) in MWI	waste management	inert material landfill	No	CH	disposal, glass, 0%water, to inert material landfill	144E+01	kg		1	122	(13,13,15,6)
	→	Disposal, glass, in incineration		waste management	municipal incineration	No	CH	disposal, glass, 0%water, to municipal incineration	3.60E+00	kg		1	122	(13,13,15,6)
	→	Disposal, rubber, in incineration		waste management	municipal incineration	No	CH	disposal, rubber, unspecified, 0%water, to municipal incineration	149E+00	kg		1	122	(13,13,15,6)
	→	Disposal, glazing, in incineration		waste management	building demolition	No	RER	disposal, glazing 2-IV, U<1.1W/m2K, LSG, in municipal incineration	100E+00	m2				

Fig. 5.5 Flows for “Disposal, glazing 2-IV, U<1.1W/m²K, LSG, in municipal incineration” and their representation in the ecoinvent database

5.6 Disposal, glazing 3-IV, $U < 0.5 \text{ W/m}^2\text{K}$, in municipal incineration

The meta information of “Disposal, glazing 3-IV, $U < 0.5 \text{ W/m}^2\text{K}$, in municipal incineration” (Tab. 5.6) includes all important and necessary information to use the data correctly.

Tab. 5.6 ecoinvent meta information of the product “Disposal, glazing 3-IV, $U < 0.5 \text{ W/m}^2\text{K}$, in municipal incineration”

Name	Disposal, glazing 3-IV, $U < 0.5 \text{ W/m}^2\text{K}$, in municipal incineration
Location	RER
Infrastructure Process	0
Unit	m^2
Data Set Version	2.0
Included Processes	The dataset describes the disposal of 1.06 m^2 triple glazing (3-IV) (not laminated safety glass) in a municipal incineration. 80% of the glass goes for landfill, 20% to MSWI. 1.06 m^2 corresponds to 1 m^2 visible glass.
Amount	1
Local Name	Entsorgung, Verglasung 3-IV, $U < 0.5 \text{ W/m}^2\text{K}$, in Kehrrechtverbrennung
Synonyms	
General Comment to reference function	This dataset is applied for the disposal of 3-IV glazing of windows.
Start Date	1996
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data estimated according to Swiss practice.
Technology text	The technology described involves the disposal of aluminium, rubber and glass of a 3-IV glazing in a municipal incineration plant (glass is partly landfilled).
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Calculation from waste composition
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this dataset is representative for the whole Europe is taken into account in the uncertainties.

Fig. 5.6 shows all in- and output flows of the product “Disposal, glazing 3-IV, $U < 0.5 \text{ W/m}^2\text{K}$, in municipal incineration”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Disposal, glazing 3-IV, U<0.5 W/m ² K, in MSWI	→	Disposal, aluminium and Zeolithe, in incineration	Assumption: 80%in landfill, 20%(edges) in MWI	waste management	municipal incineration	No	CH	disposal, aluminium, 0% water, to municipal incineration	2.45E+00	kg	Calculation based on composition	1	122	(13,13,15,6)
		Disposal, glass, in landfill		waste management	inert material landfill	No	CH	disposal, glass, 0%water, to inert material landfill	2.16E+01	kg		1	122	(13,13,15,6)
		Disposal, glass, in incineration		waste management	municipal incineration	No	CH	disposal, glass, 0%water, to municipal incineration	5.40E+00	kg		1	122	(13,13,15,6)
		Disposal, rubber, in incineration		waste management	municipal incineration	No	CH	disposal, rubber, unspecified, 0%water, to municipal incineration	6.57E-01	kg		1	122	(13,13,15,6)
		Disposal, glazing, in incineration		waste management	building demolition	No	RER	disposal, glazing 3-IV, U<0.5W/m ² K, in municipal incineration	1.00E+00	m ²				

Fig. 5.6 Flows for “Disposal, glazing 3-IV, U<0.5W/m²K, in municipal incineration” and their representation in the ecoinvent database

5.7 Window frame, wood, $U=1.5 \text{ W/m}^2\text{K}$, at plant

The meta information of “Window frame, wood, $U=1.5 \text{ W/m}^2\text{K}$, at plant” (Tab. 5.7) includes all important and necessary information to use the data correctly.

Tab. 5.7 ecoinvent meta information of the product “Window frame, wood, $U=1.5 \text{ W/m}^2\text{K}$, at plant”

Name	Window frame, wood, $U=1.5 \text{ W/m}^2\text{K}$, at plant
Location	RER
Infrastructure Process	0
Unit	m^2
Data Set Version	2.0
Included Processes	Included processes are timber sawing, varnishing (primer, solvents, paint), section bar rolling for steel fittings, joining, fitting, all the road transport at different production phases and the disposal of paint remains.
Amount	1
Local Name	Fensterrahmen, Holz, $U=1.5 \text{ W/m}^2\text{K}$, ab Werk
Synonyms	
General Comment to reference function	This dataset describes all the processes and material inputs needed to produce a wooden window frame with 1 m^2 visible area. 1 m^2 of visible wooden window frame weighs 80.2 kg.
Start Date	1996
End Date	2004
Data Valid For Entire Period	
Other Period Text	
Geography text	Window frames produced and sold in Switzerland and Germany.
Technology text	The dataset describes highly automated technology processes in window frame manufacturing.
Representativeness [%]	
Production Volume	
Sampling Procedure	Company documentation, measurements
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this dataset is representative for whole Europe is taken into account in the uncertainties.

Fig. 5.7 shows all in- and output flows of the product “Window frame, wood, $U=1.5 \text{ W/m}^2\text{K}$, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Electricity	Window frame, wood, U=1.5W/m ² K, part 1		raw cut	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	7.11E-01	kWh	various Empa Studies	1	122	(13,13,15,2)
Sawn timber, softwood				wooden materials	extraction	No	RER	sawn timber, softwood, planed, kiln dried, at plant	2.11E-01	m ³	various Empa Studies	1	122	(13,13,15,4)
Transport, road			sawn timber to window factory	transport systems	road	No	RER	transport, lorry >16t, fleet average	3.82E+01	tkm	various Empa Studies	1	2.05	(13,13,15,5)
Electricity			Profiling wood	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	128E+01	kWh	various Empa Studies	1	122	(13,13,15,2)
Electricity			Varnishing: wood preservative, primer, solvents,	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	7.53E+00	kWh	various Empa Studies	1	122	(13,13,15,2)
Benzimidazole				agricultural means of production	pesticides	No	RER	benzimidazole-compounds, at regional storehouse	3.96E-03	kg	various Empa Studies	1	122	(13,13,15,4)
Butanol				chemicals	organics	No	RER	butanol, 1-, at plant	190E-02	kg	various Empa Studies	1	122	(13,13,15,4)
Water				water supply	production	No	RER	water, completely softened, at plant	3.77E-01	kg	various Empa Studies	1	122	(13,13,15,12)
Toluene				chemicals	organics	No	RER	toluene, liquid, at plant	3.10E-02	kg	various Empa Studies	1	122	(13,13,15,4)
Acetone				chemicals	organics	No	RER	acetone, liquid, at plant	103E-02	kg	various Empa Studies	1	122	(13,13,15,4)
Methyl ethyl ketone				chemicals	organics	No	RER	methyl ethyl ketone, at plant	2.38E-04	kg	various Empa Studies	1	122	(13,13,15,4)
Propylene glycol				chemicals	organics	No	RER	propylene glycol, liquid, at plant	2.38E-04	kg	various Empa Studies	1	122	(13,13,15,4)
Titanium dioxide				chemicals	inorganics	No	RER	titanium dioxide, production mix, at plant	5.95E-04	kg	various Empa Studies	1	122	(13,13,15,4)
Butanol				chemicals	organics	No	RER	butanol, 1-, at plant	7.14E-04	kg	various Empa Studies	1	122	(13,13,15,4)
Toluene				chemicals	organics	No	RER	toluene, liquid, at plant	1.19E-04	kg	various Empa Studies	1	122	(13,13,15,4)
Isopropanol				chemicals	organics	No	RER	isopropanol, at plant	4.76E-04	kg	various Empa Studies	1	122	(13,13,15,4)
Zinc coating				metals	processing	No	RER	zinc coating, pieces	169E-02	m ²	various Empa Studies	1	122	(13,13,15,3)
Steel				metals	extraction	No	RER	steel, low-alloyed, at plant	1.77E-01	kg	various Empa Studies	1	122	(13,13,15,3)

Fig. 5.7 Flows for “Window frame, wood, U=1.5 W/m²K, at plant” (Part 1) and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Section bar rolling	Window frame, wood, U=1.5W/m2K, part 2			metals	processing	No	RER	section bar rolling, steel	1.77E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Alkyd paint				paintings	production	No	RER	alkyd paint, white, 60%in H2O, at plant	5.49E+00	kg	various Empa Studies	1	122	(13,13,15,4)
Acetone				chemicals	organics	No	RER	acetone, liquid, at plant	7.00E-03	kg	various Empa Studies	1	122	(13,13,15,4)
White spirit				paintings	production	No	RER	white spirit, at plant	7.00E-03	kg	various Empa Studies	1	122	(13,13,15,4)
Heat from waste wood incineration				wood energy	heating systems	No	CH	pellets, mixed, burned in furnace 50kW	5.40E+01	MJ	various Empa Studies	1	122	(13,13,15,1)
Heat from waste wood incineration				wood energy	fuels	No	RER	wood pellets, u=10% at storehouse	-7.83E-03	m3	various Empa Studies	1	122	(13,13,15,1)
Transport, road, 50 km				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	4.84E-01	tkm	various Empa Studies	1	2.05	(13,13,15,5)
				waste management	municipal incineration	No	CH	disposal, paint, 0%water, to municipal incineration	2.79E-01	kg	various Empa Studies	1	122	(13,13,15,6)
Electricity		Disposal paint	Joining	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	9.03E-01	kWh	various Empa Studies	1	122	(13,13,15,2)
Alkyd resin				paintings	production	No	RER	alkyd resin, long oil, 70%in white spirit, at plant	2.44E-02	kg	various Empa Studies	1	122	(13,13,15,4)
Melamine formaldehyde resin				paintings	production	No	RER	melamine formaldehyde resin, at plant	7.33E-02	kg	various Empa Studies	1	122	(13,13,15,4)
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	8.54E-02	kWh	various Empa Studies	1	122	(13,13,15,2)
Sawn hardwood				wooden materials	extraction	No	RER	sawn timber, hardwood, planed, kiln dried, u=10% at transport, lorry 20-28t, fleet average	1.71E-03	m3	various Empa Studies	1	122	(13,13,15,4)
Transport, road, 50 km				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	6.65E-02	tkm	various Empa Studies	1	2.05	(13,13,15,5)
				waste management	municipal incineration	No	CH	disposal, paint, 0%water, to municipal incineration	6.67E-03	kg	various Empa Studies	1	122	(13,13,15,6)
Electricity		Disposal paint	Fitting	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.57E+01	kWh	various Empa Studies	1	122	(13,13,15,2)
EPDM rubber				plastics	polymers	No	RER	synthetic rubber, at plant	1.14E+00	kg	various Empa Studies	1	122	(13,13,15,3)
Nylon, glass fibre reinforced				plastics	polymers	No	RER	nylon 66, glass-filled, at plant	3.49E-01	kg	various Empa Studies	1	122	(13,13,15,3)

Fig. 5.7 Flows for “Window frame, wood, U=1.5 W/m2K, at plant” (Part 1) and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Anodising aluminium sheet	Window frame, wood, U=1.5W/m2K, part 3	Aluminium		metals	processing	No	RER	anodising, aluminium sheet	8.10E-01	m2	various Empa Studies	1	122	(13,13,15,3)
Section bar extrusion aluminium				metals	processing	No	RER	section bar extrusion, aluminium	3.06E+00	kg	various Empa Studies	1	122	(13,13,15,3)
Aluminium				metals	extraction	No	RER	aluminium, production mix, at plant	3.06E+00	kg	various Empa Studies	1	122	(13,13,15,3)
			recycled --> cut-off						7.11E-02	kg	various Empa Studies			
POM				plastics	polymers	No	RER	polyvinylchloride, at regional storage	1.36E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Polypropylen				plastics	polymers	No	RER	polypropylene, granulate, at plant	2.33E-02	kg	various Empa Studies	1	122	(13,13,15,3)
Polyethylene				plastics	polymers	No	RER	polyethylene, LDPE, granulate, at plant	2.33E-02	kg	various Empa Studies	1	122	(13,13,15,3)
Zinc coating				metals	processing	No	RER	zinc coating, pieces	4.76E-01	m2	various Empa Studies	1	122	(13,13,15,3)
Steel				metals	extraction	No	RER	steel, low-alloyed, at plant	5.00E+00	kg	various Empa Studies	1	122	(13,13,15,3)
Section bar rolling, steel				metals	processing	No	RER	section bar rolling, steel	5.00E+00	kg	various Empa Studies	1	122	(13,13,15,3)
Copper for AlZn-cast alloy				metals	extraction	No	RER	copper, at regional storage	6.23E-03	kg	various Empa Studies	1	122	(13,13,15,3)
Zinc for AlZn-cast alloy				metals	extraction	No	RER	zinc, primary, at regional storage	2.90E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Aluminium for AlZn-cast alloy				metals	extraction	No	RER	aluminium, production mix, cast alloy, at plant	1.56E-02	kg	various Empa Studies	1	122	(13,13,15,3)
Transport, road, 50 km			Materials for fitting to window frame plant (50 km)	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	5.02E-01	tkm	various Empa Studies	1	2.05	(13,13,15,5)
Infrastructure			Metal working plant as proxy: Output 43675 t/a --> 4.57E-10	metals	general manufacturing	Yes	RER	metal working factory	3.67E-08	unit	various Empa Studies	1	3.06	(2,4,13,15,9)
		Heat waste	total	air	unspecified			Heat, waste	2.08E+02	MJ	various Empa Studies	1	126	(3,4,13,15,13)
		Window frame, wood	80.2kg/m2	building components	windows	No	RER	window frame, wood, U=15 W/m2K, at plant	1.00E+00	m2				

Fig. 5.7 Flows for “Window frame, wood, U=1.5 W/m2K, at plant” (Part 1) and their representation in the ecoinvent database (cont)

5.8 Window frame, wood-metal, $U=1.6 \text{ W/m}^2\text{K}$, at plant

The meta information of “Window frame, wood-metal, $U=1.6 \text{ W/m}^2\text{K}$, at plant” (Tab. 5.8) includes all important and necessary information to use the data correctly.

Tab. 5.8 ecoinvent meta information of the product “Window frame, wood-metal, $U=1.6 \text{ W/m}^2\text{K}$, at plant”

Name	Window frame, wood-metal, $U=1.6 \text{ W/m}^2\text{K}$, at plant
Location	RER
Infrastructure Process	0
Unit	m^2
Data Set Version	2.0
Included Processes	Included processes are timber sawing, varnishing (primer, solvents, paint), section bar rolling for steel fittings, joining, fitting, all the road transport at different production phases and the disposal of the paint remains.
Amount	1
Local Name	Fensterrahmen, Holz-Metall, $U=1.6 \text{ W/m}^2\text{K}$, ab Werk
Synonyms	
General Comment to reference function	This dataset describes all the processes and material inputs needed to produce a wood-metal (steel) window frame with 1 m^2 visible area. 1 m^2 of visible wooden window frame weighs 83.4 kg.
Start Date	1996
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	Window frames produced and sold in Switzerland and Germany.
Technology text	The dataset describes highly automated technology processes in window frame manufacturing.
Representativeness [%]	
Production Volume	
Sampling Procedure	Company documentation, measurements
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this dataset is representative for whole Europe is taken into account in the uncertainties.

Fig. 5.8 shows all in- and output flows of the product “Window frame, wood-metal, $U=1.6 \text{ W/m}^2\text{K}$, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Electricity	Window frame, wood-metal, U=1.6W/m ² K, part 1		raw cut	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	7.11E-01	kWh	various Empa Studies	1	122	(13,13,15,2)
Sawn timber				wooden materials	extraction	No	RER	sawn timber, softwood, planed, kiln dried, at plant	1.95E-01	m ³	various Empa Studies	1	122	(13,13,15,4)
Transport, road			sawn timber to window factory	transport systems	road	No	RER	transport, lorry >16t, fleet average	3.53E+01	tkm	various Empa Studies	1	2.05	(13,13,15,5)
Electricity			Profiling wood	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.28E+01	kWh	various Empa Studies	1	122	(13,13,15,2)
Electricity			Varnishing: wood preservative, primer, solvents,	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.20E+01	kWh	various Empa Studies	1	122	(13,13,15,2)
Benzimidazole				agricultural means of production	pesticides	No	RER	benzimidazole-compounds, at regional storehouse	8.64E-03	kg	various Empa Studies	1	122	(13,13,15,4)
Butanol				chemicals	organics	No	RER	butanol, 1-, at plant	4.15E-02	kg	various Empa Studies	1	122	(13,13,15,4)
Water				water supply	production	No	RER	water, completely softened, at plant	8.23E-01	kg	various Empa Studies	1	122	(13,13,15,12)
Toluene				chemicals	organics	No	RER	toluene, liquid, at plant	2.44E-02	kg	various Empa Studies	1	122	(13,13,15,4)
Acetone				chemicals	organics	No	RER	acetone, liquid, at plant	8.13E-03	kg	various Empa Studies	1	122	(13,13,15,4)
Methyl ethyl ketone				chemicals	organics	No	RER	methyl ethyl ketone, at plant	1.87E-04	kg	various Empa Studies	1	122	(13,13,15,4)
Propylene glycol				chemicals	organics	No	RER	propylene glycol, liquid, at plant	1.87E-04	kg	various Empa Studies	1	122	(13,13,15,4)
Titanium dioxide				chemicals	inorganics	No	RER	titanium dioxide, production mix, at plant	4.68E-04	kg	various Empa Studies	1	122	(13,13,15,4)
Butanol				chemicals	organics	No	RER	butanol, 1-, at plant	5.62E-04	kg	various Empa Studies	1	122	(13,13,15,4)
Toluene				chemicals	organics	No	RER	toluene, liquid, at plant	9.36E-05	kg	various Empa Studies	1	122	(13,13,15,4)
Isopropanol				chemicals	organics	No	RER	isopropanol, at plant	3.75E-04	kg	various Empa Studies	1	122	(13,13,15,4)
Zinc coating				metals	processing	No	RER	zinc coating, pieces	1.16E-02	m ²	various Empa Studies	1	122	(13,13,15,3)
Steel				metals	extraction	No	RER	steel, low-alloyed, at plant	1.21E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Section bar rolling				metals	processing	No	RER	section bar rolling, steel	1.21E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Alkyd paint				paintings	production	No	RER	alkyd paint, white, 60%in H ₂ O, at plant	5.06E+00	kg	various Empa Studies	1	122	(13,13,15,4)
Acetone				chemicals	organics	No	RER	acetone, liquid, at plant	5.51E-03	kg	various Empa Studies	1	122	(13,13,15,4)

Fig. 5.8 Flows for “Window frame, wood-metal, U=1.6 W/m²K, at plant” (Part 1) and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
White spirit	Window frame, wood-metal, U=1.6W/m2K, part 2	Disposal paint		paintings	production	No	RER	white spirit, at plant	5.51E-03	kg	various Empa Studies	1	122	(13,13,15,4)
Heat from waste wood incineration				wood energy	heating systems	No	CH	pellets, mixed, burned in furnace 50kW	5.40E+01	MJ	various Empa Studies	1	122	(13,13,15,1)
Heat from waste wood incineration				wood energy	fuels	No	RER	wood pellets, u=10% at storehouse	-7.83E-03	m3	various Empa Studies	1	122	(13,13,15,1)
Transport, road, 50 km				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	5.95E-01	tkm	various Empa Studies	1	2.05	(13,13,15,5)
				waste management	municipal incineration	No	CH	disposal, paint, 0%water, to municipal incineration	3.25E+00	kg	various Empa Studies	1	122	(13,13,15,6)
Electricity		Joining		electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	9.03E-01	kWh	various Empa Studies	1	122	(13,13,15,2)
Alkyd resin				paintings	production	No	RER	alkyd resin, long oil, 70%in white spirit, at plant	2.44E-02	kg	various Empa Studies	1	122	(13,13,15,4)
M elamine formaldehyde resin				paintings	production	No	RER	melamine formaldehyde resin, at plant	7.33E-02	kg	various Empa Studies	1	122	(13,13,15,4)
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	8.54E-02	kWh	various Empa Studies	1	122	(13,13,15,2)
Sawn hardwood				wooden materials	extraction	No	RER	sawn timber, hardwood, planed, kiln dried, u=10% at transport, lorry 20-28t, fleet average	1.71E-03	m3	various Empa Studies	1	122	(13,13,15,3)
Transport, road, 50 km				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	6.65E-02	tkm	various Empa Studies	1	2.05	(13,13,15,5)
				waste management	municipal incineration	No	CH	disposal, paint, 0%water, to municipal incineration	6.67E-03	kg	various Empa Studies	1	122	(13,13,15,6)
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.59E+01	kWh	various Empa Studies	1	122	(13,13,15,2)
EPDM rubber				plastics	polymers	No	RER	synthetic rubber, at plant	2.91E+00	kg	various Empa Studies	1	122	(13,13,15,3)
Nylon				plastics	polymers	No	RER	nylon 66, glass-filled, at plant	1.09E+00	kg	various Empa Studies	1	122	(13,13,15,3)
Anodising aluminium sheet				metals	processing	No	RER	anodising, aluminium sheet	3.40E+00	m2	various Empa Studies	1	122	(13,13,15,3)
Section bar extrusion aluminium				metals	processing	No	RER	section bar extrusion, aluminium	1.22E+01	kg	various Empa Studies	1	122	(13,13,15,3)
Aluminium				metals	extraction	No	RER	aluminium, production mix, at plant	1.22E+01	kg	various Empa Studies	1	122	(13,13,15,3)
		Aluminium	recycled --> cut-off						7.11E-02	kg	various Empa Studies			

Fig. 5.8 Flows for “Window frame, wood-metal, U=1.6 W/m2K, at plant” (Part 1) and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
POM	Window frame, wood-metal, U=1.6W/m2K, part 3			plastics	polymers	No	RER	polyvinylchloride, at regional storage	136E-01	kg	various Empa Studies	1	122	(13,13,15,3)	
Polypropylen				plastics	polymers	No	RER	polypropylene, granulate, at plant	2.33E-02	kg	various Empa Studies	1	122	(13,13,15,3)	
Polyethylene				plastics	polymers	No	RER	polyethylene, LDPE, granulate, at plant	2.33E-02	kg	various Empa Studies	1	122	(13,13,15,3)	
Zinc coating				metals	processing	No	RER	zinc coating, pieces	4.76E-01	m2	various Empa Studies	1	122	(13,13,15,3)	
Steel				metals	extraction	No	RER	steel, low-alloyed, at plant	5.00E+00	kg	various Empa Studies	1	122	(13,13,15,3)	
Section bar rolling, steel				metals	processing	No	RER	section bar rolling, steel	5.00E+00	kg	various Empa Studies	1	122	(13,13,15,3)	
Copper for AlZn-cast alloy				Zinc die-cast	metals	extraction	No	RER	copper, at regional storage	6.23E-03	kg	various Empa Studies	1	122	(13,13,15,3)
Zinc for AlZn-cast alloy				Zinc die-cast	metals	extraction	No	RER	zinc, primary, at regional storage	2.90E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Aluminium for AlZn-cast alloy				Zinc die-cast	metals	extraction	No	RER	aluminium, production mix, cast alloy, at plant	156E-02	kg	various Empa Studies	1	122	(13,13,15,3)
Transport, road, 50 km					transport systems	road	No	CH	transport, lorry 20-28t, fleet average	109E+00	tkm	various Empa Studies	1	2.05	(13,13,15,5)
Infrastructure				Metal working plant as proxy: Output 43675 t/a --> 4.57E-10	metals	general manufacturing	Yes	RER	metal working factory	3.81E-08	unit	various Empa Studies	1	3.06	(2,4,13,15,9)
		Heat waste	total	air	unspecified			Heat, waste	2.24E+02	MJ	various Empa Studies	1	126	(3,4,13,15,13)	
		Window frame, wood-	83.4kg/m2	building components	windows	No	RER	window frame, wood-metal, U=16 W/m2K, at plant	100E+00	m2					

Fig. 5.8 Flows for “Window frame, wood-metal, U=1.6 W/m2K, at plant” (Part 1) and their representation in the ecoinvent database (cont)

5.9 Window frame, plastic (PVC), $U=1.6 \text{ W/m}^2\text{K}$, at plant

The meta information of “Window frame, plastic (PVC), $U=1.6 \text{ W/m}^2\text{K}$, at plant” (Tab. 5.9) includes all important and necessary information to use the data correctly.

Tab. 5.9 ecoinvent meta information of the product “Window frame, plastic (PVC), $U=1.6 \text{ W/m}^2\text{K}$, at plant”

Name	Window frame, plastic (PVC), $U=1.6 \text{ W/m}^2\text{K}$, at plant
Location	RER
Infrastructure Process	0
Unit	m^2
Data Set Version	2.0
Included Processes	Included processes are injection moulding and extrusion of PVC, section bar rolling for steel fittings, section bar extrusion for aluminium parts, all the road transport at different production phases and the process heat waste.
Amount	1
Local Name	Fensterrahmen, Kunststoff (PVC), $U=1.6 \text{ W/m}^2\text{K}$, ab Werk
Synonyms	
General Comment to reference function	This dataset describes all the processes and material inputs needed to produce a plastic window frame with 1 m^2 visible area. 1 m^2 of visible plastic window frame weighs 94.5 kg.
Start Date	1996
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	Window frames produced and sold in Switzerland and Germany.
Technology text	The dataset describes highly automated technology processes in window frame manufacturing.
Representativeness [%]	
Production Volume	
Sampling Procedure	Company documentation, measurements
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this dataset is representative for whole Europe is taken into account in the uncertainties.

Fig. 5.9 shows all in- and output flows of the product “Window frame, plastic (PVC), $U=1.6 \text{ W/m}^2\text{K}$, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000								Uncertainty						
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
PVC	→	Window frame, plastic (PVC), U=1.6 W/m2K, at plant, part 1				Recycled material, primary PVC as proxy	plastics	polymers	No	RER	polyvinylchloride, at regional storage	1.16E+01	kg	various Empa Studies	1	122	(13,13,15,3)	
PVC	→						plastics	polymers	No	RER	polyvinylchloride, at regional storage	4.27E+01	kg	various Empa Studies	1	122	(13,13,15,3)	
Injection moulding	→						plastics	processing	No	RER	injection moulding	190E+00	kg	various Empa Studies	1	122	(13,13,15,3)	
Extrusion	→						plastics	processing	No	RER	extrusion, plastic pipes	5.43E+01	kg	various Empa Studies	1	122	(13,13,15,3)	
Rubber	→						sealing, coextruded with profile	plastics	polymers	No	RER	synthetic rubber, at plant	5.71E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Transport, road	→						PVC, EPDM : 500 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	2.84E+01	tkm	various Empa Studies	1	2.05	(13,13,15,5)
Steel	→							metals	extraction	No	RER	steel, low-alloyed, at plant	3.30E+01	kg	various Empa Studies	1	122	(13,13,15,3)
Section bar rolling, steel	→							metals	processing	No	RER	section bar rolling, steel	3.30E+01	kg	various Empa Studies	1	122	(13,13,15,3)
Zinc coating, coils	→							metals	processing	No	RER	zinc coating, coils	2.11E+00	m2	various Empa Studies	1	122	(13,13,15,3)
Transport, road	→						Steel, 50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	157E+00	tkm	various Empa Studies	1	2.05	(13,13,15,5)
Electricity	→						Cutting / welding	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	138E+01	kWh	various Empa Studies	1	122	(13,13,15,2)
Steel (screws)	→							metals	extraction	No	RER	steel, low-alloyed, at plant	102E-01	kg	various Empa Studies	1	122	(13,13,15,3)
				→	PVC cuttings		to recycling --> cut-off						4.20E+00	kg	various Empa Studies			
				→	Steel cuttings		to recycling --> cut-off						108E-01	kg	various Empa Studies			

Fig. 5.9 Flows for “Window frame, plastic (PVC), U=1.6 W/m²K, at plant” (Part 1) and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Polymethylen	Window frame, plastic (PVC), U=1.6 W/m2K, at plant, part 2		POM; Polyvinylchloride as proxy	plastics	polymers	No	RER	polyvinylchloride, at regional storage	136E-01	kg	various Empa Studies	1	122	(13,13,15,3)
PP			Granulate	plastics	polymers	No	RER	polypropylene, granulate, at plant	5.78E-03	kg	various Empa Studies	1	122	(13,13,15,3)
LDPE				plastics	polymers	No	RER	polyethylene, LDPE, granulate, at plant	5.78E-03	kg	various Empa Studies	1	122	(13,13,15,3)
Zinc coating			Pieces	metals	processing	No	RER	zinc coating, pieces	4.63E-01	m2	various Empa Studies	1	122	(13,13,15,3)
Steel				metals	extraction	No	RER	steel, low-alloyed, at plant	4.86E+00	kg	various Empa Studies	1	122	(13,13,15,3)
Section bar rolling, steel				metals	processing	No	RER	section bar rolling, steel	4.86E+00	kg	various Empa Studies	1	122	(13,13,15,3)
Transport, road			50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	2.31E-01	tkm	various Empa Studies	1	2.05	(13,13,15,5)
Copper for AlZn-cast alloy				metals	extraction	No	RER	copper, at regional storage	6.98E-03	kg	various Empa Studies	1	122	(13,13,15,3)
Zinc for AlZn-cast alloy				metals	extraction	No	RER	zinc, primary, at regional storage	3.25E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Aluminium for AlZn-cast alloy				metals	extraction	No	RER	aluminium, production mix, cast alloy, at plant	1.75E-02	kg	various Empa Studies	1	122	(13,13,15,3)
Transport, road			50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	1.75E-02	tkm	various Empa Studies	1	2.05	(13,13,15,5)
Aluminium				metals	extraction	No	RER	aluminium, production mix, at plant	1.10E+00	kg	various Empa Studies	1	122	(13,13,15,3)
Section bar extrusion, aluminium				metals	processing	No	RER	section bar extrusion, aluminium	1.10E+00	kg	various Empa Studies	1	122	(13,13,15,3)
Polystyrene foam slab				insulation materials	production	No	RER	polystyrene foam slab, at plant	1.84E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Polyvinyl-chloride				plastics	polymers	No	RER	polyvinylchloride, at regional storage	3.90E+00	kg	various Empa Studies	1	122	(13,13,15,3)

Fig. 5.9 Flows for “Window frame, plastic (PVC), U=1.6 W/m2K, at plant” (Part 1) and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Transport, road	Window frame, plastic (PVC), U=1.6 W/m2K, at plant, part 3			transport systems	road	No	CH	transport, lorry 20-28t, fleet average	195E-01	tkm	various Empa Studies	1	2.05	(13,13,15,5)
Adhesive			Chemicals, organic as Proxy for Tetrahydrofuran	chemicals	organics	No	GLO	chemicals organic, at plant	2.87E-02	kg	various Empa Studies	1	122	(13,13,15,3)
PVC				plastics	polymers	No	RER	polyvinylchloride, at regional storage	5.96E-02	kg	various Empa Studies	1	122	(13,13,15,3)
Polystyrene			HIPS	plastics	polymers	No	RER	polystyrene, high impact, HIPS, at plant	2.08E-01	kg	various Empa Studies	1	122	(13,13,15,3)
PP			Granulate	plastics	polymers	No	RER	polypropylene, granulate, at plant	2.13E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Rubber EPDM				plastics	polymers	No	RER	synthetic rubber, at plant	2.27E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Transport, road, 50 km			50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	3.68E-02	tkm	various Empa Studies	1	2.05	(13,13,15,5)
Infrastructure			Metal working plant as proxy: Output 43675 t/a --> 4.57E-10	metals	general manufacturing	Yes	RER	metal working factory	4.32E-08	unit	various Empa Studies	1	3.06	(2,4,13,15,9)
		Heat waste	total	air	unspecified			Heat, waste	4.98E+01	MJ	various Empa Studies	1	126	(3,4,13,15,13)
		Window frame, plastic	94.5kg/m2	building components	windows	No	RER	window frame, plastic (PVC), U=1.6 W/m2K, at plant	1.00E+00	m2				

Fig. 5.9 Flows for “Window frame, plastic (PVC), U=1.6 W/m2K, at plant” (Part 1) and their representation in the ecoinvent database (cont)

5.10 Window frame, aluminium, $U=1.6 \text{ W/m}^2\text{K}$, at plant

The meta information of “Window frame, aluminium, $U=1.6 \text{ W/m}^2\text{K}$, at plant” (Tab. 5.10) includes all important and necessary information to use the data correctly.

Tab. 5.10 ecoinvent meta information of the product “Window frame, aluminium, $U=1.6 \text{ W/m}^2\text{K}$, at plant”

Name	Window frame, aluminium, $U=1.6 \text{ W/m}^2\text{K}$, at plant
Location	RER
Infrastructure Process	0
Unit	m^2
Data Set Version	2.0
Included Processes	Included processes are section bar rolling for steel parts and fittings, section bar extrusion for aluminium parts, extrusion of HDPE plastic, surface treatment (powder coating), all the road transport at different production phases, the heat waste and the disposal of the plastic cuttings.
Amount	1
Local Name	Fensterrahmen, Aluminium, $U=1.6 \text{ W/m}^2\text{K}$, ab Werk
Synonyms	
General Comment to reference function	This dataset describes all the processes and material inputs needed to produce a aluminium window frame with 1 m^2 visible area. 1 m^2 of visible aluminium window frame weighs 50.7 kg .
Start Date	1996
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	Window frames produced and sold in Switzerland and Germany.
Technology text	The dataset describes highly automated technology processes in window frame manufacturing.
Representativeness [%]	
Production Volume	
Sampling Procedure	Company documentation, measurements
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this dataset is representative for whole Europe is taken into account in the uncertainties.

Fig. 5.10 shows all in- and output flows of the product “Window frame, aluminium, $U=1.6 \text{ W/m}^2\text{K}$, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Aluminium profile	Window frame, aluminium, U=1.6W/m ² K, part 1	Aluminium cuttings	For frame material	metals	extraction	No	RER	aluminium, production mix, at plant	3.58E+01	kg	various Empa Studies	1	122	(13,13,15,3)
Section bar extrusion			For frame material; Section bar extrusion of aluminium	metals	processing	No	RER	section bar extrusion, aluminium	3.58E+01	kg	various Empa Studies	1	122	(13,13,15,3)
			2% recycled --> cut-off						7.02E-01	kg	various Empa Studies			
Polyamide		Polyamide cuttings	For frame material	plastics	others	No	RER	glass fibre reinforced plastic, polyamide, injection	5.21E+00	kg	various Empa Studies	1	122	(13,13,15,3)
			0.02 waste management		municipal incineration	No	CH	disposal, plastics, mixture, 15.3%water, to municipal	102E-01	kg	various Empa Studies	1	122	(13,13,15,6)
Nylon PA6		Steel cuttings	For frame material	plastics	polymers	No	RER	nylon 6, at plant	146E-02	kg	various Empa Studies	1	122	(13,13,15,3)
Steel			For frame material	metals	extraction	No	RER	chromium steel 18/8, at plant	1.10E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Section bar rolling			For frame material; Section bar rolling of steel	metals	processing	No	RER	section bar rolling, steel	1.13E-01	kg	various Empa Studies	1	122	(13,13,15,3)
			2% recycled --> cut-off						2.21E-03	kg	various Empa Studies			
Die-casting alloy		Aluminium cuttings	For frame material	metals	extraction	No	RER	aluminium, production mix, at plant	174E+00	kg	various Empa Studies	1	122	(13,13,15,3)
Transport, road, 100 km			Transport for frame materials (100 km)	transport systems	road	No	RER	transport, lorry >16t, fleet average	4.21E+00	tkm	various Empa Studies	1	2.05	(13,13,15,5)
Aluminium profile			For fittings	metals	extraction	No	RER	aluminium, production mix, at plant	2.17E+00	kg	various Empa Studies	1	122	(13,13,15,3)
Section bar extrusion			For fittings	metals	processing	No	RER	section bar extrusion, aluminium	2.17E+00	kg	various Empa Studies	1	122	(13,13,15,3)
			2% recycled --> cut-off						4.25E-02	kg	various Empa Studies			
Polyamide			For fittings	plastics	others	No	RER	glass fibre reinforced plastic, polyamide, injection	6.46E-02	kg	various Empa Studies	1	122	(13,13,15,3)
Steel high-alloyed			For fittings	metals	extraction	No	RER	chromium steel 18/8, at plant	3.46E-01	kg	various Empa Studies	1	122	(13,13,15,3)

Fig. 5.10 Flows for “Window frame, aluminium, U=1.6 W/m²K, at plant” (Part 1) and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Steel low-alloyed	Window frame, aluminium, U=1.6W/m2K, part 2	Steel cuttings	For fittings	metals	extraction	No	RER	reinforcing steel, at plant	5.16E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Section bar rolling			For fittings	metals	processing	No	RER	section bar rolling, steel	8.63E-01	kg	various Empa Studies	1	122	(13,13,15,3)
			2% recycled --> cut-off						169E-02	kg	various Empa Studies			
Transport, road, 100 km		Window frame, aluminium, U=1.6W/m2K, part 2	Transport for fitting materials (100 km)	transport systems	road	No	RER	transport, lorry >16t, fleet average	3.04E-01	tkm	various Empa Studies	1	2.05	(13,13,15,5)
Metal adhesive			For Production	paintings	production	No	DE	adhesive for metals, at plant	2.90E-01	kg	various Empa Studies	1	122	(13,13,15,4)
PEHD			For Production	plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	2.46E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Extrusion PEHD			For Production	plastics	processing	No	RER	extrusion, plastic film	2.46E-01	kg	various Empa Studies	1	122	(13,13,15,3)
Propanol			For Production; 2-Propanol, C3H8O; Isopropanol as proxy	chemicals	organics	No	RER	isopropanol, at plant	2.08E-02	kg	various Empa Studies	1	122	(13,13,15,3)
Electricity			For Production; Medium voltage	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	127E+00	kWh	various Empa Studies	1	122	(13,13,15,2)
Transport, road, 100 km			Transport during production phase (100 km)	transport systems	road	No	RER	transport, lorry >16t, fleet average	5.56E-02	tkm	various Empa Studies	1	2.05	(13,13,15,5)
Powder coating			For surface treatment	metals	processing	No	RER	powder coating, aluminium sheet	9.80E+00	m2	various Empa Studies	1	122	(13,13,15,3)
ABS			For material sealing	plastics	polymers	No	RER	acrylonitrile-butadiene-styrene copolymer, ABS, at	4.00E-01	kg	various Empa Studies	1	122	(13,13,15,3)
EPDM			For material sealing; EPDM profile	plastics	polymers	No	RER	synthetic rubber, at plant	4.87E+00	kg	various Empa Studies	1	122	(13,13,15,3)
Infrastructure			Metal working plant as proxy: Output 43675 t/a --> 4.57E-10	metals	general manufacturing	Yes	RER	metal working factory	2.32E-08	unit	various Empa Studies	1	3.06	(2,4,13,15,9)
		Disposal PE		waste management	building demolition	No	CH	disposal, building, polyethylene/polypropylene	2.46E-01	kg	various Empa Studies	1	122	(13,13,15,6)
		Heat, waste		air	unspecified			Heat, waste	4.57E+00	MJ	various Empa Studies	1	122	(13,13,15,13)
		Window frame,	50.7kg/m2	building components	windows	No	RER	window frame, aluminium, U=1.6 W/m2K, at plant	100E+00	m2				

Fig. 5.10 Flows for “Window frame, aluminium, U=1.6 W/m2K, at plant” (Part 1) and their representation in the ecoinvent database (cont)

5.11 Disposal, window frame, wood, in municipal incineration

The meta information of “Disposal, window frame, wood, in municipal incineration” (Tab. 5.11) includes all important and necessary information to use the data correctly.

Tab. 5.11 ecoinvent meta information of the product “Disposal, window frame, wood, in municipal incineration”

Name	Disposal, window frame, wood, in municipal incineration
Location	RER
Infrastructure Process	0
Unit	m ²
Data Set Version	2.0
Included Processes	Included processes are the disposal of wood, paint, rubber, plastic, steel, aluminium, copper and zinc parts, and the transport to the incineration site. Metal parts are too small to be efficiently collected and recycled.
Amount	1
Local Name	Entsorgung, Fensterrahmen, Holz, in Kehrrechtverbrennung
Synonyms	
General Comment to reference function	This dataset describes the disposal of a wooden window frame with 1 m ² visible area via a municipal incineration plant. Distance to incineration site is assumed to be 50 km.
Start Date	1996
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	Swiss practice of waste management of window frames.
Technology text	Waste management includes incineration of wood, paint, rubber, plastic, steel, aluminium, copper and zinc parts, and the transport to the incineration site by average 28t lorry.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Calculation from window frame material composition
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this dataset is representative for the whole Europe is taken into account in the uncertainties.

Fig. 5.11 shows all in- and output flows of the product “Disposal, window frame, wood, in municipal incineration”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Transportation to incineration → Disposal, wooden window frame, in MSWI	→	Disposal wood	inventoried as untreated wood since paint used to	waste management	municipal incineration	No	CH	disposal, wood untreated, 20%water, to municipal	6.52E+01	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal paint		waste management	municipal incineration	No	CH	disposal, paint, 0%water, to municipal incineration	5.23E+00	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal nylon		waste management	municipal incineration	No	CH	disposal, plastics, mixture, 15.3%water, to municipal	3.49E-01	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal steel	not recycled since parts are to small to be efficiently	waste management	municipal incineration	No	CH	disposal, steel, 0%water, to municipal incineration	4.93E+00	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal rubber		waste management	municipal incineration	No	CH	disposal, rubber, unspecified, 0%water, to municipal	1.12E+00	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal aluminium	not recycled since parts are to small to be efficiently	waste management	municipal incineration	No	CH	disposal, aluminium, 0% water, to municipal	1.56E-02	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal copper	not recycled since parts are to small to be efficiently	waste management	municipal incineration	No	CH	disposal, copper, 0%water, to municipal incineration	6.23E-03	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal zinc	not recycled since parts are to small to be efficiently	waste management	municipal incineration	No	CH	disposal, zinc in car shredder residue, 0%water, to	2.90E-01	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal polyethylene		waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal	2.56E-01	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal aluminium	recycled --> cut off						2.84E+00	kg	Calculated from window	1	122	(13,13,15,6)
			50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	3.87E+00	tkm	Calculated	1	2.05	(13,13,15,5)
		Disposal window frame, wood		waste management	building demolition	No	RER	disposal, window frame, wood, in municipal incineration	1.00E+00	m2				

Fig. 5.11 Flows for “Disposal, window frame, wood, in municipal incineration” and their representation in the ecoinvent database

5.12 Disposal, window frame, wood-metal, in municipal incineration

The meta information of “Disposal, window frame, wood-metal, in municipal incineration” (Tab. 5.12) includes all important and necessary information to use the data correctly.

Tab. 5.12 ecoinvent meta information of the product “Disposal, window frame, wood-metal, in municipal incineration”

Name	Disposal, window frame, wood-metal, in municipal incineration
Location	RER
Infrastructure Process	0
Unit	m ²
Data Set Version	2.0
Included Processes	Included processes are the disposal of wood, paint, rubber, plastic, steel, aluminium, copper and zinc parts, and the transport to the incineration site. Metal parts are too small to be efficiently collected and recycled.
Amount	1
Local Name	Entsorgung, Fensterrahmen, Holz-Metall, in Kehrlichtverbrennung
Synonyms	
General Comment to reference function	This dataset describes the disposal of a wood-metal window frame with 1 m ² visible area via a municipal incineration plant. Distance to incineration site is assumed to be 50 km.
Start Date	1996
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	Swiss practice of waste management of window frames.
Technology text	Waste management includes incineration of wood, paint, rubber, plastic, steel, aluminium, copper and zinc parts, and the transport to the incineration site by average 28t lorry.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Calculation from window frame material composition
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this dataset is representative for the whole Europe is taken into account in the uncertainties.

Fig. 5.12 shows all in- and output flows of the product “Disposal, window frame, wood-metal, in municipal incineration”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Transportation to incineration Disposal, wood-metal window frame, in MSWI	→	Disposal wood	inventoried as untreated wood since paint used to	waste management	municipal incineration	No	CH	disposal, wood untreated, 20%water, to municipal	5.69E+01	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal paint		waste management	municipal incineration	No	CH	disposal, paint, 0%water, to municipal incineration	4.80E+00	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal Nylon 66		waste management	municipal incineration	No	CH	disposal, plastics, mixture, 15.3%water, to municipal	109E+00	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal steel	not recycled since parts are to small to be efficiently	waste management	municipal incineration	No	CH	disposal, steel, 0%water, to municipal incineration	4.88E+00	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal rubber	not recycled since parts are to small to be efficiently	waste management	municipal incineration	No	CH	disposal, rubber, unspecified, 0%water, to municipal	2.88E+00	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal aluminium	not recycled since parts are to small to be efficiently	waste management	municipal incineration	No	CH	disposal, aluminium, 0% water, to municipal	156E-02	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal copper	not recycled since parts are to small to be efficiently collected	waste management	municipal incineration	No	CH	disposal, copper, 0%water, to municipal incineration	6.23E-03	kg	Calculated from window composition	1	122	(13,13,15,6)
		Disposal zinc		waste management	municipal incineration	No	CH	disposal, zinc in car shredder residue, 0%water, to	2.90E-01	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal polyethylene		waste management	building demolition	No	CH	disposal, building, polyethylene/polypropylene	2.56E-01	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal aluminium	recycled --> cut off						122E+01	kg	Calculated from window	1	122	(13,13,15,6)
			50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	3.56E+00	tkm	Calculated	1	122	(13,13,15,6)
		Disposal window frame, wood-metal		waste management	building demolition	No	RER	disposal, window frame, wood-metal, in municipal incineration	100E+00	m2				

Fig. 5.12 Flows for “Disposal, window frame, wood-metal, in municipal incineration” and their representation in the ecoinvent database

5.13 Disposal, window frame, plastic, in municipal incineration

The meta information of “Disposal, window frame, plastic, in municipal incineration” (Tab. 5.13) includes all important and necessary information to use the data correctly.

Tab. 5.13 ecoinvent meta information of the product “Disposal, window frame, plastic, in municipal incineration”

Name	Disposal, window frame, plastic, in municipal incineration
Location	RER
Infrastructure Process	0
Unit	m ²
Data Set Version	2.0
Included Processes	Included processes are the disposal of rubber, plastic (polypropylene, expanded polystyrene, polyvinylchloride), steel, aluminium, copper and zinc, and the transport to the incineration site. Metal parts are too small to be efficiently collected and recycled.
Amount	1
Local Name	Entsorgung, Fensterrahmen, Plastik, in Kehrlichtverbrennung
Synonyms	
General Comment to reference function	This dataset describes the disposal of a plastic window frame with 1 m ² visible area via a municipal incineration plant. Distance to incineration site is assumed to be 50 km.
Start Date	1996
End Date	2004
Data Valid For Entire Period	1
Other Period Text	
Geography text	Swiss practice of waste management of window frames.
Technology text	Waste management includes incineration of rubber, plastic, steel, aluminium, copper and zinc parts, and the transport to the incineration site by average 28t lorry.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Calculation from window frame material composition
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this dataset is representative for the whole Europe is taken into account in the uncertainties.

Fig. 5.13 shows all in- and output flows of the product “Disposal, window frame, plastic, in municipal incineration”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Transportation to incineration →	Disposal, plastic window frame, in MSWI, part 1	Disposal steel	Not separated from plastic before disposed in MSWI. Steel is recycled	waste management	municipal incineration	No	CH	disposal, steel, 0%water, to municipal incineration	3.61E+01	kg	Calculated from window composition	1	122	(13,13,15,6)
		Disposal rubber		waste management	municipal incineration	No	CH	disposal, rubber, unspecified, 0%water, to municipal	7.98E-01	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal aluminium	not recycled since parts are too small to be efficiently	waste management	municipal incineration	No	CH	disposal, aluminium, 0%water, to municipal	1.75E-02	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal copper	not recycled since parts are too small to be efficiently	waste management	municipal incineration	No	CH	disposal, copper, 0%water, to municipal incineration	6.98E-03	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal zinc	not recycled since parts are too small to be efficiently	waste management	municipal incineration	No	CH	disposal, zinc in car shredder residue, 0%water, to	3.25E-01	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal expanded		waste management	municipal incineration	No	CH	disposal, expanded polystyrene, 5%water, to	3.92E-01	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal PVC		waste management	building demolition	No	CH	disposal, building, polyvinylchloride products, disposal, building,	5.55E+01	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal PE/PP		waste management	building demolition	No	CH	disposal, building, polyethylene/polypropylene	3.68E-01	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal aluminium	not recycled since parts are too small to be efficiently	waste management	municipal incineration	No	CH	disposal, aluminium, 0%water, to municipal	1.02E+00	kg	Calculated from window	1	122	(13,13,15,6)
		Disposal window frame,	50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	4.72E+00	tkm	Calculated	1	122	(13,13,15,6)
				waste management	building demolition	No	RER	disposal, window frame, plastic, in municipal	1.00E+00	m2				

Fig. 5.13 Flows for “Disposal, window frame, plastic, in municipal incineration” and their representation in the ecoinvent database

5.14 Cladding, crossbar-pole, aluminium, at plant

The meta information of “Cladding, crossbar-pole, aluminium, at plant” (Tab. 5.14) includes all important and necessary information to use the data correctly.

Tab. 5.14 ecoinvent meta information of the product “Cladding, crossbar-pole, aluminium, at plant”

Name	Cladding, crossbar-pole, aluminium, at plant
Location	RER
Infrastructure Process	0
Unit	m ²
Data Set Version	2.0
Included Processes	This dataset describes the processes and materials required to build 1m ² (visible area) of an aluminium cladding, type crossbar-pole. No transport activities between production and construction site are taken into account.
Amount	1
Local Name	Riegel-Pfosten Fassade, Aluminium, ab Werk
Synonyms	
General Comment to reference function	The data is calculated per 1 m ² of cladding. The aluminium cladding weighs 86.7 kg per m ² .
Start Date	2005
End Date	2005
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data based on a recently built public building in Switzerland.
Technology text	The technology represented in this dataset is mainly related to the production of the various parts integrated in the cladding construction.
Representativeness [%]	
Production Volume	
Sampling Procedure	Company documentation
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this Swiss dataset is representative for the whole Europe is taken into account in the uncertainties.

Fig. 5.14 shows all in- and output flows of the product “Cladding, crossbar-pole, aluminium, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Steel, profiles	Cladding Aluminium, part 1			metals	extraction	No	RER	reinforcing steel, at plant	2.41E+01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Steel, profiling				metals	processing	No	RER	section bar rolling, steel	2.41E+01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Powder coating, steel				metals	processing	No	RER	powder coating, steel	8.86E-01	m2	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Steel, sheet				metals	extraction	No	RER	steel, low-alloyed, at plant	122E+00	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Sheet rolling, steel				metals	processing	No	RER	sheet rolling, steel	122E+00	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Galvanizing steel sheet				metals	processing	No	RER	zinc coating, coils	156E-01	m2	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
stainless steel, profile				metals	extraction	No	RER	chromium steel 18/8, at plant	9.12E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Steel, profiling				metals	processing	No	RER	section bar rolling, steel	9.12E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,4)
Aluminium profiles				metals	extraction	No	RER	aluminium, production mix, at plant	3.13E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Profiling Aluminium				metals	processing	No	RER	section bar extrusion, aluminium	3.13E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Small parts, stainless steel				metals	extraction	No	RER	chromium steel 18/8, at plant	179E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Aluminium clamping profile				metals	extraction	No	RER	aluminium, production mix, at plant	9.36E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Aluminium cover profile				metals	extraction	No	RER	aluminium, production mix, at plant	5.37E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,4)
Profiling Aluminium				metals	processing	No	RER	section bar extrusion, aluminium	147E+00	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Powder coating, aluminium				metals	processing	No	RER	powder coating, aluminium sheet	3.08E-01	m2	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
EPDM profile				plastics	polymers	No	RER	synthetic rubber, at plant	124E+00	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
PUR insulation block				plastics	polymers	No	RER	polyurethane, rigid foam, at plant	5.39E-02	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)

Fig. 5.14 Flows for “Cladding, crossbar-pole, aluminium, at plant” (Part 1) and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Aluminium connective profile	Cladding Aluminium, part 2			metals	extraction	No	RER	aluminium, production mix, at plant	100E+00	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Profiling Aluminium				metals	processing	No	RER	section bar extrusion, aluminium	100E+00	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Powder coating, aluminium				metals	processing	No	RER	powder coating, aluminium sheet	2.36E-01	m2	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Aluminium window profile				metals	extraction	No	RER	aluminium, production mix, at plant	8.21E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Profiling Aluminium				metals	processing	No	RER	section bar extrusion, aluminium	8.21E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Powder coating, aluminium				metals	processing	No	RER	powder coating, aluminium sheet	2.04E-01	m2	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,4)
Thermal insulation spacer, polyamid, 25			Polyamid with 25 weight-% glass fiber; according to	plastics	others	No	RER	glass fibre reinforced plastic, polyamide, injection	1.26E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
EPDM profile				plastics	polymers	No	RER	synthetic rubber, at plant	1.06E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Aluminium door profile				metals	extraction	No	RER	aluminium, production mix, at plant	2.14E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Profiling Aluminium				metals	processing	No	RER	section bar extrusion, aluminium	2.14E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Powder coating, aluminium				metals	processing	No	RER	powder coating, aluminium sheet	5.95E-02	m2	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Thermal insulation spacer, polyamid, 25 %glass fibre				plastics	others	No	RER	glass fibre reinforced plastic, polyamide, injection moulding, at plant	2.61E-02	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
EPDM profile				plastics	polymers	No	RER	synthetic rubber, at plant	2.18E-02	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Rock wool 60 kg/m³				insulation materials	production	No	CH	rock wool, at plant	7.20E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Glass wool 38 kg/m³				insulation materials	production	No	CH	glass wool mat, at plant	4.72E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
XPS 20 kg/m³				insulation materials	production	No	RER	polystyrene, extruded (XPS), at plant	2.02E-02	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Vapour barrier				plastics	others	No	DE	natural rubber based sealing, at plant	8.53E-02	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)

Fig. 5.14 Flows for “Cladding, crossbar-pole, aluminium, at plant” (Part 1) and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Water foil outdoor	Cladding Aluminium, part 3		natural rubber (butyl)	plastics	others	No	DE	natural rubber based sealing, at plant	5.56E-02	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
PUR adhesive				paintings	production	No	DE	adhesive for metals, at plant	198E-02	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
PUR Foam				plastics	polymers	No	RER	polyurethane, rigid foam, at plant	9.46E-02	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Plywood			600 kg/m3	wooden materials	extraction	No	RER	plywood, indoor use, at plant	136E-03	m3	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Edge band, hardwood			780 kg/m3 (20%water)	wooden materials	extraction	No	RER	sawn timber, hardwood, planed, air / kiln dried, u=10% at plant	2.79E-04	m3	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
3-IV laminated safety glass			PVB and laminating inventoried separately	building components	windows	No	RER	glazing, triple (3-IV), U<0.5 W/m2K, at plant	6.88E-01	m2	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
PVB film (for LSG)			addition according to 2-IV LSG	plastics	polymers	No	RER	polyvinylchloride, at regional storage	7.23E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Film extrusion				plastics	processing	No	RER	extrusion, plastic film	7.23E-01	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,3)
Electricity for				electricity	production mix	No	UCTE	electricity, medium voltage,	6.66E+00	kWh	Firma Metall-Bau-	1	124	(14,13,15,2)
		→ Steel	to recycling --> cut-off						5.18E-01	kg				
		→ Aluminium	to recycling --> cut-off						7.49E-02	kg				
		→ Plastics		waste management	municipal incineration	No	CH	disposal, plastics, mixture, 15.3%water, to municipal incineration	3.63E-02	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,6)
		→ mineral insulation materials		waste management	inert material landfill	No	CH	disposal, mineral wool, 0% water, to inert material landfill	2.34E-02	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,6)
		→ wood		waste management	municipal incineration	No	CH	disposal, wood untreated, 20%water, to municipal incineration	2.03E-02	kg	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,6)

Fig. 5.14 Flows for “Cladding, crossbar-pole, aluminium, at plant” (Part 1) and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
3-IV Glass	Cladding Aluminium, part 4			waste management	building demolition	No	RER	disposal, glazing 3-IV, U<0.5W/m2K, in municipal incineration	135E-02	m2	Firma Metall-Bau-Technik (2005)	1	124	(14,13,15,6)
Electricity			Same amount as for aluminium window production	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	127E+00	kWh	Own calculations and assumptions	1	124	(14,13,15,2)
Transport road			Metals & plastics 100km, wood & mineral insulation 50 km, glazing 100 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	8.59E+00	tkm	ecoinvent standard distances for metals and plastics	1	2.06	(14,13,15,5)
Transport train			Metals & plastics 200 km, wood 0 km, mineral insulation 100 km, glazing 600 km	transport systems	train	No	RER	transport, freight, rail	3.81E+01	tkm	ecoinvent standard distances for metals and plastics	1	2.06	(14,13,15,5)
Infrastructure			Metal working plant as proxy: Output 43675 t/a --> 4.57E-10 unit/kg product (50a)	metals	general manufacturing	Yes	RER	metal working factory	3.96E-08	unit	Calculated	1	3.06	(2,4,13,15,9)
		Heat, waste		air	unspecified			Heat, waste	2.85E+01	MJ	various Empa Studies	1	122	(13,13,15,13)
		Aluminium cladding	86.7kg/m2	building components	cladding	No	RER	cladding, crossbar-pole, aluminium, at plant	100E+00	m2				

Fig. 5.14 Flows for “Cladding, crossbar-pole, aluminium, at plant” (Part 1) and their representation in the ecoinvent database (cont)

5.15 Door, outer, wood-aluminium, at plant

The meta information of “Door, outer, wood-aluminium, at plant” (Tab. 5.15) includes all important and necessary information to use the data correctly.

Tab. 5.15 ecoinvent meta information of the product “Door, outer, wood-aluminium, at plant”

Name	Door, outer, wood-aluminium, at plant
Location	RER
Infrastructure Process	0
Unit	m ²
Data Set Version	2.0
Included Processes	This dataset describes the production of 1m ² (visible area) outer door, (entrance door), including the processes for the installation. Not included are transports from plant to construction site.
Amount	1
Local Name	Aussentüre, Holz-Aluminium, ab Werk
Synonyms	
General Comment to reference function	The dataset can be applied to describe an average entrance door of a one-family house. The door frame is made of steel (paint coated), the door leaf is made of wood, planked with powder coated aluminium on both sides. The lock, door handles and frame joints are fabricated from steel. Data are based on a door of 0.8x2.0 m and scaled to 1m ² . 1 m ² of the aluminium planked massive wood outer door weighs 38.8 kg. The functional life time for the door frame is 60 years, for the door leaf it is 30 years.
Start Date	1997
End Date	2005
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from producers in Switzerland.
Technology text	The technology represents average European production.
Representativeness [%]	
Production Volume	
Sampling Procedure	Published study, Internet search
Extrapolations	see Geography
Uncertainty Adjustments	none

Fig. 5.15 shows all in- and output flows of the product “Door, outer, wood-aluminium, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Steel for door frame	Outer door, wood-aluminium, background processes	Heat, waste		metals	extraction	No	RER	reinforcing steel, at plant	105E+01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Sheet rolling				metals	processing	No	RER	sheet rolling, steel	105E+01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Zinc coating				metals	processing	No	RER	zinc coating, coils	9.38E-01	m2	Werner et al. 1997	1	122	(13,4,3,1,13)
Thickness correction zinc coating				metals	processing	No	RER	zinc coating, pieces, adjustment per um	5.06E+01	m2	Werner et al. 1997	1	122	(13,4,3,1,13)
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	7.12E-02	kWh	Werner et al. 1997	1	122	(13,4,3,1,12)
Wood for packaging				air	unspecified			Heat, waste	2.56E-01	MJ	calculated	1	122	(13,4,3,1,13)
PE film for packaging				wooden materials	extraction	No	RER	sawn timber, softwood, planed, kiln dried, at plant	2.50E-04	m3	Werner et al. 1997	1	122	(13,4,3,1,14)
PE film for packaging				plastics	polymers	No	RER	polyethylene, LDPE, granulate, at plant	3.13E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Paper				plastics	processing	No	RER	extrusion, plastic film	3.13E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
EPDM				paper & plastics	packaging	No	RER	kraft paper, unbleached, at plant	3.13E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Paraxylol				plastics	polymers	No	RER	synthetic rubber, at plant	3.13E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Alkyd resin				chemicals	organics	No	GLO	chemicals organic, at plant	1.13E-04	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Organic chemicals				paintings	production	No	RER	alkyd resin, long oil, 70% in white spirit, at plant	2.50E-04	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Transport barge				chemicals	organics	No	GLO	chemicals organic, at plant	7.50E-05	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Transport lorry 16t	Outer door, wood-aluminium, manufacturing frame	Heat, waste		transport systems	ship	No	RER	transport, barge	6.06E+00	tkm	Werner et al. 1997	1	122	(13,4,3,1,13)
Transport lorry 16t				transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	1.34E+00	tkm	Werner et al. 1997	1	122	(13,4,3,1,13)
Transport lorry 40t				transport systems	road	No	CH	transport, lorry >28t, fleet average	6.25E-01	tkm	Werner et al. 1997	1	122	(13,4,3,1,13)
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.36E+00	kWh	Werner et al. 1997	1	122	(13,4,3,1,12)
Heat				air	unspecified			Heat, waste	4.91E+00	MJ	calculated	1	122	(13,4,3,1,13)
Xylole				natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace >100kW	2.51E+00	MJ	Werner et al. 1997	1	122	(13,4,3,1,11)
Heptane				chemicals	organics	No	RER	xylene, at plant	1.13E-04	kg	Werner et al. 1997	1	122	(13,13,15,4)
				chemicals	organics	No	RER	heptane, at plant	7.50E-06	kg	Werner et al. 1997	1	122	(13,13,15,4)
			Wood (packaging)	waste management	municipal incineration	No	CH	disposal, wood untreated, 20%water, to municipal incineration	1.13E-01	kg	Werner et al. 1997	1	124	(14,13,15,6)
			PE (packaging)	waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	2.81E-03	kg	Werner et al. 1997	1	122	(13,13,15,6)
			paper (Packaging)	waste management	municipal incineration	No	CH	disposal, paper, 11.2%water, to municipal incineration	2.81E-03	kg	Werner et al. 1997	1	122	(13,13,15,6)

Fig. 5.15 Flows for “Door, outer, wood-aluminium, at plant” and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent 2000								Uncertainty					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Cement mortar	→	Outer door, wood-aluminium, installation frame			mortar and plaster	production	No	CH	cement mortar, at plant			5.00E+01	kg	Werner et al. 1997	1	122	(13,13,15,3)
Water	→				Electricity consumption based on aluminium window frame (scaled with ratio of door frame area to total door area)	water supply	production	No	RER	tap water, at user			5.00E+00	kg	Werner et al. 1997	1	122
Alkyd varnish	→	Outer door, wood-aluminium, surface treatment	→ Municipal waste to incineration		paintings	production	No	RER	alkyd paint, white, 60%in H2O, at plant			2.63E-01	kg	Werner et al. 1997	1	122	(13,4,3,113)
Organic chemicals	→				chemicals	organics	No	GLO	chemicals organic, at plant			125E-03	kg	Werner et al. 1997	1	122	(13,4,3,113)
Tin can	→				metals	extraction	No	RER	steel, low-alloyed, at plant			106E-02	kg	Werner et al. 1997	1	122	(13,4,3,114)
Packaging (cardboard)	→				paper & cardboard	cardboard & corrugated board	No	RER	packaging, corrugated board, mixed fibre, single wall, at plant			125E-01	kg	Werner et al. 1997	1	122	(13,4,3,114)
					waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9%water, to municipal incineration			138E-01	kg	Werner et al. 1997	1	122	(13,4,3,116)
Transport lorry 16t	→				transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average			9.38E-03	tkm	Werner et al. 1997	1	2.05	(13,4,3,115)
Transport Lorry 40t	→				transport systems	road	No	CH	transport, lorry >28t, fleet average			136E-01	tkm	Werner et al. 1997	1	2.05	(13,4,3,115)

Fig. 5.15 Flows for “Door, outer, wood-aluminium, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Aluminium	Outer door, aluminium planked wooden door leaf, part 1		2 sheets of 0.8x2x0.0005 m	metals	extraction	No	RER	aluminium, production mix, at plant	2.70E+00	kg	Calculated from Dimension	1	122	(13,4,3,1,13)
Sheet rolling				metals	processing	No	RER	sheet rolling, aluminium	2.70E+00	kg	Calculated from Dimension	1	122	(13,4,3,1,13)
Powder coating, aluminium			coated after cutting to size	metals	processing	No	RER	powder coating, aluminium sheet	2.00E+00	m2	Calculated from Dimension	1	124	(14,13,15,3)
Softwood			Board of 0.86x2.05x0.045 m	wooden materials	extraction	No	RER	sawn timber, softwood, planed, kiln dried, at plant	4.96E-02	m3	Calculated from Dimension	1	122	(13,4,3,1,13)
Melamine formaldehyde resin			for gluing aluminium to wood	paintings	production	No	RER	melamine formaldehyde resin, at plant	1.13E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Paper				paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	1.69E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Steel for fittings				metals	extraction	No	RER	reinforcing steel, at plant	4.50E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Galvanising fittings				metals	processing	No	RER	zinc coating, pieces	3.94E-02	m2	Werner et al. 1997	1	122	(13,4,3,1,13)
Polyamide				plastics	polymers	No	RER	nylon 66, at plant	2.00E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Aluminium				metals	extraction	No	RER	aluminium, production mix, at plant	1.50E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Anodising aluminium				metals	processing	No	RER	anodising, aluminium sheet	1.38E-02	m2	Werner et al. 1997	1	122	(13,4,3,1,13)
Polyethylene				plastics	polymers	No	RER	polyethylene, LDPE, granulate, at plant	6.25E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Internal transports				construction	machinery	No	GLO	diesel, burned in building	2.47E+00	MJ	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	1.34E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,14)

Fig. 5.15 Flows for “Door, outer, wood-aluminium, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Electricity	Outer door, aluminium planked wooden door leaf, part 2	Heat, waste		electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	4.08E+00	kWh	Werner et al. 1997	1	122	(13,4,3,1,12)
Heat (gas)				air	unspecified			Heat, waste	1.47E+01	MJ	calculated	1	122	(13,4,3,1,13)
				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	3.96E-02	MJ	Werner et al. 1997	1	122	(13,4,3,1,11)
		Industrial residual wood, hardwood	not used for heat production.						4.58E-03	m3	calculated based on Werner et al. 1997	1	0.00	(13,4,3,1,11)
Allocation correction			This flow subtracts the biomass energy, the biogenic CO2 and the resource stored in the industrial residue wood.	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-4.58E-03	m3	calculated	1	122	(13,4,3,1,11)
Transport lorry 40t				transport systems	road	No	CH	transport, lorry >28t, fleet average	5.72E+00	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Transport lorry 16t				transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	1.88E-01	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
		Municipal waste to PE to incineration waste to sanitary		waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9%water, to disposal, polyethylene, 0.4% water, to municipal	2.44E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,16)
				waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal	6.25E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,16)
				waste management	sanitary landfill	No	CH	disposal, inert material, 0% water, to sanitary landfill	6.06E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,16)
Infrastructure		Outer door, wood-aluminium	Metal working plant as proxy: Output 43675 t/a --> 4.57E-10 Entrance door, massive wood, aluminium planked. 27.5 kg/m2 (without mortar for installation)	metals	general manufacturing	Yes	RER	metal working factory	1.26E-08	unit	Calculated	1	3.06	(2,4,13,15,9)
				building components	doors	No	RER	door, outer, wood-aluminium, at plant	1.00E+00	m2				

Fig. 5.15 Flows for “Door, outer, wood-aluminium, at plant” and their representation in the ecoinvent database (cont)

5.16 Door, outer, wood-glass, at plant

The meta information of “Door, outer, wood-glass, at plant” (Tab. 5.16) includes all important and necessary information to use the data correctly.

Tab. 5.16 ecoinvent meta information of the product “Door, outer, wood-glass, at plant”

Name	Door, outer, wood-glass, at plant
Location	RER
Infrastructure Process	0
Unit	m ²
Data Set Version	2.0
Included Processes	This dataset describes the production of 1m ² (visible area) outer door, (entrance door), including the processes for the installation. Not included are transports from plant to construction site.
Amount	1
Local Name	Aussentüre, Holz-Glas, ab Werk
Synonyms	
General Comment to reference function	The dataset can be applied to describe an average entrance door of a one-family house. The door frame is made of steel (paint coated), the door leaf is made of wood, planked with powder coated aluminium on both sides, with a glass insert (62 % wood, 38% glass). The lock, door handles and frame joints are fabricated from steel. Data are based on a door of 0.8x2.0 m and scaled to 1m ² . 1 m ² of the aluminium planked massive wood-glass outer door weighs 36.5 kg. The functional life time for the door frame is 60 years, for the door leaf it is 30 years.
Start Date	1997
End Date	2005
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from producers in Switzerland.
Technology text	The technology represents average European production.
Representativeness [%]	
Production Volume	
Sampling Procedure	Published study, internet search
Extrapolations	see Geography
Uncertainty Adjustments	none

Fig. 5.16 shows all in- and output flows of the product “Door, outer, wood-glass, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Steel for door frame	Outer door, wood-glass, background processes			metals	extraction	No	RER	reinforcing steel, at plant	105E+01	kg	Werner et al. 1997	1	122	(13,4,3,1,1,3)
Sheet rolling				metals	processing	No	RER	sheet rolling, steel	105E+01	kg	Werner et al. 1997	1	122	(13,4,3,1,1,3)
Zinc coating				metals	processing	No	RER	zinc coating, coils	9.38E-01	m2	Werner et al. 1997	1	122	(13,4,3,1,1,3)
Thickness correction zinc coating				metals	processing	No	RER	zinc coating, pieces, adjustment per um	5.06E+01	m2	Werner et al. 1997	1	122	(13,4,3,1,1,3)
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	7.12E-02	kWh	Werner et al. 1997	1	122	(13,4,3,1,1,2)
		Heat, waste		air	unspecified			Heat, waste	2.56E-01	MJ	calculated	1	122	(13,4,3,1,1,3)
Wood for packaging				wooden materials	extraction	No	RER	sawn timber, softwood, planed, kiln dried, at plant	2.50E-04	m3	Werner et al. 1997	1	122	(13,4,3,1,1,4)
PE film for packaging				plastics	polymers	No	RER	polyethylene, LDPE, granulate, at plant	3.13E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,1,4)
PE film for packaging				plastics	processing	No	RER	extrusion, plastic film	3.13E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,1,4)
Paper				paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	3.13E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,1,4)
EPDM				plastics	polymers	No	RER	synthetic rubber, at plant	3.13E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,1,3)
Paraxylol				chemicals	organics	No	GLO	chemicals organic, at plant	1.13E-04	kg	Werner et al. 1997	1	122	(13,4,3,1,1,3)
Alkyd resin				paintings	production	No	RER	alkyd resin, long oil, 70% in white spirit, at plant	2.50E-04	kg	Werner et al. 1997	1	122	(13,4,3,1,1,3)
Organic chemicals				chemicals	organics	No	GLO	chemicals organic, at plant	7.50E-05	kg	Werner et al. 1997	1	122	(13,4,3,1,1,3)
Transport barge				transport systems	ship	No	RER	transport, barge	6.06E+00	tkm	Werner et al. 1997	1	122	(13,4,3,1,1,3)
Transport lorry 16t				transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	134E+00	tkm	Werner et al. 1997	1	122	(13,4,3,1,1,3)
Transport Lorry 40t				transport systems	road	No	CH	transport, lorry >28t, fleet average	6.25E-01	tkm	Werner et al. 1997	1	122	(13,4,3,1,1,3)

Fig. 5.16 Flows for “Door, outer, wood-glass, at plant” and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Electricity	Outer door, wood-glass, manufacturing frame	Heat, waste		electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	136E+00	kWh	Werner et al. 1997	1	122	(13,4,3,1,12)
Heat				air	unspecified			Heat, waste	4.91E+00	MJ	calculated	1	122	(13,4,3,1,113)
Xylole				natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace >100kW	2.51E+00	MJ	Werner et al. 1997	1	122	(13,4,3,1,11)
Heptane				chemicals	organics	No	RER	xylene, at plant	1.13E-04	kg	Werner et al. 1997	1	122	(13,1,3,1,5,4)
		Wood (packaging) PE (packaging) paper (Packaging)		chemicals	organics	No	RER	heptane, at plant	7.50E-06	kg	Werner et al. 1997	1	122	(13,1,3,1,5,4)
				waste management	municipal incineration	No	CH	disposal, wood untreated, 20%water, to municipal incineration	1.13E-01	kg	Werner et al. 1997	1	124	(14,1,3,1,5,6)
				waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	2.81E-03	kg	Werner et al. 1997	1	122	(13,1,3,1,5,6)
				waste management	municipal incineration	No	CH	disposal, paper, 11.2%water, to municipal incineration	2.81E-03	kg	Werner et al. 1997	1	122	(13,1,3,1,5,6)
				mortar and plaster	production	No	CH	cement mortar, at plant	5.00E+01	kg	Werner et al. 1997	1	122	(13,1,3,1,5,3)
			Electricity consumption based on aluminium window frame (scaled with ratio of door frame area to total door area)	water supply	production	No	RER	tap water, at user	5.00E+00	kg	Werner et al. 1997	1	122	(13,1,3,1,5,4)
Cement mortar	Outer door, wood-glass, installation frame			paintings	production	No	RER	alkyd paint, white, 60%in H2O, at plant	2.63E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Water				chemicals	organics	No	GLO	chemicals organic, at plant	1.25E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Alkyd varnish	Outer door, wood-glass, surface treatment	Municipal waste to incineration		metals	extraction	No	RER	steel, low-alloyed, at plant	1.06E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Organic chemicals				paper & cardboard	cardboard & corrugated board	No	RER	packaging, corrugated board, mixed fibre, single wall, at plant	1.25E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Tin can				waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9%water, to municipal incineration	1.38E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,16)
Packaging (cardboard)				transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	9.38E-03	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Transport lorry 16t				transport systems	road	No	CH	transport, lorry >28t, fleet average	1.36E-01	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Transport Lorry 40t														

Fig. 5.16 Flows for “Door, outer, wood-glass, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Glass	Outer door, aluminium planked wooden door leaf with glass insert. 62 % of wooden door leaf + 0.38 m2 2-IV glazing, part 1			building components	windows	No	RER	glazing, double (2-IV), U<1.1 W/m2K, laminated safety glass, at plant	3.80E-01	m2	Calculated	1	122	(13,13,15,4)
Aluminium			2 sheets of 0.8x2x0.0005 m	metals	extraction	No	RER	aluminium, production mix, at plant	167E+00	kg	Calculated from Dimension	1	122	(13,4,3,11,3)
Sheet rolling				metals	processing	No	RER	sheet rolling, aluminium	167E+00	kg	Calculated from Dimension	1	122	(13,4,3,11,3)
Powder coating, aluminium			coated after cutting to size	metals	processing	No	RER	powder coating, aluminium sheet	124E+00	m2	Calculated from Dimension	1	124	(14,13,15,3)
Softwood			Board of 0.86x2.05x0.045 m	wooden materials	extraction	No	RER	sawn timber, softwood, planed, kiln dried, at plant	3.07E-02	m3	Calculated from Dimension	1	122	(13,4,3,11,3)
Melamine formaldehyde resin			for gluing aluminium to wood	paintings	production	No	RER	melamine formaldehyde resin, at plant	6.98E-02	kg	Werner et al. 1997	1	122	(13,4,3,11,3)
Paper				paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	105E-01	kg	Werner et al. 1997	1	122	(13,4,3,11,4)
Steel for fittings			total amount of wooden door leaf	metals	extraction	No	RER	reinforcing steel, at plant	4.50E-01	kg	Werner et al. 1997	1	122	(13,4,3,11,3)
Galvanising fittings			total amount of wooden door leaf	metals	processing	No	RER	zinc coating, pieces	3.94E-02	m2	Werner et al. 1997	1	122	(13,4,3,11,3)
Polyamide			total amount of wooden door leaf	plastics	polymers	No	RER	nylon 66, at plant	2.00E-03	kg	Werner et al. 1997	1	122	(13,4,3,11,3)
Aluminium			total amount of wooden door leaf	metals	extraction	No	RER	aluminium, production mix, at plant	150E-01	kg	Werner et al. 1997	1	122	(13,4,3,11,3)
Anodising aluminium			total amount of wooden door leaf	metals	processing	No	RER	anodising, aluminium sheet	138E-02	m2	Werner et al. 1997	1	122	(13,4,3,11,3)
Polyethylene			total amount of wooden door leaf	plastics	polymers	No	RER	polyethylene, LDPE, granulate, at plant	6.25E-01	kg	Werner et al. 1997	1	122	(13,4,3,11,4)

Fig. 5.16 Flows for “Door, outer, wood-glass, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Internal transports	Outer door, aluminium planked wooden door leaf with glass insert. 62 % of wooden door leaf + 0.38 m2 2-IV glazing, part 2	Heat, waste	total amount of wooden door leaf	construction processes	machinery	No	GLO	diesel, burned in building machine	2.47E+00	MJ	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	8.31E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Electricity			total amount of wooden door leaf	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	4.08E+00	kWh	Werner et al. 1997	1	122	(13,4,3,1,12)
			total amount of wooden door leaf	air	unspecified			Heat, waste	1.47E+01	MJ	calculated	1	122	(13,4,3,1,13)
Heat (gas)			total amount of wooden door leaf	natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	3.96E-02	MJ	Werner et al. 1997	1	122	(13,4,3,1,11)
		Industrial residual wood, hardwood	not used for heat production.						2.84E-03	m3	calculated based on Werner et al. 1997	1	0.00	(13,4,3,1,11)
Allocation correction			This flow subtracts the biomass energy, the biogenic CO2 and the resource stored in the industrial residue wood.	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-2.84E-03	m3	calculated	1	122	(13,4,3,1,11)
Transport lorry 40t		Municipal waste to incineration	600 km assumed for glazing	transport systems	road	No	CH	transport, lorry >28t, fleet average	8.36E+00	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Transport lorry 16t			100 km assumed for glazing	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	9.18E-01	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
			total amount of wooden door leaf since waste from packaging is the same	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9%water, to municipal incineration	2.44E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,16)
			total amount of wooden door leaf since waste from packaging is the same	waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	6.25E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,16)
			total amount of wooden door leaf since waste from packaging is the same	waste management	sanitary landfill	No	CH	disposal, inert material, 0% water, to sanitary landfill	6.06E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,16)
Infrastructure		Outer door, wood-glass	Metal working plant as proxy: Output 43675 t/a --> 4.57E-10 unit/kg product (50a)	metals	general manufacturing	Yes	RER	metal working factory	1.35E-08	unit	Calculated	1	3.06	(2,4,13,15,9)
			Entrance door, massive wood (aluminium planked) and glass. 35.9 kg/m2 (without mortar for installation)	building components	doors	No	RER	door, outer, wood-glass, at plant	1.00E+00	m2				

Fig. 5.16 Flows for “Door, outer, wood-glass, at plant” and their representation in the ecoinvent database (cont)

5.17 Door, inner, wood, at plant

The meta information of “Door, inner, wood, at plant” (Tab. 5.17) includes all important and necessary information to use the data correctly.

Tab. 5.17 ecoinvent meta information of the product “Door, inner, wood, at plant”

Name	Door, inner, wood, at plant
Location	RER
Infrastructure Process	0
Unit	m ²
Data Set Version	2.0
Included Processes	This dataset describes the production of 1m ² (visible area) inner door, including the processes for the installation. Not included are transports from plant to construction site.
Amount	1
Local Name	Innentüre, Holz, ab Werk
Synonyms	
General Comment to reference function	The dataset can be applied to describe an average room door. The door frame is made of massive wood, the door leaf is made of wood. The lock, door handles and frame joints are fabricated from steel. Data are based on a door of 0.8x2.0 m and scaled to 1m ² . 1 m ² of the wooden inner door weighs 27.6 kg. The functional life time for the door frame is 50 years, for the door leaf it is 30 years.
Start Date	1997
End Date	2005
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from producers in Switzerland.
Technology text	The technology represents average European production.
Representativeness [%]	
Production Volume	
Sampling Procedure	Published study, Internet search
Extrapolations	see Geography
Uncertainty Adjustments	none

Fig. 5.17 shows all in- and output flows of the product “Door, inner, wood, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Soft wood for door frame	Inner door, wood, frame			wooden materials	extraction	No	RER	sawn timber, softwood, planed, kiln dried, at plant	3.13E-02	m3	Werner et al. 1997	1	122	(1,3,4,3,1,13)
Transport lorry 16t				wood for frame to plant	transport systems	No	CH	transport, lorry 3.5-20t, fleet average	3.46E+00	tkm	Werner et al. 1997	1	2.05	(1,3,4,3,1,15)
Steel for fittings	Inner door, wood, fittings			metals	extraction	No	RER	reinforcing steel, at plant	2.13E-01	kg	Werner et al. 1997	1	122	(1,3,4,3,1,13)
Galvanising fittings				metals	processing	No	RER	zinc coating, pieces	1.25E-02	m2	Werner et al. 1997	1	122	(1,3,4,3,1,13)
EPDM				plastics	polymers	No	RER	synthetic rubber, at plant	1.56E-01	kg	Werner et al. 1997	1	122	(1,3,4,3,1,13)
Polyamid				plastics	polymers	No	RER	nylon 66, at plant	2.50E-02	kg	Werner et al. 1997	1	122	(1,3,4,3,1,13)
Solder				metals	extraction	No	RER	brazing solder, cadmium free, at plant	3.75E-04	kg	Werner et al. 1997	1	122	(1,3,4,3,1,13)
Ammonia				chemicals	inorganics	No	RER	ammonia, liquid, at regional storehouse	3.98E-05	kg	Werner et al. 1997	1	122	(1,3,4,3,1,14)
HCl				chemicals	inorganics	No	RER	hydrochloric acid, 30% in H ₂ O, at plant	6.88E-04	kg	Werner et al. 1997	1	122	(1,3,4,3,1,14)
Zinc				metals	extraction	No	RER	zinc, primary, at regional storage	5.40E-04	kg	Werner et al. 1997	1	122	(1,3,4,3,1,14)
Ethylene dichloride				chemicals	organics	No	RER	ethylene dichloride, at plant	3.67E-05	kg	Werner et al. 1997	1	122	(1,3,4,3,1,14)
Lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	4.93E-03	kg	Werner et al. 1997	1	122	(1,3,4,3,1,14)
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.07E-01	kWh	Werner et al. 1997	1	122	(1,3,4,3,1,12)
Heat				air	unspecified			Heat, waste	7.44E-01	MJ	calculated	1	122	(1,3,4,3,1,13)
Internal transports				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	1.14E-02	MJ	Werner et al. 1997	1	122	(1,3,4,3,1,11)
				construction processes	machinery	No	GLO	diesel, burned in building machine	1.06E-01	MJ	Werner et al. 1997	1	2.05	(1,3,4,3,1,15)

Fig. 5.17 Flows for “Door, inner, wood, at plant” and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Transport lorry 40t	Inner door, wood, fittings	Plastics to incineration		transport systems	road	No	CH	transport, lorry >28t, fleet average	3.63E-01	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Transport lorry 16t				transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	1.13E-02	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Transport rail				transport systems	train	No	RER	transport, freight, rail	2.50E-03	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
				waste management	municipal incineration	No	CH	disposal, plastics, mixture, 15.3%water, to municipal incineration	7.50E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,16)
				waste management	hazardous waste incineration	No	CH	disposal, hazardous waste, 25%water, to hazardous waste incineration	3.30E-05	kg	Werner et al. 1997	1	122	(13,4,3,1,16)
				waste management	sanitary landfill	No	CH	disposal, inert material, 0% water, to sanitary landfill	1.88E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,16)
				waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10%water, to hazardous waste incineration	2.50E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,16)
				air	unspecified			NM VOC, non-methane volatile organic compounds, unspecified origin	4.88E-04	kg	Werner et al. 1997	1	2.05	(13,4,3,1,123)
				air	unspecified			Ethane, 1,2-dichloro-	3.67E-06	kg	Werner et al. 1997	1	2.05	(13,4,3,1,123)

Fig. 5.17 Flows for “Door, inner, wood, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Lubricating oil	Inner door, wood, manufacturing raw frame	Heat, waste		chemicals	organics	No	RER	lubricating oil, at plant	8.94E-05	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.13E+00	kWh	Werner et al. 1997	1	122	(13,4,3,1,12)
				air	unspecified			Heat, waste	1.13E+01	MJ	calculated	1	122	(13,4,3,1,13)
Transport lorry 16t				transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	1.25E-03	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Heat from burning of residual wood from process			35% of residual wood = 0.011 m ³ --> wood needed for energy production = 0.0204 m ³ = 175 MJ (8545 MJ/m ³ at 10% water content)	wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 1000kW	1.09E+02	MJ	calculated based on Werner et al. 1997	1	122	(13,4,3,1,11)
Correction of wood input		Industrial residual wood, softwood	0.000316 m ³ / KJ	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40% at plant	3.45E-02	m ³	calculated based on Werner et al. 1997	1	122	(13,4,3,1,11)
Correction of transport of wood chips			0.00283 tkm/MJ	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	3.09E-01	tkm	calculated based on Werner et al. 1997	1	122	(13,4,3,1,11)
			not used for heat production.						6.88E-03	m ³	calculated based on Werner et al. 1997			(13,4,3,1,11)
Allocation correction			This flow subtracts the biomass energy, the biogenic CO ₂ and the resource stored in the industrial residue wood.	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-6.88E-03	m ³	calculated	1	122	(13,4,3,1,11)

Fig. 5.17 Flows for “Door, inner, wood, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000								Uncertainty				
Input		Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
Acrylic filler	→	Inner door, wood, frame, surface treatment	Heat, waste		mortar and plaster	production	No	RER	acrylic filler, at plant	125E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,13)	
Acrylic dispersion	→				paintings	production	No	RER	acrylic dispersion, 65%in H2O, at plant	188E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)	
Acrylic varnish	→				paintings	production	No	RER	acrylic varnish, 87.5%in H2O, at plant	188E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)	
Tin can	→				metals	extraction	No	RER	steel, low-alloyed, at plant	156E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,13)	
Packaging (cardboard)	→				paper & cardboard	cardboard & corrugated board	No	RER	packaging, corrugated board, mixed fibre, single wall, at plant	5.61E-04	kg	Werner et al. 1997	1	122	(13,4,3,1,13)	
Mineral wool	→				insulation materials	production	No	CH	glass wool mat, at plant	8.00E-05	kg	Werner et al. 1997	1	122	(13,4,3,1,13)	
Organic chemicals	→				chemicals	organics	No	GLO	chemicals organic, at plant	101E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,13)	
Electricity	→				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	7.92E+00	kWh	Werner et al. 1997	1	122	(13,4,3,1,12)	
					air	unspecified			Heat, waste	2.85E+01	MJ	calculated	1	122	(13,4,3,1,13)	
Transport lorry 40t	→					transport systems	road	No	CH	transport, lorry >28t, fleet average	2.00E-01	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Transport lorry 16t	→			transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	125E-02	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)		
Transport rail	→			transport systems	train	No	RER	transport, freight, rail	8.75E-03	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)		
			Aromatic hydrocarbons to air	Not inventoried sincetoday water based coatings are used					2.44E-02		Werner et al. 1997			(13,4,3,1,123)		
			Heptane to air						1.19E-01		Werner et al. 1997			(13,4,3,1,123)		
			Municipal waste to incineration		waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9%water, to municipal incineration	7.06E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,16)	
Screws	→	Inner door, wood frame, installation	Heat, waste		metals	extraction	No	RER	reinforcing steel, at plant	8.75E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,13)	
Galvanising screws	→				metals	processing	No	RER	zinc coating, pieces	8.13E-03	m2	Werner et al. 1997	1	122	(13,4,3,1,13)	
Dowel (hardwood)	→				Density = 715 kg/m3	wooden materials	extraction	No	RER	sawn timber, hardwood, planed, kiln dried, u=10% at plant	5.24E-06	m3	various Empa Studies	1	122	(13,4,3,1,13)
Lubricating oil	→				chemicals	organics	No	RER	lubricating oil, at plant	1.10E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,14)	
Electricity	→				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.88E-03	kWh	Werner et al. 1997	1	122	(13,4,3,1,12)	
					air	unspecified			Heat, waste	104E-02	MJ	calculated	1	122	(13,4,3,1,13)	

Fig. 5.17 Flows for “Door, inner, wood, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Particle board	Inner door, wood, door leaf, part 1			wooden materials	extraction	No	RER	particle board, indoor use, at plant	181E-02	m3	Werner et al. 1997	1	122	(13,4,3,1,13)
Fibre board				wooden materials	extraction	No	RER	fibreboard hard, at plant	6.88E-03	m3	Werner et al. 1997	1	122	(13,4,3,1,13)
Softwood				wooden materials	extraction	No	RER	sawn timber, softwood, planed, kiln dried, at plant	8.13E-03	m3	Werner et al. 1997	1	122	(13,4,3,1,13)
PVAc glue				plastics	monomers	No	RER	vinyl acetate, at plant	2.13E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Paper				paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	1.69E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Melamine formaldehyde resin				paintings	production	No	RER	melamine formaldehyde resin, at plant	1.13E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Urea formaldehyde resin				paintings	production	No	RER	urea formaldehyde resin, at plant	2.31E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Ammonia				chemicals	inorganics	No	RER	ammonia, liquid, at regional storehouse	7.50E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
HCl				chemicals	inorganics	No	RER	hydrochloric acid, 30% in H2O, at plant	1.56E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
wheat flour				agricultural means of production	feed	No	CH	wheat IP, at feed mill	5.69E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Steel for fittings				metals	extraction	No	RER	reinforcing steel, at plant	4.50E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Galvanising fittings				metals	processing	No	RER	zinc coating, pieces	3.94E-02	m2	Werner et al. 1997	1	122	(13,4,3,1,13)
Polyamide				plastics	polymers	No	RER	nylon 66, at plant	2.00E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Aluminium				metals	extraction	No	RER	aluminium, production mix, at plant	1.50E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)

Fig. 5.17 Flows for “Door, inner, wood, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Anodising aluminium	Inner door, wood, door leaf, part 2	Heat, waste		metals	processing	No	RER	anodising, aluminium sheet	138E-02	m2	Werner et al. 1997	1	122	(13,4,3,1,13)
Alkyd varnish				paintings	production	No	RER	alkyd paint, white, 60%in H2O, at plant	8.44E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Organic chemicals				chemicals	organics	No	GLO	chemicals organic, at plant	2.59E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Tin can				metals	extraction	No	RER	steel, low-alloyed, at plant	3.88E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Mineral wool				insulation materials	production	No	CH	glass wool mat, at plant	2.07E-04	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Packaging (cardboard)				paper & cardboard	cardboard & corrugated board	No	RER	packaging, corrugated board, mixed fibre, single wall, at plant	2.72E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Polyethylene				plastics	polymers	No	RER	polyethylene, LDPE, granulate, at plant	6.25E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Internal transports				construction processes	machinery	No	GLO	diesel, burned in building machine	2.47E+00	MJ	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	134E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	4.08E+00	kWh	Werner et al. 1997	1	122	(13,4,3,1,12)
Heat (gas)				air	unspecified			Heat, waste	147E+01	MJ	calculated	1	122	(13,4,3,1,13)
Heat (residual wood)				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	3.96E-02	MJ	Werner et al. 1997	1	122	(13,4,3,1,11)
Correction of wood input				Biogenic CO2 emission = 2.62 kg --> 0.715 kg C in residual wood --> 144 kg residual wood (atro) --> 3.21E+03 m3 = 27.4 MJ (8545 MJ/m3 at 10%water content)										
			0.000316 m3 / KJ	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40% at plant	5.41E-03	m3	calculated based on Werner et al. 1997	1	122	(13,4,3,1,11)

Fig. 5.17 Flows for “Door, inner, wood, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Correction of transport of wood chips	Inner door, wood, door leaf, part 3	Industrial residual wood, hardwood	0.00283 tkm/MJ	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	4.85E-02	tkm	calculated based on Werner et al. 1997	1	122	(13,4,3,1,1,1)
			not used for heat production.						2.19E-05	m3	calculated based on Werner et al. 1997			(13,4,3,1,1,1)
Allocation correction		Industrial residual wood, hardwood	This flow subtracts the biomass energy, the biogenic CO2 and the resource stored in the industrial residue wood.	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	-2.19E-05	m3	calculated	1	122	(13,4,3,1,1,1)
			not used for heat production.						2.44E-05	m3	calculated based on Werner et al. 1997			(13,4,3,1,1,1)
Allocation correction			This flow subtracts the biomass energy, the biogenic CO2 and the resource stored in the industrial residue wood.	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-2.44E-05	m3	calculated	1	122	(13,4,3,1,1,1)
Transport lorry 40t				transport systems	road	No	CH	transport, lorry >28t, fleet average	5.72E+00	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,1,5)
Transport lorry 16t				transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	1.88E-01	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,1,5)
		Municipal waste to incineration		waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9%water, to municipal incineration	2.44E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,1,6)
		PE to incineration		waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	6.25E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,1,6)
		waste to sanitary landfill		waste management	sanitary landfill	No	CH	disposal, inert material, 0% water, to sanitary landfill	6.06E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,1,6)
Infrastructure			Metal working plant as proxy: Output 43675 t/a --> 4.57E-10 unit/kg product (50a)	metals	general manufacturing	Yes	RER	metal working factory	1.26E-08	unit	Calculated	1	3.06	(2,4,13,15,9)
		Inner door, wood	27.5 kg/m2	building components	doors	No	RER	door, inner, wood, at plant	1.00E+00	m2				

Fig. 5.17 Flows for “Door, inner, wood, at plant” and their representation in the ecoinvent database (cont)

5.18 Door, inner, glass-wood, at plant

The meta information of “Door, inner, glass-wood, at plant” (Tab. 5.18) includes all important and necessary information to use the data correctly.

Tab. 5.18 ecoinvent meta information of the product “Door, inner, glass-wood, at plant”

Name	Door, inner, glass-wood, at plant
Location	RER
Infrastructure Process	0
Unit	m ²
Data Set Version	2.0
Included Processes	This dataset describes the production of 1m ² (visible area) inner door, including the processes for the installation. Not included are transports from plant to construction site.
Amount	1
Local Name	Innentüre, Glas-Holz, ab Werk
Synonyms	
General Comment to reference function	The dataset can be applied to describe an average room door. The door frame is made of massive wood, the door leaf is made of wood with a glass insert (62 % wood, 38% glass). The lock, door handles and frame joints are fabricated from steel. Data are based on a door of 0.8x2.0 m and scaled to 1m ² . 1 m ² of the wooden inner door weighs 27.6 kg. The functional life time for the door frame is 50 years, for the door leaf it is 30 years.
Start Date	1997
End Date	2005
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from producers in Switzerland.
Technology text	The technology represents average European production.
Representativeness [%]	
Production Volume	
Sampling Procedure	Published study, Internet search
Extrapolations	see Geography
Uncertainty Adjustments	none

Fig. 5.18 shows all in- and output flows of the product “Door, inner, glass-wood, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Soft wood for door frame	Inner door, wood-glass, wood-glass, frame part 1			wooden materials	extraction	No	RER	sawn timber, softwood, planed, kiln dried, at plant	3.13E-02	m3	Werner et al. 1997	1	122	(13,4,3,1,13)
Transport lorry 16t				wood for frame to plant	transport systems	No	CH	transport, lorry 3.5-20t, fleet average	3.46E+00	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Steel for fittings				metals	extraction	No	RER	reinforcing steel, at plant	2.13E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Galvanising fittings				metals	processing	No	RER	zinc coating, pieces	125E-02	m2	Werner et al. 1997	1	122	(13,4,3,1,13)
EPDM				plastics	polymers	No	RER	synthetic rubber, at plant	156E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Polyamid				plastics	polymers	No	RER	nylon 66, at plant	2.50E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Solder				metals	extraction	No	RER	brazing solder, cadmium free, at plant	3.75E-04	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Ammonia				chemicals	inorganics	No	RER	ammonia, liquid, at regional storehouse	3.98E-05	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
HCl				chemicals	inorganics	No	RER	hydrochloric acid, 30% in H2O, at plant	6.88E-04	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Zinc				metals	extraction	No	RER	zinc, primary, at regional storage	5.40E-04	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Ethylene dichloride				chemicals	organics	No	RER	ethylene dichloride, at plant	3.67E-05	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	4.93E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.07E-01	kWh	Werner et al. 1997	1	122	(13,4,3,1,12)
		Heat, waste		air	unspecified			Heat, waste	7.44E-01	MJ	calculated	1	122	(13,4,3,1,13)

Fig. 5.18 Flows for “Door, inner, glass-wood, at plant” and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment

Heat	Inner door, wood-glass, fittings, part 2				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	1.14E-02	MJ	Werner et al. 1997	1	1.22	(13,4,3,1,1,1)
Internal transports					construction processes	machinery	No	GLO	diesel, burned in building machine	1.06E-01	MJ	Werner et al. 1997	1	2.05	(13,4,3,1,1,5)
Transport lorry 40t					transport systems	road	No	CH	transport, lorry >28t, fleet average	3.63E-01	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,1,5)
Transport lorry 16t					transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	1.13E-02	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,1,5)
Transport rail					transport systems	train	No	RER	transport, freight, rail	2.50E-03	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,1,5)
		Plastics to incineration			waste management	municipal incineration	No	CH	disposal, plastics, mixture, 15.3%water, to municipal incineration	7.50E-03	kg	Werner et al. 1997	1	1.22	(13,4,3,1,1,6)
		Hazardous waste			waste management	hazardous waste incineration	No	CH	disposal, hazardous waste, 25%water, to hazardous waste incineration	3.30E-05	kg	Werner et al. 1997	1	1.22	(13,4,3,1,1,6)
		waste to sanitary landfill			waste management	sanitary landfill	No	CH	disposal, inert material, 0% water, to sanitary landfill	1.88E-03	kg	Werner et al. 1997	1	1.22	(13,4,3,1,1,6)
		waste oil			waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10%water, to hazardous waste incineration	2.50E-03	kg	Werner et al. 1997	1	1.22	(13,4,3,1,1,6)
		NM VOC			air	unspecified			NM VOC, non-methane volatile organic compounds, unspecified origin	4.88E-04	kg	Werner et al. 1997	1	2.05	(13,4,3,1,1,23)
		Ethylene dichloride			air	unspecified			Ethane, 1,2-dichloro-	3.67E-06	kg	Werner et al. 1997	1	2.05	(13,4,3,1,1,23)

Fig. 5.18 Flows for “Door, inner, glass-wood, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Lubricating oil	Inner door, wood-glass, manufacturing raw frame	Heat, waste		chemicals	organics	No	RER	lubricating oil, at plant	8.94E-05	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.13E+00	kWh	Werner et al. 1997	1	122	(13,4,3,1,12)
				air	unspecified			Heat, waste	1.13E+01	MJ	calculated	1	122	(13,4,3,1,13)
Transport lorry 16t		Industrial residual wood, softwood		transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	1.25E-03	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Heat from burning of residual wood from process			35% of residual wood = 0.011 m3 --> wood needed for energy production = 0.0204 m3 = 175 MJ (8545 MJ/m3 at 10% water content)	wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 1000kW	1.09E+02	MJ	calculated based on Werner et al. 1997	1	122	(13,4,3,1,11)
Correction of wood input			0.000316 m3 / KJ	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40% at plant	3.45E-02	m3	calculated based on Werner et al. 1997	1	122	(13,4,3,1,11)
Correction of transport of wood chips			0.00283 tkm/MJ	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	3.09E-01	tkm	calculated based on Werner et al. 1997	1	122	(13,4,3,1,11)
			not used for heat production.						6.88E-03	m3	calculated based on Werner et al. 1997			(13,4,3,1,11)
Allocation correction			This flow subtracts the biomass energy, the biogenic CO2 and the resource stored in the industrial residue wood.	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-6.88E-03	m3	calculated	1	122	(13,4,3,1,11)

Fig. 5.18 Flows for “Door, inner, glass-wood, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Acrylic filler	Inner door, wood-glass, frame, surface treatment	Heat, waste		mortar and plaster	production	No	RER	acrylic filler, at plant	125E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Acrylic dispersion				paintings	production	No	RER	acrylic dispersion, 65%in H ₂ O, at plant	188E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Acrylic varnish				paintings	production	No	RER	acrylic varnish, 87.5%in H ₂ O, at plant	188E-01	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Tin can				metals	extraction	No	RER	steel, low-alloyed, at plant	156E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Packaging (cardboard)				paper & cardboard	cardboard & corrugated board	No	RER	packaging, corrugated board, mixed fibre, single wall, at plant	5.61E-04	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Mineral wool				insulation materials	production	No	CH	glass wool mat, at plant	8.00E-05	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Organic chemicals				chemicals	organics	No	GLO	chemicals organic, at plant	1.01E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	7.92E+00	kWh	Werner et al. 1997	1	122	(13,4,3,1,12)
				air	unspecified			Heat, waste	2.85E+01	MJ	calculated	1	122	(13,4,3,1,13)
Transport lorry 40t				transport systems	road	No	CH	transport, lorry >28t, fleet average	2.00E-01	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Transport lorry 16t	Inner door, wood-glass frame, installation	Aromatic hydrocarbons to air Heptane to air Municipal waste to incineration		transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	1.25E-02	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
Transport rail				transport systems	train	No	RER	transport, freight, rail	8.75E-03	tkm	Werner et al. 1997	1	2.05	(13,4,3,1,15)
			Not inventoried since today water based coatings are used						2.44E-02		Werner et al. 1997			(13,4,3,1,123)
									1.19E-01		Werner et al. 1997			(13,4,3,1,123)
				waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9%water, to municipal incineration	7.06E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,16)
Screws				metals	extraction	No	RER	reinforcing steel, at plant	8.75E-02	kg	Werner et al. 1997	1	122	(13,4,3,1,13)
Galvanising screws				metals	processing	No	RER	zinc coating, pieces	8.13E-03	m2	Werner et al. 1997	1	122	(13,4,3,1,13)
Dowel (hardwood)			Density = 715 kg/m3	wooden materials	extraction	No	RER	sawn timber, hardwood, planed, kiln dried, u=10% at plant	5.24E-06	m3	various Empa Studies	1	122	(13,4,3,1,13)
Lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	1.10E-03	kg	Werner et al. 1997	1	122	(13,4,3,1,14)
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.88E-03	kWh	Werner et al. 1997	1	122	(13,4,3,1,12)
		Heat, waste		air	unspecified			Heat, waste	1.04E-02	MJ	calculated	1	122	(13,4,3,1,13)

Fig. 5.18 Flows for “Door, inner, glass-wood, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty					
Input		Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
Glass	→	Inner door, wood-glass, door leaf. 62 % of wooden door leaf + 0.38 m2 2-IV glazing, part 1			building components	windows	No	RER	glazing, double (2-IV), U<1.1 W/m2K, laminated safety glass, at plant	3.80E-01	m2	Calculated	1	122	(13,13,15,4)	
Particle board	→				wooden materials	extraction	No	RER	particle board, indoor use, at plant	1.12E-02	m3	calculated, based on Werner et al. 1997	1	122	(13,4,3,113)	
Fibre board	→				wooden materials	extraction	No	RER	fibreboard hard, at plant	4.26E-03	m3	calculated, based on Werner et al. 1997	1	122	(13,4,3,113)	
Softwood	→				wooden materials	extraction	No	RER	sawn timber, softwood, planed, kiln dried, at plant	5.04E-03	m3	calculated, based on Werner et al. 1997	1	122	(13,4,3,113)	
PVAc glue	→				plastics	monomers	No	RER	vinyl acetate, at plant	132E-01	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,113)	
Paper	→				paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	105E-01	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,114)	
M elamine formaldehyde resin	→				paintings	production	No	RER	melamine formaldehyde resin, at plant	6.98E-02	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,113)	
Urea formaldehyde resin	→				paintings	production	No	RER	urea formaldehyde resin, at plant	143E-01	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,113)	
Ammonia	→				chemicals	inorganics	No	RER	ammonia, liquid, at regional storehouse	4.65E-03	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,114)	
HCl	→				chemicals	inorganics	No	RER	hydrochloric acid, 30%in H2O, at plant	9.69E-03	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,114)	
wheat flour	→				agricultural means of production	feed	No	CH	wheat IP, at feed mill	3.53E-02	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,114)	
Steel for fittings	→				total amount of wooden door leaf	metals	extraction	No	RER	reinforcing steel, at plant	4.50E-01	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,113)
Galvanising fittings	→				total amount of wooden door leaf	metals	processing	No	RER	zinc coating, pieces	3.94E-02	m2	calculated, based on Werner et al. 1997	1	122	(13,4,3,113)

Fig. 5.18 Flows for “Door, inner, glass-wood, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Polyamide	Inner door, wood-glass, door leaf. 62 % of wooden door leaf + 0.38 m2 2-IV glazing, part 2		total amount of wooden door leaf	plastics	polymers	No	RER	nylon 66, at plant	2.00E-03	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,13)
Aluminium			total amount of wooden door leaf	metals	extraction	No	RER	aluminium, production mix, at plant	150E-01	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,13)
Anodising aluminium			total amount of wooden door leaf	metals	processing	No	RER	anodising, aluminium sheet	138E-02	m2	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,13)
Alkyd varnish				paintings	production	No	RER	alkyd paint, white, 60%in H2O, at plant	5.23E-01	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,13)
Organic chemicals				chemicals	organics	No	GLO	chemicals organic, at plant	160E-03	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,13)
Tin can				metals	extraction	No	RER	steel, low-alloyed, at plant	2.40E-02	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,14)
Mineral wool				insulation materials	production	No	CH	glass wool mat, at plant	128E-04	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,14)
Packaging (cardboard)			total amount of wooden door leaf since packaging is the same	paper & cardboard	cardboard & corrugated board	No	RER	packaging, corrugated board, mixed fibre, single wall, at plant	2.72E-03	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,14)
Polyethylene			total amount of wooden door leaf since packaging is the same	plastics	polymers	No	RER	polyethylene, LDPE, granulate, at plant	6.25E-01	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,14)
Internal transports			total amount of wooden door leaf	construction processes	machinery	No	GLO	diesel, burned in building machine	2.47E+00	MJ	calculated, based on Werner et al. 1997	1	2.05	(13,4,3,1,15)
Lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	8.31E-03	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,14)
Electricity			total amount of wooden door leaf	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	4.08E+00	kWh	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,12)
		Heat, waste	total amount of wooden door leaf	air	unspecified			Heat, waste	147E+01	MJ	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,13)

Fig. 5.18 Flows for “Door, inner, glass-wood, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000								Uncertainty						
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
Heat (gas)	→	Inner door, wood-glass, door leaf. 62 % of wooden door leaf + 0.38 m2 2-IV glazing, part 3				total amount of wooden door leaf	natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	3.96E-02	MJ	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,1,1)	
Heat (residual wood)	→					Biogenic CO2 emission = 2.62 kg --> 0.715 kg C in residual wood --> 144 kg residual wood (atro) --> 3.21E-3 m3 = 27.4 MJ (8545 MJ/m3 at 10%water content)	wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 1000kW	1.06E+01	MJ	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,1,1)	
Correction of wood input	→					0.000316 m3 / KJ	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40% at plant	3.36E-03	m3	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,1,1)	
Correction of transport of wood chips	→					0.00283 tkm/MJ	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	3.01E-02	tkm	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,1,1)	
				→	Industrial residual wood, hardwood		not used for heat production.						1.36E-05	m3	calculated, based on Werner et al. 1997			(13,4,3,1,1,1)
Allocation correction	→					This flow subtracts the biomass energy, the biogenic CO2 and the resource stored in the industrial residue wood.	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	-1.36E-05	m3	calculated, based on Werner et al. 1997	1	122	(13,4,3,1,1,1)	
				→	Industrial residual wood, hardwood		not used for heat production.						1.52E-05	m3	calculated, based on Werner et al. 1997			(13,4,3,1,1,1)

Fig. 5.18 Flows for “Door, inner, glass-wood, at plant” and their representation in the ecoinvent database (cont)

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Allocation correction	Inner door, wood-glass, door leaf. 62 % of wooden door leaf + 0.38 m2 2-IV glazing, part 4		This flow subtracts the biomass energy, the biogenic CO2 and the resource stored in the industrial residue wood.	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-1.52E-05	m3	calculated, based on Werner et al. 1997	1	122	(13,4,3,11,1)
Transport lorry 40t			600 km assumed for glazing	transport systems	road	No	CH	transport, lorry >28t, fleet average	8.36E+00	tkm	calculated, based on Werner et al. 1997 and assumption	1	2.05	(13,4,3,11,5)
Transport lorry 16t			100 km assumed for glazing	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	9.18E-01	tkm	calculated, based on Werner et al. 1997 and assumption	1	2.05	(13,4,3,11,5)
		Municipal waste to incineration	total amount of wooden door leaf since waste from packaging is the same	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9%water, to municipal incineration	2.44E-02	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,11,6)
		PE to incineration	total amount of wooden door leaf since waste from packaging is the same	waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	6.25E-01	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,11,6)
		waste to sanitary landfill	total amount of wooden door leaf since waste from packaging is the same	waste management	sanitary landfill	No	CH	disposal, inert material, 0% water, to sanitary landfill	6.06E-03	kg	calculated, based on Werner et al. 1997	1	122	(13,4,3,11,6)
Infrastructure			Metal working plant as proxy: Output 43675 t/a --> 4.57E-10 unit/kg product (50a)	metals	general manufacturing	Yes	RER	metal working factory	1.26E-08	unit	Calculated	1	3.06	(2,4,13,15,9)
		Inner door, glass wood	27.5 kg/m2	building components	doors	No	RER	door, inner, glass-wood, at plant	1.00E+00	m2				

Fig. 5.18 Flows for “Door, inner, glass-wood, at plant” and their representation in the ecoinvent database (cont)

5.19 Disposal, door, outer, wood-aluminium, in municipal incineration

The meta information of “Disposal, door, outer, wood-aluminium, in municipal incineration” (Tab. 5.19) includes all important and necessary information to use the data correctly.

Tab. 5.19 ecoinvent meta information of the product “Disposal, door, outer, wood-aluminium, in municipal incineration”

Name	Disposal, door, outer, wood-aluminium, in municipal incineration
Location	RER
Infrastructure Process	0
Unit	m ²
Data Set Version	2.0
Included Processes	The dataset describes the disposal of a wood-aluminium outer door in a municipal incineration. The door frame out of steel is recycled at 100%. The rest of the materials (rest of steel parts, aluminium, rubber, plastics, mortar, paints and wood) is disposed in a municipal solid waste incineration.
Amount	1
Local Name	Entsorgung, Aussentüre, Holz-Aluminium, in Kehrrechtverbrennungsanlage
Synonyms	
General Comment to reference function	The dataset can be applied for a standard outer door (wood-aluminium) disposal process in Switzerland.
Start Date	2005
End Date	2005
Data Valid For Entire Period	1
Other Period Text	
Geography text	Dataset represents standard in Europe and/or Switzerland.
Technology text	The technology in this dataset represents standard incineration technology in Europe.
Representativeness [%]	
Production Volume	
Sampling Procedure	Calculation from waste composition
Extrapolations	see Geography
Uncertainty Adjustments	none

Fig. 5.19 shows all in- and output flows of the product “Disposal, door, outer, wood-aluminium, in municipal incineration”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000								Uncertainty			
Input		Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Transportation to incineration Disposal, outer door, wood-aluminium, in MSWI	→	Steel frame	Steel to recycling --> cut off							105E+01	kg	Calculated from composition	1	122	(13,13,15,6)
	→	Steel (fittings, frame, leaf and installation)		waste management	municipal incineration	No	CH	disposal, steel, 0%water, to municipal incineration	4.75E-01	kg	Werner et al. 1998	1	122	(13,4,3,116)	
	→	Aluminium parts	sheets not recycled because glued	waste management	municipal incineration	No	CH	disposal, aluminium, 0% water, to municipal incineration	2.85E+00	kg	Werner et al. 1998	1	122	(13,4,3,116)	
	→	Wood		waste management	municipal incineration	No	CH	disposal, wood untreated, 20%water, to municipal incineration	2.43E+01	kg	Werner et al. 1997	1	122	(13,4,3,116)	
	→	EPDM		waste management	municipal incineration	No	CH	disposal, rubber, unspecified, 0%water, to municipal incineration	3.13E-01	kg	Werner et al. 1999	1	122	(13,4,3,116)	
	→	mortar		waste management	building demolition	No	CH	disposal, building, mineral plaster, to final disposal	5.00E+01	kg	Werner et al. 1999	1	122	(13,4,3,116)	
	→	Plastics		waste management	municipal incineration	No	CH	disposal, plastics, mixture, 15.3%water, to municipal incineration	1.15E-01	kg	Werner et al. 1999	1	122	(13,4,3,116)	
	→	Acrylic filler and paint, alkyde paint		waste management	municipal incineration	No	CH	disposal, emulsion paint, 0% water, to municipal incineration	2.63E-01	kg	Werner et al. 2001	1	122	(13,4,3,116)	
			50 km for materials to disposal. No transport if material is recycled.	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	3.92E+00	tkm	Own calculations	1	2.05	(13,13,15,5)	
	→	Disposal outer door, wood-aluminium		waste management	building demolition	No	RER	disposal, door, outer, wood-aluminium, in municipal incineration	100E+00	m2					

Fig. 5.19 Flows for “Disposal, door, outer, wood-aluminium, in municipal incineration” and their representation in the ecoinvent database

5.20 Disposal, door, outer, wood-glass, in municipal incineration

The meta information of “Disposal, door, outer, wood-glass, in municipal incineration” (Tab. 5.20) includes all important and necessary information to use the data correctly.

Tab. 5.20 ecoinvent meta information of the product “Disposal, door, outer, wood-glass, in municipal incineration”

Name	Disposal, door, outer, wood-glass, in municipal incineration
Location	RER
Infrastructure Process	0
Unit	m ²
Data Set Version	2.0
Included Processes	The dataset describes the disposal of a wood-glass outer door in a municipal incineration. The door frame out of steel is recycled at 100%. The rest of the materials (rest of steel parts, aluminium, rubber, plastics, mortar, paints and wood) is disposed in a municipal solid waste incineration. Glass also goes into a MSWI, but can be recovered there.
Amount	1
Local Name	Entsorgung, Aussentüre, Holz-Glas, in Kehrichtverbrennungsanlage
Synonyms	
General Comment to reference function	The dataset can be applied for a standard outer door (wood-glass) disposal process in Switzerland.
Start Date	2005
End Date	2005
Data Valid For Entire Period	1
Other Period Text	
Geography text	Dataset represents standard in Europe and/or Switzerland.
Technology text	The technology in this dataset represents standard incineration technology in Europe.
Representativeness [%]	
Production Volume	
Sampling Procedure	Calculation from waste composition
Extrapolations	see Geography
Uncertainty Adjustments	none

Fig. 5.20 shows all in- and output flows of the product “Disposal, door, outer, wood-glass, in municipal incineration”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Transportation to incineration Disposal, outer door, wood-glass, in MSWI		→ Steel frame	Steel to recycling --> cut off						105E+01	kg	Calculated from composition	1	122	(13,13,15,6)
		→ Steel (fittings, frame, leaf and installation)		waste management	municipal incineration	No	CH	disposal, steel, 0%water, to municipal incineration	4.75E-01	kg	Werner et al. 1998	1	122	(13,4,3,11,6)
		→ Aluminium parts	sheets not recycled because glued	waste management	municipal incineration	No	CH	disposal, aluminium, 0% water, to municipal incineration	182E+00	kg	Werner et al. 1998	1	122	(13,4,3,11,6)
		→ Wood		waste management	municipal incineration	No	CH	disposal, wood untreated, 20%water, to municipal incineration	151E+01	kg	Werner et al. 1997	1	122	(13,4,3,11,6)
		→ Glazing	211kg/m2	waste management	building demolition	No	RER	disposal, glazing 2-IV, U<1.1W/m2K, LSG, in municipal incineration	3.80E-01	m2	Calculated	1	122	(13,4,3,11,6)
		→ EPDM		waste management	municipal incineration	No	CH	disposal, rubber, unspecified, 0%water, to municipal incineration	3.13E-01	kg	Werner et al. 1999	1	122	(13,4,3,11,6)
		→ mortar		waste management	building demolition	No	CH	disposal, building, mineral plaster, to final disposal	5.00E+01	kg	Werner et al. 1999	1	122	(13,4,3,11,6)
		→ Plastics		waste management	municipal incineration	No	CH	disposal, plastics, mixture, 15.3%water, to municipal incineration	7.18E-02	kg	Werner et al. 1999	1	122	(13,4,3,11,6)
		→ Acrylic filler and paint, alkyde paint		waste management	municipal incineration	No	CH	disposal, emulsion paint, 0% water, to municipal incineration	2.63E-01	kg	Werner et al. 2001	1	122	(13,4,3,11,6)
			50 km for materials to disposal. No transport if material is recycled.	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	3.42E+00	tkm	Own calculations	1	2.05	(13,13,15,5)
		→ Disposal, outer door, wood-glass		waste management	building demolition	No	RER	disposal, door, outer, wood-glass, in municipal incineration	100E+00	m2				

Fig. 5.20 Flows for “Disposal, door, outer, wood-glass, in municipal incineration” and their representation in the ecoinvent database

5.21 Disposal, door, inner, wood, in municipal incineration

The meta information of “Disposal, door, inner, wood, in municipal incineration” (Tab. 5.21) includes all important and necessary information to use the data correctly.

Tab. 5.21 ecoinvent meta information of the product “Disposal, door, inner, wood, in municipal incineration”

Name	Disposal, door, inner, wood, in municipal incineration
Location	RER
Infrastructure Process	0
Unit	m ²
Data Set Version	2.0
Included Processes	The dataset describes the disposal of a wood-aluminium inner door in a municipal incineration. All the materials (steel parts, aluminium, rubber, plastics, paints and wood) are disposed in a municipal solid waste incineration.
Amount	1
Local Name	Entsorgung, Innentüre, Holz, in Kehrrechtverbrennungsanlage
Synonyms	
General Comment to reference function	The dataset can be applied for a standard inner door (wood) disposal process in Switzerland.
Start Date	2005
End Date	2005
Data Valid For Entire Period	1
Other Period Text	
Geography text	Dataset represents standard in Europe and/or Switzerland.
Technology text	The technology in this dataset represents standard incineration technology in Europe.
Representativeness [%]	
Production Volume	
Sampling Procedure	Calculation from waste composition
Extrapolations	see Geography
Uncertainty Adjustments	none

Fig. 5.21 shows all in- and output flows of the product “Disposal, door, inner, wood, in municipal incineration”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Transportation to incineration → Disposal, inner door, wood, in MSWI	→	Wood		waste management	municipal incineration	No	CH	disposal, wood untreated, 20%water, to municipal incineration	7.57E+00	kg	Werner et al. 1997	1	122	(13,4,3,1,16)
	→	Particle board / fibre board		waste management	building demolition	No	CH	disposal, building, fibre board, to final disposal	174E+01	kg	Werner et al. 1998	1	122	(13,4,3,1,16)
	→	Steel (fittings, frame, leaf and installation)		waste management	municipal incineration	No	CH	disposal, steel, 0%water, to municipal incineration	4.78E-01	kg	Werner et al. 1998	1	122	(13,4,3,1,16)
	→	Aluminium parts		waste management	municipal incineration	No	CH	disposal, aluminium, 0% water, to municipal incineration	153E-01	kg	Werner et al. 1998	1	122	(13,4,3,1,16)
	→	Plastics		waste management	municipal incineration	No	CH	disposal, plastics, mixture, 15.3%water, to municipal incineration	7.47E-01	kg	Werner et al. 1999	1	122	(13,4,3,1,16)
	→	EPDM		waste management	municipal incineration	No	CH	disposal, rubber, unspecified, 0%water, to municipal incineration	143E-01	kg	Werner et al. 1999	1	122	(13,4,3,1,16)
	→	Acrylic filler and paint, alkyde paint		waste management	municipal incineration	No	CH	disposal, emulsion paint, 0% water, to municipal incineration	112E+00	kg	Werner et al. 2001	1	122	(13,4,3,1,16)
	→	Disposal inner door, wood	50 km for materials to disposal. No transport if material is recycled.	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	138E+00	tkm	Calculated	1	2.05	(13,13,15,5)
				waste management	building demolition	No	RER	disposal, door, inner, wood, in municipal incineration	100E+00	m2				

Fig. 5.21 Flows for “Disposal, door, inner, wood, in municipal incineration” and their representation in the ecoinvent database

5.22 Disposal, door, inner, glass-wood, in municipal incineration

The meta information of “Disposal, door, inner, glass-wood, in municipal incineration” (Tab. 5.22) includes all important and necessary information to use the data correctly.

Tab. 5.22 ecoinvent meta information of the product “Disposal, door, inner, glass-wood, in municipal incineration”

Name	Disposal, door, inner, glass-wood, in municipal incineration
Location	RER
Infrastructure Process	0
Unit	m ²
Data Set Version	2.0
Included Processes	The dataset describes the disposal of a glass-wood inner door in a municipal incineration. All the materials (steel parts, aluminium, rubber, plastics, paints and wood) are disposed in a municipal solid waste incineration. Glass also goes into a MSWI, but can be recovered there.
Amount	1
Local Name	Entsorgung, Innentüre, Glas-Holz, in Kehrlichtverbrennungsanlage
Synonyms	
General Comment to reference function	The dataset can be applied for a standard inner door (wood-glass) disposal process in Switzerland.
Start Date	2005
End Date	2005
Data Valid For Entire Period	1
Other Period Text	
Geography text	Dataset represents standard in Europe and/or Switzerland.
Technology text	The technology in this dataset represents standard incineration technology in Europe.
Representativeness [%]	
Production Volume	
Sampling Procedure	Calculation from waste composition
Extrapolations	see Geography
Uncertainty Adjustments	none

Fig. 5.22 shows all in- and output flows of the product “Disposal, door, inner, glass-wood, in municipal incineration”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000								Uncertainty			
Input		Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Transportation to incineration →	Disposal, inner door, glass-wood, in MSWI	→	Wood	211kg/m2	waste management	municipal incineration	No	CH	disposal, wood untreated, 20%water, to municipal incineration	7.02E+00	kg	Werner et al. 1997	1	122	(13,4,3,116)
		→	Particle board / fibre board		waste management	building demolition	No	CH	disposal, building, fibre board, to final disposal	108E+01	kg	Werner et al. 1998	1	122	(13,4,3,116)
		→	Glazing		waste management	building demolition	No	RER	disposal, glazing 2-IV, U<1.1W/m2K, LSG, in municipal incineration	3.80E-01	m2	Calculated	1	122	(13,4,3,116)
		→	Steel (fittings, frame, leaf and installation)		waste management	municipal incineration	No	CH	disposal, steel, 0%water, to municipal incineration	3.24E-01	kg	Werner et al. 1998	1	122	(13,4,3,116)
		→	Aluminium parts		waste management	municipal incineration	No	CH	disposal, aluminium, 0% water, to municipal incineration	9.46E-02	kg	Werner et al. 1998	1	122	(13,4,3,116)
		→	Plastics		waste management	municipal incineration	No	CH	disposal, plastics, mixture, 15.3%water, to municipal incineration	4.64E-01	kg	Werner et al. 1999	1	122	(13,4,3,116)
		→	EPDM		waste management	municipal incineration	No	CH	disposal, rubber, unspecified, 0%water, to municipal incineration	143E-01	kg	Werner et al. 1999	1	122	(13,4,3,116)
		→	Acrylic filler and paint, alkyde paint		waste management	municipal incineration	No	CH	disposal, emulsion paint, 0% water, to municipal incineration	7.98E-01	kg	Werner et al. 2001	1	122	(13,4,3,116)
		→	Disposal inner door, glass-wood		50 km for materials to disposal. No transport if material is recycled.	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	2.00E+01	tkm	Calculated	1	2.05
				waste management	building demolition	No	RER	disposal, door, inner, glass-wood, in municipal incineration	100E+00	m2					

Fig. 5.22 Flows for “Disposal, door, inner, glass-wood, in municipal incineration” and their representation in the ecoinvent database

5.23 Polysulphide, sealing compound, at plant

The meta information of “Polysulphide, sealing compound, at plant” (Tab. 5.23) includes all important and necessary information to use the data correctly.

Tab. 5.23 ecoinvent meta information of the product “Polysulphide, sealing compound, at plant”

Name	Polysulphide, sealing compound, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	The dataset describes the raw materials and processes required to produce a polysulphide sealing compound. Included is heat , electricity, and the infrastructure, but no extra transport activities apart from the ones related to the raw materials. Input materials are crushed limestone (filling material), chemicals, a hardening paste and rigid foam (polyurethane).
Amount	1
Local Name	Polysulfid, Dichtungsmasse, ab Werk
Synonyms	
General Comment to reference function	This dataset is primarily used in the glazing production.
Start Date	1996
End Date	2003
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data based on a number of confidential reports.
Technology text	Polysulphide sealing compound is produced by heating a mixture of filling material, chemicals, polyurethane and a hardening paste.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Company documentation
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that this dataset is representative for the whole Europe is taken into account in the uncertainties.

Fig. 5.23 shows all in- and output flows of the product “Polysulphide, sealing compound, at plant”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000								Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment	
Limestone	Polysulphide, sealing compound		Limestone, crushed	construction materials	others	No	CH	limestone, milled, packed, at plant	4.70E-01	kg	Richter 1996	1	122	(1,3,13,15,4)	
Chemicals			including polysulfide-polymer	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	3.70E-01	kg	Richter 1996	1	122	(1,3,13,15,4)	
Hardening paste			contains M nO --> M n as proxy	metals	extraction	No	RER	manganese, at regional storage	150E-02	kg	Richter 1996	1	122	(1,3,13,15,3)	
Heat, oil			Light oil	oil	heating systems	No	RER	heat, light fuel oil, at industrial furnace 1MW	3.63E-01	MJ	Richter 1996	1	122	(1,3,13,15,11)	
Polyurethane			Amount corrected to balance in- and outputs. Richter 1996 gives 0.137 kg	plastics	polymers	No	RER	polyurethane, rigid foam, at plant	145E-01	kg	calculated	1	122	(1,3,13,15,3)	
Electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	164E-01	kWh	Richter 1996	1	122	(1,3,13,15,2)	
Infrastructure			Chemical plant as proxy	chemicals	organics	Yes	RER	chemical plant, organics	4.00E-10	unit	-	1	3.06	(2,4,13,15,9)	
				air	unspecified			Heat, waste	5.89E-01	MJ	Calculation	1	126	(3,4,13,15,13)	
		Heat waste Sealing compound, polysulphide		plastics	others	No	RER	polysulphide, sealing compound, at plant	100E+00	kg					

Fig. 5.23 Flows for “Polysulphide, sealing compound, at plant” and their representation in the ecoinvent database

5.24 Natural rubber based sealing (caoutchouc)

The meta information of “Sealing compound, rubber-based (caoutchouc)” (Tab. 5.24) includes all important and necessary information to use the data correctly.

Tab. 5.24 ecoinvent meta information of the product “natural rubber based sealing, at plant”

Name	Sealing compound, rubber-based (caoutchouc)
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	1.0
Included Processes	The dataset describes the production of a rubber-based sealing compound to be used e.g. in the aluminium cladding production.
Amount	1
Local Name	Dichtmasse auf Kautschukbasis
Synonyms	
General Comment to reference function	Caoutchouc is approximated by the dataset "Chemicals, organic". Included are white spirit and the required transportation by rail and road, and hydrocarbon emissions to air. This dataset does not include any energy consumption during the production phase.
Start Date	1996
End Date	2003
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data retrieved from a safety data sheet by a German producer. Applicable for Europe.
Technology text	Simple mass balance including transport. No process energy included.
Representativeness [%]	
Production Volume	
Sampling Procedure	Company information
Extrapolations	see Geography
Uncertainty Adjustments	Representative for Germany.

Fig. 5.24 shows all in- and output flows of the product “Sealing compound, rubber-based (caoutchouc)”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Chemicals, organic	Sealing compound, rubber-based		Organic chemicals as proxy for caoutchouc	chemicals	organics	No	GLO	chemicals organic, at plant	7.44E-01	kg	Personal communication 1997, Teroson AG	1	127	(2,4,3,2,15,3)
White spirit				paintings	production	No	RER	white spirit, at plant	4.86E-01	kg		1	127	(2,4,3,2,15,3)
Transport 32 t lorry			100 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	100E-01	tkm	Based on ecoinvent guidelines	1	2.11	(4,4,1,11,5,5)
Transport rail			600 km	transport systems	train	No	RER	transport, freight, rail	6.00E-01	tkm		1	2.11	(4,4,1,11,5,5)
Infrastructure				Chemical plant as proxy	chemicals	organics	Yes	RER	chemical plant, organics	4.00E-10	unit	-	1	3.06
		Sealing compound rubber-based (caoutchouc)	Sealing compound for window frame; 'chemicals, organic' as proxy for caoutchouc; balanced without energy	plastics	others	No	DE	natural rubber based sealing, at plant	100E+00	kg				
		Hydrocarbons		air	unspecified			NM VOC, non-methane volatile organic compounds, unspecified origin	2.30E-01	kg	Personal communication 1997, Teroson AG (GER)	1	2.08	(2,4,3,11,5,23)

Fig. 5.24 Flows for “Sealing compound, rubber-based (caoutchouc)” and their representation in the ecoinvent database

5.25 Metal adhesive compound

The meta information of “adhesive for metals” (Tab. 5.25) includes all important and necessary information to use the data correctly.

Tab. 5.25 ecoinvent meta information of the product “Metal adhesive compound”

Name	Metal adhesive compound
Location	DE
Infrastructure Process	0
Unit	kg
Data Set Version	1.0
Included Processes	The dataset describes the production of a metal adhesive compound used e.g. in the aluminium window frame production.
Amount	1
Local Name	Metallkleber
Synonyms	1
General Comment to reference function	Basic materials applied in the production are organic chemicals, liquid epoxy resin, hydrated lime and silica sand. This dataset does not include any energy consumption in the production phase. Road and rail transport are included.
Start Date	1996
End Date	2003
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data retrieved from a safety data sheet by a German producer. Applicable for Europe.
Technology text	Simple mass balance. No process energy included.
Representativeness [%]	
Production Volume	
Sampling Procedure	Company information
Extrapolations	see Geography
Uncertainty Adjustments	None

Fig. 5.25 shows all in- and output flows of the product “adhesive for metals”, the allocation to the datasets in ecoinvent database V2.0, the sources and the standard deviation of all in- and output flows.

General Flow information				Representation in ecoinvent 2000								Uncertainty					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Chemicals, organic	→	Metal adhesive compound					chemicals	organics	No	GLO	chemicals organic, at plant	4.11E-02	kg	Personal communication Mr. Geissler, 1997 safety data sheet by Agomet Klebstoffe GmbH (D); Data scaled to	1	127	(2,4,3,2,15,3)
Epoxy resin	→				Basis Bisphenol A-diglycidylether	plastics	monomers	No	RER	epoxy resin, liquid, at plant	6.16E-01	kg	1		127	(2,4,3,2,15,3)	
Lime Ca(OH)2	→					construction materials	binder	No	CH	lime, hydrated, packed, at plant	2.74E-01	kg	1		127	(2,4,3,3,15,3)	
Silica sand SiO2	→					construction materials	additives	No	DE	silica sand, at plant	6.85E-02	kg	1	127	(2,4,3,3,15,12)		
Transport 32 t lorry	→				100 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	100E-01	tkm	Based on ecoinvent guidelines	1	2.11	(4,4,1,1,15,5)	
Transport rail	→				600 km	transport systems	train	No	RER	transport, freight, rail	6.00E-01	tkm	Based on ecoinvent guidelines	1	2.11	(4,4,1,1,15,5)	
Infrastructure	→				Chemical plant as proxy	chemicals	organics	Yes	RER	chemical plant, organics	4.00E-10	unit	-	1	3.06	(2,4,13,15,9)	
				→	Metal adhesive compound			paintings	production	No	DE	adhesive for metals, at plant	100E+00	kg			

Fig. 5.25 Flows for “Metal adhesive compound” and their representation in the ecoinvent database

6 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007b)). For detailed results compare the different flows and the ecoinvent database meta information.

The data quality has to be looked at from the point of view of each individual dataset:

- The different glazing datasets are of rather good quality. The data origins from information of private companies. Hardly any proxies have been used in the three glazing datasets. The same applies to the disposal (glazing) datasets. Though it has to be considered that the glazing materials are partly being disposed of via municipal solid waste incineration (MSWI). This is not always the case.
- The window frame datasets are of good quality. The list of the input materials (metals/plastics/chemicals) is very detailed and accurate. The electricity consumption is derived from confidential data by private companies. The disposal data for the window frames are as accurate as possible, taking into account that the standard disposal scenario is a disposal via MSWI. Only big metal pieces and parts (aluminium or steel) are assumed to be recycled at 100%. Smaller metal pieces are assumed to be incinerated, as they are often glued to other materials or too small to be (cost-) efficiently recycled. This might not represent the real situation everywhere.
- The information for the aluminium cladding dataset is again very detailed and represents an existing cladding of a school-building in Switzerland (personal communication Metall-Bau-Technik Company). This dataset is of good quality.
- The inner and outer door datasets represent simply structured door types (wood, wood-metal, and wood-glass). The data are mainly based on an EMPA study carried out by Werner et al. in 1997. The datasets are very detailed and of good quality. Some adaptations were made to account for important developments in the past 10 years. The average distance of transport for all components is calculated via an assumed standard transport distance (50 km). The disposal datasets of the different doors were established similar to the window frame disposal datasets.

7 Literature

- | | |
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